Threshold Primitives and Proof Techniques for Module Lattices

By

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Лізі Франціївні.

To my grandmother Lisa.

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Abstract

This thesis explores a number of topics in lattice-based cryptography ranging from proof techniques to primitive design. In the first part of the thesis, we present a modular unifying framework for proving Leftover Hash Lemmas (LHLs) in cyclotomic rings. We showcase the power of the framework by proving new statements and covering mathematical settings not considered before. We further prove LHLs in the presence of leakage and provide concrete bounds for wide families of distributions and leakage functions. We then prove a first "Gaussian" Leftover Hash Lemma over rings with a sublinear dependency of the ring degree. To do that we give a tighter analysis of when a discrete Gaussian matrix over rings is surjective with high probability, bound the shortest basis norm of its kernel and establish when the resulting distribution is not affected by the geometry of the matrix. Using these techniques, we prove the hardness of k-MSIS and k-MLWE assumptions, which were previously used without proof.

In the second part of this thesis, we build a range of lattice-based threshold primitives. First, we use k-MSIS and k-MLWE to design a non-interactive trapdoor sharing mechanism. It allows for thresholdising a broad range of trapdoor-based cryptographic primitives which we exemplify with a threshold signature and a threshold identity based encryption schemes. Even though the public parameters of our construction still grow as T^2 (where T is a signing, resp. decryption threshold), it allows for a 1-round, simple algebraic signing (resp. decryption) procedure. In the last two chapters, we present a family of CCA-secure threshold key encapsulation mechanisms (TKEMs). Both constructions are built on a transform we dub BCHK+ combining a threshold identity based encryption (TIBE) scheme, a one-time signature and a number of random oracles. We build a TIBE from scratch achieving a three-round protocol, with parameter sizes only logarithmically dependent on T. In the second construction we significantly reduce the concrete public key and ciphertext sizes with a simpler TIBE design achieving the first practical CCA-secure lattice-based TKEM.

Declaration

These doctoral studies were conducted under the supervision of Dr. Rachel Player and Prof. Martin R. Albrecht.

The work presented in this thesis is the result of original research carried out by myself, in collaboration with others, whilst enrolled in the Information Security Group as a candidate for the degree of Doctor of Philosophy. This work has not been submitted for any other degree or award in any other university or educational establishment.

Oleksandra Lapiha
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Chapter 1

Introduction

This thesis investigates a number of question in the area of lattice-based cryptography. We study the properties of lattices as mathematical objects and use them for building secure cryptographic primitives such as threshold signatures and encryption schemes.

Even though working with lattices simply satisfies our scientific curiosity, the primitives constructed from lattice problems are also valued in the wider cryptographic community. This is because, unlike the discrete log and factoring problems, they are conjectured secure even against adversaries with quantum capabilities. For example, the NIST post-quantum standardisation competition run between 2016 and 2025 [NIS25] has chosen to standardise key encapsulation and digital signature algorithms (ML-KEM [NIS24a], ML-DSA [NIS24b], SLH-DSA [NIS24c], FN-DSA and HQC) where three out of five are based on lattice problems.

If digital signatures and key encapsulation mechanisms (KEM) may be considered ready for standardisation, less is known in the area of more advanced post-quantum constructions. In recent years we have seen an increase in the number of new lattice-based assumptions that allow for more efficient and expressive primitives [ACL+22, WW23, BLNS23]. Generally, they extend the standard SIS and LWE problems with additional leakage on secret values. Some of the new assumptions have been reduced to standard assumptions such as k-SIS, k-LWE [BF11a, LPSS14] or Hint-MLWE [BF11a, KLSS23] and others remain to be analysed. In this thesis we benefit from extended lattice assumptions to build a Threshold Trapdoor Gadget and Threshold Key Encapsulation Mechanisms. We also make progress on analysing these assumptions by giving a reduction from a standard MSIS assumption for a ring version of k-SIS called k-MSIS.

Furthermore, in the last 20 years we have seen a development in the algebraic constructions of the field. At first [Ajt96, GPV08], lattice-based cryptography was built

using lattices generated by kernels of a random integer matrices. However, later it was discovered that matrices over cyclotomic rings of degree larger than 1 often result in more compact cryptographic schemes and faster computations, e.g. [SSTX09, LPR10, LS15]. Cyclotomic rings require new algebraic arguments to prove the same constructions secure and we build some of them in this thesis i.e. (Gaussian) Leftover Hash Lemmas.

At a high level, in this thesis we explored and made progress on the following questions

1. How do Leftover Hash Lemmas work for cyclotomics number fields? What ring and entropy conditions are sufficient? What are the effects of secret vector leakage?

2. How does the security of Module SIS change given a number of short vectors in the kernel? Can we exploit the ring structure to improve the reduction?

3. Can we secret share lattice trapdoors by giving out short preimages of given vectors? How do we set the preimages for a threshold T-of-N secret sharing?

4. Can we efficiently provide CCA security for lattice-based threshold KEMs? What minimal guarantees do we need on the integrity of the ciphertexts?

1.1 Thesis Overview

We discuss the structure of each chapter in more detail.

Background. In Chapter 2 we discuss the fundamentals of Algebra, Probability, Lattices and the results from the literature that we need in this thesis.

LHL Survey. In Chapter 3 we work with a cyclotomic ring $\mathcal{R}$ of degree $d > 1$ and an integer modulus q, defining the quotient ring $\mathcal{R}_q \coloneqq \mathcal{R}/q\mathcal{R}$. Given a vector $\mathbf{x}$ from some input distribution χ over $\mathcal{R}^m$ (or $\mathcal{R}_q^m$) with guaranteed "good" properties, we map it to a nearly uniform vector over $\mathcal{R}_q^n$, for $n \leq m$ by multiplying $\mathbf{x}$ from the left with a wide uniform matrix $\mathbf{A}$ from $\mathcal{R}_q^{n \times m}$. In other words, our goal is to show that $\mathbf{A}\mathbf{x} \bmod q$ (given $\mathbf{A}$) is "close" to a uniformly sampled vector $\mathbf{u}$. Formally,

$$\Big\{(\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m}), \mathbf{A} \cdot \mathbf{x} \bmod q) \mid \mathbf{x} \leftarrow \chi^m\Big\} \sim \Big\{(\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m}), \mathbf{u} \leftarrow \mathcal{U}(\mathcal{R}_q^n))\Big\}.$$

This type of statement is commonly called the Leftover Hash Lemma (LHL). Most commonly, the closeness of distributions is measured using the statistical distance, but in the context of lattice-based cryptography, we also use Rényi divergence. The

LHL over cyclotomic rings has been widely studied and used over the past 15 years [Mic07, SSTX09, SS11].

In Chapter 3 we group and generalise versions of this statement available in the literature with a range of conditions on matrix $\mathbf{A}$ and vector $\mathbf{x}$. We also systematise and generalise previous approaches to leakage on the sampled vector $\mathbf{x}$. The published version of this chapter is available as

> "Leftover Hash Lemma(s) Over Cyclotomic Rings", by Katharina Boudgoust and Oleksandra Lapiha, EUROCRYPT 2026

The author of this thesis contributed to all sections of this chapter leading for Section 3.2 (excluding Section 3.2.4) and Section 3.3. The output of this paper is based on extensive joint discussions.

GLHL and the k-MSIS reduction. Then in Chapter 4 we consider a similar statement except the matrix and the vector follow a discrete Gaussian distribution and the product is not reduced modulo q. Hence, this version has been dubbed the "Gaussian" Leftover Hash Lemma in the literature. In a simplified form it states

$$\left\{(\mathbf{X} \leftarrow \mathcal{D}^{n\times m}_{\mathcal{R},\sigma_x}, \mathbf{X}\cdot\mathbf{v}) \mid \mathbf{v} \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma_v}\right\} \sim \left\{(\mathbf{X} \leftarrow \mathcal{D}^{n\times m}_{\mathcal{R},\sigma_x}, \mathbf{u} \leftarrow \mathcal{D}_{\mathcal{R}^n,\sigma_u})\right\}.$$

The algebraic setting of this chapter is more general, we consider all number fields and state the versions for cyclotomic number fields as corollaries. In addition, we apply this lemma as the central argument of MSIS to k-MSIS reduction. The published version of this chapter is available as

> "A Gaussian Leftover Hash Lemma for Modules over Number Fields", by Martin R. Albrecht, Joël Felderhoff, Russell W. F. Lai, Oleksandra Lapiha, and Ivy K. Y. Woo, EUROCRYPT 2026

The author of this thesis contributed to Section 4.2, 4.3, and 4.4 excluding Lemma 76. The proof of Lemma 77 was written by Joël Felderhoff as a result of discussions with the author. The author was also a part of discussions and edited the remaining sections of the work.

Threshold Trapdoors. In Chapter 5 we use k-MSIS to build a trapdoor sharing mechanism. Intuitively, we secret share the target image such that each share belongs to a specified subspace of the image space. Then the participants are given modified Gadget-type trapdoors that only allow them to sample preimages of a given image subspace. Lastly, we showcase our trapdoor sharing gadget by building a threshold signature and a threshold identity based encryption schemes. The published version of this chapter is available as

> "Partial Lattice Trapdoors: How to Split Lattice Trapdoors, Literally", by Martin R. Albrecht, Russell W. F. Lai, Oleksandra Lapiha, and Ivy K. Y. Woo, ASIACRYPT 2025

The author of this thesis contributed to all sections of the work excluding Theorem 12, the proof of Theorem 14 and Section 5.7. The proofs have been produced in collaboration between all authors of this work and the text has been edited by all authors.

Threshold Key Encapsulation. Lastly, Chapters 6 and 7 build two CCA secure threshold KEMs where Chapter 7 presents an improved and more efficient version of the scheme in Chapter 6. The scheme achieves CCA security using the BCHK transform [BCHK07] with an element of the Fujisaki-Okamoto transform [FO13] to provide consistency of decapsulated keys between protocol executions. The BCHK transform requires as input a passively-secure identity based encryption (IBE) with threshold key extraction and a one-time signature. We use recent advances in lattice-based threshold signatures [EEN+24, DKM+24] to build an efficient threshold IBE. The published versions of Chapter 6 and Chapter 7 are available as

> "A Simple IND-CCA Lattice-Based Threshold KEM from the BCHK+ Transform", by Oleksandra Lapiha and Thomas Prest, ASIACRYPT 2025

> "IND-CCA Lattice Threshold KEM under 30 KiB" by Katharina Boudgoust, Rafaël del Pino, Oleksandra Lapiha, and Thomas Prest, PKC 2026

The author of this thesis contributed to all sections in Chapter 6 except for Theorem 23 and Section 7.5. The text has been edited by all members of the project. In Chapter 7 the author contributed ideas for the improved construction such as replacing the matrix

extension with $H_{\mathsf{id}}(\mathsf{id})$, and switching to NTRU from ModNTRU, etc. The author has also written the security proof in Section 7.3 in collaboration with others. The ideas concerning the new secret sharing, improving the correctness analysis and other parameter optimisations are the contribution of the other authors in this project.

Other work published during the doctorate:

> "SLAP: Succinct Lattice-Based Polynomial Commitments from Standard Assumptions", by Martin R. Albrecht, Giacomo Fenzi, Oleksandra Lapiha, and Ngoc Khanh Nguyen, EUROCRYPT 2024

Chapter 2

Preliminaries

Notations. We denote matrices in uppercase boldface letters $\mathbf{A}$ and vectors in lowercase $\mathbf{x}$. We write $[\mathbf{A} \mid \mathbf{B}]$ for concatenating matrices horisontally and $[\mathbf{A} \| \mathbf{B}] \coloneqq [\mathbf{A}^\intercal \mid \mathbf{B}^\intercal]^\intercal$ for stacking them vertically. Given two vectors $\mathbf{x}, \mathbf{y}$ we denote their concatenation as $(\mathbf{x}, \mathbf{y})$. For $a, b \in \mathbb{N} \cup \{0\}$ with $a < b$, write $[a, b] \coloneqq \{a, \ldots, b-1\}$, $[a] \coloneqq [0, a]$. We denote **assert** the function that aborts the procedure and returns $\perp$ if the input condition is false. We note ln the natural logarithm and log the logarithm in base 2. The cardinality of a finite set S is denoted by $|S|$. We denote the disjoint union of two sets S and T as $S \sqcup T$. For $k \geq 1$, and matrices $\mathbf{C}_j \in \mathbb{R}^{n \times m}$, $j \in [k]$ we write $\mathsf{diag}(\{\mathbf{C}_j\}_{j \in [k]}) \in \mathbb{R}^{nk \times mk}$ to signify the block matrix with $\mathbf{C}_j$ along the main diagonal and 0 elsewhere. If the matrix on the diagonal is the same we denote it as $\mathsf{diag}(\mathbf{C}, k) \in \mathbb{R}^{nk \times mk}$ where $k \geq 1$ is the number of repetitions. We assume familiarity with Landau's asymptotic notations: $O(n)$, $o(n)$, $\omega(n)$, $\Omega(n)$, $a(n) \sim b(n)$, and Knuth's theta notation $\Theta(n)$. We use Hardy's asymptotic notation $a \asymp b$ as shorthand for $a = \Theta(b)$.

Norms. For a vector $\mathbf{x} = (x_0, \ldots, x_{n-1}) \in \mathbb{R}^n$ and an integer $p \geq 1$ we define the ℓ_p norm as $\|\mathbf{x}\|_p = (\sum_i |x_i|^p)^{1/p}$ for $p < \infty$ and $\|\mathbf{x}\|_\infty \coloneqq \max_i |x_i|$. Unless otherwise specified, we use $\|\cdot\|$ for the ℓ_2 norm. For a matrix $\mathbf{M} = (m_{ij})_{i,j} \in \mathbb{R}^{n \times m}$ with columns $\mathbf{m}_i \in \mathbb{R}^n$ we denote $\|\mathbf{M}\| = \max_i \|\mathbf{m}_i\|$ the max-column norm of $\mathbf{M}$. In particular, for any $\mathbf{x} \in \mathbb{R}^m$, it holds that $\|\mathbf{A} \cdot \mathbf{x}\| \leq \sqrt{m} \cdot \|\mathbf{A}\| \cdot \|\mathbf{x}\|$. We write $\mathfrak{s}_{\max}(\mathbf{M}) \coloneqq \max_{\mathbf{x} \neq 0} \frac{\|\mathbf{M} \cdot \mathbf{x}\|}{\|\mathbf{x}\|}$ for its spectral norm. The value $\mathfrak{s}_{\max}(\mathbf{M})$ is equal to the largest singular value of $\mathbf{M}$, it is also equal to the square root of the largest eigenvalue of $\mathbf{M}^\intercal \cdot \mathbf{M}$. We denote the smallest singular value of $\mathbf{M}$, formally defined in Section 2.1, as $\mathfrak{s}_{\min}(\mathbf{M})$. We write $\|\mathbf{M}\|_F \coloneqq \sqrt{\sum_{i,j} m_{ij}^2}$ for the Frobenius norm of the matrix. It holds that $\mathfrak{s}_{\max}(\mathbf{M}) \leq \|\mathbf{M}\|_F$. Lastly, we denote the Gram-Schmidt orthogonalisation of $\mathbf{M} \in \mathbb{R}^{n \times m}$ as matrix $\tilde{\mathbf{M}} \in \mathbb{R}^{n \times m}$ with columns $\tilde{\mathbf{m}}_i$, $i \in [m]$. Then the Gram-Schmidt (or GS) norm of a matrix $\mathbf{M} \in \mathbb{R}^{n \times m}$ is defined as

$\|\mathbf{M}\|_{GS} \coloneqq \max_i \|\tilde{\mathbf{m}}_i\|$.

Matrix square roots. A matrix $\mathbf{\Sigma} \in \mathbb{R}^{n\times n}$ is said to be positive semi-definite if it is symmetric and for every $\mathbf{x} \in \mathbb{R}^n$ it holds $\mathbf{x}^\intercal \cdot \mathbf{\Sigma} \cdot \mathbf{x} \geq 0$ and positive definite when $\mathbf{x}^\intercal \cdot \mathbf{\Sigma} \cdot \mathbf{x} > 0$. Every positive definite matrix can be written as $\mathbf{\Sigma} = \mathbf{S} \cdot \mathbf{S}^\intercal$ for some $\mathbf{S} \in \mathbb{R}^{n\times n}$. We write $\sqrt{\mathbf{\Sigma}}$ to denote any fixed matrix $\mathbf{S}$ satisfying $\mathbf{\Sigma} = \mathbf{S} \cdot \mathbf{S}^\intercal$. We say that two full rank matrices $\mathbf{S}_0, \mathbf{S}_1 \in \mathbb{R}^{n\times m}$ satisfy $\mathbf{S}_0 \geq \mathbf{S}_1$ if $\mathbf{S}_0 \cdot \mathbf{S}_0^\intercal - \mathbf{S}_1 \cdot \mathbf{S}_1^\intercal$ is positive semi-definite. We also say $\mathbf{S}_0 \geq \sigma$ for some $\sigma > 0$ if $\mathbf{S}_0 \cdot \mathbf{S}_0^\intercal - \sigma^2 \cdot \mathbf{I}$ is positive semi-definite.

Security Parameter. We denote the security parameter by λ, which is implicitly given to all algorithms unless specified otherwise. Further, we write $\mathsf{negl}(\lambda)$ (resp. $\mathsf{poly}(\lambda)$) to denote a negligible function (resp. polynomial) in λ. In this work, we implicitly assume that the vast majority of the key parameters, e.g. the ring dimension, and the dimensions of matrices and vectors, are $\mathsf{poly}(\lambda)$. However, the modulus used in this work may be superpolynomial in λ.

Lemma 1. *Let $A, B > 3$, then for any $x \geq 2(A + B\log(B))$, it holds that $x \geq A + B \cdot \log(x)$.*

2.1 Matrices and Singular Value Decomposition

For any matrix $\mathbf{M} \in \mathbb{R}^{n\times m}$ of rank r, there exist real values $0 < s_r(\mathbf{M}) \leq \ldots \leq s_1(\mathbf{M})$, and two orthogonal matrices $\mathbf{U}_M \in \mathbb{R}^{n\times n}, \mathbf{V}_M \in \mathbb{R}^{m\times m}$ such that

$$\mathbf{M} = \mathbf{U}_M \cdot \begin{bmatrix} \mathsf{diag}(\{s_i(\mathbf{M})\}_{i=1}^r) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \cdot \mathbf{V}_M.$$

The $s_i(\mathbf{M})$ are called the singular values of $\mathbf{M} \in \mathbb{R}^{n\times m}$. For any $\mathbf{x} \in \mathbb{R}^m$, it holds that $\|\mathbf{M} \cdot \mathbf{x}\| \leq \mathfrak{s}_{\max}(\mathbf{M}) \cdot \|\mathbf{x}\|$. If $\mathbf{M}$ is square and full rank, it also holds that $\mathfrak{s}_{\min}(\mathbf{M}) \cdot \|\mathbf{x}\| \leq \|\mathbf{M} \cdot \mathbf{x}\|$.

Lemma 2 (Corollary of Weyl's inequality.)**.** *Let $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n\times m}$ with $0 < n \leq m$. If $\mathfrak{s}_{\max}(\mathbf{Y}) \leq \delta \cdot \mathfrak{s}_{\min}(\mathbf{X})$ for some $\delta < 1$ then*

$$\mathfrak{s}_{\max}(\mathbf{X} \pm \mathbf{Y}) \in [1-\delta, 1+\delta] \cdot \mathfrak{s}_{\max}(\mathbf{X}) \ ,$$
$$\mathfrak{s}_{\min}(\mathbf{X} \pm \mathbf{Y}) \in [1-\delta, 1+\delta] \cdot \mathfrak{s}_{\min}(\mathbf{X}) \ .$$

Proof. Follows directly from the following inequalities $\mathfrak{s}_{\max}(\mathbf{X}) - \mathfrak{s}_{\max}(\mathbf{Y}) \leq \mathfrak{s}_{\max}(\mathbf{X}+\mathbf{Y}) \leq \mathfrak{s}_{\max}(\mathbf{X}) + \mathfrak{s}_{\max}(\mathbf{Y})$ and $\mathfrak{s}_{\min}(\mathbf{X}) - \mathfrak{s}_{\max}(\mathbf{Y}) \leq \mathfrak{s}_{\min}(\mathbf{X}+\mathbf{Y}) \leq \mathfrak{s}_{\min}(\mathbf{X}) + \mathfrak{s}_{\max}(\mathbf{Y})$ □

Lemma 3 ([LPSS14, Adapted Lemma 16 (eprint)])**.** *Let $1 \leq n \leq m$ be integers and $\sigma > 0$. Let $\mathbf{M} \in \mathbb{R}^{n \times m}$ be a full-rank matrix. There is a polynomial-time algorithm in the size of its input constructing a matrix $\mathbf{S} \in \mathbb{R}^{m \times m}$ with $\mathfrak{s}_{\max}(\mathbf{S}) \leq \sigma / \mathfrak{s}_{\min}(\mathbf{M})$ such that*

$$\sigma^2 \cdot \mathbf{I}_n = \mathbf{M} \cdot \mathbf{S} \cdot \mathbf{S}^{\mathsf{T}} \cdot \mathbf{M}^{\mathsf{T}}.$$

Proof. Assume that $\mathbf{M} = \mathbf{U}_M \cdot \left[\mathsf{diag}(\{s_i(\mathbf{M})\}_{i=1}^n) \quad \mathbf{0}\right] \cdot \mathbf{V}_M$ is a singular value decomposition of $\mathbf{M}$ with $\mathbf{U}_M \in \mathbb{R}^{n \times n}$, $\mathbf{V}_M \in \mathbb{R}^{m \times m}$ orthogonal. Set $\mathbf{S} = \mathbf{V}_M^{\mathsf{T}} \cdot \mathsf{diag}(\{s_i'\}_{i=1}^m)$ where $s_i' \coloneqq \sigma / s_i(\mathbf{M})$ for $i \leq n$ and $s_i' = 1$ for $n < i \leq m$. Computing the product we obtain the statement. □

2.2 Number Theory

For $q \in \mathbb{N}$ write $\mathbb{Z}_q$ for the integers modulo q. A number field $\mathcal{K} = \mathbb{Q}(\zeta)$ of degree d is a finite field extension of the rationals $\mathbb{Q}$ obtained by adjoining an algebraic number ζ. We denote its ring of integers by $\mathcal{R}$. We call $\mathcal{K}$ a $\mathfrak{f}$th cyclotomic number field if ζ is a $\mathfrak{f}$th primitive root of unity. Its degree is given by $d = \varphi(\mathfrak{f})$, where φ is Euler's totient function. We denote its continuous completion $\mathcal{K}_{\mathbb{R}} := \mathcal{K} \otimes \mathbb{R}$ and refer to [LPR13, Sec. 2] for more details.

We can identify $\mathcal{K} = \mathbb{Q}[X]/\langle \Phi(X) \rangle$, where $\Phi(X)$ is the minimal polynomial of ζ. Every element $x \in \mathcal{K}$ can then be written with respect to the $\mathbb{Q}$-basis $\left\{1, \zeta, \ldots, \zeta^{d-1}\right\}$, thus $x = \sum_{i=0}^{d-1} x_i \zeta^i$ with $x_i \in \mathbb{Q}$. The isomorphism $\tau \colon \mathcal{K} \to \mathbb{Q}^d$ which maps x to its coefficient vector $\tau(x) = (x_0, \ldots, x_{d-1})^{\mathsf{T}}$ is called the coefficient embedding. By associating the norm of an element x in $\mathcal{K}$ with the norm of its corresponding $\tau(x) \in \mathbb{Q}^d$, it is possible to equip $\mathcal{K}$ with a geometry. For a positive integer η, we define $S_\eta \coloneqq \tau^{-1}\left(\{-\eta, \ldots, \eta\}^d\right) = \{y \in \mathcal{R} \mid \|\tau(y)\|_\infty \leq \eta\}$, which corresponds to the set of polynomials in $\mathcal{R}$ with coefficients in $\{-\eta, \ldots, \eta\}$.

Another way of equipping $\mathcal{K}$ with a geometry is the canonical embedding. More precisely, for a number field $\mathcal{K}$ of degree d, let $\theta_1, \ldots, \theta_d$ denote the embeddings of $\mathcal{K}$ into $\mathbb{C}$. The function $\theta(x) \coloneqq (\theta_1(x), \ldots, \theta_d(x))$ is an injective map from $\mathcal{K}$ into $\mathbb{C}^d$ called the canonical (or Minkowski) embedding $\theta : \mathcal{K} \to \mathbb{C}^d$. It also helps us define the discriminant of the number field $\mathcal{K}$ as $\Delta_{\mathcal{K}} \coloneqq |\det(\theta_i(r_j))_{i,j}|^2$ for an arbitrary $\mathbb{Z}$-basis $\{r_j\}_j$ of its ring of integers $\mathcal{R}$. Its value is independent of the specific choice of the basis.

Let $d_{\mathbb{R}} + 2d_{\mathbb{C}} = d$ such that when $i \leq d_{\mathbb{R}}$ we have $\theta_i(\zeta_{\mathfrak{f}}) \in \mathbb{R}$ and when $d_{\mathbb{R}} < i \leq d_{\mathbb{R}} + d_{\mathbb{C}}$

we have $\theta_i(\zeta_\mathfrak{f}) \in \mathbb{C} \setminus \mathbb{R}$ and $\theta_{d_\mathbb{C}+i}(\zeta_\mathfrak{f}) = \overline{\theta_i(\zeta_\mathfrak{f})}$. Define

$$\mathbf{U}_H = \frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2}\mathbf{I}_{d_\mathbb{R}} & 0 & 0 \\ 0 & \mathbf{I}_{d_\mathbb{C}} & \mathbf{I}_{d_\mathbb{C}} \\ 0 & -i\mathbf{I}_{d_\mathbb{C}} & i\mathbf{I}_{d_\mathbb{C}} \end{bmatrix}.$$

Then $\theta_H(x) \coloneqq \mathbf{U}_H \cdot \theta(x) \in \mathbb{R}^d$. Since $\mathbf{U}_H$ is an orthogonal transformation, it does not affect the geometric lattice properties we consider in this thesis. Hence, we abuse the notations and use $\theta_H(\cdot)$ and $\theta(\cdot)$ interchangeably. We extend $\theta(\cdot)$ to vectors $\mathbf{x} = (x_0, \ldots, x_{k-1}) \in \mathcal{K}^k$ as $\theta(\mathbf{x}) \coloneqq (\theta(x_1), \ldots, \theta(x_k))$ and to matrices $\mathbf{U} = (u_{i,j})_{i,j} \in \mathcal{K}^{m \times k}$ as $\tilde{\theta}(\mathbf{U}) \coloneqq (\mathbf{B}_{i,j})_{i,j} \in \mathbb{R}^{d \cdot m \times d \cdot k}$ is a block matrix with blocks of the form $\mathbf{B}_{i,j} \coloneqq \mathsf{diag}(\{\theta_1(u_{i,j}), \ldots, \theta_d(u_{i,j})\})$. The matrix embedding is designed to verify $\theta(\mathbf{U} \cdot \mathbf{x}) = \tilde{\theta}(\mathbf{U}) \cdot \theta(\mathbf{x})$.

In the canonical embedding, for any $a, b \in \mathcal{K}$, it holds that $\|\theta(a) \cdot \theta(b)\| \leq \|\theta(a)\| \cdot \|\theta(b)\|$. Then for the matrix norm in canonical embedding we have the following.

Lemma 4. *Let* $\mathbf{U} = (\mathbf{u}_j)_{j \in [k]} \in \mathcal{K}^{m \times k}$. *Then* $\mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U})) \leq \sqrt{\sum_{j \in [k]} \|\theta(\mathbf{u}_j)\|^2}$.

Proof. Fix any non-zero $\mathbf{x} \in \mathcal{K}^k$. For the matrix-vector product we have $\left\|\sum_j \theta(\mathbf{u}_j x_j)\right\|^2 \leq \left(\sum_j \|\theta(\mathbf{u}_j x_j)\|\right)^2 \leq \left(\sum_j \|\theta(\mathbf{u}_j)\| \|\theta(x_j)\|\right)^2 \leq \left(\sum_j \|\theta(\mathbf{u}_j)\|^2\right) \cdot \left(\sum_j \|\theta(x_j)\|^2\right)$ due to the triangle inequality, the submultiplicativity of $\|\cdot\|$, linearity of $\theta(\cdot)$, and the Cauchy-Schwartz inequality respectively. Taking into account that $\|\theta(\mathbf{x})\|^2 = \sum_j \|\theta(x_j)\|^2$, we have $\|\theta(\mathbf{U} \cdot \mathbf{x})\| / \|\theta(\mathbf{x})\| \leq \sqrt{\sum_j \|\theta(\mathbf{u}_j)\|^2}$. □

Note that the two geometries induced by the coefficient and canonical embeddings are in general not the same. We can measure their distortion using the Vandermode matrix. Let $(\alpha_i)_{i \in \{1,\ldots,d\}} = (\theta_i(\zeta))_{i \in \{1,\ldots,d\}}$ be the roots of the defining polynomial $\Phi(X)$ of $\mathcal{K} = \mathbb{Q}(\zeta)$. They form the Vandermonde matrix $\mathbf{V}_\Phi$ defined as

$$\mathbf{V}_\Phi \coloneqq \begin{bmatrix} 1 & \alpha_1 & \cdots & \alpha_1^{d-1} \\ 1 & \alpha_2 & \cdots & \alpha_2^{d-1} \\ \vdots & \vdots & & \vdots \\ 1 & \alpha_d & \cdots & \alpha_d^{d-1} \end{bmatrix}.$$

Then $\forall x \in \mathcal{K} : \theta(x) = \mathbf{U}_H \cdot \mathbf{V}_\Phi \cdot \tau(x)$. Denote $\mathfrak{s}_{\max}(\mathfrak{f})$ the spectral norm of the Vandermonde matrix $\mathbf{V}_\Phi$ of the $\mathfrak{f}$th cyclotomic field and $\mathfrak{s}_{\min}(\mathfrak{f})$ the smallest singular value of $\mathbf{V}_\Phi$, or equivalently $1/\mathfrak{s}_{\min}(\mathfrak{f}) = \mathfrak{s}_{\max}((\mathbf{V}_\Phi)^{-1})$. Since $\mathfrak{s}_{\max}(\mathbf{U}_H) = \mathfrak{s}_{\min}(\mathbf{U}_H) = 1$, for every $x \in \mathcal{K}$ it holds

$$\|\tau(x)\| \cdot \mathfrak{s}_{\min}(\mathfrak{f}) \leq \|\theta(x)\| \leq \|\tau(x)\| \cdot \mathfrak{s}_{\max}(\mathfrak{f}). \tag{2.1}$$

In the case of power-of-two cyclotomics, i.e., $\mathfrak{f} = 2^{k+1}$ for a positive integer k and $d = 2^k$, we know that $\mathfrak{s}_{\max}(\mathfrak{f}) = \mathfrak{s}_{\min}(\mathfrak{f}) = \sqrt{d}$. Thus, τ and θ are isometric up to a factor of $\sqrt{d}$. The distortion between embeddings for other number fields was studied in depth in [BC22] and [RSW18]. However for cyclotomic fields, we have the following simple lower and upper bound on $\mathfrak{s}_{\max}(\mathfrak{f})$.

Lemma 5 ([ACX19, Proposition 2])**.** *For all positive integers* $\mathfrak{f}$

$$\sqrt{\varphi(\mathfrak{f})} \leq \mathfrak{s}_{\max}(\mathfrak{f}) \leq \sqrt{t(\mathfrak{f})},$$

where $t(\mathfrak{f}) = \mathfrak{f}$ *when* $\mathfrak{f}$ *is odd and* $t(\mathfrak{f}) = \mathfrak{f}/2$ *when it is even. The inequality turns into an equality on both sides when* $\mathfrak{f}$ *is a power of 2.*

We recall a few properties of the discriminant and the conductor of a cyclotomic number field.

Lemma 6 ([LPR13])**.** *For* $\mathfrak{f} \in \mathbb{N}$*, let* $\mathcal{K} = \mathbb{Q}(\zeta_{\mathfrak{f}})$ *be the* $\mathfrak{f}$*th cyclotomic number field. Denote* $d = \varphi(\mathfrak{f})$*. Then*

$$|\Delta_{\mathcal{K}}| = \frac{\mathfrak{f}^d}{\prod_{d|\mathfrak{f}} d^{\frac{d}{d-1}}} \leq d^d.$$

Lemma 7 ([KLNO24, Lemma 2])**.** *For any* $f \geq 4$*, a number field* $\mathcal{K}$ *containing a* f*th root of unity* $\zeta_{\mathfrak{f}}$*, and* $a \in \mathbb{Z}_f \setminus \{0\}$*, it holds that* $(1 - \zeta_{\mathfrak{f}}^a) \,|f$ *and* $\left\| f/(1 - \zeta_{\mathfrak{f}}^a) \right\|_\infty \leq f^2/2$*.*

Using the fact that every cyclotomic field $\mathcal{K}$ is a Galois field extension of $\mathbb{Q}$, we have a good grasp of the splitting behaviour of the minimal polynomial $\Phi(X)$ into irreducible factor over finite fields. More precisely, for an unramified prime q (this is guaranteed if $q \nmid \Delta_{\mathcal{K}}$), the polynomial $\Phi(X)$ uniquely factors as $\Phi(X) = \prod_{j=1}^{f} \Phi_j(X) \bmod q$, such that $f|d$, all $\Phi_j(X)$ are distinct, have degree $\delta|d$ and are irreducible modulo q, where $d = f\delta$. For cyclotomic fields $\mathcal{R} = \mathbb{Z}[X]/\langle \Phi(X) \rangle$ and an unramified prime q the above implies $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ of degree d factors into f finite fields as

$$\mathcal{R}_q = \mathbb{Z}_q[X]/\langle \Phi(X) \rangle = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\ \langle \Phi_j(X) \rangle.$$

When d grows as a function of security parameter we call $\mathcal{R}_q$ fully-splitting if $f = d$, high-splitting if δ is a constant and low-splitting if f is a constant as a function of the security parameter. For an element $x \in \mathcal{R}_q$ and $j \in \{1, \ldots, f\}$, we call $x \bmod \Phi_j(X)$ the jth CRT slot. An element x is invertible in $\mathcal{R}_q$ (denoted $x \in \mathcal{R}_q^\times$) if and only if all of its CRT-slots are non-zero.

Putting significantly stronger constraints on the relationship between q and $\Phi(X)$, the Φ_j are guaranteed to have a special shape. This special shape in turn implies that short ring elements are invertible.

Lemma 8 ([LS18, Theorem 1.1])**.** *Let $\mathcal{K} = \mathbb{Q}[X]/\langle\Phi(X)\rangle$ be the $\mathfrak{f}$th cyclotomic field of degree $d = \varphi(\mathfrak{f})$, with $\mathcal{R}$ its ring of integers. Further, let $\mathfrak{f} = \prod p_i^{e_i}$ for $e_i \geq 1$ and let $\mu = \prod p_i^{f_i}$ for $1 \leq f_i \leq e_i$. If q is a prime such that $q = 1 \bmod \mu$ and the multiplicative order modulo $\mathfrak{f}$ of q is $\mathfrak{f}/\mu$, then the polynomial $\Phi(X)$ factors modulo q as $\Phi(X) = \prod_{j=1}^{\varphi(\mu)}(X^{\mathfrak{f}/\mu} - r_j) \bmod q$, for distinct $r_j \in \mathbb{Z}_q^\times$, where $X^{\mathfrak{f}/\mu} - r_j$ are irreducible in the ring $\mathbb{Z}_q[X]$. Further, for any element y of $\mathcal{R}_q$ satisfying $0 < \|\tau(y)\|_\infty < q^{1/\varphi(\mu)}/\mathfrak{s}_{\max}(\mu)$, it holds that y is a unit in $\mathcal{R}_q$.*

Corollary 1 ([LS18, Corollary 1.2])**.** *Let $2d > k > 0$ be powers of 2. If q is a prime such that $q = 2k + 1 \bmod 4k$ then there exist distinct $r_i \in \mathbb{Z}_q^\times$ such that: (i) $X^{d/k} - r_i$ are irreducible in $\mathbb{Z}_q[X]$, and (ii) $X^d + 1$ factors as:*

$$X^d + 1 = \prod_{i=1}^{k}(X^{d/k} - r_i) \bmod q$$

An integral ideal $\mathcal{I}$ of a ring $\mathcal{R}$ is a subgroup of $\mathcal{R}$ such that $\mathcal{I} \cdot \mathcal{R} = \mathcal{I}$. The definition also applies to the quotient ring $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ for an integer modulus q. We denote a principal ideal generated by $a \in \mathcal{R}$ as $\langle a\rangle = a\mathcal{R}$. Lastly, we define factional ideals of $\mathcal{R}$ as $\mathcal{J} \coloneqq \frac{1}{a} \cdot \mathcal{I}$ for an integral ideal $\mathcal{I} \subset \mathcal{R}$ and an element $a \in \mathcal{R}$.

The set of fractional ideals forms a group, where the product of two ideals I, J is the ideal $I \cdot J = \langle\{a \cdot b, a \in I, b \in J\}\rangle$, and the inverse of an ideal I is the ideal $I^{-1} = \{x \in \mathcal{K}, x \cdot I \subseteq \mathcal{R}\}$. It is known that the ring of integers $\mathcal{R}$ of any number field is a Dedekind domain. First this implies that every prime ideal of $\mathcal{R}$ is maximal (i.e. an ideal $\mathfrak{p} \subset \mathcal{R}$ is prime iff $\mathcal{R}/\mathfrak{p}$ is a field). Second, its ideals have a unique prime factorisation property. For every ideal $I \subset \mathcal{K}$, there exist $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ prime ideals and $e_1, \ldots, e_k \in \mathbb{Z}$ such that $I = \prod_{i=1}^k \mathfrak{p}_i^{e_i}$. Up to reordering $(\mathfrak{p}_i)_i$ and $(e_i)_i$ are unique. Two elements $a, b \in \mathcal{R}$ are called coprime if together they generate the ring as $\langle a, b\rangle = \mathcal{R}$.

The norm of an ideal $\mathcal{I} \subseteq \mathcal{R}$ is defined as $N(\mathcal{I}) = |\mathcal{R}/\mathcal{I}|$ and correspondingly $N(\mathcal{J}) = |\mathcal{R}_q/\mathcal{J}|$ for an ideal $\mathcal{J}$ of $\mathcal{R}_q$. For any subideal $\mathcal{I}' \subseteq \mathcal{I} \subseteq \mathcal{R}$ the norm satisfies $N(\mathcal{I}) = |(\mathcal{R}/\mathcal{I}')/(\mathcal{I}/\mathcal{I}')|$. In particular, for any $\mathcal{I}$ that contains $q\mathcal{R}$ we have $N(\mathcal{I}) = |\mathcal{R}_q|/|\mathcal{I}/q\mathcal{R}|$. Some of our results rely on the Generalised Riemann Hypothesis (GRH) [BS96, Conjecture 8.7.3], which implies in particular that for any number field $\mathcal{K}$ and any $\varepsilon > 0$, the prime counting function $\pi_{\mathcal{K}}(x) = |\{\mathfrak{p} \text{ prime ideal}, \mathcal{N}(\mathfrak{p}) \leq x\}|$

satisfies,

$$\pi_{\mathcal{K}}(x) = \int_2^x 1/\ln(t)dt + O(x^{1/2+\varepsilon}).$$

The ideals of $\mathcal{R}_q$ allow for a much simpler characterisation.

Lemma 9 (Ideals in $\mathcal{R}_q$)**.** *Consider a cyclotomic number field $\mathcal{K}$ of degree $d = \delta \cdot f$, its ring of integers $\mathcal{R}$ and an unramified prime number q. Let $\Phi(X) = \prod_{j=1}^{f} \Phi_j(X) \bmod q$ be the factorisation of the corresponding cyclotomic polynomial in $\mathcal{R}_q$. Then every ideal $\mathcal{I} \subseteq \mathcal{R}_q$ is of the form*

$$\mathcal{I} = \langle \prod_{j \in G} \Phi_j(X) \rangle, \text{ with } N(\mathcal{I}) = q^{\delta|G|},$$

for some subset $G \subseteq \{1, \ldots, f\}$. The set $G = \varnothing$ corresponds to the ideal $\langle 1 \rangle$.

We upper bound the sum of norms of all ideals in $\mathcal{R}_q$.

Lemma 10 ([LPR13, Claim 7.3] adapted)**.** *Let $\mathcal{K}$ be a cyclotomic number field of degree d and $\mathcal{R}$ its ring of integers, let $k > 1$ be an integer and $q \geq 2$ an unramified prime number. Let $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ factor into f fields of size q^δ, for some $f \cdot \delta = d$. Let $\mathbb{I}$ be the set of all integral ideals of $\mathcal{R}_q$. Then*

$$\sum_{\mathcal{I} \in \mathbb{I}} |\mathcal{I}|^k = \sum_{\mathcal{I} \in \mathbb{I}} N(\mathcal{I})^k \leq e^{f \cdot q^{-\delta k}} \cdot q^{dk}.$$

if in addition $f \cdot q^{-\delta k} \leq 1$ we get

$$\sum_{\mathcal{I} \in \mathbb{I}} |\mathcal{I}|^{-k} = \sum_{\mathcal{I} \in \mathbb{I}} N(\mathcal{I})^{-k} \leq 1 + 2f \cdot q^{-\delta k}.$$

Proof. Let $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$ be the factorisation of the ring. Since q is an unramified prime by Lemma 9 every ideal of $\mathcal{R}_q$ is of the form $\mathcal{I} = \langle \prod_{j \in G} \Phi_j(X) \rangle$ for some $G \subseteq \{1, \ldots, f\}$. Then $\mathcal{I}' = \langle \prod_{j \notin G} \Phi_j(X) \rangle$ satisfies $|\mathcal{I}| = N(\mathcal{I}')$. Hence, we can reorder the sum over all ideals and obtain $\sum_{\mathcal{I} \in \mathbb{I}} |\mathcal{I}|^k = \sum_{\mathcal{I}' \in \mathbb{I}} N(\mathcal{I}')^k$ for $k \in \mathbb{Z} \setminus \{0\}$. Hence, we only argue about the norms of the ideals. Then

$$\begin{aligned} \sum_{\mathcal{I} \in \mathbb{I}} N(\mathcal{I})^k &= \sum_{G \subseteq \{1,\ldots,f\}} q^{k\delta \cdot |G|} = \sum_{i=0}^{f} \binom{f}{i} \cdot q^{k\delta \cdot i} \\ &= (1 + q^{\delta k})^f. \end{aligned}$$

For $k > 1$ we have $(1 + q^{\delta k})^f = q^{dk} \cdot (1 + q^{-\delta k})^f \leq q^{dk} e^{f \cdot q^{-\delta k}} \leq q^{dk} \cdot (1 + 2f \cdot q^{-\delta k})$ for $f \cdot q^{-\delta k} \leq 1$ using $(1 + x)^n \leq e^{nx}$ for $|x| \leq 1$ and $(1 + x)^n \leq 1 + 2nx$ for $0 < nx \leq 1$, $n \geq 1$. The same argument applies for $k < -1$ □

The set of ideals $\mathcal{J} \subseteq \mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ are in one to one correspondence with ideals $\mathcal{I} \subseteq \mathcal{R}$ containing $q\mathcal{R}$ via a map $\mathcal{I} \mapsto \mathcal{I}/q\mathcal{R} \subseteq \mathcal{R}_q$. Alternatively, we write $\mathcal{J} \mapsto q\mathcal{R} + \mathcal{J} \subseteq \mathcal{R}$ taking an arbitrary representative of each element of $\mathcal{J}$ in $\mathcal{R}$.

For any element a in the ring we define its algebraic norm as $N(a) \coloneqq \prod_{i=1}^{d} \theta_i(a)$, it verifies $|N(a)| = N(\langle a \rangle)$. Then for the fractional ideals $\frac{1}{a}\mathcal{I}$ where $a \in \mathcal{R}, \mathcal{I} \subseteq \mathcal{R}$ we define the norm as $N(\frac{1}{a}\mathcal{I}) \coloneqq \frac{N(\mathcal{I})}{|N(a)|}$. We denote $\mathcal{R}^* \coloneqq \{x \in \mathcal{K} \mid x \cdot \mathcal{R} \subseteq \mathcal{R}\}$ the dual of $\mathcal{R}$ as an $\mathcal{R}$-module. The following lemma allows us to compute the norms of ideals inside $\mathcal{R}^*$.

Lemma 11 ([LPR13, Section 2.5.4])**.** *Let $\mathcal{K}$ be a cyclotomic number field and $\mathcal{R}$ be its ring of integers. Then there exists an element $t \in \mathcal{R}$ such that $\mathcal{R}^* = \frac{1}{t} \cdot \mathcal{R}$. Moreover, $N(\mathcal{R}^*) = |N(t)|^{-1} = \Delta_{\mathcal{K}}^{-1}$.*

As consequence, for an ideal $\mathcal{J} \subseteq \mathcal{R}^*$ we have $N(\mathcal{J}) = N(t \cdot \mathcal{J}) \cdot \Delta_{\mathcal{K}}^{-1}$ where $t \cdot \mathcal{J}$ is an integral ideal in $\mathcal{R}$.

2.3 Probability Theory

Distributions. Let $\mathcal{P}_0, \mathcal{P}_1$ be two (discrete or absolutely continuous w.r.t. the Lebesgue measure over $\mathbb{R}^n$ for some $n \geq 1$) probability distribution defined over the same σ-algebras. By $\mathcal{U}(S)$, we denote the uniform distribution over a finite set S, which is defined as $\Pr[x \leftarrow \mathcal{U}(S)] = 1/|S|$ for every $x \in S$. We use a notation shorthand $x \leftarrow S$ for $x \leftarrow \mathcal{U}(S)$. By $\mathbb{E}(\mathcal{P})$ we denote the expected value of $\mathcal{P}$.

We define three measures of distance between probability distributions we use in this work. If $\Omega = \mathrm{Supp}(\mathcal{P}_0) \cup \mathrm{Supp}(\mathcal{P}_1)$, then the statistical distance between $\mathcal{P}_0$ and $\mathcal{P}_1$ is $\mathrm{SD}(\mathcal{P}_0, \mathcal{P}_1) \coloneqq \int_{t \in \Omega} |\mathcal{P}_0(t) - \mathcal{P}_1(t)| dt/2$ which simplifies to $\mathrm{SD}(\mathcal{P}_0, \mathcal{P}_1) = \frac{1}{2} \sum_{x \in \Omega} |\mathcal{P}_0(x) - \mathcal{P}_1(x)|$ for discrete distributions. We write $\mathcal{P}_0 \approx_s \mathcal{P}_1$ when $\mathrm{SD}(\mathcal{P}_0, \mathcal{P}_1) = \mathsf{negl}(\lambda)$ and $\mathcal{P}_0 \approx_\varepsilon \mathcal{P}_1$ when $\mathrm{SD}(\mathcal{P}_0, \mathcal{P}_1) \leq \varepsilon$ for some $\varepsilon > 0$. The statistical distance has the probability preservation property, meaning that for any event $E \subseteq \mathrm{Supp}(\mathcal{P}_1)$

$$\mathcal{P}_1(E) \geq \mathcal{P}_0(E) - \mathrm{SD}(\mathcal{P}_0, \mathcal{P}_1). \tag{2.2}$$

When $\mathrm{Supp}(\mathcal{P}_0) \subseteq \mathrm{Supp}(\mathcal{P}_1)$, we define the order-2 Rényi divergence as $\mathrm{RD}(\mathcal{P}_0 || \mathcal{P}_1) \coloneqq \int_{t \in \mathrm{Supp}(\mathcal{P}_0)} \mathcal{P}_0(t)^2 / \mathcal{P}_1(t) dt$ or $\mathrm{RD}(\mathcal{P}_0; \mathcal{P}_1) = \sum_{x \in \mathrm{Supp}(\mathcal{P}_0)} \frac{\mathcal{P}_0(x)^2}{\mathcal{P}_1(x)}$ for discrete distributions. We also define the Rényi divergence of infinite order for discrete distributions as $\mathrm{RD}_\infty(\mathcal{P}_0; \mathcal{P}_1) = \sup_{x \in \mathrm{Supp}(\mathcal{P}_0)} \frac{\mathcal{P}_0(x)}{\mathcal{P}_1(x)}$. For any two distributions $\mathcal{P}, \mathcal{Q}$ it holds $\mathrm{RD}(\mathcal{P}; \mathcal{Q}) \leq \mathrm{RD}_\infty(\mathcal{P}; \mathcal{Q})$. This and other properties of the Rényi divergence, such as data processing inequality, have been shown in [vEH14]. For distributions $\mathcal{P}_0$ and $\mathcal{P}_1$ parametrised by

λ we call $\mathcal{P}_0$ and $\mathcal{P}_1$ *Rényi close* if $\mathrm{RD}(\mathcal{P}_0;\mathcal{P}_1) \in \mathcal{O}(1)$ as a function of λ. Probability preservation property for Rényi divergence is of the form

$$\mathcal{P}_1(E) \geq \mathcal{P}_0(E)^2/\mathrm{RD}(\mathcal{P}_0\|\mathcal{P}_1) \tag{2.3}$$

for any event $E \subseteq \mathrm{Supp}(\mathcal{P}_1)$. For $\varepsilon > 0$ we extend the notion above to smooth Rényi divergence that excludes anomalous points that occur with a bounded probability

$$\mathrm{RD}^{\varepsilon}(\mathcal{P}_0||\mathcal{P}_1) = \inf_{\substack{S\subset\mathrm{Supp}(\mathcal{P}_1)\\ \mathcal{P}_0(S)>1-\varepsilon}} \sum_{x\in S} \frac{\mathcal{P}_0(x)^2}{\mathcal{P}_1(x)}. \tag{2.4}$$

Lastly, given a PPT adversary algorithm $\mathcal{A}$, we define the distinguishing advantage of $\mathcal{A}$ as $\mathsf{Adv}_{\mathcal{A}}(\mathcal{P}_0,\mathcal{P}_1) \coloneqq |\Pr(\mathcal{A}(\mathcal{P}_0)=1) - \Pr(\mathcal{A}(\mathcal{P}_1)=1)|$.

The collision probability of $\mathcal{P}$ is defined as $\mathrm{Coll}(\mathcal{P}) = \Pr[x = x' | x, x' \leftarrow \mathcal{P}]$. If $\mathcal{P}$ is a discrete probability distribution over S, the collision probability equals $\mathrm{Coll}(\mathcal{P}) = \sum_{x\in S} \mathcal{P}(x)^2$ and the expected value equals $\mathbb{E}(\mathcal{P}) = \sum_{x\in S} x \cdot \mathcal{P}(x)$. The min-entropy of $\mathcal{P}$ is defined as $\mathrm{H}_\infty(\mathcal{P}) = -\log_2(\max_{x\in S}\mathcal{P}(x))$. It is true that $\mathrm{Coll}(\mathcal{P}) \leq \sum_{x\in S}\mathcal{P}(x)\cdot \max_{y\in S}\mathcal{P}(y) = \max_{y\in S}\mathcal{P}(y)$, which implies $\mathrm{Coll}(\mathcal{P}) \leq 2^{-\mathrm{H}_\infty(\mathcal{P})}$. For two probability distributions $\mathcal{P}$ (over S) and $\mathcal{Q}$ (over T) that possibly depends on $\mathcal{P}$ we define the *average conditional collision probability* as

$$\widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) \coloneqq \mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr[x = x' \mid x, x' \leftarrow \mathcal{P}, y].$$

We remark that there are different flavours of how to define average conditional collision probability in the literature [FB14]. We opted for this one as it allows for equalities when working with the Rényi divergence (cf. Lemma 13 below). If both $\mathcal{P}$ and $\mathcal{Q}$ are discrete, this equals $\widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) = \sum_{y\in T}\mathcal{Q}(y)\sum_{x\in S}\Pr[\mathcal{P}=x \mid y]^2$. If $\mathcal{Q}$ is independent of $\mathcal{P}$, we recover the ordinary collision probability, i.e., $\widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) = \mathrm{Coll}(\mathcal{P})$. The average conditional collision probability is upper bounded by the average conditional min-entropy introduced in [DORS08, Sec. 2.4],

$$\widetilde{\mathrm{H}_\infty}(\mathcal{P} \mid \mathcal{Q}) \coloneqq -\log_2\left(\mathbb{E}_{y\leftarrow\mathcal{Q}} \max_{x\in S} \Pr[\mathcal{P}=x \mid y]\right).$$

More precisely, $\widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) \leq \mathbb{E}_{y\leftarrow\mathcal{Q}}\max_{x\in S}\Pr[\mathcal{P}=x\mid y] = 2^{-\widetilde{\mathrm{H}_\infty}(\mathcal{P}|\mathcal{Q})}$.

The statistical distance to uniform, even in the presence of leakage, can be upper bounded in terms of the (average conditional) collision probability of $\mathcal{P}$ under leakage $\mathcal{Q}$.

Lemma 12 ([ILL89],[DORS08, Lem. 2.4])**.** *Let $\mathcal{P}$ be a discrete probability distribution over a finite set S and $\mathcal{Q}$ be a discrete probability distribution over a finite set T. Then,*

i) $\mathrm{SD}(\mathcal{P},\mathcal{U}(S)) \leq \frac{1}{2}\sqrt{|S| \cdot \mathrm{Coll}(\mathcal{P}) - 1}$,

ii) $\mathrm{SD}((\mathcal{P},\mathcal{Q}),(\mathcal{U}(S),\mathcal{Q})) \leq \frac{1}{2}\sqrt{|S| \cdot \widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) - 1}$.

Simply expanding the definition, the Rényi divergence to uniform distribution can be expressed in terms of average conditional collision probability too.

Lemma 13. *Let $\mathcal{P}$ be a discrete probability distribution over a finite set S and $\mathcal{Q}$ be a discrete probability distribution over a finite set T. Then,*

i) $\mathrm{RD}(\mathcal{P};\mathcal{U}(S)) = |S| \cdot \mathrm{Coll}(\mathcal{P})$.

ii) $\mathrm{RD}((\mathcal{P},\mathcal{Q}),(\mathcal{U}(S),\mathcal{Q})) = |S| \cdot \widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q})$.

We recall a general result, lower bounding the average conditional collision probability by the min-entropy and the bit-size of the leakage.

Lemma 14 ([DORS08, Lem. 2.2])**.** *Let $\mathcal{P}$ and $\mathcal{Q}$ be two discrete probability distributions, where $\mathcal{Q}$ has at most $|T|$ possible values. Then, $\widetilde{\mathrm{H}_\infty}(\mathcal{P} \mid \mathcal{Q}) \geq \mathrm{H}_\infty(\mathcal{P}) - \log_2|T|$. This implies that $\widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) \leq 2^{-\mathrm{H}_\infty(\mathcal{P})} \cdot |T|$.*

We recall the notion of well-spread distributions over $\mathcal{R}_q$ as defined in [AAB+24]. This is an additional characteristic for the entropy of distributions complementing the collision probability and min-entropy.

Definition 1 (Well-Spreadness)**.** *Let $\mathcal{R}$ be the ring of integers of a number field and q an unramified prime such that the factorisation of $\mathcal{R}_q$ into subfields is given by $\mathcal{R}_q = \mathbb{Z}_q[X]/\langle\Phi(X)\rangle = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle\Phi_j(X)\rangle$, denote $\delta \coloneqq d/f$. Let $\mathcal{P}$ be a distribution over $\mathcal{R}_q$ and $B \in [0,1]$ be a constant. We say that $\mathcal{P}$ is B-well-spread if for all $j \in [f]$ and for all $y \in \mathbb{Z}_q[X]/\langle\Phi_j(X)\rangle$ it holds*

$$\Pr[x \bmod \Phi_j(X) = y | x \leftarrow \mathcal{P}] \leq B.$$

The closer B gets to $1/q^\delta$, the more $\mathcal{P} \bmod \Phi_j(X)$ acts like the uniform distribution over a given CRT-slot (not necessarily independent of the other $\mathcal{P} \bmod \Phi_i(X)$ for $i \neq j$).

The next useful lemma was proven in [Mic07, Lem. 4.3] for ideals and generalised to modules in [BJRW23, Lem. B.3].

Lemma 15. *Let $\mathcal{A}$ be an arbitrary finite commutative ring and n, m be positive integers. Consider an arbitrary vector $\mathbf{z} = (z_i)_{i=1}^m \in \mathcal{A}^m$. We define the ideal $\mathcal{I}_\mathbf{z} \coloneqq \langle z_1, \ldots, z_m\rangle$. For $\mathbf{A} \leftarrow \mathcal{U}(\mathcal{A}^{n\times m})$, the vector $\mathbf{Az}$ is uniformly distributed over the module $\mathcal{I}_\mathbf{z}^n$. In particular, $\Pr_{\mathbf{A}\leftarrow\mathcal{U}(\mathcal{A}^{n\times m})}[\mathbf{Az} = \mathbf{0}] = \frac{1}{|\mathcal{I}_\mathbf{z}|^n}$.*

Lemma 16 ([BJRW23, Lemma 2.6])**.** *Let $\mathcal{K}$ be a cyclotomic field with $\mathcal{R}$ its ring of integers of degree d. Further, let q be a non-ramified prime modulus such that $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$ for $\Phi_j(X)$ of degree δ. Let $0 < n \leq m$ be integers. Then*

$$\Pr_{\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m})}(\mathbf{A} \cdot \mathcal{R}_q^m = \mathcal{R}_q^n) \geq \prod_{i=0}^{n-1} \left(1 - \frac{1}{q^{\delta(m-i)}}\right)^f .$$

Remark 1. When $nf > 1$, and $q^\delta > \mathsf{poly}(\lambda)$ the above probability is at least $1 - \frac{nf}{q^{\delta \cdot (m-n+1)}}$ and overwhelming in λ.

Lemma 17 ([LM00, Lemma 1 (p. 1325)])**.** *Let χ^2_m be a Chi-squared random variable of degree $m > 0$ then*

$$\Pr(\chi^2_m \geq m + 2\sqrt{mk} + 2k) \leq \exp(-k) \ ,$$
$$\Pr(\chi^2_m \leq m - 2\sqrt{mk}) \leq \exp(-k) \ .$$

2.4 Lattices

A lattice Λ is a discrete additive subgroup of $\mathbb{R}^m$. It is usually defined as all integer linear combinations of a set of basis vectors that are linearly independent in $\mathbb{R}^m$. Formally, a set of $n \leq m$ basis vectors $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_n] \in \mathbb{R}^{m \times n}$ defines a lattice $\Lambda(\mathbf{B}) \coloneqq \{\sum_{i=1}^n \mathbf{b}_i x_i | x_i \in \mathbb{Z}\}$. The parameter m here is the lattice dimension and the parameter n is its rank. We say that a lattice is full-rank when $n = m$. The determinant of a lattice is defined as $\det(\Lambda(\mathbf{B})) \coloneqq \sqrt{\det(\mathbf{B}^\intercal \mathbf{B})}$. We further define the dual of a lattice Λ as $\Lambda^* = \{\mathbf{y} \in \mathrm{span}_{\mathbb{R}}(\Lambda) \colon \langle \mathbf{y}, \mathbf{x} \rangle \in \mathbb{Z} \ \forall \mathbf{x} \in \Lambda\}$. The first minimum of a lattice Λ with respect to the ℓ_2-norm is defined by $\lambda_1(\Lambda) = \min_{\mathbf{x} \in \Lambda \setminus \{\mathbf{0}\}} \|\mathbf{x}\|$. We write $\lambda_1^\infty(\Lambda)$ for the first minimum with respect to the infinity norm. Denote by $\mathcal{C}_n$ a closed ball of dimension n and radius 1. Then for $i \geq 1$ the ith minimum of the lattice is defined as $\lambda_i(\Lambda) = \min\{r > 0 : \dim \mathrm{span}(\Lambda \cap r\mathcal{C}_n) \geq i\}$.

Module lattices. An $\mathcal{R}$-module $\mathcal{M} \subset \mathcal{K}^m$ is an additive subgroup of $\mathcal{K}^m$ that is stable under multiplication by any element of $\mathcal{R}$ (fractional ideals of $\mathcal{R}$ are rank-1 modules). Any $\mathcal{R}$-module $\mathcal{M}$ of rank r can be represented as a (non-unique) pseudobasis $((I_i, \mathbf{b}_i))_{i=1,\ldots,r}$, where I_i are fractional ideals of $\mathcal{R}$ and $\mathbf{b}_i \in \mathcal{K}^m$ are linearly independent over $\mathcal{K}$. When $r = m$, we say that $\mathcal{M}$ is full-rank.

Any $\mathcal{R}$-module $\mathcal{M} \subset \mathcal{K}^r$ of rank r defines module lattices $\tau(\mathcal{M})$ and $\theta(\mathcal{M})$ in $\mathbb{R}^{dr}$ through the coefficient embedding τ or the canonical embedding θ. We write $\mathcal{D}^\tau_{\mathcal{M},\sqrt{\mathbf{\Sigma}},\mathbf{c}}$ (respectively $\mathcal{D}^\theta_{\mathcal{M},\sqrt{\mathbf{\Sigma}},\mathbf{c}}$) to denote the discrete Gaussian distribution obtained by embedding

the module $\mathcal{M}$ and the center $\mathbf{c} \in \mathcal{R}^r$ through τ (resp. θ). Any ideal $\mathcal{I}$ of ring $\mathcal{R}$ defines a lattice in $\mathbb{R}^d$ through the same embeddings.[1] Such lattices are called ideal lattices. Under the canonical embedding we have $\det(\theta(\mathcal{I})) = \mathcal{N}(I) \cdot \sqrt{\Delta_{\mathcal{K}}}$.

Let $\mathcal{M} \subset \mathcal{K}^r$ be a full-rank module given by a pseudobasis $((\mathbf{b}_i, I_i))_{1 \le i \le r}$. Its determinantal ideal is the fractional ideal $\det(\mathcal{M}) := \det_{\mathcal{K}}(\mathbf{B}) \cdot \prod_i I_i$, and it is independent of the choice of the pseudobasis. For any submodule $\mathcal{M}' \subseteq \mathcal{M}$, it holds that $\det(\mathcal{M}') \subset \det(\mathcal{M})$ with equality if and only if $\mathcal{M} = \mathcal{M}'$.

Q-ary lattices. An important class of lattices we consider in this work are q-ary lattices. For a matrix $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ we define

- $\Lambda_q^{\perp}(\mathbf{A}) \coloneqq \{\mathbf{y} \in \mathcal{R}^m \colon \mathbf{A} \cdot \mathbf{y} = \mathbf{0} \bmod q\}$,
- $\Lambda_q^{\mathbf{v}}(\mathbf{A}) \coloneqq \{\mathbf{y} \in \mathcal{R}^m : \mathbf{A} \cdot \mathbf{y} = \mathbf{v} \bmod q\}$,
- $\Lambda_q(\mathbf{A}^{\intercal}) \coloneqq \{\mathbf{y} \in \mathcal{R}^m \colon \exists \mathbf{s} \in \mathcal{R}^n \text{ s.t } \mathbf{A}^{\intercal} \cdot \mathbf{s} = \mathbf{y} \bmod q\}$,
- $\Lambda(\mathbf{A}^{\intercal}) \coloneqq \{\mathbf{y} \in \mathcal{R}^m : \exists\ \mathbf{s} \in \mathcal{R}^n,\ \mathbf{A}^{\intercal} \cdot \mathbf{s} = \mathbf{y}\}$.

Both $\Lambda_q^{\perp}(\mathbf{A})$ and $\Lambda_q(\mathbf{A}^{\intercal})$ are an $\mathcal{R}$-module of rank m, which via an embedding defines a module lattice in $\mathbb{R}^{dm}$. The lattice $\Lambda(\mathbf{A}^{\intercal})$ is rank deficient and its rank as an $\mathcal{R}$-module is equal to the rank of $\mathbf{A}$. The set $\Lambda_q^{\mathbf{v}}(\mathbf{A})$ denotes a coset of the lattice $\Lambda_q^{\perp}(\mathbf{A})$. For $a \in \mathcal{R}_q = \mathbb{Z}_q[X]/\langle \Phi(X) \rangle$ define $\mathrm{Rot}(a) \coloneqq [\tau(a) \mid \cdots \mid \tau(X^{d-1} \cdot a)] \in \mathbb{Z}_q^{d \times d}$ and for $\mathbf{A} = (a_{ij})_{i,j} \in \mathcal{R}_q^{n \times m}$ define a block matrix $\mathrm{Rot}(\mathbf{A}) \coloneqq (\mathrm{Rot}(a_{ij}))_{i,j} \in \mathbb{Z}_q^{dn \times dm}$. We note that for the coefficient embedding $\tau(\Lambda_q^{\perp}(\mathbf{A})) = \Lambda_q^{\perp}(\mathrm{Rot}(\mathbf{A}))$.

For $\mathbf{X} \in \mathcal{K}^{r \times m}$ we define its kernel module as $\Lambda^{\perp}(\mathbf{X}) := \{\mathbf{y} \in \mathcal{R}^m \mid \mathbf{X} \cdot \mathbf{y} = \mathbf{0}\}$. When $\mathbf{X}$ has $\mathcal{K}$-rank l, it is a $\mathcal{R}$-module of rank $m - l$.

Gaussians. Let $\mathbf{\Sigma} \in \mathbb{R}^{m \times m}$ be a symmetric positive-definite matrix, let $\mathbf{c} \in \mathbb{R}^m$ and Λ be a full-rank m-dimensional lattice. For any vector $\mathbf{x} \in \mathbb{R}^m$ we define the Gaussian mass function $\rho_{\sqrt{\mathbf{\Sigma}},\mathbf{c}}(\mathbf{x})$ as $\rho_{\sqrt{\mathbf{\Sigma}},\mathbf{c}}(\mathbf{x}) \coloneqq \exp\left(-\pi(\mathbf{x}-\mathbf{c})^{\intercal}\mathbf{\Sigma}^{-1}(\mathbf{x}-\mathbf{c})\right)$. Then, the discrete Gaussian distribution $\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}}$ over Λ is given by

$$\forall \mathbf{x} \in \Lambda : \mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}}(\mathbf{x}) = \frac{\rho_{\sqrt{\mathbf{\Sigma}},\mathbf{c}}(\mathbf{x})}{\sum_{\mathbf{z} \in \Lambda} \rho_{\sqrt{\mathbf{\Sigma}},\mathbf{c}}(\mathbf{z})}.$$

When $\mathbf{c} = \mathbf{0}$, we omit it from the notation. When $\mathbf{\Sigma} = \sigma^2 \mathbf{I}_m$, we say the discrete Gaussian distribution is spherical and simply write $\mathcal{D}_{\Lambda,\sigma,\mathbf{c}}$ (resp. $\rho_{\sigma,\mathbf{c}}$). When the lattice is not

[1] The formal definition of the discrete Gaussian distributions is given below.

specified, for any matrix $\mathbf{\Sigma} \in \mathbb{R}^{n\times n}$ and center $\mathbf{c} \in \mathbb{R}^n$, the notation $\mathcal{D}_{\sqrt{\mathbf{\Sigma}},\mathbf{c}}$ means the continuous Gaussian distribution of covariance $\mathbf{\Sigma}$ and center $\mathbf{c}$.

Gaussians in a rank-deficient lattice. Let $\mathbf{c} \in \mathbb{R}^m$, $\Lambda \subset \mathbb{R}^m$ be a lattice of rank $n < m$ and $\mathbf{S} \in \mathbb{R}^{m\times n}$ be a full-rank matrix with columns in $\mathbb{R}$-span(Λ). Note that the matrix $\mathbf{S}\cdot\mathbf{S}^\intercal$ is now positive semi-definite and does not have an inverse. Let $(\mathbf{S}\cdot\mathbf{S}^\intercal)^+$ denote the Moore-Penrose pseudoinverse of $\mathbf{S}\cdot\mathbf{S}^\intercal$. We define the Gaussian mass function with respect to $\mathbf{S} \in \mathbb{R}^{m\times n}$ as $\rho_{\mathbf{S},\mathbf{c}}(\mathbf{x}) \coloneqq \exp\left(-\pi(\mathbf{x}-\mathbf{c})^\intercal(\mathbf{S}\mathbf{S}^\intercal)^+(\mathbf{x}-\mathbf{c})\right)$. Then the Discrete Gaussian Distribution over Λ is defined accordingly.

We often use the following lemma when arguing about discrete Gaussian distributions in rank-deficient lattices.

Lemma 18 ([GMPW20, Lemma 2.3])**.** *Let $m \geq n > 0$ and $\mathbf{\Sigma} > 0$. For any n-dimensional lattice Λ, center $\mathbf{c} \in \mathbb{R}^n$ and a full-rank matrix $\mathbf{U} \in \mathbb{R}^{m\times n}$*

$$\mathbf{U}\cdot\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}} = \mathcal{D}_{\mathbf{U}\cdot\Lambda,\sqrt{\mathbf{U}\cdot\mathbf{\Sigma}\cdot\mathbf{U}^\intercal},\mathbf{U}\cdot\mathbf{c}}$$

Remark 2. Let $\mathbf{U} \in \mathbb{R}^{m\times n}$, $m \geq n$ be full-rank. Denote $\mathbf{S}_U$ the orthonormal basis of $\mathbb{R}$-span$(\mathbf{U}) \subset \mathbb{R}^m$. Let $\mathbf{U} = \mathbf{V}_0\cdot[\mathbf{D}||\mathbf{0}_{(m-n)\times n}]\cdot\mathbf{V}_1$ for orthogonal matrices $\mathbf{V}_0, \mathbf{V}_1$ and a diagonal matrix $\mathbf{D}$ be the SVD of $\mathbf{U}$. Even though $\mathbf{S}_U$ only depends on $\mathbb{R}$-span$(\mathbf{U})$, it can be computed as $\mathbf{S}_U = \mathbf{V}_0\cdot[\mathbf{I}_n||\mathbf{0}_{(m-n)\times n}]$. Indeed, $\mathbf{U}$ and $\mathbf{S}_U$ have the same span and the vectors in $\mathbf{S}_U$ are of norm 1 and pairwise orthogonal. For $\mathbf{S} \in (\mathbb{R}^{n\times n})^\times$ by definition and by Lemma 18 we have $\mathcal{D}_{\Lambda(\mathbf{U}),\mathbf{U}\cdot\mathbf{S}} = \mathbf{U}\cdot\mathcal{D}_{\mathbb{Z}^n,\mathbf{S}}$. Hence, when working with a discrete Gaussian in lattice $\Lambda(\mathbf{U})$ we take the distribution $\mathcal{D}_{\Lambda(\mathbf{U}),\mathbf{S}_U\cdot\mathbf{S}} = \mathbf{U}\cdot\mathcal{D}_{\mathbb{Z}^n,(\mathbf{V}_1^\intercal\cdot\mathbf{D}^{-1})\cdot\mathbf{S}}$. Note that $\mathcal{D}_{\Lambda(\mathbf{U}),\mathbf{S}_U\cdot\mathbf{S}}$ only depends on $\Lambda(\mathbf{U})$ and not a specific basis $\mathbf{U}$. This will become relevant in the construction of Chapter 5.

Smoothing. An important quantity related to discrete Gaussians over lattices is the smoothing parameter [MR07]. For some $\varepsilon > 0$ and a lattice Λ, we define $\eta_\varepsilon(\Lambda)$ as the smallest $s > 0$ such that $\rho_{1/s}(\Lambda^*) \leq 1+\varepsilon$. For any $\varepsilon > 0$ and $k \geq 1$, we denote $\eta_{\varepsilon,k} \coloneqq \sqrt{\ln(2k\cdot(1+1/\varepsilon))/\pi}$.

The notion of the smoothing parameter was adapted to elliptical Gaussians for example in [Pei10]. For a symmetric positive definite matrix $\mathbf{\Sigma} \in \mathbb{R}^{m\times m}$ the inequality $\mathbf{\Sigma} \geq \eta_\varepsilon^2(\Lambda) \coloneqq s^2$ correctly implies $\rho_{\sqrt{\mathbf{\Sigma}^{-1}}}(\Lambda^*) = \sum_{\mathbf{x}\in\Lambda^*}\exp(-\pi\cdot\mathbf{x}^\intercal\mathbf{\Sigma}\mathbf{x}) \leq \sum_{\mathbf{x}\in\Lambda^*}\exp(-\pi\cdot\mathbf{x}^\intercal s^2\mathbf{I}_m\mathbf{x}) = \rho_{1/s}(\Lambda^*) \leq 1+\varepsilon$. We bound the value of the smoothing parameter for given lattices using statements below.

Lemma 19 ([Pei08])**.** *For any $\varepsilon > 0$ and n-dimensional lattice Λ it holds*

$$\eta_\varepsilon(\Lambda) \leq \frac{1}{\lambda_1^\infty(\Lambda^*)} \cdot \eta_{\varepsilon,n}.$$

Lemma 20 ([MR07, Lemma 3.3] adapted)**.** *For any $\varepsilon \in (0, 1/2)$ and a rank-n lattice Λ it holds*

$$\eta_\varepsilon(\Lambda) \leq \lambda_n(\Lambda) \cdot \eta_{\varepsilon,n}.$$

In particular for any function in $\omega(\sqrt{\log n})$ there exist a negligible function $\varepsilon(n)$ such that $\eta_\varepsilon(\Lambda) \leq \omega(\sqrt{\log n}) \cdot \lambda_n(\Lambda)$.

Lemma 21 ([GPV08, Lemma 3.1])**.** *For $\varepsilon > 0$ and a full-rank basis $\mathbf{B} \in \mathbb{R}^{n \times n}$:*

$$\eta_\varepsilon(\Lambda(\mathbf{B})) \leq \|\mathbf{B}\|_{GS} \cdot \eta_{\varepsilon,n}.$$

Lemma 22 ([LPR13, Lemma 2.6])**.** *For any n-dimensional lattice Λ*

$$\eta_{2^{-2n}}(\Lambda) \leq \sqrt{n}/\lambda_1(\Lambda^*).$$

Bounds on norm of Gaussians. Prior work has developed numerous approaches to upper bound norms of a discrete Gaussian vector and the norm of its inner product with a given value.

Lemma 23 ([Ban93, Lemma 1.5])**.** *Let $m > 0$ be an integer, $c > 1/\sqrt{2\pi}$, $\sigma > 0$ and Λ be an m-dimensional lattice. Let $\mathbf{x} \leftarrow \mathcal{D}_{\Lambda,\sigma}$. Then*

$$\Pr(\|\mathbf{x}\| > c \cdot \sigma \cdot \sqrt{m}) \leq C^m$$

where $C \coloneqq c \cdot \sqrt{2\pi e} \cdot e^{-\pi c^2} < 1$. In particular for $c = 1$: $\Pr(\|\mathbf{x}\| > \sigma \cdot \sqrt{m}) \leq 2^{-2m}$.

Corollary 2. *Let $\Lambda \subset \mathbb{R}^m$ be an m-dimensional lattice and $c > 1/\sqrt{2\pi}$ and $\mathbf{S} \in \mathbb{R}^{m \times m}$ a non-singular matrix, then*

$$\Pr(\|\mathcal{D}_{\Lambda,\mathbf{S}}\| \geq \mathfrak{s}_{\max}(\mathbf{S}) \cdot c \cdot \sqrt{m}) \leq \beta_m(c),$$

where $\beta_m(c) = (c \cdot \sqrt{2\pi e} \cdot e^{-\pi c^2})^m$.

Proof. Since $\mathbf{S}$ is square and non-singular, by Lemma 18 it holds that $\mathbf{S}^{-1} \cdot \mathcal{D}_{\Lambda,\mathbf{S}}$ is the same distribution as $\mathcal{D}_{\mathbf{S}^{-1}\Lambda,1}$. Now, by Lemma 23, we have $\Pr(\|\mathcal{D}_{\mathbf{S}^{-1}\Lambda,1}\| > c \cdot \sqrt{m}) \leq (c \cdot \sqrt{2\pi e} \exp(-\pi \cdot c^2))^m$. Then $\|\mathcal{D}_{\Lambda,\mathbf{S}}\| = \|\mathbf{S} \cdot \mathbf{S}^{-1} \cdot \mathcal{D}_{\Lambda,\mathbf{S}}\| \leq \mathfrak{s}_{\max}(\mathbf{S}) \cdot \|\mathbf{S}^{-1} \cdot \mathcal{D}_{\Lambda,\mathbf{S}}\| \leq \mathfrak{s}_{\max}(\mathbf{S}) \cdot c \cdot \sqrt{m}$ with probability at least $1 - (c \cdot \sqrt{2\pi e} \exp(-\pi \cdot c^2))^m$. □

Lemma 24 (Adapted from [MP12, Lem. 2.6])**.** *Let $\Lambda \subseteq \mathbb{R}^m$ be a lattice, $\mathfrak{s}_{\min}(\sqrt{\boldsymbol{\Sigma}}) \geq \eta_\varepsilon(\Lambda)$ for some $\varepsilon > 0$, and $\mathbf{c} \in \mathrm{span}(\Lambda)$. We have*

$$\Pr\left[\left\|\mathcal{D}_{\Lambda+\mathbf{c},\sqrt{\boldsymbol{\Sigma}}}\right\| \geq \mathfrak{s}_{\max}(\boldsymbol{\Sigma}) \cdot \sqrt{m}\right] \leq 2^{-m} \cdot \frac{1+\varepsilon}{1-\varepsilon}.$$

Lemma 25 ([Pei08, Corollary 5.3] adapted)**.** *Let Λ be an m-dimensional lattice and $\sigma > 0$. Then, for $\mathbf{x} \leftarrow \mathcal{D}_{\Lambda,\sigma}$*

$$\Pr[\|\mathbf{x}\|_\infty \geq \sigma \cdot t] \leq 2em \cdot e^{-\pi t^2}.$$

We use the following results by Lyubashevsky [Lyu12] and implications thereof.

Lemma 26 (Lemma 4.3 in [Lyu12])**.** *Let $\mathbf{v} \in \mathbb{R}^d$ and $\mathbf{z} \leftarrow D_{\mathbb{Z}^d,\sigma}$ then*

$$\Pr\left(\left|\mathbf{z}^\mathsf{T} \cdot \mathbf{v}\right| > \tau\sigma\|\mathbf{v}\| \cdot (2\pi)^{-1/2}\right) \leq 2e^{-\tau^2/2}. \tag{2.5}$$

In particular, if $\tau \geq \sqrt{2(\lambda+1)\ln 2}$, then the RHS of Equation (2.5) is at most $2^{-\lambda}$.

Lemma 27 (Lemma 4.4 in [Lyu12], extended)**.** *For any $\tau > 1$ and any lattice $\Lambda \subseteq \mathbb{R}^d$,*

$$\Pr\left(\|\mathbf{z}\| > \tau\sigma\sqrt{d} \cdot (2\pi)^{-1/2} \mid \mathbf{z} \leftarrow D_{\Lambda,\sigma}\right) \leq C^d. \tag{2.6}$$

*where $C = \tau \cdot e^{\frac{1}{2}(1-\tau^2)} < 1$. If $\tau \geq 1 + \frac{\lambda \ln 2}{3d}$ **or** $\tau \geq 1 + \sqrt{\frac{4\lambda\ln 2}{d}}$, we have $C^d \leq 2^{-\lambda}$.*

The original formulation of [Lyu12, Lemma 4.4] takes $\Lambda = \mathbb{Z}^d$, but we can see that its proof goes through for any lattice $\Lambda \subset \mathbb{R}^d$.

Lemma 28. *Let $\mathbf{u} \in \mathcal{R}^k$ and $\mathbf{v} \leftarrow \mathcal{D}^k_{\tau(\mathcal{R}),\sigma}$ where $\mathcal{R}$ is embedded into the real space via the coefficient embedding $\tau(\mathcal{R})$. Then for any $t > 0$*

$$\mathbb{P}\left[\left\|\mathbf{u}^\mathsf{T}\mathbf{v}\right\|_\infty > \|\mathbf{u}\|\sigma\sqrt{\frac{1}{\pi}\ln\left(\frac{2d}{t}\right)}\right] \leq t$$

Proof. Denote $\mathbf{z} \coloneqq \mathbf{u}^\mathsf{T}\mathbf{v}$. Then $\tau(\mathbf{z}) = \mathrm{Rot}(\mathbf{u}^\mathsf{T}) \cdot \tau(\mathbf{v})$. Hence, all coefficients of $\mathbf{z}$ are of the form $z_i = \tilde{\mathbf{u}}_i^\mathsf{T} \cdot \tau(\mathbf{v})$, $i \in [d \cdot k]$ where $\tilde{\mathbf{u}}_i \in \mathbb{Z}^{d\cdot k}$ is a permutation of $\tau(\mathbf{u})$ with some coefficients multiplied by -1. From Lemma 26, we have:

$$\mathbb{P}\left[|\langle\tilde{\mathbf{u}}_i, \tau(\mathbf{v})\rangle| > r\right] \leq 2e^{-\frac{\pi r^2}{\|\mathbf{u}_i\|^2\sigma^2}}$$

By union bound and since $\forall i \in [d], \|\tilde{\mathbf{u}}_i\| = \|\mathbf{u}\|$:

$$\mathbb{P}\left[\left\|\mathbf{u}^\mathsf{T}\mathbf{v}\right\|_\infty > r\right] \leq 2d \cdot e^{-\frac{\pi r^2}{\|\mathbf{u}\|^2\sigma^2}}.$$

Given a fixed $t > 0$ solving $t = 2d \cdot e^{-\frac{\pi r^2}{\|\mathbf{u}\|^2\sigma^2}}$ for r gives the desired result. □

Corollary 3 follows directly from Lemma 26.

Corollary 3. *Let $\mathbf{a} \in \mathbb{Z}^d$ be an arbitrary vector. For $\mathbf{x} \leftarrow D_{\tau(\mathcal{R}),\sigma}$ where $\mathcal{R}$ is embedded into the real space via the coefficient embedding $\tau(\mathcal{R})$, the following holds with probability at least $1 - 2^{-\lambda}$:*

$$\|\mathbf{a}+\mathbf{x}\|^2 \leq \|\mathbf{a}\|^2 + 2 \cdot \sqrt{(\lambda+1)\ln 2 \cdot \pi^{-1}}\,\|\mathbf{a}\|\,\sigma + \|\mathbf{x}\|^2 \tag{2.7}$$

$$\sim \|\mathbf{a}\|^2 + \|\mathbf{x}\|^2 \qquad \text{when } \lambda = o(d), \tag{2.8}$$

in the sense of Landau symbols $\sim$ and $o(\cdot)$.

Proof. We start from the identity $\|\mathbf{a}+\mathbf{x}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{x}\|^2 + 2 \cdot \tau(\mathbf{a})^{\mathsf{T}}\tau(\mathbf{x})$, then apply Lemma 26 on the last summand. □

Gaussian weight of lattice cosets. In this thesis it will often be useful to approximate the Gaussian weight of a lattice coset as discussed below.

Lemma 29 ([MP12, Lemma 2.5])**.** *Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice, $\mathbf{\Sigma} > 0$, and $\mathbf{c} \in \mathbb{R}^n$. It holds that $\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda+\mathbf{c}) \leq \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda)$. Furthermore, if $\sqrt{\mathbf{\Sigma}} \geq \eta_\varepsilon(\Lambda)$ for some $\varepsilon > 0$ and $\mathbf{c} \in \operatorname{span}(\Lambda)$, then $\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda+\mathbf{c}) \geq \frac{1-\varepsilon}{1+\varepsilon} \cdot \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda)$.*

The following is a direct corollary of Lemma 29.

Corollary 4 (Generalisation of [BF11b, Lemma 4.4])**.** *Let $\mathbf{A} \in \mathcal{R}_q^{n\times m}$ be surjective and $\sqrt{\mathbf{\Sigma}} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ for some $\varepsilon > 0$ for the given embedding of the lattice. For any $\mathbf{v} \in \mathcal{R}_q^n$, it holds that*

$$\frac{1-\varepsilon}{1+\varepsilon} \cdot \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_q^\perp(\mathbf{A})) \leq \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_q^{\mathbf{v}}(\mathbf{A})) \leq \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_q^\perp(\mathbf{A})).$$

Furthermore, if $\varepsilon = \mathsf{negl}(\lambda)$, then

$$\rho_{\sqrt{\mathbf{\Sigma}}}(\mathcal{R}^m) = \left|\mathcal{R}_q^n\right| \cdot \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_q^\perp(\mathbf{A})) \cdot (1 - \mathsf{negl}(\lambda)).$$

Then by Corollary 4 and $\rho_{\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}})}(\mathcal{R}^m) \leq \rho_{\sqrt{\mathbf{\Sigma}}}(\mathcal{R}^m)$ the following holds.

Corollary 5. *Let $\mathbf{A} \in \mathcal{R}_q^{n\times m}$ be surjective and $\sqrt{\mathbf{\Sigma}} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ for some $\varepsilon > 0$ for the chosen embedding of the lattice. Let $\sigma^* = \mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}})$. If $\varepsilon = \mathsf{negl}(\lambda)$, then*

$$\rho_{\sigma^*}(\mathcal{R}^m) \leq \left|\mathcal{R}_q^n\right| \cdot \rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_q^\perp(\mathbf{A})) \cdot (1 - \mathsf{negl}(\lambda)).$$

Lemma 30 ([Reg05, Claim 3.8])**.** *For any n-dimensional lattice Λ, $\varepsilon > 0$, $\sigma \geq \eta_\varepsilon(\Lambda)$, and $\mathbf{c} \in \mathbb{R}^n$, it holds that $\rho_\sigma(\Lambda+\mathbf{c}) \in [1 \pm \varepsilon] \cdot \sigma^n \cdot \det(\Lambda)^{-1}$. Furthermore, when $\mathbf{c} = \mathbf{0}$, $\rho_\sigma(\Lambda) \geq \sigma^n \cdot \det(\Lambda)^{-1}$.*

Adapting this statement to non-spherical discrete Gaussians we get the following.

Corollary 6. *Let Λ be an n-dimensional lattice and $\sqrt{\mathbf{\Sigma}} \geq \eta_\varepsilon(\Lambda)$. Then*

$$\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda) \in [1-\varepsilon, 1+\varepsilon] \cdot \frac{\det(\sqrt{\mathbf{\Sigma}})}{\det(\Lambda)}$$

Proof. If $\sqrt{\mathbf{\Sigma}} \geq \eta_\varepsilon(\Lambda)$ then $1 \geq \eta_\varepsilon(\sqrt{\mathbf{\Sigma}}^{-1} \cdot \Lambda)$. By Lemma 30 the statement follows. □

Then we state one more direct corollary of Lemma 30.

Corollary 7 (Generalisation of [ABB10b, Full version, Lemma 21])**.** *For any $i \geq 0$, $\varepsilon > 0$, $\sigma \geq \eta_\varepsilon(\mathcal{R}^i)$ for a given choice of embedding, it holds that*

$$\sigma^{di} \cdot \det(\mathcal{R}^i)^{-1} \leq \rho_\sigma(\mathcal{R}^i) \leq (1+\varepsilon) \cdot \sigma^{di} \cdot \det(\mathcal{R}^i)^{-1}.$$

The following lemma gives a lower bound on min-entropy of a discrete gaussian sample.

Lemma 31 (Adapted from [PR06, Lem. 2.10])**.** *Let Λ be a rank k lattice in $\mathbb{R}^n$. Let $\varepsilon > 0$ and $\sqrt{\mathbf{\Sigma}} \geq 2 \cdot \eta_\varepsilon(\Lambda)$. Let $\mathbf{y} \in \Lambda$ then*

$$\Pr_{\mathbf{y}' \leftarrow \mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}}}[\mathbf{y}' = \mathbf{y}] \leq 2^{-k} \cdot \frac{1+\varepsilon}{1-\varepsilon} \ .$$

Transformations of discrete Gaussians. In the following we argue about the distribution of a linear transformation of a discrete Gaussian, a convolution of two discrete Gaussian vectors and a convolution between a discrete Gaussian and a continuous Gaussian.

Lemma 32 ([GMPW20, Theorem 3.1 (eprint)])**.** *Let $\varepsilon \in [0,1)$ and $1 \leq n \leq m$ be integers. Let $\Lambda \subset \mathbb{R}^m$ be an m-dimensional lattice, $\mathbf{c} \in \mathbb{R}^m$ and $\mathbf{\Sigma} \in \mathbb{R}^{m \times m}$ be positive definite. For a matrix $\mathbf{T} \in \mathbb{R}^{n \times m}$ such that $\mathrm{span}_{\mathbb{R}}(\Lambda \cap \ker(\mathbf{T})) = \mathrm{span}_{\mathbb{R}}(\ker(\mathbf{T}))$ and $\eta_\varepsilon(\Lambda \cap \ker(\mathbf{T})) \leq \sqrt{\mathbf{\Sigma}}$. It holds*

$$\mathrm{SD}(\mathbf{T} \cdot \mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}}, \mathcal{D}_{\mathbf{T}\cdot\Lambda,\sqrt{\mathbf{\Sigma}'},\mathbf{T}\cdot\mathbf{c}}) \leq \frac{\varepsilon}{1-\varepsilon}$$

where $\mathbf{\Sigma}' = \mathbf{T} \cdot \mathbf{\Sigma} \cdot \mathbf{T}^{\mathrm{T}}$.

Lemma 33 ([Reg09, Claim 3.9])**.** *Let Λ be an n-dimensional lattice, $\varepsilon \in (0, 1/2)$, and $r, s > 0$ such that $\frac{r \cdot s}{\sqrt{r^2+s^2}} \geq \eta_\varepsilon(\Lambda)$. Then*

$$\mathrm{SD}(\mathcal{D}_{\Lambda,r} + \mathcal{D}^n_s, \mathcal{D}^n_{\sqrt{r^2+s^2}}) \leq 4\,\varepsilon.$$

Lemma 34 (Adapted [Pei10])**.** *Let $m, \mathbf{\Sigma}_0, \mathbf{\Sigma}_1 > 0$, $\varepsilon \leq 1/2$ s.t. $\sqrt{\mathbf{\Sigma}_0} \geq \eta_\varepsilon(\mathcal{R}^m)$ for the chosen lattice embedding. Denote $\mathbf{\Sigma}_2^{-1} \coloneqq \mathbf{\Sigma}_0^{-1} + \mathbf{\Sigma}_1^{-1}$ and assume that $\sqrt{\mathbf{\Sigma}_2} \geq \eta_\varepsilon(\mathcal{R}^m)$ and $\mathbf{x}_i \leftarrow \mathcal{D}_{\mathcal{R}^m, \sqrt{\mathbf{\Sigma}_i}}$, $i \in [2]$. Then*

$$\mathrm{SD}(\mathbf{x}_0 + \mathbf{x}_1, \mathcal{D}_{\mathcal{R}^m, \sqrt{\mathbf{\Sigma}_0 + \mathbf{\Sigma}_1}}) \leq 8\varepsilon.$$

We prove that sampling a discrete Gaussian vector over a lattice coset can be done in two stages. In the first step we sample a Gaussian over a full ring denote $\mathcal{R}^{m_0}$ below and in the second step we sample the remaining part from a modified coset.

Lemma 35 (Implicit in [CHKP10])**.** *Let module lattices be embedded using the coefficient embedding. Let $\varepsilon \in (0, 1/2)$, and $n, m_0, m_1 > 0$ be integers such that $n < m_0 + m_1$. Let $\mathbf{F} = [\mathbf{F}_0 \mid \mathbf{F}_1] \in \mathcal{R}_q^{n \times (m_0 + m_1)}$, $\mathbf{t} \in \mathcal{R}_q^n$. Let $\mathbf{\Sigma}_0 \in \mathbb{R}^{dm_0 \times dm_0}, \mathbf{\Sigma}_1 \in \mathbb{R}^{dm_1 \times dm_1}$ be positive definite such that $\mathfrak{s}_{\min}(\mathbf{\Sigma}_1) \geq \eta_\varepsilon(\mathbf{F}_1)$. Denote $\mathbf{\Sigma} \coloneqq \begin{bmatrix} \mathbf{\Sigma}_0 & \mathbf{0} \\ \mathbf{0} & \mathbf{\Sigma}_1 \end{bmatrix}$. Then the following distributions are 2ε-statistically close:*

$$\mathcal{D}_0 = \left\{ (\mathbf{z}, \mathbf{x}) \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{t}}(\mathbf{F}), \mathbf{\Sigma}} \right\}$$
$$\mathcal{D}_1 = \left\{ (\mathbf{z}, \mathbf{x}) \mid \mathbf{z} \leftarrow \mathcal{D}_{\mathcal{R}^{m_0}, \mathbf{\Sigma}_0}, \mathbf{x} \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{t} - \mathbf{F}_0 \mathbf{z}}(\mathbf{F}_1), \mathbf{\Sigma}_1} \right\}$$

Lastly, we prove that the distance between a discrete Gaussian sample and its shift is small for both the statistical distance and Rényi divergence with the corresponding conditions.

Lemma 36 (Noise drowning (adapted from [GKPV10]))**.** *For any $\mathbf{c} \in \mathcal{R}^m$, it holds that*

$$\mathrm{SD}\left(\mathcal{D}_{\mathcal{R}^m, \sqrt{\mathbf{\Sigma}}},\ \mathcal{D}_{\mathcal{R}^m, \sqrt{\mathbf{\Sigma}}} + \mathbf{c}\right) \leq O\left(\|\mathbf{c}\| / \mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}})\right).$$

In particular, if $\sqrt{\mathbf{\Sigma}} \geq \lambda^{\omega(1)} \cdot \|\mathbf{c}\|$, then $\mathcal{D}_{\mathcal{R}^m, \sqrt{\mathbf{\Sigma}}} \approx_s \mathcal{D}_{\mathcal{R}^m, \sqrt{\mathbf{\Sigma}}} + \mathbf{c}$ for any choice of the ring embedding.

Lemma 37 ([LSS14, Lemma 4.2])**.** *For any m-dimensional lattice $\Lambda \subseteq \mathbb{R}^m$ and rank m matrix $\sqrt{\mathbf{\Sigma}} \in \mathbb{R}^{m \times m}$, let $P = \mathcal{D}_{\Lambda, \sqrt{\mathbf{\Sigma}}, \mathbf{w}}$ and $Q = \mathcal{D}_{\Lambda, \sqrt{\mathbf{\Sigma}}, \mathbf{z}}$ for some fixed $\mathbf{w}, \mathbf{z} \in \mathbb{R}^m$. If $\mathbf{w}, \mathbf{z} \in \Lambda$, let $\varepsilon = 0$. Otherwise, fix $\varepsilon \in (0, 1)$ and assume that $\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}}) \geq \eta_\varepsilon(\Lambda)$. Then $\mathrm{RD}(P \| Q) \leq (\frac{1+\varepsilon}{1-\varepsilon})^2 \cdot \exp(2\pi \|\mathbf{w} - \mathbf{z}\|^2 / \mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}})^2)$.*

Lemma 38. *Let $\mathcal{K}$ be a number field of degree d. Let $n, m, k \in \mathbb{N}$. Let $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ be such that $\Lambda_q^\perp(\mathbf{A})$ is full-rank, and let $\sqrt{\mathbf{\Sigma}} \in \mathbb{R}^{dm \times dm}$ satisfy $\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}}) \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ for some $0 < \varepsilon \leq 2^{-2k}$ for the chosen embedding. Let $\mathbf{e}_i \in \mathcal{R}^m$ be the ith unit vector*

for $i \in [k]$. The Rényi divergence between the distributions $P := \prod_{i\in[k]} \mathcal{D}_{\Lambda_q^\perp(\mathbf{A}),\sqrt{\mathbf{\Sigma}}}$ and $Q := \prod_{i\in[k]} \mathcal{D}_{\Lambda_q^\perp(\mathbf{A}),\sqrt{\mathbf{\Sigma}},\mathbf{e}_i}$ is upper-bounded by

$$\mathrm{RD}(P\|Q) \le 5\exp\left(\frac{2\pi\cdot dk}{\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}})^2}\right).$$

Proof. Denote by $1+\delta := (1+\varepsilon)/(1-\varepsilon)$, equivalently $\delta = 2\varepsilon/(1-\varepsilon)$. Using the multiplicative property of Rényi divergence, we have

$$\begin{aligned}\mathrm{RD}(P\|Q) &\le \prod_{i\in[k]} (1+\delta)^2 \cdot \exp\left(2\pi\|\mathbf{e}_i\|^2/\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}})^2\right)\\ &\le (1+(2^{2k}-1)\cdot\delta)\cdot\exp\left(\frac{2\pi\cdot dk}{\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}})^2}\right) \quad \le 5\cdot\exp\left(\frac{2\pi\cdot dk}{\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}})^2}\right),\end{aligned}$$

where the first inequality is by Lemma 37, the second uses $(1+x)^r \le 1+(2^r-1)x$ for any $x\in[0,1]$ and $r\in\mathbb{R}\setminus(0,1)$, and the last is by $2^{2k}\cdot\delta = 2^{2k}\cdot 2\varepsilon/(1-\varepsilon) \le 4$.

□

The following result argues about the distribution of discrete Gaussians module sublattices. It is usually referenced as [GPV08, Corollary 2.8], but was already implicit in [MR07]. The version for non-spherical discrete Gaussians we state is for instance proven in [Jeu24, Lemma 1.25].

Lemma 39. *Let Λ, Λ' be m-dimensional lattices such that $\Lambda' \subseteq \Lambda$, let $\varepsilon \in (0,1)$, $\mathbf{\Sigma} \in \mathbb{R}^{m\times m}$ be symmetric positive-definite and $\mathbf{c} \in \mathbb{R}^m$. Let $(\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}} \bmod \Lambda')$ denote the distribution on Λ/Λ' obtained by sampling a vector from $\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}}$ and reducing the sample modulo Λ'. Then for any $\sqrt{\mathbf{\Sigma}} \ge \eta_\varepsilon(\Lambda')$ and $\forall \mathbf{x} \in \Lambda/\Lambda'$:*

$$\Pr[\mathbf{x} \leftarrow \mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}} \bmod \Lambda'] \in \left[\frac{1-\varepsilon}{1+\varepsilon}, \frac{1+\varepsilon}{1-\varepsilon}\right]\frac{1}{|\Lambda/\Lambda'|}.$$

Corollary 8. *Let Λ, Λ' be m-dimensional lattices such that $\Lambda' \subseteq \Lambda$, let $\varepsilon \in (0,1/2)$, $\mathbf{\Sigma} \in \mathbb{R}^{m\times m}$ be symmetric positive-definite and $\mathbf{c} \in \mathbb{R}^m$. Then for any $\sqrt{\mathbf{\Sigma}} \ge \eta_\varepsilon(\Lambda')$*

$$\mathrm{SD}((\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}} \bmod \Lambda'), \mathcal{U}(\Lambda/\Lambda')) \le \varepsilon/(1-\varepsilon).$$

Note that for $\varepsilon \in (0,1/2)$ we have $\varepsilon/(1-\varepsilon) \le 2\varepsilon$.

Corollary 9 ([BLL+15, Lemma 2.10] adapted)**.** *Let Λ, Λ' be m-dimensional lattices such that $\Lambda' \subseteq \Lambda$, let $\varepsilon \in (0,1)$, $\mathbf{\Sigma} \in \mathbb{R}^{m\times m}$ be symmetric positive-definite and $\mathbf{c} \in \mathbb{R}^m$. Then for any $\sqrt{\mathbf{\Sigma}} \ge \eta_\varepsilon(\Lambda')$*

$$\mathrm{RD}((\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}} \bmod \Lambda'); \mathcal{U}(\Lambda/\Lambda')) \le \mathrm{RD}_\infty((\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}},\mathbf{c}} \bmod \Lambda'); \mathcal{U}(\Lambda/\Lambda')) \le \frac{1+\varepsilon}{1-\varepsilon}.$$

We remark that [Pre17, GJK24] propose a different formulation of the above corollary, making use of the notion of relative error. However, their bound reflects an asymptotic behaviour, whereas the one we use holds in the worst case.

Properties of module lattices. We state a number of useful properties of module and ideal lattices.

Lemma 40 (Adapted from [MG02, Lemma 7.1])**.** *Let $\mathcal{M} \subset \mathcal{K}^r$ be a full-rank module, and $\mathbf{v}_1, \ldots, \mathbf{v}_r \in \mathcal{M}$ be a $\mathcal{K}$ linearly independent set of vectors. Then there exists a matrix $\mathbf{B} \in \mathbb{R}^{dr \times dr}$ that generates $\theta(\mathcal{M})$ as a $\mathbb{Z}$-lattice satisfying*

$$\|\mathbf{B}\| \leq \sqrt{rd} \cdot \delta_{\mathcal{K}} \cdot \max \|\mathbf{v}_i\|.$$

Lemma 41. *Let $\mathcal{M} \subset \mathcal{K}^r$ be a full-rank module, let $\mathbf{b}_1, \ldots, \mathbf{b}_r \in \mathcal{M}$, $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_r]$ and $\mathfrak{p}$ a prime ideal of $\mathcal{R}$ with residue field $F = \mathcal{R}/\mathfrak{p}$. Then $\det_{\mathcal{K}}(\mathbf{B}) \in \det(\mathcal{M})$, and furthermore it holds that $\det_F(\mathbf{B} \bmod \mathfrak{p}) = \det_{\mathcal{K}}(\mathbf{B}) \bmod (\det(\mathcal{M}) \cdot \mathfrak{p})$.*

Proof. The fact that $\det_{\mathcal{K}}(\mathbf{B}) \in \det(\mathcal{M})$ comes from the fact that $\mathbf{B} \cdot \mathcal{R}^r \subseteq \mathcal{M}$ and that $\det_{\mathcal{K}}(\mathbf{B}) \cdot \mathcal{R} = \det(\mathbf{B} \cdot \mathcal{R}^r)$. This implies that $\det(\mathbf{B}) \bmod \det(\mathcal{M}) \cdot \mathfrak{p}$ is well-defined. Now let us define the function $\phi(\mathbf{b}_1, \ldots, \mathbf{b}_r) := \det_{\mathcal{K}}(\mathbf{B}) \bmod \det(\mathcal{M}) \cdot \mathfrak{p}$. This function is a multilinear map satisfying that for any $\mathbf{m} \in \mathcal{M}$ and $p \in \mathfrak{p}$, $\phi(\mathbf{b}_1 + p \cdot \mathbf{m}, b_2, \ldots, b_r) = \phi(\mathbf{B}) + p \cdot \det_{\mathcal{K}}(\mathbf{m}, \mathbf{b}_2, \ldots, \mathbf{b}_r) = \phi(\mathbf{B}) \bmod \mathfrak{p} \det(\mathcal{M})$, so it can be lifted as a multilinear map $\phi' : (\mathcal{M}/\mathfrak{p}\mathcal{M})^r \to F$. The multilinear map ϕ' is alternating and clearly satisfies $\phi'(\mathbf{I}_r) = 1$, it is then equal to the determinant over $(\mathcal{M}/\mathfrak{p}\mathcal{M})^r$. □

Lemma 42. *Let $a, b \in \mathcal{R}$ be two coprime elements. Then there exist $u, v \in \mathcal{R}$ with $\|u\|, \|v\| \leq \max(\|a\|_\infty, \|b\|_\infty) \cdot \sqrt{d} \cdot \lambda_d(\mathcal{R})$ such that $a \cdot u + b \cdot v = 1$.*

Proof. Without loss of generality, assume that $\|a\|_\infty \leq \|b\|_\infty$. Let $b_1, \ldots, b_d$ be a $\mathbb{Z}$-basis of $\mathcal{R}$ satisfying $\|b_i\| \leq \max(1, \sqrt{i}/2) \cdot \lambda_i(\mathcal{R})$. Let M be the rank-1 module in $\mathcal{K}^2$ generated by $(b_i \cdot (-b, a)^\mathsf{T})_{1 \leq i \leq d}$. The norm of the elements of its basis are bounded by $\lambda_d(\mathcal{R}) \cdot \|a\|_\infty$, hence the covering radius of M is at most $\sqrt{d} \cdot \lambda_d(\mathcal{R}) \cdot \|a\|_\infty$. Let $u_0, v_0 \in \mathcal{R}$ such that $a \cdot u_0 + b \cdot v_0 = 1$ and let (u, v) be the closest element of M to (u_0, v_0), then it holds that $a \cdot (u_0 - u) + b \cdot (v_0 - v) = 1$ and that $\|(u_0 - u, v_0 - v)\| \leq \sqrt{d} \cdot \lambda_d(\mathcal{R}) \cdot \|a\|_\infty$, hence the result. □

We can bound the shortest vector and the smoothing parameter of an ideal lattice using the algebraic norm of the corresponding ideal.

Lemma 43 ([LPR13, Lemma 2.14])**.** *Let $\mathcal{K}$ be a number field of degree d. For any fractional ideal $\mathcal{I}$ of $\mathcal{K}$*

$$\sqrt{d} \cdot N(\mathcal{I})^{1/d} \leq \lambda_1(\theta(\mathcal{I})).$$

Using Equation 2.1, we obtain

$$\sqrt{d} \cdot N(\mathcal{I})^{1/d} / \mathfrak{s}_{\max}(\mathfrak{f}) \leq \lambda_1(\tau(\mathcal{I})).$$

Lemma 44 ([PR07, Lemma 6.5] adapted)**.** *Let $\varepsilon = 2^{-d}$. For any ideal $\mathcal{I}$ in degree d ring $\mathcal{R}$ we have*

$$\eta_\varepsilon(\theta(\mathcal{I})) \leq (N(\mathcal{I}) \cdot |\Delta_\mathcal{K}|)^{1/d}.$$

Lemma 45 ([LPR13, Claim 7.1])**.** *Let $r, \varepsilon > 0$. For any n-dimensional lattice Λ*

$$\rho_{1/r}(\Lambda) \leq \max\left(1, \left(\frac{\eta_\varepsilon(\Lambda^*)}{r}\right)^n\right)(1+\varepsilon)$$

In particular for any fractional ideal $\mathcal{I} \subset \mathcal{K}$ we have

$$\rho_{1/r}(\theta(\mathcal{I})) \leq \max\left(1, N(\mathcal{I})^{-1} r^{-d}\right)(1+2^{-2d})$$
$$\rho_{1/r}(\tau(\mathcal{I})) \leq \max\left(1, N(\mathcal{I})^{-1} \mathfrak{s}_{\max}(\mathfrak{f})^d \cdot r^{-d}\right)(1+2^{-2d})$$

Remark 3. To obtain the bound for the ideals we apply Lemma 22 and Lemma 43 to the general statement.

For both embeddings we know a useful description for the dual of the q-ary kernel lattice.

Lemma 46 ([RSW18, Lemma 5.1] adapted)**.** *Let $\mathcal{K}$ be a number field with $\mathcal{R}$ its ring of integers. Further, let $q \geq 2$ be a modulus defining $\mathcal{R}_q$. Let $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ for integers $0 < n < m$. Then with respect to the canonical embedding θ we have*

$$\theta(\Lambda_q^\perp(\mathbf{A}))^* = \theta\left((\mathcal{R}^*)^m + \left\{1/q \cdot \mathbf{A}^\mathsf{T} \cdot \mathbf{s} \mid \mathbf{s} \in (\mathcal{R}^*)^n\right\}\right),$$

With respect to the coefficient embedding τ using the transformation to integer lattices we get

$$\tau(\Lambda_q^\perp(\mathbf{A}))^* = \Lambda_q^\perp(\mathrm{Rot}(\mathbf{A}))^* = 1/q \cdot \Lambda_q(\mathrm{Rot}(\mathbf{A})^\mathsf{T}).$$

2.5 Hard Problems

The constructions in this rely on cryptographic assumptions we introduce in this section.

Definition 2 (MSIS Problem)**.** *Let the parameters* $\mathsf{par} = (\mathcal{R}, q, n, m, \beta)$ *be functions* λ*, where* $\mathcal{R}$ *is the ring of integers of a number field* $\mathcal{K}$*,* q, n, m *are positive integers and* $\beta > 0$*. An* $\mathsf{MSIS}_{\mathsf{par}}$ *problem asks the following: Given a uniform matrix* $\mathbf{A} \leftarrow \mathcal{R}_q^{n \times m}$*, find a vector* $\mathbf{u} \in \mathcal{R}^m$ *such that*

$$(1) \quad \mathbf{A} \cdot \mathbf{u} = \mathbf{0} \bmod q \qquad \text{and} \qquad (2) \quad 0 < \|\mathbf{u}\| \leq \beta.$$

When the target vector is $\mathbf{t} \leftarrow \mathcal{U}(\mathcal{R}_q^n)$ *as in the adversary solves* $\mathbf{A} \cdot \mathbf{u} = \mathbf{t} \bmod q$ *we get the inhomogeneous* MSIS *problem denoted* $\mathsf{MISIS}_{\mathsf{par}}$*. For any PPT* $\mathcal{A}$*, we denote the probability of* $\mathcal{A}$ *solving the problem as* $\mathsf{Adv}_{\mathcal{A}}^{\mathsf{MSIS}_{\mathsf{par}}}(\lambda)$ *(resp.* $\mathsf{Adv}_{\mathcal{A}}^{\mathsf{MISIS}_{\mathsf{par}}}(\lambda)$*).*

The hardness of MISIS problem follows from that of MSIS. The latter was studied in [LS15] and is considered hard at least for q is polynomial in nd and $\beta \leq q/\sqrt{nd} \cdot \omega(\log(nd))$.

Definition 3 (MLWE Problem)**.** *Let the parameters* $\mathsf{par} = (\mathcal{R}, q, n, m, \chi_s, \chi_e)$ *be functions of* λ*, where* $\mathcal{R}$ *is the ring of integers of a number field* $\mathcal{K}$*,* q, n, m *are positive integers and* χ_s, χ_e *are distributions over* $\mathcal{R}$*. An* $\mathsf{MLWE}_{\mathsf{par}}$ *problem asks the following: Given a uniform matrix* $\mathbf{A} \leftarrow \mathcal{R}_q^{n \times m}$*, and a vector* $\mathbf{b} \in \mathcal{R}_q^m$ *distinguish if*

$$\begin{aligned} &(1) \quad \mathbf{b} = \mathbf{A}^{\mathsf{T}}\mathbf{s} + \mathbf{e} \bmod q \text{ where } \mathbf{s} \leftarrow \chi_s^n,\ \mathbf{e} \leftarrow \chi_e^m, \\ \text{or} \quad &(2) \quad \mathbf{b} \leftarrow \mathcal{R}_q^m \text{ is uniformly random.} \end{aligned}$$

For any PPT $\mathcal{A}$*, its advantage equals* $\mathsf{Adv}_{\mathcal{A}}^{\mathsf{MLWE}_{\mathsf{par}}}(\lambda) \coloneqq \mathsf{Adv}_{\mathcal{A}}((\mathbf{A}, \mathbf{A}^{\mathsf{T}}\mathbf{s} + \mathbf{e}), (\mathbf{A}, \mathcal{U}(\mathcal{R}_q^m)))$*. When* $n = 1$ *this problem is called Ring LWE and we denote it as* $\mathsf{RLWE}_{\mathcal{R},q,m,\chi_s,\chi_e} = \mathsf{MLWE}_{\mathcal{R},q,1,m,\chi_s,\chi_e}$*.*

The MLWE problem is considered hard at least for $\chi_s, \chi_e \coloneqq \mathcal{D}_{\mathcal{R},\sigma}^2$ when σ and q are polynomial in d [SSTX09, LPR10]. By a hybrid argument we also have that $\{a \cdot s_j + e_j\}_{j \in \mathsf{poly}(\lambda)}$ for a fixed a and varying $s_j, e_j \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^2$ is indistinguishable from uniform, cf. [PW08]. We refer to this setting as multi-instance $\mathsf{RLWE}_{\mathcal{R},q,m,\mathcal{D}_{\mathcal{R},\sigma},\mathcal{D}_{\mathcal{R},\sigma}}$.

Definition 4 (k-MSIS Problem [BF11b])**.** *Fix an arbitrary choice of embedding for the ring elements and module lattices. Let the parameters* $\mathsf{par} = (\mathcal{R}, q, n, m, k, \sqrt{\mathbf{\Sigma}}, \beta)$ *be functions of* λ*, where* $\mathcal{R}$ *is the ring of integers of a number field* $\mathcal{K}$ *of degree* d*,* q, n, m, k *are positive integers with* $k < m$*,* $\mathbf{\Sigma} \in \mathbb{R}^{dm \times dm}$ *is positive definite, and* $\beta > 0$*. Let* $\mathcal{R}_q \coloneqq \mathcal{R}/(q \cdot \mathcal{R})$*. A* $\mathsf{k\text{-}SIS}_{\mathsf{par}}$ *problem asks the following: Given a uniform matrix* $\mathbf{A} \leftarrow \mathcal{R}_q^{n \times m}$ *and* k *vectors* $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_k) \in \mathcal{R}^{m \times k}$ *where* $\mathbf{x}_i \leftarrow \mathcal{D}_{\Lambda_q^{\perp}(\mathbf{A}), \sqrt{\mathbf{\Sigma}}}$*,* $i \in [k]$*, find a vector* $\mathbf{u} \in \mathcal{R}^m$ *such that*

$$(1) \quad \mathbf{A} \cdot \mathbf{u} = \mathbf{0} \bmod q, \qquad (2) \quad 0 < \|\mathbf{u}\| < \beta, \qquad \text{and} \qquad (3) \quad \mathbf{u} \notin \mathcal{K}\text{-span}(\mathbf{X}).$$

For any PPT $\mathcal{A}$, its advantage $\mathsf{Adv}^{\mathsf{k\text{-}SIS}_{\mathsf{par}}}_{\mathcal{A}}(\lambda)$ is the probability of finding a correct solution.

The work of [BF11b] defines and proves hardness of the integer version of this assumption. In Chapter 4 we prove that the assumption above has a reduction from MSIS for a selected range of parameters. Similarly to k-MSIS we introduce k-MLWE.

Definition 5 (k-MLWE Problem [LPSS14])**.** *Fix an arbitrary choice of embedding for the ring elements and module lattices. Let $\mathsf{par} = (\mathcal{R}, q, n, m, k, \mathbf{\Sigma}, \chi_s, \chi_e)$ be functions of λ, where $\mathcal{R}$ is the ring of integers of a number field $\mathcal{K}$ of degree d,q, n, m, k are positive integers with $k < m$, $\mathbf{\Sigma} \in \mathbb{R}^{dm \times dm}$ is positive definite, and χ_s, χ_e are distribution over $\mathcal{R}$. A $\mathsf{k\text{-}MLWE}_{\mathsf{par}}$ problem asks the following: Given a uniform matrix $\mathbf{A} \leftarrow \mathcal{R}_q^{n \times m}$, k vectors $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_k) \in \mathcal{R}^{m \times k}$ where $\mathbf{x}_i \leftarrow \mathcal{D}_{\Lambda_q^\perp(\mathbf{A}), \sqrt{\mathbf{\Sigma}}}$ for each $i \in [k]$, and a vector $\mathbf{b} \in \mathcal{R}_q^m$ distinguish if*

$$\begin{aligned} &(1) \quad \mathbf{b} = \mathbf{A}^{\mathsf{T}}\mathbf{s} + \mathbf{e} \bmod q \text{ where } \mathbf{s} \leftarrow \chi_s^n,\ \mathbf{e} \leftarrow \chi_e^m, \\ \text{or} \quad &(2) \quad \mathbf{b} = \mathbf{d} + \mathbf{e} \bmod q \text{ where } \mathbf{d} \leftarrow \left\{\mathbf{d} \in \mathcal{R}^m : \mathbf{d}^{\mathsf{T}} \cdot \mathbf{X} = \mathbf{0} \bmod q\right\},\ \mathbf{e} \leftarrow \chi_e^m. \end{aligned}$$

For any PPT $\mathcal{A}$, its advantage is $\mathsf{Adv}^{\mathsf{k\text{-}MLWE}_{\mathsf{par}}}_{\mathcal{A}}(\lambda) \coloneqq \mathsf{Adv}_{\mathcal{A}}((\mathbf{A}, \mathbf{A}^{\mathsf{T}}\mathbf{s} + \mathbf{e}), (\mathbf{A}, \mathbf{d} + \mathbf{e}))$.

Similarly, the integer version of this assumption was defined and proved standard in [LPSS14]. The module version above is discussed in Chapter 4. Parts of the work rely on the NTRU and Module-NTRU assumptions.

Definition 6 (NTRU)**.** *Let $q \geq 2 \in \mathbb{Z}$ and $\beta > 0$ a real number. A $(\mathcal{R}_q, \chi_f, \chi_g)$-NTRU instance is an element $h \in \mathcal{R}_q$ computed as follows*

i Sample $(f, g) \leftarrow (\chi_f, \chi_g)$.

ii If there exists non-zero $h \in \mathcal{R}_q$ s.t. $g \cdot h = f \bmod q$ return h.

iii Else restart the procedure.

The $(\mathcal{R}_q, \chi_f, \chi_g)$-NTRU problem, denoted $\mathsf{NTRU}_{\mathcal{R}, q, \chi_f, \chi_g}$, asks to distinguish an NTRU instance from uniform. For a PPT adversary $\mathcal{A}$ we define $\mathsf{Adv}^{\mathsf{NTRU}}(\mathcal{A}) \coloneqq \mathsf{Adv}_{\mathcal{A}}(f/g \bmod q, \mathcal{U}(\mathcal{R}_q))$.

When $\chi_f = \chi_g = \mathcal{D}_{\mathcal{R}, \sigma}$ and when $\sigma < \sqrt{q}$ the problem is considered to be computationally hard for some choices of parameters [HPS98, ABD16, CJL16, KF17, DvW21]. In particular, if $\sigma \ll \sqrt{q}$ then distinguishing an NTRU instance from uniform is exponential in $\tilde{\Theta}(d \cdot \log(\sigma)/\log^2(q))$ under known algorithms.

Lemma 47 (NTRU Trapdoor quality)**.** *Let $\sigma = O(\sqrt{q})$, there exists a randomised algorithm* $\mathsf{TrapGenNTRU}(1^d, q)$ *that outputs a vector* $\mathbf{a} \coloneqq \begin{bmatrix}1 & a'\end{bmatrix} \in \mathcal{R}_q^2$ *and a full rank matrix* $\mathbf{T_a} \in \mathcal{R}^{2\times 2}$*, where* $\mathbf{T_a}$ *is a basis for* $\Lambda_q^\perp(\mathbf{a})$*. Moreover,* $\|\tau(\mathbf{T_a})\|_{GS} = O(\sqrt{q})$ *and* a' *is indistinguishable from random based on the* $\mathsf{NTRU}_{\mathcal{R},q,\mathcal{D}_{\mathcal{R},\sigma},\mathcal{D}_{\mathcal{R},\sigma}}$ *assumption. In practice we use* $\|\tau(\mathbf{T_a})\|_{GS} = 1.17\sqrt{q}$ *as in [CPS+20].*

We recall the Module-NTRU assumption by [CPS+20]. While [CPS+20] did not formally define it, Definition 7 formalises a number of assumptions made in [CPS+20].

Definition 7 ($\mathsf{ModNTRU}$, adapted from [CPS+20])**.** *Let* $\text{GS_SLACK} \geq 1$*. Let $\mathcal{D}$ be the distribution over $\mathcal{R}^m \times (\mathcal{R}^{m\times m})^\times$ defined as follows: $\forall 1 \leq i \leq m$, the ith row of $\begin{bmatrix}\mathbf{g} & \mathbf{F}\end{bmatrix} \sim \mathcal{D}$ is sampled from a discrete Gaussian $\mathcal{D}_{R^{m+1},\sigma_i}$ with $\sigma_i = \frac{\text{GS_SLACK}}{\sqrt{2\pi\, d(m+2-i)}}\, q^{1/(m+1)}$, conditioned on $\mathbf{F}$ being invertible. A $\mathsf{ModNTRU}_{\mathcal{R},q,m}$ instance is a vector $\mathbf{h} \in \mathcal{R}_q^m$ s.t. $\mathbf{h} = \mathbf{F}^{-1} \cdot \mathbf{g} \bmod q$ for $(\mathbf{g}, \mathbf{F}) \leftarrow \mathcal{D}$. The $\mathsf{ModNTRU}_{\mathcal{R},q,m}$ assumption stipulates:*

1. *A $\mathsf{ModNTRU}_{\mathcal{R},q,m}$ instance is pseudorandom over R_q^m.*
2. *With probability $\Omega(1)$, a pair $(\mathbf{g}, \mathbf{F}) \leftarrow \mathcal{D}$ can be efficiently completed to a basis $\mathbf{T_h}$ of $\Lambda_q^\perp\left(\begin{bmatrix}1 & \mathbf{h}^\mathsf{T}\end{bmatrix}\right)$ that satisfies:*

$$\|\tau(\mathbf{T_h})\|_{GS} \leq \text{GS_SLACK} \cdot q^{1/(m+1)}, \tag{2.9}$$

$$\tag{2.10}$$

In particular, this implies the existence of a PPT algorithm $\mathsf{TrapGenNTRU}(\lambda)$ *that generates a pseudorandom* $\mathsf{ModNTRU}_{\mathcal{R},q,m}$ *instance along with a trapdoor satisfying Item 2. Note that [CPS+20] (implicitly) assumes that the* $\mathsf{ModNTRU}_{\mathcal{R},q,m}$ *assumption holds for* $(m, \text{GS_SLACK}) \in \{(1, 1.17), (2, 1.17), (3, 1.24)\}$.

Remark 4. Combined with Lemma 51, the $\mathsf{ModNTRU}_{\mathcal{R},q,m}$ assumption implies that $\mathbf{T}_h$ is an Ajtai trapdoor for the matrix $\mathbf{H} = \begin{bmatrix}1 & \mathbf{h}^\mathsf{T}\end{bmatrix}$ and the $\mathsf{SamplePre}(\cdot)$ algorithm allows one to efficiently sample $\mathbf{x} \sim_{O(\varepsilon)} \mathcal{D}_{\Lambda_q^t(\mathbf{H}),\mathbf{\Sigma}}$ for any $t \in \mathcal{R}_q$ and $\mathbf{\Sigma}$ such that $\mathfrak{s}_{\min}(\mathbf{\Sigma}) \geq \text{GS_SLACK} \cdot q^{1/(m+1)} \cdot \eta_\varepsilon(\mathcal{R}^{m+1})$

Lastly, we rely on the Hint-MLWE assumption, introduced by Kim et al. [KLSS23]. More specifically, we use a variant by Esgin et al. [EEN+24], recalled in Definition 8. While the distribution of the $\mathbf{c}_i$ in Definition 8 may seem artificial, it is a natural byproduct of the signing procedure in [EEN+24] that we re-use in Chapters 6 and 7 of this thesis.

$\mathcal{P}(\mathbf{s}, \mathbf{e}, \ell)$

$i \in [\ell] : u_i \leftarrow \mathcal{R}_q$

$(c_{0,i}, c_{1,i}) \coloneqq \mathsf{Decomp}_\beta(u_i)$

$\mathbf{p}_i \leftarrow \mathcal{D}_{\tau(\mathcal{R}^{n+m}),\sigma_p}$

$\mathbf{z}_i = \begin{bmatrix} \mathbf{s} \\ \mathbf{e} \end{bmatrix} \cdot c_{0,i} + \mathbf{p}_i$

return $\{(\mathbf{c}_i = (c_{0,i}, c_{1,i}), \mathbf{z}_i)\}_i$

Figure 2.1: Hint distribution for Hint-MLWE

Definition 8 (Hint-MLWE)**.** *Let $m > 0$, $\ell \geq 0$ and q be integers, and $\sigma, \sigma_p > 0$. Let $\mathbf{A} \leftarrow \mathcal{R}_q^{m \times n}$. The $\mathsf{Hint\text{-}MLWE}^{\ell,\sigma_p,\beta}_{\mathcal{R},n,m,q,\sigma}$ problem requires the Adversary to distinguish distributions $\mathcal{D}_{\mathsf{real}}, \mathcal{D}_{\mathsf{ideal}}$ where $\mathbf{s} \leftarrow \mathcal{D}_{\mathcal{R}^n,\sigma}, \mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma}$ and:*

$$\mathcal{D}_{\mathsf{real}} = \Big\{(\mathbf{A}, \mathbf{b} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e}, \{\mathbf{c}_i, \mathbf{z}_i\}_{i=0}^{\ell-1}) \mid \{\mathbf{c}_i, \mathbf{z}_i\}_i \leftarrow \mathcal{P}(\mathbf{s}, \mathbf{e}, \ell)\Big\},$$
$$\mathcal{D}_{\mathsf{ideal}} = \Big\{(\mathbf{A}, \mathbf{b}, \{\mathbf{c}_i, \mathbf{z}_i\}_{i=0}^{\ell-1}) \mid \mathbf{b} \leftarrow \mathcal{R}_q^m, \{\mathbf{c}_i, \mathbf{z}_i\}_i \leftarrow \mathcal{P}(\mathbf{s}, \mathbf{e}, \ell)\Big\},$$

where the distribution $\mathcal{P}(\mathbf{s}, \mathbf{e})$ is described in Figure 2.1.

When the adversary receives no hints ($\ell = 0$) in Definition 8, we recover the $\mathsf{MLWE}_{\mathcal{R},n,m,q,\sigma}$ problem. We note $\mathsf{Hint\text{-}MLWE}^{\ell,\sigma_p,\beta}_{\mathcal{R},1,m,q,\sigma} = \mathsf{Hint\text{-}RLWE}^{\ell,\sigma_p,\beta}_{\mathcal{R},m,q,\sigma}$. When $(\ell, n) = (0, 1)$, we recover the Ring-LWE problem $\mathsf{RLWE}_{\mathcal{R},m,q,\sigma}$. In [KLSS23, EEN+24] the standard hardness of Hint-RLWE is proved for $\ell \geq 1$. We recall the reduction in Lemma 48. Note that there is a factor 2 in the formula for $\frac{1}{\sigma_0^2}$, which seems to be an artifact of the proof.

Lemma 48 (Theorem 1 and Lemma 2 from [EEN+24])**.** *Assume $\ell = \omega(\lambda\, d \log d)^2$, and let $B_{\mathsf{hint}} = \frac{\ell\, d\, M^2}{12}$, where $M = 2\left\lceil \frac{q-1}{2\beta} \right\rfloor + 1$. Let $\sigma_0, \sigma, \sigma_p > 0$ such that $\frac{1}{\sigma_0^2} = 2\left(\frac{1}{\sigma^2} + \frac{B_{\mathsf{hint}}(1+o(1))}{\sigma_p^2}\right)$. If $\sigma_0 \geq \sqrt{2}\eta_\varepsilon(\mathbb{Z}^d)$ for $0 < \varepsilon \leq 1/2$, where $\eta_\varepsilon(\mathbb{Z}^d)$ is the smoothing parameter of $\mathbb{Z}^d$, then there is an efficient reduction from $\mathsf{RLWE}_{\mathcal{R},m,q,\sigma_0}$ to $\mathsf{Hint\text{-}RLWE}^{\ell,\sigma_p,\beta}_{\mathcal{R},m,q,\sigma}$ that reduces the advantage by at most 4ε.*

2.6 Trapdoor Sampling and Other Algorithms

In this thesis we use supplementary algorithms for example for secret sharing or efficient preimage sampling. We present them in detail in this subsection.

Decomposition function. For an odd prime q and β that is a power of 2 we define the function $\mathsf{Decomp}_\beta : \mathbb{Z}_q \to \mathbb{Z} \times \mathbb{Z}$ that identifies $x \in \mathbb{Z}_q$ with its representative in

$\left\{-\frac{q-1}{2}, \ldots, \frac{q-1}{2}\right\}$ and returns the unique pair (c_0, c_1) such that $x = c_0 \cdot \beta + c_1$, $|c_1| \leq \beta/2$ and $|c_0| \leq \left\lceil \frac{q-1}{2\beta} \right\rfloor$. For a cyclotomic number field $\mathcal{K}$, its ring of integers $\mathcal{R}$ and its quotient $\mathcal{R}_q \coloneqq \mathcal{R}/q\mathcal{R}$ we extend the function to $\mathsf{Decomp}_\beta : \mathcal{R}_q \to \mathcal{R} \times \mathcal{R}$ by decomposing the elements coefficientwise. Then $\|c_1\|_\infty \leq \beta/2$ and $\|c_0\|_\infty \leq \left\lceil \frac{q-1}{2\beta} \right\rfloor$ in the coefficient embedding.

Shamir Secret Sharing. Let $\mathbb{F}$ be a finite field. Given $a \in \mathbb{F}$, $1 \leq T$ and a set $\mathcal{S} \subseteq \mathbb{F}^*$, a $(T, \mathcal{S})$-sharing of a is obtained by doing the following:

1. Generate $P \in \mathbb{F}_{<T}[x]$ uniformly at random, conditioned to $P(0) = a$.
2. For $s \in \mathcal{S}$, compute $[\![a]\!]_s^P \coloneqq P(s)$.
3. The output is the indexed tuple $[\![a]\!]_\mathcal{S}^P = ([\![a]\!]_s^P)_{s \in \mathcal{S}}$.

Given $(a, P, \mathcal{S})$, $[\![a]\!]_\mathcal{S}^P$ is uniquely defined. We say that an indexed tuple $[\![a]\!]$ is a valid T-sharing of a if there exists a polynomial $P \in \mathbb{F}_{<T}[x]$ and an evaluation set $\mathcal{S} \subseteq \mathbb{F}^*$ such that $[\![a]\!] = [\![a]\!]_\mathcal{S}^P$. If $|\mathcal{S}| \geq T$, then for any indexed tuple $b = (b_s)_{s \in \mathcal{S}}$ there exists at most a single pair (a, P) such that $b = [\![a]\!]_\mathcal{S}^P$.

When P and/or $\mathcal{S}$ are clear from context, we may omit them and write $[\![a]\!]^P$, $[\![a]\!]_\mathcal{S}$ or $[\![a]\!]$. The set of valid $(T, \mathcal{S})$-sharings is a $\mathbb{F}$-linear space of dimension T. Indeed, given $\lambda, \mu \in \mathbb{F}$:

$$\lambda\, [\![a]\!]_\mathcal{S}^P + \mu\, [\![b]\!]_\mathcal{S}^Q = [\![\lambda\, a]\!]_\mathcal{S}^{\lambda P} + [\![\mu\, b]\!]_\mathcal{S}^{\mu Q} = [\![\lambda\, a + \mu\, b]\!]_\mathcal{S}^{\lambda P + \mu Q} \tag{2.11}$$

Consider a $(T, \mathcal{S})$-sharing $[\![a]\!]_\mathcal{S}$. If $|\mathcal{S}| < T$, then $[\![a]\!]_\mathcal{S}$ leaks no information about the underlying shared value. If $|\mathcal{S}| = T$, then there exists a unique $a \in \mathbb{F}$ that reconstructs from this sharing, and we can recover it using Lagrange interpolation:

$$a = \sum_{s \in \mathcal{S}} \lambda_{s,\mathcal{S}}\, [\![a]\!]_s, \quad \text{where} \quad \lambda_{s,\mathcal{S}} = \prod_{s' \in \mathcal{S} \setminus s} \frac{s'}{s' - s}. \tag{2.12}$$

For a prime q we extend the secret sharing to elements $a = \sum_{i \in [d]} a_i \cdot X^i \in \mathcal{R}_q \sim \mathbb{Z}_q[X]/(X^d + 1)$ by sharing them coefficientwise over the field $\mathbb{Z}_q^\times$.

Vandermonde Secret Sharing of [BCdP+25]. We first recall some definitions related to dictionaries. A dictionary, or associative array, is a collection of ordered (key, value) pairs with unique keys. We note $\{:\}$ the empty dictionary, and $\mathsf{Dict}[k]$ the value of Dict corresponding to the key k. We note $\mathsf{Dict}_1 \cup \mathsf{Dict}_2$ the union of two dictionaries.

When a key k is present in both Dict_1 and Dict_2 and $\mathsf{Dict}_1[k], \mathsf{Dict}_2[k]$ are either both sets or both dictionaries, the new value $(\mathsf{Dict}_1 \cup \mathsf{Dict}_2)[k]$ sets as $\mathsf{Dict}_1[k] \cup \mathsf{Dict}_2[k]$.

We present the algorithms for sharing and recovering a secret value with Vandermonde secret sharing in Algorithms 1 and 2. The high level intuition of the algorithms below is optimising the additive replicated secret sharing from e.g. [dENP25]. The main secret $\mathbf{s}$ is still shared additively $\mathbf{s} = \sum_{i=0}^{T-2} \mathbf{s}_i + (\mathbf{s} - \sum_{i=0}^{T-2} \mathbf{s}_i)$ for $\mathbf{s}_i \leftarrow \chi$ multiple times. However, [BCdP+25] optimises the number of shares per party by placing the users in general user set $\mathcal{P}$ at the leafs of a binary tree. Then for every possible reconstruction set $|\mathsf{act}| = T$ the secret is shared in two additive shares at multiple levels of the tree. If a given party in act is the only one on its side of the tree the share is added to its dictionary under a corresponding key. Otherwise, the share is shared in two again for the next level of the tree.

Algorithm 1 $\mathsf{VandShare}(\mathbf{s}, \mathcal{P}, T, \mathbf{i} = \varnothing) \to \mathsf{Dict}$

$N = |\mathcal{P}|$
if $T = 1$ **then**
  return $\mathsf{Dict} \coloneqq \{\mathcal{U} : \{\mathbf{i} : \mathbf{s}\} \mid \mathcal{U} \in \mathcal{P}\}$
else
  $\mathsf{Dict} = \{\mathcal{U} : \{:\} \mid \mathcal{U} \in \mathcal{P}\}$, $b = \lfloor N/2 \rfloor$
  Parse $\mathcal{P} = \mathcal{P}_L \sqcup \mathcal{P}_R$, with $\mathcal{P}_L$ the b smallest elements of $\mathcal{P}$
  for $k = \max(0, T - N + b), \ldots, \min(b, T)$ **do**
    $\mathbf{i}_L \coloneqq \mathbf{i} \parallel \texttt{":L:"} \parallel k$, $\mathbf{i}_R \coloneqq \mathbf{i} \parallel \texttt{":R:"} \parallel (T - k)$
    if $k = 0$ **then**
      $\mathsf{Dict} \coloneqq \mathsf{Dict} \cup \mathsf{VandShare}(\mathbf{s}, \mathcal{P}_R, T, \mathbf{i}_R)$
    else if $k = T$ **then**
      $\mathsf{Dict} \coloneqq \mathsf{Dict} \cup \mathsf{VandShare}(\mathbf{s}, \mathcal{P}_L, T, \mathbf{i}_L)$
    else
      $\mathbf{s}_0 \leftarrow \chi$ ▷ This guarantees short shares
      $\mathbf{s}_1 \coloneqq (\mathbf{s} - \mathbf{s}_0) \bmod q$
      $\mathsf{Dict}_L \coloneqq \mathsf{VandShare}(\mathbf{s}_0, \mathcal{P}_L, k, \mathbf{i}_L)$
      $\mathsf{Dict}_R \coloneqq \mathsf{VandShare}(\mathbf{s}_1, \mathcal{P}_R, T - k, \mathbf{i}_R)$
      $\mathsf{Dict} \coloneqq \mathsf{Dict} \cup \mathsf{Dict}_L \cup \mathsf{Dict}_R$
  return Dict ▷ The values of Dict are also dictionaries.

Lemma 49. *Let $\varepsilon \in (0, 1/2)$, $m, N, T \geq 1$, $\sigma \geq \sqrt{2}\eta_\varepsilon(\mathcal{R}^m)$, $\mathbf{s} \leftarrow D_{\mathcal{R}^m,\sigma}$, and $\mathsf{Shares} \leftarrow \mathsf{Share}(\mathbf{s}, [N], T)$. In Algorithm 1, let $\chi = D_{\mathcal{R}^m,\sigma}$. For any user $i \in [N]$ and any share $\mathbf{s}^{(\mathbf{i})} \coloneqq \mathsf{Shares}[i][\mathbf{i}]$, we have $\mathrm{SD}(\mathbf{s}^{(\mathbf{i})}, D_{\mathcal{R}^m,\sqrt{k}\cdot\sigma}) \leq (k-1) \cdot 8\varepsilon$ for a value $k \in \mathbb{N}, 1 \leq k \leq \log N + 1$ determined by Algorithm 1.*

Algorithm 2 $\mathsf{VandRecover}(\mathcal{P}, \mathsf{act}, \mathbf{i} = \varnothing) \to \mathsf{Indices}$

1: $N = |\mathcal{P}|, T = |\mathsf{act}|$
2: **if** $T = 1$ **then**
3: **return** $\mathsf{Indices} \coloneqq \{\mathcal{U} : \mathbf{i} \mid \mathcal{U} \in \mathcal{P}\}$
4: **else**
5: $b = \lfloor N/2 \rfloor$
6: Parse $\mathcal{P} = \mathcal{P}_L \sqcup \mathcal{P}_R$, with $\mathcal{P}_L$ the b smallest elements of $\mathcal{P}$
7: $k = |\mathcal{P}_L|, \mathsf{act}_L = \mathsf{act} \cap \mathcal{P}_L, \mathsf{act}_R = \mathsf{act} \cap \mathcal{P}_R$
8: $\mathbf{i}_L \coloneqq \mathbf{i} \parallel \texttt{":L:"} \parallel k,\ \ \mathbf{i}_R \coloneqq \mathbf{i} \parallel \texttt{":R:"} \parallel (T - k)$
9: **if** $k = 0$ **then**
10: **return** $\mathsf{VandRecover}(\mathcal{P}_R, \mathsf{act}_R, \mathbf{i}_R)$
11: **else if** $k = T$ **then**
12: **return** $\mathsf{VandRecover}(\mathcal{P}_L, \mathsf{act}_L, \mathbf{i}_L)$
13: **else**
14: $\mathsf{Indices}_L \coloneqq \mathsf{VandRecover}(\mathcal{P}_L, \mathsf{act}_L, \mathbf{i}_L)$
15: $\mathsf{Indices}_R \coloneqq \mathsf{VandRecover}(\mathcal{P}_R, \mathsf{act}_R, \mathbf{i}_R)$
16: **return** $\mathsf{Indices} \coloneqq \mathsf{Indices}_L \sqcup \mathsf{Indices}_R$

Proof. Depending on the index i of the user and the group $\mathsf{act} \subset [N]$, $|\mathsf{act}| = T$ the share $\mathbf{s}^{(\mathbf{i})}$ follows the distribution $\mathbf{s}^{(\mathbf{i})} \leftarrow \left\{\sum_{j\in[k]} \mathbf{x}_j \mid \mathbf{x}_j \leftarrow D_{\mathcal{R}^m,\sigma}\right\}$. The value $1 \le k \le \log N + 1$ is the depth of the tree path where the party i is the only party in act on its side of the tree. This corresponds to the base case of the recursive algorithm (i.e. Line 3 of Algorithm 1). Applying Lemma 34 $k-1$ times to $\mathbf{x}_j \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma}$, for $j \in [k]$ we get

$$\mathrm{SD}(\sum_{j\in[k]} \mathbf{x}_j, \mathcal{D}_{\mathcal{R}^m,\sqrt{k}\sigma}) \le (k-1) \cdot 8\varepsilon$$

where σ has to satisfy $\sigma \ge \sqrt{1 + 1/(k-1)} \cdot \eta_\varepsilon(\mathcal{R}^m)$. The worst case for this condition is $k = 2$ when $\sigma \ge \sqrt{2} \cdot \eta_\varepsilon(\mathcal{R}^m)$ as stated above. □

We recall the syntax and properties we expect from lattice trapdoor algorithms.

Definition 9 (Lattice Trapdoors)**.** *Fix $\mathcal{R}, q$ parametrised by λ. A lattice trapdoor scheme over $\mathcal{R}_q$ is a tuple of PPT algorithms* $(\mathsf{TrapGen}, \mathsf{SamplePre})$*:*

- $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m)$*: On input dimensions $n, m \in \mathbb{N}$, generate a matrix $\mathbf{A} \in \mathcal{R}_q^{n\times m}$ together with a trapdoor* td*.*
- $\mathbf{u} \leftarrow \mathsf{SamplePre}(\mathsf{td}, \mathbf{v}, \mathbf{S})$*: On input a trapdoor* td*, a target image $\mathbf{v} \in \mathcal{R}_q$-span$(\mathbf{A})$, and a Gaussian parameter $\mathbf{S}^{dm\times dm} > 0$, output a vector $\mathbf{u} \in \mathcal{R}^m$. The syntax is extended naturally to matrix images.*

Let $n, m \in \mathbb{N}$ and $\tilde{s} > 0$ be parametrised by λ. The scheme is said to be $(n, m, \tilde{s})$-correct if, for $\mathfrak{s}_{\min}(\mathbf{S}) \geq \tilde{s}$, the following hold:

- *The distribution $\left\{\mathbf{A} : (\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m)\right\}$ is statistically close to the uniform distribution over $\mathcal{R}_q^{n \times m}$.*
- *For any $\mathbf{v} \in \mathcal{R}_q^n$ and $(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m)$, the distributions $\mathcal{D}_{\Lambda_q^{\mathbf{v}}(\mathbf{A}), \mathbf{S}}$ and $\mathsf{SamplePre}(\mathsf{td}, \mathbf{v}, \mathbf{S})$ are statistically close except with negligible probability.*

One way to instantiate the syntax above is using the trapdoor generation from [MP12] adapted to the module setting.

Gadget matrix. Let $\delta \geq 2$ be an integer. We set $\tilde{q} \coloneqq \lceil \log_\delta q \rceil$, $\mathbf{g}^\intercal \coloneqq [1, \delta, \ldots, \delta^{\tilde{q}-1}] \in \mathcal{R}_q^{1 \times \tilde{q}}$ and $\mathbf{G}_n \coloneqq \mathbf{I}_n \otimes \mathbf{g}^\intercal \in \mathcal{R}_q^{n \times (n \cdot \tilde{q})}$. When the dimensions are clear from context we simply write $\mathbf{G}$. Write $\mathbf{G}_n^{-1} : \mathcal{R}_q^{n \times t} \to \mathcal{R}_q^{(n \cdot \tilde{q}) \times t}$ for the inverse function that takes a matrix of entries in $\mathcal{R}_q$, and decomposes each entry w.r.t. the base δ. We also write $\mathbf{g}^{-1}$ for $\mathbf{G}_1^{-1}$. As an immediate corollary of [MP12, Theorem 4.1] and Lemma 20 there exists a negligible function $\varepsilon(d \cdot n \cdot \tilde{q})$ such that $\eta_\varepsilon(\Lambda_q^\perp(\mathbf{G}_n)) \leq \sqrt{5} \cdot \omega(\sqrt{\log(d \cdot n \cdot \tilde{q})})$.

Lemma 50 (Adapted from [MP12, Theorem. 5.5 (eprint)])**.** *Let $\mathcal{R}_q$ have degree d and split into fields of superpolynomial size in the security parameter. Let $\delta > 1$, $m \geq n + \tilde{m}$ for $\tilde{m} \coloneqq n \cdot \lceil \log_\delta(q) \rceil$ and $\mathbf{S} \geq \delta \cdot \sigma \cdot \sqrt{2\,\tilde{m} \cdot (m - \tilde{m}) \cdot d} \cdot \omega(\sqrt{\log(m \cdot d)})$ where $\sigma > 2\,d \cdot q^{n/m + 2/(d \cdot m)}$. The following properties are satisfied:*

- *There exist an efficient algorithm $\mathsf{TrapGen}(n, m, \mathbf{H})$ constructing a matrix $\mathbf{A} = [\mathbf{A}_0 \mid -\mathbf{A}_0 \cdot \mathbf{R} + \mathbf{H} \cdot \mathbf{G}]$ and a "$\mathbf{G}_n$-trapdoor", i.e. $\mathsf{td} = (\mathbf{R}, \mathbf{H}) \in \mathcal{R}_q^{m \times \tilde{m}} \times (\mathcal{R}_q^{n \times n})^\times$ satisfying $\mathbf{A} \cdot \mathbf{R} \equiv \mathbf{H} \cdot \mathbf{G}_n \bmod q$, where $\mathbf{H}$ is an arbitrary invertible matrix and $\mathbf{R}$ satisfies $\|\mathbf{R}\| \leq \sigma \cdot \sqrt{2\,\tilde{m} \cdot (m - \tilde{m}) \cdot d}$. And the distribution of $\mathbf{A}$ is statistically close to uniform. We omit the tag when $\mathbf{H} = \mathbf{I}_n$.*
- *There exist the second efficient algorithm $\mathsf{SamplePreMP}(\mathsf{td}, \mathbf{t}, \mathbf{S})$ returning $\mathbf{x}$ such that*

$$\mathrm{SD}(\mathbf{x}, \mathcal{D}_{\Lambda_q^{\mathbf{t}}(\mathbf{F}), \sigma}) \leq 4\varepsilon.$$

  for any $\mathsf{td} = (\mathbf{R}, \mathbf{H})$, $\mathbf{t} \in \mathcal{R}_q^n$ and $\mathbf{S} \geq \delta \cdot \mathfrak{s}_{\max}(\mathbf{R}) \cdot \eta_\varepsilon(\mathcal{R}^m)$.

Lemma 51 ([GPV08, Theorem 4.1 (eprint)])**.** *Let $0 < n < m$ be integers, $\varepsilon \in (0, 1/2)$ and q be a prime number. Let $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ and $\mathbf{T}_A \in \mathbb{Z}^{dm \times dm}$ s.t. $\mathrm{Rot}(\mathbf{A}) \cdot \mathbf{T}_A = \mathbf{0} \bmod q$. Let $\mathbf{t} \in \mathcal{R}_q^n$. And*

$$\sigma > \|\mathbf{T}_A\|_{GS} \cdot \eta_\varepsilon(\mathcal{R}^m).$$

Then there exists an efficient algorithm we denote $\mathsf{SamplePre}(\mathbf{A}, \mathbf{T}_A, \mathbf{t}, \sigma)$ *that returns* $\mathbf{x}$ *such that*

$$\mathrm{SD}(\mathbf{x}, \mathcal{D}_{\Lambda_q^{\mathbf{t}}(\mathbf{F}),\sigma}) \leq O(\varepsilon).$$

The following result is immediate from linear algebra for fields, but requires a bit more care for more general $\mathcal{R}_q$.

Proposition 1. *Let* $\mathcal{R}_q \coloneqq \mathcal{R}/(q\mathcal{R})$*. Let* $\mathbf{A} \in \mathcal{R}_q^{m\times k}$ *and* $\mathbf{B} \in \mathcal{R}_q^{n\times k}$*. There exists a PPT algorithm sampling* $\mathbf{X} \in \mathcal{R}_q^{m\times n}$ *uniformly at random subject to the condition that* $\mathbf{A} \equiv \mathbf{X} \cdot \mathbf{B} \bmod q$*.*

Proof. We first note that $\mathcal{R}_q$ is a principal ideal ring (PIR) by Lemma 9.[2] In [Sto00] an algorithm is given for computing the Howell Form of matrices over PIRs. Using the Howell Form of $\mathbf{B}^{\mathbf{T}}$, we can compute an arbitrary solution $\tilde{\mathbf{X}}$ satisfying $\mathbf{A} \equiv \tilde{\mathbf{X}} \cdot \mathbf{B} \bmod q$ [Sto00, p.27]. Moreover, using the Howell Form of $\mathbf{B}^{\mathbf{T}}$, we can compute a basis for the kernel of $\mathbf{B}^{\mathbf{T}}$ [Sto00, p.70]. Picking a random element from the kernel (which is well-defined as a linear combination of its basis with coefficients uniform in $\mathcal{R}_q$), we sample $\mathbf{X}'$ s.t. $\mathbf{X}' \cdot \mathbf{B} \equiv \mathbf{0} \bmod q$. The final solution is $\tilde{\mathbf{X}} + \mathbf{X}'$. □

Lemma 52 (Adapted [KY16, Lemma 1])**.** *Let* $m > 0$ *be an integer,* $\varepsilon \in (0, 1/2)$ *and* q *be a prime number. Let* $\mathbf{b} \in \mathcal{R}^m$ *and* $\mathbf{R} \in \mathcal{R}^{m\times m}$*. Let* $\sigma > \sqrt{2} \cdot \eta_\varepsilon(\mathcal{R}^m)$*,* $\sigma' > 2 \cdot \mathfrak{s}_{\max}(\mathbf{R}) \cdot \sigma$ *and* $\mathbf{x} \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma}$*. Then there exists an efficient algorithm* $\mathsf{ReRand}(\mathbf{R}, \mathbf{b} + \mathbf{x}, \sigma', \sigma)$ *that outputs* $\mathbf{b}' = \mathbf{R}\mathbf{b} + \mathbf{x}'$ *where* $\mathrm{SD}(\mathbf{x}', \mathcal{D}_{\mathcal{R}^m,\sigma'}) \leq 6\varepsilon$*.*

2.7 Cryptographic Primitives

In this section we define the syntax and the security properties of the cryptographic primitive used in the thesis.

Definition 10 (Digital Signature Scheme)**.** *A digital signature scheme a triple of PPT algorithms* $\mathsf{SIG} = (\mathsf{SIG.Keygen}, \mathsf{SIG.Sign}, \mathsf{SIG.Verify})$ *where*

1. $\mathsf{SIG.Keygen}(1^\lambda) \to (\mathsf{sk}, \mathsf{vk})$ *takes as input the security parameter and outputs the signing key* sk *and the verification key* vk*.*

2. $\mathsf{SIG.Sign}(\mathsf{sk}, \mathsf{msg}) \to \mathsf{sig}$ *takes as input a message* $\mathsf{msg} \in \{0,1\}^*$ *and the signing key* sk *and outputs a signature* sig*.*

[2] Following, [Sto00], we do not require PIRs to not have zero divisors, in contrast to principal ideal domains (PIDs).

3. $\mathsf{SIG.Verify}(\mathsf{vk}, \mathsf{msg}, \mathsf{sig}) \to b$ *takes as input the verification key* vk*, a message* $\mathsf{msg} \in \{0,1\}^*$ *and a signature* sig*, and outputs a bit* $b = 1$ *if the signature and message pass the verification and* $b = 0$ *if they do not.*

Remark 5. Below we define several protocols for distributed computation. In this thesis we only consider protocols with centralised setup executed by a trusted party. We also assume that signing (resp. decrypting) parties know which other signers (resp. decryptors) are participating in the current session, we capture this assumption by passing the set of active parties $|\mathsf{act}| = T$ as input to the signing (reps. decrypting) algorithm. They also implicitely keep the state and an id for the current open sessions and check the consistency of the state between rounds. We assume the syncronous broadcast network model, hence all sent messages are received by all other parties within a bounded timeframe. Lastly, the $\mathsf{Combine}$ operation that completes the signature (resp. decryption) can be executed by a dedicated combiner or by every participating party on the bradcasted partial signatures (resp. decryptions). We do not protect the protocol against the adversary that witholds their final signature (resp. decryption) share.

Definition 11. *(Threshold Signature Scheme) A* T*-out-of-*N *threshold signature scheme, or* (T, N)*-signature scheme, for the number of signers* N*, threshold* $T \leq N$ *and message space* $\mathcal{M}$ *consists of PPT algorithms* $\mathsf{TSIG} = (\mathsf{TSIG.MKeygen}, \mathsf{TSIG.Keygen}, \mathsf{TSIG.Sign}, \mathsf{TSIG.Combine}, \mathsf{TSIG.Verify})$ *where*

1. $\mathsf{TSIG.MKeygen}(1^\lambda) \to (\mathsf{mpk}, \mathsf{msk})$*: The master key generation algorithm generates a master public key* mpk *and a master secret key* msk*.*

2. $\mathsf{TSIG.Keygen}(\mathsf{mpk}, \mathsf{msk}, j) \to \mathsf{sk}_j$*: The key generation algorithm, on input* mpk *and* msk *and an index* $j \in [N]$*, generates a signing key for signer* j*.*

3. $\mathsf{TSIG.Sign}(\mathsf{mpk}, \mathsf{sk}_j, \mathsf{act}, \mathsf{msg}) \to \mathsf{sig}_j$*: The signing algorithm, on input* mpk*, a signing key* sk_j *from signer* j*, a set* $\mathsf{act} \subseteq [N]$ *of size* $|\mathsf{act}| = T$*, and a message* $\mathsf{msg} \in \mathcal{M}$*, generates a partial signature* sig_j *for* msg*.*

4. $\mathsf{TSIG.Combine}(\{\mathsf{sig}_j\}_{j \in \mathsf{act}}) \to \mathsf{sig}$*: The combining algorithm inputs a tuple of partial signatures* sig_j *from a set* act *of signers, and reconstructs a full signature* sig*.*

5. $\mathsf{TSIG.Verify}(\mathsf{mpk}, \mathsf{sig}, \mathsf{msg}) \to b$*: The verification algorithm inputs* mpk*, a full signature* sig *and a message* msg*, and outputs a bit* $b \in \{0,1\}$*.*

Definition 12 (IBE with Threshold Key Generation)**.** *A threshold identity-based encryption scheme is a tuple of PPT algorithms* $\mathsf{TIBE} = (\mathsf{TIBE.Setup}, \mathsf{TIBE.Encrypt}, (\mathsf{TIBE.ShareExtract}_r)_r, \mathsf{TIBE.ShareVerify}, \mathsf{TIBE.Combine})$ *s.t.:*

1. $\mathsf{TIBE.Setup}(1^\lambda) \to (\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N})$ *takes as input the security parameter. It outputs the encryption key* ek *and* N *decryption key shares* $\{\mathsf{dk}_i\}_{i\in N}$*.*

2. $\mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{id}, \mathsf{msg}) \to \mathsf{ct}$ *takes as input an encryption key* ek*, identity* id *and message* msg*. It outputs a ciphertext* ct*.*

3. *For* $r \in \{0, \ldots, \mathsf{rounds} - 1\}$*,* $\mathsf{TIBE.ShareExtract}_r(\mathsf{id}, \mathsf{act}, \mathsf{dk}_i, \{\mathsf{contrib}_{k,j}\}_{j<r}^{k\in\mathsf{act}}) \to \mathsf{contrib}_{i,r}$ *takes as input an identity* id*, the set of active participants* act*, a decryption key share* dk_i *and contributions from previous rounds* $\{\mathsf{contrib}_{k,j}\}_{j<r}^{k\in\mathsf{act}}$*. It outputs a contribution of the current round* $\mathsf{contrib}_{i,r}$*.*

4. $\mathsf{TIBE.ShareVerify}(\mathsf{ek}, \mathsf{id}, \mathsf{act}, \{\mathsf{contrib}_{j,r}\}_{j,r}, i) \to b \in \{1, \perp\}$ *takes as input an encryption key* ek*, an identity* id*, a set of active participants* act*, their key share contributions* $\{\mathsf{contrib}_{j,r}\}_{r\le\mathsf{rounds}}^{j\in\mathsf{act}}$ *and the index of one party* $i \in \mathsf{act}$*. It outputs either* 1 *(meaning accept) or* $\perp$ *(meaning reject).*

5. $\mathsf{TIBE.Combine}(\mathsf{ek}, \mathsf{ct}, \mathsf{id}, \mathsf{act}, \{\mathsf{contrib}_{i,r}\}_{i,r}) \to \mathsf{msg}/\perp$ *takes as input a set of active participants* act*, their key share contributions* $\{\mathsf{contrib}_{i,r}\}_{r\le\mathsf{rounds}}^{i\in\mathsf{act}}$*, an identity* id *and a ciphertext* ct*. It outputs a message* msg *or an error* $\perp$*.*

Definition 13 (Threshold Key Encapsulation Mechanism)**.** *A threshold key encapsulation mechanism (TKEM) is a tuple of PPT algorithms* $\mathsf{TKEM} = (\mathsf{TKEM.Keygen}, \mathsf{TKEM.Encaps}, (\mathsf{TKEM.ShareDecaps}_r)_r, \mathsf{TKEM.Combine})$ *s.t.:*

1. $\mathsf{TKEM.Keygen}(1^\lambda) \to (\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N})$ *takes as input the security parameter. It outputs the encapsulation key* ek *and* N *decapsulation key shares* $\{\mathsf{dk}_i\}_{i\in N}$*.*

2. $\mathsf{TKEM.Encaps}(\mathsf{ek}) \to \mathsf{ct}, K$ *takes as input an encapsulation key* ek*. It outputs a ciphertext* ct *and a session key from the key space* $K \in \mathcal{K}$*.*

3. *For* $r \in [1, \mathsf{rounds}+1]$*,* $\mathsf{TKEM.ShareDecaps}_r(\mathsf{ct}, \mathsf{act}, \mathsf{dk}_i, \{\mathsf{contrib}_{i,j}\}_{j<r}) \to \mathsf{contrib}_{i,r}$ *takes as input a ciphertext* ct*, a set of active parties* act*, a decapsulation key share* dk_i *and all contributions by parties in* act *up to the current round* r*. It outputs a contribution* $\mathsf{contrib}_{i,r}$ *for the current round.*

4. $\mathsf{TKEM.Combine}(\mathsf{ct}, \mathsf{act}, \{\mathsf{contrib}_{i,r}\}_{i,r}) \to K/\perp$ *takes as input a ciphertext* ct*, a set of active parties* act *and their contributions* $\{\mathsf{contrib}_{i,r}\}_{i\in\mathsf{act},r\leq\mathsf{rounds}}$*. It outputs a key* K *or an error message* $\perp$*.*

Definition 14 ((Q,ε)-sEU-CMA Security)**.** *A signature scheme* SIG *satisfies strong* $\mathsf{EU\text{-}CMA}$ *if any PPT adversary* $\mathcal{A}$ *with at most* Q *signing queries wins* $\mathsf{Exp}^{\mathsf{sEU\text{-}CMA}}_{\mathcal{A},\mathsf{SIG}}(1^\lambda)$ *(see Figure 2.2a) with probability at most* ε*. When* $Q = 1$ *we say that* SIG *is an* ε*-*$\mathsf{sEU\text{-}CMA}$ *one-time signature scheme.*

In the Definition 15 below we assume that the adversary declares the set corrupt parties as well as all their signing queries before receiving the public key information. We call this notion Highly Selective Security of Threshold Signatures.

Definition 15 (TSIG Highly Selective Security)**.** *A* (T,N)*-signature scheme* TSIG *is said to be highly-selectively secure, if for any PPT* $\mathcal{A}$ *it holds that* $\Pr\left[\mathsf{Exp}_{\mathsf{TSIG},\mathcal{A}}(1^\lambda) = 1\right] \leq \mathsf{negl}(\lambda)$*, where* $\mathsf{Exp}_{\mathsf{TSIG},\mathcal{A}}$ *is defined in Figure 2.3a.*

In the Definition 16 we also assume the adversary declares the set corrupt parties as well as all their signing queries at the start of the security experiment. We call this notion Highly Selective Security of Threshold IBE.

Definition 16 (TIBE Highly Selective Security)**.** *A one-round Threshold IBE scheme* $\mathsf{TIBE} = (\mathsf{TIBE.Setup}, \mathsf{TIBE.Encrypt}, \mathsf{TIBE.ShareExtract}_0, \mathsf{TIBE.ShareVerify}, \mathsf{TIBE.Combine})$ *is said to be highly-selectively secure, if for any PPT* $\mathcal{A}$ *it holds that*

$$\left|\Pr\left[\mathsf{Exp}^{\mathsf{hs},0}_{\mathsf{TIBE},\mathcal{A}}(1^\lambda) = 1\right] - \Pr\left[\mathsf{Exp}^{\mathsf{hs},1}_{\mathsf{TIBE},\mathcal{A}}(1^\lambda) = 1\right]\right| \leq \mathsf{negl}(\lambda),$$

where $\mathsf{Exp}^{\mathsf{hs},b}_{\mathsf{TIBE},\mathcal{A}}$ *is defined in Figure 2.3b.*

Remark 6. In Definitions 17 and 21 we assume the set of corrupt parties satisfies $|\mathsf{cor}| = T - 1$. We believe our security arguments can be extended to a more general constraint $|\mathsf{cor}| \leq T - 1$, however we defer this extension to future work.

Definition 17 (Selective-ID/CCA2 Security)**.** *We say that a* TIBE *(resp. a* TKEM*) scheme achieves selective-ID indistinguishability, selective-ID one-wayness (resp. selective-CCA2 security) if no efficient adversary against the game in Figure 2.2b, Figure 2.2d (resp. Figure 2.2c) has a non-negligible advantage. We denote the probability of winning this game for an adversary* $\mathcal{A}$ *as* $\mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}IND}}_{\mathcal{A},\mathsf{TIBE}}$*,* $\mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}OW}}_{\mathcal{A},\mathsf{TIBE}}$ *(resp.* $\mathsf{Adv}^{\mathsf{CCA2}}_{\mathcal{A},\mathsf{TKEM}}$*).*

$\mathsf{Exp}^{\mathsf{sEU\text{-}CMA}}_{\mathcal{A},\mathsf{SIG}}(1^\lambda)$

$(\mathsf{sk}, \mathsf{vk}) \leftarrow \mathsf{SIG.Keygen}(1^\lambda)$
$(\mathsf{msg}^*, \mathsf{sig}^*) \leftarrow \mathcal{A}^{\mathsf{SIG.Sign}(\mathsf{sk},\cdot)}(\mathsf{vk})$
Let $\{(\mathsf{msg}_i, \mathsf{sig}_i)\}_{i=0}^{Q-1}$ be the list of $\mathcal{A}$'s oracle queries
return $(\mathsf{SIG.Verify}(\mathsf{vk}, \mathsf{msg}^*, \mathsf{sig}^*) = 1) \wedge (\mathsf{msg}^*, \mathsf{sig}^*) \notin \{(\mathsf{msg}_i, \mathsf{sig}_i)\}_{i=0}^{Q-1}$

(a) The (Q, ε)-sEU-CMA security game

$\mathsf{Exp}^{\mathsf{sel\text{-}ID\text{-}IND}}_{\mathcal{A},\mathsf{TIBE}}(1^\lambda)$

$(\mathsf{id}^*, \mathsf{cor}) \leftarrow \mathcal{A}$
assert $|\mathsf{cor}| = T - 1$
$(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N}) \leftarrow \mathsf{TIBE.Setup}(1^\lambda)$
$(\mathsf{msg}_0, \mathsf{msg}_1)$
$\leftarrow \mathcal{A}^{\mathsf{TIBE.ShareExtract}(\cdot,\mathsf{id}),\mathsf{id}\neq\mathsf{id}^*}(\{\mathsf{dk}_i\}_{\mathsf{cor}})$
$b \leftarrow \{0,1\}$
$\mathsf{ct}^* = \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{id}^*, \mathsf{msg}_b)$
$b' \leftarrow \mathcal{A}^{\mathsf{TIBE.ShareExtract}(\cdot,\mathsf{id}),\mathsf{id}\neq\mathsf{id}^*}(\{\mathsf{dk}_i\}_{\mathsf{cor}})$
return $(b = b')$

(b) TIBE selective-ID security

$\mathsf{Exp}^{A}_{\mathcal{A},\mathsf{TKEM}}(1^\lambda)$

$\mathsf{cor} \leftarrow \mathcal{A}$
assert $|\mathsf{cor}| = T - 1$
$(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N}) \leftarrow \mathsf{TKEM.Keygen}(1^\lambda)$
$1 \leftarrow \mathcal{A}^{\mathsf{TKEM.ShareDecaps}(\cdot)}(\{\mathsf{dk}_i\}_{\mathsf{cor}})$
$(\mathsf{ct}^*, K_0) = \mathsf{TKEM.Encaps}(\mathsf{ek})$
$K_1 \leftarrow \mathcal{K}$
$b \leftarrow \{0,1\}$
$b' \leftarrow \mathcal{A}^{\mathsf{TKEM.ShareDecaps}(\mathsf{ct},\cdot),\mathsf{ct}\neq\mathsf{ct}^*}((\mathsf{ct}^*, K_b))$
return $(b = b')$

(c) TKEM selective security

$\mathsf{Exp}^{\mathsf{sel\text{-}ID\text{-}OW}}_{\mathcal{A},\mathsf{TIBE}}(1^\lambda)$

$(\mathsf{id}^*, \mathsf{cor}) \leftarrow \mathcal{A}$
assert $|\mathsf{cor}| = T - 1$
$(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N}) \leftarrow \mathsf{TIBE.Setup}(1^\lambda)$
$\mathcal{A}^{\mathsf{TIBE.ShareExtract}(\cdot,\mathsf{id}),\mathsf{id}\neq\mathsf{id}^*}(\{\mathsf{dk}_i\}_{\mathsf{cor}})$
$\mathsf{msg}^* \leftarrow \{0,1\}^d$
$\mathsf{ct}^* = \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{id}^*, \mathsf{msg}^*)$
$\mathsf{msg} \leftarrow \mathcal{A}^{\mathsf{TIBE.ShareExtract}(\cdot,\mathsf{id}),\mathsf{id}\neq\mathsf{id}^*}(\{\mathsf{dk}_i\}_{\mathsf{cor}}, \mathsf{ct}^*)$
return $(\mathsf{msg} = \mathsf{msg}^*)$

(d) TIBE Selective-ID One-Way Security

$\mathsf{Exp}^{\mathsf{Robust}}_{\mathcal{A},\mathsf{TIBE}}(1^\lambda)$

$(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N}) \leftarrow \mathsf{TIBE.Setup}(1^\lambda)$
$(\mathsf{msg}, \mathsf{id}, \mathsf{rand}, \mathsf{ct}, \mathsf{act}, S) \leftarrow \mathcal{A}(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in[N]})$
assert $\mathsf{rand} \in \mathsf{RSpace}$
assert $\mathsf{ct} = \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{id}, \mathsf{msg}; \mathsf{rand})$
for $i \in \mathsf{act}$
  assert $\mathsf{TIBE.ShareVerify}(\mathsf{ek}, \mathsf{act}, S, i)$
return $\mathsf{msg} \neq \mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{act}, S)$

(e) TIBE Extraction Share Robustness

Figure 2.2: Security experiments

$\mathsf{Exp}^{\mathsf{hs}}_{\mathsf{TSIG},\mathcal{A}}(1^\lambda)$

$\left\{\mathsf{cor}, \{\mathsf{act}_\ell, \mathsf{msg}_\ell\}_{\ell\in[L]}, (\mathsf{act}^*, i^*, \mathsf{msg}^*)\right\} \leftarrow \mathcal{A}$

// set cor of corrupt signers, signature queries $\{\mathsf{act}_\ell, \mathsf{msg}_\ell\}_{\ell\in[L]}$, challenge $(\mathsf{act}^*, i^*, \mathsf{msg}^*)$

assert $|\mathsf{cor}| \leq T-1$; **assert** $i^* \in \mathsf{act}^* \setminus \mathsf{cor}$

if $\exists\, \ell, \ell' \in [L] : \mathsf{msg}_\ell = \mathsf{msg}_{\ell'}$: **return** $\perp$

$(\mathsf{mpk}, \mathsf{msk}) \leftarrow \mathsf{TSIG.MKeygen}(1^\lambda)$

for $j \in [N] : \mathsf{sk}_j \leftarrow \mathsf{TSIG.Keygen}(\mathsf{mpk}, \mathsf{msk}, j)$

for $\ell \in [L] : \mathsf{sig}_{\ell,j} \leftarrow \mathsf{TSIG.Sign}\left\{\mathsf{mpk}, \mathsf{sk}_j, \mathsf{act}_\ell, \mathsf{msg}_\ell\right\}\ \forall j \in \mathsf{act}_\ell$

$\mathsf{sig}^*_j \leftarrow \mathsf{TSIG.Sign}\left\{\mathsf{mpk}, \mathsf{sk}_j, \mathsf{act}^*, \mathsf{msg}^*\right\}\ \forall j \in \mathsf{act}^* \setminus \{i^*\}$

$\mathsf{sig}^* \leftarrow \mathcal{A}\left\{\mathsf{mpk}, (\mathsf{sk}_j)_{j\in\mathsf{cor}}, \left\{\mathsf{sig}_{\ell,j}\right\}_{\ell\in[L], j\in\mathsf{act}_\ell}, \left\{\mathsf{sig}^*_j\right\}_{j\in\mathsf{act}^*\setminus\{i^*\}}\right\}$

return $(1 \leftarrow \mathsf{TSIG.Verify}(\mathsf{mpk}, \mathsf{sig}^*, \mathsf{msg}^*))$

(a) TSIG Highly Selective Security.

$\mathsf{Exp}^{\mathsf{hs},b}_{\mathsf{TIBE},\mathcal{A}}(1^\lambda)$

$\left\{\mathsf{cor}, \{\mathsf{act}_\ell, \mathsf{id}_\ell\}_{\ell\in[L]}, (\mathsf{act}^*, i^*, \mathsf{id}^*)\right\} \leftarrow \mathcal{A}$

// set cor of corrupt authorities, key queries $\{\mathsf{act}_\ell, \mathsf{id}_\ell\}_{\ell\in[L]}$, challenge $(\mathsf{act}^*, i^*, \mathsf{id}^*)$

assert $|\mathsf{cor}| \leq T-1$; **assert** $i^* \in \mathsf{act}^* \setminus \mathsf{cor}$

if $\exists \ell, \ell' \in [L] : \mathsf{id}_\ell = \mathsf{id}_{\ell'}$: **return** $\perp$

$\left\{\mathsf{ek}, \{\mathsf{dk}_j\}_{j\in[N]}\right\} \leftarrow \mathsf{TIBE.Setup}(1^\lambda)$

for $\ell \in [L] : \mathsf{sk}_{\ell,j} \leftarrow \mathsf{TIBE.ShareExtract}_0(\mathsf{ek}, \mathsf{dk}_j, \mathsf{act}_\ell, \mathsf{id}_\ell)\ \forall j \in \mathsf{act}_\ell$

$\mathsf{sk}^*_j \leftarrow \mathsf{TIBE.ShareExtract}_0(\mathsf{ek}, \mathsf{dk}_j, \mathsf{act}^*, \mathsf{id}^*)\ \forall j \in \mathsf{act}^* \setminus \{i^*\}$

$(\mathsf{msg}_0, \mathsf{msg}_1) \leftarrow \mathcal{A}(\mathsf{ek}, (\mathsf{dk}_j)_{j\in\mathsf{cor}}, \left\{\mathsf{sk}_{\ell,j}\right\}_{\ell\in[L], j\in\mathsf{act}_\ell}, \left\{\mathsf{sk}^*_j\right\}_{j\in\mathsf{act}^*\setminus\{i^*\}})$

$\mathsf{ctxt} \leftarrow \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{id}^*, \mathsf{msg}_b)$; $b' \leftarrow \mathcal{A}(\mathsf{ctxt})$

return b'

(b) TIBE Highly Selective Security.

Figure 2.3: Security experiments

Definition 18 (TSIG Correctness)**.** *A (T, N)-signature scheme is said to be δ-correct if, for any $(\mathsf{mpk}, \mathsf{msk}) \in \mathsf{TSIG.MKeygen}(1^\lambda)$, $j \in [N]$, $\mathsf{sk}_j \in \mathsf{TSIG.Keygen}(\mathsf{mpk}, \mathsf{msk}, j)$, $\mathsf{msg} \in \mathcal{M}$, and any set $\mathsf{act} \subseteq [N]$ of size $|\mathsf{act}| = T$, it holds that*

$$\Pr\left[1 \leftarrow \mathsf{TSIG.Verify}(\mathsf{mpk}, \mathsf{sig}, \mathsf{msg}) \;\middle|\; \begin{array}{l} \left\{\mathsf{sig}_j \leftarrow \mathsf{TSIG.Sign}\left\{\mathsf{mpk}, \mathsf{sk}_j, \mathsf{act}, \mathsf{msg}\right\}\right\} \\ \mathsf{sig} \leftarrow \mathsf{TSIG.Combine}\left\{\left\{\mathsf{sig}_j\right\}_{j\in\mathsf{act}}\right\} \end{array}\right] \geq 1-\delta.$$

$\mathsf{Exp}^{\mathsf{CD}}_{\mathcal{A},\mathsf{TKEM}}(1^\lambda)$

$\mathsf{cor} \leftarrow \mathcal{A}$
assert $|\mathsf{cor}| = T - 1$
$(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N}) \leftarrow \mathsf{TKEM.Keygen}(1^\lambda)$
$(\mathsf{ct}, \mathsf{act}, \mathsf{act}', S, S')$
$\quad \leftarrow \mathcal{A}^{\mathsf{TKEM.ShareDecaps}(\cdot)}(\{\mathsf{dk}_i\}_{i\in\mathsf{cor}})$
$K_0 = \mathsf{TKEM.Combine}(\mathsf{ct}, \mathsf{act}, S)$
$K_1 = \mathsf{TKEM.Combine}(\mathsf{ct}, \mathsf{act}', S')$
return $(\forall i : K_i \neq \perp) \wedge (K_0 \neq K_1)$

(a) Weak Decapsulation Consistency

$\mathsf{Exp}^{\mathsf{strong\text{-}CD}}_{\mathcal{A},\mathsf{TKEM}}(1^\lambda)$

$(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N}) \leftarrow \mathsf{TKEM.Keygen}(1^\lambda)$
$(\mathsf{ct}, \mathsf{act}, \mathsf{act}', S, S') \leftarrow \mathcal{A}(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in[N]})$
for $i \in \mathsf{act}$, $j \in \mathsf{act}'$
$\quad$ **assert** $\mathsf{TKEM.ShareVerify}(\mathsf{ek}, \mathsf{ct}, \mathsf{act}, S, i)$
$\quad$ **assert** $\mathsf{TKEM.ShareVerify}(\mathsf{ek}, \mathsf{ct}, \mathsf{act}', S', j)$
$K = \mathsf{TKEM.Combine}(\mathsf{ct}, \mathsf{act}, S)$
$K' = \mathsf{TKEM.Combine}(\mathsf{ct}, \mathsf{act}', S')$
return $K \neq K'$

(b) Strong Decapsulation Consistency

Figure 2.4: Consistency experiments.

Definition 19 (TIBE Correctness)**.** *A* TIBE *scheme is δ-correct if for every message* msg*, identity* id*, every key set* $(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N})$ *and every set* act *of decryptors over the randomness of the honest encryption and key extraction procedures*

$$\Pr_{\mathsf{ct},\mathsf{contrib}}(\mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{act}, \mathsf{contrib}) = \mathsf{msg}) \geq 1 - \delta$$

where we denote $\mathsf{ct} \coloneqq \mathsf{TKEM.Encaps}(\mathsf{ek}, \mathsf{msg})$ *and* $\mathsf{contrib} \coloneqq \{\mathsf{contrib}_{i,j}\}^{j\leq r}_{i\in\mathsf{act}}$ *from the key share extraction procedure.*

Definition 20 (TKEM Correctness)**.** *A* TKEM *scheme is δ-correct if for every message* msg*, every key set* $(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N})$ *and every set* act *of decapsulators over the randomness of the honest encapsulation and decapsulation procedures*

$$\Pr_{\mathsf{ct},\mathsf{contrib}}(\mathsf{TKEM.Combine}(\mathsf{ct}, \mathsf{act}, \mathsf{contrib}) = \mathsf{msg}) \geq 1 - \delta$$

where we denote $\mathsf{ct} \coloneqq \mathsf{TKEM.Encaps}(\mathsf{ek}, \mathsf{msg})$ *and* $\mathsf{contrib} \coloneqq \{\mathsf{contrib}_{i,j}\}^{j\leq r}_{i\in\mathsf{act}}$ *from the partial decapsulation procedure.*

Definition 21 (Decapsulation Consistency [BBH06])**.** *We say that a* TKEM *scheme satisfies Weak (resp. Strong) Decapsulation Consistency if no PPT adversary can win the* CD *(resp.* strong-CD*) experiment in Figure 2.4a (resp. Figure 2.4b) with a non-negligible probability. We write* $\mathsf{Adv}^{\mathsf{CD}}_{\mathcal{A},\mathsf{TKEM}}$ *(resp.* $\mathsf{Adv}^{\mathsf{strong\text{-}CD}}_{\mathcal{A},\mathsf{TKEM}}$*) for the probability of winning the game for an adversary $\mathcal{A}$.*

The difference between the strong and weak consistency notions in Definition 21 is that the Weak Decapsulation Consistency does not prevent the adversary from sending invalid decapsulation shares and forcing the decapsulation to fail. However, Strong Decapsulation Consistency assumes existence of share verification algorithms TKEM.ShareVerify which guarantee a unique decapsulation outcome for both honest and malicious ciphertexts.

Definition 22 (Extraction Share Correctness)**.** *A* TIBE *scheme has δ'-correct extraction shares if for every message* msg*, key* $(\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in N})$*, set* act *of key extractors of size at least T and every party $i \in$* act *over the randomness of the honest encryption and key extraction algorithms*

$$\Pr_{\mathsf{ct},\mathsf{contrib}}(\mathsf{TIBE.ShareVerify}(\mathsf{ek}, \mathsf{id}, \mathsf{act}, \mathsf{contrib}, i) = 1) \geq 1 - \delta'.$$

Here we denote $\mathsf{ct} \coloneqq \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{id}, \mathsf{msg})$ *and* $\mathsf{contrib} \coloneqq \{\mathsf{contrib}_{j,r}\}_{r\in\mathsf{rounds}}^{j\in\mathsf{act}}$ *from the shared key extraction procedure.*

Definition 23 (Extraction Share Robustness)**.** *Let* TIBE *be a threshold identity-based encryption scheme with* RDistr *beings the distribution of its encryption randomness. Further, let* RSpace *be a set such that $R \leftarrow$* RDistr *lies in* RSpace *with overwhelming probability. A* TIBE *scheme is extraction share robust for randomness space* RSpace *if no PPT adversary can win the* Robust *experiment in Figure 2.2e with a non-negligible probability. We write* $\mathsf{Adv}^{\mathsf{Robust}}_{\mathcal{A},\mathsf{TIBE}}$ *for the winning probability of adversary $\mathcal{A}$ in the* Robust *experiment.*

Definition 24 (γ-spreadness adapted from [HHK17])**.** *Let* TIBE *be a Threshold IBE scheme defined in Definition 12 and let $\mathcal{K}, \mathcal{M}, \mathcal{I}, \mathcal{C}$ denote the sets of valid encryption keys, messages, identities, and ciphertexts respectively. For $\gamma > 0$ the* TIBE *is γ-spread if for every tuple* $(\mathsf{msg} \in \mathcal{M}, \mathsf{id} \in \mathcal{I}, \mathsf{ek} \in \mathcal{E})$

$$-\log \max_{\mathsf{ct}\in\mathcal{C}} \Pr_{\mathsf{rand}}(\mathsf{ct} = \mathsf{TIBE.Encrypt}(\mathsf{msg}, \mathsf{id}, \mathsf{ek}; \mathsf{rand})) \geq \gamma.$$

Chapter 3

Leftover Hash Lemma(s) over Cyclotomic Rings

The leftover hash lemma (LHL) is a versatile tool in cryptography, allowing to extract uniform looking randomness from a random variable which is itself not uniformly random, but has "good" properties, such as large min-entropy [ILL89, HILL99]. It can be applied in various contexts, but the one we consider in this work is its use in lattice-based cryptography.

Here, the LHL was first used over the special ring $\mathcal{R} = \mathbb{Z}$, i.e., the cyclotomic ring of degree 1. When q is a prime and hence $\mathbb{Z}_q$ a field, the original LHL [ILL89, HILL99] immediately holds. In more details, to apply the LHL it is enough to show that the function $\mathbf{A}\colon \mathbf{x} \mapsto \mathbf{A}\mathbf{x} \bmod q$ defines a universal hash function. That is, for a randomly chosen $\mathbf{A}$ and arbitrary inputs $\mathbf{x} \neq \mathbf{x}'$, the probability of $\mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{x}'$ equals one over the size of the image. Put differently, if we fix all but one columns of $\mathbf{A}$ and $\mathbf{x} \neq \mathbf{x}'$, the remaining column is uniquely defined. This is true over $\mathbb{Z}_q$ with prime q as every non-zero element in a field is invertible. For example, the standard LHL over $\mathbb{Z}$ is used in Regev's original public-key encryption scheme [Reg05], the hardness of LWE with binary secrets [GKPV10, BLP+13], or with deterministic noise [AKPW13]. In parallel, an alternative strategy to prove the LHL was deployed in the special case where $\mathbf{x}$ follows a discrete Gaussian distribution over $\mathbb{Z}$. Such distributions arise naturally when working over lattices, hence meriting a special approach. Here, the so-called smoothing property [MR07] of discrete Gaussians is used to show that for a large enough Gaussian width, $\mathbf{A}\mathbf{x} \bmod q$ (given $\mathbf{A}$) is statistically close to a uniformly sampled vector $\mathbf{u}$. Note that the min-entropy of the Gaussian distribution scales with its width. This variant of the LHL is for instance used for proving the security of Regev's dual

public-key encryption scheme [GPV08, BLR+18] and the re-randomisation of LWE samples [ACPS09, GMPW20]. As far as we are aware, [BLR+18] were the first to use the Rényi divergence as an alternative closeness measure for the LHL over $\mathbb{Z}$.

Leakage Version. The standard LHL [ILL89, HILL99] can be extended to allow for leakage on the vector $\mathbf{x}$, as proven in [DORS08]. This is commonly referred to as *generalised* LHL. More precisely, the goal is now to show that $\mathbf{Ax} \bmod q$, given $\mathbf{A}$ as well as some additional leakage y on $\mathbf{x}$, is close to a uniformly sampled vector $\mathbf{u}$. It requires that the distribution of $\mathbf{x}$ given leakage y still has "good" properties. Concretely, it is sufficient to show that the average conditional min-entropy of $\mathbf{x}$ given y is high. The generalised LHL can be applied to the same use cases over $\mathbb{Z}$ described above. It served for instance to demonstrate the hardness of LWE with entropic secrets [BD20] or to argue the circuit privacy of FHE [BdPMW16].

The General Case. When using cyclotomic rings of degrees larger than 1, one cannot apply the LHL in the same way as before. In general, $\mathcal{R}_q$ is not a field, even if q is prime.[1] Hence, not every non-zero ring element is invertible. This observation marked the start of a long sequence of works, many claiming a novel LHL over cyclotomic rings. To the best of our knowledge, the first LHL over (not necessarily cyclotomic) rings is due to Micciancio [Mic07]. It was originally restricted to cyclic rings of the form $\mathbb{Z}[X]/\langle X^d - 1\rangle$ for any integer d and matrices $\mathbf{A}$ with one row. It was subsequently generalised to cyclotomic rings [SSTX09] and matrices of multiple rows [BJRW23]. The latter work was also the first to use the Rényi divergence measure in the general case. In [KY16], limiting the setting to power-of-2 cyclotomic rings with a two-splitting prime modulus[2] allowed for tighter parameters. All quoted works only consider the uniform distribution over elements of small norm as the input distribution. Later, [LW20] claimed a different proof approach leading to more general conditions on the input distribution. Their LHL applied to the ring of integers of any number field and arbitrary hash input distributions, as long as they had a small enough collision probability (or large enough min-entropy). It can be seen as a generalisation of [MM11] to number fields. Note that all the results listed until now do not allow for any leakage. The work of [LW20] was generalised to cover a fixed leakage function (namely, leaking $\mathbf{x} + \mathbf{e}$, where $\mathbf{e}$ follows a continuous Gaussian distribution) in [LWZW24]. In a recent follow-up work [JLWG25], the LHL of [LWZW24]

[1] Requiring $\mathcal{R}_q$ to be a field is a very restrictive condition. For power-of-2 cyclotomic rings, predominant in practice, there exists no integer q meeting the condition.

[2] That is, the ring $\mathcal{R}_q$ is the product of two finite fields.

was generalised to arbitrary leakage functions. In [BI22], another version of the LHL is proven tailored to the setting of power-of-2 cyclotomics with power-of-2 moduli and noisy linear leakage.

In parallel to generalising the original LHL proof to the structured setting, another line of work [SS11, LPR13, RSW18] aimed to extend the smoothing-based argument to higher degree cyclotomics. Even though the high level approach stays the same, the main challenge lies in computing the smoothing parameter of a kernel lattice corresponding to the matrix $\mathbf{A}$. The computations are significantly easier for integer matrices compared to matrices over higher-degree cyclotomic rings. Additionally, there are two ways of sampling a discrete Gaussian distribution over $\mathcal{R}$ depending on the ring embedding, and the smoothing analysis depends on this choice. Whereas the generalised LHL can handle arbitrary leakage, allowing for leakage in the smoothing-based approach is less obvious. So far, it had only been addressed for specific leakage functions [DGKS21, SSE+24].

Ideally, one aims for an LHL over cyclotomic rings as versatile and general as the standard LHL over finite fields. Moreover, the application of such an LHL should not require the user to be an expert on cyclotomic rings. This brings us the research questions of this chapter:

> *Is it possible to 1) generalise the seemingly independent flavours of the LHL over cyclotomic rings into few single results, which 2) allow an easy adaptation to new mathematical settings and 3) new leakage functions?*

Our Results. We positively answer the above questions. First, we show that the large set of LHLs over cyclotomic rings can be reduced to two main theorems, addressing point 1) of our research question. One approach leads us to the LHL *via counting ideals*, and the other one to the LHL *via smoothing*. The previous results can then be expressed as corollaries of one of the two theorems. We further generalise both theorems to the leakage context, hence providing *generalised* LHLs over cyclotomic rings. We highlight that the counting approach applies to a large class of input distributions (including discrete Gaussians), whereas the smoothing approach is so far limited to discrete Gaussian distributions. Still, the latter approach has its benefits as it allows for parameter configurations not covered by the counting proof technique, a wider range of matrix distributions, and a tailored analysis in the case of linear leakage. With this new perspective at hand, we note that many of the previous results extend to broader settings, addressing point 2) and 3) of our research question. As a concrete example, our results in Section 4.2 provide a smoothing-based LHL in the coefficient embedding

for *any* cyclotomic ring, whereas before the state-of-the-art was limited to power-of-two cyclotomics. Moreover, still for the smoothing-based approach we prove a tailored LHL allowing for *exact* linear leakage, which to the best of our knowledge has not appeared in prior works.

Chapter structure. In Section 3.1 we discuss the first family of results that relies on the counting-based approach. We first state the general theorem and then prove various leftover hash lemmas as its corollaries. Then Section 3.2 we prove the general theorem for the smoothing-based approach and analyse the smoothing parameter for a range of lattices. In both sections we also discuss the effect of leakage on the LHL statements. Lastly, in Section 3.3 we analyse a number concrete distributions for their entropic properties useful for the LHL.

Open problems. The last missing step to systemise the study of leftover hash lemmas over cyclotomic rings, would be to find a general theorem encompassing both the counting and the smoothing approach. Moreover, tighter min-entropy bounds modulo ideals and under specific leakage distributions would yield better LHL results. Extending the smoothing-based approach to distributions other than discrete Gaussian and to larger classes of leakage, beyond linear leakage, could be an interesting future direction. Finally, a curious algebraic problem would be to analyse the probability of $\langle x_1 - x'_1, \ldots, x_m - x'_m \rangle = \mathcal{I}$ for random vectors $\mathbf{x}, \mathbf{x}'$ that appears in Theorem 1 rather than bounding it by $\Pr[\forall i : (x_i - x'_i) \in \mathcal{I}]$ as in Corollary 10.

3.1 LHL via Counting Ideals

Let $\mathcal{K}$ be a number field with $\mathcal{R}$ its ring of integers. Let m, n, q be positive integers and let $\mathcal{P}$ be a probability distribution over $\mathcal{R}_q^m$ with $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$. We define the distribution $\mathcal{P}'$ as follows: Sample $\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m})$ and $\mathbf{x} \leftarrow \mathcal{P}$, then output $(\mathbf{A}\mathbf{x} \bmod q, \mathbf{A}) \in \mathcal{R}_q^n \times \mathcal{R}_q^{n \times m}$. To ease readability, we sometimes assume the modulo q operation without writing it explicitly. Our overall goal is to show that, under clearly specified conditions, the distribution $\mathcal{P}'$ is (statistically or Rényi) close to the uniform distribution over $\mathcal{R}_q^n \times \mathcal{R}_q^{n \times m}$.

We start by bounding the statistical distance and the Rényi divergence using algebraic properties of ideals over the ring $\mathcal{R}_q$ in Section 3.1.1. In Section 3.1.2 we subsequently generalise the result to allow for leakage y on the hash input $\mathbf{x}$. Lastly, in Section 3.1.3

we derive different corollaries which replace the algebraic properties by more concrete terms already studied in the literature, e.g. collision probability or well-spreadness. Those corollaries both recover flavours of the leftover hash lemma over cyclotomic rings previously proposed in the literature, as well as provide new interpretations. Note that the statements of the main results Theorems 1 and 2 are purely over the ring and hence agnostic to the choice of the embedding. They actually apply to any number field, not necessarily cyclotomic. In contrast, the probabilistic terms such as collision probability and well-spreadness in Section 3.1.3 require specifying which embedding is used.

In Section 3.3 we give bounds with detailed proofs on well-spreadness and collision probability for various distributions over cyclotomic rings proposed in the lattice-based literature.

3.1.1 General Theorem

For $\mathbf{x} = (x_i)_{i=1}^m, \mathbf{x}' = (x_i')_{i=1}^m \in \mathcal{R}_q^m$, we define the ideal $\mathcal{I}_{\mathbf{x},\mathbf{x}'}$ as $\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \langle x_1 - x_1', \ldots, x_m - x_m' \rangle \subset \mathcal{R}_q$. This ideal has already implicitly appeared in previous LHLs [Mic07, MM11]. The notions of greatest common divisor ideal [Con24] or maximally belonging [LW20] are also used in the literature.

Theorem 1. *Let $\mathcal{K}$ be a number field with $\mathcal{R}$ its ring of integers. Let m, n, q be positive integers and $\mathcal{P}$ be a distribution over $\mathcal{R}_q^m$. We define*

$$\mathcal{P}' = \left\{ (\mathbf{A}\mathbf{x} \bmod q, \mathbf{A}) \mid \mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m}), \mathbf{x} \leftarrow \mathcal{P} \right\}, \text{ and}$$

$$\mathcal{U}' = \left\{ (\mathbf{u}, \mathbf{A}) \mid \mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m}), \mathbf{u} \leftarrow \mathcal{U}(\mathcal{R}_q^{n}) \right\}.$$

Denote by $\mathbb{I}$ the set of all ideals in $\mathcal{R}_q$. Then, it holds

$$\mathrm{SD}(\mathcal{P}', \mathcal{U}') \leq \frac{1}{2} \sqrt{\sum_{\substack{\mathcal{I}\in\mathbb{I} \\ \mathcal{I}\neq\mathcal{R}_q}} N(\mathcal{I})^n \cdot \Pr_{\mathbf{x},\mathbf{x}'\leftarrow\mathcal{P}}[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I}]},$$

$$\mathrm{RD}(\mathcal{P}'; \mathcal{U}') = \sum_{\mathcal{I}\in\mathbb{I}} N(\mathcal{I})^n \cdot \Pr_{\mathbf{x},\mathbf{x}'\leftarrow\mathcal{P}}[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I}].$$

We highlight that the formula is exact in the Rényi divergence case.

Proof. Our goal is to bound $\mathrm{Coll}(\mathcal{P}')$, and then apply Lemma 12 and Lemma 13. Over the randomness of $\mathbf{A}, \mathbf{A}' \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m})$ and $\mathbf{x}, \mathbf{x}' \leftarrow \mathcal{P}$ it yields

$$\begin{aligned}\mathrm{Coll}(\mathcal{P}') &= \Pr[\mathbf{A} = \mathbf{A}' \wedge \mathbf{A}\cdot\mathbf{x} = \mathbf{A}'\cdot\mathbf{x}'] = \Pr[\mathbf{A} = \mathbf{A}'] \cdot \Pr[\mathbf{A}\cdot\mathbf{x} = \mathbf{A}\cdot\mathbf{x}' | \mathbf{A} = \mathbf{A}'] \\ &= \frac{1}{|\mathcal{R}_q|^{n\cdot m}} \cdot \Pr[\mathbf{A}(\mathbf{x} - \mathbf{x}') = \mathbf{0}].\end{aligned}$$

Let $\mathbf{x} = (x_1, \ldots, x_m)$ and $\mathbf{x}' = (x'_1, \ldots, x'_m)$. We condition the remaining term on the value of ideal $\mathcal{I}_{\mathbf{x},\mathbf{x}'}$ generated by $\mathbf{x}$ and $\mathbf{x}'$, thus

$$\begin{aligned}\Pr[\mathbf{A}(\mathbf{x}-\mathbf{x}')=\mathbf{0}] &= \sum_{\mathcal{I}\in\mathbb{I}} \Pr[\mathbf{A}(\mathbf{x}-\mathbf{x}')=\mathbf{0} | \mathcal{I}_{\mathbf{x},\mathbf{x}'}=\mathcal{I}] \cdot \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'}=\mathcal{I}] \\ &= \sum_{\mathcal{I}\in\mathbb{I}} \frac{\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'}=\mathcal{I}]}{|\mathcal{I}|^n}.\end{aligned}$$

The last equality holds by Lemma 15. To obtain the final statement for the Rényi divergence, we put everything together and use Lemma 13 with $\mathrm{Supp}(\mathcal{P}') \subseteq \mathcal{R}_q^n \times \mathcal{R}_q^{n\times m} := S$, where $|S| = |\mathcal{R}_q|^{n\cdot(m+1)}$. It holds

$$\begin{aligned}\mathrm{RD}(\mathcal{P}';\mathcal{U}') &= |\mathcal{R}_q|^{n\cdot(m+1)} \cdot \mathrm{Coll}(\mathcal{P}') = |\mathcal{R}_q|^{n\cdot(m+1)} \cdot \frac{1}{|\mathcal{R}_q|^{n\cdot m}} \cdot \sum_{\mathcal{I}\in\mathbb{I}} \frac{\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'}=\mathcal{I}]}{|\mathcal{I}|^n} \\ &= \sum_{\mathcal{I}\in\mathbb{I}} N(\mathcal{I})^n \cdot \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'}=\mathcal{I}].\end{aligned}$$

The last equality uses the fact that $N(\mathcal{I}) = |\mathcal{R}_q|/|\mathcal{I}|$. The statement for the statistical distance is obtained analogously using Lemma 12, the fact that $N(\mathcal{R}_q) = 1$ and that $\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{R}_q] \leq 1$.

□

3.1.2 General Theorem in the Presence of Leakage

In this section, besides the hash input distribution $\mathcal{P}$ over $\mathcal{R}_q^m$, we also consider a leakage distribution $\mathcal{Q}$ over a domain T. The distribution $\mathcal{P}'$ is as before: Sample $\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m})$ and $\mathbf{x} \leftarrow \mathcal{P}$, then output $(\mathbf{A}\mathbf{x} \bmod q, \mathbf{A})$. Our overall goal is to show that, under specific conditions, the distribution $\mathcal{P}'$ is (statistically or Rényi) close to the uniform distribution $\mathcal{U}(\mathcal{R}_q^n \times \mathcal{R}_q^{n\times m})$, even in the presence of leakage sampled from $\mathcal{Q}$, that depends on $\mathbf{x}$.

Recall the definition of the ideal $\mathcal{I}_{\mathbf{x},\mathbf{x}} = \langle x_1 - x'_1, \ldots, x_m - x'_m \rangle \subset \mathcal{R}_q$ for $\mathbf{x} = (x_i)_{i=1}^m$, $\mathbf{x}' = (x'_i)_{i=1}^m \in \mathcal{R}_q^m$. Theorem 2 below is similar to Theorem 1, where $\Pr_{\mathbf{x},\mathbf{x}'\leftarrow\mathcal{P}}[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I}]$ is replaced by $\mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr_{\mathbf{x},\mathbf{x}'\leftarrow\mathcal{P}}[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y]$.

Theorem 2. *Let $\mathcal{K}$ be a number field with $\mathcal{R}$ its ring of integers. Let m, n, q be positive integers, let $\mathcal{P}$ be a distribution over $\mathcal{R}_q^m$ and let $\mathcal{Q}$ be another distribution over a set T. We define*

$$\mathcal{P}' = \left\{(\mathbf{A}\mathbf{x} \bmod q, \mathbf{A}) \mid \mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m}), \mathbf{x} \leftarrow \mathcal{P}\right\}, \text{ and}$$
$$\mathcal{U}' = \left\{(\mathbf{u}, \mathbf{A}) \mid \mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m}), \mathbf{u} \leftarrow \mathcal{U}(\mathcal{R}_q^n)\right\}.$$

Denote by $\mathbb{I}$ the set of all ideals in $\mathcal{R}_q$. Then, it holds

$$\mathrm{SD}\left((\mathcal{P}', \mathcal{Q}), (\mathcal{U}', \mathcal{Q})\right) \leq \frac{1}{2}\sqrt{\sum_{\substack{\mathcal{I}\in\mathbb{I}\\ \mathcal{I}\neq\mathcal{R}_q}} N(\mathcal{I})^n \cdot \mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr_{\mathbf{x},\mathbf{x}'\leftarrow\mathcal{P}}[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y]},$$

$$\mathrm{RD}\left((\mathcal{P}', \mathcal{Q}); (\mathcal{U}', \mathcal{Q})\right) = \sum_{\mathcal{I}\in\mathbb{I}} N(\mathcal{I})^n \cdot \mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr_{\mathbf{x},\mathbf{x}'\leftarrow\mathcal{P}}[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y].$$

Proof. Similar to the proof of Theorem 1, we bound the average conditional collision probability of $\mathcal{P}'$ given leakage $\mathcal{Q}$ and then apply Lemma 12 and Lemma 13. Over the random choices of $\mathbf{A}, \mathbf{A}' \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m})$ and $\mathbf{x}, \mathbf{x}' \leftarrow \mathcal{P}$ it holds

$$\begin{aligned}
\widetilde{\mathrm{Coll}}(\mathcal{P}' \mid \mathcal{Q}) &= \mathbb{E}_{y\leftarrow\mathcal{Q}}\left[\Pr\left[(\mathbf{A}, \mathbf{A}\mathbf{x}) = (\mathbf{A}', \mathbf{A}'\mathbf{x}') \mid y\right]\right] \\
&= \mathbb{E}_{y\leftarrow\mathcal{Q}}\left[\Pr[\mathbf{A} = \mathbf{A}' \mid y] \cdot \Pr\left[\mathbf{A}\mathbf{x} = \mathbf{A}'\mathbf{x}' \mid y, \mathbf{A} = \mathbf{A}'\right]\right] \\
&= \mathbb{E}_{y\leftarrow\mathcal{Q}}\left[\Pr[\mathbf{A} = \mathbf{A}'] \cdot \Pr\left[\mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{x}' \mid y\right]\right] \\
&= \frac{1}{|\mathcal{R}_q|^{n\cdot m}} \cdot \mathbb{E}_{y\leftarrow\mathcal{Q}}\left[\Pr\left[\mathbf{A}(\mathbf{x} - \mathbf{x}') = \mathbf{0} \mid y\right]\right],
\end{aligned}$$

where line 2 to 3 we used that $y \leftarrow \mathcal{Q}$ is independent of $\mathbf{A}, \mathbf{A}' \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m})$. Let $\mathbf{x} = (x_1, \ldots, x_m)$ and $\mathbf{x}' = (x'_1, \ldots, x'_m)$. We condition the term inside the expected value on the ideal $\mathcal{I}_{\mathbf{x},\mathbf{x}'}$ generated by the difference of $\mathbf{x}$ and $\mathbf{x}'$, that is

$$\begin{aligned}
\Pr\left[\mathbf{A}(\mathbf{x} - \mathbf{x}') = \mathbf{0} \mid y\right] &= \sum_{\mathcal{I}\in\mathbb{I}} \Pr\left[\mathbf{A}(\mathbf{x} - \mathbf{x}') = \mathbf{0} \mid y \wedge \mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I}\right] \cdot \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y] \\
&= \sum_{\mathcal{I}\in\mathbb{I}} \frac{\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y]}{|\mathcal{I}|^n},
\end{aligned}$$

where we used that Lemma 15 applies even in the presence of leakage y on $\mathbf{x}$ and $\mathbf{x}'$ as the result holds for arbitrary hash inputs. Using Lemma 12 and Lemma 13 together with $\mathrm{Supp}(\mathcal{P}') \subseteq \mathcal{R}_q^{n\times m} \times \mathcal{R}_q^n$ concludes the proof. We write the argument in full for the Rényi divergence

$$\begin{aligned}
\mathrm{RD}\left((\mathcal{P}', \mathcal{Q}); (\mathcal{U}', \mathcal{Q})\right) &= |\mathcal{R}_q|^{n\cdot(m+1)} \cdot \widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) \\
&= |\mathcal{R}_q|^{n\cdot(m+1)} \cdot \frac{1}{|\mathcal{R}_q|^{n\cdot m}} \cdot \mathbb{E}_{y\leftarrow\mathcal{Q}} \sum_{\mathcal{I}\in\mathbb{I}} \frac{\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y]}{|\mathcal{I}|^n} \\
&= \sum_{\mathcal{I}\in\mathbb{I}} \frac{|\mathcal{R}_q|^n}{|\mathcal{I}|^n} \mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y] \\
&= \sum_{\mathcal{I}\in\mathbb{I}} N(\mathcal{I})^n \cdot \mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y].
\end{aligned}$$

For the statistical distance we proceed in the same way, except we further use that $N(\mathcal{R}_q) = 1$ and $\mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{R}_q \mid y] \leq 1$. □

3.1.3 Corollaries

The first corollary provides an LHL with respect to the (average conditional) collision probability. For a distribution $\mathcal{P}$ over $\mathcal{R}_q^m$ and ideal $\mathcal{I}$ of $\mathcal{R}_q$, we denote by $\mathcal{P} \bmod \mathcal{I}$ the distribution which first samples an element from $\mathcal{P}$ and then computes its value modulo $\mathcal{I}$ coordinatewise.

Corollary 10. *Let $\mathcal{K}$ be a number field with $\mathcal{R}$ its ring of integers. Let m, n, q be positive integers and $\mathcal{P}$ be a distribution over $\mathcal{R}_q^m$, defining the distributions $\mathcal{P}'$ and $\mathcal{U}'$ as in Theorem 2. Let $\mathcal{Q}$ be the leakage distribution. Denote $\mathbb{I}$ the set of all ideals in $\mathcal{R}_q$. Then, it holds*

$$\mathrm{SD}\left((\mathcal{P}', \mathcal{Q}), (\mathcal{U}', \mathcal{Q})\right) \leq \frac{1}{2}\sqrt{\sum_{\substack{\mathcal{I}\in\mathbb{I} \\ \mathcal{I}\neq\mathcal{R}_q}} N(\mathcal{I})^n \cdot \widetilde{\mathrm{Coll}}(\mathcal{P} \bmod \mathcal{I} \mid \mathcal{Q})],$$

$$\mathrm{RD}\left((\mathcal{P}', \mathcal{Q}); (\mathcal{U}', \mathcal{Q})\right) \leq \sum_{\mathcal{I}\in\mathbb{I}} N(\mathcal{I})^n \cdot \widetilde{\mathrm{Coll}}(\mathcal{P} \bmod \mathcal{I} \mid \mathcal{Q}).$$

If $\mathcal{Q}$ is independent of $\mathcal{P}$ then

$$\mathrm{SD}(\mathcal{P}', \mathcal{U}') \leq \frac{1}{2}\sqrt{\sum_{\substack{\mathcal{I}\in\mathbb{I} \\ \mathcal{I}\neq\mathcal{R}_q}} N(\mathcal{I})^n \cdot \mathrm{Coll}(\mathcal{P} \bmod \mathcal{I})},$$

$$\mathrm{RD}(\mathcal{P}'; \mathcal{U}') \leq \sum_{\mathcal{I}\in\mathbb{I}} N(\mathcal{I})^n \cdot \mathrm{Coll}(\mathcal{P} \bmod \mathcal{I}).$$

Proof. Let $\mathbf{x} = (x_1, \ldots, x_m)$ and $\mathbf{x}' = (x'_1, \ldots, x'_m)$, defining $\mathcal{I}_{\mathbf{x},\mathbf{x}'}$. The corollary follows from Theorem 2 as for $\mathbf{x}, \mathbf{x} \leftarrow \mathcal{P}$ and $\mathcal{I} \in \mathbb{I}$, it holds

$$\begin{aligned}\mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y] &\leq \mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr[\forall i : x_i - x'_i \in \mathcal{I} \mid y] \\ &= \mathbb{E}_{y\leftarrow\mathcal{Q}} \mathrm{Coll}(\mathcal{P} \bmod \mathcal{I} \mid y) = \widetilde{\mathrm{Coll}}(\mathcal{P} \bmod \mathcal{I} \mid \mathcal{Q}).\end{aligned}$$

□

The corollary recovers the recent LHL from [JLWG25, Lem. 5.1], generalising the LHLs in [LW20, Thm. 5.7] and [LWZW24, Thm. 4]. Note that the inequality $\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I}] \leq \Pr[\forall i : x_i - x'_i \in \mathcal{I}]$ above is loose and computing the probability of $\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I}$ is an interesting open question.

The LHL proven in [BI22, Lem. 12] can be seen as a special case of Corollary 10, where both d and q are powers of 2, the leakage takes at most $|\mathcal{R}_q|$ values and the hash input distribution $\mathcal{P}$ is a spherical discrete Gaussian (over a non-trivial lattice). The main

part of the proof consists of providing a global bound H for the min-entropy modulo any ideal and using Lemma 14. To do so, they make use of the special shape of $\mathbb{I}$ when both d and q are powers of 2.

In the case of a uniform bounded hash input distribution, one can compute the sum of norms and collision probabilities, as given in the following corollary. It covers the LHL versions proved in [Mic07, Thm. 4.2], [SSTX09, Thm. 3.2], and [BJRW23, Lem. 2.8].

Corollary 11 (Bounded Uniform)**.** *Let $\mathcal{K}$ be a cyclotomic field with $\mathcal{R}$ its ring of integers. Further, let q be an unramified prime modulus such that $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$ for distinct, irreducible $\Phi_j(X)$ of degree δ. Let m, n, η be positive integers such that $\eta < q/2$. Define $S_\eta = \{x \in \mathcal{R} \colon \|\tau(x)\|_\infty \leq \eta\}$, and let $\mathcal{P}$ be the uniform distribution over $S_\eta^m \subset \mathcal{R}_q^m$, defining the distributions $\mathcal{P}'$, and $\mathcal{U}'$ as in Theorem 1. Then, it holds*

$$\mathrm{SD}(\mathcal{P}', \mathcal{U}') \leq \frac{1}{2}\sqrt{\left(\left(\frac{q^n}{(2\eta+1)^m}\right)^\delta + 1\right)^f - 1},$$

$$\mathrm{RD}(\mathcal{P}'; \mathcal{U}') \leq \left(\left(\frac{q^n}{(2\eta+1)^m}\right)^\delta + 1\right)^f.$$

Note that one can simplify (but loosen) the bounds using

$$\left(\left(\frac{q^n}{(2\eta+1)^m}\right)^\delta + 1\right)^f \leq \left(\frac{q^n}{(2\eta+1)^m} + 1\right)^d.$$

Proof. We use the statement of Corollary 10 as a starting point. Recall from Lemma 9 that every ideal $\mathcal{I}$ of $\mathcal{R}_q$ is of the form $\mathcal{I} = \langle \prod_{j \in G} \Phi_j(X) \rangle$ with norm $N(\mathcal{I}) = q^{\delta|G|}$, for some subset of indices $G \subseteq \{1, \ldots, f\}$. Using Lemma 60,

$$\begin{aligned}\sum_{\mathcal{I} \in \mathbb{I}} N(\mathcal{I})^n \cdot \mathrm{Coll}(\mathcal{P} \bmod \mathcal{I}) &\leq \sum_{G \subseteq \{1,\ldots,f\}} q^{n\delta|G|} \cdot \frac{1}{(2\eta+1)^{m\delta|G|}} \\ &= \sum_{j=0}^{f} \binom{f}{j} \left(\frac{q^{n\delta}}{(2\eta+1)^{m\delta}}\right)^j = \left(\left(\frac{q^n}{(2\eta+1)^m}\right)^\delta + 1\right)^f\end{aligned}$$

here for the first summand we used $N(\mathcal{R}_q) \cdot \mathrm{Coll}(\mathcal{P} \bmod \mathcal{R}_q) = 1 \cdot 1 = 1$

□

Similarly, we obtain a corollary for discrete Gaussian distributions in the coefficient embedding, implicit in [JLWG25].

Corollary 12 (Discrete Gaussian)**.** *Let $\mathcal{K}$ be a cyclotomic field with $\mathcal{R}$ its ring of integers. Further, let q be an unramified prime modulus such that $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$ for distinct, irreducible $\Phi_j(X)$ of degree δ. Let m, n be positive integers, σ be a positive real and $\mathbf{c} \in \mathcal{R}^m$, and set $\mathcal{P} = \mathcal{D}^{\tau}_{\mathcal{R}^m,\sigma,\mathbf{c}}$, defining the distributions $\mathcal{P}'$ and $\mathcal{U}'$ as in Theorem 1. If* [3] *$\sigma \leq \frac{\sqrt{\pi}}{2} \cdot \frac{q}{\sqrt{\ln(4md)}}$ it holds*

$$\mathrm{SD}(\mathcal{P}', \mathcal{U}') \leq \frac{1}{\sqrt{2}} \sqrt{\left(\left(\frac{q^n}{\sigma^m}\right)^{\delta} + 1\right)^{f} - 1},$$

$$\mathrm{RD}(\mathcal{P}'; \mathcal{U}') \leq 2 \left(\left(\frac{q^n}{\sigma^m}\right)^{\delta} + 1\right)^{f}.$$

Proof. The proof is very similar to the one of Corollary 11. For the special case $G = \varnothing$ we have $\mathcal{I} = \mathcal{R}_q$ then $N(\mathcal{I}) = 1$ and for any $x, y \in \mathcal{R}_q$ we have $\Pr(x = y \bmod \mathcal{I}) = 1$. Otherwise, using Lemma 62 with $\sigma \leq \frac{\sqrt{\pi}}{2} \cdot \frac{q}{\sqrt{\ln(4md)}} \leq \frac{\sqrt{\pi}}{2} \cdot \frac{q}{\sqrt{\ln(4m\delta|G|)}}$ for all $G \subseteq \{1, \ldots, f\}$, $G \neq \varnothing$, as well as $\mathrm{Coll}(\mathcal{P} \bmod \mathcal{I}) \leq 2^{-\mathrm{H}_\infty(\mathcal{P} \bmod \mathcal{I})}$ it holds

$$\begin{aligned}
\sum_{\mathcal{I} \in \mathbb{I}} N(\mathcal{I})^n \cdot \mathrm{Coll}(\mathcal{P} \bmod \mathcal{I}) &\leq \sum_{\mathcal{I} \in \mathbb{I}} N(\mathcal{I})^n \cdot 2^{-\mathrm{H}_\infty(\mathcal{P} \bmod \mathcal{I})} \\
&\leq 1 + \sum_{G \subseteq \{1,\ldots,f\}, G \neq \varnothing} q^{n\delta|G|} \cdot 2^{-(m\delta|G| \log \sigma - 1)} \\
&= 2 \sum_{j=0}^{f} \binom{f}{j} \left(\frac{q^{n\delta}}{2^{m\delta \log \sigma}}\right)^{j} = 2 \left(\left(\frac{q^n}{\sigma^m}\right)^{\delta} + 1\right)^{f},
\end{aligned}$$

concluding the proof for the Rényi divergence. For the statistical distance, we have a similar reasoning

$$\begin{aligned}
\sum_{\mathcal{I} \in \mathbb{I}, \mathcal{I} \neq \mathcal{R}_q} N(\mathcal{I})^n \cdot 2^{-\mathrm{H}_\infty(\mathcal{P} \bmod \mathcal{I})} &\leq \sum_{G \subseteq \{1,\ldots,f\}, G \neq \varnothing} q^{n\delta|G|} \cdot 2^{-(m\delta|G| \log \sigma - 1)} \\
&= 2 \left(\sum_{j=0}^{f} \binom{f}{j} \left(\frac{q^{n\delta}}{2^{m\delta \log \sigma}}\right)^{j} - 1\right) \\
&= 2 \left(\left(\left(\frac{q^n}{\sigma^m}\right)^{\delta} + 1\right)^{f} - 1\right).
\end{aligned}$$

□

For general distributions, it might be difficult to compute the product of norms and collision probabilities for each ideal. Then, one can upper bound either the norms or the collision probabilities separately, as in the two corollaries below.

[3] We believe the upper bound on the Gaussian parameter may be a proof artefact of the argument in [JLWG25].

The following statement generalises the LHL in [KY16, Lem. 4] from power-of-two cyclotomics to general cyclotomics, and additionally allows for arbitrarily shaped leakage. Essentially, it says that if $\mathcal{R}_q$ splits into few subfields and $\mathcal{P}$ provides elements of short norm with high probability, the ideal $\mathcal{I}_{\mathbf{x},\mathbf{x}'}$ generates the full ring unless $\mathbf{x}$ and $\mathbf{x}'$ collide. In other words, in the low-splitting regime, the function $\mathbf{A}\colon \mathbf{x} \mapsto \mathbf{A}\mathbf{x} \bmod q$ defines a universal hash function and we recover the standard (generalised) LHL [ILL89, HILL99, DORS08]. To the best of our knowledge, this work is the first to derive a generalisation of the LHL in the low-splitting regime that allows for leakage.

Corollary 13 (Low-Splitting)**.** *Let $\mathcal{K} = \mathbb{Q}[X]/\langle\Phi(X)\rangle$ be the νth cyclotomic field of degree $d = \varphi(\nu)$, with $\mathcal{R}$ its ring of integers. Further, let $\nu = \prod p_i^{e_i}$ for $e_i \geq 1$ and let $\mu = \prod p_i^{f_i}$ for $1 \leq f_i \leq e_i$. Let q be a prime such that $q = 1 \bmod \mu$ and the multiplicative order modulo ν of q is ν/μ. Let $\mathfrak{s}_1(\mu)$ be the spectral norm of the Vandermonde matrix of $\mathbb{Q}(\zeta_\mu)$ and let η be a positive integer such that $2\eta < q^{1/\varphi(\mu)}/\mathfrak{s}_1(\mu)$, defining $S_\eta = \{x \in \mathcal{R}_q\colon \|\tau(x)\|_\infty \leq \eta\}$. Further, let $\mathcal{P}$ be a distribution over $\mathcal{R}_q^m$ such that $\Pr[\forall i \in [m] : x_i \in S_\eta \mid \sum_{i\in[m]} x_i \cdot X^i = x \leftarrow \mathcal{P}] \geq 1 - C$ for some real $C \in [0,1]$. Moreover, let $\mathcal{Q}$ be the leakage distribution. We define the distributions $\mathcal{P}'$ and $\mathcal{U}'$ as in Theorem 2. Then, it holds*

$$\mathrm{SD}((\mathcal{P}',\mathcal{Q}),(\mathcal{U}',\mathcal{Q})) \leq \frac{1}{2}\sqrt{|\mathcal{R}_q|^n \cdot \left(\widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) + 2C\right)}$$

$$\mathrm{RD}((\mathcal{P}',\mathcal{Q});(\mathcal{U}',\mathcal{Q})) \leq 1 + |\mathcal{R}_q|^n \cdot \left(\widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) + 2C\right).$$

If $\mathcal{Q}$ is independent of $\mathcal{P}$, this simplifies to

$$\mathrm{SD}(\mathcal{P}',\mathcal{U}') \leq \frac{1}{2}\sqrt{|\mathcal{R}_q|^n \cdot (\mathrm{Coll}(\mathcal{P}) + 2C)},$$

$$\mathrm{RD}(\mathcal{P}';\mathcal{U}') \leq 1 + |\mathcal{R}_q|^n \cdot (\mathrm{Coll}(\mathcal{P}) + 2C).$$

Proof. First, we recall that $N(\mathcal{I}) \leq |\mathcal{R}_q|$ for every ideal $\mathcal{I}$ of $\mathcal{R}_q$. By the linearity of expectation, it is thus enough to show that for $\mathbf{x}, \mathbf{x}' \leftarrow \mathcal{P}$ it yields

$$\begin{aligned}\sum_{\substack{\mathcal{I}\in\mathbb{I}\\ \mathcal{I}\neq\mathcal{R}_q}} \mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y] &= \mathbb{E}_{y\leftarrow\mathcal{Q}} \sum_{\substack{\mathcal{I}\in\mathbb{I}\\ \mathcal{I}\neq\mathcal{R}_q}} \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} = \mathcal{I} \mid y] \\ &= \mathbb{E}_{y\leftarrow\mathcal{Q}} \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} \neq \mathcal{R}_q \mid y] \leq \widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}) + 2C.\end{aligned}$$

By the law of total probabilities and the linearity of the expectation we have

$$\begin{aligned}\mathbb{E}_y \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} \neq \mathcal{R}_q \mid y] &= \mathbb{E}_y \left(\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} \neq \mathcal{R}_q \mid \mathbf{x} \neq \mathbf{x}', y] \cdot \Pr[\mathbf{x} \neq \mathbf{x}' \mid y]\right. \\ &\quad \left. + \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} \neq \mathcal{R}_q \mid \mathbf{x} = \mathbf{x}', y] \cdot \Pr[\mathbf{x} = \mathbf{x}' \mid y]\right) \\ &\leq \mathbb{E}_y(\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} \neq \mathcal{R}_q \mid \mathbf{x} \neq \mathbf{x}', y]) + \mathbb{E}_y(\mathrm{Coll}(\mathcal{P} \mid y)) \\ &= \mathbb{E}_y(\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} \neq \mathcal{R}_q \mid \mathbf{x} \neq \mathbf{x}', y]) + \widetilde{\mathrm{Coll}}(\mathcal{P} \mid \mathcal{Q}).\end{aligned}$$

It remains to bound the first probability term by $2C$. It holds

$$\begin{aligned}\Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} \neq \mathcal{R}_q \mid \mathbf{x} \neq \mathbf{x}', y] &\leq \Pr[\mathcal{I}_{\mathbf{x},\mathbf{x}'} \neq \mathcal{R}_q \mid \mathbf{x} \neq \mathbf{x}' \wedge \|\mathbf{x} - \mathbf{x}'\|_\infty \leq 2\eta, y] \\ &\quad + \Pr[\|\mathbf{x} - \mathbf{x}'\|_\infty > 2\eta] \leq 0 + 2C,\end{aligned}$$

where we used the properties of $\mathcal{P}$ and Lemma 8 to prove $\langle \mathbf{x} - \mathbf{x}' \rangle = \mathcal{R}_q$ for any $\|\mathbf{x} - \mathbf{x}'\| \leq 2\eta$. Note that the latter result holds for arbitrary elements fulfilling the conditions, so even in the presence of leakage. This concludes the proof.

□

Remark 7. To apply the corollary above in the presence of leakage one has different possibilities. Most available literature uses the chain rule from Lemma 14. Informally, it lower bounds the incurred entropy loss by the size of the leakage space. Alternatively, one can use more specific chain rules for concrete choices of distributions. As an example, [BD20, Sec. 5] proved a lower bound where, given $\mathbf{x} \leftarrow \mathcal{P}$ for arbitrary distribution $\mathcal{P}$, the distribution $\mathcal{Q}$ samples a Gaussian error $\mathbf{e}$ and outputs $\mathbf{x} + \mathbf{e}$.

Remark 8. Note that the above result enforces a relationship between the norm bound η of the hash input and the splitting behaviour of the ring $\mathcal{R}$ modulo q. In particular, the larger is μ (and thus $\mathfrak{s}_1(\mu)$ and $\varphi(\mu)$), the larger q needs to be for a fixed η to fulfil the condition. For example, if we are in the high-splitting regime, where ν/μ is a small constant, the modulus q would need to be exponentially large, even for the smallest $\eta = 1$. Thus, for practical parameter sets, this approach requires a low-splitting regime.

Next, we provide an alternative corollary, involving the well-spreadness of the hash input distribution $\mathcal{P}$. Even though it generally leads to looser bounds than Corollary 13, it can be used in the high-splitting regime with practical parameters. Well-spreadness serves as an upper bound on $\mathrm{Coll}(\mathcal{P} \bmod \mathcal{I})$ for any possible ideal $\mathcal{I}$. Note that we do not propose a leakage variant for it, as well-spreadness does depend on the distribution. Such generalisation would require a notion of well-spreadness under leakage, which we leave as future work.

Corollary 14 (Well-Spread)**.** *Let $\mathcal{K}$ be a cyclotomic field such that $\mathcal{R}$ is its ring of integers. Let m, n, q be positive integers, where q is an unramified prime s.t. $\mathcal{R}_q$ splits on a product of f subfields of degree q^δ. For $B \in [0,1]$, let $\mathcal{X}$ be a B-well-spread distribution over $\mathcal{R}_q$, defining the distribution $\mathcal{P} = (\mathcal{X})^m$ and $\mathcal{P}'$ and $\mathcal{U}'$ as in Theorem 1. Then, it holds*

$$\mathrm{SD}(\mathcal{P}',\mathcal{U}') \leq \frac{1}{2}\sqrt{B^m \cdot |\mathcal{R}_q|^n \cdot e^{f \cdot q^{-\delta n}}}, \quad \textit{and} \quad \mathrm{RD}(\mathcal{P}';\mathcal{U}') \leq B^m \cdot |\mathcal{R}_q|^n \cdot e^{f \cdot q^{-\delta n}}.$$

Proof. Let $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$ and for $j \in [f]$, denote $\mathcal{I}_j = \langle \Phi_j(X) \rangle$. Let $\mathcal{I} \in \mathbb{I}$ s.t. $\mathcal{I} \neq \mathcal{R}_q$. Then by Lemma 9, $\mathcal{I} = \prod_{j \in G} \mathcal{I}_j$ for some $G \subseteq \{1, \ldots, f\}$, $G \neq \varnothing$. Then for an arbitrary $j \in G$

$$\mathrm{Coll}(\mathcal{P} \bmod \mathcal{I}) = \Pr_{\mathbf{x},\mathbf{y} \leftarrow \mathcal{P}}[\mathbf{x} = \mathbf{y} \bmod \mathcal{I}] \leq \Pr_{\mathbf{x},\mathbf{y} \leftarrow \mathcal{P}}[\mathbf{x} = \mathbf{y} \bmod \mathcal{I}_j] \leq B^m.$$

Here the last inequality follows from the well-spreadness of $\mathcal{X}$ and $\mathcal{P}$ being a product of m independent distributions. By Lemma 10 for q prime, $\sum_{\substack{\mathcal{I} \in \mathbb{I} \\ \mathcal{I} \neq \mathcal{R}_q}} N(\mathcal{I})^n \leq \sum_{\mathcal{I} \in \mathbb{I}} N(\mathcal{I})^n \leq |\mathcal{R}_q|^n \cdot e^{f \cdot q^{-\delta n}}$ and the corollary follows. □

3.2 LHL via Smoothing

In this section, we consider an alternative approach to obtain an LHL over cyclotomic rings. It focuses on the distribution of a discrete Gaussian vector $\mathbf{x}$ over $\mathcal{R}$ multiplied with a matrix $\mathbf{A}$ sampled from some distribution $\mathcal{P}_A$ over $\mathcal{R}_q$. We prove that the output distribution is independent of $\mathcal{P}_A$ and close to uniform with respect to statistical distance and Rényi divergence. Note that in Section 3.1 we only looked at $\mathcal{P}_A$ being the uniform distribution. In this section, we take a more general perspective as slightly different distributions have been studied in the literature. The only works we are aware of that use the Rényi divergence in combination with a smoothing-based LHL (in slightly different contexts), are [BLR+18, Cor. 4.4] and [GJK24, Lem. 14].

We start by providing the general result in Section 3.2.1. It guarantees closeness to uniform as long as the width of $\mathbf{x}$ lies above the smoothing parameter of the kernel lattice of $\mathbf{A}$ and the image of $\mathbf{A}$ is of full rank, both with overwhelming probability over $\mathbf{A} \leftarrow \mathcal{P}_A$. Moreover, we show in Corollary 15 that once we can guarantee both conditions with respect to one embedding, we obtain an LHL for the other embedding too, hence allowing to switch between canonical and coefficient embedding easily. To the best of our knowledge, this observation leads to the first smoothing-based LHL in the coefficient embedding for arbitrary cyclotomic rings.

In Section 3.2.2, we give a concrete analysis of the smoothing parameter of the kernel lattice $\Lambda_q^\perp(\mathbf{A})$ for different choices of $\mathcal{P}_A$. The section both recalls existing results on non-uniform $\mathcal{P}_A$ and shows that we can generalise them to statements for $\mathbf{A}$ that comes from the uniform distribution. In Section 3.2.3 and Section 3.2.4 we look at smoothing in the presence of exact and noisy linear leakage, respectively.

3.2.1 General Theorem

In this section, we formally describe the general *smoothing-based* argument, leaving the analysis of the smoothing parameter for later. We emphasise that the proof itself is not novel, but separating the general argument from the concrete smoothing analysis allows for a more modular perspective. The theorem is stated for cyclotomic fields, but the proof generalises to any number field.

Theorem 3. *Let $\mathcal{K}$ be a cyclotomic number field with $\mathcal{R}$ its ring of integers. For positive integer q, we define the finite ring $\mathcal{R}_q$. Let $\mathcal{P}_A$ be a distribution on the set of matrices $\mathcal{R}_q^{n\times m}$ for $n \le m$. Let $\varepsilon \in (0,1)$, and $\beta, \delta_1, \delta_2 > 0$ and $\pi \in \{\tau, \theta\}$ represent either lattice embedding. Suppose that*

$$\Pr\Big[\eta_\varepsilon(\pi(\Lambda_q^\perp(\mathbf{A}))) > \beta | \mathbf{A} \leftarrow \mathcal{P}_A\Big] < \delta_1,$$
$$\Pr\Big[\mathrm{im}(\mathbf{A}) \neq \mathcal{R}_q^n | \mathbf{A} \leftarrow \mathcal{P}_A\Big] < \delta_2.$$

Let $\mathbf{c} \in \mathcal{R}^m$, $\mathbf{\Sigma} \in \mathbb{R}^{dm\times dm}$ positive definite s.t. $\mathbf{\Sigma} \ge \beta^2$. Define

$$\mathcal{P}' = \Big\{(\mathbf{A}\mathbf{x} \bmod q, \mathbf{A}) \mid \mathbf{A} \leftarrow \mathcal{P}_A, \mathbf{x} \leftarrow \mathcal{D}^\pi_{\mathcal{R}^m, \sqrt{\mathbf{\Sigma}}, \mathbf{c}}\Big\},$$
$$\mathcal{U}' = \Big\{(\mathbf{u}, \mathbf{A}) \mid \mathbf{A} \leftarrow \mathcal{P}_A, \mathbf{u} \leftarrow \mathcal{U}(\mathcal{R}_q^n)\Big\}.$$

Then

$$\mathrm{SD}(\mathcal{P}', \mathcal{U}') < \varepsilon/(1-\varepsilon) + \delta_1 + \delta_2, \qquad \mathrm{RD}^{(\delta_1+\delta_2)}(\mathcal{P}'; \mathcal{U}') < \frac{1+\varepsilon}{1-\varepsilon}.$$

Proof. We partition the set $\mathcal{R}_q^{n\times m}$ into $\mathcal{S} = \{\mathbf{A} \in \mathcal{R}_q^{n\times m} | \eta_\varepsilon(\pi(\Lambda_q^\perp(\mathbf{A}))) \le \beta, \mathrm{im}(\mathbf{A}) = \mathcal{R}_q^n\}$, that is, matrices that satisfy both constrains, and its complement. We first prove the statements for an arbitrary fixed matrix $\mathbf{A} \in \mathcal{S}$.

The mapping $\mathbf{A} : \mathbf{x} \in \mathcal{R}^m \mapsto \mathbf{A}\mathbf{x} \bmod q \in \mathcal{R}_q^n$ is a homomorphism between $\mathcal{R}^m$ and $\mathcal{R}_q^n$ as additive groups. Hence, it defines an isomorphism between $\mathcal{R}^m / \ker(\mathbf{A})$ and $\mathrm{im}(\mathbf{A})$ by the first isomorphism theorem. Note that $\ker(\mathbf{A}) = \Lambda_q^\perp(\mathbf{A})$ and $\mathrm{im}(\mathbf{A}) = \mathcal{R}_q^n$. Therefore, the image $\mathbf{A}\mathbf{x} \bmod q$ is uniform in $\mathcal{R}_q^n$ if and only if $\mathbf{x} \bmod \Lambda_q^\perp(\mathbf{A})$ is uniform

in $\mathcal{R}^m/\Lambda_q^\perp(\mathbf{A})$. Since $\mathbf{A} \in \mathcal{S}$, we know that $\mathbf{x} \leftarrow \mathcal{D}^\pi_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}}$ has the standard deviation parameter $\mathbf{\Sigma} \geq \beta^2$ exceeding the smoothing parameter of $\pi(\Lambda_q^\perp(\mathbf{A}))$. Then Corollary 8 for lattices $\Lambda = \pi(\mathcal{R}^m)$ and $\Lambda' = \pi(\Lambda_q^\perp(\mathbf{A}))$ implies

$$\mathrm{SD}(\mathbf{A}\mathbf{x} \bmod q, \mathcal{U}(\mathcal{R}_q^n)) = \mathrm{SD}(\mathbf{x} \bmod \Lambda_q^\perp(\mathbf{A}), \mathcal{U}(\mathcal{R}^m/\Lambda_q^\perp(\mathbf{A}))) \leq \varepsilon/(1-\varepsilon).$$

Similarly, by Corollary 9 for the Rényi divergence

$$\mathrm{RD}(\mathbf{A}\mathbf{x} \bmod q; \mathcal{U}(\mathcal{R}_q^n)) = \mathrm{RD}(\mathbf{x} \bmod \Lambda_q^\perp(\mathbf{A}); \mathcal{U}(\mathcal{R}^m/\Lambda_q^\perp(\mathbf{A}))) \leq \frac{1+\varepsilon}{1-\varepsilon}.$$

Now consider a matrix $\mathbf{A}$ sampled at random. The probability that $\mathbf{A} \leftarrow \mathcal{P}_A$ is not in $\mathcal{S}$ is at most $\delta_1 + \delta_2$. For the smooth Rényi divergence we conclude immediately according to Equation (2.4). The statistical distance changes as

$$\mathrm{SD}(\mathcal{P}', \mathcal{U}') = \mathbb{E}_{\mathbf{A}}(\mathrm{SD}(\mathbf{A}\mathbf{x} \bmod q, \mathcal{U}(\mathcal{R}_q^n))) \leq \varepsilon/(1-\varepsilon) + \delta_1 + \delta_2,$$

since $\mathrm{SD}(\mathbf{A}\mathbf{x}, \mathcal{U}(\mathcal{R}_q^n)) \leq 1$. □

The theorem holds for both the canonical embedding θ and the coefficient embedding τ of the module lattices, as long as we have good bounds for the smoothing parameter in the respective kernel lattice.

We show that it is enough to know a bound for a smoothing parameter in one embedding, as it already allows us to apply the theorem to the other embedding using the Vandermonde transformation map between embeddings.

Corollary 15. *For $\nu \geq 1$ let $\mathcal{R}$ be the ν-th cyclotomic ring of degree d. Suppose that the conditions of Theorem 3 on distribution $\mathcal{P}_A$ are satisfied in the canonical embedding θ, then the statistical and Rényi closeness holds for $\mathbf{x} \leftarrow \mathcal{D}^\tau_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}}$ with $\mathbf{\Sigma} \geq \beta^2/\mathfrak{s}_d(\nu)^2$. Similarly, if the conditions are satisfied in the coefficient embedding τ, they imply the statement for $\mathbf{x} \leftarrow \mathcal{D}^\theta_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}}$ with $\mathbf{\Sigma} \geq \beta^2 \cdot \mathfrak{s}_1(\nu)^2$.*

Remark 9. An alternative proof strategy would be to use Equation 2.1 in combination with Lemma 19. However, it would only apply to proofs that bound the smoothing parameter through the first minimum of the dual lattice, as in [SS11, RSW18]. Our proof strategy leads to a more general result, also covering any other proof approach for bounding the smoothing parameter as for example in [LPR13].

Proof. Let $\mathbf{V} = \mathbf{I}_m \otimes (\mathbf{U}_H \cdot \mathbf{V}_\Phi) \in \mathbb{R}^{dm \times dm}$ be the embedding distortion map. Then for any subset $\Lambda \subset \mathcal{R}^m$ we have $\mathbf{V} \cdot \tau(\Lambda) = \theta(\Lambda)$ and $\mathbf{V}^{-1} \cdot \theta(\Lambda) = \tau(\Lambda)$. Since $\mathbf{U}_H$ is an

isometry we have $\mathfrak{s}_1(\mathbf{V}) = \mathfrak{s}_1(\nu)$ and $\mathfrak{s}_{dm}(\mathbf{V}) = \mathfrak{s}_d(\nu)$. Rewrite the distribution as

$$\begin{aligned}\mathcal{D}^{\tau}_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}} := \mathcal{D}_{\tau(\mathcal{R}^m),\sqrt{\mathbf{\Sigma}},\tau(\mathbf{c})} &= \mathbf{V}^{-1} \cdot \mathcal{D}_{\mathbf{V}\cdot\tau(\mathcal{R}^m),\sqrt{\mathbf{V}\mathbf{\Sigma}\mathbf{V}^T},\mathbf{V}\tau(\mathbf{c})} \\ &= \mathbf{V}^{-1} \cdot \mathcal{D}_{\theta(\mathcal{R}^m),\sqrt{\mathbf{V}\mathbf{\Sigma}\mathbf{V}^T},\theta(\mathbf{c})} =: \mathbf{V}^{-1} \cdot \mathcal{D}^{\theta}_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}},\end{aligned}$$

which holds by Lemma 18 since $\mathbf{V}$ is injective. Besides, for any $\mathbf{x} \in \theta(\mathcal{R}^m)$

$$\begin{aligned}(\mathbf{V}^{-1} \cdot \mathbf{x}) \bmod \tau(\Lambda_q^{\perp}(\mathbf{A})) &= \mathbf{V}^{-1} \cdot \mathbf{x} \bmod \mathbf{V}^{-1} \cdot \theta(\Lambda_q^{\perp}(\mathbf{A})) \\ &= \mathbf{V}^{-1} \cdot \mathbf{x} + \mathbf{V}^{-1} \cdot \theta(\Lambda_q^{\perp}(\mathbf{A})) \\ &= \mathbf{V}^{-1} \cdot (\mathbf{x} \bmod \theta(\Lambda_q^{\perp}(\mathbf{A}))).\end{aligned}$$

One of the conditions in Theorem 3 states $\Pr\left[\eta_{\varepsilon}(\theta(\Lambda_q^{\perp}(\mathbf{A}))) < \beta | \mathbf{A} \leftarrow \mathcal{P}_A\right] \geq 1 - \delta_1$. Say for a fixed matrix $\mathbf{A} \in \mathcal{R}_q^{n\times m}$ this condition is true. Then since we require $\mathfrak{s}_{\min}(\mathbf{V})^2 \cdot \mathbf{\Sigma} \geq \beta^2$, we have $\mathbf{V}\mathbf{\Sigma}\mathbf{V}^T \geq \eta_{\varepsilon}(\theta(\Lambda_q^{\perp}(\mathbf{A})))^2$. For such matrix we apply Corollary 8 as follows

$$\begin{aligned}\mathcal{D}^{\tau}_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}} \bmod \tau(\Lambda_q^{\perp}(\mathbf{A})) &= \mathbf{V}^{-1} \cdot \left(\mathcal{D}^{\theta}_{\mathcal{R}^m,\sqrt{\mathbf{V}\mathbf{\Sigma}\mathbf{V}^T},\mathbf{c}} \bmod \theta(\Lambda_q^{\perp}(\mathbf{A}))\right) \\ &\approx_{\varepsilon/(1-\varepsilon)} \mathbf{V}^{-1} \cdot \mathcal{U}(\theta(\mathcal{R}^m)/\theta(\Lambda_q^{\perp}(\mathbf{A}))) \\ &= \mathcal{U}(\tau(\mathcal{R}^m)/\tau(\Lambda_q^{\perp}(\mathbf{A}))).\end{aligned}$$

Thus $\mathrm{SD}(\mathcal{D}^{\theta}_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}} \bmod \tau(\Lambda_q^{\perp}(\mathbf{A})), \mathcal{U}(\tau(\mathcal{R}^m)/\tau(\Lambda_q^{\perp}(\mathbf{A})))) \leq \varepsilon/(1-\varepsilon)$ holds for any $\mathbf{\Sigma} \geq \beta^2/\mathfrak{s}_d(\nu)^2$. Taking into account the remaining conditions of Theorem 3 independent of the embedding for a random $\mathbf{A} \leftarrow \mathcal{P}_A$ we obtain

$$\mathrm{SD}((\mathbf{A}\mathbf{x} \bmod q, \mathbf{A}), (\mathbf{u}, \mathbf{A})) \leq \varepsilon/(1-\varepsilon) + \delta_1 + \delta_2.$$

The statements for the Rényi divergence and the implication from coefficient embedding to canonical embedding are obtained in the same way.

□

3.2.2 Smoothing Analysis

The quality of Theorem 3 heavily depends on the distribution $\mathcal{P}_A$, as well as the bound β on the smoothing parameter, determining the probabilities δ_1 and δ_2. Overall, we distinguish two different strategies to analyse the smoothing parameter of $\Lambda_q^{\perp}(\mathbf{A})$. The first strategy was given in [LPR13], it directly computes the expected Gaussian weight of the dual lattice. The second strategy was detailed in [SS11] and its follow-up work [RSW18]. It focuses on lower bounding the length of the shortest vector in the dual lattice.

We first present statements that appeared in prior work where the distribution of $\mathbf{A}$ is not truly uniform and $\mathrm{im}(\mathbf{A}) = \mathcal{R}_q^n$ with probability 1 by construction. Then we give generalisations of both strategies to matrices uniform in $\mathcal{R}_q^{n\times m}$. We showcase the proofs in both canonical and coefficient embedding even though both can be applied to an arbitrary embedding. Note that when $\mathbf{A}$ is uniform, Lemma 16 provides bounds on the probability δ_2 that the image of $\mathbf{A}$ has full rank. The first classic result comes from [SS11] and considers a uniform matrix with invertible entries which ensures the full rank condition.

Lemma 53 ([SS11, Theorem 2])**.** *Let $\mathcal{K}$ be a cyclotomic number field of degree d with $\mathcal{R}$ its ring of integers, where $d \geq 8$ is a power of 2. Let $q \geq 5$ be a prime, $m \geq 2$ be an integer, $\varepsilon \in (0, 1/2)$ and $\varepsilon' > 0$. Let $r = \sqrt{\ln(2dm(1+1/\varepsilon))/\pi} \cdot \min(\sqrt{d} \cdot q^{\frac{1}{m}+\varepsilon'}, q^{\frac{1}{m}+f\cdot\varepsilon'})$. Then*

$$\Pr[\eta_\varepsilon(\tau(\Lambda_q^\perp(\mathbf{a}))) > r \mid \mathbf{a} \leftarrow \mathcal{U}((\mathcal{R}_q^\times)^m)] \leq 2^{3d(m+1)} \cdot q^{-\varepsilon' md}.$$

Its canonical embedding version is described in [RSW18].

Lemma 54 ([RSW18, Lemma 5.2])**.** *Let $\mathcal{K}$ be a cyclotomic number field of degree d with $\mathcal{R}$ its ring of integers. Let $q \geq 2$ be a prime not dividing $\Delta_\mathcal{K}$. Let $m \geq 2$, $\varepsilon \in (0, 1/2)$ and $\varepsilon' > 0$. Then for $r = \sqrt{\ln(2dm(1+1/\varepsilon))/\pi} \cdot \Delta_\mathcal{K}^{1/d} \cdot q^{\frac{1}{m}+\varepsilon'}$ it holds*

$$\Pr[\eta_\varepsilon(\theta(\Lambda_q^\perp(\mathbf{a}))) > r \mid \mathbf{a} \leftarrow \mathcal{U}((\mathcal{R}_q^\times)^m)] \leq 2^{3d(m+1)} \cdot q^{-\varepsilon' md}.$$

We also state the original [LPR13] statement that ensure the full rank constraint by prepending identity to a uniform matrix.

Lemma 55 ([LPR13, Theorem 7.4])**.** *Let $\mathcal{K}$ be a cyclotomic number field of degree d with $\mathcal{R}$ its ring of integers. Let $q > 2$, $0 < n \leq m$ be integers, $\varepsilon = q^2 \cdot m \cdot 2^{1-2d}$ and $s > 0$ s.t. $2^{2s} > 2 \cdot (2^{-d(m-2)} + q^2 \cdot m)$. Let $\mathcal{P}_A = \left\{\mathbf{A} = [\mathbf{I}_n \mid \bar{\mathbf{A}}] \mid \bar{\mathbf{A}} \leftarrow \mathcal{U}\left(\mathcal{R}_q^{n\times(m-n)}\right)\right\}$. Then for every $r > 2d$*

$$\mathbb{E}_{\mathbf{A}\leftarrow\mathcal{P}_A}\left[\rho_{1/r}(\theta(\Lambda_q^\perp(\mathbf{A}))^*)\right] \leq 1 + 2(r/d)^{-dm} \cdot q^{dn+2} + \varepsilon.$$

If $r > 2d \cdot q^{n/m+2/(dm)}$ then

$$\Pr\left[\eta_{2^{-d+s}}(\theta(\Lambda_q^\perp(\mathbf{A}))) > r \mid \mathbf{A} \leftarrow \mathcal{P}_A\right] \leq 2^{-d+s}.$$

Lemma 56 (Explicit form of Lemma 55)**.** *Let $\mathbf{F} = \begin{bmatrix}1 & a & b\end{bmatrix}$ where $(a, b) \leftarrow \mathcal{R}_q^2$, $q \geq 2$ and $\delta > 0$ s.t. $2^{2\delta} \geq 2^{-d+1} + 6q^2$. Then with probability at least $1 - 2^{-d+\delta}$:*

$$\eta_{2^{-d+\delta}}(\Lambda_q^\perp(\mathbf{F})) \leq 2d \cdot q^{1/3+1/3d}.$$

The following lemma can be seen as a generalisation of [LPR13] to completely uniform matrices. Unfortunately, considering this less structured distribution makes the parameters depend on the factorisation of the integer modulus q. Contrary to the original work we restrict q to be a prime and delegate lifting this restriction to future work.

Lemma 57 ([LPR13] for Uniform Matrices)**.** *Let $\mathcal{K}$ be a cyclotomic field of degree d with $\mathcal{R}$ its ring of integers. Further, let $q > 2$ be an unramified prime modulus such that $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$ for distinct, irreducible $\Phi_j(X)$ of degree δ. Let $0 < n \leq m, 2 \leq m$ be integers and $s > 0$, $\varepsilon = m \cdot 2^{1-2d} + 2f \cdot q^{-\delta(m-n)} \cdot (1 + m \cdot 2^{1-2d})$ such that $\delta(m-n)\log q > 2d$, $2^{2s} > 2 \cdot (2^{-d(m-2)} + m + f \cdot (1 + m \cdot 2^{1-2d}))$, $2f \cdot q^{-\delta(m-n)} \leq 1$. Then for every $r > 2d$*

$$\mathbb{E}_{\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m})} \left[\rho_{1/r}(\theta(\Lambda_q^{\perp}(\mathbf{A}))^*) \right] \leq 2\exp(1) \cdot (r/d)^{-dm} \cdot q^{dn} + 1 + \varepsilon.$$

If $r > 2\exp(1/dm) \cdot d \cdot q^{n/m}$ then

$$\Pr \left[\eta_{2^{-d+s}}(\theta(\Lambda_q^{\perp}(\mathbf{A}))) > r \mid \mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m}) \right] \leq 2^{-d+s}.$$

Proof. First, we identify the exact shape of the dual lattice of $\Lambda_q^{\perp}(\mathbf{A})$ from Lemma 46 for the canonical embedding

$$\theta(\Lambda_q^{\perp}(\mathbf{A}))^* = \theta\left((\mathcal{R}^*)^m + \left\{ \frac{1}{q} \mathbf{A}^T \cdot \mathbf{s} \mid \mathbf{s} \in (\mathcal{R}^*)^n \right\} \right).$$

For cleaner notations we drop mentioning the embedding $\theta(\cdot)$ in the rest of the proof and denote the expected value we analyse as $E \coloneqq \mathbb{E}_{\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m})} \left[\rho_{1/r}(\Lambda_q^{\perp}(\mathbf{A})^*) \right]$. For arbitrary rings $\{\mathcal{R}_i\}_{i=0,1,2}$ where $\mathcal{R}_2 = \mathcal{R}_0/\mathcal{R}_1$ is a quotient of $\mathcal{R}_0$ we define a function $\mathrm{rep} : \mathcal{R}_2 \to \mathcal{R}_0$ that maps an element of $\mathcal{R}_2$ to its arbitrary representative in $\mathcal{R}_0$. For the expectation we have

$$\begin{aligned} E &\leq \mathbb{E}_{\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m})} \left[\sum_{\mathbf{s} \in (\mathcal{R}_q^*)^n} \rho_{1/r} \left(1/q \cdot ((q\mathcal{R}^*)^m + \mathrm{rep}(\mathbf{A}^T \cdot \mathbf{s})) \right) \right] \\ &= \sum_{\mathbf{s} \in (\mathcal{R}_q^*)^n} \mathbb{E}_{\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m})} \left[\rho_{1/r} \left(1/q \cdot ((q\mathcal{R}^*)^m + \mathrm{rep}(\mathbf{A}^T \cdot \mathbf{s})) \right) \right] \\ &= \sum_{\mathbf{s} \in (\mathcal{R}_q^*)^n} \left(\mathbb{E}_{\mathbf{a} \leftarrow \mathcal{U}(\mathcal{R}_q^{n})} \left[\rho_{1/r} \left(1/q \cdot (q\mathcal{R}^* + \mathrm{rep}(\mathbf{a}^T \cdot \mathbf{s})) \right) \right] \right)^m, \end{aligned}$$

where the first equality holds by linearity of the expectation and the second holds since the rows of $\mathbf{A}$ are iid. Note that since we add $q\mathcal{R}^*$ to a representative of $\mathbf{a}^T \cdot \mathbf{s} \in \mathcal{R}^*/q\mathcal{R}^*$

the value above is independent of the specific choice of the representative. By Lemma 15, the value $\mathbf{a}^T \cdot \mathbf{s} \bmod q$ is distributed uniformly in the ideal $\mathcal{I}_{\mathbf{s}} \subset \mathcal{R}_q^*$. Then

$$\begin{aligned}\mathbb{E}\left[\rho_{1/r}\left(1/q \cdot (q\mathcal{R}^* + \mathrm{rep}(\mathbf{a}^T \cdot \mathbf{s}))\right)\right] &= \sum_{x \in \mathcal{I}_{\mathbf{s}}} \frac{1}{|\mathcal{I}_{\mathbf{s}}|} \cdot \rho_{1/r}\left(1/q \cdot (q\mathcal{R}^* + \mathrm{rep}(x))\right) \\ &= \frac{1}{|\mathcal{I}_{\mathbf{s}}|} \cdot \rho_{1/r}\left(1/q \cdot (q\mathcal{R}^* + \mathrm{rep}(\mathcal{I}_{\mathbf{s}}))\right).\end{aligned}$$

Now we group the summands in $\mathbb{E}_{\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m})}\left[\rho_{1/r}(\Lambda_q^{\perp}(\mathbf{A})^*)\right]$ based on the ideal $\mathbf{s}$ generates. Denote $\mathcal{I}' = q\mathcal{R}^* + \mathrm{rep}(\mathcal{I}) \subset \mathcal{R}^*$ an ideal of $\mathcal{R}^*$ containing $q\mathcal{R}^*$ that corresponds to an ideal $\mathcal{I} \subset \mathcal{R}_q^*$. Following Lemma 11 we have another bijection $\mathcal{R} = t \cdot \mathcal{R}^*$ for a ring element t. It extends to the ideals of $\mathcal{R}^*$ denoting $\mathcal{J}' \coloneqq t \cdot \mathcal{I}' = t \cdot \mathrm{rep}(\mathcal{I}) + q\mathcal{R}$ and to the ideals of $\mathcal{R}_q^*$ denoting $\mathcal{J} \coloneqq t \cdot \mathcal{I} \subset \mathcal{R}_q$. Then $\mathcal{I} = \mathcal{I}' \bmod q\mathcal{R}$, $\mathcal{J} = \mathcal{J}' \bmod q\mathcal{R}$ and $|\mathcal{J}| = |\mathcal{I}|$. Then we have

$$\begin{aligned}N(1/q \cdot \mathcal{I}') = \frac{N(\mathcal{I}')}{|N(q)|} &= \frac{N(\mathcal{J}')}{|N(t)| \cdot |N(q)|} = \frac{N(\mathcal{J})}{|N(t)| \cdot |N(q)|} \\ &= \frac{|\mathcal{R}_q|}{|\mathcal{J}| \cdot \Delta_{\mathcal{K}} \cdot |\mathcal{R}_q|} = \frac{1}{|\mathcal{J}| \cdot \Delta_{\mathcal{K}}}. \end{aligned} \tag{3.1}$$

In the transitions above we use multiplicativity of norms, the fact that for $\mathcal{J} = \mathcal{J}' \bmod q\mathcal{R}$ we have $N(\mathcal{J}) = N(\mathcal{J}')$ and the definition of the norm in finite rings. Then if T is the set of all ideals $\mathcal{I}' \subset \mathcal{R}^*$ that contain $q\mathcal{R}^*$ using the notation above we have

$$\begin{aligned}E &\le \sum_{\mathcal{I}' \in T} \left(\frac{1}{|\mathcal{I}' \bmod q\mathcal{R}^*|} \cdot \rho_{1/r}(1/q \cdot \mathcal{I}')\right)^m \sum_{\mathbf{s}: \mathcal{I}_{\mathbf{s}} = \mathcal{I}' \bmod q\mathcal{R}^*} 1 \\ &\le \sum_{\mathcal{I}' \in T} \left(\frac{1}{|\mathcal{I}' \bmod q\mathcal{R}^*|} \cdot \rho_{1/r}(1/q \cdot \mathcal{I}')\right)^m \cdot |\mathcal{I}' \bmod q\mathcal{R}^*|^n \\ &= \sum_{\mathcal{I}' \in T} \frac{1}{|\mathcal{I}|^{m-n}} \cdot \rho_{1/r}(1/q \cdot \mathcal{I}')^m = \sum_{\mathcal{I}' \in T} \frac{1}{|\mathcal{J}|^{m-n}} \cdot \rho_{1/r}(1/q \cdot \mathcal{I}')^m.\end{aligned}$$

Now for each ideal $\mathcal{I}' \in T$ by Lemma 45 and Equation (3.1)

$$\begin{aligned}\rho_{1/r}(1/q \cdot \mathcal{I}')^m &\le \max(1, r^{-dm} \cdot N(1/q \cdot \mathcal{I}')^{-m})(1 + 2^{-2d})^m \\ &\le \max(1, r^{-dm} \cdot (|\mathcal{J}| \cdot \Delta_{\mathcal{K}})^m)(1 + 2^{-2d})^m \\ &\le \underbrace{1 + m2^{1-2d}}_{\alpha} + \underbrace{2r^{-dm}\Delta_{\mathcal{K}}^m}_{\beta} \cdot |\mathcal{J}|^m.\end{aligned}$$

where $\mathcal{J} \subset \mathcal{R}_q$ is defined as above. In the last inequality $(1 + 2^{-2d})^m \le \frac{1}{1 - m \cdot 2^{-2d}} \le 1 + 2 \cdot m \cdot 2^{-2d}$ since $m \cdot 2^{-2d} \le 1/2$. For the case when $r^{-dm} \cdot (|t \cdot \mathcal{J} \bmod q\mathcal{R}| \cdot \Delta_{\mathcal{K}})^m > 1$

we use $\frac{1}{1-m\cdot 2^{-2d}} \leq 2$. To bound the larger of two values we take the sum of the bounds. Denote $\mathbb{I}$ the set of ideals in $\mathcal{R}_q$. Putting the arguments together we get

$$\begin{aligned} E &\leq \alpha \cdot \sum_{\mathcal{J}\in\mathbb{I}} |\mathcal{J}|^{n-m} + \beta \cdot \sum_{\mathcal{J}\in\mathbb{I}} |\mathcal{J}|^{n} \\ &\leq \alpha \cdot (1 + 2fq^{-\delta(m-n)}) + 2\exp(1) \cdot r^{-dm} d^{dm} q^{dn} \end{aligned}$$

where in the last inequality we use Lemma 10. Lastly we apply the Markov inequality. Define arbitrary positive values $\varepsilon', \varepsilon''$ such that $\varepsilon' \cdot \varepsilon'' > \varepsilon$. Then for any $r > \exp(1/dm) \cdot d \cdot q^{n/m} \cdot (2/(\varepsilon'' \cdot \varepsilon' - \varepsilon))^{1/dm}$ we have

$$\begin{aligned} \Pr(\rho_{1/r}(\Lambda_q^\perp(\mathbf{A})^*) - 1 > \varepsilon') &\leq \frac{\mathbb{E}_{\mathbf{A}}\left[\rho_{1/r}(\Lambda_q^\perp(\mathbf{A})^*)\right] - 1}{\varepsilon'} \\ &\leq \frac{m2^{1-2d} + 2fq^{-\delta(m-n)} \cdot (1 + m2^{1-2d}) + \varepsilon''\varepsilon' - \varepsilon}{\varepsilon'} \\ &\leq \varepsilon''. \end{aligned}$$

Here in the last inequality we used the definition of ε from the lemma statement. Set $\varepsilon' = \varepsilon'' = 2^{-d+s}$. Then for $\delta(m-n)\log q > 2d$ and $2^{2s} > 2\cdot(2^{-d(m-2)} + m + f\cdot(1+m\cdot 2^{1-2d}))$ we have $\varepsilon' \cdot \varepsilon'' > \varepsilon$ and $\left(\frac{2}{\varepsilon''\cdot\varepsilon' - \varepsilon}\right)^{1/dm} \leq 2$, and the statement follows.

□

We also extend [SS11, RSW18] to the setting where $\mathcal{P}_A$ is the uniform distribution over $\mathcal{R}_q^{n\times m}$, proven in the coefficient embedding.

Lemma 58 ([SS11, RSW18] for Uniform Matrices)**.** *Let $\mathcal{K}$ be a cyclotomic field of degree d with $\mathcal{R}$ its ring of integers. Further, let $q \geq 5$ be an unramified prime modulus such that $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X)\rangle$ for distinct, irreducible $\Phi_j(X)$ of degree δ. Let $d \geq 8$ be a power of 2. Let $\max(n, 2) \leq m$ be integers, $\varepsilon \in (0, 1/2)$ and $\varepsilon' > 0$.*

Let $r = \sqrt{\ln(2dm(1 + 1/\varepsilon))/\pi} \cdot \sqrt{d} \cdot q^{n/m} \cdot 2^{\varepsilon'}$. Then

$$\Pr[\eta_\varepsilon(\tau(\Lambda_q^\perp(\mathbf{A}))) > r \mid \mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m})] \leq 2^{-dm(\varepsilon'-3)}$$

Proof. Following Lemma 46 the dual lattice to $\Lambda_q^\perp(\mathbf{A})$ with respect to the coefficient embedding is of the form

$$\begin{aligned} \tau(\Lambda_q^\perp(\mathbf{A}))^* &= \frac{1}{q}\Lambda_q(\mathrm{Rot}(\mathbf{A})^T) \\ &= \frac{1}{q}\Big\{\tau(\mathbf{t}) \mid \mathbf{t} \in \mathcal{R}^m \text{ s.t. } \exists \mathbf{s} \in \mathcal{R}_q^n : \tau(\mathbf{t}) = \mathrm{Rot}(\mathbf{A})^T \cdot \tau(\mathbf{s}) \bmod q\Big\} \\ &= \frac{1}{q}\tau(\Lambda_q(\bar{\mathbf{A}}^T)). \end{aligned}$$

By properties of power-of-2 cyclotomics, the matrix $\mathrm{Rot}(\mathbf{A})^T$ is in bijective correspondence with $\mathrm{Rot}(\bar{\mathbf{A}}^T)$ for some $\bar{\mathbf{A}} \in \mathcal{R}_q^{n\times m}$. The bijection maps every coordinate $a_{ij} = \sum_{k=0}^{d-1} \alpha_k X^k \in \mathcal{R}_q$ of $\mathbf{A}$ to a corresponding coordinate of $\bar{\mathbf{A}}$ equal $\bar{a}_{ij} = \alpha_0 - \sum_{k=1}^{d-1} \alpha_k X^{d-k}$. Therefore, $\bar{\mathbf{A}}$ follows the uniform distribution and it suffices for us to lower bound the first minimum of the lattice $\lambda_1^\infty(\frac{1}{q}\tau(\Lambda_q(\mathbf{A}^T)))$ for $\mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n\times m})$.

We claim that $\lambda_1^\infty(\frac{1}{q}\tau(\Lambda_q(\mathbf{A}^T))) \geq 1/(\sqrt{d}\cdot q^{\varepsilon'/\log q + n/m})$ except with probability $\leq 2^{-dm(\varepsilon'-3)}$. Let $B > 0$ and then apply Lemma 19. For every $\mathbf{t} \in \mathcal{R}^m$ of norm $0 < \|\mathbf{t}\|_\infty < B$ and $\mathbf{s} \in \mathcal{R}_q^n$ define

$$p(\mathbf{t}, \mathbf{s}) = \Pr_{\{\mathbf{a}_i\}_i \leftarrow (\mathcal{R}_q^n)^m}(\forall i : t_i = \mathbf{a}_i^T \cdot \mathbf{s} \bmod q) = \prod_{i=1}^{m} \Pr_{\mathbf{a}_i \leftarrow \mathcal{R}_q^n}(t_i = \mathbf{a}_i^T \cdot \mathbf{s} \bmod q).$$

Denote $p_i(t_i, \mathbf{s}) \coloneqq \Pr_{\mathbf{a}_i \leftarrow \mathcal{R}_q^n}(t_i = \mathbf{a}_i^T \cdot \mathbf{s} \bmod q)$. By Lemma 15 the value $\mathbf{a}_i^T \cdot \mathbf{s} \bmod q$ is distributed uniformly in $\mathcal{I}_\mathbf{s}$. Then for any fixed $t_i \in \mathcal{I}_\mathbf{s}$

$$p_i(t_i, \mathbf{s}) = \frac{1}{|\mathcal{I}_\mathbf{s}|}.$$

And the probability p that the lattice contains a short infinity norm vector is bounded as

$$\begin{aligned}
p &= \Pr_{\mathbf{A}}(\exists \mathbf{t} : \mathbf{t} \in \tau(\Lambda_q(\mathbf{A}^T)), 0 < \|\mathbf{t}\|_\infty < B) \\
&\leq \sum_{0<\|\mathbf{t}\|_\infty<B, \mathbf{s}\in\mathcal{R}_q^n} p(\mathbf{t}, \mathbf{s}) \\
&= \sum_{0<\|\mathbf{t}\|_\infty<B, \mathbf{s}\in\mathcal{R}_q^n} \prod_i p_i(\mathbf{t}_i, \mathbf{s}) \\
&\leq \sum_{k=0}^{f} \sum_{\substack{\mathcal{I}\in\mathbb{I} \\ |\mathcal{I}|=q^{\delta k}}} \sum_{\mathbf{s}\in\mathcal{I}^n} \sum_{\substack{0<\|\mathbf{t}\|_\infty<B \\ \mathbf{t}\in\mathcal{I}^m}} \frac{1}{q^{m\cdot\delta k}} \\
&\leq |\{\mathcal{I} \in \mathbb{I}\}| \cdot \max_{k\in[f]} q^{n\cdot\delta k} \cdot \frac{N(B,k)^m}{q^{m\cdot\delta k}} = 2^f \cdot \max_{k\in[f]} \frac{N(B,k)^m}{q^{(m-n)\cdot\delta k}}
\end{aligned}$$

The first inequality comes from the union bound. The second inequality conditions on the ideal generated by coefficients of $\mathbf{s}$. We enumerate over all ideals of $\mathcal{R}_q$ according to their norm. We also substitute the generating condition by $\mathbf{s} \in \mathcal{I}^n$. In the third inequality $N(B,k)$ denotes the number of elements $t \in \mathcal{R}_q$ such that $0 < \|t\|_\infty < B, t \in \mathcal{I}$ for $\mathcal{I}$ of cardinality $q^{\delta k}$. Then $|\mathbf{s} \in \mathcal{I}^n| = q^{n\cdot\delta k}$.

We now analyse $N(B,k)$. For convenience, let $S = \{j_1, \ldots, j_{f-k}\}$ be an arbitrary subset of $\{1, \ldots, f\}$ of cardinality $f - k$. The bound we obtain in the end only depends on the cardinality of S. We proceed by a volume argument.

Denote $\lambda_\infty = \lambda_1^\infty(\tau(\mathcal{I}))$ for $\mathcal{I} = \langle q, \prod_{j\in S}\Phi_j(X)\rangle$. Then $N(B,k)$ is upper bounded by the ratio between volumes of two hypercubes $\left\{\mathbf{v}\in\mathbb{R}^d \mid \|\mathbf{v}\|_\infty \le B + \lambda_\infty/2\right\}$ and $\left\{\mathbf{v}\in\mathbb{R}^d \mid \|\mathbf{v}\|_\infty \le \lambda_\infty\right\}$ as follows

$$N(B,k) \le \left(\frac{2B}{\lambda_\infty}+1\right)^d$$

By Lemma 43 and inequalitites between norms $\frac{1}{\mathfrak{s}_1(\nu)}N(\mathcal{I})^{1/d} \le \frac{1}{\sqrt{d}}\cdot\lambda_1(\tau(\mathcal{I})) \le \lambda_\infty$. For the power-of-2 cyclotomics $\mathfrak{s}_1(\nu) = \sqrt{d}$. Computing the norm we get $1/\sqrt{d}\cdot q^{\delta(f-k)/d} = 1/\sqrt{d}\cdot q^{1-k/f} \le \lambda_\infty$. Take $B = \frac{1}{\sqrt{d}}q^\beta$ then for $\beta < 1-k/f$ we have $\lambda_\infty > B$ and $N(B,k) = 0$. Otherwise, for large enough q we have $2q^{\beta-(1-k/f)} > 1$ and

$$N(B,k) \le (2q^{\beta-1+k/f}+1)^d \le 2^{2d}\cdot q^{d\beta-d+\delta k}.$$

Here in the first expression we insert the computed bounds and in the second inequality we use $(x+1)^d \le (2x)^d$ for $x > 1$. Hence,

$$\begin{aligned} p &\le 2^f \cdot \max_{k\in[f]} \frac{2^{2dm}q^{dm\beta-dm+\delta km}}{q^{(m-n)\cdot\delta k}} \\ &\le 2^{d(2m+1)}q^{dm\beta-dm+dn} \le 2^{-dm(\varepsilon'-3)}. \end{aligned}$$

In the first inequality use that the maximum is attained for $k = f$ and the inequality $f < d$ for the other term. In the last equality we set $\beta = 1-n/m-\varepsilon'/\log q$ and bound $2^{d(2m+1)} \le 2^{3dm}$. It follows that the shortest vector of $1/q\cdot\tau(\Lambda_q(\mathbf{A}^T))$ is at least $\frac{1}{\sqrt{d}}q^{\beta-1} = \frac{1}{\sqrt{d}}\cdot q^{-n/m-\varepsilon'/\log q}$ with probability in $1-2^{-dm(\varepsilon'-3)}$. Using the conventional smoothing bound from Lemma 19 and $q^{1/\log q} = 2$ we obtain the result. □

3.2.3 Smoothing in the Presence of Exact Linear Leakage

In the following, we study the smoothing-based approach under linear leakage on the hash input vector $\mathbf{x}$. We distinguish between *exact* and *noisy* leakage. The first type of leakage is addressed in this section and the proof technique is inspired by a recent LHL variant [SSE+24], lifting it to a more general leakage setting. The second class of leakage is treated in Section 3.2.4, using recent results [ENP24] on the distribution of discrete Gaussians under noisy linear leakage. It can be seen as a generalization of the first type of leakage studied in [DGKS21].

In the exact case we consider leakage of the form $(\mathbf{E}, \mathbf{E}\cdot\mathbf{x})$ for a given wide matrix $\mathbf{E} = (\mathbf{I}_l \mid \bar{\mathbf{E}}) \in \mathcal{R}^{l\times m}$ with $l < m$. The matrix $\bar{\mathbf{E}}$ can be sampled from a short distribution or be fixed as long as it satisfies the required conditions.

We rely heavily on the approach of [SSE+24]. The original work defined matrices $\mathbf{A}$ and $\mathbf{E}$ so that the leakage is not exact and corresponds to our Section 3.2.4. However, their techniques are so powerful that they cover the case of exact leakage we present here. The original work also covers composite modulus q which we omit for simplicity.

Theorem 4. *Let $\mathcal{K}$ be a cyclotomic number field with d a power of 2. Let q be a prime, $0 < n + l \leq m$ be integers. Let $\varepsilon \in (0, 1/2)$, $\delta_E, \delta_2 \in (0, 1)$, $\varepsilon' > 3$, $B_{\max} > 0$, and let $\mathbf{\Sigma} \in \mathbb{R}^{dm \times dm}$ be positive-definite. Assume*

$$\mathbf{\Sigma} \geq dm \cdot \max\left(2B_{\max}, \sqrt{d}e^{\varepsilon'} \cdot q^{(n+l)/m}\right) \cdot \sqrt{\ln(2d(m-l)(1+1/\varepsilon))/\pi},$$
$$q \geq 2 \cdot B_{\max} \cdot dm \cdot \max\left(2B_{\max}, \sqrt{d}e^{\varepsilon'} \cdot q^{(n+l)/m}\right).$$

Let $\mathcal{P}_E$ be a distribution on the set of matrices $\mathcal{R}^{l \times m}$ s.t. with probability at least $1 - \delta_E$ for $\mathbf{E} = [\mathbf{I}_l \mid \bar{\mathbf{E}}] \leftarrow \mathcal{P}_E$ it holds

1. $\mathfrak{s}_1(\mathrm{Rot}(\mathbf{E})) < B_{\max}$,
2. $\Pr\left[\mathbf{A} \cdot \Lambda^\perp(\mathbf{E}) \bmod q = \mathcal{R}_q^n \mid \mathbf{A} \leftarrow \mathcal{U}(\mathcal{R}_q^{n \times m})\right] \geq 1 - \delta_2$.

Then for distributions defined as

$$\mathcal{P}' = \left\{\mathbf{A}\mathbf{x} \bmod q, \mathbf{E}\mathbf{x}, \mathbf{A}, \mathbf{E} \mid \mathbf{x} \leftarrow \mathcal{D}^\tau_{\mathcal{R}^m, \mathbf{\Sigma}}\right\},$$
$$\mathcal{U}' = \left\{\mathbf{u}, \mathbf{E}\mathbf{x}, \mathbf{A}, \mathbf{E} \mid \mathbf{x} \leftarrow \mathcal{D}^\tau_{\mathcal{R}^m, \mathbf{\Sigma}}, \mathbf{u} \leftarrow \mathcal{U}(\mathcal{R}_q^n)\right\},$$

we have

$$\mathrm{SD}(\mathcal{P}', \mathcal{U}') \leq \varepsilon/(1-\varepsilon) + 2^{-dm(\varepsilon'-3)} + \delta_E + \delta_2,$$
$$\mathrm{RD}^{(2^{-dm(\varepsilon'-3)} + \delta_E + \delta_2)}(\mathcal{P}'; \mathcal{U}') < \frac{1+\varepsilon}{1-\varepsilon}.$$

Proof. Similarly to Theorem 3 let $\mathcal{S}$ represent the set of all matrices $(\mathbf{A}, \mathbf{E})$ for which conditions (1, 2) hold. Consider an arbitrary pair $(\mathbf{A}, \mathbf{E}) \in \mathcal{S}$.

For any leakage $\mathbf{w} \in \mathrm{im}(\mathbf{E})$ $(= \mathbf{E} \cdot \mathbf{t}'$ for some $\mathbf{t}')$ there exists a vector $\mathbf{t} \in \mathcal{R}^m$ s.t. $\mathbf{A} \cdot \mathbf{t} = \mathbf{0} \bmod q$ and $\mathbf{E} \cdot \mathbf{t} = \mathbf{w}$. We construct it as $\mathbf{t} = \mathbf{t}' + \mathbf{t}''$ where we denote $\mathbf{r} \coloneqq \mathbf{A}\mathbf{t}' \bmod q$ and find $\mathbf{t}'' \in \Lambda^\perp(\mathbf{E})$ s.t. $\mathbf{A} \cdot \mathbf{t}'' = -\mathbf{r} \bmod q$. Such $\mathbf{t}''$ exists by Condition 2 of the theorem. Then, given the leakage $\mathbf{w}$ the distribution of $\mathbf{x}$ is

$$\left\{\mathbf{x} \leftarrow \mathcal{D}_{\mathcal{R}^m, \mathbf{\Sigma}} \mid \mathbf{x} \in (\Lambda^\perp(\mathbf{E}) + \mathbf{t})\right\} = \mathcal{D}_{\Lambda^\perp(\mathbf{E}) + \mathbf{t}, \mathbf{\Sigma}} = \mathbf{t} + \mathcal{D}_{\Lambda^\perp(\mathbf{E}), \mathbf{\Sigma}, -\mathbf{t}},$$

and it is enough to analyse $\mathbf{A} \cdot (\mathbf{t} + \mathcal{D}_{\Lambda^\perp(\mathbf{E}), \mathbf{\Sigma}, -\mathbf{t}}) = \mathbf{A} \cdot \mathcal{D}_{\Lambda^\perp(\mathbf{E}), \mathbf{\Sigma}, -\mathbf{t}} \bmod q$. The matrix $\mathbf{A} : \Lambda^\perp(\mathbf{E}) \to \mathcal{R}_q^n$ represents a linear map. It generates an isomorphism $\Lambda^\perp(\mathbf{E}) \bmod (\Lambda_q^\perp(\mathbf{A}) \cap$

$\Lambda^\perp(\mathbf{E})) \sim \mathbf{A} \cdot \Lambda^\perp(\mathbf{E}) \bmod q = \mathcal{R}_q^n$. Here the last equality follows again from Condition 2. Hence, it remains to prove that $\mathcal{D}_{\Lambda^\perp(\mathbf{E}),\mathbf{\Sigma},-\mathbf{t}} \bmod (\Lambda_q^\perp(\mathbf{A}) \cap \Lambda^\perp(\mathbf{E}))$ is uniformly distributed. By Corollary 8 and Corollary 9 the distribution is close to uniform if $\mathbf{\Sigma} \geq \eta_\varepsilon(\tau(\Lambda_q^\perp(\mathbf{A}) \cap \Lambda^\perp(\mathbf{E})))$. In the remainder of the proof we analyse the smoothing parameter of this lattice.

Since $\mathbf{E} = [\mathbf{I} \mid \bar{\mathbf{E}}]$ contains the identity, the matrix $\mathrm{Rot}(\mathbf{E})$ is full rank hence the lattice $\tau(\Lambda^\perp(\mathbf{E}))$ is of rank $d(m-l)$ in $\mathbb{R}^{dm}$. We know that $\tau(\Lambda_q^\perp(\mathbf{A}))$ is full rank in the same dimension. Hence, their intersection is a lattice with the same rank as $\tau(\Lambda^\perp(\mathbf{E}))$. It remains to upper bound $\lambda_{d(m-l)}(\tau(\Lambda_q^\perp(\mathbf{A}) \cap \Lambda^\perp(\mathbf{E})))$ and apply the classic smoothing parameter bound in Lemma 20.

Consider a simpler lattice $\Lambda = \Lambda_q^\perp(\mathbf{A}) \cap \Lambda_q^\perp(\mathbf{E})$ of rank dm where the kernel of $\mathbf{E}$ is taken modulo q. By the transference theorems

$$\lambda_{d(m-l)}(\tau(\Lambda)) \leq \frac{dm}{\lambda_{dl+1}(\tau(\Lambda)^*)} = \frac{dm}{1/q \cdot \lambda_{dl+1}(\Lambda_q([\mathrm{Rot}(\mathbf{A})^T \mid \mathrm{Rot}(\mathbf{E})^T]))},$$

where we know the shape of the dual lattice by Lemma 46. Denote $\mathbf{M} = [\mathrm{Rot}(\mathbf{A}) \| \mathrm{Rot}(\mathbf{E})]$. Since $\mathrm{Rot}(\mathbf{E})^T$ is only of rank dl, the $(dl+1)$st minimum of $\Lambda_q(\mathbf{M}^T)$ has to be larger than some vector outside the span of $\mathrm{Rot}(\mathbf{E})^T$. Formally, $\lambda_{dl+1}(\Lambda_q(\mathbf{M}^T)) \geq \lambda_1(\Lambda_q(\mathbf{M}^T) \setminus \mathrm{Rot}(\mathbf{E})^T \cdot \mathbb{Z}^{dl})$.

We represent $\tau(\mathbf{u}) \in \Lambda_q(\mathbf{M}^T) \setminus \mathrm{Rot}(\mathbf{E})^T \cdot \mathbb{Z}^{dl}$ as $\mathbf{u} = \mathrm{Rot}(\mathbf{A})^T \cdot \tau(\mathbf{v}_A) + \mathrm{Rot}(\mathbf{E})^T \cdot \tau(\mathbf{v}_E) \bmod q$. We analyse $\mathbf{u}$ reduced modulo q since is has the shortest norm in the lattice. Consider the following cases. Firstly, assume $\mathbf{v}_A \neq 0 \bmod q$. The proof of [SS11] from Lemma 58[4] still applies with the shift $\mathrm{Rot}(\mathbf{E})^T \cdot \tau(\mathbf{v}_E) \bmod q$ and only adds another union bound over $\mathbf{v}_E \in \mathcal{R}_q^l$. Therefore, for any $\varepsilon' > 0$ in this case we have,

$$\Pr\left(\|\tau(\mathbf{u})\| \geq \frac{q}{\sqrt{d}e^{\varepsilon'} \cdot q^{(n+l)/m}}\right) \geq 1 - 2^{-dm(\varepsilon'-3)}.$$

Otherwise, if $\mathbf{v}_A = 0 \bmod q$ and $0 < \|\tau(\mathbf{v}_E) \bmod q\| \leq q/2B_{\max}$

$$\|\tau(\mathbf{u})\| = \left\|\mathrm{Rot}(\mathbf{E})^T \cdot \tau(\mathbf{v}_E)\right\| \leq \mathfrak{s}_1(\mathrm{Rot}(\mathbf{E})^T) \cdot \|\tau(\mathbf{v}_E)\| < B_{\max} \cdot q/2B_{\max} = q/2.$$

This means that $\mathrm{Rot}(\mathbf{E})^T \cdot \tau(\mathbf{v}_E)$ is never reduced modulo q and it belongs to the set $\mathrm{Rot}(\mathbf{E})^T \cdot \mathbb{Z}^{dl}$ we excluded from the lattice. Lastly, if $\mathbf{v}_A = 0 \bmod q$ and $q/2B_{\max} < \|\tau(\mathbf{v}_E) \bmod q\|$, we lower bound the norm

$$\|\tau(\mathbf{u})\| = \left\|\begin{bmatrix}\mathbf{I}_{dl} \\ \mathrm{Rot}(\bar{\mathbf{E}})^T\end{bmatrix} \cdot \tau(\mathbf{v}_E) \bmod q\right\| \geq \|\tau(\mathbf{v}_E) \bmod q\| > \frac{q}{2B_{\max}}.$$

[4]Here the proof strategy only applies to the power-of-2 cyclotomics in the coefficient embedding.

Overall, the shortest vector is larger than $q \cdot \min\left(\frac{1}{2B_{\max}}, (\sqrt{d}e^{\varepsilon'} \cdot q^{(n+l)/m})^{-1}\right)$. Combining this bound with the arguments above we get

$$\begin{aligned}\lambda_{d(m-l)}(\tau(\Lambda)) &\leq \frac{dm}{1/q \cdot \lambda_{dl+1}(\Lambda_q(\mathbf{M}^T))} \\ &\leq dm \cdot \max\left(2B_{\max}, \sqrt{d}e^{\varepsilon'} \cdot q^{(n+l)/m}\right). \end{aligned} \tag{3.2}$$

For vectors $\mathbf{x}$ with norm bounded this way it holds

$$\begin{aligned}\|\mathrm{Rot}(\mathbf{E}) \cdot \tau(\mathbf{x})\| &\leq \mathfrak{s}_1(\mathrm{Rot}(\mathbf{E})) \cdot \|\tau(\mathbf{x})\| \\ &< B_{\max} \cdot dm \cdot \max\left(2B_{\max}, \sqrt{d}e^{\varepsilon'} \cdot q^{(n+l)/m}\right) \leq q/2\end{aligned}$$

by our conditions on q. In other words, $\mathbf{Ex} = \mathbf{0}$ also holds over the integers. Therefore, $\lambda_{d(m-l)}(\tau(\Lambda_q^{\perp}(\mathbf{A}) \cap \Lambda^{\perp}(\mathbf{E}))) \leq \lambda_{d(m-l)}(\tau(\Lambda))$ from Equation (3.2). Finally, when $(\mathbf{A}, \mathbf{E})$ are not necessarily in $\mathcal{S}$ we adapt the statistical distance in the same way as in Theorem 3, concluding the proof. □

Remark 10. Since the proof relies on the smoothing argument and Corollaries 8+9, the transformation between embeddings from Corollary 15 still applies here. For the canonical embedding statement we require the same conditions and

$$\boldsymbol{\Sigma} \geq \mathfrak{s}_1(\nu) \cdot dm \cdot \max\left(2B_{\max}, \sqrt{d}e^{\varepsilon'} \cdot q^{(n+l)/m}\right) \cdot \sqrt{\ln(2d(m-l)(1+1/\varepsilon))/\pi}.$$

Remark 11. For Condition 1 need $\mathfrak{s}_1(\mathrm{Rot}(\mathbf{E})) < B_{\max}$ so the distribution of $\bar{\mathbf{E}}$ has to have short outputs. For Condition 2, every vector $\mathbf{w} = (\mathbf{w}_1, \mathbf{w}_2)$ in $\Lambda^{\perp}(\mathbf{E})$ can be represented as $\mathbf{w}_1 = -\bar{\mathbf{E}} \cdot \mathbf{w}_2$ for some $\mathbf{w}_2 \in \mathcal{R}^{m-l}$. Hence, the matrix product is equal to

$$[\mathbf{A}_1 \mid \mathbf{A}_2] \cdot \begin{bmatrix} -\bar{\mathbf{E}} \cdot \mathbf{w}_2 \\ \mathbf{w}_2 \end{bmatrix} = (-\bar{\mathbf{E}} \cdot \mathbf{A}_1 + \mathbf{A}_2) \cdot \mathbf{w}_2.$$

The matrix $\mathbf{A}_2 \in \mathcal{R}_q^{n \times (m-l)}$ is uniform and independent of the other summand, so the distribution of the matrix sum is still uniform. Then by Lemma 16 we can take $1 - \delta_2 = \prod_{i=0}^{n-1}\left(1 - \frac{1}{q^{\delta(m-l-i)}}\right)^f$.

3.2.4 Smoothing in the Presence of Noisy Linear Leakage

Lastly, we consider leakage of the form $(\mathbf{E}, \mathbf{E} \cdot \mathbf{x} + \mathbf{y})$ for a given matrix $\mathbf{E}$ and a hidden noise vector $\mathbf{y}$. The matrix $\mathbf{E}$ can be sampled from a short distribution or be fixed as long as it satisfies the conditions below. Moreover, we allow for Q many leaked hints

defined by different matrices $(\mathbf{E}_i)_{i\in[Q]}$ and noise vectors $(\mathbf{y}_i)_{i\in[Q]}$ coming from potentially different noise distributions.

Our key observation is that recent advancements in studying Module Learning With Errors under noisy leakage are also useful in the context of our smoothing LHL under leakage. We recall the following result.

Lemma 59 ([ENP24, Lem. 1])**.** *Let m, Q be positive integers, and $\sigma_{\mathbf{x}}, (\sigma_{\mathbf{y},i})_{i\in[Q]}$ be positive reals. Take $\mathbf{E}_0, \ldots, \mathbf{E}_{Q-1} \in \mathbb{Z}^{m\times m}$ and let $\mathbf{I}$ denote the identity matrix of dimension m. We define $\mathbf{\Sigma} = \left(\frac{1}{\sigma_{\mathbf{x}}^2}\mathbf{I} + \sum_{i\in[Q]} \frac{1}{\sigma_{\mathbf{y},i}^2} \cdot \mathbf{E}_i^T\mathbf{E}_i\right)^{-1}$. Then the following two distributions over $\mathbb{Z}^{m(Q+1)}$ are identical:*

$$D_1 = \left\{(\mathbf{x}, \mathbf{z}_0, \ldots, \mathbf{z}_{Q-1}) | \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma_{\mathbf{x}}}, \mathbf{y}_i \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma_{\mathbf{y}_i}}, \mathbf{z}_i = \mathbf{E}_i \cdot \mathbf{x} + \mathbf{y}_i\right\}$$

$$D_2 = \left\{ (\widehat{\mathbf{x}}, \mathbf{z}_0, \ldots, \mathbf{z}_{Q-1}) \middle| \begin{array}{l} \mathbf{x} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma_{\mathbf{x}}}, \mathbf{y}_i \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma_{\mathbf{y}_i}}, \mathbf{z}_i = \mathbf{E}_i \cdot \mathbf{x} + \mathbf{y}_i \\ \mathbf{c} = \mathbf{\Sigma} \cdot \sum\limits_{i\in[Q]} \dfrac{1}{\sigma_{\mathbf{y},i}^2} \mathbf{E}_i^T \mathbf{z}_i, \quad \widehat{\mathbf{x}} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}} \end{array} \right\}$$

The above lemma allows us to derive the following smoothing-based LHL with noisy linear leakage.

Theorem 5. *Let $\mathcal{K}$ be a cyclotomic field of degree d. Let q be an integer. Let $\mathcal{P}_A$ be a distribution on the set of matrices $\mathcal{R}_q^{n\times m}$ for $n \le m$. Let $\varepsilon \in (0,1)$, and $\beta, \delta_1, \delta_2 > 0$. Suppose that*

$$\Pr\left[\eta_\varepsilon(\Lambda_q^\perp(\mathbf{A})) > \beta | \mathbf{A} \leftarrow \mathcal{P}_A\right] < \delta_1,$$
$$\Pr\left[\mathrm{im}(\mathbf{A}) \neq \mathcal{R}_q^n | \mathbf{A} \leftarrow \mathcal{P}_A\right] < \delta_2.$$

Let $Q > 0$ be an integer, and $\sigma_{\mathbf{x}}, (\sigma_{\mathbf{y},i})_{i\in[Q]}$ be positive reals. For $i \in [Q]$ take $\mathbf{E}_i \in \mathcal{R}^{m\times m}$, defining its corresponding rotation matrix $\mathrm{Rot}(\mathbf{E}_i) \in \mathbb{Z}^{md\times md}$, and let $\mathbf{I}$ denote the identity matrix of dimension dm. We define

$$\mathbf{\Sigma} = \left(\frac{1}{\sigma_{\mathbf{x}}^2}\mathbf{I} + \sum_{i\in[Q]} \frac{1}{\sigma_{\mathbf{y},i}^2} \cdot \mathrm{Rot}(\mathbf{E}_i)^T \mathrm{Rot}(\mathbf{E}_i)\right)^{-1},$$

and assume that $\mathbf{\Sigma} \ge \beta^2$. Define

$$\mathcal{P}' = \left\{(\mathbf{A}\mathbf{x} \bmod q, \mathbf{A}, (\mathbf{E}_i, \mathbf{E}_i\mathbf{x} + \mathbf{y}_i)_i) \mid \mathbf{A} \leftarrow \mathcal{P}_A, \mathbf{x} \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma_{\mathbf{x}}}, \mathbf{y}_i \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma_{\mathbf{y}_i}}\right\}$$
$$\mathcal{U}' = \left\{(\mathbf{u} \leftarrow \mathcal{U}(\mathcal{R}_q^n), \mathbf{A}, (\mathbf{E}_i, \mathbf{E}_i\mathbf{x} + \mathbf{y}_i)_i) \mid \mathbf{A} \leftarrow \mathcal{P}_A, \mathbf{x} \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma_{\mathbf{x}}}, \mathbf{y}_i \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma_{\mathbf{y}_i}}\right\}$$

where $i \in [Q]$. Then

$$\mathrm{SD}(\mathcal{P}',\mathcal{U}') < \varepsilon/(1-\varepsilon) + \delta_1 + \delta_2, \qquad \mathrm{RD}^{(\delta_1+\delta_2)}(\mathcal{P}';\mathcal{U}') < \frac{1+\varepsilon}{1-\varepsilon}.$$

Proof. Let D_1 and D_2 be the two distributions from Lemma 59 using the rotation matrices $\mathrm{Rot}(\mathbf{E}_0), \ldots, \mathrm{Rot}(\mathbf{E}_{Q-1})$. Let us rewrite $\mathcal{P}'$ and introduce $\mathcal{P}''$ as follows.

$$\mathcal{P}' = \Big\{(\mathbf{A}\mathbf{x} \bmod q, \mathbf{A}, (\mathbf{E}_i, \mathbf{z}_i)_{i\in[Q]}) \mid \mathbf{A} \leftarrow \mathcal{P}_A, \tau(\mathbf{x}, \mathbf{z}_0, \ldots, \mathbf{z}_{Q-1}) \leftarrow D_1\Big\},$$

$$\mathcal{P}'' = \Big\{(\mathbf{A}\hat{\mathbf{x}} \bmod q, \mathbf{A}, (\mathbf{E}_i, \mathbf{z}_i)_{i\in[Q]}) \mid \mathbf{A} \leftarrow \mathcal{P}_A, \tau(\hat{\mathbf{x}}, \mathbf{z}_0, \ldots, \mathbf{z}_{Q-1}) \leftarrow D_2\Big\}.$$

By Lemma 59, the two distributions are identical, hence not changing the SD or RD value. In $\mathcal{P}''$, the distribution of the hash input $\hat{\mathbf{x}}$ comes from a discrete Gaussian distribution with covariance matrix $\mathbf{\Sigma}$ defined in the lemma statement and with center $\tau^{-1}(\mathbf{c})$, where $\mathbf{c} = \mathbf{\Sigma} \cdot \sum_{i\in[Q]} \frac{1}{\sigma^2_{\mathbf{y},i}} \mathrm{Rot}(\mathbf{E_i})^T \tau(\mathbf{z})_i$. As by assumption $\mathbf{\Sigma} \geq \beta^2$, we can now invoke Theorem 3, concluding the proof. □

3.3 Properties of Some Concrete Distributions

The different LHL flavours we presented in Section 3.1.3 make use of the collision probability or well-spreadness of a given hash input distribution $\mathcal{P}$ over R_q^m for a ring R, modulus q and dimension m. In some cases, we might assume that $\mathcal{P}$ is obtained by independently and identically sampling every of the m ring elements over some distribution $\mathcal{X}$ over R_q, that is, $\mathcal{P} = \mathcal{X}^m$. In this section, we give bounds on well-spreadness and collision probability for different distributions over cyclotomic rings which appeared in the lattice literature. Some bounds have been (implicitly) proven before, some are generalised results and some are new bounds. We will detail our contribution for every distribution. All results are stated with respect to the coefficient embedding.

3.3.1 Bounded Uniform Distribution

The following lemma gives concrete formulas for the collision probability and the well-spreadness of the bounded uniform distribution over R_q. All were already implicitly proved in [Mic07]. The bounded uniform distribution is used in many places in lattice-based cryptography. A popular example is the signature scheme Dilithium [LDK+22].

Lemma 60 (Adapted from [Mic07])**.** *Let $\mathcal{R}$ be a cyclotomic ring of degree d and let q be an unramified prime number such that $\Phi(X) = \prod_{j=1}^{f} \Phi_j(X) \bmod q$ is the factorisation of*

the cyclotomic polynomial in $\mathcal{R}_q$ with every Φ_j being of degree δ and $d = f\delta$. Let $\eta < q/2$ be a positive integer. Let $\mathcal{I} = \langle \prod_{j\in G} \Phi_j(X) \rangle$ for some non-empty subset $G \subseteq \{1, \ldots, f\}$ be an arbitrary ideal of $\mathcal{R}_q$ (this representation of $\mathcal{I}$ is correct by Lemma 9). Let $\mathcal{X}$ be the uniform distribution over $S_\eta = \{c \in \mathcal{R}_q \colon \|\tau(c)\|_\infty \leq \eta\}$, defining the hash input distribution $\mathcal{P} = \mathcal{X}^m$ for some dimension m. Then

- $\mathrm{Coll}(\mathcal{X} \bmod \mathcal{I}) \leq 1/(2\eta+1)^{\delta|G|}$,
- $\mathrm{Coll}(\mathcal{P} \bmod \mathcal{I}) \leq 1/(2\eta+1)^{\delta|G|m}$,
- $\mathrm{Coll}(\mathcal{P}) = 1/(2\eta+1)^{dm}$.

In particular the distribution $\mathcal{X}$ is $1/(2\eta+1)^{\delta}$-well-spread.

Proof. Firstly, $\mathrm{Coll}(\mathcal{P}) = \mathrm{Coll}(\mathcal{X})^m$, where

$$\mathrm{Coll}(\mathcal{X}) = \sum_{c\in S_\eta} \Pr[c \leftarrow \mathcal{X}]^2 = (2\eta+1)^d \cdot \frac{1}{(2\eta+1)^{2d}} = \frac{1}{(2\eta+1)^d}.$$

For given non-empty set $G \subseteq \{1, \ldots, f\}$, ideal $\mathcal{I} = \langle \prod_{j\in G} \Phi_j(X) \rangle$ and $c(X) \leftarrow \mathcal{X}$ denote

$$c(X) = \underbrace{\sum_{i=0}^{\delta|G|-1} \alpha_i X^i}_{c_0} + \underbrace{\sum_{i=\delta|G|}^{d-1} \alpha_i X^i}_{c_1},$$

where all $\alpha_i \leftarrow \mathcal{U}(\{-\eta, \ldots, \eta\})$. Note that for any fixed value of c_1, the sum $c_0 + c_1 = c$ is distributed uniformly in a subset of $\mathcal{R}_q$ of size $(2\eta+1)^{\delta|G|}$. Then for two samples $c_0 + c_1 \leftarrow \mathcal{X}, c_0' + c_1' \leftarrow \mathcal{X}$ it holds

$$\begin{aligned}
\Pr(c = c' \bmod \mathcal{I}) &= \Pr(c_0 + (c_1 \bmod \prod_{j\in G} \Phi_j(X)) = c_0' + (c_1' \bmod \prod_{j\in G} \Phi_j(X))) \\
&= \Pr(c_0 = c_0' + (c_1' - c_1 \bmod \prod_{j\in G} \Phi_j(X))) \\
&\leq (2\eta+1)^{-\delta|G|}.
\end{aligned}$$

□

Remark 12. We can see that for $\delta \neq d$, Corollary 13 provides tighter bounds than Corollary 14. Hence, which statement gives the better results depends on the splitting behavior of the ring.

3.3.2 Discrete Gaussian Distribution

Second, we look at the maybe most natural distribution arising in lattice-based cryptography: discrete Gaussian distributions. Using the smoothing properties of Gaussians, we compute in Lemma 61 a novel well-spreadness bound. As Section 3.1 applies to distributions over $\mathcal{R}_q$, we consider the discrete Gaussian distribution reduced modulo q.

Lemma 61. *Let $\mathcal{K}$ be a cyclotomic field with $\mathcal{R}$ its ring of integers. Further, let q be an unramified prime modulus such that $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$ for distinct, irreducible $\Phi_j(X)$ of degree δ. Let $\varepsilon = 2^{-d}$, where d is the ring degree of $\mathcal{R}$. Let $\sigma > d \cdot q^{\delta/d}$. Then $\mathcal{D}^{\theta}_{\mathcal{R},\sigma} \bmod q$ is B-well-spread with*

$$B = \frac{1}{q^{\delta}} \left(1 + 2^{-d+2}\right).$$

Proof. By Lemma 44 and Lemma 6, for all $j \in [f]$,

$$\eta_{\varepsilon}(\langle q, \Phi_j(X) \rangle) \leq (d(\langle q, \Phi_j(X) \rangle) \cdot |\Delta_{\mathcal{K}}|)^{1/d} \leq q^{\delta/d} \cdot d.$$

Thus $\sigma > \max_j(\eta_{\varepsilon}(\langle q, \Phi_j(X) \rangle))$ and by Lemma 39 $\forall y \in \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$

$$\max_{j \in [f]} \left(\Pr[x \bmod (q, \Phi_j(X)) = y | x \leftarrow \mathcal{D}_{\mathcal{R},\sigma}]\right) \leq \frac{1}{q^{\delta}}(1 + 2^{-d+2}).$$

□

We further state a recent result on the min-entropy of discrete Gaussian distributions modulo ideals of $\mathcal{R}_q$ in the coefficient embedding. We state a slightly generalised version considering discrete Gaussians over $\mathcal{R}^m$ for $m \geq 1$.

Lemma 62 (Adapted from [JLWG25, Cor. 4])**.** *Let $\mathcal{K}$ be a cyclotomic field with $\mathcal{R}$ its ring of integers. Further, let q be an unramified prime modulus such that $\mathcal{R}_q = \prod_{j=1}^{f} \mathbb{Z}_q[X]/\langle \Phi_j(X) \rangle$ for distinct, irreducible $\Phi_j(X)$ of degree δ. Let $\mathcal{I}$ be an ideal of $\mathcal{R}_q$ of norm $q^{\delta k}$ for some $k \in \{0, \ldots, f\}$ (this is correct by Lemma 9). For $m \in \mathbb{N}$, let σ be a positive real and $\mathbf{c} \in \mathcal{R}^m$ be a centre. If $\sigma \leq \frac{\sqrt{\pi}}{2} \cdot \frac{q}{\sqrt{\ln(4m\delta k)}}$ it holds*

$$\mathrm{H}_{\infty}(\mathcal{D}^{\tau}_{\mathcal{R}^m,\sigma,\mathbf{c}} \bmod \mathcal{I}) \geq m\delta k \log \sigma - 1.$$

3.3.3 Biased Ternary Distribution

Next we consider biased ternary distributions. Such distributions are for instance relevant in compact proof systems and were used in [ALS20]. Let $\mathcal{D}$ be the ternary distribution with

bias $p \in [0,1]$ over $\{\pm 1, 0\}$, which samples 0 with probability p and ± 1 with probability $(1-p)/2$. The following lemma provides a formula for the collision probability as well as for well-spreadness. The proof of the latter follows the proof technique of [ALS20], and generalises their result form power-of-2 to arbitrary cyclotomics. It requires the special relation from Lemma 8 between modulus q and cyclotomic polynomial $\Phi(X)$. We include the full proof below to showcase the proof technique.

Lemma 63 (Generalised from [ALS20, Lemma 3.2])**.** *Let $\mathcal{K} = \mathbb{Q}[X]/\langle \Phi(X) \rangle$ be the νth cyclotomic field of degree $d = \varphi(\nu)$, with $\mathcal{R}$ its ring of integers. Further, let $\nu = \prod p_i^{e_i}$ for $e_i \geq 1$ and let $\mu = \prod p_i^{f_i}$ for $1 \leq f_i \leq e_i$. Let q be a prime such that $q = 1 \bmod \mu$ and the multiplicative order modulo ν of q is ν/μ. By Lemma 8, $\Phi(X) = \prod_{j=1}^{f}(X^\delta - r_j) \bmod q$ for distinct $r_j \in \mathbb{Z}_q^\times$ with $d = f\delta$. Let $\mathcal{X}$ be the distribution over $S_1 = \{c \in \mathcal{R}_q \colon \|\tau(c)\|_\infty \leq 1\}$ where each coefficient of the polynomial is sampled from $\mathcal{D}$, that is, $\mathcal{X} = \mathcal{D}^d$. This defines the hash input distribution $\mathcal{P} = \mathcal{X}^m = \mathcal{D}^{dm}$ for some dimension m. The distribution $\mathcal{X}$ is B-well-spread with*

$$B = \max_{j \in [f]} \left(\frac{1}{q} + \frac{1}{q} \sum_{k=1}^{q-1} \prod_{l=0}^{f-1} |p + (1-p)\cos(2\pi k r_j^l / q)| \right)^\delta, \tag{3.3}$$

and the collision probability of $\mathcal{P}$ is given by

$$\mathrm{Coll}(\mathcal{P}) = \left(\frac{1 - 2p + 3p^2}{2} \right)^{dm}.$$

Proof. We start by proving the collision probability formula. Note that $\mathrm{Coll}(\mathcal{P}) = \mathrm{Coll}(\mathcal{D}^{dm}) = \mathrm{Coll}(\mathcal{D})^{dm}$, where

$$\mathrm{Coll}(\mathcal{D}) = \sum_{j \in \{\pm 1, 0\}} \mathcal{D}(j)^2 = p^2 + 2 \cdot \left(\frac{p-1}{2} \right)^2 = \frac{3p^2 - 2p + 1}{2}.$$

We now prove the well-spreadness, which requires to analyze the distribution of the CRT-slots $\bmod\ (X^\delta - r_j)$. We start by introducing some notations. Let $c(X) = \sum_{i=0}^{d-1} c_i X^i \leftarrow \mathcal{X}$. Let $S_1^{(\delta)} \subset \mathcal{R}_q$ be the set given as

$$S_1^{(\delta)} = \left\{ c(X) = c_0 + c_\delta X^\delta + \cdots + c_{\delta(f-1)} X^{(f-1)\delta} \mid \|c\|_\infty \leq 1 \right\},$$

where δ divides the degree of each monomial. Then we can write S_1 as

$$S_1 = \left\{ c(X) = \tilde{c}_0(X) + \cdots + \tilde{c}_{\delta-1}(X) \cdot X^{\delta-1} \mid \tilde{c}_i(X) = \sum_{l=0}^{f-1} c_{\delta l + i} X^{\delta l} \in S_1^{(\delta)} \right\}.$$

Computing $c(X) \bmod (X^\delta - r_j)$ is equivalent to evaluating δ-powers of X as $X^\delta = r_j$. Or evaluating polynomials $\bar{c}_i(X)$ in r_j where

$$\bar{c}_i(X) \coloneqq \tilde{c}_i(X^{1/\delta}) = c_i + c_{\delta+i}X + \cdots + c_{\delta(f-1)+i}X^{f-1}.$$

Overall we obtain

$$c(X) \bmod (X^\delta - r_j) = \bar{c}_0(r_j) + \bar{c}_1(r_j) \cdot X + \cdots + \bar{c}_{\delta-1}(r_j) \cdot X^{\delta-1}.$$

Note that the distributions of each coefficient $\bar{c}_i(r_j)$ are i.i.d. and equal to $\sum_{l=0}^{f-1} c_{\delta l+i} r_j^l$, a convolution of f independent random variables $c_{\delta l+i} r_j^l$. Since distributions of $\bar{c}_i(r_j)$ are identical we prove the statement only for $i = 0$.

Denote P_1 the distribution of $\bar{c}_0(r_j) = \sum_{l=0}^{f-1} c_{\delta l} r_j^l$ with values in $\mathbb{Z}_q$. By the Fourier transform inversion property for finite fields $\forall y \in \mathbb{Z}_q$:

$$P_1(y) = \frac{1}{q} \sum_{k=0}^{q-1} \hat{P}_1(k) \cdot e^{-2\pi i y \frac{k}{q}},$$

where by $\hat{F}$ we denote the Fourier transform of a function F. For $l \in [f]$ let μ_l denote the distribution $\mu_l = \left\{ c_{\delta l} r_j^l \mid c_{\delta l} \leftarrow \mathcal{D} \right\}$. Then $P_1 = \sum_{l=0}^{f-1} \mu_l$ and $\hat{P}_1 = \prod_{l=0}^{f-1} \hat{\mu}_l$. For any $k \in \mathbb{Z}_q$

$$\begin{aligned} \hat{\mu}_l(k) &= p + \frac{1-p}{2} \cdot e^{-2\pi i k \frac{r_j^l}{q}} + \frac{1-p}{2} \cdot e^{+2\pi i k \frac{r_j^l}{q}} \\ &= p + (1-p) \cdot \cos(2\pi k r_j^l / q), \end{aligned}$$

where the transformation uses the Euler formula and the odd and even properties of the sine and cosine functions accordingly. Then

$$\begin{aligned} P_1(y) &= \frac{1}{q} \sum_{k=0}^{q-1} \prod_{l=0}^{f-1} \left(p + (1-p) \cdot \cos(2\pi k r_j^l / q) \right) \cdot e^{-2\pi i y \frac{k}{q}} \\ &\leq \frac{1}{q} + \frac{1}{q} \sum_{k=1}^{q-1} \prod_{l=0}^{f-1} \left| p + (1-p) \cdot \cos(2\pi k r_j^l / q) \right|, \end{aligned}$$

where the first inequality follows from taking the absolute value on both sides and applying the triangle inequality and the last equality only separates the term corresponding to $k = 0$. As all coefficients are identically and independently distributed, the same bound holds for all of the δ coefficients. Given that the bound on P_1 depends on the underlying r_j, the final bound on the formula of the well-spreadness B takes the maximum over r_j. □

Remark 13. As already observed in [ALS20, Lemma 3.3], in the special case of power-of-2 cyclotomic rings, we can exploit symmetries in order to simplify the well-spreadness formula. More precisely, for any $j \in [f]$, the element r_j generates the same group. That is, $\forall j : \left\{\pm 1, \pm r_j, \ldots, \pm r_j^{f-1}\right\} = \langle r_j \rangle = G_\mu$ where G_μ is the only subgroup of order μ in $\mathbb{Z}_q^\times$. This group contains a subgroup $\langle 1, -1 \rangle$ therefore the following set $\forall j : \left\{1, r_j, \ldots, r_j^{f-1}\right\} = G_\mu / \langle -1 \rangle$ also forms the same quotient group for every j. This is exactly the set that appears in the product of the well-spreadness formula. Therefore, the product contains the same values for every r_j only multiplied in a different order and we do not need to take the maximum over $j \in [f]$ to obtain the bound on well-spreadness.

Similarly, we do not have to sum over all $k \in 1 \ldots q-1$. We obtain the same product value for every $k' = \alpha \cdot k$ with $\alpha \in G_\mu$. Then for all k, α we have

$$\left\{k \cdot \alpha, k \cdot \alpha \cdot r_j, \ldots, k \cdot \alpha \cdot r_j^{f-1}\right\} = k \cdot (\alpha \cdot G_\mu)/\langle -1 \rangle = k \cdot G_\mu / \langle -1 \rangle.$$

Then the sum becomes

$$P_1(y) = \frac{1}{q} + \frac{2f}{q} \sum_{k \in \mathbb{Z}_q^\times / G_\mu} \prod_{l=0}^{f-1} \left| p + (1-p) \cdot \cos(2\pi k r_j^l / q) \right|,$$

and numerical estimations can be computed $2f$ times faster. The same optimisations apply to Lemma 64 and Lemma 65 in the case of power-of-2 cyclotomics. The group arguments above are not true for arbitrary cyclotomic rings, as for example -1 may not even belong to G_μ.

3.3.4 Fixed Hamming Weight Ternary Distribution

Another distribution that appears in the literature, is the uniform bounded distribution with an additional restriction on the Hamming weight of the vectors. Similar to the biased ternary distribution, it appears in the context of compact proof systems. It allows for even more efficient proofs as it generates elements of smaller ℓ_1-norm. Concretely, we consider the uniform distribution over

$$S_{1,w} = \left\{c(X) = \tilde{c}_0(X) + \cdots + \tilde{c}_{\delta-1}(X) \cdot X^{\delta-1} \mid \tilde{c}_i \in S_1^{(\delta)} \wedge \mathrm{HW}(\tilde{c}_i) = \tilde{w}\right\}.$$

Note that every element $c(X)$ of $S_{1,w}$ has the same Hamming weight $\mathrm{HW}(c) = \delta \cdot \tilde{w} \coloneqq w$ and coefficients equal to ± 1 or 0. We remark that the non-zero coefficients are somewhat regularly distributed. This tweak simplifies the analysis of well-spreadness significantly.

The same proof technique applies when the non-zero coefficients are distributed in the set $S_\eta \setminus \{0\}$, although it makes the Fourier Transform of the coefficient distributions more complex. A similar distribution was analysed in [ESLR23].

Below, we provide a formula for the collision probability and the well-spreadness. The latter has been proven for power-of-2 cyclotomics in the work of [ESZ22], where they built upon the approach of [ALS20] detailed above. The main obstacle the proof deals with is that the Hamming weight restriction introduces dependencies into some of the distributions. The authors first remove this dependency and apply the Fourier analysis tools and then prove that the distribution analysed is close enough to the original one. We generalise the result to arbitrary cyclotomics and provide a proof for completeness. As for the biased ternary case, we require the special relation from Lemma 8 between modulus q and cyclotomic polynomial $\Phi(X)$.

Lemma 64 (Generalised from [ESZ22, Lemma 1])**.** *Let $\mathcal{K} = \mathbb{Q}[X]/\langle\Phi(X)\rangle$ be the νth cyclotomic field of degree $d = \varphi(\nu)$, with $\mathcal{R}$ its ring of integers. Further, let $\nu = \prod p_i^{e_i}$ for $e_i \geq 1$ and let $\mu = \prod p_i^{f_i}$ for $1 \leq f_i \leq e_i$. Let q be a prime such that $q = 1 \bmod \mu$ and the multiplicative order modulo ν of q is ν/μ. By Lemma 8, $\Phi(X) = \prod_{j=1}^{f}(X^\delta - r_j) \bmod q$ for distinct $r_j \in \mathbb{Z}_q^\times$ with $d = f\delta$. Let $\mathcal{X}$ be the uniform distribution over $S_{1,w}$, defining the hash input distribution $\mathcal{P} = \mathcal{X}^m$ for some dimension m. The distribution $\mathcal{X}$ is B-well-spread with*

$$B = \max_{j\in[f]} \left(\frac{\beta}{q} + \frac{\beta}{q} \sum_{k=1}^{q-1} \left| \frac{1}{f} \sum_{l=0}^{f-1} \cos(2\pi k r_j^l / q) \right|^{\tilde{w}} \right)^{\delta}, \tag{3.4}$$

where $\beta = \frac{f^{\tilde{w}}(f-\tilde{w})!}{f!}$ and the collision probability of $\mathcal{P}$ is given by

$$\mathrm{Coll}(\mathcal{P}) = \left(\binom{f}{\tilde{w}}^{-\delta} 2^{-w} \right)^m.$$

Proof. We start with the collision probability of $\mathcal{P}$, where $\mathrm{Coll}(\mathcal{P}) = \mathrm{Coll}(\mathcal{X})^m$. We compute $|S_{1,\tilde{w}}| = \binom{f}{\tilde{w}}^{\delta} 2^w$. Then, the collision probability of $\mathcal{X}$, which is the uniform distribution over $S_{1,\tilde{w}}$, is $\binom{f}{\tilde{w}}^{-\delta} 2^{-w}$.

It remains to show the well-spreadness. To obtain a uniformly random element in $S_{1,\tilde{w}}$ each polynomial $\tilde{c}_i(X)$ can be sampled independently in the following way. First, we choose the non-zero monomials $(t_1, \ldots, t_{\tilde{w}})$ as an ordered $\tilde{w}$-tuple of distinct elements from $[f]$ uniformly at random. This ensures the Hamming weight condition. Then we choose the signs of the non-zero monomials $c_{\delta t_\alpha + i} \leftarrow \mathcal{U}(\{1, -1\})$ for $\alpha = 1, \ldots, \tilde{w}$. Finally, we set $\tilde{c}_i(X) = \sum_{\alpha=1}^{\tilde{w}} c_{\delta t_\alpha + i} \cdot X^{\delta \cdot t_\alpha}$, defining $c(X) = \sum_{i=0}^{\delta-1} \tilde{c}_i(X) \cdot X^i$. Note that this method samples each possible $\tilde{c}_i(X)$ polynomial $\tilde{w}!$ times, but this does not skew the distribution.

Now reduce the polynomial $c(X)$ modulo $X^\delta - r_j$ for some $j \in [f]$. Similar to the proof of Lemma 63, the distributions of each coefficient of the resulting polynomial are

independent and equal to $\bar{c}_0(r_j) = \sum_{\alpha=1}^{\tilde{w}} c_{\delta t_\alpha} \cdot r_j^{t_\alpha} \in \mathbb{Z}_q$. Denote this distribution P_2. We aim to prove that $\max_{y\in\mathbb{Z}_q}(P_2(y))$ is bounded.

Define a similar distribution P_1 where instead of choosing a tuple of distinct powers of X they are allowed to repeat. In this distribution, we sample $t_\alpha \leftarrow \mathcal{U}([f]), c_{\delta t_\alpha} \leftarrow \mathcal{U}(\{1,-1\})$ for each $\alpha = 1,\ldots,\tilde{w}$. Then, P_1 is the distribution of $\bar{c}_0(r_j) = \sum_{\alpha=1}^{\tilde{w}} c_{\delta t_\alpha} \cdot r_j^{t_\alpha} \in \mathbb{Z}_q$. This simplification allows us to represent the distribution as a sum of $\tilde{w}$ independent random variables and easily compute corresponding Fourier transforms. Using the inversion property of Fourier transforms over finite fields $\forall y \in \mathbb{Z}_q$

$$P_1(y) = \frac{1}{q} + \frac{1}{q}\sum_{k=1}^{q-1} \hat{P}_1(k) \cdot e^{-2\pi i\, y\frac{k}{q}}.$$

The distribution P_1 is a convolution of $\tilde{w}$ distributions μ_α where

$$\mu_\alpha = \left\{c_{\delta t_\alpha} \cdot r_j^{t_\alpha} \mid c_{\delta t_\alpha} \leftarrow \{1,-1\}, t_\alpha \leftarrow [f]\right\}.$$

Note that the distribution is the same for all α. Then $\hat{P}_1(k) = \prod_{\alpha=1}^{\tilde{w}} \hat{\mu}_\alpha(k) = \hat{\mu}_1(k)^{\tilde{w}}$. Define a set $S = \left\{\pm 1, \pm r_j, \ldots, \pm r_j^{f-1}\right\}$ then $\mu_1 = \mathcal{U}(S)$ and for any $k \in \mathbb{Z}_q$

$$\begin{aligned}\hat{\mu}_1(k) &= \sum_{x\in S} \mu_1(x) \cdot e^{-2\pi i k\frac{x}{q}} \\ &= \sum_{l=0}^{f-1} \frac{1}{2f} \cdot (e^{-2\pi i k\frac{r_j^l}{q}} + e^{+2\pi i k\frac{r_j^l}{q}}) = \frac{1}{2f}\sum_{l=0}^{f-1} 2\cos(2\pi k r_j^l/q).\end{aligned} \tag{3.5}$$

In the above, the first equality uses the shape of S and $\mu_1 = \mathcal{U}(S)$, the second equality uses the Euler formula and that cosine and sine are even and odd functions accordingly. Substituting this expression into the main formula we get

$$\begin{aligned}P_1(y) &= \frac{1}{q} + \frac{1}{q}\sum_{k=1}^{q-1}\left(\frac{1}{f}\sum_{l=0}^{f-1}\cos(2\pi k r_j^l/q)\right)^{\tilde{w}} \cdot e^{-2\pi i\, y\frac{k}{q}} \\ &\le \frac{1}{q} + \frac{1}{q}\sum_{k=1}^{q-1}\left|\frac{1}{f}\sum_{l=0}^{f-1}\cos(2\pi k r_j^l/q)\right|^{\tilde{w}}.\end{aligned}$$

For the inequality we take the absolute value on both sides and apply the triangle inequality on the right. The left side does not change since $P_1(y)$ is a positive real number. To obtain a statement about $P_2(y)$ we compute an upper bound on $\mathrm{RD}_\infty(P_2;P_1) := \max_{y\in\mathbb{Z}_q} \frac{P_2(y)}{P_1(y)}$. Note that we take the Rényi divergence of the infinite order as for uniform distributions this is optimal.

Denote as $\mathcal{T}$ the set of ordered $\tilde{w}$-tuples with non-repeating coordinates in $[f]$. Then $|\mathcal{T}| = \frac{f!}{(f-\tilde{w})!}$. The distributions P_1, P_2 can be seen as computing the same function on the following distributions of ordered tuples

$$D_1 = \{((t_1, \ldots, t_{\tilde{w}}), (c_{\delta t_1}, \ldots, c_{\delta t_{\tilde{w}}})) \mid t_\alpha \leftarrow \mathcal{U}([f]), c_{\delta t_\alpha} \leftarrow \mathcal{U}(\{1, -1\})\},$$
$$D_2 = \{((t_1, \ldots, t_{\tilde{w}}), (c_{\delta t_1}, \ldots, c_{\delta t_{\tilde{w}}})) \mid (t_1, \ldots, t_{\tilde{w}}) \leftarrow \mathcal{U}(\mathcal{T}), c_{\delta t_\alpha} \leftarrow \mathcal{U}(\{1, -1\})\}.$$

Note that $\mathcal{T} \subset [f]^{\tilde{w}}$ so $\mathrm{Supp}(D_2) \subset \mathrm{Supp}(D_1)$ and divergence $\mathrm{RD}_\infty(D_2; D_1)$ is well defined. Then by the data processing inequality for Rényi divergence

$$\begin{aligned}\mathrm{RD}_\infty(P_2; P_1) &\leq \mathrm{RD}_\infty(D_2; D_1) \\ &= \frac{|\mathrm{Supp}(D_1)|}{|\mathrm{Supp}(D_2)|} = \frac{2^{\tilde{w}} f^{\tilde{w}} (f-\tilde{w})!}{2^{\tilde{w}} f!} = \frac{f^{\tilde{w}}(f-\tilde{w})!}{f!} = \beta,\end{aligned}$$

where the first equality holds since D_1 and D_2 are uniform distributions on their support. Therefore, by the probability preservation property of RD_∞, $\forall y : P_2(y) \leq \beta \cdot P_1(y)$ which implies the main statement. □

3.3.5 Centred Binomial Uniform Distribution

Let $\mathcal{D}$ be the centred binomial distribution with parameter η, defined over $j \in \{-\eta, \ldots, \eta\}$ by

$$\Pr[j \leftarrow \mathcal{D}] = 2^{-2\eta} \cdot \binom{2\eta}{\eta + j}.$$

By sampling each coefficient independently over $\mathcal{D}$, we obtain a distribution $\mathcal{X}$ over the ring R of degree d, where $\mathcal{X} = \mathcal{D}^d$. Similar to the bounded uniform distribution, it has found widespread use in lattice-based cryptography. For instance, it is used in the encryption scheme Kyber [SAB+22]. We provide a formula of collision probability and well-spreadness in the lemma below, applying to arbitrary cyclotomic rings. The latter has been analysed in [CLS16] for the special case of power-of-2 cyclotomics.

Lemma 65. *Let $\mathcal{K} = \mathbb{Q}[X]/\langle \Phi(X) \rangle$ be the νth cyclotomic field of degree $d = \varphi(\nu)$, with $\mathcal{R}$ its ring of integers. Further, let $\nu = \prod p_i^{e_i}$ for $e_i \geq 1$ and let $\mu = \prod p_i^{f_i}$ for $1 \leq f_i \leq e_i$. Let q be a prime such that $q = 1 \bmod \mu$ and the multiplicative order modulo ν of q is ν/μ. By Lemma 8, $\Phi(X) = \prod_{j=1}^{f} (X^\delta - r_j) \bmod q$ for distinct $r_j \in \mathbb{Z}_q^\times$ with $d = f\delta$. Define $\mathcal{X} = \left\{ c(X) = \sum_{i=0}^{d-1} c_i X^i \mid c_i \leftarrow \mathcal{D} \right\}$, that is, $\mathcal{X} = \mathcal{D}^d$. This defines the hash input distribution $\mathcal{P} = \mathcal{X}^m = \mathcal{D}^{dm}$ for some dimension m.*

The collision probability of $\mathcal{P}$ is given by

$$\mathrm{Coll}(\mathcal{P}) = \left(2^{-4\eta}\binom{4\eta}{2\eta}\right)^{dm},$$

and the distribution $\mathcal{X}$ is B-well spread with

$$B = \max_{j\in[f]}\left(\frac{1}{q} + \frac{1}{q}\sum_{k=1}^{q-1}\prod_{l=0}^{f-1}|\cos(\pi k r_j^l/q)|^{2\eta}\right)^{\delta}. \tag{3.6}$$

Proof. By the definition of collision probability and by the Vandermonde identity

$$\mathrm{Coll}(\mathcal{P}) = 2^{-4\eta}\sum_{i=-\eta}^{\eta}\binom{2\eta}{\eta+i}^2 = 2^{-4\eta}\binom{4\eta}{2\eta}.$$

Similarly to the proof of Lemma 63 denote P_1 the distribution of $\bar{c}_0(r_j) = \sum_{l=0}^{f-1} c_{\delta l} r_j^l$ with values in $\mathbb{Z}_q$. By the Fourier transform inversion property for finite fields $\forall y \in \mathbb{Z}_q$:

$$P_1(y) = \frac{1}{q}\sum_{k=0}^{q-1}\hat{P}_1(k)\cdot e^{-2\pi i y\frac{k}{q}}.$$

For $l \in [f]$ denote as μ_l the distribution $\mu_l = \left\{c_{\delta l} r_j^l \mid c_{\delta l} \leftarrow \mathcal{D}\right\}$. Then $P_1 = \sum_{l=0}^{f-1}\mu_l$ and $\hat{P}_1 = \prod_{l=0}^{f-1}\hat{\mu}_l$. For $k \in \mathbb{Z}_q$

$$\begin{aligned}
\hat{\mu}_l(k) &= \sum_{s=-\eta}^{\eta} 2^{-2\eta}\cdot\binom{2\eta}{\eta+s}e^{-2\pi i k\frac{s r_j^l}{q}} = 2^{-2\eta}\cdot e^{2\pi i k\frac{\eta r_j^l}{q}}\cdot\sum_{s=0}^{2\eta}\binom{2\eta}{s}e^{-2\pi i k\frac{s r_j^l}{q}} \\
&= 2^{-2\eta}\cdot e^{2\pi i k\frac{\eta r_j^l}{q}}\cdot\left(1+e^{-2\pi i k\frac{r_j^l}{q}}\right)^{2\eta} \\
&= 2^{-2\eta}\cdot\left(e^{\pi i k\frac{r_j^l}{q}}\right)^{2\eta}\cdot\left(1+e^{-2\pi i k\frac{r_j^l}{q}}\right)^{2\eta} = \left(\frac{e^{\pi i k\frac{r_j^l}{q}}+e^{-\pi i k\frac{r_j^l}{q}}}{2}\right)^{2\eta} = \cos(\pi k r_j^l/q)^{2\eta},
\end{aligned}$$

where the transformations above use the binomial and Euler formulas and the odd and even property of the sine and cosine functions accordingly. Then

$$\begin{aligned}
P_1(y) &= \frac{1}{q}\sum_{k=0}^{q-1}\prod_{l=0}^{f-1}\cos(\pi k r_j^l/q)^{2\eta}\cdot e^{-2\pi i y\frac{k}{q}} \\
&\le \frac{1}{q}\sum_{k=0}^{q-1}\prod_{l=0}^{f-1}\left|\cos(\pi k r_j^l/q)\right|^{2\eta} = \frac{1}{q} + \frac{1}{q}\sum_{k=1}^{q-1}\prod_{l=0}^{f-1}\left|\cos(\pi k r_j^l/q)\right|^{2\eta},
\end{aligned}$$

where the first inequality comes from taking the absolute value on both sides and applying the triangle inequality, the second equality separates the term $k = 0$. □

Remark 14. When $\eta = 1$ we recover the result in Lemma 63 for $p = 1/2$, using $\cos(x)^2 = \frac{1}{2} + \frac{1}{2}\cos(2x)$.

Chapter 4

Gaussian Leftover Hash Lemma for Modules

The classic Leftover Hash Lemma (LHL) covered in Chapter 3 is an argument about randomness extraction. However, it only applies when the output space of the function, in our case $\mathbf{x} \in \mathcal{R}^m \mapsto \mathbf{A} \cdot \mathbf{x} \bmod q \in \mathcal{R}_q^n$, is a finite group.

In [AGHS13], Agrawal, Gentry, Halevi, and Sahai prove an LHL-type statement that holds over lattices rather than in finite groups. For $\mathbf{X} \in \mathbb{Z}^{r \times m}$ with columns sampled independently from $\mathcal{D}_{\mathbb{Z}^r,\sigma}$ – a discrete Gaussian over $\mathbb{Z}^r$ – and $\mathbf{v}$ sampled from $\mathcal{D}_{\mathbb{Z}^m,\sigma'}$, they upper bound the statistical distance between $(\mathbf{X}, \mathbf{X} \cdot \mathbf{v})$ and $(\mathbf{X}, \mathcal{D}_{\mathbb{Z}^r,\sigma' \cdot \sqrt{\mathbf{X}\mathbf{X}^\intercal}})$ for any $m \geq r \ln(r\sigma)$ (constant factors are omitted for this introduction). In follow-up works, Aggrawal and Regev [AR16] improve this result with an alternative proof. In 2020, Kirshanova, Nguyen, Stehlé and Wallet [KNSW20] further improve the result with a better lower bound $m \geq r \log(\sigma)$. Hereafter, we refer to statements of this type as *Gaussian LHLs*.

As stated the output distribution of $\mathbf{X} \cdot \mathbf{v}$ depends on the matrix $\mathbf{X}$ and might be far from a spherical Gaussian. In order to have a spherical output, the solution is usually to sample $\mathbf{X}$ and then to sample $\mathbf{v}$ from the Gaussian distribution with parameter $\sigma' \cdot \widetilde{\mathbf{X}}$ where $\widetilde{\mathbf{X}}$ is a pseudoinverse of $\mathbf{X}$. However, in some applications this choice of the distribution of $\mathbf{v}$ is not available. The 'sphericity' of the distribution $\mathcal{D}_{\mathcal{R}^r,\sqrt{\mathbf{X}\mathbf{X}^\intercal}}$ is controlled by the singular values of $\mathbf{X}$. In [AGHS13, Lemma 8], the authors argue that with high probability for a wide enough Gaussian matrix $\mathbf{X} \in \mathbb{Z}^{r \times m}$ of parameter σ, its singular values are within a constant factor of $\sqrt{m}\sigma$. Together, this establishes when $\mathbf{X} \cdot \mathbf{v}$ follows a distribution close to independent from both of its inputs, enabling various applications discussed below.

A limitation of existing Gaussian LHLs is that they do not take advantage of the algebraic structure of the lattice, while most practical primitives based on lattices currently use module lattices: if $\mathcal{M}$ is a rank-r module over the ring of integers $\mathcal{R}$ of a degree-d number field $\mathcal{K}$, it is a rank-$d \cdot r$ lattice when considered through, e.g. the canonical embedding of $\mathcal{K}$. In [KNSW20], the authors give a bound of $m \geq dr \ln(\sigma)$ independent vectors required for the Gaussian LHL result to work over $\mathcal{R}^r$. However, a better bound for the number of independent vectors is to be expected, since any single vector $\mathbf{v} \in \mathcal{R}^r$ gives $d-1$ other vectors (over $\mathbb{Z}$) 'for free' (think $X^i \cdot \mathbf{v}$ in the cyclotomic case). We would then expect a lower bound on m that grows linearly in r (the rank of the module) and sublinear in d (the degree of the number field).

The main contribution of this chapter is proving that only $m \geq r \cdot (\log(dr\sigma))^{1+o(1)} + \lambda/\log(d)$ vectors are necessary to get a Gaussian LHL in the context of module lattices extending the techniques of [AR16]. Event though [KNSW20] has a better bounds for the dimension and the gaussian parameter, we were only able to adapt the approach of [AR16] to the ring setting.

In Section 4.3, we prove that if $m \geq r \cdot (\log(dr\sigma))^{1+o(1)} + \lambda/\log(d)$, $\mathbf{X} \in \mathcal{R}^{r \times m}$ is a Gaussian matrix and $\mathbf{v} \in \mathcal{R}^m$ is a Gaussian vector with parameter $\sqrt{\mathbf{\Sigma}}$ then the distributions $(\mathbf{X}, \mathbf{X} \cdot \mathbf{v})$ and $(\mathbf{X}, \mathcal{D}_{\mathcal{R}^r, \sqrt{\mathbf{X} \cdot \mathbf{\Sigma} \cdot \mathbf{X}^T}})$ have negligible statistical distance. In particular, in Section 4.1 we establish when $\mathbf{X}$ generates $\mathcal{R}^r$ and in Section 4.2 we upper bound $\eta_\varepsilon(\Lambda^\perp(\mathbf{X}))$. The composition of these two results establishes the claim, but these results may be of independent interest.

Moreover, in Section 4.4 we establish a bound when we may expect $\mathbf{X} \cdot \mathbf{v}$ to be close to spherical for any ring of integers of a number field: if $\mathbf{X} \in \mathcal{R}^{r \times m}$ is a Gaussian matrix with $m \geq (rd)^3$ with parameter σ, then the Rényi divergence between the distribution $\mathcal{D}_{\mathcal{R}^r, \sqrt{\mathbf{X}\mathbf{X}^T}}$ and a spherical Gaussian of parameter $\approx \sigma\sqrt{m}$ is $O(1)$.

Finally, as a first explicit application, in Section 4.5 we give a reduction from MSIS to k-MSIS that supports a linear number of hints and discuss how a similar reduction from MLWE to k-MLWE can be obtained.

Applications. A direct application of our Gaussian Leftover Hash Lemma over modules are reductions from MSIS and MLWE to k-MSIS and k-MLWE respectively that support $k = \mathcal{O}(m)$ following the blueprint from [LPSS14].[1] In this chapter, we only give the reduction for MSIS to k-MSIS since the MLWE to k-MLWE reduction proceeds analogously.

[1]The reduction from SIS to k-SIS in [BF11b] which supports $k = \mathcal{O}(1)$ translates without our Gaussian LHL generalisation.

This allows us to lift constructions based on k-SIS and k-LWE to the module setting, such as linearly homomorphic signature schemes [BF11b], traitor tracing [LPSS14] and partial GPV-style trapdoors in Chapter 5. Moreover, k-SIS may be considered as the oldest and most well-founded (due to the reduction from SIS) member of a family of "SIS with hints" assumptions that have been proposed recently. Indeed, works such as [ACL+22] prove the existence of hard instances of their newly introduced assumption by proving these instances as hard as k-MSIS. However, the hardness of k-MSIS is only assumed and not established.

Our generalisation of the Gaussian Leftover Hash Lemma to modules also enables the translation of various applications where [AGHS13, AR16] was used to rerandomise ciphertexts to achieve a form of confidentiality, such as the rerandomisation of GSW ciphertexts [BdPMW16] or the speculative suggestion in [ACL+22] to use similar techniques to make their construction zero-knowledge. Similarly, the special 1×2 case of a Gaussian LHL over rings had been established and used in [LSS14] to prove that a given distribution does not leak secret values. More generally, our result implies a generalisation of Gaussian convolution theorems such as [GMPW20, Theorem 3] to the module setting. Such tools have been used to give tighter reductions for SIS and LWE [MP13].

Moreover, the 1×2 case was generalised to more rings and used in [PS21] to establish that NTRU instances can be rerandomised. This in turn establishes a reduction to the decision NTRU problem. This chapter gives a building block to generalise such results to larger ranks $r > 1$. Similarly, in [SS13] the primitivity of the matrix of the special case 1×2 was considered to prove that NTRU signing keys can be efficiently sampled. This chapter is a building block for generalising such reductions to a module setting.

Finally, in LWE-based encryption schemes, showing correctness often involves upper bounding the norm of the product $\mathbf{X} \cdot \mathbf{e}$ between a short matrix $\mathbf{X}$ from the decryption key and a short vector $\mathbf{e}$ from the encryption randomness. A naive upper bound is $\|\mathbf{X}\| \cdot \|\mathbf{e}\|$. This can be further upper bounded by $\sqrt{r} \cdot d \cdot m \cdot s^2$ if both $\mathbf{X}$ follow a discrete Gaussian distribution $\mathcal{D}_{\mathcal{R},s}^{r \times m}$ and $\mathbf{e}$ follows $\mathcal{D}_{\mathcal{R},s}^{m}$. A better approach is to consider the product for a fixed $\mathbf{X}$ with $\mathbf{X} \cdot \mathcal{R}^m = \mathcal{R}^r$ and $s \geq \eta_\varepsilon(\Lambda^\perp(\mathbf{X}))$. In this case, $\mathbf{X} \cdot \mathbf{e}$ is statistically close to a sample from $\mathcal{D}^r_{\mathcal{R},s\cdot\sqrt{\mathbf{X}^\mathsf{T}\cdot\mathbf{X}}}$. Further upper bounding $\|\mathbf{X}\|$ and $\left\|\mathcal{D}_{\mathcal{R}^r,\sqrt{\mathbf{X}^\mathsf{T}\cdot\mathbf{X}}\cdot s}\right\|$ then yields a tighter bound of $\sqrt{r \cdot d \cdot m} \cdot s^2$ at the cost of an additional correctness error from the Gaussian LHL. In the literature, the second approach seems to have been applied in the setting where $\mathcal{R} = \mathbb{Z}$ or when $r = 1$ but not in more general settings, probably due to the difficulty of upper bounding $\eta_\varepsilon(\Lambda^\perp(\mathbf{X}))$, which we take first steps to address in Section 4.2.

Open problems. Even if our Gaussian LHL gives a polynomially bounded norm growth between the norm of the matrix $\mathbf{X}$ and the norm of the vector $\mathbf{v}$, the degree of the norm growth is large (see Theorem 8). This blowup comes from the fact that the upper bound on the smoothing parameter of $\Lambda^{\perp}(\mathbf{X})$ is very loose. We think that this is a proof artefact, and can be improved in future work.

The looseness of this bound comes from the fact that the proof of Lemma 77 is fundamentally "worst-case". A first way to improve it would be to use more probabilistic arguments instead of pigeonhole-like ones (see Section 4.2.3). Another blowup factor in the proof of Lemma 77 is that when constructing a set of size 2^d of elements coprime to some $a \in \mathcal{R}$, Lemma 76 give element of size $\simeq \|a\|^2$. We think that restricting Lemma 76 to specific number fields (e.g. prime-power cyclotomics, where the prime ideal distribution is better known) will make our result more usable in practice.

Lastly, we could improve the bound in Lemma 77 would be to find a trade-off relation between the size of the smoothing parameter of $\Lambda^{\perp}(\mathbf{X})$ and the width of $\mathbf{X}$. Currently, our proof shows a threshold effect: if the width of $\mathbf{X}$ is less than a certain m_0, then the smoothing parameter is large, if it is above m_0, we have an upper bound on the smoothing parameter that does not depend much on m. Finding more subtle relations between the two would allow to improve the bound, at the cost of increasing the number of columns in $\mathbf{X}$.

Even though in $m \geq r \cdot (\log(dr\sigma))^{1+o(1)} + \lambda/\log(d)$ we remove the linear dependency of m on the ring degree d we still have the $\lambda/\log(d)$. In practice, the $\log(d)$ in the denominator makes the value a small power of 2, ≈ 16. However, the dependency on λ seems to be inherent to [AR16] proof technique. It would an interesting problem to extend the alternative proof techniques of [AGHS13, KNSW20]. The particular argument we were not able to prove is that for any integer vector $\mathbf{z}$ with a lower bound on its norm and the matrix $\mathbf{X}$ sampled above we have $\Pr(\|\mathbf{X}^{\mathsf{T}} \cdot \mathbf{z}\| \geq B) \leq 2^{-dr}$ for a well chosen bound B.

Chapter structure. In Section 4.1 we prove that a random wide discrete Gaussian matrix is surjective with overwhelming probability under selected parameters, this guarantees the support of the distribution $\mathbf{X} \cdot \mathbf{v}$ is the full ring $\mathcal{R}^r$. Then in Section 4.2 we prove an upper bound on the smoothing parameter of $\Lambda^{\perp}(\mathbf{X})$. In Section 4.3 we combine the arguments into a Gaussian LHL statement and give to approaches for making $\mathbf{X} \cdot \mathbf{v}$ have spherical covariance parameter. Lastly, in Section 4.5 we use our Gaussian LHL to prove a MSIS-to-k-MSIS reduction.

Chapter specific notations. In this chapter, we denote integral ideals by gothic letters ($\mathfrak{a}, \mathfrak{p} \ldots$) and general ideals by uppercase letters ($I, J, \ldots$). As an exception we denote the number field conductor as $\mathfrak{f}$ and we sometimes treat it as an integer. We omit specifying the embedding for module lattices since we always use the *canonical embedding*. We assume that we know a basis $\mathbf{B}^{\mathcal{R}} \in \mathbb{R}^{d \times d}$ of $\Phi(\mathcal{R})$ as a $\mathbb{Z}$-module. We denote by δ_K the maximal ℓ_∞ norm of its columns (in particular $\delta_K = 1$ for cyclotomics fields with the power basis). The results of this work are existential instead of constructive, so we use the 'best' basis of $\mathcal{R}$. According to [MG02, Corollary 7.2] there always exists a basis of $\mathcal{R}$ such that $1 \leq \delta_K \leq \sqrt{d} \cdot \lambda_d(\mathcal{R})/2$.

4.1 Generating module lattices with Gaussian vectors

In this section, we study the surjectivity of Gaussian matrices. For a matrix where each column is sampled from an $\mathcal{R}$-module $\mathcal{M}$, we say that a matrix is surjective if $\mathcal{R}$-linear combinations of its columns generate the module $\mathcal{M}$.

4.1.1 Surjective matrices over finite fields

We first establish some facts about random matrices over finite fields $\mathbb{F}$ to be used in subsequent proofs. First, extending the above notion of surjectivity to $\mathbb{F}$-vector spaces, we show that a random matrix over $\mathbb{F}$ is surjective with overwhelming probability in its dimensions if each entry of the matrix is independently distributed and has constant min-entropy.

Lemma 66. *Let $r, m \geq 0$ be integers. Let $\mathbb{F}$ be a finite field, and $(D_{i,j})_{1 \leq i \leq r, 1 \leq j \leq m}$ be distributions over $\mathbb{F}$ such that there exists $\alpha \in (0,1)$ such that for any i, j, $\max_{x \in \mathbb{F}} D_{i,j}(x) \leq \alpha$. If $\mathbf{M} \in \mathbb{F}^{r \times m}$ is a random matrix such that*

- *all the entries of $\mathbf{M}$ are independent random variables, and*
- *the (i,j)-th entry of $\mathbf{M}$ is sampled from $D_{i,j}$,*

then it holds that

$$\Pr(\mathbf{M} \cdot \mathbb{F}^m = \mathbb{F}^r) \geq 1 - r \cdot \alpha^{m-r+1}.$$

Proof. Let H be a subspace of $\mathbb{F}^m$ of dimension k and $i = 1, \ldots, d$, it holds that the probability of a vector in $\mathbb{F}^m$ sampled from $D_{i,1} \times \ldots \times D_{i,m}$ belonging to this subspace is

upper bounded by α^{m-k} [NP20, Generalisation of Lemma 2.3]. As $\mathbb{F}$ is a field, the matrix $\mathbf{M}$ is surjective if and only if $\mathbf{M}^T$ is injective. We now lower-bound the probability of $\mathbf{M}^T$ being injective when $\mathbf{M}$ is built row by row. Let $\mathbf{M}_i$ be the matrix composed of the first i rows of $\mathbf{M}$. For $i = 1, \ldots, r$, the matrix $\mathbf{M}_i^T$ is injective if and only if $\mathbf{M}_{i-1}^T$ is injective and the ith row of $\mathbf{M}$ is not in the span of the $i-1$ previous ones, which has dimension $i-1$. The probability of $\mathbf{M}^T$ being injective is then lower-bounded by

$$\prod_{i=1}^{r}(1-\alpha^{m-i+1}) = \prod_{i=m-r+1}^{m}(1-\alpha^{i}) \geq (1-\alpha^{m-r+1})^r \geq 1 - r \cdot \alpha^{m-r+1}.$$

□

The next lemma shows the equivalence between a matrix $\mathbf{B}$ with columns in $\mathcal{M}$ being surjective and ($\mathbf{B}$ mod $\mathfrak{p}$) being surjective for prime ideals $\mathfrak{p}$ of $\mathcal{R}$.

Lemma 67 (Adaptated from [NP20, Lemma 2.1])**.** *Let $r, m \geq 1$ be integers and $\mathcal{M}$ be an $\mathcal{R}$-module of rank r where $\mathcal{R}$ is a Dedekind domain. A matrix $\mathbf{B} \in \mathcal{M}^m$ satisfies $\mathbf{B} \cdot \mathcal{R}^m = \mathcal{M}$ if and only if ($\mathbf{B}$ mod $\mathfrak{p}$) is surjective in $\mathcal{M}/\mathfrak{p}\mathcal{M}$ for any prime ideal $\mathfrak{p}$ of $\mathcal{R}$.*

Proof. If $\mathbf{B}$ is surjective in $\mathcal{M}$, then it is clearly surjective modulo any prime $\mathfrak{p}$. Now we assume that $\mathbf{B}$ mod $\mathfrak{p}$ is surjective modulo every $\mathfrak{p}$. Let $\mathcal{M}' = \mathbf{B} \cdot \mathcal{R}^m$. Let $\mathfrak{p}$ be any ideal, we will show that $\mathfrak{p}$ does not divide $\det(\mathcal{M}')/\det(\mathcal{M})$. Since the matrix $\mathbf{B}$ mod $\mathfrak{p}$ is surjective, there exists a square submatrix $\mathbf{B}'$ of $\mathbf{B}$ which is invertible modulo $\mathfrak{p}$, and hence full-rank over $\mathcal{K}$. Let $\mathcal{M}'' = \mathbf{B}' \cdot \mathcal{R}^r \subset \mathcal{M}'$. Assume that the ideal $\mathfrak{p}$ divides $\det(\mathcal{M}')/\det(\mathcal{M})$, then it also divides $\det(\mathcal{M}'')/\det(\mathcal{M})$ by inclusion, which is equivalent to that $\det_{\mathcal{K}}(\mathbf{B}') \in \det(\mathcal{M}) \cdot \mathfrak{p}$, which contradicts the fact that $\mathbf{B}'$ is invertible in $\mathcal{M}/\mathfrak{p}\mathcal{M}$ by Lemma 41. Since $\mathfrak{p}$ can be any ideal, it holds that $\det(\mathcal{M}') = \det(\mathcal{M})$, and hence that $\mathbf{B}$ is surjective in $\mathcal{M}$. □

4.1.2 Surjective matrices over rings

In this subsection, we prove that Gaussian matrices of size $r \times m$ over some full rank modules $\mathcal{M}$ are surjective over $\mathcal{M}$ with high probability when $m \geq 2r$. The main idea of the proof is to consider the matrix modulo all the primes $\mathfrak{p}$ of the number field, showing that with high probability (using Lemmas 66 and 67), the matrix modulo $\mathfrak{p}$ is surjective over $\mathcal{R}/\mathfrak{p}$. We first prove it in the case of free modules in Lemma 71, where the main contribution of this subsection is, and then generalise it in **??**. We first prove that the

Gaussian distribution over some ideal I modulo the ideal $I \cdot \mathfrak{p}$ satisfies the distribution hypothesis of Lemma 66.

Lemma 68. *Let $\mathbf{\Sigma} = \mathbf{S}\mathbf{S}^{\mathsf{T}} - \sigma^2\mathbf{I}$ with $\mathbf{S} \in \mathbb{R}^{dr \times dr} \geq x \cdot \sigma\mathbf{I}$ for some $x > 1$. Let $(\mathbf{\Sigma}')^{-1} = (\mathbf{\Sigma}^{-1} + \sigma^{-2}\mathbf{I})$. Then $\mathbf{\Sigma}' \geq \sigma^2 \cdot (1 - 1/x^2)\,\mathbf{I}$.*

Proof. By definition, $\mathbf{\Sigma}$ is diagonalisable with eigenvalues $\geq (x^2 - 1) \cdot \sigma^2$. The eigenvalues of $\mathbf{\Sigma}'$ are exactly the $(s^{-1} + \sigma^{-2})^{-1}$ where the s are the eigenvalues of $\mathbf{\Sigma}$. We then have, for every $s \geq (x^2 - 1) \cdot \sigma^2$:

$$\frac{1}{\frac{1}{s} + \frac{1}{\sigma^2}} \geq \frac{\sigma^2}{\frac{1}{x^2 - 1} + 1} = \sigma^2 \cdot \left(1 - \frac{1}{x^2}\right).$$

□

Lemma 69. *For any integer $N \geq 2$, the number of prime ideals dividing N is at most $d \cdot \log_2(N)$.*

Proof. If $\mathfrak{p}$ is a prime ideal dividing N, then it must be above a integral prime dividing N. Since prime numbers split in at most d prime ideals, and that the number of prime integer dividing a number N is always less than $\log_2(N)$, the result follows. □

Lemma 70 (Generalised from [JLWG25, Corollary 7])**.** *Let $\varepsilon \in (0, 1/\sqrt{2})$, I be a (possibly fractional) ideal of $\mathcal{R}$, $B \geq 1$, $c \in \mathcal{K}_{\mathbb{R}}$ be a centre and $\sigma \geq B \cdot \Delta_{\mathcal{K}}^{1/d} \cdot \mathcal{N}(I)^{1/d} \cdot \eta_{\varepsilon}^{(d)}$. For any integral ideal $\mathfrak{a} \subset \mathcal{R}$, we define $\mathcal{D}_{I,\sigma,c}^{(\mathfrak{a})}$ the distribution $\mathcal{D}_{I,\sigma,c}$ modulo $\mathfrak{a} \cdot I$. Then for any $a \in I/(I \cdot \mathfrak{a})$, it holds that*

$$\mathcal{D}_{I,\sigma,c}^{(\mathfrak{a})}(a) \leq (1 + 4\varepsilon) \cdot \begin{cases} \frac{1}{\mathcal{N}(\mathfrak{a})} & \text{if } \mathcal{N}(\mathfrak{a}) \leq B^d \\ \frac{1}{B^d} & \text{else .} \end{cases}$$

Lemma 71 (Assuming GRH)**.** *Let $r \geq 1$ and $m \geq 2r$ be integers. Let $\mathbf{C} = [\mathbf{c}_1, \ldots, \mathbf{c}_m] \in \mathbb{R}^{rd \times md}$ a centre matrix and $\sigma > 0$. Let $\varepsilon \in (0, 1/4)$. Let $B > (\log \Delta_{\mathcal{K}})^{\Omega(1)/d}$. Assuming that the following two inequalities hold*

$$B^d \geq r \cdot d^2 \ln(dr \cdot \sigma) \qquad \text{and} \qquad \sigma \geq B \cdot \Delta_{\mathcal{K}}^{1/d} \cdot \eta_{\varepsilon}(\mathcal{R}^r),$$

and $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_m] \in \mathcal{K}^{r \times m}$ such that $\mathbf{x}_i \leftarrow \mathcal{D}_{\mathcal{R}^r,\sigma,\mathbf{c}_i}$, then

$$\begin{aligned} \Pr_{\mathbf{X}}[\mathbf{X} \cdot \mathcal{R}^m \neq \mathcal{M}] \leq \frac{m \cdot \varepsilon}{1 - \varepsilon} + r \cdot \left(\frac{B}{2}\right)^{-(r+1)d} + r \cdot 2^{-dr} \\ + (1 + 4\varepsilon)^{m-2r+1} \cdot r \cdot P_{\mathcal{K}}(m - 2r + 1), \end{aligned}$$

where $P_{\mathcal{K}}(s) = \sum_{\mathfrak{p}} \mathcal{N}(\mathfrak{p})^{-s}$ is the prime zeta function of $\mathcal{K}$.

Proof. We abuse the notation and write the distribution of $\mathbf{X}$ as $\mathcal{D}_{\mathcal{R}^{r\times m},\sigma,\mathbf{C}}$. Note that the coefficients of $\mathbf{X}$ are all independent and sampled from $\mathcal{D}_{\mathcal{R},\sigma,c}$ for some $c \in \mathcal{K}$, so by Lemma 70, Lemma 66 holds for any submatrix of $\mathbf{X}$. Write $\mathbf{X} = [\mathbf{X}_1|\mathbf{X}_2]$ with $\mathbf{X}_1 \in \mathcal{R}^{r\times r}$, and let $\mathfrak{p}$ be a prime ideal of norm $\geq B^d$. Then by Lemma 67, $\mathbf{X}_1$ is invertible modulo $\mathfrak{p}$ with probability $\geq 1 - r(B/2)^{-(r+1)d}$ and in that case, in particular $\det_K(\mathbf{X}_1) \neq 0$, we assume this is the case for the rest of the proof. By Lemma 23, with probability $\geq 1 - r \cdot 2^{-dr}$, the norm of every column of $\mathbf{X}_1$ is bounded by $\sqrt{rd} \cdot \sigma$, and hence its determinant is bounded by $(\sqrt{rd}\sigma)^{rd}$. Let $\mathcal{P}$ be the set of prime ideals dividing $\det_K(\mathbf{X}_1)$, by Lemma 69 it holds that $|\mathcal{P}| \leq rd^2 \cdot \log(\sqrt{rd}\sigma) =: l$. We now prove that $\mathbf{X}$ is full-rank modulo every prime ideal. For any prime $\mathfrak{p}$ not in $\mathcal{P}$, $\mathbf{X}_1$ is invertible modulo $\mathfrak{p}$, and so $\mathbf{X}$ is full-rank modulo $\mathfrak{p}$. For all $\mathfrak{p} \in \mathcal{P}$, by Lemma 66 it holds that $\mathbf{X}_2$ is full-rank mod $\mathfrak{p}$ with probability $\geq 1 - r(1-4\varepsilon)^{m-2r+1} \cdot \alpha$ for $\alpha = B^{-(m-2r+1)d}$ if $\mathcal{N}(\mathfrak{p}) \geq B^d$ and $\alpha = \mathcal{N}(\mathfrak{p})^{-(m-2r+1)}$ else.

Let us order the primes of $\mathcal{R}$ as $\{\mathfrak{p}_1, \mathfrak{p}_2, \ldots\}$ and $\mathcal{P} = \{\mathfrak{q}_1, \ldots, \mathfrak{q}_l\}$, both in increasing norm order. It holds that $\mathcal{N}(\mathfrak{p}_i) \leq \mathcal{N}(\mathfrak{q}_i)$ for any $i = 1, \ldots, l$. We let B be a real such that the first l prime ideals of $\mathcal{R}$ have their norm $\leq B^d$. The existence of B will be proven at the end of the proof. Under the condition that $\mathbf{X}_1$ is invertible and has bounded columns, by the union bound it holds that

$$\begin{aligned}
&\Pr(\mathbf{X} \text{ not surjective}) \leq \sum_{i=1}^{l} \Pr(\mathbf{X}_2 \text{ not surjective mod } \mathfrak{q}_i) \\
&\leq r \cdot (1+4\varepsilon)^{m-2r+1} \cdot \left(\sum_{i=1}^{a} \frac{1}{\mathcal{N}(\mathfrak{q}_i)^{m-2r+1}} + \sum_{i=a+1}^{l} \frac{1}{B^{d(m-2r+1)}} \right) \\
&\leq r \cdot (1+4\varepsilon)^{m-2r+1} \cdot \sum_{i=1}^{l} \frac{1}{\mathcal{N}(\mathfrak{p}_i)^{m-2r+1}} \leq r \cdot (1+4\varepsilon)^{m-2r+1} \cdot P_{\mathcal{K}}(m-2r+1).
\end{aligned}$$

Now let us prove that B satisfying our condition exists and give a lower bound on it. The condition is that B is such that the first l primes of $\mathcal{R}$ are of norm less than B^d. By [BS96, Theorem 8.7.4], the GRH implies that for $X \geq \log(\Delta_{\mathcal{K}})^{\Omega(1)}$, the number of prime ideals $\pi_{\mathcal{K}}(X)$ of norm less than X is less than $1.1 \cdot X/\ln(X)$. This implies that as long as $B^d \geq \log(\Delta_{\mathcal{K}})^{\Omega(1)}$, then $\pi_{\mathcal{K}}(B^d) \leq B^d$, then as long as

$$l = r \cdot d^2 \ln(dr \cdot \sigma) \leq B^d,$$

the norm of the first l prime ideals is less than B^d.

□

In the ring case ($r = 1$), Lemma 71 can be compared to [SS11, Lemma 4.4], where the probability of generating $\mathcal{R}$ with two elements is proven close to $1/\zeta_{\mathcal{K}}(2)$ for power-of-two

cyclotomic number fields (note that for $m = 2, r = 1$, our upper bound is equal to ∞). A similar proof can be used to prove that for any number field, for suitable parameters, the probability of generating $\mathcal{R}$ with m elements of $\mathcal{R}$ is close to $1/\zeta_{\mathcal{K}}(m)$. We note that when m tends to infinity, we have $1/\zeta_{\mathcal{K}}(m) \sim P_{\mathcal{K}}(m)$.

4.1.3 Asymptotic behavior of $P_{\mathcal{K}}$

The upper bound in Lemma 71 is described in terms of the prime zeta function $P_{\mathcal{K}}$. In the following, we upper bound $P_{\mathcal{K}}$ in terms of the smallest prime ideal norm $N_{\mathcal{K}}$, for which we know explicit value when $\mathcal{K}$ is a cyclotomic field of prime-power conductor.

Lemma 72. *Let $\mathcal{K}$ be a number field and $\mathcal{R}$ its ring of integers. Let $N_{\mathcal{K}}$ denote the norm for the smallest prime ideal in $\mathcal{R}$ and $a_{\mathcal{K}}$ denote the number of different prime ideals of norm $N_{\mathcal{K}}$. Let $\kappa_{\mathcal{K}} \in (0, 1)$ be the ratio between the norm of the smallest ideal and the norm of the second-smallest ideal in $\mathcal{R}$. Then for any $x \geq 2$, it holds that*

$$P_{\mathcal{K}}(x) \leq \frac{a_{\mathcal{K}}}{N_{\mathcal{K}}^x} \cdot \left(1 + \frac{2d}{a_{\mathcal{K}}} \cdot N_{\mathcal{K}} \cdot \kappa_{\mathcal{K}}^{x-1}\right) \leq \frac{3d}{N_{\mathcal{K}}^{x-1}} \ .$$

Proof. Let N be the norm of the second-smallest prime ideal of $\mathcal{R}$, i.e. $\kappa_{\mathcal{K}} = N_{\mathcal{K}}/N$. For any $n \geq N$, let a_n be the number of prime ideals of norm n. For any $x \geq 1$, let $\pi_{\mathcal{K}}(x)$ be the number of prime ideals of norm $\leq x$. We use Abel's summation formula and the fact that $\pi_{\mathcal{K}}(x) \leq d \cdot x$ for any $x \geq 2$:

$$\begin{aligned} P_{\mathcal{K}}(x) - a_{\mathcal{K}} \cdot N_{\mathcal{K}}^{-x} = \sum_{n \geq N} \frac{a_n}{n^x} &= \lim_{u \to \infty} \left(\frac{\pi_{\mathcal{K}}(u)}{u^x}\right) + \int_N^{\infty} \pi_{\mathcal{K}}(u) \cdot \frac{x \cdot \mathrm{d}u}{u^{x+1}} \\ &\leq d \cdot x \cdot \int_N^{\infty} u^{-x} \mathrm{d}u = d \cdot \frac{x}{x-1} \cdot N^{-x+1} \leq 2d \cdot N^{-x+1}. \end{aligned}$$

Substituting $N = N_{\mathcal{K}}/\kappa_{\mathcal{K}}$ and rearranging gives the first inequality

$$P_{\mathcal{K}}(x) \leq \frac{a_{\mathcal{K}}}{N_{\mathcal{K}}^x} \cdot \left(1 + \frac{2d}{a_{\mathcal{K}}} \cdot N_{\mathcal{K}} \cdot \kappa_{\mathcal{K}}^{x-1}\right) \ .$$

The second inequality follows from $a_{\mathcal{K}} \leq d$ and $\kappa_{\mathcal{K}} \leq 1$.

□

Lemma 73. *Let $q = p^k$ be a power of a prime and let $\mathcal{K}_q$ be the qth cyclotomic field. Then $N_{\mathcal{K}_q} = p$ and the norm of the second-smallest ideal is at least $q + 1$.*

Proof. First, it holds that p totally ramifies in $\mathcal{K}_q$, so an ideal of norm p exists in $\mathcal{K}_q$, now let us prove that there is no smaller ideal in this field. Let $N \neq p$ be a prime and $\mathfrak{p}$ a prime of $\mathcal{K}_q$ above N. The norm of $\mathfrak{p}$ is N^l where l is the smallest integer such that $N^l = 1 \bmod q$, in particular, since $N^l \neq 1$, it holds that $N^l \geq q + 1$. □

4.1.4 Specialised bounds

We conclude by giving concrete bounds for Gaussian matrices being surjective for general number fields and for the special case of prime-power cyclotomic fields.

Theorem 6 (General Number Field, Assuming GRH)**.** *Let $\mathcal{K}$ be a number field of degree $d \geq 2$ and of smallest ideal norm $N_{\mathcal{K}}$. Let $0 < \varepsilon < 2^{-6}$. Then for any $\sigma > 0$, $m \geq r \geq 1$ and centre $\mathbf{C} \in \mathcal{K}_{\mathbb{R}}^{r \times m}$ such that*

$$\begin{aligned}
\sigma &\geq 9 \cdot r^{\frac{1}{d}} \cdot \log\left(r \cdot \Delta_K^{\frac{1}{d}} \cdot \eta_\varepsilon(\mathcal{R}^r)\right)^{\frac{1}{d}} \cdot \Delta_{\mathcal{K}}^{\frac{1}{d}} \cdot \eta_\varepsilon(\mathcal{R}^r), \\
m &\geq 2r + \frac{\log(1/\varepsilon) + 1 + \log(2d)}{\log(N_{\mathcal{K}}/1.4)} \text{ and} \\
9 \cdot r^{1/d} &\cdot \log(r\Delta_K^{1/d} \cdot \eta_\varepsilon(\mathcal{R}^r))^{1/d} > (\log \Delta_{\mathcal{K}})^{\Omega(1)/d},
\end{aligned}$$

it holds that

$$\Pr_{\mathbf{X} \leftarrow \mathcal{D}_{\mathcal{R}^{r \times m}, \sigma, \mathbf{C}}}(\mathbf{X} \cdot \mathcal{R}^m \neq \mathcal{R}^r) \leq 2m \cdot (\varepsilon + 2^{-dr}).$$

Proof. We show below that Lemma 71 holds for some $B \geq 4$. The values of the parameters and Lemma 72 give

$$\begin{aligned}
&\frac{m \cdot \varepsilon}{1 - \varepsilon} + r\left(\frac{B}{2}\right)^{-(r+1)d} + r2^{-dr} \ + (1 + 4\varepsilon)^{m-2r+1} \cdot r \cdot P_{\mathcal{K}}(m - 2r + 1) \\
&\leq 1.2 \cdot m \cdot \varepsilon + r \cdot 2^{-d} \cdot 2^{-dr} + r2^{-dr} + r(1.4)^{m-2r+1} \cdot 3d \cdot N_{\mathcal{K}}^{-(m-2r)} \\
&\leq 1.2 \cdot m \cdot \varepsilon + r \cdot (2^{-d} \cdot 2^{-dr} + 2^{-dr} + 1.4 \cdot \varepsilon) \\
&\leq \varepsilon \cdot (1.2m + 1.4r) + 2r \cdot 2^{-dr} \leq 2m \cdot (\varepsilon + 2^{-dr}).
\end{aligned}$$

Now we argue for the existence of B. The hypothesis of Lemma 71 are satisfied if $B^d \geq r \cdot d^2 \ln(dr \cdot \sigma)$ and $\sigma \geq B \cdot \Delta_{\mathcal{K}}^{1/d} \cdot \eta_\varepsilon(\mathcal{R}^r)$, that is to say if B satisfies

$$B^d \geq rd^2 \cdot \log(dr \cdot \Delta_{\mathcal{K}}^{1/d} \cdot \eta_\varepsilon(\mathcal{R}^r) \cdot B) = rd \log(B^d) + rd^2 \log(dr \cdot \Delta_{\mathcal{K}}^{1/d} \cdot \eta_\varepsilon(\mathcal{R}^r)).$$

By Lemma 1, this inequality holds for any B satisfying

$$B^d \geq 2\left(rd^2 \log(dr \cdot \Delta_{\mathcal{K}}^{1/d} \cdot \eta_\varepsilon(\mathcal{R}^r)) + rd \log(rd)\right).$$

It holds that

$$\begin{aligned}
&\left(2\left(rd^2\log(dr\cdot\Delta_K^{1/d}\cdot\eta_\varepsilon(\mathcal{R}^r))+rd\log(rd)\right)\right)^{1/d}\\
\leq\ &\left(2r\cdot\left((d^2+d)\log(d)+d^2\log(r\Delta_K^{1/d}\cdot\eta_\varepsilon(\mathcal{R}^r))+d\log(r)\right)\right)^{1/d}\\
\leq\ &(2r)^{1/d}\cdot\left(\left((d^2+d)\log(d)\right)^{1/d}+\left(d^2\log(r\Delta_K^{1/d}\cdot\eta_\varepsilon(\mathcal{R}^r))\right)^{1/d}+(d\log(r))^{1/d}\right)\\
\leq\ &(2r)^{1/d}\cdot\left(2.5+2\log(r\Delta_K^{1/d}\cdot\eta_\varepsilon(\mathcal{R}^r))^{1/d}+\sqrt{2}\log(r)\right)\\
\leq\ &9\cdot r^{1/d}\cdot\log\left(r\Delta_K^{1/d}\cdot\eta_\varepsilon(\mathcal{R}^r)\right)^{\frac{1}{d}},
\end{aligned}$$

hence taking $B = 9\cdot r^{1/d}\cdot\log(r\Delta_K^{1/d}\cdot\eta_\varepsilon(\mathcal{R}^r))^{1/d}$ gives us the desired result. □

Corollary 16 (Prime-Power Cyclotomics, Assuming GRH)**.** *Let $p^k > 2$ a prime power, and $\mathcal{K} = \mathbb{Q}(\zeta_{p^k})$ be the p^k-cyclotomic number field of degree $d = (p-1)\cdot p^{k-1}$. Let $0 < \varepsilon < 2^{-6}$. Then for any $\sigma > 0$, $m \geq r \geq 1$ and centre $\mathbf{C} \in \mathcal{K}_{\mathbb{R}}^{r\times m}$ such that*

$$\begin{aligned}
\sigma &\geq 9\cdot r^{\frac{1}{d}}\cdot\log\left(rd\cdot\eta_\varepsilon(\mathcal{R}^r)\right)^{\frac{1}{d}}\cdot d\cdot\eta_\varepsilon(\mathcal{R}^r),\\
m &\geq 2r+\frac{\log(1/\varepsilon)+1+\log(2d)}{\log(p/1.4)}\ \textit{and}\\
9\cdot r^{1/d}\cdot\log(r\Delta_K^{1/d}\cdot\eta_\varepsilon(\mathcal{R}^r))^{1/d} &> (\log\Delta_{\mathcal{K}})^{\Omega(1)/d},
\end{aligned}$$

it holds that

$$\Pr_{\mathbf{X}\leftarrow\mathcal{D}_{\mathcal{R}^{r\times m},\sigma,\mathbf{C}}}(\mathbf{X}\cdot\mathcal{R}^m \neq \mathcal{R}^r) \leq 2m\cdot(\varepsilon+2^{-dr}).$$

Proof. Follows from Theorem 6 and Lemmas 6 and 73. □

Remark 15. Note that in a security proof ε is typically set to $2^{-\lambda}$ where λ is the security parameter. Then the bound on m becomes $m \geq 2r + \frac{\lambda+1+\log(2d)}{\log(p/1.4)}$

Remark 16. In Theorem 6 we state $B > (\log\Delta_{\mathcal{K}})^{\Omega(1)/d}$ as an additional condition that must hold for an unspecified constant. We inherit this constant from the previous work in [BS96, Theorem 8.7.4] and more research is required to compute its exact value. We note that for cyclotomic number fields $(\log\Delta_{\mathcal{K}})^{1/d} \leq (d\cdot\log(d))^{1/d}$ approaches 1 when d tends to infinity. Hence, we believe $9\cdot r^{1/d}\cdot\log(r\Delta_K^{1/d}\cdot\eta_\varepsilon(\mathcal{R}^r))^{1/d} > (\log\Delta_{\mathcal{K}})^{\Omega(1)/d}$ is likely to hold for cryptographic parameters.

4.2 Smoothing parameter of kernel lattices

In this section we upper bound $\eta_\varepsilon(\Lambda^\perp(\mathbf{X}))$ for $\mathbf{X} = [\mathbf{x}_1 \mid \cdots \mid \mathbf{x}_m] \in \mathcal{R}^{r\times m}$, where $\mathbf{x}_i \leftarrow \mathcal{D}_{\mathcal{R}^r,\sigma}$. We emphasise that the lattice $\Lambda^\perp(\mathbf{X})$ of interest here is the rank-deficient $\mathcal{R}$-kernel lattice of $\mathbf{X}$ but not the full-rank q-ary lattice $\Lambda_q^\perp(\mathbf{X}) := \{\mathbf{v}\in\mathcal{R}^m, \mathbf{X}\cdot\mathbf{v} = \mathbf{0} \bmod q\}$.

4.2.1 Baseline: Exponential bound for $m > r$

We first write down an approach which gives an upper bound of $\eta_\varepsilon(\Lambda^\perp(\mathbf{X}))$ which is exponential in r. It uses the short vectors in $\mathbf{X}$ to construct $m-r$ short linearly independent kernel vectors. Via the real basis of $\mathcal{R}$ we transform them into $d \cdot (m-r)$ short linearly independent kernel vectors in $\mathbb{R}^{dm}$. This way we obtain a bound on $\lambda_{d(m-r)}(\Lambda^\perp(\mathbf{X}))$ and the smoothing parameter is exponential in r. This can be seen as extending the approach of [LSS14] to $r > 1, m > 2$. The proof also relies on ideas from [BF11b, Thm 4.2 (eprint)]. As a reminder, we use a shorthand $\eta_{\varepsilon,k} \coloneqq \sqrt{\ln(2k \cdot (1+1/\varepsilon))/\pi}$ for $\varepsilon > 0$ and $k \geq 1$.

Lemma 74. *Let $\sigma, t > 0$ and $0 < r < m$ be integers. Define a matrix distribution $\mathcal{P} = \{\mathbf{X} = [\mathbf{x}_1 \mid \cdots \mid \mathbf{x}_m] \leftarrow (\mathcal{D}_{\mathcal{R}^r,\sigma})^m \mid [\mathbf{x}_1 \mid \cdots \mid \mathbf{x}_r] \in GL_r(\mathcal{K})\}$ Then*

$$\Pr_{\mathbf{X} \leftarrow \mathcal{P}} \left(\eta_\varepsilon(\Lambda^\perp(\mathbf{X})) \leq \delta_\mathcal{K} \cdot \sqrt{rd} \cdot (\sqrt{r} \cdot \sigma \cdot t)^r \cdot \eta_{\varepsilon,(d\cdot(m-r))} \right) \geq 1 - 2me \cdot rd \cdot e^{-\pi t^2}.$$

Proof. The inverse of $\mathbf{X}_1$ can be written as $\mathbf{X}_1^{-1} = \frac{1}{\det(\mathbf{X}_1)} \cdot \mathrm{adj}(\mathbf{X}_1)$ where $\mathrm{adj}(\mathbf{X}_1)$ has entries in $\mathcal{R}$. Then $\det(\mathbf{X}_1) \cdot \mathbf{X}_1^{-1}$ is a matrix over the ring and

$$[\mathbf{X}_1 \mid \mathbf{X}_2] \cdot \begin{bmatrix} \det(\mathbf{X}_1) \cdot \mathbf{X}_1^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{m-r} \end{bmatrix} = [\det(\mathbf{X}_1) \cdot \mathbf{I}_r \mid \mathbf{X}_2].$$

The kernel of $[\det(\mathbf{X}_1) \cdot \mathbf{I}_r \mid \mathbf{X}_2]$ contains $\mathcal{K}$ linearly independent vectors $(\mathbf{x}_{r+i} \mid\mid \det(\mathbf{X}_1) \cdot \mathbf{e}_i)$ for $i \in [m-r]$. Hence, vectors $\det(\mathbf{X}_1) \cdot (\mathbf{X}_1^{-1}\mathbf{x}_{r+i} \mid\mid \mathbf{e}_i) \in \mathcal{R}^m$ for $i \in [m-r]$ are also linearly independent and in the kernel of $\mathbf{X}$. It remains to bound the norm of these vectors. Denote $\mathbf{y}_i \coloneqq \det(\mathbf{X}_1) \cdot \mathbf{X}_1^{-1}\mathbf{x}_{r+i} \in \mathcal{R}^r$ and let y_{ij} be the jth coefficient of $\mathbf{y}_i$. Then $\mathbf{X}_1 \cdot \mathbf{y}_i = \det(\mathbf{X}_1) \cdot \mathbf{x}_{r+i}$. By the Cramer's rule the only solution to this linear equation over $\mathcal{K}$ can be expressed as $y_{ij} = \det(\tilde{\mathbf{X}}_j)/\det(\mathbf{X}_1)$. Here $\tilde{\mathbf{X}}_j$ stands for the matrix $\mathbf{X}_1$ with its jth column replaced by $\det(\mathbf{X}_1) \cdot \mathbf{x}_{r+i}$. Cancelling the $\det(\mathbf{X}_1)$ factor we get $y_{ij} = \det\begin{bmatrix} \mathbf{x}_1 & \mid & \ldots \mathbf{x}_{r+i} \ldots & \mid & \mathbf{x}_r \end{bmatrix}$. Hence,

$$\begin{aligned}
\|\theta(\mathbf{y}_i)\|_\infty &= \max_{j\in[r]}(\left\|\theta(\det\begin{bmatrix} \mathbf{x}_1 & \mid & \ldots \mathbf{x}_{r+i} \ldots & \mid & \mathbf{x}_r \end{bmatrix})\right\|_\infty) \\
&= \max_{j\in[r]} \max_{k\in[d]} \det\begin{bmatrix} \sigma_k(\mathbf{x}_1) & \mid & \ldots \sigma_k(\mathbf{x}_{r+i}) \ldots & \mid & \sigma_k(\mathbf{x}_r) \end{bmatrix} \\
&\leq \max_{j\in[r]} \max_{k\in[d]} \|\sigma_k(\mathbf{x}_{r+i})\| \cdot \prod_{\ell\in[r]\setminus\{j\}} \|\sigma_k(\mathbf{x}_\ell)\| \\
&\leq \max_{j\in[r]} \left(\sqrt{r} \cdot \|\mathbf{x}_{r+i}\|_\infty \cdot \prod_{\ell\in[r]\setminus\{j\}} \sqrt{r} \cdot \|\mathbf{x}_\ell\|_\infty \right) \\
&\leq (\sqrt{r} \cdot \sigma \cdot t)^r
\end{aligned}$$

with probability at least $1-r\cdot 2edr\cdot e^{-\pi t^2}$. Here we apply the Hadamard's inequality for the first upper bound, the norm inequalities for the second and the tail bound from Lemma 25 for the last transition. Then for the Euclidean norm we get $\|\mathbf{y}_i\| \leq \sqrt{dr}\cdot(\sqrt{r}\cdot\sigma\cdot t)^r$ for all $i \in [m-r]$. This holds with probability $1 - m\cdot 2edr\cdot e^{-\pi t^2}$, since the tail bound must be true for all $\mathbf{x}_\ell$, $\ell \in [m]$.

As a result, we obtain $m-r$ vectors of bounded norm that are linearly independent over $\mathcal{K}$. Multiplying by the ring basis $\mathbf{B}_\mathcal{R}$ we extend this set to $d\cdot(m-r)$ linearly independent vectors in $\theta(\Lambda^\perp(\mathbf{X})) \subset \mathbb{R}^{dm}$ with norm bounded by $\delta_\mathcal{K}\cdot\sqrt{rd}\cdot(\sqrt{r}\cdot\sigma\cdot t)^r$. Applying a classic bound on smoothing from Lemma 20, we get the statement.

□

4.2.2 Polynomial bound for $m \geq \Omega(\log(\varepsilon^{-1})/\log(\mathfrak{f}))$ in extensions of $\mathfrak{f}$th cyclotomic field

When $r \geq \omega(1)$, which is typically the case when the ring degree d is small, the strategy in Section 4.2.1 leads to a superpolynomial bound on $\lambda_{d(m-r)}(\Lambda^\perp(\mathbf{X}))$ which is often not usable in applications. For a more useful bound we turn to the approach of [AR16] which focuses on the setting $\mathcal{K} = \mathbb{Q}$. We extend their strategy to any number field $\mathcal{K} \supseteq \mathbb{Q}(\zeta_\mathfrak{f})$ containing the $\mathfrak{f}$th cyclotomic field where the quality of the obtained bound depends on $\mathfrak{f}$. For readability, we chose to name all of the implied absolute constants of the theorems and lemmas of this subsection C. We highlight that those refer to possibly different constants. We detail the results starting with a few supporting statements.

Proposition 2 (Gaussian weight of a symmetric set.)**.** *Let $S \subset \mathcal{R}^r$ be a set such that $S = -S$ and $\sigma > 0$. Then for any $\mathbf{c} \in \mathcal{R}^r$*

$$\rho_\sigma(\mathbf{c}+S) \geq \rho_\sigma(\mathbf{c})\cdot\rho_\sigma(S)$$

Proof. For any $\mathbf{s} \in S$, it holds that $\rho_\sigma(\mathbf{c}+\mathbf{s}) + \rho_\sigma(\mathbf{c}-\mathbf{s}) = \rho_\sigma(\mathbf{s})\cdot\rho_\sigma(\mathbf{c})\cdot(\rho_\sigma(-2\mathbf{s}^\mathsf{T}\mathbf{c}) + \rho_\sigma(2\mathbf{s}^\mathsf{T}\mathbf{c})) = 2\rho_\sigma(\mathbf{s})\cdot\rho_\sigma(\mathbf{c})\cdot\cosh(\pi\cdot\mathbf{s}^\mathsf{T}\mathbf{c}/\sigma^2) = (\rho_\sigma(\mathbf{s})+\rho_\sigma(-\mathbf{s}))\cdot\rho_\sigma(\mathbf{c})\cdot\cosh(\pi\cdot\sigma^{-2}\cdot\mathbf{s}^\mathsf{T}\mathbf{c}) \geq (\rho_\sigma(\mathbf{s})+\rho_\sigma(-\mathbf{s}))\cdot\rho_\sigma(\mathbf{c})$ and then by adding over all pairs $-\mathbf{s}, \mathbf{s}$ in S, the result follows. □

Proposition 3 (Counting lattice points.)**.** *Let $\Lambda \subset \mathbb{R}^n$ be a full rank lattice. Let $R \geq \sqrt{n}\cdot\eta_{1/2}(\Lambda)$ be a real number and $B(0,R) \coloneqq \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq R\}$. Then*

$$|\Lambda \cap B(0,R)| \leq 1.5\cdot\exp(\pi\cdot n)\cdot\left(\frac{R}{\sqrt{n}}\right)^n\cdot\frac{1}{\det(\Lambda)}.$$

Proof. Let $\sigma = R/\sqrt{n}$. By Corollary 6, it holds that

$$\rho_\sigma(\Lambda \cap B(0,R)) \leq \rho_\sigma(\Lambda) \leq 1.5\frac{\sigma^n}{\det \Lambda}.$$

Then, since we have $\min_{\mathbf{x}\in\Lambda\cap B(0,R)} \rho_\sigma(\mathbf{x}) \cdot |\Lambda \bigcap B(0,R)| \leq \rho_\sigma(\Lambda \bigcap B(0,R))$ and $\min_{\mathbf{x}\in\Lambda\cap B(0,R)} \rho_\sigma(\mathbf{x}) \leq \exp(-\pi R^2/\sigma^2) \leq \exp(-\pi n)$, we get the statement. □

Now we are ready to proceed to the main theorem.

Theorem 7. *Let $\mathcal{K}$ be a number field of degree d containing the cyclotomic field of conductor $\mathfrak{f}$. Let $\sigma \geq \max(\eta_{1/2}(\mathcal{R}^r), \sqrt{2\pi d})$, and $\mathbf{X} \in \mathcal{R}^{r\times m}$ be a matrix with columns sampled from $\mathcal{D}_{\mathcal{R}^r,\sigma}$ for some $m \geq 1$. Let $\varepsilon > 0$, and $T_{\mathcal{K},r,\sigma} = r \cdot \log(dr \cdot \Delta_{\mathcal{K}}^{1/(2d)} \cdot \sigma)$. There exists an absolute constant $C > 2$ such that for*

$$m \geq C \cdot T_{\mathcal{K},r,\sigma} \cdot \log(T_{\mathcal{K},r,\sigma}) + \frac{2\log(r/\varepsilon)}{\log(\mathfrak{f}/1.8)},$$

we have with probability $\geq 1-\varepsilon$ that

$$\eta_\varepsilon(\Lambda^\perp(\mathbf{X})) = O\left(\eta_{\varepsilon,((m-r)d)} \cdot m^{1.5} \cdot d^{12} \cdot \delta_{\mathcal{K}}^{14} \cdot \mathfrak{f}^2 \cdot \Delta_{\mathcal{K}}^{4/d} \cdot \sigma\right).$$

To prove Theorem 7, we state in Lemma 75 that for a large enough set $A \subseteq \mathcal{R}$ there exists a preimage of a small multiple of the ith unit vector $\mathbf{e}_i \in \mathcal{R}^r$ for any i. This almost generalises Lemma 4.2 in [AR16] to the ring setting except that instead of obtaining a preimage of $\mathbf{e}_i$ we only obtain a preimage of a small multiple of $\mathfrak{f} \cdot \mathbf{e}_i$. To not break the flow, we defer the proof of Lemma 75 to Section 4.2.3.

Definition 25. *We say that a set $A \subset \mathcal{R}$ is B-admissible if it is symmetric ($-A = A$); it contains 0 and 1; its size is at least 2^d; all of $x \in A$ satisfy $\|x\|_\infty \leq B$. If a set A is B-admissible, we define*

$$A_\Pi = \{a \cdot b \mid a, b \in A\}, \quad A_{\Sigma\Pi} = \{a + b \mid a, b \in A_\Pi\}.$$

Lemma 75. *Let $\mathcal{K} \supseteq \mathbb{Q}(\zeta_\mathfrak{f})$ be a number field of degree d containing the cyclotomic field of conductor $\mathfrak{f}$. Let $\sigma \geq \max(\eta_{1/2}(\mathcal{R}^r), \sqrt{2\pi d})$ and let $\mathbf{X} \in \mathcal{R}^{r\times m}$ be a matrix with columns sampled from $\mathcal{D}_{\mathcal{R}^r,\sigma}$ for some $m \geq 1$.*

Let $\varepsilon > 0$, $1 \leq i \leq r$, and $A \subset \mathcal{R}$ be B-admissible set for some $B > 0$. Let $T = T_{\mathcal{K},r,B,\sigma} = r \cdot \log(r \cdot B \cdot \sigma/\Delta_K^{1/(2d)})$, then there exists an absolute constant $C > 2$ such that for any

$$m \geq C \cdot T \cdot \log(T) + \frac{2\log(1/\varepsilon)}{\log(\mathfrak{f}/1.8)},$$

there exists $\mathbf{c} \in A_{\Sigma\Pi}^m, u \in \mathcal{R}$ with $\|u\|_\infty \leq \mathfrak{f}^2/2$ and $a \in A_\Pi$ such that $\mathbf{X} \cdot u \cdot \mathbf{c} = \mathfrak{f} \cdot a \cdot \mathbf{e}_i$ with probability $\geq 1-\varepsilon$ over the randomness of $\mathbf{X}$.

Lemma 75 gives a probabilistic guarantee for the existence of a short preimage of $\mathfrak{f} \cdot a \cdot \mathbf{e}_i$ for some short $a \in A_\Pi$ defined relative to a set A. To obtain preimages of $\mathfrak{f} \cdot \mathbf{e}_i$, our strategy is to invoke Lemma 75 again on a different B'-admissible set A' whose elements are coprime with a. This gives a short preimage of $\mathfrak{f} \cdot a' \cdot \mathbf{e}_i$ for a short $a' \in A'_\Pi$. We then use the co-primality of a and a' and an effective version of Bezout's identity (Lemma 42) to create a preimage of $\mathfrak{f} \cdot \mathbf{e}_i$. We discuss why preimages of $\mathfrak{f} \cdot \mathbf{I}_r$ are sufficient in the statement on the smoothing parameter in Lemma 78. We did not find a formula for the size of the set A' in the literature, so we provide one in Lemma 76, which we think might have application beyond this work.

Lemma 76. *Let $\mathcal{K}$ be a number field of degree d with ring of integer $\mathcal{R}$. Let $\mathfrak{a}$ be an integral ideal of $\mathcal{R}$ with prime factorisation $\mathfrak{a} = \prod_{i=1}^{l} \mathfrak{p}_i^{e_i}$ and $\varepsilon = \min(2^{-d}, 3^{-l}/10)$. Then for*

$$R \geq \frac{\mathcal{N}(\mathfrak{a})^{1/d} \cdot \Delta_{\mathcal{K}}^{1/d} \cdot \ln(1/\varepsilon^{1/d})}{2\sqrt{d}},$$

we have

$$\left| \left\{ x \in \mathcal{R},\ \|x\| \leq R,\ \langle x \rangle \text{ coprime with } \mathfrak{a} \right\} \right| \geq \frac{R^d}{4\sqrt{\Delta_{\mathcal{K}}} \cdot d^{d/2} \cdot 2^{d+l}}.$$

Proof. Let D denote the Gaussian distribution over $\mathcal{R}$ with parameter $\sigma \geq \mathcal{N}(a)^{1/d} \cdot \Delta_{\mathcal{K}}^{1/d} \cdot \ln(1/\varepsilon)/d$ tail-cut for $\|x\| \geq R := 2 \cdot \sqrt{d} \cdot \sigma$. By Lemma 23 and [MR07, Lemmas 3.3 and 4.4] and some computations [FPS22, Lemma 2.7], it holds that for any ideal $\mathfrak{a}$ such that $\sigma \geq \eta_\varepsilon(\mathfrak{a})$, $D(\mathfrak{a}) \in [1 - 5\varepsilon, 1 + 5\varepsilon] \cdot 1/\mathcal{N}(\mathfrak{a})$.

We write $\mathfrak{a} = \prod_{i=1}^{l} \mathfrak{p}_i^{e_i}$. An element $x \in \mathcal{R}$ is coprime with $\mathfrak{a}$ if and only if it does not belong to any of the $\mathfrak{p}_i$. We now compute the probability of sampling an element coprime to $\mathfrak{a}$ from D. By the inclusion-exclusion principle and the fact that for any prime ideals $\mathfrak{p}, \mathfrak{q}$, $\mathfrak{p} \cap \mathfrak{q} = \mathfrak{p}\mathfrak{q}$, it holds that

$$\begin{aligned}
D(\mathcal{R} \setminus \bigcup_{i=1}^{l} \mathfrak{p}_i) &= \sum_{k=0}^{l} (-1)^k \sum_{|I|=k} D(\prod_{i \in I} \mathfrak{p}_i) \geq \sum_{k=0}^{l} (-1)^k \sum_{|I|=k} \frac{1 - (-1)^k \cdot 5\varepsilon}{\mathcal{N}(\prod_{i \in I} \mathfrak{p}_i)} \\
&= \sum_{k=0}^{l} (-1)^k \sum_{|I|=k} \frac{1}{\mathcal{N}(\prod_{i \in I} \mathfrak{p}_i)} - 5\varepsilon \sum_{k=0}^{l} \sum_{|I|=k} \frac{1}{\mathcal{N}(\prod_{i \in I} \mathfrak{p}_i)} \\
&= \prod_{i=1}^{l} \left(1 - \frac{1}{\mathcal{N}(\mathfrak{p}_i)}\right) - 5\varepsilon \cdot \prod_{i=1}^{l} \left(1 + \frac{1}{\mathcal{N}(\mathfrak{p}_i)}\right)
\end{aligned}$$

Setting $\varepsilon = \min(2^{-d}, 0.1 \cdot \prod_i (\mathcal{N}(\mathfrak{p}_i) - 1)/(\mathcal{N}(\mathfrak{p}_i) + 1))$, we get that $D(\mathcal{R} \setminus \bigcup_{i=1}^{l} \mathfrak{p}_i) \geq 0.5 \cdot \prod_i (1 - 1/\mathcal{N}(\mathfrak{p}_i))$. To make this into a counting argument it suffices to say that the

maximal value of $D(x)$ is equal to $D(0) \leq (1+5\varepsilon)\cdot\sqrt{\Delta_\mathcal{K}}/\sigma^d \leq 2\sqrt{\Delta_\mathcal{K}}/\sigma^d$. Then the number of ring elements of ℓ_2 norm less than R coprime with $\mathfrak{a}$ is greater than

$$\frac{D(\mathcal{R}\setminus\bigcup_{i=1}^{l}\mathfrak{p}_i)}{D(0)} \geq \frac{\prod_i(1-1/\mathcal{N}(\mathfrak{p}_i))\cdot\sigma^d}{4\sqrt{\Delta_\mathcal{K}}} = \frac{\prod_i(1-1/\mathcal{N}(\mathfrak{p}_i))\cdot R^d}{4\sqrt{\Delta_\mathcal{K}}\cdot d^{d/2}\cdot 2^d}.$$

Note that since $\mathcal{N}(\mathfrak{p}_i)\geq 2$ for all i, then $\prod_i(1-1/\mathcal{N}(\mathfrak{p}_i)) \geq 2^{-l}$ and $\prod_i(\mathcal{N}(\mathfrak{p}_i)-1)/(\mathcal{N}(\mathfrak{p}_i)+1) \geq 3^{-l}$, which concludes the proof. □

Lemma 77. *Let $\mathcal{K}\supseteq\mathbb{Q}(\zeta_\mathfrak{f})$ be a number field of degree d containing the cyclotomic field of conductor $\mathfrak{f}$. Let $\sigma \geq \max(\eta_{1/2}(\mathcal{R}^r), \sqrt{2\pi d})$ and $\mathbf{X}\in\mathcal{R}^{r\times m}$ be a matrix with columns sampled from $\mathcal{D}_{\mathcal{R}^r,\sigma}$ for some $m\geq 1$.*

Let $\varepsilon>0$, and $T_{\mathcal{K},r,\sigma} = r\cdot\log(dr\cdot\delta_K\cdot\Delta_\mathcal{K}^{1/(2d)}\cdot\sigma)$. There exists an absolute constant $C>2$ such that for

$$m \geq C\cdot T_{\mathcal{K},r,\sigma}\cdot\log(T_{\mathcal{K},r,\sigma}) + \frac{2\log(r/\varepsilon)}{\log(\mathfrak{f}/1.8)},$$

there exists a matrix $\mathbf{U}\in\mathcal{R}^{m\times r}$ with $\|\mathbf{U}\| \leq O(\sqrt{md}\cdot\mathfrak{f}^2\cdot\Delta_\mathcal{K}^{4/d}\cdot d^{10.5}\cdot\delta_\mathcal{K}^{13})$ and $\mathbf{X}\cdot\mathbf{U} = \mathfrak{f}\cdot\mathbf{I}_r$, with probability $\geq 1-\varepsilon$ over the randomness of $\mathbf{X}$.

Proof. Let $A = \mathbf{B}_\mathcal{R}\cdot\{\pm1,0\}^d \subset \mathcal{R}$, which is bounded in infinity norm by $d\cdot\delta_\mathcal{K}$, symmetric and contains 3^d elements. As long as m is large enough by Lemma 75, there exists a short preimage to a short multiple of $\mathfrak{f}\cdot\mathbf{e}_1$ with probability $1-\varepsilon/r$. Namely, there exists $\mathbf{c}\in A_{\Sigma\Pi}^m, u\in\mathcal{R}$ with $\|u\|_\infty \leq \mathfrak{f}^2/2$, $\|\mathbf{c}\|_\infty \leq 2d^2\delta_\mathcal{K}^2$ and $a'\in A_\Pi$ such that $\mathbf{X}\cdot u\cdot\mathbf{c} = \mathfrak{f}\cdot a'\cdot\mathbf{e}_1$. Note that by construction $\|a'\|_\infty \leq d^2\delta_\mathcal{K}^2$ and that a' belongs to a maximum of $l = \log_2(\mathcal{N}(a')) \leq d\log(d^2\delta_\mathcal{K}^2)$ different prime ideals.

Let $\widetilde{A}$ be the set of elements of $\mathcal{R}$ which are coprime with a' of norm bounded by $R = 4\cdot\Delta_\mathcal{K}^{1/d}\cdot d^{2.5}\cdot\delta_\mathcal{K}^3$. One can check that R follows the hypothesis of Lemma 76, and that in that case $\left|\widetilde{A}\right| \geq 2^d$. The set $\widetilde{A}$ is then R-admissible. Then by Lemma 75 there exists $\widetilde{\mathbf{c}}\in\widetilde{A_{\Sigma\Pi}}^m, \widetilde{u}\in\mathcal{R}$ with $\|\widetilde{u}\|_\infty \leq \mathfrak{f}^2/2$, $\|\widetilde{\mathbf{c}}\|_\infty \leq 2R^2$ and $\widetilde{a'}\in\widetilde{A_\Pi}$, $||\widetilde{a'}||_\infty \leq R^2$ such that $\mathbf{X}\cdot\widetilde{u}\,\widetilde{\mathbf{c}} = \mathfrak{f}\cdot\widetilde{a'}\cdot\mathbf{e}_1$ with probability $1-\varepsilon/(2r)$.

By construction, a' and $\widetilde{a'}$ are coprime, then by Lemma 42 there exist bounded norm $\alpha,\beta\in\mathcal{R}$ with $\|\alpha\|_\infty, \|\beta\|_\infty \leq R^2\cdot\sqrt{d}\cdot\lambda_d(\mathcal{R})$ such that $a'\alpha + \widetilde{a'}\beta = 1$, and finally it holds that

$$\mathbf{X}\cdot(\alpha\cdot u\cdot\mathbf{c} + \beta\cdot\widetilde{u}\cdot\widetilde{\mathbf{c}}) = \mathfrak{f}\cdot\mathbf{e}_1.$$

We name $\mathbf{u}_1 := \alpha\cdot u\cdot\mathbf{c} + \beta\cdot\widetilde{u}\cdot\widetilde{\mathbf{c}}$. It holds that $\|\mathbf{u}_1\|_\infty \leq 2\sqrt{d}\cdot\mathfrak{f}^2\cdot\lambda_d(\mathcal{R})\cdot R^4$. Running the same argument r times for all the $\mathbf{e}_j$ and using the union bound, we get that with probability $1-\varepsilon$, we can construct $\mathbf{U} = [\mathbf{u}_1,\ldots,\mathbf{u}_r]$ such that $\mathbf{X}\cdot\mathbf{U} = \mathfrak{f}\cdot\mathbf{I}_r$ and

$\|\mathbf{U}\| \leq O(\sqrt{md} \cdot \mathfrak{f}^2 \cdot \Delta_{\mathcal{K}}^{4/d} \cdot d^{10.5} \cdot \delta_{\mathcal{K}}^{13})$. The choice of lower bound on m follows from the parameters of Lemma 75 when we used it during the proof. □

Corollary 17 (Cyclotomics version of Theorem 7)**.** *Let $\mathfrak{f} > 2$ and $\mathcal{K} = \mathbb{Q}(\zeta_{p^k})$ be the cyclotomic number field of conductor $\mathfrak{f}$ and degree d. Let $1 \leq r \leq m$, $\varepsilon > 0$ and $T = r \cdot \log(r \cdot d^{3/2} \cdot \sigma)$. Let $\sigma > 0$, and $\mathbf{X} \in \mathcal{R}^{r \times m}$ be a matrix with columns sampled from $\mathcal{D}_{\mathcal{R}^r,\sigma}$ and $\mathbf{c} \in \mathcal{R}^m$. There exists an absolute constant $C > 2$ such that for*

$$\sigma \geq \max(\eta_{1/2}(\mathcal{R}^r), \sqrt{2\pi d}), \qquad m \geq C \cdot T \cdot \log(T) + \frac{2\log(r/\varepsilon)}{\log(p^k/1.8)},$$

with probability $\geq 1 - \varepsilon$ we have

$$\eta_\varepsilon(\Lambda^\perp(\mathbf{X})) = O\left(\eta_{\varepsilon,((m-r)d)} \cdot m^{1.5} \cdot \mathfrak{f}^2 \cdot d^{16} \cdot \sigma\right).$$

We now use the short preimage computed in Lemma 77 to compute a short basis of $\Lambda^\perp(\mathbf{X})$, and then an upper bound on the smoothing parameter.

Lemma 78. *Let $\mathbf{U} \in \mathcal{R}^{m \times r}$ such that $\mathbf{X} \cdot \mathbf{U} = \mathfrak{f} \cdot \mathbf{I}_r$, then there exists a basis $\mathbf{B} \in \mathbb{R}^{dm \times d(m-r)}$ of $\Phi(\Lambda^\perp(\mathbf{X}))$ such that*

$$\|\mathbf{B}\| \leq \sqrt{(m-r)\,d} \cdot \delta_{\mathcal{K}} \cdot (\|\mathbf{U}\| \cdot \|\mathbf{X}\| + \mathfrak{f}),$$

in particular, for any $\varepsilon > 0$, it holds that

$$\eta_\varepsilon(\Lambda^\perp) \leq \eta_{\varepsilon,((m-r)\,d)} \cdot \sqrt{(m-r)\,d} \cdot \delta_{\mathcal{K}} \cdot (\|\mathbf{U}\| \cdot \|\mathbf{X}\| + \mathfrak{f}).$$

Proof. One can check that the column vectors of matrix $\mathbf{Y} = \mathfrak{f} \cdot \mathbf{I}_m - \mathbf{U} \cdot \mathbf{X} \in \mathcal{R}^{m \times m}$ belong to $\Lambda^\perp(\mathbf{X})$ and that their norm is bounded by $\|\mathbf{U}\| \cdot \|\mathbf{X}\| + \mathfrak{f}$. Since $\mathbf{U} \cdot \mathbf{X}$ is of rank at most r (over $\mathcal{K}$), the rank of $\mathbf{Y}$ is at least $m - r$. Hence, $\mathbf{Y}$ contains a free subset of size $m - r$ generating $\mathrm{span}_{\mathcal{K}}(\Lambda^\perp(\mathbf{X}))$. Then Lemma 40 implies the existence of a basis $\mathbf{B}$ with the corresponding norm bound. Lastly, we get a bound on the smoothing parameter by noting that $\lambda_{(m-r)d}(\Lambda^\perp(\mathbf{X})) \leq \|\mathbf{B}\|$. □

4.2.3 Proof of Lemma 75

In this subsection, we fix $\mathcal{K} \supseteq \mathbb{Q}(\zeta_{\mathfrak{f}})$ be a number field of degree d containing the cyclotomic field of conductor $\mathfrak{f}$, $\sigma \geq \max(\sqrt{2\pi d}, \eta_{1/2}(\mathcal{R}^r))$, and $\mathbf{X} \in \mathcal{R}^{r \times m}$ a matrix with columns sampled from $\mathcal{D}_{\mathcal{R}^r,\sigma}$ for some $m \geq 1$.

It remains to prove Lemma 75. Without loss of generality we take $i = 1$. We fix a B-admissible set A and define two set sequences, for $1 \leq j \leq m$:

$$S_j = \left\{ \sum_{i=1}^{j} a_i \cdot \mathbf{x}_i,\ a_i \in A \right\}, \quad \hat{S}_j = \{\mathbf{s}/a,\ \mathbf{s} \in S_j, a \in A \setminus \{0\}\}$$

For $1 \leq j \leq m$ we define the following events:

$$\begin{aligned} \texttt{Win}_j &: \text{Two sets in } \left\{\hat{S}_j, \hat{S}_j + \zeta_{\mathfrak{f}}^x \cdot \mathbf{e_1} \mid x \in [\mathfrak{f}]\right\} \text{ have non-empty intersection,} \\ \texttt{Gain}_j &: \mathbf{x}_{j+1} \notin \hat{S}_j \text{ and } \|\mathbf{x}_{j+1}\| \leq \sqrt{dr} \cdot \sigma. \end{aligned}$$

The following Lemma 79 shows that the event $\texttt{Win}_j$ implies the existence of a short preimage of a multiple of $\mathbf{e}_1$.

Lemma 79. *Let $j \geq 1$. If $\texttt{Win}_j$ is true, then there exists $\mathbf{c} \in (A_{\Sigma\Pi})^j$, $u \in \mathcal{R}$ with $\|u\|_\infty \leq \mathfrak{f}^2/2$ and $a' \in A_\Pi$ such that $\mathbf{X}_{[1:j]} \cdot u \cdot \mathbf{c} = \mathfrak{f} \cdot a' \cdot \mathbf{e}_1$, where A_Π and $A_{\Sigma\Pi}$ are defined as in Lemma 75.*

Proof. Since $\texttt{Win}_j$ is true, two cases can happen: (1) $(\hat{S}_j + \zeta_{\mathfrak{f}}^x \cdot \mathbf{e_1}) \cap (\hat{S}_j + \zeta_{\mathfrak{f}}^y \cdot \mathbf{e_1}) \neq \varnothing$ for some $x, y \in [\mathfrak{f}]$; (2) $\hat{S}_j \cap (\hat{S}_j + \zeta_{\mathfrak{f}}^y \cdot \mathbf{e_1}) \neq \varnothing$ for some $y \in [\mathfrak{f}]$. We focus on Case (1) and note that Case (2) can be treated similarly. In Case (1), there exist $\mathbf{a}_1, \mathbf{a}_2 \in A^j$ and $b_1, b_2 \in A$ such that $\mathbf{X} \cdot (\mathbf{a}_1/b_1 - \mathbf{a}_2/b_2) = (\zeta_{\mathfrak{f}}^x - \zeta_{\mathfrak{f}}^y) \cdot \mathbf{e}_1$. Multiplying the previous equality by $b_1 \cdot b_2 \cdot \mathfrak{f}/(\zeta_{\mathfrak{f}}^x - \zeta_{\mathfrak{f}}^y)$ and applying Lemma 7 gives the result. □

Lemma 80 ([AR16, Clm. 4.1 adapted])**.** *There exists an absolute constant $c_1 > 1$ such that, for any sequence $1 \leq i_1 < \ldots < i_k$ of integers of length $k \geq c_1 \cdot r \log(r \cdot B \cdot \sigma/\Delta_K^{1/(2d)})$, if $\texttt{Gain}_{i_l}$ is true for $l \in [1, k+1]$ then $\texttt{Win}_{i_k}$ is true.*

Proof. Assume that $\texttt{Win}_{i_k}$ is not true, we prove the statement by contradiction. Without loss of generality, we also remove all $\mathbf{x}_j$ for which $\texttt{Gain}_j$ is false, and consider $i_l = l$ for $l \in [1, k+1]$. For all $l \in [1, k+1]$ and $a \in A$, we have that $a \cdot \mathbf{x}_{l+1} \notin S_l$, since else $\mathbf{x}_{l+1} \in \hat{S}_l$ which is forbidden by $\texttt{Gain}_l$. This implies that the size of S_l is at least $|A|^l \geq 2^{dl}$. Then, since the sets $\hat{S}_j$ are non-intersecting, we have $|S_l| + \sum_a \left|S_l + \zeta_{\mathfrak{f}}^a \cdot \mathbf{e}_1\right| \geq (\mathfrak{f}+1)2^{dl}$.

Furthermore, note that every element of S_l and $S_l + \zeta_{\mathfrak{f}}^a \cdot \mathbf{e}_1$ has norm bounded by $2 \cdot \sqrt{rd} \cdot B \cdot \sigma \cdot l$. By Proposition 3 and the condition on σ, the set $\mathcal{R}^r \cap B(0, C)$ is of size at most $1.5 \cdot (2\exp(\pi)B \cdot l \cdot \sigma/\Delta_K^{1/(2d)})^{rd} \leq C^{rd}$ for $C = O(B \cdot l \cdot \sigma/\Delta_K^{1/(2d)})$. This implies that for k large enough we have $(\mathfrak{f}+1)2^{dk} > C^{rd}$, and a pair of sets in $\left\{S_j, S_j + \zeta_{\mathfrak{f}}^a \cdot \mathbf{e_1} \mid a \in [\mathfrak{f}]\right\}$ must have a non-empty intersection by the pigeonhole principle.

This means that $\texttt{Win}_k$ is true, and we have a contradiction. Using Lemma 1, it can be computed that there exists an absolute $c' > 1$ such that $(\mathfrak{f}+1)\cdot 2^{dk} > C^{rd}$ holds for any $k \geq c' \cdot r\log(r \cdot B \cdot \sigma/\Delta_K^{1/(2d)})$. □

We now show that as long as $\texttt{Win}_j$ is false, the event $\texttt{Gain}_j$ happens with high probability.

Lemma 81. *Let $\sigma \geq \sqrt{2\pi d}$ and $j \geq 1$. It holds that* $\Pr(\texttt{Gain}_j \mid \neg\texttt{Win}_j) \geq 1 - 1.8/\mathfrak{f}$.

Proof. For clarity, we omit the "$\mid \neg\texttt{Win}_j$" for the probabilities in this proof. Note that

$$\Pr(\neg\texttt{Gain}_j) = \frac{\rho_\sigma(\hat{S}_j \cup \mathcal{R}^r \setminus B(0,R))}{\rho_\sigma(\mathcal{R}^r)} \leq \frac{\rho_\sigma(\hat{S}_j)}{\rho_\sigma(\mathcal{R}^r)} + \beta_{dr}(\sqrt{2\pi d}),$$

where $R = \sqrt{dr}\sigma$ and the inequality follows from Corollary 2. One can prove that it holds that $\beta_{rd}(\sqrt{2\pi d}) \leq 1/(10d)$ for any $d \geq 2$, $r \geq 1$. For any $i \in [1, \mathfrak{f}]$, by Proposition 2 and the bound on σ, we have that $\rho_\sigma(\hat{S}_j + \zeta_\mathfrak{f}^i \mathbf{e}_1) \geq \rho_\sigma(\hat{S}_j)/\sqrt{e}$. Now since $\texttt{Win}_j$ is false, it holds that $\rho_\sigma(\hat{S}_j) + \sum_i \rho_\sigma(\hat{S}_j + \zeta_\mathfrak{f}^i \mathbf{e}_1) \leq \rho_\sigma(\mathcal{R}^r)$, and then $\rho_\sigma(\hat{S}_j)/\rho_\sigma(\mathcal{R}^r) \leq 1/(1+\mathfrak{f}/\sqrt{e}) \leq \sqrt{e}/\mathfrak{f}$. Finally, $\Pr(\neg\texttt{Gain}_j) \leq 1.8/\mathfrak{f}$ □

Lemma 82. *Let $Y_1, Y_2, \ldots$ be a sequence of not necessarily independent random Boolean variables such that for any $i \geq 1$ and any $\{y_1, \ldots y_{i-1}\} \in \{0,1\}^{i-1}$ it holds that*

$$\Pr(Y_i = 1 \mid Y_{i-1} = y_{i-1}, \ldots Y_1 = y_1) \geq 1 - \alpha \tag{4.1}$$

for some $\alpha \in (0, 2/3)$. Let $T \geq 3$ be a target value. Then exists an absolute constant $c_2 > 2$ such that for any $\varepsilon \in (0, 1/2)$ and any

$$m \geq c_2 \cdot T\log(T) + 2 \cdot \frac{\log(1/\varepsilon)}{\log(1/\alpha)},$$

it holds that

$$\Pr(\sum_{i=1}^{m} Y_i \leq T) \leq \varepsilon.$$

Furthermore, if we assume $T \geq 13$ and $\alpha \leq 0.1$, one can choose $c_2 = 4$.

Lemma 83. *Let $Y_1, Y_2, \ldots$ be a sequence of not necessarily independent random Boolean variables such that for any $i \geq 1$ and any $\{y_1, \ldots y_{i-1}\} \in \{0,1\}^{i-1}$ it holds that*

$$\Pr(Y_i = 1 \mid Y_{i-1} = y_{i-1}, \ldots Y_1 = y_1) \geq 1 - \alpha$$

for some $\alpha \in (0,1)$. Let $\tau \in \{0,1\}^m$ and $|\tau|$ its hamming weight. It holds that

$$\Pr((Y_1, \ldots, Y_m) = \tau) \leq \alpha^{m-|\tau|}.$$

Proof. It holds that

$$\Pr((Y_1,\ldots,Y_m)=\tau)=\prod_{i=1}^{m}\Pr(Y_i=\tau_i|(Y_1,\ldots,Y_{i-1})=\tau_{[1:i-1]}),$$

we upper-bound each $\Pr(Y_i=\tau_i|(Y_1,\ldots,Y_{i-1})=\tau_{[1:i-1]})$ by 1 if $\tau_i=1$, and by α if $\tau_i=0$, giving the result. □

Proof. (of Lemma 82) By Lemma 83, it holds that

$$\begin{aligned}\Pr(\sum_{i=1}^{m}Y_i\le T)=&\sum_{\substack{\tau\in\{0,1\}^m\\|\tau|\le T}}\Pr((Y_1,\ldots,Y_m)=\tau)\\ \le&\sum_{k=0}^{T}\binom{m}{k}\alpha^{m-k}\le\alpha^{m-T}\cdot\binom{m}{T}\cdot(T+1)\\ \le&\alpha^{m-T}\cdot\frac{e^T\cdot(T+1)\cdot m^T}{T^T}\\ \le&7\cdot\alpha^{m-T}\cdot m^T,\end{aligned}$$

where we used the fact that $T\le m/2$, by assumption. Now, note that

$$7\cdot\alpha^{m-T}\cdot m^T\le\varepsilon\Leftrightarrow m\ge T+\frac{\log(1/\varepsilon)+\log(7)}{\log(1/\alpha)}+\frac{T}{\log(1/\alpha)}\cdot\log(m),$$

which, by Lemma 1 holds for any

$$m\ge 2\cdot\left(T+\frac{\log(1/\varepsilon)+\log(7)}{\log(1/\alpha)}+\frac{T}{\log(1/\alpha)}\cdot\log\left(\frac{T}{\log(1/\alpha)}\right)\right),$$

a proper analysis proves that for the range of parameters $T\ge 13$ and $\alpha\le 0.1$, the above bound is less than

$$4T\log(T)+\frac{2\log(1/\varepsilon)}{\log(1/\alpha)},$$

which concludes the proof. □

With all of these result, we can now prove Lemma 75.

Proof. (of Lemma 75) Combining Lemmas 79, 80 and 82 with $\alpha=1.8/\mathfrak{f}$ ($\mathfrak{f}\ge 2$ since $\zeta_2=-1\in K$) by Lemma 81 proves Lemma 75. □

4.3 Gaussian LHL over rings

We combine the arguments above to obtain a Gaussian Leftover Hash Lemma.

Theorem 8 (Gaussian LHL, Assuming GRH)**.** *Let $\mathcal{K}$ be a number field of degree $d \geq 2$ containing the cyclotomic field of conductor $\mathfrak{f}$ and of smallest ideal norm $N_{\mathcal{K}}$. Let $\mathbf{c} \in \mathcal{R}^m$, $\sigma \in \mathbb{R}^{dr \times dr}$ positive definite, $1 \leq r \leq m$, $0 < \varepsilon < 2^{-6}$, and $T = T_{\mathcal{K},r,\sigma} = r \cdot \log(dr \cdot \Delta_{\mathcal{K}}^{1/(2d)} \cdot \sigma)$. There exists an absolute constant $C > 2$ such that for*

$$\sigma \geq \max\Bigg(\underbrace{\sqrt{2\pi d}}_{\text{for full image}}, \underbrace{9 \cdot r^{\frac{1}{d}} \cdot \log\left(r \cdot \Delta_K^{\frac{1}{d}} \cdot \eta_\varepsilon(\mathcal{R}^r)\right)^{\frac{1}{d}} \cdot \Delta_{\mathcal{K}}^{\frac{1}{d}} \cdot \eta_\varepsilon(\mathcal{R}^r)}_{\text{for the smoothing bound}} \Bigg),$$

$$m \geq \max\left(\underbrace{2r + \frac{\log(1/\varepsilon) + 1 + \log(2d)}{\log(N_{\mathcal{K}}/1.4)}}_{\text{for full image}}, \underbrace{C \cdot T \cdot \log(T) + \frac{2\log(r/\varepsilon)}{\log(\mathfrak{f}/1.8)}}_{\text{for the smoothing bound}} \right),$$

$$\sqrt{\boldsymbol{\Sigma}} \in \omega\left(\eta_{\varepsilon,((m-r)d)} \cdot m^{1.5} \cdot d^{12} \cdot \delta_{\mathcal{K}}^{14} \cdot \mathfrak{f}^2 \cdot \Delta_{\mathcal{K}}^{4/d} \cdot \sigma\right),$$

it holds

$$\mathrm{SD}((\mathbf{X} \leftarrow (\mathcal{D}_{\mathcal{R}^r,\sigma})^m, \mathbf{X} \cdot \mathcal{D}_{\mathcal{R}^m,\sqrt{\boldsymbol{\Sigma}},\mathbf{c}}), (\mathbf{X} \leftarrow (\mathcal{D}_{\mathcal{R}^r,\sigma})^m, \mathcal{D}_{\mathcal{R}^r,\sqrt{\boldsymbol{\Sigma}'},\mathbf{X}\cdot\mathbf{c}})) \leq \varepsilon'$$

where $\boldsymbol{\Sigma}' = \tilde{\theta}(\mathbf{X}) \cdot \boldsymbol{\Sigma} \cdot \tilde{\theta}(\mathbf{X})^{\mathsf{T}}$, $\varepsilon' = \frac{\varepsilon}{1-\varepsilon} + m \cdot (\varepsilon + 2^{-dr}) + \varepsilon$.

Proof. Denote $\mathcal{S} \subset \mathcal{R}^{r \times m}$ the set of matrices $\mathbf{X}$ satisfying $\sqrt{\boldsymbol{\Sigma}} \geq \eta_\varepsilon(\Lambda^\perp(\mathbf{X}))$ and $\mathbf{X} \cdot \mathcal{R}^m = \mathcal{R}^r$. Consider a fixed element $\mathbf{X} \in \mathcal{S}$. We verify conditions of Lemma 32 for the real lattice embedding of $\mathcal{R}^m$ and the $\tilde{\theta}(\cdot)$ embedding of $\mathbf{X}$. For matrix $\tilde{\theta}(\mathbf{X})$ as a linear map over $\mathbb{R}$ the set $\ker(\tilde{\theta}(\mathbf{X}))$ is indeed equal to $\mathrm{span}_{\mathbb{R}}(\theta(\Lambda^\perp(\mathbf{X})))$ and $\ker(\tilde{\theta}(\mathbf{X})) \cap \theta(\mathcal{R}^m) = \theta(\Lambda^\perp(\mathbf{X}))$. Hence,

$$\mathrm{SD}(\mathbf{X} \cdot \mathcal{D}_{\mathcal{R}^m,\sqrt{\boldsymbol{\Sigma}},\mathbf{c}}, \mathcal{D}_{\mathbf{X}\cdot\mathcal{R}^m,\sqrt{\boldsymbol{\Sigma}'},\mathbf{X}\cdot\mathbf{c}}) \leq \frac{\varepsilon}{1-\varepsilon}.$$

Lastly, we apply $\mathbf{X} \cdot \mathcal{R}^m = \mathcal{R}^r$ for the support. Now for the statistical distance between tuples $E \coloneqq \mathrm{SD}((\mathbf{X}, \mathbf{X} \cdot \mathcal{D}_{\mathcal{R}^m,\sqrt{\boldsymbol{\Sigma}},\mathbf{c}}), (\mathbf{X}, \mathcal{D}_{\mathcal{R}^r,\sqrt{\boldsymbol{\Sigma}'},\mathbf{X}\cdot\mathbf{c}}))$ we have

$$\begin{aligned}
E &= \mathbb{E}_{\mathbf{X}}(\mathbf{X} \cdot \mathcal{D}_{\mathcal{R}^m,\sqrt{\boldsymbol{\Sigma}},\mathbf{c}}, \mathcal{D}_{\mathbf{X}\cdot\mathcal{R}^m,\sqrt{\boldsymbol{\Sigma}'},\mathbf{X}\cdot\mathbf{c}}) \\
&\leq \Pr(\mathbf{X} \in \mathcal{S}) \cdot \max_{\mathbf{X}\in\mathcal{S}}(\mathrm{SD}(\mathbf{X} \cdot \mathcal{D}_{\mathcal{R}^m,\sqrt{\boldsymbol{\Sigma}},\mathbf{c}}, \mathcal{D}_{\mathcal{R}^r,\sqrt{\boldsymbol{\Sigma}'},\mathbf{X}\cdot\mathbf{c}})) \\
&+ \Pr(\mathbf{X} \notin \mathcal{S}) \cdot \max_{\mathbf{X}\notin\mathcal{S}}(\mathrm{SD}(\mathbf{X} \cdot \mathcal{D}_{\mathcal{R}^m,\sqrt{\boldsymbol{\Sigma}},\mathbf{c}}, \mathcal{D}_{\mathcal{R}^r,\sqrt{\boldsymbol{\Sigma}'},\mathbf{X}\cdot\mathbf{c}})) \\
&\leq 1 \cdot \frac{\varepsilon}{1-\varepsilon} + (m \cdot (\varepsilon + 2^{-dr}) + \varepsilon) \cdot 1
\end{aligned}$$

where the first equality holds by the definition of statistical distance. □

Corollary 18 (Prime-Power Cyclotomics version of Theorem 8)**.** *Let $p^k > 2$ a prime power and $\mathcal{K} = \mathbb{Q}(\zeta_{p^k})$ be the p^k-cyclotomic number field of degree $d = (p-1) \cdot p^{k-1}$. Let $1 \leq r \leq m$, $0 < \varepsilon < 2^{-6}$, and $T = r \cdot \log(r \cdot d^{3/2} \cdot \sigma)$. Let $\sigma \in \mathbb{R}^{dr \times dr}$ be positive definite, and $\mathbf{X} \in \mathcal{R}^{r \times m}$ be a matrix with columns sampled from $\mathcal{D}_{\mathcal{R}^r,\sigma}$ and $\mathbf{c} \in \mathcal{R}^m$. There exists an absolute constant $C > 2$ such that for*

$$\sigma \geq 9 \cdot r^{1+\frac{1}{d}} \cdot d^{3/2} \cdot \eta_{\varepsilon,rd} \cdot \log^{1/d}\left((rd)^{3/2} \cdot \eta_{\varepsilon,rd}\right),$$
$$m \geq \max\left(2r + \frac{\log(1/\varepsilon) + 1 + \log(2d)}{\log(p/1.4)}, C \cdot T \cdot \log(T) + \frac{2\log(r/\varepsilon)}{\log(p^k/1.8)}\right),$$

and for $\sqrt{\mathbf{\Sigma}} \in \omega\left(\eta_{\varepsilon,((m-r)d)} \cdot m^{1.5} \cdot p^{18k} \cdot \sigma\right)$, $\mathbf{\Sigma}' = \tilde{\theta}(\mathbf{X}) \cdot \mathbf{\Sigma} \cdot \tilde{\theta}(\mathbf{X})^{\mathsf{T}}$ it holds

$$\mathrm{SD}((\mathbf{X}, \mathbf{X} \cdot \mathcal{D}_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}},\mathbf{c}}), (\mathbf{X}, \mathcal{D}_{\mathcal{R}^r,\sqrt{\mathbf{\Sigma}'},\mathbf{X}\cdot\mathbf{c}})) \leq \frac{\varepsilon}{1-\varepsilon} + m \cdot (\varepsilon + 2^{-dr}) + \varepsilon$$

4.4 Spherical covariance

One way to make the distribution close to a spherical discrete Gaussian is to adapt the covariance parameter of the input vector as in the corollary below. Besides this, we set the centre of the distribution to $\mathbf{c} = 0$ to remove the dependence on $\mathbf{X}$ in the second distribution completely.

Corollary 19 (of Theorem 8)**.** *Let $1 \leq r \leq m$, and $\sigma > 0$. Following Lemma 3 and using that $\mathbf{M} = \tilde{\theta}(\mathbf{X})$ is full rank we can efficiently construct a matrix $\mathbf{S} \in \mathbb{R}^{dm \times dm}$ s.t. $\sigma^2 \cdot \mathbf{I}_{dr} = \tilde{\theta}(\mathbf{X}) \cdot \mathbf{S} \cdot \mathbf{S}^{\mathsf{T}} \cdot \tilde{\theta}(\mathbf{X})^{\mathsf{T}}$. Letting $\mathbf{\Sigma} = \mathbf{S} \cdot \mathbf{S}^{\mathsf{T}}$ and $\mathbf{c} = \mathbf{0}$ we obtain*

$$\mathrm{SD}((\mathbf{X}, \mathbf{X} \cdot \mathcal{D}_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}}}), (\mathbf{X}, \mathcal{D}_{\mathcal{R}^r,\sigma})) \leq \frac{\varepsilon}{1-\varepsilon} + m \cdot (\varepsilon + 2^{-dr}) + \varepsilon$$

Furthermore, $\sigma/\mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{X})) \leq \mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}}) \leq \mathfrak{s}_{\max}(\sqrt{\mathbf{\Sigma}}) \leq \sigma/\mathfrak{s}_{\min}(\tilde{\theta}(\mathbf{X}))$.

We also consider a different approach for removing the dependence on $\mathbf{X}$ in the case when the covariance of the input cannot be adjusted. We prove that when the number of columns m is large enough the singular values of $\mathbf{X} \cdot \mathbf{X}^{\mathsf{T}}$ are so close that the distribution has a constant Rényi Divergence to a spherical discrete Gaussian. The ideas are inspired by the arguments in [AGHS13, Lemma 8]. First we detail an adaptation of the proof of [Sil85] for continuous Gaussian matrices. To adapt it we tighten the bounds on the convergence speed of the singular values.

Lemma 84 (Continuous Gaussians. Adapted from [Sil85])**.** *Let* $\mathbf{Z} \leftarrow \mathcal{D}_\sigma^{r\times m}$ *for* $1 \le r \le m$*,* $\sigma > 0$ *and parameters* $d, k \ge 1$ *s.t* $rd \ge 4$*. If* $m \ge 49\cdot\max(r,k)\cdot(rd)^2$ *then with probability at least* $1 - 4r\cdot\exp(-k)$ *we have:*

$$m\cdot\left(1-\frac{1}{rd}\right) \le s_{\min}^2(\mathbf{Z}) \le s_{\max}^2(\mathbf{Z}) \le m\cdot\left(1+\frac{1}{rd}\right).$$

Proof. Denote the first row of $\mathbf{Z}$ as $\mathbf{z}_0 \leftarrow \mathcal{D}_\sigma^m$. Define $\mathbf{O}_0 \in \mathbb{R}^{m\times m}$ a matrix where the first column is equal to $\mathbf{z}_0/\|\mathbf{z}_0\|$ and the remaining columns are a non-random completion to an orthogonal matrix (e.g. taking the unit vectors and running the Gram-Shmidt orthogonalisation starting from $\mathbf{z}_0/\|\mathbf{z}_0\|$). Then

$$\mathbf{Z}\cdot\mathbf{O}_0 = \begin{bmatrix} X_m & 0 & \dots & 0 \\ & & \tilde{\mathbf{Z}}_0 & \end{bmatrix}$$

where $X_m \coloneqq \|\mathbf{z}_0\| \leftarrow \chi_m$. The distribution of the rows in $\tilde{\mathbf{Z}}_0$ remains independent Gaussian. This is because $\mathbf{O}_0$ is orthogonal and the Gaussian Distribution is rotationally invariant. Then we define a rotation $\mathbf{O}_1 \in \mathbb{R}^{r\times r}$ on the right in a similar way such that

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & \tilde{\mathbf{O}}_1 & \\ 0 & & & \end{bmatrix}\cdot\mathbf{Z}\cdot\mathbf{O}_0 = \begin{bmatrix} X_m & 0 & \dots & 0 \\ Y_{r-1} & & & \\ \vdots & & \tilde{\mathbf{Z}}_1 & \\ 0 & & & \end{bmatrix}$$

for $Y_{r-1} \leftarrow \chi_{r-1}$. Here the first row of $\tilde{\mathbf{O}}_1$ is a normalisation of the first column in $\tilde{\mathbf{Z}}_0$ and the rest is completed to an orthogonal matrix. We repeat the algorithm until

$$\prod_{j=0}^{r-2}\mathbf{O}_{2j+1}\cdot\mathbf{Z}\cdot\prod_{i=0}^{r-1}\mathbf{O}_{2i} = \begin{bmatrix} X_m & 0 & 0 & \dots & 0 \\ Y_{r-1} & X_{m-1} & 0 & \dots & 0 \\ \vdots & Y_{r-2} & & & \\ 0 & \vdots & & \dots & \end{bmatrix}$$

As consequence, there exists an orthogonal transformation $\mathbf{O} \in \mathbb{R}^{r\times r}$ s.t.
$\mathbf{O}\cdot\mathbf{Z}\cdot\mathbf{Z}^{\mathsf{T}}\cdot\mathbf{O}^{\mathsf{T}} =$

$$\begin{bmatrix} X_m^2 & X_mY_{r-1} & 0 & & & 0 \\ X_mY_{r-1} & X_{m-1}^2+Y_{r-1}^2 & X_{m-1}Y_{r-2} & 0 & & 0 \\ & & \dots & \dots & & \\ 0 & & & 0 & X_{m-r+2}Y_1 & X_{m-r+1}^2+Y_1^2 \end{bmatrix}$$

where $X_i \leftarrow \chi_i, Y_i \leftarrow \chi_i$, and the variables denoted by different letter or index are pairwise independent. By the concentration inequalities of Lemma 17 and the union

bound, $\forall i \in [m-r+1, m+1]$, $j \in [1, r]$ it holds $\sqrt{i - 2\sqrt{ik}} \leq X_i \leq \sqrt{i + 2\sqrt{ik} + 2k}$ and $\sqrt{j - 2\sqrt{jk}} \leq Y_j \leq \sqrt{j + 2\sqrt{jk} + 2k}$ with overall probability $\geq 1 - 4r \cdot \exp(-k)$. Consider $r > 1$ and denote $Y_r = Y_0 = 0$. Then by the Gershgorin circle theorem the eigen values of $\mathbf{O} \cdot \mathbf{Z} \cdot \mathbf{Z}^\mathsf{T} \cdot \mathbf{O}^\mathsf{T}$ (and hence the eigen values of $\mathbf{Z} \cdot \mathbf{Z}^\mathsf{T}$) are bounded as

$$\lambda_{\min}(\mathbf{Z} \cdot \mathbf{Z}^\mathsf{T}) \geq \min_{i \in [0,r]} (X_{m-i}^2 + Y_{r-i}^2 - X_{m-i+1} Y_{r-i} - X_{m-i} Y_{r-i-1})$$
$$\lambda_{\max}(\mathbf{Z} \cdot \mathbf{Z}^\mathsf{T}) \leq \max_{i \in [0,r]} (X_{m-i}^2 + Y_{r-i}^2 + X_{m-i+1} Y_{r-i} + X_{m-i} Y_{r-i-1})$$

Substituting in the concentration inequalities and simplifying the expressions we get that for $m \geq c^2 \cdot \max(r, k) \cdot x^2$

$$\frac{\lambda_{\min}(\mathbf{Z} \cdot \mathbf{Z}^\mathsf{T})}{m} \geq 1 - \frac{2}{c \cdot x} - \frac{2}{c^2 \cdot x^2} - 2\sqrt{5} \cdot \frac{1}{c \cdot x} \cdot \sqrt{1 + \frac{2}{c \cdot x} + \frac{2}{c^2 \cdot x^2}}, \tag{4.2}$$
$$\frac{\lambda_{\max}(\mathbf{Z} \cdot \mathbf{Z}^\mathsf{T})}{m} \leq 1 + \frac{2}{c \cdot x} + \frac{7}{c^2 \cdot x^2} + 2\sqrt{5} \cdot \frac{1}{c \cdot x} \cdot \sqrt{1 + \frac{2}{c \cdot x} + \frac{2}{c^2 \cdot x^2}}. \tag{4.3}$$

Then for $c = 7, x = rd$ for $rd \geq 4$ we have[2]

$$\Pr(1 - \frac{1}{rd} \leq \lambda_{\min}/m \leq \lambda_{\max}/m \leq 1 + \frac{1}{rd}) \geq 1 - 4r \cdot \exp(-k).$$

Lastly, we consider the special case when $r = 1$. Then using $\mathbf{z}^\mathsf{T} \cdot \mathbf{z} \sim \chi_m^2$, $k/m \leq 1/(7\,rd)^2$ and Lemma 17 we get an even tighter bound

$$1 - \frac{1}{7\,rd} \leq 1 - \sqrt{\frac{k}{m}} \leq \mathfrak{s}_{\min}^2(\mathbf{z})/m = \mathfrak{s}_{\max}^2(\mathbf{z})/m \leq 1 + \sqrt{\frac{k}{m}} + 2 \cdot \frac{k}{m} \leq 1 + \frac{1}{7\,rd}$$

with probability $\geq 1 - 2\exp(-k)$. □

As a side-result, we give a statement of [AGHS13, Lemma 8] with explicit constants, in the general case of ring of integers.

Lemma 85 (Discrete Gaussians)**.** *Let $1 < r \leq m$ be integers and $k \geq 1$ s.t. $m \geq 64 \cdot \max(r, k)$ let $\sigma_x \geq 10 \cdot \eta_\varepsilon(\mathcal{R})$ for $\varepsilon \in (0, 1/2)$. Let $\mathbf{X} \leftarrow (\mathcal{D}_{\mathcal{R}^r, \sigma_x})^m$ then it holds that*

$$\frac{\sqrt{m} \cdot \sigma_x}{7} \leq \mathfrak{s}_{\min}(\tilde{\theta}(\mathbf{X})) \leq \mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{X})) \leq 2.2 \cdot \sqrt{m} \cdot \sigma_x,$$

each inequality holding with probability at least $1 - 4rm \cdot \varepsilon - 4rd \cdot \exp(-k)$ over the randomness of $\mathbf{X}$.

[2] See the included SageMath script or Section A.1 for the corresponding calculations.

Proof. We follow a proof similar as the one of Theorem 9. We work with diagonal by block matrices of dimension $dr \times dm$ with blocks of dimension d. For matrix $\mathbf{M}$ we denote $\mathbf{M}_{ij}$, $i \in [r], j \in [m]$ the corresponding $d \times d$ block. Let $\sigma_y \coloneqq \sigma_x/\sqrt{96}$, and consider $\mathbf{Z} = \tilde{\theta}(\mathbf{X}) + \mathbf{Y}$ where $\mathbf{X} \leftarrow (\mathcal{D}_{\mathcal{R}^r,\sigma_x})$ and $\mathbf{Y}$ is diagonal by block with $\mathbf{Y}_{ij} \leftarrow \mathsf{diag}\left(\mathcal{D}^d_{\sigma_y}\right)$, $i \in [r], j \in [m]$. One can compute that $\sigma_x \cdot \sigma_y/\sqrt{\sigma_x^2 + \sigma_y^2} \geq \sigma_x/10$, and then by the condition on σ_x we can use Lemma 33 for $\mathbf{Z}$. We then have $\mathrm{SD}(\mathbf{Z}_{ij}, \mathsf{diag}(\mathcal{D}^d_{\sqrt{\sigma_x^2+\sigma_y^2}})) \leq 4\varepsilon$.

We analyse the singular values of $\mathbf{Z}$. First we multiply by unitary matrices on the right and on the left to reorder the rows and columns and obtain a **block-diagonal** matrix. Now the on the diagonal we have matrices we denote $\mathbf{Z}_i$ sampled independently from $\mathbf{Z}_i \leftarrow \mathcal{D}^{r\times m}_{\sqrt{\sigma_x^2+\sigma_y^2}}$ and similarly $\mathbf{Y}_i \leftarrow \mathcal{D}^{r\times m}_{\sigma_y}$ for $i \in [d]$. We can use Equations (4.2) and (4.3) of the proof of Lemma 84 with $c \cdot x = 8$ (which matches the condition on m) to obtain, for any $i \in [d]$:

$$\begin{aligned}
\mathfrak{s}_{\max}(\mathbf{Y}_i)^2 &\leq 2 \cdot m \cdot \sigma_y^2, \\
\mathfrak{s}_{\min}(\mathbf{Z}_i)^2 &\geq \frac{m}{12} \cdot (\sigma_x^2 + \sigma_y^2), \\
\mathfrak{s}_{\max}(\mathbf{Z}_i)^2 &\leq 2 \cdot m \cdot (\sigma_x^2 + \sigma_y^2)
\end{aligned}$$

with probability $\geq 1 - 4rd\exp(-k)$. Now, one can compute that with our choice of σ_y, it holds that $\mathfrak{s}_{\max}(\mathbf{Y}_i)/\mathfrak{s}_{\min}(\mathbf{Z}_i) \leq 1/2$. Then using Lemma 2 and the fact that $\tilde{\theta}(\mathbf{X}) = \mathbf{Z} - \mathbf{Y}$, we get that for any $i \in [d]$:

$$\begin{aligned}
\mathfrak{s}_{\min}(\mathbf{X}_i) \geq \frac{1}{2} \cdot \mathfrak{s}_{\min}(\mathbf{Z}_i) \geq \frac{\sqrt{m}}{2\sqrt{12}} \cdot \sqrt{\sigma_x^2 + \sigma_y^2} \geq \frac{\sqrt{m} \cdot \sigma_x}{7}, \\
\mathfrak{s}_{\max}(\mathbf{X}_i) \leq \frac{3}{2} \cdot \mathfrak{s}_{\max}(\mathbf{Z}_i) \leq 2.2 \cdot \sqrt{m} \cdot \sigma_x,
\end{aligned}$$

for any $i \in [d]$ with probability $\geq 1 - 4rm \cdot \varepsilon - 4rd\exp(-k)$. The union bound give the desired result. □

Now we provide an adaptation of [PS21, Lemma 2.3] on the Rényi Divergence between discrete Gaussian distributions with different covariance parameters. We provide the proof for completeness.

Lemma 86. *Let Λ be an n-dimensional lattice $\varepsilon \in (0,1)$ and $\mathbf{\Sigma}_0, \mathbf{\Sigma}_1 \in \mathbb{R}^{n\times n}$ be positive definite s.t. $\sqrt{\mathbf{\Sigma}_i} \geq \eta_\varepsilon(\Lambda), i = 0, 1$ and $\mathfrak{s}_{\max}(\sqrt{\mathbf{\Sigma}_0}) \leq \mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}_1})$. Then*

$$\mathrm{RD}(\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}_0}}; \mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}_1}}) \leq \frac{1+\varepsilon}{1-\varepsilon} \cdot \frac{\det(\sqrt{\mathbf{\Sigma}_1})}{\det(\sqrt{\mathbf{\Sigma}_0})}$$

Proof. By definition

$$\begin{aligned}
&\mathrm{RD}(\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}_0};\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}_1}) = \sum_{\mathbf{s}\in\Lambda}\Pr(\mathbf{s}\leftarrow\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}_0})\cdot\frac{\Pr(\mathbf{s}\leftarrow\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}_0})}{\Pr(\mathbf{s}\leftarrow\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}_1})}\\
&= \mathbb{E}_{\mathbf{s}\leftarrow\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}_0}}\left(\frac{\rho_{\sqrt{\mathbf{\Sigma}}_0}(\mathbf{s})}{\rho_{\sqrt{\mathbf{\Sigma}}_0}(\Lambda)}\cdot\frac{\rho_{\sqrt{\mathbf{\Sigma}}_1}(\Lambda)}{\rho_{\sqrt{\mathbf{\Sigma}}_1}(\mathbf{s})}\right) = \frac{\rho_{\sqrt{\mathbf{\Sigma}}_1}(\Lambda)}{\rho_{\sqrt{\mathbf{\Sigma}}_0}(\Lambda)}\cdot\mathbb{E}_{\mathbf{s}\leftarrow\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}_0}}\left(\frac{\rho_{\sqrt{\mathbf{\Sigma}}_0}(\mathbf{s})}{\rho_{\sqrt{\mathbf{\Sigma}}_1}(\mathbf{s})}\right)\\
&\le \frac{1+\varepsilon}{1-\varepsilon}\cdot\frac{\det(\sqrt{\mathbf{\Sigma}}_1)}{\det(\sqrt{\mathbf{\Sigma}}_0)}\cdot\mathbb{E}_{\mathbf{s}\leftarrow\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}_0}}\exp\left(\pi\cdot\left(\left\|\sqrt{\mathbf{\Sigma}}_1^{-1}\cdot\mathbf{s}\right\|^2-\left\|\sqrt{\mathbf{\Sigma}}_0^{-1}\cdot\mathbf{s}\right\|^2\right)\right)
\end{aligned}$$

where the last transition follows from Corollary 6. Denote $\tilde{\mathbf{s}} \coloneqq \sqrt{\mathbf{\Sigma}}_0^{-1}\cdot\mathbf{s}$, then

$$\begin{aligned}
\left\|\sqrt{\mathbf{\Sigma}}_1^{-1}\cdot\mathbf{s}\right\|^2-\left\|\sqrt{\mathbf{\Sigma}}_0^{-1}\cdot\mathbf{s}\right\|^2 &= \left\|\sqrt{\mathbf{\Sigma}}_1^{-1}\cdot\sqrt{\mathbf{\Sigma}}_0\cdot\tilde{\mathbf{s}}\right\|^2-\|\tilde{\mathbf{s}}\|^2\\
&\le\left(\frac{s^2_{\max}(\sqrt{\mathbf{\Sigma}}_0)}{s^2_{\min}(\sqrt{\mathbf{\Sigma}}_1)}-1\right)\cdot\|\tilde{\mathbf{s}}\|^2\le 0
\end{aligned}$$

since $\mathfrak{s}_{\max}(\sqrt{\mathbf{\Sigma}}_0)\le\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}}_1)$. Combining the above we get the statement. □

Finally, we prove the target statement. On high level, we first add continuous noise to the discrete Gaussian matrix $\tilde{\theta}(\mathbf{X})$ to make it approach a continuous Gaussian sample. Then we apply the bounds from Lemma 84 to the continuous matrices. Lastly, we obtain bounds on the singular values of their difference equal to $\tilde{\theta}(\mathbf{X})$ and use Lemma 86.

Theorem 9. *Let $1<r\le m$ be integers and $k\ge 1$ s.t. $m\ge 49\cdot\max(r,k)\cdot(rd)^2$, $rd\ge 4$ let $\sigma_x\ge 2rd\cdot\eta_\varepsilon(\mathcal{R})$ for $\varepsilon\in(0,1/2)$. Let $\mathbf{X}\leftarrow(\mathcal{D}_{\mathcal{R}^r,\sigma_x})^m$ then for $\sigma=\sqrt{m\cdot(\sigma_x^2+2\cdot\eta_\varepsilon^2(\mathcal{R}))}\cdot\left(1-\frac{1}{rd}\right)^{3/2}$ we have*

$$\mathrm{RD}(\mathcal{D}_{\mathcal{R}^r,\sigma};\mathcal{D}_{\mathcal{R}^r,\sqrt{\tilde{\theta}(\mathbf{X})\cdot\tilde{\theta}(\mathbf{X})^{\mathrm{T}}}})\le 4\cdot\exp(3).$$

with probability at least $1-4rm\cdot\varepsilon-4rd\cdot\exp(-k)$ over the randomness of $\mathbf{X}$.

Proof. In the proof we work with diagonal by block matrices of dimension $dr\times dm$ with blocks of dimension d. For matrix $\mathbf{M}$ we denote $\mathbf{M}_{ij}$, $i\in[r],j\in[m]$ the corresponding $d\times d$ block. Consider $\mathbf{Z}=\tilde{\theta}(\mathbf{X})+\mathbf{Y}$ where $\mathbf{X}\leftarrow(\mathcal{D}_{\mathcal{R}^r,\sigma_x})$ and $\mathbf{Y}$ is diagonal by block with $\mathbf{Y}_{ij}\leftarrow\mathsf{diag}\left(\mathcal{D}^d_{\sigma_y}\right)$, $i\in[r],j\in[m]$. By Lemma 33 for $\mathbf{Z}$ we have $\mathrm{SD}(\mathbf{Z}_{ij},\mathsf{diag}(\mathcal{D}^d_{\sqrt{\sigma_x^2+\sigma_y^2}}))\le 4\varepsilon$ for $\sigma_y\coloneqq\sqrt{2}\cdot\eta_\varepsilon(\mathcal{R})$.

We analyse the singular values of $\mathbf{Z}$. First we multiply by unitary matrices on the right and on the left to reorder the rows and columns and obtain a **block-diagonal**

matrix. Now the on the diagonal we have matrices we denote $\mathbf{Z}_i$ sampled independently from $\mathbf{Z}_i \leftarrow \mathcal{D}^{r\times m}_{\sqrt{\sigma_x^2+\sigma_y^2}}$ and similarly $\mathbf{Y}_i \leftarrow \mathcal{D}^{r\times m}_{\sigma_y}$ for $i \in [d]$. By Lemma 84 for $i \in [d]$

$$m\cdot(\sigma_x^2+\sigma_y^2)\cdot\left(1-\frac{1}{rd}\right) \le s^2_{\min}(\mathbf{Z}_i) \le s^2_{\max}(\mathbf{Z}_i) \le m\cdot(\sigma_x^2+\sigma_y^2)\cdot\left(1+\frac{1}{rd}\right)$$
$$m\cdot\sigma_y^2\cdot\left(1-\frac{1}{rd}\right) \le s^2_{\min}(\mathbf{Y}_i) \le s^2_{\max}(\mathbf{Y}_i) \le m\cdot\sigma_y^2\cdot\left(1+\frac{1}{rd}\right)$$

with probability at least $1-8dr\cdot\exp(-k)$. Note that $\mathbf{X}_i = \mathbf{Z}_i - \mathbf{Y}_i$ and one can verify that $\mathfrak{s}_{\max}(\mathbf{Y}_i) \le \delta\cdot\mathfrak{s}_{\min}(\mathbf{Z}_i)$ for $\sigma_x \ge \sigma_y\cdot\sqrt{2}\cdot rd$, $\delta = 1/rd$ and $rd \ge 3$, see below.

$$\begin{aligned}
m\cdot\sigma_y^2\cdot\left(1+\frac{1}{rd}\right) &\le \delta^2\cdot m\cdot(\sigma_x^2+\sigma_y^2)\cdot\left(1-\frac{1}{rd}\right)\\
\frac{\sigma_x^2}{2\cdot(rd)^2}\cdot\left(1+\frac{1}{rd}\right) &\le \delta^2\cdot\sigma_x^2\cdot\left(1-\frac{1}{rd}\right)\\
\frac{1}{2\cdot(rd)^2}\cdot\left(1+\frac{1}{rd}\right) &\le \frac{1}{(rd)^2}\cdot\left(1-\frac{1}{rd}\right)\\
1+\frac{1}{rd} &\le 2-\frac{2}{rd}\\
\frac{3}{rd} &\le 1
\end{aligned}$$

Then using Lemma 2 we get

$$\begin{aligned}
m\cdot(\sigma_x^2+\sigma_y^2)\cdot\left(1-\frac{1}{rd}\right)\cdot(1-\delta)^2 &\le s^2_{\min}(\mathbf{X}_i)\\
m\cdot(\sigma_x^2+2\cdot\eta_\varepsilon^2(\mathcal{R}))\cdot\left(1-\frac{1}{rd}\right)^3 &\le s^2_{\min}(\mathbf{X}_i)
\end{aligned}$$

and

$$\begin{aligned}
s^2_{\max}(\mathbf{X}_i) &\le m\cdot(\sigma_x^2+\sigma_y^2)\cdot\left(1+\frac{1}{rd}\right)\cdot(1+\delta)^2\\
s^2_{\max}(\mathbf{X}_i) &\le m\cdot(\sigma_x^2+2\cdot\eta_\varepsilon^2(\mathcal{R}))\cdot\left(1+\frac{1}{rd}\right)^3
\end{aligned}$$

Then for $\sigma = \sqrt{m\cdot(\sigma_x^2+2\cdot\eta_\varepsilon^2(\mathcal{R}))}\cdot\left(1-\frac{1}{rd}\right)^{3/2}$ by Lemma 86

$$\begin{aligned}
\mathrm{RD}(\mathcal{D}_{\mathcal{R}^r,\sigma};\mathcal{D}_{\mathcal{R}^r,\sqrt{\tilde\theta(\mathbf{X})\cdot\tilde\theta(\mathbf{X})^{\mathrm{T}}}}) &\le \frac{1+\varepsilon}{1-\varepsilon}\cdot\frac{\left(1+\frac{1}{rd}\right)^{3rd/2}}{\left(1-\frac{1}{rd}\right)^{3rd/2}}\\
&\le \frac{1+\varepsilon}{1-\varepsilon}\cdot\frac{\exp(3/2)}{\exp(-3/2)\cdot\left(1-\frac{1}{rd}\right)} \le 4\exp(3)
\end{aligned}$$

with probability at least $1-4rm\cdot\varepsilon-4rd\cdot\exp(-k)$ over the randomness of $\mathbf{X}$. The last inequality used $\exp(x)\cdot(1-\frac{x^2}{n}) \le (1+\frac{x}{n})^n \le \exp(x)$ for $n \ge 1$, $|x| \le n$ and also $\varepsilon \le 1/2, rd \ge 4$. □

$\mathsf{HintTG}_{m,k}(\sigma_1, \sigma_2) \to (\mathbf{X} \in \mathcal{R}^{k\times(m+k)}, \mathbf{U} \in \mathcal{R}^{(m+k)\times(m+k)})$

1 : $\mathbf{X}_1 \leftarrow \mathcal{D}^m_{\mathcal{R}^k,\sigma_1}$

2 : $\mathbf{\Sigma} \coloneqq \sqrt{\mathbf{S}\cdot\mathbf{S}^\mathsf{T}}$ s.t. $\sigma_2^2 \cdot \mathbf{I}_{dr} = \tilde{\theta}(\mathbf{X}_1)\cdot\mathbf{S}\cdot\mathbf{S}^\mathsf{T}\cdot\tilde{\theta}(\mathbf{X}_1)^\mathsf{T}$ from Corollary 19

3 : $\forall : i \in [k] : \mathbf{r}_i \leftarrow \mathcal{D}_{\mathcal{R}^m,\sqrt{\mathbf{\Sigma}}}$. Let $\mathbf{R} \coloneqq (\mathbf{r}_1, \ldots, \mathbf{r}_k)$

4 : $\mathbf{X}_2 \coloneqq \mathbf{X}_1 \cdot \mathbf{R} + \mathbf{I}_k \in \mathcal{R}^{k\times k}$

5 : $\mathbf{U} \coloneqq \begin{bmatrix} -\mathbf{R} & -\mathbf{I}_m - \mathbf{R}\mathbf{X}_1 \\ \mathbf{I}_k & \mathbf{X}_1 \end{bmatrix} \in \mathcal{R}^{(m+k)\times(m+k)}$

6 : **return** $((\mathbf{X}_1, \mathbf{X}_2), \mathbf{U})$

Figure 4.1: HintTG procedure.

Remark 17. We expect an equivalent statement for $\mathsf{negl}(\lambda)$ statistical distance requires a number of columns exponential in rd. We leave this as an open problem.

4.5 Hardness of k-MSIS and k-MLWE problems

In this section we formalise a major application of our Gaussian LHL result: a reduction from the SIS problem to the k-SIS problem in the module setting. Based on our new Gaussian LHL (Theorem 8), we obtain the following generalisation and adaption of [LPSS14, Theorem 17 (eprint)], which is the core of our MSIS-to-k-MSIS reduction.

Lemma 87. *Let $\mathcal{K}$ be a number field of degree $d \geq 2$ containing the cyclotomic field of conductor f. Let $k = r \geq 1$, $m \geq 64\max(\lambda/\log(dk), k)$, $0 < \varepsilon \leq \mathsf{negl}(\lambda)$, $d \cdot k \geq \lambda$ and $\sigma_1 \cdot \mathbf{I} = \mathbf{S}_X \in \mathbb{R}^{dk\times dk}$ follow constraints of Theorem 8 and satisfy $\sigma_1 \geq 10 \cdot \eta_\varepsilon(\mathcal{R})$. Additionally, let $\sigma_2 \geq \sigma_1 d\sqrt{mk} \cdot \omega\left(\eta_{\varepsilon,((m-r)d)} \cdot m^{1.5} \cdot d^{12} \cdot \delta_\mathcal{K}^{14} \cdot f^2 \cdot \Delta_\mathcal{K}^{4/d} \cdot \sigma_1\right)$. Denote by $\mathbf{e}_i \in \mathcal{R}^k$ the i-th unit vector. Then there exists a PPT algorithm $\mathsf{HintTG}_{m,k}$ (Figure 4.1) which, on input parameters (σ_1, σ_2), outputs $((\mathbf{X}_1, \mathbf{X}_2), \mathbf{U}) \in \mathcal{R}^{k\times(m+k)} \times \mathcal{R}^{(m+k)\times(m+k)}$ satisfying*

1. $\mathrm{SD}\left((\mathbf{X}_1, \mathbf{X}_2),\ ((\mathcal{D}_{\mathcal{R}^k,\sigma_1})^m, \prod_{i\in[k]} \mathcal{D}_{\mathcal{R}^k,\sigma_2,\mathbf{e}_i})\right) \leq \mathsf{negl}(\lambda)$.
2. *We have* $\det(\mathbf{U}) = 1$, *and letting* $\bar{\mathbf{U}} \in \mathcal{R}^{(m+k)\times m}$ *be the last m columns of* $\mathbf{U}$, *then* $(\mathbf{X}_1, \mathbf{X}_2) \cdot \bar{\mathbf{U}} = \mathbf{0}_{k\times m}$.
3. *Lastly, it holds that* $\left\|\bar{\mathbf{U}}\right\| = \Omega(\sigma_2 \cdot d \cdot \sqrt{m})$ *with overwhelming probability in λ.*

Proof. We generate required values using $\mathsf{HintTG}_{m,k}$ described in Figure 4.1. We show that the output from HintTG satisfies all Items 1 to 3 in the lemma.

For Item 1 applying Theorem 8 and Corollary 19 k times over all $\mathbf{X}_1 \cdot \mathbf{r}_i$ together with the union bound, we have

$$\mathrm{SD}\left((\mathbf{X}_1, \mathbf{X}_1 \cdot \mathbf{R}),\ ((\mathcal{D}_{\mathcal{R}^k,\sigma_1})^m, (\mathcal{D}_{\mathcal{R}^k,\sigma_2})^k)\right) \leq \mathsf{negl}(\lambda).$$

The only difference between $\mathbf{X}_1 \cdot \mathbf{R}$ and the output $\mathbf{X}_2 = (\mathbf{x}_{2,1}, \ldots, \mathbf{x}_{2,k})$ from HintTG is that each of the ith column $\mathbf{x}_{2,i}$ of $\mathbf{X}_2$ is offset by $\mathbf{e}_i \in \mathcal{R}^k$, the ith unit vector, respectively. The claim follows by noting that for any $\mathbf{y} \in \mathcal{R}^k$ following $\mathcal{D}^k_{\mathcal{R},\sigma_2}$ and any fixed $\mathbf{e}_i \in \mathcal{R}^k$, it holds that $\mathbf{y} + \mathbf{e}_i$ follows $\mathcal{D}_{\mathcal{R}^k,\sigma_2,\mathbf{e}_i}$. For Item 2, observe that it holds

$$\mathbf{U} = \begin{bmatrix} \mathbf{I}_m & -\mathbf{R} \\ & \mathbf{I}_k \end{bmatrix} \cdot \begin{bmatrix} & -\mathbf{I}_m \\ \mathbf{I}_k & \mathbf{X}_1 \end{bmatrix}.$$

Noting that both matrices are triangular with diagonal entries ± 1 implies $\det(\mathbf{U}) = 1$. For the kernel condition we note

$$(\mathbf{X}_1, \mathbf{X}_2) = (\mathbf{X}_1, \mathbf{X}_1\mathbf{R} + \mathbf{I}_k) = (\mathbf{X}_1, \mathbf{I}_k) \cdot \begin{bmatrix} \mathbf{I}_m & \mathbf{R} \\ & \mathbf{I}_k \end{bmatrix},$$

thus

$$\begin{aligned} (\mathbf{X}_1, \mathbf{X}_2) \cdot \mathbf{U} =& (\mathbf{X}_1, \mathbf{I}_k) \cdot \begin{bmatrix} & -\mathbf{I}_m \\ \mathbf{I}_k & \mathbf{X}_1 \end{bmatrix} = (\mathbf{I}_k, \mathbf{0}_{k\times m}), \\ \text{implying} \quad (\mathbf{X}_1, \mathbf{X}_2) \cdot \bar{\mathbf{U}} =& \mathbf{0}_{k\times m}. \end{aligned}$$

Finally, for Item 3, we bound the norm of $\bar{\mathbf{U}}$, obtaining

$$\begin{aligned} \left\|\bar{\mathbf{U}}\right\| &= \left\| \begin{bmatrix} -\mathbf{I}_m - \mathbf{R}\mathbf{X}_1 \\ -\mathbf{X}_1 \end{bmatrix} \right\| \\ &\leq (1 + \|\mathbf{R}\|^2 \cdot \|\mathbf{X}_1\|^2 + \|\mathbf{X}_1\|^2)^{1/2} \\ &\leq 1 + (\|\mathbf{R}\| + 1) \cdot \|\mathbf{X}_1\| \\ &\leq 1 + (\sigma_2 \cdot \sqrt{d \cdot m}/\mathfrak{s}_{\min}(\tilde{\theta}(\mathbf{X}_1)) + 1) \cdot \sigma_1 \cdot \sqrt{d \cdot m} \\ &\leq 1 + (\sigma_2 \cdot \sqrt{d \cdot m} \cdot 7/(\sqrt{m} \cdot \sigma_1) + 1) \cdot \sigma_1 \cdot \sqrt{d \cdot m} \\ &= \Omega(\sigma_2 \cdot d \cdot \sqrt{m}) \end{aligned}$$

with probability overwhelming in $d \cdot k \geq \lambda$. Here, the third inequality follows from the Gaussian tail bound together with the bound on $\mathfrak{s}_{\max}(\sqrt{\mathbf{\Sigma}})$ from Corollary 19, the fourth inequality holds by Lemma 85, and the last equality holds since $\sigma_2 \cdot d$ dominates $\sigma_1 \cdot \sqrt{d}$. □

In the remainder of the chapter, we focus on the case where $\mathcal{K}$ is a cyclotomic field. From Lemma 87, we have the following generalisation of [LPSS14, Lemma 19 (eprint)] without the restriction of $\mathcal{R} = \mathbb{Z}$.

The original proof relies on [BF11b, Lemma 4.5 and Theorem 4.3] which we replace by its generalisation to cyclotomic rings in Lemmas 35 and 94.

Lemma 88 (Generalisation of [LPSS14, Lemma 19 (eprint)])**.** *Let $\mathcal{K}$ be a cyclotomic field of degree $d \geq 2$ with conductor f. Let $k = r \geq 1$, $m \geq 64\max(\lambda/\log(dk), k)$, $0 < \varepsilon \leq \mathsf{negl}(\lambda)$, and $\sigma_1 \cdot \mathbf{I} = \mathbf{S}_X$ follow constraints of Theorem 8 and satisfy $\sigma_1 \geq 10 \cdot \eta_\varepsilon(\mathcal{R})$. Additionally, let $\sigma_2 \geq \sigma_1 d\sqrt{mk} \cdot \omega\left(\eta_{\varepsilon,((m-r)d)} \cdot m^{1.5} \cdot d^{12} \cdot \delta_{\mathcal{K}}^{14} \cdot f^2 \cdot \Delta_{\mathcal{K}}^{4/d} \cdot \sigma_1\right)$. Denote $\sqrt{\mathbf{\Sigma}} \coloneqq \mathsf{diag}(\sigma_1\mathbf{I}_m, \sigma_2\mathbf{I}_k)$, and suppose the following holds:*

- *q is an unramified prime that splits into g ideals in $\mathcal{R}$;*
- *$1 \leq n \leq m$, $d \cdot (m-k) \geq \Omega(\lambda)$, and $ng/q^{d(m-n+1)/g} \leq \mathsf{negl}(\lambda)$;*
- *letting $\eta_A \geq 8\, d\sqrt{m} \cdot q^{n/m+2/(d\cdot m)}$, then there exists $\varepsilon \leq \mathsf{negl}(\lambda), a \geq 1$, such that it holds $\max\left\{\eta_A, 2\sqrt{d} \cdot (a^k q^n)^{1/(m-k)}\right\} \leq \sigma_1$ and $\sigma_2 \leq \min\left\{q^{1/g}/\sqrt{m}, a \cdot \sigma_1\right\}$.*

Denote by $\mathbf{e}_i \in \mathcal{R}^{(m+k)}$ the $(m+i)$-th unit vector. Let $\mathsf{HintTG}_{m,k}$ be the PPT algorithm in Figure 4.1. Then, the following distributions are statistically close in λ:

$$\left\{ \left(\mathbf{B}, \mathbf{X}^{\mathsf{T}}\right) \;\middle|\; \begin{array}{l} \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m} \\ (\mathbf{X}, \mathbf{U}) \leftarrow \mathsf{HintTG}_{m,k}(\sigma_1, \sigma_2) \\ \bar{\mathbf{U}}^{\mathsf{T}} \coloneqq \mathbf{U} \cdot (\mathbf{0}_{(m+k)\times k}, \mathbf{I}_m) \\ \mathbf{B} \coloneqq \mathbf{A} \cdot \bar{\mathbf{U}} \bmod q \end{array} \right\} \approx_s \left\{ (\mathbf{B}, \mathbf{X}) \;\middle|\; \begin{array}{l} \mathbf{B} \leftarrow \mathcal{R}_q^{n\times(m+k)} \\ \mathbf{X} \leftarrow \prod\limits_{i\in[k]} \mathcal{D}_{\Lambda_q^\perp(\mathbf{B}), \sqrt{\mathbf{\Sigma}}, \mathbf{e}_i} \end{array} \right\}.$$

Proof. We consider a sequence of hybrid distributions.

Hyb_0: The RHS distribution in the statement above.

Hyb_1: We first sample $\mathbf{X} \leftarrow \prod_{i\in[k]} \mathcal{D}_{\mathcal{R}^{m+k}, \sqrt{\mathbf{\Sigma}}, \mathbf{e}_i}$, then sample $\mathbf{B} \leftarrow \mathcal{R}_q^{n\times(m+k)}$ subject to $\mathbf{B} \cdot \mathbf{X} = \mathbf{0} \bmod q$.

Hyb_2: Same as Hyb_1, except we sample $\mathbf{X}$ via $(\mathbf{X}^{\mathsf{T}}, \mathbf{U}) \leftarrow \mathsf{HintTG}_{m,k}(\sigma_1, \sigma_2)$.

Hyb_3: Same as Hyb_2, except that we sample $\mathbf{A} \leftarrow \mathcal{R}_q^{n\times m}$ and set $\mathbf{B} \coloneqq \mathbf{A} \cdot \bar{\mathbf{U}} \bmod q$.

It holds RHS $\approx_s \mathsf{Hyb}_1$ by Theorem 12. Then it holds $\mathsf{Hyb}_1 \approx_s \mathsf{Hyb}_2$ by Lemma 87 item 1. In the rest we show $\mathsf{Hyb}_2 \approx_s \mathsf{Hyb}_3$, which completes the proof since $\mathsf{Hyb}_3 \equiv$ LHS.

By Lemma 94, the columns of $\mathbf{X}$ sampled in Hyb_0 are $\mathcal{R}_q$-linearly independent with overwhelming probability in λ. In both Hyb_2 and Hyb_3, the sampling of $(\mathbf{X}^\intercal = (\mathbf{X}_1, \mathbf{X}_2), \mathbf{U})$ is identical. Since we have shown RHS $\approx_s \mathsf{Hyb}_2$, we have that columns of $\mathbf{X}$ from Hyb_2 and Hyb_3 are also $\mathcal{R}_q$-linearly independent with overwhelming probability in λ. Conditioned on the latter, we argue that the distribution of $\mathbf{B}$ in Hyb_2 and Hyb_3 are identical.

Consider an arbitrary sample $(\mathbf{X}, \mathbf{U})$ where $\mathbf{X}$ is $\mathcal{R}_q$-injective. In Hyb_2, the matrix $\mathbf{B}$ is sampled uniformly from $S \coloneqq \left\{\mathbf{M} \in \mathcal{R}_q^{n\times(m+k)} : \mathbf{M}\cdot\mathbf{X} = \mathbf{0}_{n\times k} \bmod q\right\}$. Since $\mathbf{X}$ is injective, we have $|S| = q^{dn(m+k)}/q^{dnk} = q^{dnm}$. Next, observe that for $\bar{\mathbf{U}}$ in Hyb_3 we have $\bar{\mathbf{U}}\cdot\mathbf{X} = \mathbf{0}_{m\times k}$ by Lemma 87 Item 2. Therefore, for any $\mathbf{A} \in \mathcal{R}_q^{n\times m}$ it holds $\mathbf{A}\cdot\bar{\mathbf{U}}\cdot\mathbf{X} = \mathbf{0}_{m\times k} \bmod q$, implying $\mathcal{R}_q^{n\times m}\cdot\bar{\mathbf{U}} \subseteq S$. Further, by Lemma 87 Item 2, $\det(\mathbf{U}) = 1$, thus (right-multiplication by) $\mathbf{U}$ is a bijective map, and (right-multiplication by) $\bar{\mathbf{U}}$ is injective. Thus, $|\mathcal{R}_q^{n\times m}\cdot\bar{\mathbf{U}}| = |\mathcal{R}_q^{n\times m}| = q^{dnm}$. Therefore, $\mathcal{R}_q^{n\times m}\cdot\bar{\mathbf{U}} = S$, and the two ways of sampling $\mathbf{B}$ are the same. This implies $\mathsf{Hyb}_2 \approx_s \mathsf{Hyb}_3$ concluding the proof. □

For a random matrix $\mathbf{A}$ and k Gaussian samples $(\mathbf{x}_i)_{i\in[k]}$ from $\Lambda_q^\perp(\mathbf{A})$, the k-SIS problem asks to find a short non-zero $\mathbf{u}$ which is not in the $\mathcal{K}$-span of $(\mathbf{x}_i)_{i\in[k]}$ and s.t. $\mathbf{A}\cdot\mathbf{u} = \mathbf{0} \bmod q$. See Definition 4 for a formal definition. The following theorem provides a reduction from MSIS to k-MSIS.

Theorem 10 (SIS-to-k-SIS reduction over cyclotomic rings)**.** *Let $\mathcal{K}$ be a cyclotomic field of degree $d \geq 2$ with conductor f. Take the parameters $k, n, m, q \in \mathbb{N}$, $0 < \varepsilon \leq \mathsf{negl}(\lambda)$, $\sigma_1 > 0$, $\sqrt{\mathbf{\Sigma}} \in \mathbb{R}^{d(m+k)\times d(m+k)}$ specified by Lemma 88. Further suppose that $\sigma_2, \sigma_1 \geq 8d\sqrt{m}\cdot q^{n/m+2/(d\cdot m)}$.*

Let $\beta_1 > 0$ and $\beta_0 \geq \Omega(\sigma_2 \cdot d \cdot \sqrt{m})\cdot\beta_1$. Denote $\mathsf{par}_0 = (\mathcal{R}, q, n, m, \beta_0)$, $\mathsf{par}_1 = (\mathcal{R}, q, n, m+k, k, \sqrt{\mathbf{\Sigma}}, \beta_1)$. For any PPT adversary $\mathcal{B}$, there is a PPT adversary $\mathcal{A}$ such that

$$\mathsf{Adv}_{\mathcal{A}}^{\mathsf{SIS}_{\mathsf{par}_0}}(\lambda) \geq (\mathsf{Adv}_{\mathcal{B}}^{\mathsf{k\text{-}SIS}_{\mathsf{par}_1}}(\lambda))^2/O(1) - \mathsf{negl}(\lambda).$$

Remark 18. Firstly, here we consider the hints with simple covariance matrices, but this approach extends directly to $\sqrt{\mathbf{\Sigma}} = \mathsf{diag}(\mathbf{S}_{X1}, \mathbf{S}_{X2})$. Secondly, the constant 64 in $m \geq 64\max(\lambda/\log(dk), k)$ can be reduced depending on the parametrisation of the problem. For example, we can get a suitable singular value lower bound for $\lambda = 128$, $k = 4\lambda$, $m = 10\max(\lambda/\log(dk), k)$. We leave more specific parameter settings for the applications.

Proof. We consider an intermediate search problem, which asks a PPT algorithm $\mathcal{B}$ to solve Shift-k-SIS, defined below:

$\underline{\mathsf{Shift\text{-}k\text{-}SIS}_{\mathcal{B}}(1^\lambda)}$

$\mathbf{B} \leftarrow \mathcal{R}_q^{n\times(m+k)};\ \mathbf{X} \leftarrow (\prod_{i\in[k]} \mathcal{D}_{\Lambda_q^\perp(\mathbf{B}),\sqrt{\mathbf{\Sigma}},\mathbf{e}_i})$

$\mathbf{u} \leftarrow \mathcal{B}(\mathbf{B}, \mathbf{X})$

return $(\mathbf{B}\cdot\mathbf{u} = \mathbf{0} \bmod q) \wedge (\|\mathbf{u}\| \le \beta_1) \wedge (\mathbf{u} \notin \mathcal{K}\text{-span}(\mathbf{X}))$

The only difference between Shift-k-SIS and k-SIS is the Gaussian parameters of $\mathbf{X}$, where in Shift-k-SIS the ith preimage is offset by the ith unit vector $\mathbf{e}_i$. The theorem follows from combining Lemmas 89 and 90 below.

Lemma 89. *Given the parameter constrains of Theorem 10, for any PPT algorithm $\mathcal{B}$, it holds* $\mathsf{Adv}^{\mathsf{Shift\text{-}k\text{-}SIS}}_{\mathcal{B}}(\lambda) \ge (\mathsf{Adv}^{\mathsf{k\text{-}SIS}_{\mathsf{par}_1}}_{\mathcal{B}}(\lambda))^2/O(1) - \mathsf{negl}(\lambda)$.

Proof. We define a sequence of hybrids:

Hyb_0: Identical to the $\mathsf{k\text{-}SIS}_{\mathsf{par}_1}$ experiment. We denote the problem instance as $(\mathbf{A}, \mathbf{X}) \in \mathcal{R}_q^{n\times(m+k)} \times \mathcal{R}^{(m+k)\times k}$.

Hyb_1: Same as Hyb_0, except that we sample $\mathbf{A} \leftarrow \mathcal{R}_q^{n\times(m+k)}$ subject to $\eta_\varepsilon(\Lambda_q^\perp(\mathbf{A})) \le 2d\cdot q^{n/m+2/(d\cdot m)}$ for some $\varepsilon \le 2^{-2k}$.

Hyb_2: Same as Hyb_1, except we shift the distribution $\mathbf{X} \leftarrow \prod_{i\in[k]} \mathcal{D}_{\Lambda_q^\perp(\mathbf{A}),\sqrt{\mathbf{\Sigma}}}$ to $\mathbf{X} \leftarrow \prod_{i\in[k]} \mathcal{D}_{\Lambda_q^\perp(\mathbf{A}),\sqrt{\mathbf{\Sigma}},\mathbf{e}_i}$.

Hyb_3: Same as Hyb_2, except that we sample $\mathbf{A} \leftarrow \mathcal{R}_q^{n\times(m+k)}$ without the constraint on $\eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$, i.e. we undo the change introduced in Hyb_1. Note that Hyb_3 is identical to Shift-k-SIS.

Let $\mathcal{B}$ be a PPT solver against $\mathsf{Hyb}_0 = \mathsf{k\text{-}SIS}_{\mathsf{par}_1}$ with advantage $\mathsf{Adv}^{\mathsf{k\text{-}SIS}}_{\mathcal{B}}(\lambda) \in \mathsf{poly}(\lambda)$. We claim:

1. $\mathcal{B}$ solves Hyb_1 with advantage $\delta_1 \ge \mathsf{Adv}^{\mathsf{SIS}}_{\mathcal{B}}(\lambda) - 2^{-\lambda}$.
2. $\mathcal{B}$ solves Hyb_2 with advantage $\delta_2 \ge \delta_1^2/O(1)$.
3. $\mathcal{B}$ solves $\mathsf{Hyb}_3 = \mathsf{Shift\text{-}k\text{-}SIS}$ with advantage $\delta_3 \ge \delta_2 - 2^{-\lambda}$.

Items 1 and 3 hold by Lemma 57, which says that the probability of $\eta_\varepsilon(\Lambda_q^\perp(\mathbf{A})) \le 2\exp(1/dm)\cdot d\cdot q^{n/m}$ for any $\varepsilon \le 2^{-2k}$ is at least $1 - 2^{-\Omega(d)} \ge 1 - 2^{-\lambda}$. In the rest we argue that Item 2 holds, which concludes the proof.

We notice that the Rényi divergence between the distributions $P := \mathsf{k\text{-}SIS}_{\mathsf{par}_1}$ and $Q := \mathsf{Shift\text{-}k\text{-}SIS}$ is identical to the one between the distributions of $\mathbf{X}$ in Hyb_1 and Hyb_2. Thus we use Lemma 38 to obtain $R(P\|Q) \le 5\exp\left(\frac{2\pi\cdot dk}{s_{\min}(\sqrt{\mathbf{\Sigma}})^2}\right) \in O(1)$. Note that

the conditions required by Lemma 38 are satisfied since, for any $\varepsilon \leq 2^{-2k}$, we have $\mathfrak{s}_{\min}(\sqrt{\boldsymbol{\Sigma}}) \geq 2d \cdot q^{n/m+2/(d\cdot m)} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ in Hyb_2, where the first inequality is by design of Hyb_1. From here, Item 2 follows from Equation (2.3). □

Lemma 90. *Following the parameter constraints of Theorem 10, for any PPT adversary $\mathcal{B}$, there is a PPT adversary $\mathcal{A}$ such that* $\mathsf{Adv}_{\mathcal{A}}^{\mathsf{SIS}_{\mathsf{par}_0}}(\lambda) \geq \mathsf{Adv}_{\mathcal{B}}^{\mathsf{Shift\text{-}k\text{-}SIS}}(\lambda) - \mathsf{negl}(\lambda)$.

Proof. We construct the PPT algorithm $\mathcal{A}$. On input a $\mathsf{SIS}_{\mathsf{par}_0}$ instance $\mathbf{A}$, let $\mathcal{A}$ proceed as follows:

Sample $(\mathbf{X}^\intercal, \mathbf{U}) \leftarrow \mathsf{HintTG}_{m,k}(\sigma_1, \sigma_2)$ using $\mathsf{HintTG}_{m,k}$ from Figure 4.1.
Let $\bar{\mathbf{U}}^\intercal \in \mathcal{R}^{(m+k)\times m}$ be the last m columns of $\mathbf{U}$ and $\mathbf{B} \coloneqq \mathbf{A} \cdot \bar{\mathbf{U}} \bmod q$.
Send $(\mathbf{B}, \mathbf{X})$ to $\mathcal{B}$, and receive a vector $\mathbf{u}$ from $\mathcal{B}$.
Return $\mathbf{u}^* \coloneqq \bar{\mathbf{U}} \cdot \mathbf{u}$.

By Lemma 88, the distribution of $(\mathbf{B}, \mathbf{X})$ is statistically close in λ to that from $\mathsf{Shift\text{-}k\text{-}SIS}$. It remains to show that $\mathbf{u}^*$ returned by $\mathcal{A}$ is a SIS solution for $\mathbf{A}$. Suppose that $\mathbf{u}$ returned by $\mathcal{B}$ is a valid solution for $\mathsf{Shift\text{-}k\text{-}SIS}$, meaning that $\mathbf{B} \cdot \mathbf{u} = \mathbf{0} \bmod q$, $0 < \|\mathbf{u}\| \leq \beta_1$, and $\mathbf{u} \notin \mathcal{K}\text{-span}(\mathbf{X})$. Then, we have

$$\mathbf{A} \cdot \mathbf{u}^* = \mathbf{A} \cdot \bar{\mathbf{U}} \cdot \mathbf{u} = \mathbf{B} \cdot \mathbf{u} = \mathbf{0} \bmod q.$$

Moreover, $\bar{\mathbf{U}} \cdot \mathbf{X} = \mathbf{0}$ by Lemma 87 Item 2, and $\mathbf{X}$ is $\mathcal{R}_q$-linearly independent with overwhelming probability by Lemma 94. Conditioned on the latter, $\mathbf{X}$ is an $\mathcal{R}_q$-basis of the (right-)kernel of $\bar{\mathbf{U}}$, and thus $\bar{\mathbf{U}} \cdot \mathbf{u} = \mathbf{0}$ would imply $\mathbf{u} \in \mathcal{R}_q\text{-span}(\mathbf{X})$, therefore also $\mathbf{u} \in \mathcal{K}\text{-span}(\mathbf{X})$, a contradiction. We thus conclude $\mathbf{u}^* = \bar{\mathbf{U}} \cdot \mathbf{u} \neq \mathbf{0}$ with overwhelming probability. Finally, the norm of $\mathbf{u}^*$ is bounded by

$$\|\mathbf{u}^*\| \leq \left\|\bar{\mathbf{U}}\right\| \cdot \|\mathbf{u}\| \leq \Omega(\sigma_2 \cdot d \cdot \sqrt{m}) \cdot \beta_1 \leq \beta_0$$

with overwhelming probability, where the second inequality follows from Lemma 87 Item 3. Putting everything together, we conclude that $\mathbf{u}^*$ returned by $\mathcal{A}$ is a SIS solution for $\mathsf{SIS}_{\mathsf{par}_0}$ with overwhelming probability. □

Combining the arguments above we get the statement. □

LWE to k-LWE reduction over cyclotomic rings. Analogous to the above, using Theorem 8, we obtain a reduction from the LWE problem to the k-LWE problem over general cyclotomic rings. The reduction is similar to the above, we provide a sketch and refer to [LPSS14] for the details in the special case of $\mathcal{R} = \mathbb{Z}$.

To recall, in the k-LWE problem (formal definition given in Definition 5), an adversary is given as challenge instance $(\mathbf{B}, \mathbf{X}, \mathbf{c})$, where $\mathbf{B} \leftarrow \mathcal{R}_q^{n\times(m+k)}$ is uniform, $\mathbf{X} \in \mathcal{R}^{(m+k)\times k}$ are Gaussian preimages satisfying $\mathbf{BX} = \mathbf{0} \bmod q$, and it is asked to distinguish whether $\mathbf{c} \in \mathcal{R}^{m+k}$ is an LWE sample, i.e. $\mathbf{c}^\intercal = \mathbf{s}^\intercal\mathbf{B} + \mathbf{e}^\intercal \bmod q$ for some secret $\mathbf{s}$ and error $\mathbf{e}$, or $\mathbf{c} = \mathbf{d} + \mathbf{e} \bmod q$ where $\mathbf{d}$ is a random sample over $\{\mathbf{d} : \mathbf{d}^\intercal\mathbf{X} = \mathbf{0} \bmod q\}$. In the LWE to k-LWE reduction, given an LWE instance $(\mathbf{A}, \hat{\mathbf{c}})$, we simulate $\mathbf{B}, \mathbf{X}$ using Lemma 88, in the same way as in the reduction in Theorem 10. It remains to simulate the LWE sample $\mathbf{c}$, for which we let $\mathbf{c}^\intercal \coloneqq \hat{\mathbf{c}}^\intercal \cdot \bar{\mathbf{U}} + \bar{\mathbf{e}} \bmod q$ for some freshly sampled error $\bar{\mathbf{e}} \in \mathcal{R}^{m+k}$. By picking the Gaussian parameters of $\bar{\mathbf{e}}$ appropriately and appealing to standard Gaussian convolution lemmas, we can arrive at that, if $\hat{\mathbf{c}}^\intercal = \mathbf{s}^\intercal\mathbf{A} + \hat{\mathbf{e}}^\intercal \bmod q$ then $\mathbf{c}^\intercal = (\mathbf{s}^\intercal\mathbf{A} + \hat{\mathbf{e}}^\intercal) \cdot \bar{\mathbf{U}} + \bar{\mathbf{e}} \approx \mathbf{s}^\intercal\mathbf{B} + \mathbf{e} \bmod q$ for $\mathbf{e}$ of the desired shape (independent of $\bar{\mathbf{U}}$), else if $\hat{\mathbf{c}}$ is uniformly random then $\mathbf{c}^\intercal \approx \mathbf{d} + \mathbf{e} \bmod q$.

Chapter 5

Partial Lattice Trapdoors

Lattice-based cryptographic constructions commonly involve generating some public matrix $\mathbf{A}$ and a secret matrix $\mathbf{U}$ with short elements, the latter called a *trapdoor* of $\mathbf{A}$, which satisfy some non-trivial relation such as $\mathbf{A} \cdot \mathbf{U} = \mathbf{0} \bmod q$. Many cryptographic tasks ranging from signatures to advanced encryption are then accomplished by using $\mathbf{U}$ to sample a short preimage $\mathbf{x}$ with respect to some target image vector $\mathbf{y}$, so that $\mathbf{A} \cdot \mathbf{x} = \mathbf{y} \bmod q$ holds.

Since the first lattice trapdoor scheme [GPV08], lattice trapdoors have been all-or-nothing: Either finding short preimages w.r.t. any target is easy with a trapdoor, or such a problem w.r.t. any target is believed to be hard. In this chapter, we ask and answer the natural question: *can we partition a lattice trapdoor?* By this, we mean a non-interactive distribution of the preimage-sampling procedure using a trapdoor divided among N parties:

- With a *partial trapdoor* (possibly generated from a master (full) trapdoor), any party j can *locally* compute a partial preimage $\mathbf{x}_j$ of $\mathbf{y}$.
- After collecting enough partial preimages $(\mathbf{x}_j)_{j \in \mathsf{act}}$ for some T-subset $\mathsf{act} \subseteq [N]$, anyone can locally recover a full preimage $\mathbf{x}$ of $\mathbf{y}$.

Such a partial trapdoor primitive would allow trivial and non-interactive thresholdisation of (the preimage sampling operations in) any trapdoor-based lattice-based primitives, such as (hash-and-sign) signatures [GPV08], homomorphic signatures [BF11b, GVW15], identity-based encryption [GPV08, ABB10a], attribute-based encryption [BGG+14, Wee22], to name a few.

Although generic tools such as the universal thresholdiser [BGG+18] offer a solution to the above problem, it requires homomorphically evaluating the highly non-linear

trapdoor preimage sampling algorithm over a fully homomorphic encryption ciphertext of the trapdoor, making it mostly a theoretical feasibility result. In this chapter, we focus on exploring more direct and algebraic approach to the problem.

A natural partial lattice trapdoor. Without size constraints, the above task may seem easy. Let us consider the trapdoor generation algorithm in [MP12] which outputs, say, two matrices $\mathbf{A}, \mathbf{U} = (\mathbf{U}_0\ \mathbf{U}_1)$ satisfying

$$\begin{bmatrix} \mathbf{G} & \mathbf{0} \\ \mathbf{0} & \mathbf{G} \end{bmatrix} = \mathbf{A} \cdot (\mathbf{U}_0\ \mathbf{U}_1) \bmod q$$

where $\mathbf{G} \coloneqq \mathbf{I} \otimes \mathbf{g}^\intercal$, with $\mathbf{I}$ denoting the identity matrix and $\mathbf{g}^\intercal = \begin{bmatrix} 1 & 2 & \dots & 2^{\lceil \log q \rceil - 1} \end{bmatrix}$ the gadget vector. If we hand $\mathbf{U}_j$ to the jth party ($j = 0, 1$), this immediately allows each party to compute a partial preimage $\mathbf{x}_j$ for given target $\mathbf{y} \coloneqq (\mathbf{y}_0 \| \mathbf{y}_1)$, i.e. $(\mathbf{y}_0 \| \mathbf{0}) = \mathbf{A} \cdot \mathbf{x}_0 \bmod q$ and $(\mathbf{0} \| \mathbf{y}_1) = \mathbf{A} \cdot \mathbf{x}_1 \bmod q$. We then have $(\mathbf{y}_0 \| \mathbf{y}_1) = \mathbf{A} \cdot (\mathbf{x}_0 + \mathbf{x}_1) \bmod q$. This basic idea can be extended to more parties in the obvious way, and the "public parameters" $\mathbf{A}$ grow linearly in N. Moreover, with some effort, this can be generalised to the T-out-of-N setting by applying Shamir secret sharing on the image space.

(In)security? We inspect security of the toy-example above instantiated with integer lattices. For a traditional full trapdoor, we require that the (inhomogeneous) Short Integer Solutions SIS and/or Learning With Errors (LWE) problems w.r.t. $\mathbf{A}$ remain hard so long as the trapdoor is not made available. Generalising to the partial trapdoor setting, we may wish to argue that the inhomogeneous[1] SIS (ISIS) problem remains hard even when given the partial trapdoors of up to $T - 1$ corrupt parties and many partial preimages generated by honest parties.

Adapting the conventional proof strategy to our setting reveals multiple challenges. In the original security proof of [GPV08], a key step is to sample a short Gaussian vector $\mathbf{x}$ from a public lattice, e.g. $\mathbb{Z}^m$, and to compute a challenge image $\mathbf{y} = \mathbf{A} \cdot \mathbf{x} \bmod q$ for which the adversary must provide a short preimage. This is made possible by two convenient facts:

1. The tuples $(\mathbf{x}, \mathbf{y})$ sampled as above are statistically indistinguishable from those sampled in the real scheme, i.e. first sample a random $\mathbf{y}$, then sample $\mathbf{x}$ using a full trapdoor.

[1] The homogeneous SIS problem becomes trivial given a partial trapdoor.

2. The preimages $\mathbf{x} \in \mathbb{Z}^m$ can be easily sampled without any secret information: they are just Gaussian vectors over the public lattice $\mathbb{Z}^m$ which admits a public short basis $\mathbf{I}_m$, the identity matrix. In particular, this implies that after seeing many preimages $\mathbf{x} \in \mathbb{Z}^m$ for random images, an adversary can only learn the trivial lattice where these preimages are from: $\mathbb{Z}^m$.

Neither of the above hold in the partial trapdoor setting. First, the distribution of a partial preimage $\mathbf{x}_j$ generated by a possibly rank-deficient secret trapdoor $\mathbf{U}_j$ is a priori unclear: it does not come from $\mathbb{Z}^m$, but some (rank-deficient) random sublattice $\Lambda_j \subset \mathbb{Z}^m$. Indeed, it is not obvious what the "distribution of the lattice Λ_j" is. Second and critical, it seems difficult to sample even a single vector from any such sublattice Λ_j without knowing any secrets about $\mathbf{A}$: if we sample a short vector $\mathbf{x}_j$ over $\mathbb{Z}^m$, it is highly unlikely that $\mathbf{A} \cdot \mathbf{x}_j = (\mathbf{y}_0 \| \mathbf{0}) \bmod q$ for any $\mathbf{y}_0$. Third, adding to this list of challenges, it is not clear how to simulate the partial trapdoors for corrupt parties without already solving the ISIS problem. Finally, in the case of LWE based constructions, how can we still meaningfully consider the LWE hardness, when any single party j is now able to distinguish an LWE sample $\mathbf{s}^\intercal \cdot \mathbf{A} + \mathbf{e}^\intercal \bmod q$ from uniform by computing a preimage of $\mathbf{0}$ using its partial trapdoor $\mathbf{U}_j$?

Our contributions. In this chapter, we show that a T-out-of-N generalisation of the above natural partial trapdoor indeed allows us to sample a matrix $\mathbf{A}$ together with n partial trapdoors, such that the ISIS and (an appropriate variant of) the LWE problem are hard w.r.t. $\mathbf{A}$ even when given at most $T - 1$ partial trapdoors, under the standard SIS and LWE assumptions, respectively.

A formal notion. We introduce the notion of "partial lattice trapdoors", that allow sampling preimages of vectors from a designated subspace. To enable plug-and-play thresholdisation of typical constructions based on lattice trapdoors, we endow partial trapdoors with two security properties: one-wayness and indistinguishability, which are natural analogues of the hardness of the ISIS and LWE problems for standard (fully) trapdoored lattices.

A construction and new proof techniques. We provide a simple and algebraic construction of partial lattice trapdoors and formally prove that our construction achieves both security properties, one-wayness and indistinguishability. Our construction produces matrices $\mathbf{A}$ of size $T^2 \cdot \mathsf{poly}(\lambda, \log N)$ and (partial) preimages of size $T \cdot \mathsf{poly}(\lambda, \log N)$.

Notably, these quantities are quadratic and linear in the threshold T, respectively, but are otherwise polylogarithmic in the number of shareholders N.

Our security proofs use a variety of techniques which may be of independent interest. In particular, the proofs require non-trivial analysis of lattice subspaces which is, to our knowledge, a setting not commonly seen in the literature. Moreover, in order to simulate partial preimages, we borrow ideas of the BASIS trapdoor sampling technique (but, critically, not the BASIS assumption) from [WW23] originally used for constructing functional commitments, demonstrating the usefulness of such techniques also in security proofs.

Related work. This work presents an algebraic method for non-interactively distributing lattice preimage sampling. To the best of our knowledge, the only existing technique allowing the latter is the universal thresholdiser [BGG+18], which requires homomorphically evaluating the trapdoor preimage sampling algorithm.

The work of [BKP13] considers sharing a lattice trapdoor using techniques from multi-party computation (MPC). This work exploits the linearity of trapdoor sampling in [MP12], which enables linear-secret sharing if certain perturbation-correction vectors are pre-computed (and consumed per signature).[2] However, the approach requires the secret dealer to pre-compute as many perturbation vectors as the number of preimages to be sampled in a distributed way, meaning that either the number of preimages to be given out is bounded or the secret dealer needs to participate periodically. Their second protocol removes the trusted setup, but requires generic MPC for some heavy offline interactive computation.

A recent line of work [GKS24, DKM+24, EKT24, BKL+25] studied constructions of lattice-based threshold signatures. The constructions thresholdising hash-then-sign signatures can be seen as threshold preimage sampling. They may provide smaller parameters, but all of them require two, three rounds of interaction.

On the high level, [Wee21] proposed the notion of half trapdoors of the form $\mathbf{A} \cdot \mathbf{T}_{1/2} = [\mathbf{0} \| \mathbf{G}]$; thus, $\mathbf{T}_{1/2}$ matches $\mathbf{T}_1$ in our introduction. It was shown in [Wee21] that LWE w.r.t $\mathbf{A}$ remains hard in the presence of an oracle outputting certain preimages sampled using $\mathbf{T}_{1/2}$. In this work, we consider stronger properties: we hand out the partial trapdoor $\mathbf{T}_1$ and also simulate preimage queries for the image space of $\mathbf{T}_0$. Lastly, in [PT24] "subspace trapdoors" are considered, but in the context of cryptanalysis.

[2] This is somewhat analogous to Beaver triples in generic MPC applications.

Open problems. For both of our threshold signature and IBE, we only prove a notion that we dub "highly-selective security" where the adversary has to provide all of its queries upfront. This is because, for query message μ (resp. identity id), our way of programming the random oracle value $\mathsf{H}(\mu)$ (resp. $\mathsf{H}(\mathsf{id})$) depends on the set of participating parties T in the corresponding partial signature (T, μ) (resp. identity key (T, id)) query.

Moreover, similar to GPV we do not prove security against the same message (resp. identity) being queried more than once. As in GPV we may address this limitation by de-randomising the signing algorithm when the same message is queried again for the same signing set T. If the same message is queried but for different signing sets T_1, T_2 then we are unable to argue security (and our security experiment forbids it).

In the setting of signatures, an easy fix to both issues is to sample preimages of $\mathsf{H}(\mu, T)$ and to derandomise the sampling algorithm. This change would recover the usual notion of adaptive security with static corruptions for threshold signatures. A similar fix does not apply in IBE since in general the encrypter does not know which set of parties T will be decrypting the message.

In the current protocol, the shares of each party can be checked individually and parties with invalid shares can be banned. This property guarantees a type of robustness known as Identifiable Aborts. An interesting direction for future work is achieving full robustness of the signature scheme, i.e. being able to handle participating sets of $> T$ parties and guarantee correct reconstruction when at least T parties are honest. We could plausibly use techniques of error correction for Shamir secret sharing schemes and subtractive sets [AL21].

Chapter structure. We start with Section 5.1 we formally define the syntax and security of a threshold partial lattice trapdoor, together with its instantiation and correctness analysis in Section 5.2. Sections 5.3 and 5.4 are devoted to a number of lemmas and their proofs, which will be useful for proving security (one-wayness and indistinguishability) in Sections 5.5 and 5.6. Lastly, in Section 5.7 we showcase the gadget by building a threshold signature and a threshold IBE scheme.

Chapter specific notations. For a set $\mathsf{act} \subset [N]$ and any $j \in \mathsf{act}$, denote by $\mathbf{e}_{\mathsf{act},j} \in \{0,1\}^T$ the unit-vector that corresponds to the index of j in act when ordered in increasing order. We denote a matrix $(1, \ldots, 1)^\intercal \in \mathcal{R}^T$ as $\mathbf{1}_T$. For a T-subset $S \subset \mathcal{R}_q$ we write $\mathbf{V}_S$ for the "Vandermonde-style" matrix where the jth column is $(v_{c_j}^0, \ldots, v_{c_j}^{T-1})^\intercal$ and v_{c_j} is the jth entry in S. When the modulus q is clear from context we use $a \equiv b$ as

shorthand for $a = b \bmod q$. Recall that Gaussians over rank-deficient lattices are defined as in Remark 2. For full-rank $\mathbf{U} \in \mathbb{R}^{m \times n}$, $m \geq n$ with SVD $\mathbf{U} = \mathbf{V}_0 \cdot [\mathbf{D} || \mathbf{0}_{(m-n)\times n}] \cdot \mathbf{V}_1$ and $\mathbf{S} \in (\mathbb{R}^{n \times n})^{\times}$ we have $\mathcal{D}_{\Lambda(\mathbf{U}),\mathbf{S}_U \cdot \mathbf{S}} \coloneqq \mathbf{U} \cdot \mathcal{D}_{\mathbb{Z}^n,(\mathbf{V}_1^{\intercal} \cdot \mathbf{D}^{-1}) \cdot \mathbf{S}}$.

Algebraic structure. In this chapter we assume that, $\mathcal{R}_q$ splits into fields $\mathcal{R}_{\mathfrak{q}_j} \coloneqq \mathcal{R}/\mathfrak{q}_j$ with $|\mathcal{R}_{\mathfrak{q}_j}| \in \lambda^{\omega(1)}$, where λ is the security parameter. We note that this is a non-trivial assumption. We make use of this assumption, to apply (a) a regularity lemma for uniformly random matrices $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ (Theorem 3 and Lemma 57) and to argue that (b) some $\mathbf{A}, \{\mathbf{C}_j\}$ sampled below are surjective with high probability. In both cases, our assumption can be waived by conditioning on such that $\mathbf{A}, \{\mathbf{C}_j\}$ contain invertible submatrices. The module lattices we consider in this chapter are embedded into the real space with respect to canonical embedding denoted $\theta(\cdot)$.

5.1 Partial Lattice Trapdoors: Definitions

In this section we define the syntax and the security properties of partial trapdoors. However, we first define the cryptographic assumptions we use for our scheme. The security of our trapdoor sharing and its applications rely on versions of k-MSIS and k-MLWE defined in Definitions 4 and 5 and mentioned in Chapter 4. We slightly generalise these assumptions to cover hints with a varied gaussian parameter.

Intuitively, the security arguments in this chapter require one half of the hints to be more 'narrow' than the other to cover for the subsequent norm blow-up.

Definition 26 (k-MSIS Assumption With Varied Hints.)**.** *Let $d, n, m, \log q, k, \{\mathfrak{s}_{\max}(\mathbf{S}_i)\}_i$, $\beta \in \mathsf{poly}(\lambda)$ with $n, k \leq m$. An instance of $\mathsf{k\text{-}MSIS}_{\mathcal{R},q,n,m,\{\mathbf{S}_i\}_i,\beta}$ problem is a uniform matrix $\mathbf{A} \leftarrow \mathcal{R}_q^{n \times m}$ and k vectors $\mathbf{e}_i \leftarrow \mathcal{D}_{\Lambda_q^{\perp}(\mathbf{A}),\mathbf{S}_i}$ for $i \in [k]$ and arbitrary full-rank matrices $\mathbf{S}_i \in \mathbb{R}^{dm \times dm}$. A solution to this problem is a vector $\mathbf{v} \in \mathcal{R}_q^m$ such that: (1) $\mathbf{A} \cdot \mathbf{v} \equiv \mathbf{0} \bmod q$, (2) $0 < \|\mathbf{v}\| \leq \beta$, (3) $\mathbf{v} \notin \mathcal{K}\text{-span}(\mathbf{e}_0, \ldots, \mathbf{e}_{k-1})$. The $\mathsf{k\text{-}MSIS}_{\mathcal{R},q,n,m,k,\{\mathbf{S}_i\}_i,\beta}$ assumption states that any PPT $\mathcal{A}$ finds a solution with probability $\leq \mathsf{negl}(\lambda)$.*

Definition 27 (k-MLWE Assumption With Varied Hints.)**.** *Let $d, n, m, \log q, k, \sigma$, $\{\mathfrak{s}_{\max}(\mathbf{S}_i)\}_i \in \mathsf{poly}(\lambda)$ with $n, k \leq m$. Let $\{\mathbf{S}_i\}_i$ be a set of arbitrary full-rank matrices $\mathbf{S}_i \in \mathbb{R}^{dm \times dm}$. The $\mathsf{k\text{-}MLWE}_{\mathcal{R},q,n,m,k,\{\mathbf{S}_i\}_i,\mathcal{U}(\mathcal{R}_q),\mathcal{D}_{\mathcal{R},\sigma}}$ assumption states that for any*

PPT $\mathcal{A}$, the following distributions over $(\mathbf{A}, \mathbf{U} = [\mathbf{u}_0| \ldots, |\mathbf{u}_{k-1}], \mathbf{b})$ are indistinguishable:

$$\left\{\begin{array}{r} \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m} \\ \{\mathbf{u}_i\}_i \leftarrow \left\{\mathcal{D}_{\Lambda_q^\perp(\mathbf{A}),\mathbf{S}_i}\right\}_{i=0}^{k-1} \\ \mathbf{s} \leftarrow \mathcal{R}_q^n,\ \mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^m \\ \mathbf{b}^\intercal \coloneqq \mathbf{s}^\intercal \cdot \mathbf{A} + \mathbf{e}^\intercal \bmod q \end{array}\right\} \quad \textit{and} \quad \left\{\begin{array}{r} \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m} \\ \{\mathbf{u}_i\}_i \leftarrow \left\{\mathcal{D}_{\Lambda_q^\perp(\mathbf{A}),\mathbf{S}_i}\right\}_{i=0}^{k-1} \\ \mathbf{b} \leftarrow \mathcal{U}(\left\{\mathbf{x}|\mathbf{x}^\intercal \cdot \mathbf{U} = \mathbf{0} \bmod q\right\}) + \mathcal{D}_{\mathcal{R},\sigma}^m \end{array}\right\}.$$

Remark 19. The reduction in Chapter 4 proves that k-MSIS and k-MLWE have a reduction from the respective standard assumptions when $\mathbf{S}_i = \mathbf{S} = \mathsf{diag}(\sigma \cdot \mathbf{I}, \sigma' \cdot \mathbf{I})$. We believe it is possible to adapt arguments in Chapter 4 to two sets of hints with parameters $\mathbf{S}_{\mathtt{Hon}} = \mathsf{diag}(\sigma_0 \cdot \mathbf{I}, \sigma_0' \cdot \mathbf{I})$ and $\mathbf{S}_{\mathtt{Cor}} = \mathsf{diag}(\sigma_1 \cdot \mathbf{I}, \sigma_1' \cdot \mathbf{I})$. However, we leave this adaptation for future work.

Definition 28 (Partial Lattice Trapdoors)**.** *A (T, N)-threshold partial lattice trapdoor scheme over $\mathcal{R}_q$ is an extension of a (full) lattice trapdoor scheme* $(\mathsf{TrapGen}, \mathsf{SamplePreMP})$ *over $\mathcal{R}_q$ with the PPT algorithms* $(\mathsf{PTrapGen}, \mathsf{PSampPre}, \mathsf{TSIG.Combine})$*:*

$\mathsf{ptd}_j \leftarrow \mathsf{PTrapGen}(\mathbf{A}, \mathsf{td}, j)$*: The partial trapdoor generation algorithm inputs a matrix $\mathbf{A}$, its (full) trapdoor* td *and an index $j \in [N]$, and generates a partial trapdoor* ptd_j *of $\mathbf{A}$ for j.*

$\mathbf{x}_j \leftarrow \mathsf{PSampPre}\left(\mathbf{A}, \mathsf{ptd}_j, \mathsf{act}, \mathbf{y}, \sigma\right)$*: The partial preimage sampling algorithm, given a partial trapdoor* ptd_j *for index j, a set* $\mathsf{act} \subset [N]$*, a target image $\mathbf{y}$, and a Gaussian parameter σ, samples a partial preimage $\mathbf{x}_j$ for j.*

$\mathbf{x} \leftarrow \mathsf{TSIG.Combine}\left((\mathbf{x}_j)_{j\in\mathsf{act}}\right)$*: The reconstruction algorithm, given a tuple of partial preimages $(\mathbf{x}_j)_{j\in\mathsf{act}}$, reconstructs a preimage $\mathbf{x}$.*

Definition 29 (Correctness)**.** *Let $N, n, m, \tilde{m}, T \in \mathbb{N}$ with $T \leq N$ and $n \leq m$, and $\tilde{s}, \beta, \sigma > 0$. A (T, N)-threshold partial lattice trapdoor scheme is said to be $(n, m, \tilde{s}, \beta, \sigma)$-correct if the underlying lattice trapdoor scheme* $(\mathsf{TrapGen}, \mathsf{SamplePreMP})$ *is $(n, m, \tilde{s})$-correct, and for any T-subset* $\mathsf{act} \subset [N]$ *and vector $\mathbf{y} \in \mathcal{R}_q^n$, it holds that*

$$\Pr\left[\begin{array}{r|l} \mathbf{A} \cdot \mathbf{x} \equiv \mathbf{y} & (\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m) \\ \wedge\ \|\mathbf{x}\| \leq \beta & \mathsf{ptd}_j \leftarrow \mathsf{PTrapGen}(\mathbf{A}, \mathsf{td}, j)\ \forall j \in [N] \\ & \mathbf{x}_j \leftarrow \mathsf{PSampPre}\left(\mathbf{A}, \mathsf{ptd}_j, \mathsf{act}, \mathbf{y}, \sigma\right)\ \forall j \in \mathsf{act} \\ & \mathbf{x} \leftarrow \mathsf{TSIG.Combine}\left((\mathbf{x}_j)_{j\in\mathsf{act}}\right) \end{array}\right] \geq 1 - \mathsf{negl}(\lambda).$$

$\underline{\mathsf{Exp}^{\mathtt{x},b}_{\mathcal{A},\mathsf{par}_{\mathtt{x}}}(1^\lambda)}$

$\mathsf{cor} \leftarrow \mathcal{A}(1^\lambda) \quad /\!/$ set cor of corrupt parties
assert $|\mathsf{cor}| \leq T-1$
$(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m)$
for $j \in [N]:\ \mathsf{ptd}_j \leftarrow \mathsf{PTrapGen}(\mathbf{A}, \mathsf{td}, j)$
$(\mathsf{act}^*, i^*) \leftarrow \mathcal{A}^{\mathsf{PSampPreO}(\cdot)}\left(\mathbf{A}, \left(\mathsf{ptd}_j\right)_{j\in\mathsf{cor}}\right)$
$(\mathsf{hint}, \mathbf{y}^*) \leftarrow \mathsf{SISChalO}(\mathsf{act}^*, i^*)$
if $\mathtt{x} = \mathtt{OW}$:
  $\mathbf{x}^* \leftarrow \mathcal{A}^{\mathsf{PSampPreO}(\cdot)}(\mathsf{hint}, \mathbf{y}^*)$
  $b_0 \coloneqq (\mathbf{A}\cdot\mathbf{x}^* = \mathbf{y}^* \bmod q)$
  $b_1 \coloneqq (\|\mathbf{x}^*\| \leq \beta)$
  return $b_0 \wedge b_1$
if $\mathtt{x} = \mathtt{IND}, b \in \{0,1\}$:
  $(\mathbf{c}_0, c_1) \leftarrow \mathsf{LWEChalO}(\mathbf{A}, \mathbf{y}^*, b)$
  $b' \leftarrow \mathcal{A}^{\mathsf{PSampPreO}(\cdot)}(\mathsf{hint}, \mathbf{y}^*, \mathbf{c}_0, c_1)$
  return b'

$\underline{\mathsf{SISChalO}(\mathsf{act}^* \subset [N], i^*)}$

assert $i^* \in \mathsf{act}^* \setminus \mathsf{cor}$
$\mathbf{y}^* \leftarrow \mathcal{R}_q^n$
for $j \in \mathsf{act}^* \setminus \{i^*\}$:
  $\mathbf{x}_j^* \leftarrow \mathsf{PSampPre}\left(\mathbf{A}, \mathsf{ptd}_j, \mathsf{act}^*, \mathbf{y}^*, \sigma\right)$
$\mathsf{hint} \coloneqq \left(\mathbf{x}_j^*\right)_{j\in\mathsf{act}^*\setminus\{i^*\}}$
return $(\mathsf{hint}, \mathbf{y}^*)$

$\underline{\mathsf{LWEChalO}(\mathbf{A}, \mathbf{y}^*, b)}$

$\mathbf{s} \leftarrow \mathcal{R}_q^n,\ \mathbf{e} \leftarrow \chi^m$
$\mathbf{c}_0 \coloneqq \mathbf{s}^\mathsf{T}\cdot\mathbf{A} + \mathbf{e}^\mathsf{T} \bmod q$
if $b = 0:\ e \leftarrow \tilde{\chi};\ c_1 \coloneqq \mathbf{s}^\mathsf{T}\cdot\mathbf{y}^* + e \bmod q$
if $b = 1:\ c_1 \leftarrow \mathcal{R}_q$
return $(\mathbf{c}_0, c_1)$

$\underline{\mathsf{PSampPreO}(\mathsf{act} \subset [N])}$

$\mathbf{y} \leftarrow \mathcal{R}_q^n$
for $j \in \mathsf{act}:\ \mathbf{x}_j \leftarrow \mathsf{PSampPre}\left(\mathbf{A}, \mathsf{ptd}_j, \mathsf{act}, \mathbf{y}, \sigma\right)$
return $\left(\left(\mathbf{x}_j\right)_{j\in\mathsf{act}}, \mathbf{y}\right)$

Figure 5.1: Security and indistinguishability experiments for partial lattice trapdoor.

Definition 30 (One-wayness and Indistinguishability)**.** *Let $N, n, m, T \in \mathbb{N}$ with $T \leq N$ and $n \leq m$, and $\beta, \chi_0, \chi_1, \sigma > 0$. Denote $\mathsf{par}_{\mathtt{OW}} = (n, m, \beta, \sigma)$ and $\mathsf{par}_{\mathtt{IND}} = (n, m, \chi_0, \chi_1, \sigma)$. A (T, N)-threshold partial lattice trapdoor scheme is said to be $\mathsf{par}_{\mathtt{OW}}$-one-way, if for any PPT $\mathcal{A}$ it holds that*

$$\Pr\left[\mathsf{Exp}^{\mathtt{OW}}_{\mathcal{A},\mathsf{par}_{\mathtt{OW}}}(1^\lambda) = 1\right] \leq \mathsf{negl}(\lambda),$$

and it is said to have $\mathsf{par}_{\mathtt{IND}}$-indistinguishability, if for any PPT $\mathcal{A}$

$$\left|\Pr\left[\mathsf{Exp}^{\mathtt{IND},0}_{\mathcal{A},\mathsf{par}_{\mathtt{IND}}}(1^\lambda) = 1\right] - \Pr\left[\mathsf{Exp}^{\mathtt{IND},1}_{\mathcal{A},\mathsf{par}_{\mathtt{IND}}}(1^\lambda) = 1\right]\right| \leq \mathsf{negl}(\lambda),$$

where the experiments $\mathsf{Exp}^{\mathtt{x},b}_{\Pi,\mathcal{A},\mathsf{par}_{\mathtt{x}}}$ for $\mathtt{x} \in \{\mathtt{OW}, \mathtt{IND}\}$ and $b \in \{0,1\}$ are as in Figure 5.1.

$\mathsf{PTrapGen}(\mathbf{A} \in \mathcal{R}_q^{nT \times 2mT}, \mathsf{td}, j)$

$(\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m)$
$\mathbf{U}_j \leftarrow \mathsf{SamplePreMP}(\mathbf{A}, \mathsf{td}, \mathbf{v}_j \otimes \mathbf{C}_j, \mathbf{S})$
if $\mathsf{rank}_{\mathcal{K}}(\mathbf{U}_j) \neq m$: **return** $\perp$
$\mathsf{ptd}_j \coloneqq (\mathbf{U}_j \in \mathcal{R}^{2mT \times m}, \mathbf{C}_j \in \mathcal{R}_q^{n \times m}, \mathsf{td}_{\mathbf{C}_j})$
return ptd_j

$\mathsf{PSampPre}(\mathbf{A}, \mathsf{ptd}_j, \mathsf{act}, \mathbf{y} \in \mathcal{R}_q^{nT}, \sigma)$

$\mathbf{V}_{\mathsf{act}} \coloneqq (\mathbf{v}_j)_{j \in \mathsf{act}} \in \mathcal{R}_q^{T \times T}$
$\mathbf{z}_j \coloneqq \left((\mathbf{e}_{\mathsf{act},j}^\intercal \cdot \mathbf{V}_{\mathsf{act}}^{-1}) \otimes \mathbf{I}_n\right) \cdot \mathbf{y} \bmod q \in \mathcal{R}_q^n$
// Decomp. $\mathbf{y} = \sum_{j \in \mathsf{act}} \mathbf{v}_j \otimes \mathbf{z}_j \bmod q$ is unique
$\mathbf{V}_{0,j} \cdot [\mathbf{D}_j || \mathbf{0}] \cdot \mathbf{V}_{1,j} \coloneqq \mathsf{ptd}_j$
// Singular Value Decomposition of ptd_j
$\sqrt{\mathbf{\Sigma}_j} \coloneqq \sigma \cdot \tilde{\theta}(\mathbf{V}_{1,j}^\intercal \cdot \mathbf{D}_j^{-1}) \in \mathbb{R}^{md \times md}$
$\mathbf{d}_j \leftarrow \mathsf{SamplePreMP}(\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}, \mathbf{z}_j, \sqrt{\mathbf{\Sigma}_j})$
return $\mathbf{x}_j \coloneqq \mathbf{U}_j \cdot \mathbf{d}_j \in \mathcal{R}^{2mT}$

Figure 5.2: Partial lattice trapdoor construction with $\mathsf{TSIG.Combine}((\mathbf{x}_j)_{j \in \mathsf{act}}) = \sum_{j \in \mathsf{act}} \mathbf{x}_j$.

5.2 Construction

Let $T, n, \log q \in \mathsf{poly}(\lambda)$ and $m = n \cdot (\lceil \log q \rceil + 1)$. Let $\{v_0, v_1, \ldots, v_{k-1}\} \subseteq \mathcal{R}_q^\times$ be an arbitrary set such that $v_i - v_j \in \mathcal{R}_q^\times$ for all $i \neq j$. In our setting this holds with overwhelming probability for random v_i and v_j. More generally, this can be achieved, for example, by choosing v_j with small $\|v_j\|$ for all j [ACX21]. For $j \in [N]$, let $\mathbf{v}_j \coloneqq (1, v_j, \ldots, v_j^{T-1})$. Let $\mathbf{e}_{\mathsf{act},j} \in \{0,1\}^T$ the unit vector that corresponds to the index of j in the set $\mathsf{act} \subset [N]$ when ordered in increasing order. This vector will be used in the scheme to select a column of a $T \times T$ matrix that corresponds to party j. Let $(\mathsf{TrapGen}, \mathsf{SamplePreMP})$ be a lattice trapdoor scheme, e.g. that specified in Lemma 50, which is $(nT, 2mT, \tilde{s})$-correct with some $\tilde{s} > 0$. In Figure 5.2 we construct a (T, N)-threshold partial lattice trapdoor scheme, where we let $\mathbf{S} \in \mathbb{R}^{2dmT \times 2dmT}$ be Gaussian parameters hardwired in the $\mathsf{PTrapGen}$ algorithm.

Theorem 11 (Correctness)**.** *Let parameters be as in Tables 5.2 and 5.3, in particular $\beta \geq T \cdot \sigma \cdot \sqrt{md}$. The partial lattice trapdoor in Figure 5.2 is $(nT, 2mT, \tilde{s}, \beta, \sigma)$-correct.*

Proof. With overwhelming probability, matrices $\mathbf{U}_j$ sampled in $\mathsf{PTrapGen}$ have full $\mathcal{K}$-rank by Theorem 12, hence with overwhelming probability $\mathsf{PTrapGen}$ does not abort. We assume this is the case in the rest. For each $j \in \mathsf{act}$, the output $\mathbf{x}_j$ of $\mathsf{PSampPre}(\mathbf{A}, \mathsf{ptd}_j, \mathsf{act}, \mathbf{y}, \sigma)$ satisfies

$$\mathbf{A} \cdot \mathbf{x}_j = \mathbf{A} \cdot \mathbf{U}_j \cdot \mathbf{d}_j = (\mathbf{v}_j \otimes \mathbf{C}_j) \cdot \mathbf{d}_j = \mathbf{v}_j \otimes (\mathbf{C}_j \cdot \mathbf{d}_j) = \mathbf{v}_j \otimes \mathbf{z}_j \bmod q.$$

Observe that the vertical concatenation of $\mathbf{z}_j$ (ordered according to act) yields

$$\begin{bmatrix} \vdots \\ \mathbf{z}_j \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ ((\mathbf{e}_{\mathsf{act},j}^{\mathsf{T}} \cdot \mathbf{V}_{\mathsf{act}}^{-1}) \otimes \mathbf{I}_n) \cdot \mathbf{y} \\ \vdots \end{bmatrix} = (\mathbf{V}_{\mathsf{act}}^{-1} \otimes \mathbf{I}_n) \cdot \mathbf{y} = (\mathbf{V}_{\mathsf{act}} \otimes \mathbf{I}_n)^{-1} \cdot \mathbf{y} \bmod q.$$

Hence, the reconstructed vector $\mathbf{x} = \sum_{j \in \mathsf{act}} \mathbf{x}_j$ satisfies

$$\begin{aligned} \mathbf{A} \cdot \mathbf{x} &= \sum_{j \in \mathsf{act}} \mathbf{A} \cdot \mathbf{x}_j = \sum_{j \in \mathsf{act}} \mathbf{v}_j \otimes \mathbf{z}_j = \sum_{j \in \mathsf{act}} (\mathbf{V}_{\mathsf{act}} \otimes \mathbf{I}) \cdot (\mathbf{e}_{\mathsf{act},j} \otimes \mathbf{z}_j) \\ &= (\mathbf{V}_{\mathsf{act}} \otimes \mathbf{I}) \cdot \begin{bmatrix} \vdots \\ \mathbf{z}_j \\ \vdots \end{bmatrix} = (\mathbf{V}_{\mathsf{act}} \otimes \mathbf{I}) \cdot (\mathbf{V}_{\mathsf{act}} \otimes \mathbf{I})^{-1} \cdot \mathbf{y} = \mathbf{y} \bmod q. \end{aligned}$$

It remains to prove that $\mathbf{x}$ has norm bounded by β. For any $j \in [N]$, $\mathbf{C}_j$ generated by $\mathsf{TrapGen}$ is surjective with overwhelming probability by Lemma 50 and Remark 1. Conditioned on this, the distribution of $\mathbf{x}_j$ is identical to $\mathcal{D}_{\Lambda(\mathbf{U}_j) \cap \Lambda_q^{\mathbf{v} \otimes \mathbf{z}_j}(\mathbf{A}), \sigma \cdot \mathbf{S}_U} = \mathbf{U}_j \cdot \mathcal{D}_{\mathbb{Z}^{md}, \sigma \cdot \tilde{\theta}(\mathbf{V}_{1,j}^{\mathsf{T}} \cdot \mathbf{D}_j^{-1})}$ by Lemma 91. Then by the Gaussian tail bound from Corollary 2, $\|\mathbf{x}_j\| \leq \mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U}_j)) \cdot \sigma \cdot \mathfrak{s}_{\min}(\tilde{\theta}(\mathbf{V}_{1,j}^{\mathsf{T}} \cdot \mathbf{D}_j^{-1})) \cdot \sqrt{md} = \sigma \cdot \sqrt{md}$ with overwhelming probability. Hence, after applying the union bound, $\Pr[\|\mathbf{x}\| \geq T \cdot \sigma \cdot \sqrt{md}] \leq T \cdot \mathsf{negl}(\lambda)$. □

5.3 Useful Lemmas

In this section, we provide several lemmas which will be useful in arguing security of our partial trapdoor scheme. The techniques used here may be of independent interest. Using these, in Section 5.4 we establish statistical closeness of a sequence of distributions. The distributions we consider help us simulate the security experiment in Sections 5.5 and 5.6 and prove our main theorems on the one-wayness and indistinguishability of our partial lattice trapdoor.

Lemma 91 states that, for $\mathbf{U}$ satisfying $\mathbf{A} \cdot \mathbf{U} = \mathbf{v} \otimes \mathbf{C} \bmod q$ and an appropriately distributed vector $\mathbf{d}$, the distribution $\mathbf{U} \cdot \mathbf{d}$ is a discrete Gaussian over the intersection of the lattice $\Lambda(\mathbf{U})$ and the coset $\Lambda_q^{\mathbf{v} \otimes \mathbf{z}}(\mathbf{A})$. Recall that Gaussians over rank-deficient lattices are defined as in Remark 2. Intuitively, the following lemmas build the theoretical background for working with rank-deficient trapdoors.

Lemma 91 (Sampling from the intersection)**.** *Let $\mathbf{A} \in \mathcal{R}_q^{nT \times 2mT}$, and $\sigma > 0$. Let $\mathbf{C} \in \mathcal{R}_q^{n \times m}$ and $\mathbf{v} \in \mathcal{R}_q^T$ such that $\mathbf{C}$ is surjective. Let $\mathbf{U} = \mathbf{V}_0 \cdot [\mathbf{D} || \mathbf{0}_{m(2T-1) \times m}] \cdot \mathbf{V}_1 \in \mathcal{K}^{2mT \times m}$*

(where $||$ denotes vertical stacking) for orthogonal matrices $\mathbf{V}_0, \mathbf{V}_1$ and a full-rank diagonal matrix $\mathbf{D} \in \mathcal{K}^{m\times m}$ be the SVD of $\mathbf{U}$. Let $\mathbf{U}$ also satisfy $\mathbf{A}\cdot\mathbf{U} = \mathbf{v}\otimes\mathbf{C} \bmod q$. Let $\sqrt{\mathbf{\Sigma}} \coloneqq \tilde{\theta}(\sigma\cdot(\mathbf{V}_1^\intercal\cdot\mathbf{D}^{-1})) \in \mathbb{R}^{md\times md}$. Then for any $\mathbf{z}\in\mathcal{R}_q^n$, the following distributions are identical:

$$\Big\{\mathbf{x} \;\Big|\; \mathbf{d}\leftarrow\mathcal{D}_{\Lambda_q^{\mathbf{z}}(\mathbf{C}),\sqrt{\mathbf{\Sigma}}};\ \mathbf{x}\coloneqq\mathbf{U}\cdot\mathbf{d}\Big\} \qquad \textit{and} \qquad \mathcal{D}_{\Lambda(\mathbf{U})\cap\Lambda_q^{\mathbf{v}\otimes\mathbf{z}}(\mathbf{A}),\sigma\cdot\mathbf{S}_U}.$$

Proof. Since $\mathbf{U}$ is full-rank over $\mathcal{K}$ it is injective over the set $\Lambda_q^{\mathbf{z}}(\mathbf{C})$. Then by Lemma 18 for $\mathbf{d}\leftarrow\mathcal{D}_{\Lambda_q^{\mathbf{z}}(\mathbf{C}),\tilde{\theta}(\sigma\cdot(\mathbf{V}_1^\intercal\cdot\mathbf{D}^{-1}))}$ we have the product distribution

$$\mathbf{U}\cdot\mathbf{d} \sim \mathcal{D}_{\mathbf{U}\cdot\Lambda_q^{\mathbf{z}}(\mathbf{C}),\tilde{\theta}(\mathbf{U})\cdot\tilde{\theta}(\sigma\cdot(\mathbf{V}_1^\intercal\cdot\mathbf{D}^{-1}))} \sim \mathcal{D}_{\mathbf{U}\cdot\Lambda_q^{\mathbf{z}}(\mathbf{C}),\tilde{\theta}(\sigma\cdot\mathbf{S}_U)}.$$

It remains to show that $\mathbf{U}\cdot\Lambda_q^{\mathbf{z}}(\mathbf{C}) = \Lambda(\mathbf{U})\cap\Lambda_q^{\mathbf{v}\otimes\mathbf{z}}(\mathbf{A})$. Observe

$$\begin{aligned}\mathbf{A}\cdot\mathbf{U}\cdot\mathbf{d} = (\mathbf{v}\otimes\mathbf{C})\cdot\mathbf{d} &= \mathbf{v}\otimes(\mathbf{C}\cdot\mathbf{d}) \bmod q\\ &= \mathbf{v}\otimes\mathbf{z} \bmod q\end{aligned}$$

Hence, $\mathbf{U}\cdot\Lambda_q^{\mathbf{z}}(\mathbf{C}) \subseteq \Lambda(\mathbf{U})\cap\Lambda_q^{\mathbf{v}\otimes\mathbf{z}}(\mathbf{A})$. Similarly in the other direction, for any element $\mathbf{x}\in\Lambda(\mathbf{U})\cap\Lambda_q^{\mathbf{v}\otimes\mathbf{z}}(\mathbf{A})$ there exists $\mathbf{y}\in\mathcal{R}^m$ s.t. $\mathbf{x}=\mathbf{U}\cdot\mathbf{y}$. For $\mathbf{y}$ it holds $\mathbf{v}\otimes\mathbf{z} = \mathbf{A}\cdot\mathbf{x} = \mathbf{v}\otimes(\mathbf{C}\cdot\mathbf{y}) \bmod q$ implying $\mathbf{y}\in\Lambda_q^{\mathbf{z}}(\mathbf{C})$ and $\mathbf{x}\in\mathbf{U}\cdot\Lambda_q^{\mathbf{z}}(\mathbf{C})$. □

Proposition 4 is a regularity lemma analogous to the classic result of [GPV08].

Proposition 4 (Image distribution)**.** *Fix any surjective $\mathbf{A}\in\mathcal{R}_q^{nT\times 2mT}$, any $v\in\mathcal{R}_q^\times$, and $\sqrt{\mathbf{\Sigma}}\in\mathbb{R}^{2mT\times 2mT}$. Denote $\mathbf{v}\coloneqq(1,v,\ldots,v^{T-1})^\intercal$. If $\sqrt{\mathbf{\Sigma}}\geq\eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ for some $\varepsilon\leq\mathsf{negl}(\lambda)$ then the following distributions (parametrised by $(\mathbf{A},v)$) are statistically close in λ.*

$$\begin{aligned}\mathcal{D}_0 &= \Big\{\mathbf{U} \;\Big|\; \mathbf{U}\leftarrow(\mathcal{D}_{\mathcal{R}^{2mT},\sqrt{\mathbf{\Sigma}}})^m : \mathbf{A}\cdot\mathbf{U}\in(\mathcal{R}_q\text{-span}(\mathbf{v}\otimes\mathbf{I}))^m\Big\},\\ \mathcal{D}_1 &= \Big\{\mathbf{U} \;\Big|\; \mathbf{C}\leftarrow\mathcal{R}_q^{n\times m};\ \mathbf{U}\leftarrow(\mathcal{D}_{\mathcal{R}^{2mT},\sqrt{\mathbf{\Sigma}}})^m : \mathbf{A}\cdot\mathbf{U}=\mathbf{v}\otimes\mathbf{C} \bmod q\Big\}.\end{aligned}$$

Proof. Define the lattice Λ as follows and consider

$$\begin{aligned}\Lambda &\coloneqq \Big\{\mathbf{u}\in\mathcal{R}^{2mT} : \exists\,\mathbf{z}\in\mathcal{R}_q^n,\ \mathbf{A}\cdot\mathbf{u}=\mathbf{v}\otimes\mathbf{z} \bmod q\Big\},\\ \Lambda_q^\perp(\mathbf{A}) &\coloneqq \Big\{\mathbf{u}\in\mathcal{R}^{2mT} : \mathbf{A}\cdot\mathbf{u}=\mathbf{0} \bmod q\Big\},\\ \Lambda_q^{\mathbf{v}\otimes\mathbf{c}}(\mathbf{A}) &\coloneqq \Big\{\mathbf{u}\in\mathcal{R}^{2mT} : \mathbf{A}\cdot\mathbf{u}=\mathbf{v}\otimes\mathbf{c} \bmod q\Big\}\end{aligned}$$

for some $\mathbf{c}\in\mathcal{R}_q^n$. Sampling the columns of $\mathbf{U}$ with the span condition is equivalent to sampling them from Λ. We prove that $\mathbf{A}\cdot\mathcal{D}_{\Lambda,\sqrt{\mathbf{\Sigma}}}$ follows the uniform distribution in the

required subspace. Since $\mathbf{A}$ is surjective, the image space satisfies

$$\begin{aligned}\left\{\mathbf{A}\cdot\mathbf{u} \bmod q : \mathbf{u}\in\mathcal{R}^{2mT}\right\} &= \mathcal{R}_q^{nT},\\ \{\mathbf{A}\cdot\mathbf{u} \bmod q : \mathbf{u}\in\Lambda\} &= \left\{\mathbf{v}\otimes\mathbf{z} \bmod q : \mathbf{z}\in\mathcal{R}_q^{n}\right\}.\end{aligned}$$

Multiplication by $\mathbf{A}:\Lambda\to\mathcal{R}_q^{nT}$ is an isomorphism from $\Lambda/\Lambda_q^{\perp}(\mathbf{A})$ to the image space $\left\{\mathbf{v}\otimes\mathbf{z} \bmod q : \mathbf{z}\in\mathcal{R}_q^{n}\right\}$. Besides this, $\mathfrak{s}_{\min}(\sqrt{\boldsymbol{\Sigma}}) \geq \eta_\varepsilon(\Lambda_q^{\perp}(\mathbf{A}))$, hence by Lemma 39, $\mathcal{D}_{\Lambda,\sqrt{\boldsymbol{\Sigma}}} \bmod \Lambda_q^{\perp}(\mathbf{A})$ is statistically close to $\mathcal{U}(\Lambda/\Lambda_q^{\perp}(\mathbf{A}))$. Then, we have $\mathbf{A}\cdot(\mathcal{D}_{\Lambda,\sqrt{\boldsymbol{\Sigma}}} \bmod \Lambda_q^{\perp}(\mathbf{A})) \bmod q$ is statistically close to $\mathcal{U}(\left\{\mathbf{v}\otimes\mathbf{z} \bmod q : \mathbf{z}\in\mathcal{R}_q^{n}\right\}) = \mathbf{v}\otimes\mathcal{U}(\mathcal{R}_q^{n})$ as in $\mathcal{D}_1$. The claim follows by repeating the above argument for each column of $\mathbf{C}$. □

For a matrix $\mathbf{U}$ satisfying $\mathbf{A}\cdot\mathbf{U} = \mathbf{v}\otimes\mathbf{C} \bmod q$, Proposition 5 gives a bound on the smoothing parameter of the intersection lattice $\Lambda(\mathbf{U})\cap\Lambda_q^{\perp}(\mathbf{A})$.

Proposition 5 (Smoothing of intersection)**.** *Let $\varepsilon > 0$, $\mathbf{v}\in\mathcal{R}_q^{T}$ be a vector with unit coefficients, $\mathbf{A}\in\mathcal{R}_q^{nT\times 2mT}$, $\mathbf{C}\in\mathcal{R}_q^{n\times m}$ and $\mathbf{U}\in\mathcal{R}^{2mT\times m}$ full-rank s.t. $\mathbf{A}\cdot\mathbf{U} = \mathbf{v}\otimes\mathbf{C} \bmod q$. Then*

$$\eta_\varepsilon(\Lambda(\mathbf{U})\cap\Lambda_q^{\perp}(\mathbf{A})) \leq \mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U}))\cdot\eta_\varepsilon(\Lambda_q^{\perp}(\mathbf{C})).$$

Proof. Observe

$$\begin{aligned}\Lambda(\mathbf{U})\cap\Lambda_q^{\perp}(\mathbf{A}) &= \{\mathbf{U}\cdot\mathbf{d} \mid \mathbf{d}\in\mathcal{R}^m : \mathbf{A}\cdot\mathbf{U}\cdot\mathbf{d} = (\mathbf{v}\otimes\mathbf{C})\cdot\mathbf{d} = \mathbf{0} \bmod q\}\\ &= \{\mathbf{U}\cdot\mathbf{d} \mid \mathbf{d}\in\mathcal{R}^m : \mathbf{C}\cdot\mathbf{d} = \mathbf{0} \bmod q\} \;=\; \mathbf{U}\cdot\Lambda_q^{\perp}(\mathbf{C})\end{aligned}$$

where the second equality holds since the coefficients of $\mathbf{v}$ are invertible elements. The claim follows from a direct inspection of $\sum_{\mathbf{y}\in(\mathbf{U}\cdot\Lambda_q^{\perp}(\mathbf{C}))^*}\rho_{1/\sigma}(\mathbf{y})$. Let $\mathbf{B}\in\mathbb{R}^{md\times md}$ denote a basis of $\Lambda_q^{\perp}(\mathbf{C})$ then $\mathbf{B}_U \coloneqq \tilde{\theta}(\mathbf{U})\cdot\mathbf{B}$ is a basis of $\mathbf{U}\cdot\Lambda_q^{\perp}(\mathbf{C})$ since $\mathbf{U}$ has full rank. The matrix $\mathbf{B}_U(\mathbf{B}_U^{\mathsf{T}}\mathbf{B}_U)^{-\mathsf{T}} = \tilde{\theta}(\mathbf{U}(\mathbf{U}^{\mathsf{T}}\mathbf{U})^{-\mathsf{T}})\mathbf{B}^{-\mathsf{T}} \coloneqq \mathbf{B}_{\mathsf{dual}}$ is a basis of the corresponding dual lattice. Now we write down

$$\begin{aligned}\sum_{\mathbf{x}\in\mathbb{Z}^{md}}\rho_{1/\sigma}(\mathbf{B}_{\mathsf{dual}}\cdot\mathbf{x}) &= \sum_{\mathbf{x}\in\mathbb{Z}^{md}}\exp(-\pi\,\sigma^2\cdot\mathbf{x}^{\mathsf{T}}\cdot\mathbf{B}_{\mathsf{dual}}^{\mathsf{T}}\cdot\mathbf{B}_{\mathsf{dual}}\cdot\mathbf{x})\\ &= \sum_{\mathbf{x}\in\mathbb{Z}^{md}}\exp(-\pi\,\sigma^2\cdot\mathbf{x}^{\mathsf{T}}\cdot\mathbf{B}^{-1}\cdot\tilde{\theta}(\mathbf{U}^{\mathsf{T}}\cdot\mathbf{U})^{-1}\cdot\mathbf{B}^{-\mathsf{T}}\cdot\mathbf{x})\\ &\leq \sum_{\mathbf{x}\in\mathbb{Z}^{md}}\exp(-\pi\,\sigma^2\cdot\lambda_{\min}(\tilde{\theta}(\mathbf{U}^{\mathsf{T}}\cdot\mathbf{U})^{-1})\cdot\mathbf{x}^{\mathsf{T}}\cdot\mathbf{B}^{-1}\cdot\mathbf{B}^{-\mathsf{T}}\cdot\mathbf{x})\\ &= \sum_{\mathbf{x}\in\mathbb{Z}^{md}}\exp(-\pi\,\frac{\sigma^2}{\mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U}))^2}\cdot\mathbf{x}^{\mathsf{T}}\cdot\mathbf{B}^{-1}\cdot\mathbf{B}^{-\mathsf{T}}\cdot\mathbf{x}).\end{aligned}$$

Hence for any $\sigma' \geq \eta_\varepsilon(\Lambda_q^{\perp}(\mathbf{C}))$ taking $\sigma \geq \sigma'\cdot\mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U}))$ in the expression above is sufficient to guarantee the smoothing condition $\sum_{\mathbf{x}\in\mathbb{Z}^{md}}\rho_{1/\sigma}(\mathbf{B}_{\mathsf{dual}}\cdot\mathbf{x}) \leq 1+\varepsilon$. □

In Proposition 6 we prove a lemma concerning the $\mathcal{K}$-span of matrices $\mathbf{U}$ satisfying relation of the form $\mathbf{A} \cdot \mathbf{U} = \mathbf{v} \otimes \mathbf{C} \bmod q$. The proof relies on transforming the linear independence claim about the preimages $(\mathbf{U}_{i^*}, \tilde{\mathbf{U}})$ into one about the images $(\mathbf{v}_{i^*}, \tilde{\mathbf{V}})$, where $\tilde{\mathbf{V}} \coloneqq (\mathbf{v}_j)_{j \neq i^*}$.

Proposition 6. *Let $\mathbf{A} \in \mathcal{R}_q^{nT \times 2mT}$ be surjective. For $i \in [T]$, let $v_i \in \mathcal{R}_q^\times$, $v_i \neq v_j$ for $i \neq j$, $\mathbf{v}_i \coloneqq (1, v_i, \ldots, v_i^{T-1})^\intercal$, and let $\mathbf{U}_i \in \mathcal{R}_q^{2mT \times m}, \mathbf{C}_i \in \mathcal{R}_q^{n \times m}$ for some $m \geq 1$ be surjective matrices satisfying $\mathbf{A} \cdot \mathbf{U}_i = \mathbf{v}_i \otimes \mathbf{C}_i \bmod q$. Fix any $i^* \in [T]$ and write $\tilde{\mathbf{U}} \coloneqq (\ldots, \mathbf{U}_j, \ldots)_{j \neq i^*}$. Then $\mathcal{K}$-span $\left(\mathbf{U}_{i^*}, \tilde{\mathbf{U}}\right) \neq \mathcal{K}$-span $\left(\tilde{\mathbf{U}}\right)$.*

Proof. Denote $\tilde{\mathbf{V}} \boxtimes \tilde{\mathbf{C}} \coloneqq (\ldots, \mathbf{v}_j \otimes \mathbf{C}_j, \ldots)_{j \neq i^*}$ the concatenation of matrices $\mathbf{v}_j \otimes \mathbf{C}_j$ for all $j \in [T] \setminus \{i^*\}$. Write $\tilde{m} \coloneqq m(T-1)$. Assume towards contradiction that $\mathcal{K}$-span $(\mathbf{U}_{i^*}) \subset \mathcal{K}$-span$(\tilde{\mathbf{U}})$, meaning that for any $\mathbf{x} \in \mathcal{K}$-span $(\mathbf{U}_{i^*})$, we can write $\mathbf{x} = \tilde{\mathbf{U}} \cdot \mathbf{z}$ for some $\mathbf{z} \in \mathcal{K}^{\tilde{m}}$. Pick an arbitrary $\mathbf{y}_{i^*} \in (\mathcal{R}_q^\times)^n$. Observe that $\mathbf{v}_{i^*} \otimes \mathbf{y}_{i^*} \notin \mathcal{R}_q$-span$(\tilde{\mathbf{V}} \boxtimes \tilde{\mathbf{C}})$ since $\mathbf{V}_{[T]} \coloneqq (\mathbf{v}_i)_{i \in [T]}$ is invertible modulo q (otherwise we have $\sum_{i \in [T]} \mathbf{v}_i \otimes \mathbf{y}_i = (\mathbf{V}_{[T]} \otimes \mathbf{I}) \cdot \mathbf{y} = 0 \bmod q$ for some vector $\mathbf{y} \neq 0$). Pick an arbitrary $\mathbf{x} \in \Lambda(\mathbf{U}_{i^*}) \subset \mathcal{K}$-span $(\mathbf{U}_{i^*})$ s.t. $\mathbf{A} \cdot \mathbf{x} = \mathbf{v}_{i^*} \otimes \mathbf{y}_{i^*} \bmod q$. Such an $\mathbf{x}$ must exist because $\mathbf{C}_{i^*}$ is surjective. By assumption, there exists $\mathbf{z} \in \mathcal{K}^{\tilde{m}}$ s.t. $\mathbf{x} = \tilde{\mathbf{U}} \cdot \mathbf{z}$. We write $\mathbf{z} = \mathbf{b}/a$ with $\mathbf{b} \in \mathcal{R}^{\tilde{m}}$ and $0 \neq a \in \mathcal{R}$ and where q does not divide a and $\mathbf{b}$ simultaneously (otherwise we clear the common factor). It holds that $\mathbf{x} \cdot a = \tilde{\mathbf{U}} \cdot \mathbf{b}$. Then $(\mathbf{v}_{i^*} \otimes \mathbf{y}_{i^*}) \cdot a \equiv \mathbf{A} \cdot \mathbf{x} \cdot a = \mathbf{A} \cdot \tilde{\mathbf{U}} \cdot \mathbf{b} \equiv (\tilde{\mathbf{V}} \boxtimes \tilde{\mathbf{C}}) \cdot \mathbf{b}$. Since $\mathbf{v}_{i^*} \otimes \mathbf{y}_{i^*} \notin \mathcal{R}_q$-span$(\tilde{\mathbf{V}} \boxtimes \tilde{\mathbf{C}})$, we have $a \equiv 0$ and $\mathbf{b} \equiv \mathbf{0}$. This contradicts that q does not divide a and $\mathbf{b}$ simultaneously. □

Lemma 92 below will be useful for arguing the partial trapdoor sampling in later subsections.

Lemma 92. *Let $\mathbf{A} \in \mathcal{R}_q^{nT \times 2mT}$ be surjective, $\mathbf{v} \in \mathcal{R}_q^T$, $\sqrt{\mathbf{\Sigma}_1} \in \mathbb{R}^{2mT \times 2mT}$ and $\sqrt{\mathbf{\Sigma}_0} \in \mathbb{R}^{\tilde{m} \times \tilde{m}}$, $\tilde{m} \coloneqq n \log(q)$. If $\sqrt{\mathbf{\Sigma}_1} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$, $\sqrt{\mathbf{\Sigma}_0} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{G}))$, and $\mathbf{\Sigma}$ be such that $\mathbf{\Sigma}^{-1} = \begin{bmatrix} \mathbf{\Sigma}_1^{-1} & \\ & \mathbf{\Sigma}_0^{-1} \end{bmatrix}$, then following distributions are statistically close in λ:*

$$\mathcal{D}_0 = \left\{ \mathbf{U} \;\middle|\; \mathbf{U} \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT}, \sqrt{\mathbf{\Sigma}_1}})^m : \mathbf{A} \cdot \mathbf{U} \in (\mathcal{R}_q\text{-span}(\mathbf{v} \otimes \mathbf{I}_n))^m \right\},$$
$$\mathcal{D}_1 = \left\{ \mathbf{U} \;\middle|\; [\mathbf{U} \| \mathbf{W}] \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT+\tilde{m}}, \sqrt{\mathbf{\Sigma}}})^m : \mathbf{A} \cdot \mathbf{U} = \mathbf{v} \otimes \mathbf{G} \cdot \mathbf{W} \bmod q \right\}.$$

Proof. By Proposition 4, the distribution $\mathcal{D}_0$ is statistically close to

$$\left\{ \mathbf{U} \;\middle|\; \begin{array}{l} \mathbf{C} \leftarrow \mathcal{R}_q^{n \times m} \\ \mathbf{U} \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT}, \sqrt{\mathbf{\Sigma}_1}})^m : \mathbf{A} \cdot \mathbf{U} = \mathbf{v} \otimes \mathbf{C} \bmod q \end{array} \right\}.$$

By Lemma 39 since $\sqrt{\boldsymbol{\Sigma}_0} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{G}))$ for $\mathbf{W} \leftarrow (\mathcal{D}_{\mathcal{R}^{\tilde{m}},\sqrt{\boldsymbol{\Sigma}_0}})^m$ the product $\mathbf{G}\cdot\mathbf{W} \bmod q$ is statistically close to uniform over $\mathcal{R}_q^{n\times m}$. Hence, the above distribution is statistically close to

$$\left\{\mathbf{U} \;\middle|\; \begin{array}{l} \mathbf{W} \leftarrow (\mathcal{D}_{\mathcal{R}^{\tilde{m}},\sqrt{\boldsymbol{\Sigma}_0}})^m \\ \mathbf{U} \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT},\sqrt{\boldsymbol{\Sigma}_1}})^m : \mathbf{A}\cdot\mathbf{U} = \mathbf{v}\otimes\mathbf{G}\cdot\mathbf{W} \bmod q \end{array}\right\}.$$

It remains to show that the following distributions

$$\mathcal{D} = \left\{(\mathbf{u},\mathbf{w}) \;\middle|\; \begin{array}{l} \mathbf{w} \leftarrow \mathcal{D}_{\mathcal{R}^{\tilde{m}},\sqrt{\boldsymbol{\Sigma}_0}} \\ \mathbf{u} \leftarrow \mathcal{D}_{\mathcal{R}^{2mT},\sqrt{\boldsymbol{\Sigma}_1}} : \mathbf{A}\cdot\mathbf{u} = \mathbf{v}\otimes\mathbf{G}\cdot\mathbf{w} \bmod q \end{array}\right\},$$

$$\mathcal{D}' = \left\{(\mathbf{u},\mathbf{w}) \;\middle|\; \begin{bmatrix}\mathbf{u}\\ \mathbf{w}\end{bmatrix} \leftarrow \mathcal{D}_{\mathcal{R}^{mT+\tilde{m}},\sqrt{\boldsymbol{\Sigma}}} : \mathbf{A}\cdot\mathbf{u} = \mathbf{v}\otimes\mathbf{G}\cdot\mathbf{w} \bmod q \right\}$$

are statistically close, then the claim follows by repeating over all columns. The probability mass function of $\mathcal{D}$ is

$$\begin{aligned}\mathcal{D}(\mathbf{u},\mathbf{w}) &= \frac{\exp(-\pi\,\mathbf{w}^\intercal\cdot\boldsymbol{\Sigma}_0^{-1}\cdot\mathbf{w})}{\rho_{\sqrt{\boldsymbol{\Sigma}_0}}(\mathcal{R}^m)}\cdot\frac{\exp(-\pi\,\mathbf{u}^\intercal\cdot\boldsymbol{\Sigma}_1^{-1}\cdot\mathbf{u})}{\rho_{\sqrt{\boldsymbol{\Sigma}_1}}(\Lambda_q^{\mathbf{v}\otimes\mathbf{G}\mathbf{w}}(\mathbf{A}))} \\ &\in (1\pm\mathsf{negl}(\lambda))\frac{\exp(-\pi\,\mathbf{w}^\intercal\cdot\boldsymbol{\Sigma}_0^{-1}\cdot\mathbf{w})}{\rho_{\sqrt{\boldsymbol{\Sigma}_0}}(\mathcal{R}^m)}\cdot\frac{\exp(-\pi\,\mathbf{u}^\intercal\cdot\boldsymbol{\Sigma}_1^{-1}\cdot\mathbf{u})}{\rho_{\sqrt{\boldsymbol{\Sigma}_1}}(\Lambda_q^\perp(\mathbf{A}))}.\end{aligned}$$

The second equality holds since $\mathbf{A}$ is surjective, $\sqrt{\boldsymbol{\Sigma}_1} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ and by Corollary 4.

Denote $\Lambda \coloneqq \left\{\mathbf{x} = \begin{bmatrix}\mathbf{u}\\ \mathbf{w}\end{bmatrix} \in \mathcal{R}^{2mT+\tilde{m}} : \mathbf{A}\cdot\mathbf{u} = \mathbf{v}\otimes\mathbf{G}\cdot\mathbf{w} \bmod q\right\}$. The probability mass function of $\mathcal{D}'$ is

$$\mathcal{D}'(\mathbf{u},\mathbf{w}) = \frac{\exp(-\pi\,\mathbf{x}^\intercal\cdot\boldsymbol{\Sigma}^{-1}\cdot\mathbf{x})}{\rho_{\sqrt{\boldsymbol{\Sigma}}}(\Lambda)} = \frac{\exp(-\pi\,\mathbf{w}^\intercal\cdot\boldsymbol{\Sigma}_0^{-1}\cdot\mathbf{w})\cdot\exp(-\pi\,\mathbf{u}^\intercal\cdot\boldsymbol{\Sigma}_1^{-1}\cdot\mathbf{u})}{\rho_{\sqrt{\boldsymbol{\Sigma}}}(\Lambda)}.$$

Expanding $\rho_{\sqrt{\boldsymbol{\Sigma}}}(\Lambda)$ and denoting $\mathbf{z}(\mathbf{w}) \coloneqq \mathbf{v}\otimes\mathbf{G}\cdot\mathbf{w}$ we get

$$\begin{aligned}\rho_{\sqrt{\boldsymbol{\Sigma}}}(\Lambda) &= \sum_{\mathbf{u},\mathbf{w}:\mathbf{A}\cdot\mathbf{u}\equiv\mathbf{z}(\mathbf{w})} \exp(-\pi\,\mathbf{w}^\intercal\cdot\boldsymbol{\Sigma}_0^{-1}\cdot\mathbf{w})\cdot\exp(-\pi\,\mathbf{u}^\intercal\cdot\boldsymbol{\Sigma}_1^{-1}\cdot\mathbf{u}) \\ &= \sum_{\mathbf{w}\in\mathcal{R}^{\tilde{m}}} \exp(-\pi\,\mathbf{w}^\intercal\cdot\boldsymbol{\Sigma}_0^{-1}\cdot\mathbf{w}) \cdot \sum_{\mathbf{u}\in\Lambda_q^{\mathbf{z}(\mathbf{w})}(\mathbf{A})} \exp(-\pi\,\mathbf{u}^\intercal\cdot\boldsymbol{\Sigma}_1^{-1}\cdot\mathbf{u}) \\ &\in (1\pm\mathsf{negl}(\lambda))\cdot\sum_{\mathbf{w}\in\mathcal{R}^{\tilde{m}}} \exp(-\pi\,\mathbf{w}^\intercal\cdot\boldsymbol{\Sigma}_0^{-1}\cdot\mathbf{w}) \cdot \sum_{\mathbf{u}\in\Lambda_q^\perp(\mathbf{A})} \exp(-\pi\,\mathbf{u}^\intercal\cdot\boldsymbol{\Sigma}_1^{-1}\cdot\mathbf{u}) \\ &\in (1\pm\mathsf{negl}(\lambda))\cdot\rho_{\sqrt{\boldsymbol{\Sigma}_0}}(\mathcal{R}^{\tilde{m}})\cdot\rho_{\sqrt{\boldsymbol{\Sigma}_1}}(\Lambda_q^\perp(\mathbf{A})),\end{aligned}$$

where in the second equality holds since $\mathbf{A}$ is surjective, and the third since $\sqrt{\boldsymbol{\Sigma}_1} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ and by Corollary 4. Therefore,

$$\mathcal{D}'(\mathbf{u},\mathbf{w}) \in (1\pm\mathsf{negl}(\lambda))\cdot\frac{\exp(-\pi\,\mathbf{w}^\intercal\cdot\boldsymbol{\Sigma}_0^{-1}\cdot\mathbf{w})\cdot\exp(-\pi\,\mathbf{u}^\intercal\cdot\boldsymbol{\Sigma}_1^{-1}\cdot\mathbf{u})}{\rho_{\sqrt{\boldsymbol{\Sigma}_0}}(\mathcal{R}^{\tilde{m}})\rho_{\sqrt{\boldsymbol{\Sigma}_1}}(\Lambda_q^\perp(\mathbf{A}))}.$$

implying

$$\frac{1}{2} \sum_{\mathbf{u},\mathbf{w}:\ \mathbf{A}\cdot\mathbf{u}\equiv\mathbf{v}\otimes\mathbf{G}\cdot\mathbf{w}} |\mathcal{D}(\mathbf{u},\mathbf{w}) - \mathcal{D}'(\mathbf{u},\mathbf{w})| \leq \mathsf{negl}(\lambda)$$

and concluding the statement. □

Lastly, we upper bound the Gaussian weight of rank-deficient lattices in the lemma below. The following bound holds not only for $\Lambda_q(\mathbf{U})$ with an integer $q \geq 1$, but also for $\Lambda_{\mathfrak{q}}(\mathbf{U})$ for any ideal $\mathfrak{q} \subset \mathcal{R}$.

Lemma 93 (Generalisation of [ABB10b, Full version, Lemma 31])**.** *Let m, N be positive integers, k be a non-negative integer, $\mathbf{\Sigma} \in \mathbb{R}^{dm\times dm}$ be positive semi-definite, $\sigma = \mathfrak{s}_{\max}(\sqrt{\mathbf{\Sigma}})$, and $\mathfrak{q} \subset \mathcal{R}$ be an ideal satisfying the following:*

$$\mathcal{N}(\mathfrak{q}) = N, \qquad k \leq m, \qquad dm \geq \Omega(\lambda), \qquad \sigma \cdot \sqrt{m} \leq N^{1/d}.$$

For any $\mathbf{U} \in \mathcal{R}^{m\times k}$, it holds that

$$\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_{\mathfrak{q}}(\mathbf{U})) \leq \rho_\sigma(\mathcal{R}^k)/(1 - \mathsf{negl}(\lambda))$$

where if $k = 0$, i.e. $\mathbf{U}$ is the empty matrix, then we take $\Lambda_{\mathfrak{q}}(\mathbf{U}) = \mathfrak{q}^m$.

Proof. Below, for $\ell \geq 0$, write $\mathcal{B}_\ell \subset \mathbb{R}^\ell$ for the set of vectors in $\mathbb{R}^\ell$ with norm at most 1.

In the proof we assume there exists a k-by-k submatrix $\mathbf{W}$ of $\mathbf{U}$ so that $\mathbf{W}$ is either the empty matrix (i.e. $k = 0$) or invertible over $\mathcal{R}_{\mathfrak{q}} = \mathcal{R}/\mathfrak{q}$. Otherwise, we are able to construct a different matrix $\mathbf{U}' \in \mathcal{R}^{m\times k}$ that satisfies the condition and $\rho_\sigma(\Lambda_{\mathfrak{q}}(\mathbf{U})) \leq \rho_\sigma(\Lambda_{\mathfrak{q}}(\mathbf{U}'))$. The case $k = 0$ is clear. For $k > 0$, consider prime factorisation $\mathfrak{q} = \prod_j \mathfrak{q}_j$ and the Chinese remainder representation $\mathbf{U}_j \coloneqq \mathbf{U} \bmod \mathfrak{q}_j \in \mathcal{R}_{\mathfrak{q}_j}^{m\times k}$ of $\mathbf{U}$. For each j, let $\mathbf{U}'_j \in \mathcal{R}_{\mathfrak{q}_j}^{m\times k}$ be a matrix obtained from $\mathbf{U}_j$ by picking the maximal set of $\mathcal{R}_{\mathfrak{q}_j}$-linearly independent columns completing the set with arbitrary columns $\mathcal{R}_{\mathfrak{q}_j}$-linearly independent with the rest. Let $\mathbf{U}' \in \mathcal{R}_{\mathfrak{q}}$ be the unique matrix (modulo $\mathfrak{q}$) satisfying $\mathbf{U}' = \mathbf{U}'_j \bmod \mathfrak{q}$. It is clear that $\Lambda_{\mathfrak{q}}(\mathbf{U}) \subseteq \Lambda_{\mathfrak{q}}(\mathbf{U}')$ and that every k-by-k submatrix of $\mathbf{U}'$ is invertible modulo $\mathfrak{q}$.

Let $\mathbf{P} \in \{0,1\}^{k\times m}$ be a partial permutation matrix, i.e. each row and column contains at most a single 1, such that $\mathbf{P}\cdot\mathbf{U} = \mathbf{W}$ is a submatrix of $\mathbf{U}$ which is either the empty matrix or is invertible over $\mathcal{R}_{\mathfrak{q}}$. We observe the following two properties:

1. For all $\mathbf{x} \in \Lambda_{\mathfrak{q}}(\mathbf{U})$, since $\|\mathbf{P}\cdot\mathbf{x}\| \leq \|\mathbf{x}\|$, we have $\rho_{\sqrt{\mathbf{\Sigma}}}(\mathbf{x}) \leq \rho_\sigma(\mathbf{P}\cdot\mathbf{x})$.[3]

[3]Recall that $\sigma = \mathfrak{s}_{\max}(\sqrt{\mathbf{\Sigma}}) = \sqrt{\lambda_{\max}(\mathbf{\Sigma})}$. Then $\rho_{\sqrt{\mathbf{\Sigma}}}(\mathbf{x}) = \exp(-\pi\,\mathbf{x}^{\mathsf{T}}\mathbf{\Sigma}^{-1}\mathbf{x}) \leq \exp(-\pi\,\lambda_{\min}(\mathbf{\Sigma}^{-1})\cdot \mathbf{x}^{\mathsf{T}}\mathbf{x}) = \exp(-\pi\,\|\mathbf{x}\|^2/\sigma^2) \leq \exp(-\pi\,\|\mathbf{P}\mathbf{x}\|^2/\sigma^2) = \rho_\sigma(\mathbf{P}\mathbf{x})$.

2. The map $\mathbf{x} \mapsto \mathbf{P} \cdot \mathbf{x} \bmod \mathfrak{q}$ is injective over $\mathcal{C} \coloneqq \Lambda_\mathfrak{q}(\mathbf{U}) \cap \mathcal{R}_\mathfrak{q}^m$. We abuse the notation and denote by $\mathcal{R}_\mathfrak{q}^m$ a finite set in $\mathcal{R}^m$ where we identify the ring elements with their shortest representatives in $\mathcal{R}$. Indeed, let $\mathbf{x} = \mathbf{U} \cdot \mathbf{d} \bmod \mathfrak{q}$ and $\mathbf{x}' = \mathbf{U} \cdot \mathbf{d}' \bmod \mathfrak{q}$ for some $\mathbf{d}, \mathbf{d}' \in \mathcal{R}_\mathfrak{q}^k$. If $\mathbf{P} \cdot \mathbf{x} = \mathbf{P} \cdot \mathbf{x}' \bmod \mathfrak{q}$, then $\mathbf{W} \cdot (\mathbf{d} - \mathbf{d}') = \mathbf{0} \bmod \mathfrak{q}$, which implies $\mathbf{d} = \mathbf{d}' \bmod \mathfrak{q}$ and hence $\mathbf{x} = \mathbf{x}' \bmod \mathfrak{q}$.

Using the two properties above, we obtain

$$\rho_{\sqrt{\mathbf{\Sigma}}}(\mathcal{C}) = \sum_{\mathbf{x} \in \mathcal{C}} \rho_{\sqrt{\mathbf{\Sigma}}}(\mathbf{x}) \leq \sum_{\mathbf{x} \in \mathcal{C}} \rho_\sigma(\mathbf{P} \cdot \mathbf{x}) \leq \sum_{\mathbf{y} \in \mathcal{R}^k} \rho_\sigma(\mathbf{y}) = \rho_\sigma(\mathcal{R}^k) \tag{5.1}$$

where the first and second inequalities follow from the first and second properties respectively. Next, let $r \coloneqq N^{1/d}/(2\sigma\sqrt{m})$. We show that $\Lambda_\mathfrak{q}(\mathbf{U}) \cap r\sqrt{dm} \cdot \sigma\mathcal{B}_{dm} \subseteq \mathcal{C}$ and hence

$$\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_\mathfrak{q}(\mathbf{U}) \cap r\sqrt{dm} \cdot \sigma\mathcal{B}_{dm}) \leq \rho_{\sqrt{\mathbf{\Sigma}}}(\mathcal{C}). \tag{5.2}$$

Suppose by contradiction that $\Lambda_\mathfrak{q}(\mathbf{U}) \cap r\sqrt{dm} \cdot \sigma\mathcal{B}_{dm} \not\subseteq \mathcal{C}$. Then there exist distinct $\mathbf{a}, \mathbf{b} \in r\sqrt{dm} \cdot \sigma\mathcal{B}_{dm}$ such that $\mathbf{a} = \mathbf{b} \bmod \mathfrak{q}$ or equivalently $\mathbf{0} \neq \mathbf{a} - \mathbf{b} \in \mathfrak{q}^m$. Since $\mathbf{a}$ and $\mathbf{b}$ are contained in the ball of radius $r\sigma\sqrt{dm}$, so does $\mathbf{a} - \mathbf{b}$, and thus $\|\mathbf{a} - \mathbf{b}\| \leq 2r\sigma\sqrt{dm} = \sqrt{d} \cdot N^{1/d}$. By Lemma 43, we have

$$\lambda_1(\mathfrak{q}) \geq \sqrt{d} \cdot \mathcal{N}(\mathfrak{q})^{1/d} = \sqrt{d} \cdot N^{1/d}.$$

Therefore, $\|\mathbf{a} - \mathbf{b}\| \leq \sqrt{d} \cdot N^{1/d} \leq \lambda_1(\mathfrak{q}) = \lambda_1(\mathfrak{q}^m)$, contradicting $\mathbf{a} - \mathbf{b}$ being a non-zero vector in $\mathfrak{q}^m$. Finally, using $r = N^{1/d}/(2\sigma\sqrt{m}) > 1/\sqrt{2\pi}$ from above, let

$$\begin{aligned} C_r &= r \cdot \sqrt{2\pi e} \cdot \exp(-\pi \cdot r^2) \\ &= \sqrt{2\pi e} \cdot \frac{N^{1/d}}{2\sigma\sqrt{m}} \cdot \exp\left(-\pi \cdot \left(\frac{N^{1/d}}{2\sigma\sqrt{m}}\right)^2\right) \\ &\leq \frac{\sqrt{2\pi e}}{2} \cdot \exp(-\pi/4) \approx 0.942, \end{aligned}$$

where the last inequality holds since $\frac{N^{1/d}}{\sigma\sqrt{m}} > 1$ by the assumed parameters, and the function $\sqrt{2\pi e} \cdot x \cdot \exp(-\pi \cdot x^2)$ is decreasing for $x \geq \sqrt{1/(2\pi)}$. Applying Lemma 23 with $C = r\sqrt{2\pi e} \cdot e^{-\pi \cdot r^2}$, we have that

$$\begin{aligned} \frac{\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_\mathfrak{q}(\mathbf{U}) \cap r\sqrt{dm} \cdot \sigma\mathcal{B}_m)}{\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_\mathfrak{q}(\mathbf{U}))} &\geq \frac{\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_\mathfrak{q}(\mathbf{U}) \cap r\sqrt{dm} \cdot \sqrt{\mathbf{\Sigma}}\mathcal{B}_m)}{\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_\mathfrak{q}(\mathbf{U}))} \\ &= \frac{\rho((\sqrt{\mathbf{\Sigma}})^{-1}\Lambda_\mathfrak{q}(\mathbf{U}) \cap r\sqrt{dm} \cdot \mathcal{B}_m)}{\rho((\sqrt{\mathbf{\Sigma}})^{-1}\Lambda_\mathfrak{q}(\mathbf{U}))} \\ &\geq 1 - 2C^{dm} \quad \geq 1 - \mathsf{negl}(\lambda). \end{aligned}$$

Chaining together with Equations (5.1) and (5.2), we arrive at $\rho_{\sqrt{\mathbf{\Sigma}}}(\Lambda_{\mathfrak{q}}(\mathbf{U})) \leq \rho_\sigma(\mathcal{R}^k)/(1-\mathsf{negl}(\lambda))$. □

The statements in this section use the fact that $\mathbf{U}$ is full-rank over the field $\mathcal{K}$. Since we use such matrices in the scheme we need to ensure the full-rank condition for a randomly sampled matrix $\mathbf{U}$. In Lemma 94 we prove that columns in a discrete Gaussian matrix $\mathbf{U}$ are likely to be linearly independent (which is equivalent to the full-rank condition over $\mathcal{K}$).

We also prove Theorem 12 which states that sampling a uniform $\mathbf{A}$ and then sampling a Gaussian $\mathbf{U}$ subject to $\mathbf{A} \cdot \mathbf{U} = \mathbf{V} \bmod q$ is statistically close to first sampling $\mathbf{U}$ and then $\mathbf{A}$ subject to the same condition. This is a generalisation of previous work stated over integers to the ring setting and may be of independent interest.

Lemma 94 (Generalisation of [BF11b, Lemma 4.5])**.** *Let g, k, m, n, q be positive integers, $a \geq 1$ be a real number, $\mathbf{A} \in \mathcal{R}_q^{n \times m}$ be surjective, and $\{\mathbf{\Sigma}_i\}_{i\in[k]}$ be positive semi-definite matrices in $\mathbb{R}^{dm \times dm}$ satisfying the following constraints:*

- $k, n \leq m$,
- $d \cdot (m-k) \geq \Omega(\lambda)$,
- *q is an unramified prime factoring as $\langle q \rangle = \prod_{j=1}^{g} \mathfrak{q}_j$ with norm $\mathcal{N}(\mathfrak{q}_j) = q^{d/g}$ in $\mathcal{R}$,*
- *there exists $\varepsilon \leq \mathsf{negl}(\lambda)$ so that for all $i \in [k]$ it holds*

$$\max\left\{\eta_\varepsilon(\Lambda_q^\perp(\mathbf{A})), 2\sqrt{d}\cdot(a^k q^n)^{1/(m-k)}\right\} \leq \mathfrak{s}_{\min}\left(\sqrt{\mathbf{\Sigma}_i}\right),$$
$$\mathfrak{s}_{\max}\left(\sqrt{\mathbf{\Sigma}_i}\right) \leq \min\left\{q^{1/g}/\sqrt{m}, a \cdot \mathfrak{s}_{\min}\left(\sqrt{\mathbf{\Sigma}_i}\right)\right\}.$$

Furthermore, let $\{\mathbf{v}_i\}_{i\in[k]} \in \mathcal{R}_q^n$ be arbitrary images. For $i \in [k]$, when $i \geq 1$ let $\mathbf{u}_i \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{v}_i}(\mathbf{A}), \sqrt{\mathbf{\Sigma}_i}}$. With overwhelming probability in λ, the vectors $\{\mathbf{u}_i\}_{i\in[k]}$ are $\mathcal{R}_{\mathfrak{q}_j}$-linearly independent for all $j \in [g]$.

Proof. For $i \in \{0, \ldots, k-1\}$, let $\mathbf{U}_i \in \mathcal{R}^{m \times i}$ be the matrix with columns given by $(\mathbf{u}_j)_{j\in[i]}$, and define $\mathbf{U}_0$ to be the empty matrix. For each $j \in [g]$, define $\Lambda_{\mathfrak{q}_j}(\mathbf{U}_0) \coloneqq \mathfrak{q}_j^m$. By a union bound and Corollary 4, the probability that $\{\mathbf{u}_i\}_{i\in[k]}$ is not $\mathcal{R}_{\mathfrak{q}_j}$-linearly independent for some $j \in [g]$ is at most

$$\begin{aligned}\sum_{i\in[k]}\sum_{j\in[g]} \Pr\left[\mathbf{u}_i \in \Lambda_{\mathfrak{q}_j}(\mathbf{U}_i)\right] &\leq \sum_{i\in[k]}\sum_{j\in[g]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_{\mathfrak{q}_j}(\mathbf{U}_i))}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_q^{\mathbf{v}_i}(\mathbf{A}))} \\ &\leq (1+\mathsf{negl}(\lambda)) \cdot \sum_{i\in[k]}\sum_{j\in[g]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_{\mathfrak{q}_j}(\mathbf{U}_i))}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_q^\perp(\mathbf{A}))}.\end{aligned} \tag{5.3}$$

Below, denote by $\sigma_{i,\max} \coloneqq \mathfrak{s}_{\max}(\sqrt{\mathbf{\Sigma}_i})$ and $\sigma_{i,\min} \coloneqq \mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}_i})$ the maximum and minimum singular values of $\sqrt{\mathbf{\Sigma}_i}$ respectively. We recall the following properties proven in separate lemmas:

1. For all $j \in [g]$, the norm of $\mathfrak{q}_j$ is given by $\mathcal{N}(\mathfrak{q}_j) = q^{d/g}$. Since $\sigma_{i,\max} \leq q^{1/g}/\sqrt{m} = \mathcal{N}(\mathfrak{q}_j)^{1/d}/\sqrt{m}$, by Lemma 93, for $i \in [k]$ and $j \in [g]$,

$$\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_{\mathfrak{q}_j}(\mathbf{U}_i)) \leq \rho_{\sigma_{i,\max}}(\mathcal{R}^i)/(1 - \mathsf{negl}(\lambda)).$$

2. For all $i \in [k]$, since $\sqrt{\mathbf{\Sigma}_i} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$, by Corollary 5, we have

$$\frac{1}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_q^\perp(\mathbf{A}))} \leq \left|\mathcal{R}_q^n\right| \cdot \frac{1 - \mathsf{negl}(\lambda)}{\rho_{\sigma_{i,\min}}(\mathcal{R}^m)}.$$

3. For $i \in [k]$, since $\sigma_{i,\min} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$, we have $\sigma_{i,\min} \geq \eta_\varepsilon(\mathcal{R}^{i'})$ for all $0 \leq i' \leq m$. Then by Corollary 7,

$$\sigma_{i,\min}^{di'} \cdot \det(\mathcal{R}^{i'})^{-1} \leq \rho_{\sigma_{i,\min}}(\mathcal{R}^{i'}) \leq (1 + \mathsf{negl}(\lambda)) \cdot \sigma_{i,\min}^{di'} \cdot \det(\mathcal{R}^{i'})^{-1},$$

   and analogously for $\sigma_{i,\max}$.

Combining the first and second properties and $g \leq d$, the quantity in Equation (5.3) is upper bounded by

$$\left|\mathcal{R}_q^n\right| \cdot d \cdot \sum_{i \in [k]} \frac{\rho_{\sigma_{i,\max}}(\mathcal{R}^i)}{\rho_{\sigma_{i,\min}}(\mathcal{R}^m)} \cdot (1 + \mathsf{negl}(\lambda)). \tag{5.4}$$

Define the shorthands $\sigma \coloneqq \min\{\sigma_{i,\min}\}_{i \in [k]}$ and $r = \frac{\sigma^d}{\det(\mathcal{R})}$. By the third property, the quantity in Equation (5.4) is at most

$$\begin{aligned}
&\left|\mathcal{R}_q^n\right| \cdot d \cdot \sum_{i \in [k]} \frac{\sigma_{i,\max}^{di} \cdot \det(\mathcal{R})^{-i}}{\sigma_{i,\min}^{dm} \cdot \det(\mathcal{R})^{-m}} \cdot (1 + \mathsf{negl}(\lambda)) \\
&\leq q^{dn} \cdot d \cdot \sum_{i \in [k]} a^{di} \left(\frac{\det(\mathcal{R})}{\sigma_{i,\min}^d}\right)^{m-i} \cdot (1 + \mathsf{negl}(\lambda)) \\
&\leq q^{dn} \cdot d \cdot \sum_{i \in [k]} a^{di} \cdot r^{i-m} \cdot (1 + \mathsf{negl}(\lambda)) \\
&= q^{dn} \cdot d \cdot \sum_{i \in [k]} r^{-m} \cdot (a^d r)^i \cdot (1 + \mathsf{negl}(\lambda)) \\
&= q^{dn} \cdot d \cdot r^{-m} \cdot \frac{(a^d r)^k - 1}{(a^d r) - 1} \cdot (1 + \mathsf{negl}(\lambda))
\end{aligned}$$

where the first inequality is due to $\sigma_{i,\max} \leq a \cdot \sigma_{i,\min}$, and the second due to $\sigma \leq \sigma_{i,\min}$, for all $i \in [k]$. Recall that by Lemma 6 the discriminant $\Delta_{\mathcal{K}}$ of $\mathcal{K}$ satisfies $\Delta_{\mathcal{K}} \leq d^d$. Since $\sigma \geq 2\sqrt{d}(a^k q^n)^{1/(m-k)}$, we have

$$a^d r = a^d \cdot \frac{\sigma^d}{\det(\mathcal{R})} \geq a^d \cdot \frac{\sigma^d}{d^{d/2}} \geq a^d \cdot \frac{(2\sqrt{d}(a^k q^n)^{1/(m-k)})^d}{d^{d/2}} = 2^d a^d (a^k q^n)^{d/(m-k)} \geq 2.$$

We can therefore further upper-bound the above quantity by

$$\begin{aligned}
& q^{dn} \cdot d \cdot a^{dk} \cdot r^{k-m} \cdot (1 + \mathsf{negl}(\lambda)) \\
&= q^{dn} \cdot d \cdot a^{dk} \cdot \left(\frac{\det(\mathcal{R})}{\sigma^d}\right)^{m-k} \cdot (1 + \mathsf{negl}(\lambda)) \\
&\leq q^{dn} \cdot d \cdot a^{dk} \cdot \left(\frac{d^{d/2}}{\sigma^d}\right)^{m-k} \cdot (1 + \mathsf{negl}(\lambda)) \\
&= d \cdot \left(\sqrt{d}(a^k q^n)^{1/(m-k)}/\sigma\right)^{d\cdot(m-k)} \cdot (1 + \mathsf{negl}(\lambda)) \quad \leq \ \mathsf{negl}(\lambda)
\end{aligned}$$

where the first inequality is due to $\det(\mathcal{R}) \leq d^{d/2}$, and the second is due to $\sigma \geq 2\sqrt{d} \cdot (a^k q^n)^{1/(m-k)}$ and $2^{-d(m-k)} \leq \mathsf{negl}(\lambda)$. □

Theorem 12 (Generalisation of [BF11b, Theorem 4.3])**.** *Let g, k, m, n, q be positive integers, $a \geq 1$ be a real number, and $\{\mathbf{\Sigma}_i\}_{i\in[k]}$ be positive semi-definite matrices in $\mathbb{R}^{dm\times dm}$ satisfying the following constraints:*

- $k, n \leq m$,
- $d \cdot (m-k) \geq \Omega(\lambda)$,
- *q be an unramified prime factoring as $\langle q \rangle = \prod_{j=1}^{g} \mathfrak{q}_j$ with norm $\mathcal{N}(\mathfrak{q}_j) = q^{d/g}$ in $\mathcal{R}$,*
- $ng/q^{d(m-n+1)/g} \leq \mathsf{negl}(\lambda)$, *and*
- *there exists $\varepsilon \leq \mathsf{negl}(\lambda)$ so that for all $i \in [k]$ it holds*

$$\max\left\{\eta_A, 2\sqrt{d} \cdot (a^k q^n)^{1/(m-k)}\right\} \leq \mathfrak{s}_{\min}\left(\sqrt{\mathbf{\Sigma}_i}\right),$$
$$\mathfrak{s}_{\max}\left(\sqrt{\mathbf{\Sigma}_i}\right) \leq \min\left\{q^{1/g}/\sqrt{m}, a \cdot \mathfrak{s}_{\min}\left(\sqrt{\mathbf{\Sigma}_i}\right)\right\},$$

for some $\eta_A \geq 8\, d\sqrt{m} \cdot q^{n/m+2/(d\cdot m)}$.

For any $\mathbf{V} \in \mathcal{R}_q^{n\times k}$, the following distributions are statistically close in λ:

$$\mathcal{D}_0 = \left\{ \left[\mathbf{A}, \mathbf{U}\right] \middle| \begin{array}{l} \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m} \\ \mathbf{u}_i \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{v}_i}(\mathbf{A}), \sqrt{\mathbf{\Sigma}_i}}, i \in [k] \end{array} \right\}, \ \mathcal{D}_1 = \left\{ \left[\mathbf{A}, \mathbf{U}\right] \middle| \begin{array}{l} \mathbf{u}_i \leftarrow \mathcal{D}_{\mathcal{R}^m, \sqrt{\mathbf{\Sigma}_i}}, i \in [k] \\ \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m} : \mathbf{A}\cdot\mathbf{U} \equiv \mathbf{V} \end{array} \right\}.$$

Proof. We calculate the statistical distance directly. First, consider $\mathcal{D}_0$. Clearly, for any $\mathbf{A} \in \mathcal{R}_q^{n\times m}$ and any $\mathbf{U} \in \mathcal{R}^{m\times k}$ satisfying $\mathbf{A}\cdot\mathbf{U} \equiv \mathbf{V} \bmod q$,

$$\begin{aligned}\Pr[\mathcal{D}_0 = (\mathbf{A},\mathbf{U})] &= \frac{1}{q^{dnm}} \prod_{i\in[k]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathbf{u}_i)}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_q^{\mathbf{v}_i}(\mathbf{A}))} \\ &\in [1 \pm \mathsf{negl}(\lambda)] \cdot \frac{1}{q^{dnm}} \prod_{i\in[k]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathbf{u}_i)}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_q^{\perp}(\mathbf{A}))} \\ &\in [1 \pm \mathsf{negl}(\lambda)] \cdot \frac{q^{dnk}}{q^{dnm}} \prod_{i\in[k]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathbf{u}_i)}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathcal{R}^m)}\end{aligned}$$

where the inclusion is due to Corollary 4. Next, consider $\mathcal{D}_1$. For any $\mathbf{U} \in \mathcal{R}^{m\times k}$ having $\mathcal{R}_{\mathsf{q}_j}$-linearly independent columns for all $j \in [g]$, there are exactly $q^{dn(m-k)}$ possible choices of $\mathbf{A}$ which satisfies $\mathbf{A}\cdot\mathbf{U} \equiv \mathbf{V} \bmod q$. Therefore, for each of these choices of $(\mathbf{A},\mathbf{U})$,

$$\Pr[\mathcal{D}_1 = (\mathbf{A},\mathbf{U})] = \frac{q^{dnk}}{q^{dnm}} \prod_{i\in[k]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathbf{u}_i)}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathcal{R}^m)}.$$

Let $\mathcal{P} \subseteq \mathcal{R}_q^{n\times m}$ denote the set of all surjective matrices, and let $\mathcal{T}_{\mathbf{A},\mathbf{V}} \subseteq \mathcal{R}^{m\times k}$ denote those matrices $\mathbf{U}$ with $\mathcal{R}_q$-linearly independent columns and satisfying $\mathbf{A}\cdot\mathbf{U} = \mathbf{V} \bmod q$. The statistical distance between $\mathcal{D}_0$ and $\mathcal{D}_1$, when restricted to $\mathbf{A} \in \mathcal{P}$ and $\mathbf{U} \in \mathcal{T}_{\mathbf{A},\mathbf{V}}$, denoted by Δ, satisfies the following:

$$\begin{aligned}\Delta &\coloneqq \frac{1}{2} \sum_{\mathbf{A}\in\mathcal{P}} \sum_{\mathbf{U}\in\mathcal{T}_{\mathbf{A},\mathbf{V}}} |\Pr[\mathcal{D}_0 = (\mathbf{A},\mathbf{U})] - \Pr[\mathcal{D}_1 = (\mathbf{A},\mathbf{U})]| \\ &\le \frac{1}{2} \sum_{\mathbf{A}\in\mathcal{P}} \sum_{\mathbf{U}\in\mathcal{T}_{\mathbf{A},\mathbf{V}}} \frac{q^{dnk}}{q^{dnm}} \prod_{i\in[k]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathbf{u}_i)}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathcal{R}^m)} \cdot \mathsf{negl}(\lambda) \\ &\le \frac{1}{2} \sum_{\mathbf{A}\in\mathcal{P}} \frac{q^{dnk}}{q^{dnm}} \prod_{i\in[k]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_q^{\mathbf{v}_i}(\mathbf{A}))}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathcal{R}^m)} \cdot \mathsf{negl}(\lambda) \\ &\le \frac{1}{2} \sum_{\mathbf{A}\in\mathcal{P}} \frac{q^{dnk}}{q^{dnm}} \prod_{i\in[k]} \frac{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_q^{\perp}(\mathbf{A}))}{\rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathcal{R}^m)} \cdot \mathsf{negl}(\lambda) \\ &\le \frac{1}{2} \sum_{\mathbf{A}\in\mathcal{P}} \frac{q^{dnk}}{q^{dnm}} \prod_{i\in[k]} (|\mathcal{R}_q^n|(1-\mathsf{negl}(\lambda)))^{-1} \cdot \mathsf{negl}(\lambda) \\ &= \frac{1}{2} \sum_{\mathbf{A}\in\mathcal{P}} \frac{1}{q^{dnm}} \cdot (1-\mathsf{negl}(\lambda))^{-k}\mathsf{negl}(\lambda) \quad \le \mathsf{negl}(\lambda).\end{aligned}$$

In the above, the first inequality is clear, the second is due to $\prod_{i\in[k]} \rho_{\sqrt{\mathbf{\Sigma}_i}}(\Lambda_q^{\mathbf{v}_i}(\mathbf{A}))$ containing all terms in $\sum_{\mathbf{U}\in\mathcal{T}_{\mathbf{A},\mathbf{V}}} \prod_{i\in[k]} \rho_{\sqrt{\mathbf{\Sigma}_i}}(\mathbf{u}_i)$ plus some more, the third and fourth are

Table 5.1: Summary of distributions in Section 5.4.

	D_0	D_1	D_2	D_3	D_4
$\mathbf{A}$	TrapGen	$\mathcal{U}(\mathcal{R}_q^{nT\times 2mT})$	$\to$	Reverse-sampling from $2m(T\text{-}1)$-MSIS	$\to$
$\mathbf{C}_j, j \in \mathsf{cor}$	TrapGen	$\to$	$\to$	$\to$	$\to$
$\mathbf{U}_j, j \in \mathsf{cor}$	SamplePreMP	$\mathcal{D}_{\Lambda_q^{\mathbf{v}_j}\otimes\mathbf{C}_j(\mathbf{A}),\mathbf{S}_{\mathsf{Cor}}}$	$\to$	$\mathbf{U}_{\mathsf{Cor}}$ from $2m(T\text{-}1)$-MSIS	$\to$
$\mathbf{C}_j, j \notin \mathsf{cor}$	TrapGen	$\mathcal{U}(\mathcal{R}_q^{n\times m})$	$\mathcal{U}(\mathrm{im}(\mathbf{U}_j))$	$\to$	$\to$
$\mathbf{U}_j, j \notin \mathsf{cor}$	SamplePreMP	$\mathcal{D}_{\Lambda_q^{\mathbf{v}_j}\otimes\mathbf{C}_j(\mathbf{A}),\mathbf{S}_{\mathsf{Cor}}}$	Random m-subspace	$\to$	BASIS-style sampling from $2m(T\text{-}1)$-MSIS
$\mathbf{x}_j, j \in \mathsf{act}$	SamplePreMP	$\mathcal{D}_{\Lambda(\mathbf{U}_j)\cap\Lambda_q^{\mathbf{v}_j}\otimes\mathbf{z}_j(\mathbf{A}),\sigma\cdot\mathbf{S}_{U_j}}$	$\mathcal{D}_{\Lambda(\mathbf{U}_j),\sigma\cdot\mathbf{S}_{U_j}}$	$\to$	$\to$

Main components in the construction (Figure 5.2) and their transformation from the original to the simulated distribution. The "$\to$" denotes no change in the distribution.

due to Corollary 4, and the final one is due to $|\mathcal{P}| \le q^{dnm}$. By Remark 1, $\mathbf{A} \leftarrow \mathcal{R}_q^{n\times m}$ is surjective with probability at least

$$1 - \frac{ng}{q^{d\cdot(m-n+1)/g}} \ge 1 - \mathsf{negl}(\lambda),$$

since by assumption $ng/q^{d(m-n+1)/g} \le \mathsf{negl}(\lambda)$. By the parameter choices and Lemma 57, $\mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}_i}) \ge \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ with overwhelming probability in λ for all $i \in [k]$. Then by Lemma 94, for a surjective $\mathbf{A}$ and the parameter choices, the vectors $\{\mathbf{u}_i\}_{i\in[k]}$ where $\mathbf{u}_i \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{v}_i}(\mathbf{A}),\sqrt{\mathbf{\Sigma}_i}}$ have $\mathcal{R}_{\mathfrak{q}_j}$-linearly independent columns for all $j \in [g]$ with overwhelming probability in λ. It follows that the statistical distance between $\mathcal{D}_0$ and $\mathcal{D}_1$ (unrestricted) is also negligible in λ. □

5.4 Simulation of Partial Trapdoors and Preimages

Looking ahead, in Sections 5.5 and 5.6, we will prove the one-wayness and indistinguishability of our partial trapdoor scheme from the k-MSIS and k-MLWE assumptions respectively for $k = (m + \tilde{m}) \cdot (T - 1)$. Both proofs consist of two main steps:

1. Given a k-MSIS/MLWE instance, simulate the partial trapdoors of corrupt parties and responses to partial preimage queries to PSampPreO.

2. Use the adversary's response to solve the k-MSIS/MLWE instance.

This subsection is dedicated to Step 1, i.e. simulating partial trapdoors and partial preimages.

In more detail, for any fixed set $\mathsf{cor} \subset [N]$, $|\mathsf{cor}| = T-1$ of corrupt parties, $L \leq \mathsf{poly}(\lambda)$ number of partial preimage queries, and reconstruction sets $\mathsf{act}_\ell \subseteq [N]$ for $\ell \in [L]$, we define in Figures 5.3 and 5.4 a sequence of distributions $D_i, i \in \{0,1,2,3,4\}$ of the tuple

$$\left(\mathbf{A}, \left(\mathsf{ptd}_j = (\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j})\right)_{j\in\mathsf{cor}}, (\mathbf{x}_{\ell,j}, \mathbf{y}_\ell)_{\ell\in[L], j\in\mathsf{act}_\ell}\right)$$

which are statistically close to each other. The first distribution D_0 captures (a core part of) the view of the adversary in the one-wayness and indistinguishability experiments[4] (see Figure 5.1), where in particular partial preimages $\mathbf{x}_{\ell,j}$ are generated using genuinely distributed partial trapdoors ptd_j. After a number of modifications, in the last distribution D_4, the partial preimages and partial trapdoors are instead efficiently simulated from a varying-width k-MSIS/MLWE instance. In particular, the matrices $\mathbf{A}$ and $\mathbf{C}_j$ are no longer generated with trapdoors. For an overview of the modifications between subsequent distributions, see Table 5.1.

In this subsection, in particular for distributions D_3 and D_4, we assume w.l.o.g. that party 0 is not corrupt, i.e. $\mathsf{cor} \subset [N] \setminus \{0\}$. We abuse notations and implicitly relabel the set $\{0\} \cup \mathsf{cor}$ as $[T]$, to avoid additional layers of subscripts. Under this assumption we denote $\mathbf{V}_{\{0\}\cup\mathsf{cor}} = \mathbf{V}_{[T]} \in \mathcal{R}_q^{T\times T}$ the Vandermonde matrix with (ordered) columns $(\mathbf{v}_j)_{j\in[T]}$. For the distribution D_4, we further introduce the shorthands $\mathbf{H}_j, \hat{\mathbf{H}}_j \in \mathcal{R}_q^{nT\times nT}$ for $j \notin \mathsf{cor}$,

$$\mathbf{H}_j \coloneqq \begin{bmatrix} 1 & & & \\ & v_j^{-1} & & \\ & & \ddots & \\ & & & v_j^{-(T-1)} \end{bmatrix} \otimes \mathbf{I}_n,\ \hat{\mathbf{H}}_j \coloneqq \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & \frac{v_1}{v_j} & \dots & \frac{v_{T-1}}{v_j} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \left(\frac{v_1}{v_j}\right)^{T-1} & \dots & \left(\frac{v_{T-1}}{v_j}\right)^{T-1} \end{bmatrix} \otimes \mathbf{I}_n. \quad (5.5)$$

In this case, since $v_j \notin \{v_1, v_2, \dots, v_{t-1}\}$ for any $j \notin \mathsf{cor}$, the matrix $\hat{\mathbf{H}}_j$ is guaranteed to be invertible over $\mathcal{R}_q$.

Lemma 95 ($D_0 \approx_s D_4$)**.** *Let parameters be as in Tables 5.2 and 5.3. The distributions D_0 in Figure 5.3 and D_4 in Figure 5.4 are statistically close in λ.*

Proof. The lemma follows from Lemmas 96 to 99 below. □

We give an overview of the main ideas of the proof of Lemma 95.

[4] More precisely, in $\mathsf{Exp}^{\mathtt{OW}}_{\Pi,\mathcal{A},\mathsf{par}_{\mathtt{OW}}}$, $\mathcal{A}$ is given these together with the challenge image $\mathbf{y}^*$ and all but one partial preimages $\left(\mathbf{x}_j^*\right)_{j\in T^*\setminus\{i^*\}}$, whereas in $\mathsf{Exp}^{\mathtt{IND}}_{\Pi,\mathcal{A},\mathsf{par}_{\mathtt{IND}}}$, $\mathcal{A}$ is further given the challenge LWE sample $(\mathbf{c}_0, \mathbf{c}_1)$. We address simulating the challenges in the main security proofs, i.e. Theorems 13 and 14.

$D_0 \to (\mathbf{A}, \left(\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j \in \mathsf{cor}}, (\mathbf{x}_{\ell,j}, \mathbf{y}_\ell)_{\ell \in [L], j \in \mathsf{act}_\ell})$

$$
\begin{aligned}
& (\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^{nT}, 1^{2mT}) \\
\forall j \in [N], \quad & (\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}) \leftarrow \mathsf{PTrapGen}(\mathbf{A}, \mathsf{td}, j) \\
\forall \ell \in [L], \quad & \mathbf{y}_\ell \leftarrow \mathcal{R}_q^{nT} \\
\forall \ell \in [L], j \in \mathsf{act}_\ell, \quad & \mathbf{x}_{\ell,j} \leftarrow \mathsf{PSampPre}(\mathbf{A}, (\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}), \mathsf{act}_\ell, \mathbf{y}_\ell)
\end{aligned}
$$

$D_1 \to (\mathbf{A}, \left(\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j \in \mathsf{cor}}, (\mathbf{x}_{\ell,j}, \mathbf{y}_\ell)_{\ell \in [L], j \in \mathsf{act}_\ell})$

$$
\begin{aligned}
& \mathbf{A} \leftarrow \mathcal{R}_q^{nT \times 2mT} \\
\forall j \in [N], \quad & \begin{cases} (\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m) & j \in \mathsf{cor} \\ \mathbf{C}_j \leftarrow \mathcal{R}_q^{n \times m} & j \notin \mathsf{cor} \end{cases} \\
\forall j \in [N], \quad & \mathbf{U}_j \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT}, \mathbf{S}_{\mathsf{cor}}})^m \text{ s.t. } \mathbf{A} \cdot \mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{C}_j \bmod q \\
\forall \ell \in [L], j \in \mathsf{act}_\ell, \quad & \mathbf{z}_{\ell,j} \leftarrow \mathcal{R}_q^n; \quad \Lambda_j \coloneqq \Lambda(\mathbf{U}_j) \cap \Lambda_q^{\mathbf{v}_j \otimes \mathbf{z}_{\ell,j}}(\mathbf{A}); \quad \mathbf{x}_{\ell,j} \leftarrow \mathcal{D}_{\Lambda_j, \sigma \cdot \mathbf{S}_{U_j}} \\
\forall \ell \in [L], \quad & \mathbf{y}_\ell \coloneqq \sum_{j \in \mathsf{act}_\ell} \mathbf{v}_j \otimes \mathbf{z}_{\ell,j} \bmod q
\end{aligned}
$$

$D_2 \to (\mathbf{A}, \left(\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j \in \mathsf{cor}}, (\mathbf{x}_{\ell,j}, \mathbf{y}_\ell)_{\ell \in [L], j \in \mathsf{act}_\ell})$

$$
\begin{aligned}
& \mathbf{A} \leftarrow \mathcal{R}_q^{nT \times 2mT} \\
\forall j \in \mathsf{cor}, \quad & (\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m) \\
\forall j \in [N], \quad & \begin{cases} \mathbf{U}_j \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT}, \mathbf{S}_{\mathsf{cor}}})^m \text{ s.t. } \mathbf{A} \cdot \mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{C}_j \bmod q & j \in \mathsf{cor} \\ \mathbf{U}_j \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT}, \mathbf{S}_{\mathsf{cor}}})^m \text{ s.t. } \mathbf{A} \cdot \mathbf{U}_j \in (\mathcal{R}_q\text{-span}(\mathbf{v}_j \otimes \mathbf{I}_n))^m & j \notin \mathsf{cor} \end{cases} \\
\forall \ell \in [L], j \in \mathsf{act}_\ell, \quad & \mathbf{x}_{\ell,j} \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}_j), \sigma \cdot \mathbf{S}_{U_j}} \\
\forall \ell \in [L], \quad & \mathbf{y}_\ell \coloneqq \mathbf{A} \cdot \sum_{j \in \mathsf{act}_\ell} \mathbf{x}_{\ell,j} \bmod q
\end{aligned}
$$

Figure 5.3: Distributions D_0, D_1, D_2, with respect to a fixed set $\mathsf{cor} \subset [N]$ of corrupt parties, $L \leq \mathsf{poly}(\lambda)$ number of partial preimage queries, and reconstruction sets $\mathsf{act}_\ell \subseteq [N]$ for $\ell \in [L]$.

$\mathsf{ExtendA}(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}) \to (\mathbf{A}, \left(\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j\in\mathsf{cor}})$

$\forall j \in \mathsf{cor},\ (\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^n, 1^m)$

$\tilde{\mathbf{A}}_{\neq 0} \leftarrow \mathcal{R}_q^{n(T-1)\times 2mT}$ s.t. $\tilde{\mathbf{A}}_{\neq 0} \cdot (\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}) \equiv \left(\mathsf{diag}(\{\mathbf{C}_j\}_{j\in\mathsf{cor}}), \mathbf{G}_{n(T-1)}\right)$

$\mathbf{A} \coloneqq \left(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n\right) \cdot [\tilde{\mathbf{A}}_0 || \tilde{\mathbf{A}}_{\neq 0}] \bmod q$

$D_3 \to (\mathbf{A}, \left(\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j\in\mathsf{cor}}, (\mathbf{x}_{\ell,j}, \mathbf{y}_\ell)_{\ell\in[L], j\in\mathsf{act}_\ell})$

$\tilde{\mathbf{A}}_0 \leftarrow \mathcal{R}_q^{n\times 2mT}$

$\mathbf{U}_{\mathtt{Cor}} \leftarrow (\mathcal{D}_{\Lambda_q^\perp(\tilde{\mathbf{A}}_0), \mathbf{S}_{\mathtt{Cor}}})^{m(T-1)}$

$\mathbf{U}_{\mathtt{Hon}} \leftarrow (\mathcal{D}_{\Lambda_q^\perp(\tilde{\mathbf{A}}_0), \mathbf{S}_{\mathtt{Hon}}})^{\tilde{m}(T-1)}$

$(\mathbf{A}, \left(\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j\in\mathsf{cor}}) \leftarrow \mathsf{ExtendA}(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$

$\forall j \in [N],$ $\begin{cases} \mathbf{U}_j \coloneqq \mathbf{U}_{\mathtt{Cor}} \cdot (\mathbf{e}_{\mathsf{cor},j} \otimes \mathbf{I}_m) \quad /\!/\ (\mathbf{U}_j)_{j\in\mathsf{cor}} = \mathbf{U}_{\mathtt{Cor}} & j \in \mathsf{cor} \\ \mathbf{U}_j \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT}, \mathbf{S}_{\mathtt{Cor}}})^m \text{ s.t. } \mathbf{A}\cdot\mathbf{U}_j \in (\mathcal{R}_q\text{-span}(\mathbf{v}_j \otimes \mathbf{I}_n))^m & j \notin \mathsf{cor} \end{cases}$

$\forall \ell \in [L], j \in \mathsf{act}_\ell,\ \mathbf{x}_{\ell,j} \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}_j), \sigma\cdot\mathbf{S}_{U_j}}$

$\forall \ell \in [L],$ $\mathbf{y}_\ell \coloneqq \mathbf{A} \sum_{j\in\mathsf{act}_\ell} \mathbf{x}_{\ell,j} \bmod q$

$D_4 \to (\mathbf{A}, \left(\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j\in\mathsf{cor}}, (\mathbf{x}_{\ell,j}, \mathbf{y}_\ell)_{\ell\in[L], j\in\mathsf{act}_\ell})$

$\tilde{\mathbf{A}}_0 \leftarrow \mathcal{R}_q^{n\times 2mT}$

$\mathbf{U}_{\mathtt{Cor}} \leftarrow (\mathcal{D}_{\Lambda_q^\perp(\tilde{\mathbf{A}}_0), \mathbf{S}_{\mathtt{Cor}}})^{m(T-1)}$

$\mathbf{U}_{\mathtt{Hon}} \leftarrow (\mathcal{D}_{\Lambda_q^\perp(\tilde{\mathbf{A}}_0), \mathbf{S}_{\mathtt{Hon}}})^{\tilde{m}(T-1)}$

$(\mathbf{A}, \left(\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j\in\mathsf{cor}}) \leftarrow \mathsf{ExtendA}(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$

$\mathbf{T} \coloneqq \begin{bmatrix} \mathbf{0}_{2mT\times\tilde{m}} & \mathbf{U}_{\mathtt{Hon}} \\ -\mathbf{I}_{\tilde{m}} & \mathbf{0}_{\tilde{m}\times m(T-1)} \end{bmatrix}$

$\forall j \in [N],$ $\mathbf{B}_j \coloneqq \left[\ \mathbf{H}_j\mathbf{A} \ \middle|\ \mathbf{1}_T \otimes -\mathbf{G}_n\ \right] \quad /\!/\ \mathbf{B}_j\,\mathbf{T} \equiv \hat{\mathbf{H}}_j\,\mathbf{G}_{nT}$

$\forall j \in [k],$ $\begin{cases} \mathbf{U}_j \coloneqq \mathbf{U}_{\mathtt{Cor}} \cdot (\mathbf{e}_{\mathsf{cor},j} \otimes \mathbf{I}_n) \quad /\!/\ (\mathbf{U}_j)_{j\in\mathsf{cor}} = \mathbf{U}_{\mathtt{Cor}} & j \in \mathsf{cor} \\ \begin{bmatrix} \mathbf{U}_j \in \mathcal{R}_q^{2mT\times m} \\ \mathbf{W}_j \in \mathcal{R}_q^{\tilde{m}\times m} \end{bmatrix} \leftarrow (\mathsf{SamplePreMP}(\mathbf{B}_j, \mathbf{T}, \hat{\mathbf{H}}_j, \mathbf{0}, \mathbf{S}_{\mathtt{BASIS}}))^m & j \notin \mathsf{cor} \end{cases}$

$\forall \ell \in [L], j \in \mathsf{act}_\ell,\ \mathbf{x}_{\ell,j} \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}_j), \sigma\cdot\mathbf{S}_{U_j}}$

$\forall \ell \in [L],$ $\mathbf{y}_\ell \coloneqq \mathbf{A} \sum_{j\in\mathsf{act}_\ell} \mathbf{x}_{\ell,j} \bmod q$

Figure 5.4: Definitions of a subroutine $\mathsf{ExtendA}$ and distributions D_3, D_4, with respect to some fixed set $\mathsf{cor} \subset [N]$ of corrupt parties, $L \le \mathsf{poly}(\lambda)$ number of partial preimage queries, and reconstruction sets $\mathsf{act}_\ell \subseteq [N]$ for $\ell \in [L]$.

$D_0 \approx_s D_1$ The modifications from D_0 to D_1 are mostly about switching between running $\mathsf{TrapGen}$ and sampling uniformly random matrices, and between running $\mathsf{SamplePreMP}$ and sampling perfect Gaussian. The statistical closeness of these changes follows from standard arguments. The most interesting difference is how the $\mathbf{x}_{\ell,j}$ are sampled in D_0 and D_1, which despite the differing representations are in fact identical as we show in Lemma 91.

Lemma 96 ($D_0 \approx_s D_1$)**.** *Let parameters be as in Tables 5.2 and 5.3. The distributions D_0 and D_1 in Figure 5.3 are statistically close in λ.*

Proof. For any $\mathbf{y}_\ell \in \mathcal{R}_q^{nT}$, note that its decomposition into $\mathbf{y}_\ell = \sum_{j\in\mathsf{act}} \mathbf{v}_j \otimes \mathbf{z}_{\ell,j} \bmod q$ with respect to $(\mathbf{v}_j)_{j\in\mathsf{act}}$ is unique. Consequently, a uniformly random $\mathbf{y}_\ell \in \mathcal{R}_q^{nT}$ implies uniformly random $(\mathbf{z}_{\ell,j})_{j\in\mathsf{act}}$ and vice versa. It therefore suffices to consider the statement for any fixed $(\mathbf{y}_\ell, (\mathbf{z}_{\ell,j})_{j\in\mathsf{act}})_{\ell\in[L]}$ satisfying $\mathbf{y}_\ell = \sum_{j\in\mathsf{act}} \mathbf{v}_j \otimes \mathbf{z}_{\ell,j} \bmod q$ and the result follows from taking the average. In the below, we denote the variants of D_0 and D_1 with fixed $\mathbf{Y} \coloneqq (\mathbf{y}_\ell)_{\ell\in[L]}$ (and hence $(\mathbf{z}_{\ell,j})_{j\in\mathsf{act}}$) by $D_{0,\mathbf{Y}}$ and $D_{1,\mathbf{Y}}$ respectively. We now show that $D_{0,\mathbf{Y}}$ is statistically close to $D_{1,\mathbf{Y}}$.

In $D_{0,\mathbf{Y}}$, the tuple $(\mathbf{A}, \left(\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j\in[N]}, (\mathbf{x}_{\ell,j})_{\ell\in[L],j\in T})$ is sampled as follows: Run $\mathsf{TrapGen}$ to generate $\mathbf{A}$ and its trapdoor. Run $\mathsf{TrapGen}$ to generate all the $\mathbf{C}_j$ with their trapdoors. For $j \in [N]$, sample $\mathbf{U}_j$ from $\mathsf{SamplePreMP}(\mathbf{A}, \mathsf{td}, \mathbf{v}_j \otimes \mathbf{C}_j, \mathbf{S}_{\mathsf{Cor}})$. For $\ell \in [L]$ and $j \in \mathsf{act}_\ell$, sample $\mathbf{d}_{\ell,j}$ from $\mathsf{SamplePreMP}(\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}, \mathbf{z}_{\ell,j}, \sqrt{\mathbf{\Sigma}_j})$, and set $\mathbf{x}_{\ell,j} = \mathbf{U}_j\, \mathbf{d}_{\ell,j}$.

In $D_{1,\mathbf{Y}}$, the tuple is sampled as follows: Sample $\mathbf{A}$ uniformly at random. Run $\mathsf{TrapGen}$ to generate $\mathbf{C}_j$ with their trapdoors for $j \in \mathsf{cor}$, and sample the remaining $\mathbf{C}_j$ uniformly at random. For $j \in [N]$, sample $\mathbf{U}_j$ from $(\mathcal{D}_{\mathcal{R}^{2mT},\mathbf{S}_{\mathsf{Cor}}})^m$ subject to the constraint $\mathbf{A}\,\mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{C}_j \bmod q$. For $\ell \in [L]$ and $j \in \mathsf{act}_\ell$, sample $\mathbf{x}_{\ell,j}$ from $\mathcal{D}_{\Lambda_j,\sigma\cdot\mathbf{S}_{U_j}}$ where $\Lambda_j \coloneqq \Lambda(\mathbf{U}_j) \cap \Lambda_q^{\mathbf{v}_j\otimes\mathbf{z}_{\ell,j}}(\mathbf{A})$.

To see that the above are statistically close, we first invoke Lemma 50 to replace calls to $\mathsf{SamplePreMP}$ in $D_{0,\mathbf{Y}}$ by perfect Gaussian sampling. We then invoke Lemma 91, which states that sampling $\mathbf{x}_{\ell,j}$ from $\Lambda(\mathbf{U}_j) \cap \Lambda_q^{\mathbf{v}_j\otimes\mathbf{z}_{\ell,j}}(\mathbf{A})$ (as in $D_{1,\mathbf{Y}}$), where $\mathbf{A}\,\mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{C}_j \bmod q$, is identical to first sampling Gaussians from $\Lambda_q^{\mathbf{z}_{\ell,j}}(\mathbf{C}_j)$ and then multiplying the result by $\mathbf{U}_j$ (as in $D_{1,\mathbf{Y}}$). Finally, we invoke Lemma 50 to replace the sampling of $\mathbf{A}$ from $\mathsf{TrapGen}(1^\lambda, 1^{nT}, 1^{2mT})$ and of all $(\mathbf{C}_j)_{j\notin\mathsf{cor}}$ from $\mathsf{TrapGen}(1^\lambda, 1^n, 1^m)$ in $D_{0,\mathbf{Y}}$ to sampling them uniformly at random. □

$D_1 \approx_s D_2$ Recall that in distribution D_1 the values $\mathbf{U}_j, \mathbf{C}_j, \mathbf{x}_{\ell,j}$ are sampled subject to various constraints. In distribution D_2, the constraints for $(\mathbf{U}_j, \mathbf{C}_j)_{j\notin\mathsf{cor}}$ and $(\mathbf{x}_{\ell,j})_{\ell\in[L],j\in\mathsf{act}_\ell}$

are removed. To prove that $D_1 \approx_s D_2$, we rely on a generalised regularity lemma (Proposition 4) which states that first sampling a uniformly random $\mathbf{C}$ and then a Gaussian $\mathbf{U}$ subject to $\mathbf{A} \cdot \mathbf{U} = \mathbf{v} \otimes \mathbf{C} \bmod q$ is statistically close to first sampling a Gaussian $\mathbf{U}$ subject to $\mathbf{A} \cdot \mathbf{U} \in (\mathcal{R}_q\text{-span}(\mathbf{v} \otimes \mathbf{I}))^m$ and then computing the matrix $\mathbf{C}$.

Besides this, the only difference between D_1 and D_2 is in sampling the partial preimages $\mathbf{x}_{\ell,j}$ (and hence $\mathbf{y}_\ell$). In more detail, in D_1 they are sampled from $\Lambda(\mathbf{U}_j) \cap \Lambda_q^{\mathbf{v}_j \otimes \mathbf{z}_{\ell,j}}(\mathbf{A})$ for some random $\mathbf{z}_{\ell,j}$, while in D_2 they are simply sampled from $\Lambda(\mathbf{U}_j)$. Note that the relation between this pair of distributions is almost identical to the argument about $\mathbf{U}$, and thus we apply the same technique as in Proposition 4. The only technical question we solve is upper bounding the smoothing parameter of the intersection $\Lambda(\mathbf{U}_j) \cap \Lambda_q^{\perp}(\mathbf{A})$ detailed in Proposition 5.

Lemma 97 ($D_1 \approx_s D_2$)**.** *Let parameters be as in Tables 5.2 and 5.3. The distributions D_1 and D_2 in Figure 5.3 are statistically close in λ.*

Proof. With overwhelming probability, $\mathbf{A} \leftarrow \mathcal{R}_q^{nT \times 2mT}$ is surjective since $\mathcal{R}_q$ splits into fields of superpolynomial size. Hence, we run the arguments below assuming $\mathbf{A}$ is surjective. Since $\mathbf{S}_{\mathsf{Cor}} \geq \eta_\varepsilon(\Lambda_q^{\perp}(\mathbf{A}))$, by Proposition 4, the distribution D_2 is statistically close to a similar distribution D_2' where, for $j \notin \mathsf{cor}$, we sample $\mathbf{C}_j$ uniformly at random and then sample $\mathbf{U}_j$ subject to the constraint $\mathbf{A}\,\mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{C}_j \bmod q$, i.e. as in D_1. Now, the only difference between D_1 and D_2' is in how $\mathbf{x}_{\ell,j}$ (and hence $\mathbf{y}_\ell$) are sampled, and we can consider $(\mathbf{A}, \left(\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j \in \mathsf{cor}})$ as fixed values. Given that $\mathcal{R}_q$ splits into fields of superpolynomial size, uniform matrices $\mathbf{C}_j \in \mathcal{R}_q^{n \times m}$ are surjective with overwhelming probability. We assume this is the case for our fixed $\mathbf{C}_j$. Since in both D_1 and D_2' the distributions of $\mathbf{x}_{\ell,j}$ for different (ℓ, j) are independent, it suffices to analyse the distribution of one arbitrary $\mathbf{x} = \mathbf{x}_{\ell,j}$.

Hence, we show that for any fixed surjective $\mathbf{A} \in \mathcal{R}_q^{nT \times 2mT}$, surjective $\mathbf{C} \in \mathcal{R}_q^{n \times m}$, $\mathbf{v} = (1, v, \ldots, v^{T-1})$ with $v \in \mathcal{R}_q^{\times}$, and $\mathbf{U} \in \mathcal{R}^{2mT \times m}$ s.t. $\mathbf{A} \cdot \mathbf{U} = \mathbf{v} \otimes \mathbf{C} \bmod q$, the following distributions are statistically close in λ:

$$\left\{\mathbf{x} \;\middle|\; \mathbf{z} \leftarrow \mathcal{R}_q^n;\ \mathbf{x} \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}) \cap \Lambda_q^{\mathbf{v} \otimes \mathbf{z}}(\mathbf{A}), \sigma \cdot \mathbf{S}_U}\right\} \qquad \text{and} \qquad \left\{\mathbf{x} \;\middle|\; \mathbf{x} \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}), \sigma \cdot \mathbf{S}_U}\right\}.$$

Fix any $\mathbf{p} \in \mathcal{R}^{2mT}$ satisfying $\mathbf{A} \cdot \mathbf{p} = \mathbf{v} \otimes \mathbf{z} \bmod q$. Define a lattice $\Psi \subset \Lambda(\mathbf{U})$ and a lattice coset $\Psi + \mathbf{p}$ as

$$\Psi \coloneqq \{\mathbf{x} \in \Lambda(\mathbf{U}) : \mathbf{A} \cdot \mathbf{x} = \mathbf{0} \bmod q\},$$
$$\Psi + \mathbf{p} \coloneqq \{\mathbf{x} \in \Lambda(\mathbf{U}) : \mathbf{A} \cdot \mathbf{x} = \mathbf{v} \otimes \mathbf{z} \bmod q\}.$$

Since $\mathbf{C}$ is surjective, the set of all images equals

$$\begin{aligned}\{\mathbf{A}\cdot\mathbf{x} \bmod q : \mathbf{x}\in\Lambda(\mathbf{U})\} &= \{\mathbf{A}\cdot\mathbf{U}\cdot\mathbf{d} \bmod q : \mathbf{d}\in\mathcal{R}^m\}\\ &= \{\mathbf{v}\otimes\mathbf{C}\cdot\mathbf{d} \bmod q : \mathbf{d}\in\mathcal{R}^m\}\\ &= \left\{\mathbf{v}\otimes\mathbf{z} \bmod q : \mathbf{z}\in\mathcal{R}_q^n\right\}.\end{aligned}$$

By Proposition 5, $\eta_\varepsilon(\Lambda(\Psi)) \le \mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U}))\cdot\eta_\varepsilon(\Lambda_q^\perp(\mathbf{C}))$, and $\mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U})) \le \mathfrak{s}_{\max}(\mathbf{S}_{\mathtt{Cor}})\cdot\sqrt{2dm^2T}$ with overwhelming probability by Lemmas 4 and 24. Therefore, we have $\sigma \ge \eta_\varepsilon(\Lambda(\Psi))$ with overwhelming probability by our choice of parameters. Hence by Lemma 39, $\mathcal{D}_{\Lambda(\mathbf{U}),\sigma} \bmod \Psi$ is statistically close to $\mathcal{U}(\Lambda(\mathbf{U})/\Psi)$. Since $\Lambda(\mathbf{U})/\Psi$ is isomorphic to the image space $\left\{\mathbf{v}\otimes\mathbf{z} \bmod q : \mathbf{z}\in\mathcal{R}_q^n\right\}$, we have that $\mathbf{A}\cdot(\mathcal{D}_{\Lambda(\mathbf{U}),\sigma} \bmod \Psi) \bmod q$ is statistically close to $\mathcal{U}(\left\{\mathbf{v}\otimes\mathbf{z} \bmod q : \mathbf{z}\in\mathcal{R}_q^n\right\})$. □

$D_2 \approx_s D_3$. By hopping from D_0 to D_2, what we have accomplished so far is to let partial preimages be sampled as $\mathbf{x}_{\ell,j} \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}_j),\sigma}$. However, the matrices $\mathbf{U}_j$ are sampled inefficiently from Gaussian distributions subject to constraints. Our goal in D_3 and D_4 is to simulate the matrices $\mathbf{U}_j$ (along with $\mathbf{A}$) given a varying-width k-MSIS/MLWE instance $(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$ satisfying $\tilde{\mathbf{A}}_0\cdot(\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}) = \mathbf{0} \bmod q$, where $\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}$ are Gaussian matrices with varying Gaussian parameters $\mathbf{S}_{\mathtt{Cor}}$ and $\mathbf{S}_{\mathtt{Hon}}$ respectively. It is crucial that the Gaussian parameter $\mathbf{S}_{\mathtt{Cor}}$ associated with the corrupt parties is larger than $\mathbf{S}_{\mathtt{Hon}}$ associated with the honest parties. This is because, while we can directly use $\mathbf{U}_{\mathtt{Cor}}$ as $(\mathbf{U}_j)_{j\in\mathsf{cor}}$ for the corrupt parties, simulating partial trapdoors and partial preimages for honest parties requires rerandomising $\mathbf{U}_{\mathtt{Hon}}$ which makes the Gaussian width of the resulting partial preimages grow. Therefore, there must be a large enough gap between $\mathbf{S}_{\mathtt{Cor}}$ and $\mathbf{S}_{\mathtt{Hon}}$ so that partial preimages for both corrupt and honest parties have identical Gaussian widths.

Although being visually quite different, the closeness of D_2 and D_3 can be proven with a few arithmetic transformations and two invocations of "reverse sampling" (i.e. Theorem 12). Theorem 12 states that for fixed $\mathbf{V}$ first sampling uniformly random $\mathbf{A}$ and then Gaussian $\mathbf{U}$ subject to $\mathbf{A}\cdot\mathbf{U} = \mathbf{V} \bmod q$ is statistically close to first sampling a Gaussian $\mathbf{U}$ and then a uniformly random $\mathbf{A}$ subject to the same constraint.

Lemma 98 ($D_2 \approx_s D_3$)**.** *Let parameters be as in Tables 5.2 and 5.3. The distributions D_2 in Figure 5.3 and D_3 in Figure 5.4 are statistically close in λ.*

Proof. We argue by gradually modifying D_3 into D_2 through statistically close hybrids. Note that the only difference between D_2 and D_3 lies in how $(\mathbf{A}, (\mathbf{U}_j)_{j\in\mathsf{cor}})$ are sampled.

It therefore suffices to argue that the distributions of $(\mathbf{A}, (\mathbf{U}_j)_{j\in\mathsf{cor}})$ in D_2 and D_3 are statistically close. We denote $\mathbf{U}_{\mathtt{Cor}} \coloneqq (\mathbf{U}_j)_{j\in\mathsf{cor}} \in \mathcal{R}_q^{2mT\times m(T-1)}$ the horisontal concatenation of $\mathbf{U}_j$ for all $j \in \mathsf{cor}$. Written differently, $\mathbf{U}_j = \mathbf{U}_{\mathtt{Cor}} \cdot (\mathbf{e}_{\mathsf{cor},j} \otimes \mathbf{I}_m)$. Similarly, denote $\mathbf{U}_{\mathtt{Hon}} \coloneqq (\overline{\mathbf{U}}_j)_{j\in\mathsf{cor}} \in \mathcal{R}_q^{2mT\times \tilde{m}(T-1)}$. By Theorem 12, $(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$ sampled from

$$\left\{\begin{array}{l}\tilde{\mathbf{A}}_0 \leftarrow \mathcal{R}_q^{n\times 2mT},\\ \mathbf{U}_{\mathtt{Cor}} \leftarrow (\mathcal{D}_{\Lambda_q^\perp(\tilde{\mathbf{A}}_0),\mathbf{S}_{\mathtt{Cor}}})^{m(T-1)}\\ \mathbf{U}_{\mathtt{Hon}} \leftarrow (\mathcal{D}_{\Lambda_q^\perp(\tilde{\mathbf{A}}_0),\mathbf{S}_{\mathtt{Hon}}})^{\tilde{m}(T-1)}\end{array}\right\} \quad \text{and} \quad \left\{\begin{array}{l}\mathbf{U}_{\mathtt{Cor}} \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT},\mathbf{S}_{\mathtt{Cor}}})^{m(T-1)},\\ \mathbf{U}_{\mathtt{Hon}} \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT},\mathbf{S}_{\mathtt{Hon}}})^{\tilde{m}(T-1)},\\ \tilde{\mathbf{A}}_0 \leftarrow \mathcal{R}_q^{n\times 2mT} : \tilde{\mathbf{A}}_0 \cdot (\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}) = \mathbf{0} \bmod q\end{array}\right\}$$

are statistically close. Then, by definition of the distributions, for any fixed set $\{\mathbf{C}_j\}_{j\in\mathsf{cor}}$, the joint distribution of $(\tilde{\mathbf{A}}_0, \tilde{\mathbf{A}}_{\neq 0}, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$ from D_3 is statistically close to that sampled from

$$\left\{\begin{array}{l}\mathbf{U}_{\mathtt{Cor}} \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT},\mathbf{S}_{\mathtt{Cor}}})^{m(T-1)}, \mathbf{U}_{\mathtt{Hon}} \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT},\mathbf{S}_{\mathtt{Hon}}})^{\tilde{m}(T-1)},\\ \begin{bmatrix}\tilde{\mathbf{A}}_0\\ \tilde{\mathbf{A}}_{\neq 0}\end{bmatrix} \leftarrow \mathcal{R}_q^{nT\times 2mT} \text{ s.t. } \begin{bmatrix}\tilde{\mathbf{A}}_0\\ \tilde{\mathbf{A}}_{\neq 0}\end{bmatrix} \cdot (\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}) \equiv \underbrace{\left[\begin{array}{c|c}\mathbf{0}_{n\times(m+\tilde{m})(T-1)} & \\ \hline \mathsf{diag}(\{\mathbf{C}_j\}_{j\in\mathsf{cor}}) & \mathbf{G}_{n(T-1)}\end{array}\right]}_{\coloneqq\hat{\mathbf{C}}}\end{array}\right\}.$$

Lastly, the multiplication by $\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n$ is a bijection over $\mathcal{R}_q$, hence, the matrix $\mathbf{A} = (\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n) \cdot [\tilde{\mathbf{A}}_0 || \tilde{\mathbf{A}}_{\neq 0}]$ also follows the uniform distribution with an adjusted condition

$$\begin{aligned}\mathbf{A} \leftarrow \mathcal{R}_q^{nT\times 2mT} : \mathbf{A} \cdot (\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}) &= \left(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n\right) \cdot \hat{\mathbf{C}} \bmod q\\ &\equiv ((\ldots, \mathbf{v}_j \otimes \mathbf{C}_j, \ldots)_{j\in\mathsf{cor}}, (\ldots, \mathbf{v}_j \otimes \mathbf{G}_n, \ldots)_{j\in\mathsf{cor}}).\end{aligned}$$

Notice that the above sampling constraint is equivalent to $\mathbf{A} \cdot \mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{C}_j \bmod q$ and $\mathbf{A} \cdot \overline{\mathbf{U}}_j = \mathbf{v}_j \otimes \mathbf{G}_n \bmod q$ for all $j \in \mathsf{cor}$. Applying Theorem 12 again, $(\mathbf{A}, (\mathbf{U}_j, \overline{\mathbf{U}}_j)_{j\in\mathsf{cor}})$ from above are statistically close to

$$\left\{\begin{array}{l}\mathbf{A} \leftarrow \mathcal{R}_q^{nT\times 2mT}\\ \mathbf{U}_j \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT},\mathbf{S}_{\mathtt{Cor}}})^{m} : \mathbf{A} \cdot \mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{C}_j \bmod q \ \ \forall j \in \mathsf{cor}\\ \overline{\mathbf{U}}_j \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT},\mathbf{S}_{\mathtt{Hon}}})^{\tilde{m}} : \mathbf{A} \cdot \overline{\mathbf{U}}_j = \mathbf{v}_j \otimes \mathbf{G}_n \bmod q \ \ \forall j \in \mathsf{cor}\end{array}\right\}.$$

At this point, $(\mathbf{A}, (\mathbf{U}_j)_{j\in\mathsf{cor}})$ is sampled exactly as in D_2 which proves the statement. □

$D_3 \approx_s D_4$. The only difference between D_3 and D_4 is the distribution of $\mathbf{U}_j$ for $j \notin \mathsf{cor}$. In D_4 the matrices $\mathbf{U}_j$ for honest parties are simulated efficiently from $\mathbf{U}_{\mathtt{Hon}}$. Specifically,

the step $\mathbf{U}_j \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT},\mathbf{S}_{\mathtt{Cor}}})^m$ s.t. $\mathbf{A}\cdot\mathbf{U}_j \in (\mathcal{R}_q\text{-span}(\mathbf{v}_j \otimes \mathbf{I}_n))^m$ is replaced by the following procedure: Let $\mathbf{T} \coloneqq \begin{bmatrix} \mathbf{0}_{2mT\times\tilde{m}} & \mathbf{U}_{\mathtt{Hon}} \\ -\mathbf{I}_{\tilde{m}} & \mathbf{0}_{\tilde{m}\times\tilde{m}(T-1)} \end{bmatrix}$ be a trapdoor of the BASIS-like matrices [WW23] $\mathbf{B}_j \coloneqq [\mathbf{H}_j\cdot\mathbf{A} \mid \mathbf{1}_t \otimes -\mathbf{G}_n]$ for all $j \notin \mathsf{cor}$. Note that $\mathbf{B}_j\cdot\mathbf{T} = \hat{\mathbf{H}}_j\cdot\mathbf{G}_{nT} \bmod q$ for a unit tag matrix $\hat{\mathbf{H}}_j$. Using such a trapdoor, we can sample $\mathbf{U}_j$ as $\begin{bmatrix} \mathbf{U}_j \in \mathcal{R}_q^{2mT\times m} \\ \mathbf{W}_j \in \mathcal{R}_q^{\tilde{m}\times m} \end{bmatrix} \leftarrow (\mathsf{SamplePreMP}(\mathbf{B}_j,\mathbf{T},\hat{\mathbf{H}}_j,\mathbf{0},\mathbf{S}_{\mathtt{Cor}}))^m$, where $\mathbf{W}_j$ only determines the image of $\mathbf{U}_j$ in the required image-subspace and can be discarded.

The proof of statistical closeness between D_3 and D_4 is the most involved among all hops. It consists of two main steps. The first is to show that, if $\mathbf{T}$ is a valid $\mathbf{G}$-trapdoor of $\mathbf{B}_j$ with tag matrix $\hat{\mathbf{H}}_j$ for all $j \notin \mathsf{cor}$, then $\mathbf{U}_j$ sampled in D_3 and D_4 are statistically close. For this, we use Lemma 92 which in turn follows from the regularity lemma (Proposition 4) and direct inspection of Gaussian weights. The second step is to show that $\mathbf{T}$ is indeed a trapdoor and with the desirable quality. This follows by bounding the norm of $\mathbf{T}$ and proving that $\mathbf{B}_j\cdot\mathbf{T} = \hat{\mathbf{H}}_j\cdot\mathbf{G}_{nT} \bmod q$.

Lemma 99 ($D_3 \approx_s D_4$)**.** *Let parameters be as in Tables 5.2 and 5.3. Let* ($\mathsf{TrapGen}$, $\mathsf{SamplePreMP}$) *be that in Lemma 50. Then D_3 and D_4 in Figure 5.4 are statistically close in λ.*

Proof. It suffices to consider the distribution of $\mathbf{U}_j$ for $j \notin \mathsf{cor}$, since the distributions of other variables in D_3 and D_4 conditioning on $(\mathbf{U}_j)_{j\notin\mathsf{cor}}$ are identical.

Assume for the moment that, $\mathbf{T}$ constructed in D_4 is a valid $\mathbf{G}$-trapdoor of $\mathbf{B}_j$ with tag $\hat{\mathbf{H}}_j$ (given by Equation (5.5)), for all $j \notin \mathsf{cor}$, so that sampling $[\mathbf{U}_j||\mathbf{W}_j]$ from $\mathsf{SamplePreMP}(\mathbf{B}_j,\mathbf{T},\hat{\mathbf{H}}_j,\mathbf{0},\mathbf{S}_{\mathtt{BASIS}})$ is possible. For any $j \notin \mathsf{cor}$, we inspect the relation between $\mathbf{U}_j$ and $\mathbf{W}_j$. Write $\mathbf{A} = [\mathbf{A}_0||\ldots||\mathbf{A}_{T-1}]$, i.e. each $\mathbf{A}_i \in \mathcal{R}_q^{n\times 2mT}, i\in[T]$, is a row chunk of $\mathbf{A}$. Recall the definition of $\mathbf{H}_j$ in Equation (5.5), and observe that the constructed $\mathbf{B}_j$ is of the form

$$\begin{aligned}\mathbf{B}_j &= \left[\mathbf{H}_j\cdot\mathbf{A} \mid \mathbf{1}_T\otimes -\mathbf{G}_n\right] \\ &= \left[\begin{pmatrix} v_j^{-0}\mathbf{I}_n & & \\ & \ddots & \\ & & v_j^{-(T-1)}\mathbf{I}_n \end{pmatrix}\cdot\mathbf{A} \;\middle|\; \mathbf{1}_T\otimes -\mathbf{G}_n\right] = \left[\begin{array}{ccc|c} 1 & \cdot & \mathbf{A}_0 & -\mathbf{G}_n \\ v_j^{-1} & \cdot & \mathbf{A}_1 & -\mathbf{G}_n \\ & \vdots & & \vdots \\ v_j^{-(T-1)} & \cdot & \mathbf{A}_{T-1} & -\mathbf{G}_n \end{array}\right].\end{aligned}$$

Therefore the sampled $\mathbf{U}_j, \mathbf{W}_j$ satisfy

$$\left[\begin{array}{rcc|c} 1 & \cdot & \mathbf{A}_0 & -\mathbf{G} \\ v_j^{-1} & \cdot & \mathbf{A}_1 & -\mathbf{G} \\ & \vdots & & \vdots \\ v_j^{-(t-1)} & \cdot & \mathbf{A}_{t-1} & -\mathbf{G} \end{array}\right] \cdot \begin{bmatrix} \mathbf{U}_j \\ \mathbf{W}_j \end{bmatrix} = \mathbf{0} \bmod q.$$

Which is equivalent to $\mathbf{A}_i \cdot \mathbf{U}_j = v_j^i \cdot \mathbf{G} \cdot \mathbf{W}_j \bmod q$ for each $i \in [T]$. Concatenating vertically for all $i \in [T]$, we obtain

$$\mathbf{A} \cdot \mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{G} \cdot \mathbf{W}_j \bmod q.$$

By the choice of parameters and Lemma 50, $\mathbf{U}_j$ obtained from $\mathsf{SamplePreMP}$ is close to

$$\left\{ \mathbf{U}_j \;\middle|\; \begin{bmatrix} \mathbf{U}_j \\ \mathbf{W}_j \end{bmatrix} \leftarrow (\mathcal{D}_{\mathcal{R}^{2mT+\tilde{m}}, \mathbf{S}_{\mathtt{BASIS}}})^m : \mathbf{A} \cdot \mathbf{U}_j = \mathbf{v}_j \otimes \mathbf{G} \cdot \mathbf{W}_j \bmod q \right\}.$$

Further, with overwhelming probability, $\mathbf{A}$ is surjective by Remark 1 and $\mathbf{S}_{\mathtt{Cor}} \geq \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ by Lemma 20. Conditioned on these and applying Lemma 92 (where we set $\sqrt{\mathbf{\Sigma}_0} = \sigma_G \mathbf{I}$ and $\sqrt{\mathbf{\Sigma}_1} = \mathbf{S}_{\mathtt{Cor}}$), we have that $\mathbf{U}_j$ from above is statistically close to that in D_3.

It remains to show that, with overwhelming probability, $\mathbf{T}$ is indeed a $\mathbf{G}$-trapdoor of $\mathbf{B}_j$ with tag $\hat{\mathbf{H}}_j$ (cf. Lemma 50) under the specified parameters. By Lemmas 4 and 24, $\mathfrak{s}_{\max}(\mathbf{U}_{\mathtt{Hon}}) \leq \mathfrak{s}_{\max}(\mathbf{S}_{\mathtt{Hon}})\sqrt{2dmT \cdot \tilde{m} \cdot (T-1)}$ with overwhelming probability. Hence, $\mathfrak{s}_{\max}(\mathbf{T}) \leq \mathfrak{s}_{\max}(\mathbf{S}_{\mathtt{Hon}})\sqrt{2dmT \cdot \tilde{m} \cdot (T-1)}$, and trapdoor $\mathbf{T}$ is short enough to sample preimages with covariance parameter $\mathbf{S}_{\mathtt{BASIS}}$, see $\delta \cdot \mathfrak{s}_{\max}(\mathbf{T}) \cdot \omega(\sqrt{\log((2mT+\tilde{m}) \cdot d)}) \leq \min(\sigma_G, \mathfrak{s}_{\min}(\mathbf{S}_{\mathtt{Cor}}))$ and Lemma 50. Finally, to see that $\mathbf{B}_j \cdot \mathbf{T} = \hat{\mathbf{H}}_j \cdot \mathbf{G}_{nT} \bmod q$ holds, the following abbreviations will be handy:

$$\begin{aligned} \mathbf{H}_j \cdot \left(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n\right) &= \left[\begin{array}{c|ccc} 1 & 1 & \dots & 1 \\ \hline \frac{v_0}{v_j} & \frac{v_1}{v_j} & \dots & \frac{v_{T-1}}{v_j} \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{v_0}{v_j}\right)^{T-1} & \left(\frac{v_1}{v_j}\right)^{T-1} & \dots & \left(\frac{v_{T-1}}{v_j}\right)^{T-1} \end{array}\right] \otimes \mathbf{I}_n \\ &= \left[\begin{array}{c|ccc} \mathbf{I}_n & \mathbf{I}_n & \dots & \mathbf{I}_n \\ \hline \frac{v_0}{v_j}\mathbf{I}_n & \frac{v_1}{v_j}\mathbf{I}_n & \dots & \frac{v_{T-1}}{v_j}\mathbf{I}_n \\ \vdots & \vdots & \ddots & \vdots \\ \left(\frac{v_0}{v_j}\right)^{T-1}\mathbf{I}_n & \left(\frac{v_1}{v_j}\right)^{T-1}\mathbf{I}_n & \dots & \left(\frac{v_{T-1}}{v_j}\right)^{T-1}\mathbf{I}_n \end{array}\right] =: \left[\begin{array}{c|c} \mathbf{H}_{00} & \mathbf{H}_{01} \\ \hline \mathbf{H}_{10} & \mathbf{H}_{11} \end{array}\right], \end{aligned}$$

where the matrix dimensions are $\mathbf{H}_{00} \in \mathcal{R}_q^{n\times n}$, $\mathbf{H}_{01} \in \mathcal{R}_q^{n\times n(T-1)}$, $\mathbf{H}_{10} \in \mathcal{R}_q^{n(T-1)\times n}$, $\mathbf{H}_{11} \in \mathcal{R}_q^{n(T-1)\times n(T-1)}$. Recall that by construction, $\mathbf{A} = \left(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n\right)\cdot[\tilde{\mathbf{A}}_0||\tilde{\mathbf{A}}_{\neq 0}] \bmod q$. Using the above shorthand, we then have

$$\mathbf{H}_j \cdot \mathbf{A} = \mathbf{H}_j \cdot \left(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n\right) \cdot \begin{bmatrix}\tilde{\mathbf{A}}_0 \\ \tilde{\mathbf{A}}_{\neq 0}\end{bmatrix} = \begin{bmatrix}\mathbf{H}_{00}\cdot\tilde{\mathbf{A}}_0 + \mathbf{H}_{01}\cdot\tilde{\mathbf{A}}_{\neq 0} \\ \mathbf{H}_{10}\cdot\tilde{\mathbf{A}}_0 + \mathbf{H}_{11}\cdot\tilde{\mathbf{A}}_{\neq 0}\end{bmatrix}.$$

Consequently,

$$\begin{aligned}
\mathbf{B}_j \cdot \mathbf{T} &= \left[\begin{array}{c|c} \mathbf{H}_{00}\cdot\tilde{\mathbf{A}}_0 + \mathbf{H}_{01}\cdot\tilde{\mathbf{A}}_{\neq 0} & -\mathbf{G}_n \\ \mathbf{H}_{10}\cdot\tilde{\mathbf{A}}_0 + \mathbf{H}_{11}\cdot\tilde{\mathbf{A}}_{\neq 0} & \mathbf{1}_{T-1}\otimes -\mathbf{G}_n \end{array}\right] \cdot \begin{bmatrix} \mathbf{0}_{2mT\times\tilde{m}} & \mathbf{U}_{\mathtt{Hon}} \\ -\mathbf{I}_{\tilde{m}} & \mathbf{0}_{\tilde{m}\times m(T-1)}\end{bmatrix} \\
&= \left[\begin{array}{c|c} \mathbf{G}_n & (\mathbf{H}_{00}\cdot\tilde{\mathbf{A}}_0 + \mathbf{H}_{01}\cdot\tilde{\mathbf{A}}_{\neq 0})\cdot\mathbf{U}_{\mathtt{Hon}} \\ \mathbf{1}_{T-1}\otimes\mathbf{G}_n & (\mathbf{H}_{10}\cdot\tilde{\mathbf{A}}_0 + \mathbf{H}_{11}\cdot\tilde{\mathbf{A}}_{\neq 0})\cdot\mathbf{U}_{\mathtt{Hon}} \end{array}\right] \\
&= \left[\begin{array}{c|c} \mathbf{G}_n & \mathbf{H}_{01}\cdot(\mathbf{I}_{T-1}\otimes\mathbf{G}_n) \\ \mathbf{1}_{T-1}\otimes\mathbf{G}_n & \mathbf{H}_{11}\cdot(\mathbf{I}_{T-1}\otimes\mathbf{G}_n) \end{array}\right] \\
&= \begin{bmatrix}\mathbf{I}_n & \mathbf{H}_{01} \\ \mathbf{1}_{T-1}\otimes\mathbf{I}_n & \mathbf{H}_{11}\end{bmatrix} \cdot \begin{bmatrix}\mathbf{G}_n & \\ & \mathbf{I}_{T-1}\otimes\mathbf{G}_n\end{bmatrix} \\
&= \hat{\mathbf{H}}_j \cdot \mathbf{G}_{nT} \bmod q,
\end{aligned}$$

where the third equality follows from $\tilde{\mathbf{A}}_0 \cdot \mathbf{U}_{\mathtt{Hon}} = \mathbf{0} \bmod q$ and $\tilde{\mathbf{A}}_{\neq 0} \cdot \mathbf{U}_{\mathtt{Hon}} = \mathbf{I}_{T-1} \otimes \mathbf{G}_n \bmod q$ by construction. The proof is completed by noting that $\hat{\mathbf{H}}_j$ is invertible over $\mathcal{R}_q$, since for $j \notin \mathsf{cor}$ we have $v_j \neq v_1, \ldots, v_{T-1}$. □

5.5 Security: One-Wayness

We now prove that our construction in Figure 5.2 satisfies one-wayness. Given Section 5.4, the proof is fairly standard. We highlight some key points below.

We aim to show that, given an adversary $\mathcal{A}$ against one-wayness, we can construct an algorithm $\mathcal{B}^{\mathcal{A}}$ against the k-MSIS problem for $k = (m+\tilde{m})\cdot(T-1)$. First, we observe that the view of the adversary $\mathcal{A}$ in the one-wayness experiment essentially follows the distribution D_0 in Figure 5.3, not including partial preimages of the challenge $\mathbf{x}_{i^*}$. Therefore, we define a hybrid experiment where the adversary's view follows distribution D_4 in Figure 5.4, and Lemma 95 proves the two experiments are statistically close.

It remains to show that if $\mathcal{A}$ has non-negligible advantage in the hybrid experiment, then $\mathcal{B}^{\mathcal{A}}$ can extract a solution for a given k-MSIS instance $\left(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}\right)$. Conveniently, distribution D_4 and hence the hybrid experiment is by design efficiently simulatable

given $\left(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}\right)$. Our algorithm $\mathcal{B}^{\mathcal{A}}$ therefore simulates the hybrid experiment for $\mathcal{A}$, and upon receiving a forgery $\mathbf{x}^*$ on target $\mathbf{y}^*$ (using notations of Figure 5.1) from $\mathcal{A}$ outputs $\mathbf{x}^\dagger \coloneqq \mathbf{x}^* - \sum_{j\in\mathsf{act}^*} \mathbf{x}_j^*$ where $\sum_{j\in\mathsf{act}^*} \mathbf{x}_j^*$ is the target preimage of the Challenger. By direct inspection and tail bounds, we show that $\mathbf{x}^\dagger$ is indeed short and satisfies $\tilde{\mathbf{A}}_0 \cdot \mathbf{x}^\dagger = \mathbf{0} \bmod q$. We also show that $\mathbf{x}^\dagger \neq \mathbf{0}$ with overwhelming probability due to the entropy of a withheld preimage $\mathbf{x}_{i^*}^*$ for the chosen $i^* \in \mathsf{act}^*$. Finally, we argue that $\mathbf{x}^\dagger$ is not in the span of $(\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$ by Proposition 6.

Theorem 13. *Let parameters be as in Tables 5.2 and 5.3. The partial trapdoor scheme in Figure 5.2 is $(nT, 2mT, \beta', \sigma)$-one-way, if the $\mathsf{k\text{-}MSIS}_{\mathcal{R}_q,n,2mT,\{\tilde{\mathbf{S}}_i\}_i,\alpha}$ assumption holds for $k = (m+\tilde{m})\cdot(T-1)$.*

Proof. Let $\mathcal{A}$ be a PPT adversary winning the one-wayness experiment defined in Figure 5.1 with a non-negligible probability. We build a PPT algorithm $\mathcal{B}^{\mathcal{A}}$ solving the $\mathsf{k\text{-}MSIS}_{\mathcal{R}_q,n,2mT,\{\tilde{\mathbf{S}}_i\}_i,\alpha}$ problem with a non-negligible probability, where $k = (m+\tilde{m})\cdot(T-1)$. As before, we suppose that $\mathcal{A}$ has declared $\mathsf{cor} \subset [N]$ to be the set of $T-1$ corrupt parties in the security experiment and w.l.o.g. assume that $0 \notin \mathsf{cor}$. Consider the following hybrid experiments:[5]

- Hyb_0: The real one-wayness experiment as defined in Figure 5.1. Note that for any possible sequence of queries $(\mathsf{act}_1, \ldots, \mathsf{act}_L, (\mathsf{act}^*, i^*))$ the view

$$\left(\mathbf{A}, \left(\mathsf{ptd}_j = (\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j})\right)_{j\in\mathsf{cor}}, (\mathbf{x}_{\ell,j}, \mathbf{y}_\ell)_{\ell\in[L], j\in\mathsf{act}_\ell}, \left(\mathbf{x}_j^*\right)_{j\in\mathsf{act}^*\setminus\{i^*\}}, \mathbf{y}^*\right) \tag{5.6}$$

  of the adversary $\mathcal{A}$ almost follows the distribution D_0 defined in Figure 5.3, except that it additionally obtains the values $\left(\mathbf{x}_j^*\right)_{j\in\mathsf{act}^*\setminus\{i^*\}}, \mathbf{y}^*$. The latter can be generated in the same way as $(\mathbf{x}_{\ell,j}, \mathbf{y}_\ell)_{\ell\in[L],j\in\mathsf{act}_\ell}$, except $\mathbf{x}_{i^*}$ is not given to the adversary $\mathcal{A}$.

- Hyb_4: A modified one-wayness experiment where the view Equation (5.6) of $\mathcal{A}$ is instead generated as in distribution D_4 defined in Figure 5.4, with the same treatment to $\left(\mathbf{x}_j^*, \mathbf{y}^*\right)_{j\in\mathsf{act}^*\setminus\{i^*\}}$ mentioned above in Hyb_0.

By Lemma 95, we conclude that Hyb_0 is statistically close in λ to Hyb_4.We next show that the adversary $\mathcal{A}$'s response in Hyb_4 can be converted into a solution for k-MSIS instance $\left(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}\right)$ with non-negligible probability. First, note that $\mathcal{B}^{\mathcal{A}}$ can simulate Hyb_4 when given a k-MSIS instance $\left(\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}\right)$ by running the subroutine

[5]We only define Hyb_0 and Hyb_4 since they naturally correspond to distributions D_0 and D_4 in Figures 5.3 and 5.4 respectively.

$(\mathbf{A}, \left(\mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right)_{j\in\mathsf{cor}}) \leftarrow \mathsf{ExtendA}(\mathbf{A}, \tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$ defined in Figure 5.4 and then following the procedures in Hyb_4 (i.e. distribution D_4).

To extract a k-MSIS solution, recall that in the simulation $\mathcal{B}^{\mathcal{A}}$ samples $\mathbf{x}_j \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}_j),\sigma\cdot\mathbf{S}_{U_j}}$ for $j \in \mathsf{act}^*$, and let $\mathbf{y}^* = \mathbf{A}\cdot\sum_{j\in\mathsf{act}^*}\mathbf{x}_j \bmod q$. $\mathcal{A}$ is given $\mathbf{y}^*$ and $\mathbf{x}_j$ for $j \neq i^*$ and returns $\mathbf{x}^*$ such that $\mathbf{A}\cdot\mathbf{x}^* = \mathbf{y}^* \bmod q$. Since $(\mathbf{V}_{\{0\}\cup\mathsf{cor}}\otimes\mathbf{I}_n)$ is invertible over $\mathcal{R}_q$, we have

$$\begin{aligned}(\mathbf{V}_{\{0\}\cup\mathsf{cor}}\otimes\mathbf{I}_n)\begin{bmatrix}\tilde{\mathbf{A}}_0\\ \tilde{\mathbf{A}}_{\neq 0}\end{bmatrix} &= \mathbf{A} \bmod q,\\ (\mathbf{V}_{\{0\}\cup\mathsf{cor}}\otimes\mathbf{I}_n)\begin{bmatrix}\tilde{\mathbf{A}}_0\\ \tilde{\mathbf{A}}_{\neq 0}\end{bmatrix}(\mathbf{x}^* - \sum_{j\in\mathsf{act}^*}\mathbf{x}_j) &= \mathbf{0} \bmod q,\\ \tilde{\mathbf{A}}_0(\mathbf{x}^* - \sum_{j\in\mathsf{act}^*}\mathbf{x}_j) &= \mathbf{0} \bmod q\end{aligned}$$

i.e. we have a candidate SIS solution $\mathbf{x}^\dagger \coloneqq \mathbf{x}^* - \sum_{j\in\mathsf{act}^*}\mathbf{x}_j \bmod q$ for $\tilde{\mathbf{A}}_0$. We show that $\mathbf{x}^\dagger$ is a valid k-MSIS solution with overwhelming probability. That is, it satisfies

$$(1)\ \ \mathbf{x}^\dagger \neq \mathbf{0}, \qquad (2)\ \ \left\|\mathbf{x}^\dagger\right\| \leq \alpha, \qquad (3)\ \ \mathbf{x}^\dagger \notin \mathcal{K}\text{-span}(\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}).$$

with overwhelming probability. The vector $\mathbf{x}_{i^*} \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}_{i^*}),\sigma\cdot\mathbf{S}_{U_{i^*}}}$. By the same argument as in Proposition 5 and the norm bounds in Lemmas 4 and 24, $\eta_\varepsilon(\Lambda(\mathbf{U}_{i^*})) \leq \mathfrak{s}_{\max}(\mathbf{U}_{i^*})\cdot\eta_{\varepsilon,md} \leq B_U\cdot\eta_{\varepsilon,md}$ with overwhelming probability. Hence, $\sigma\cdot\mathbf{S}_{U_{i^*}} \geq \sigma \geq 2\,\eta_\varepsilon(\Lambda(\mathbf{U}_{i^*}))$ by the choice of parameters in Table 5.3. Since $\mathbf{x}_{i^*}$ is not revealed to the adversary the distributions of $\mathbf{x}^*$ and $\mathbf{x}_{i^*}$ are independent. Then the probability that $\mathbf{x}_{i^*} = \mathbf{x}^* - \sum_{j\in\mathsf{act}^*\setminus\{i^*\}}\mathbf{x}_j$ is bounded by Lemma 31:

$$\max_{\mathbf{x}^*}\Pr_{\mathbf{x}_{i^*}\leftarrow\mathcal{D}_{\Lambda(\mathbf{U}_{i^*}),\sigma\cdot\mathbf{S}_{U_{i^*}}}}\left(\mathbf{x}_{i^*} = \mathbf{x}^* - \sum_{j\in\mathsf{act}^*\setminus\{i^*\}}\mathbf{x}_j\right) \leq 2^{-md}\cdot\frac{1+\varepsilon}{1-\varepsilon} \leq \mathsf{negl}(\lambda).$$

By the Gaussian tail bound from Corollary 2, so $\mathbf{x}_j \leftarrow \mathbf{U}_j\cdot\mathcal{D}_{\mathbb{Z}^{md},\sigma\cdot\tilde{\theta}(\mathbf{V}_{1,j}^\intercal\cdot\mathbf{D}_j^{-1})}$ satisfies $\|\mathbf{x}_j\| \leq \mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U}_j))\cdot\sigma\cdot\mathfrak{s}_{\min}(\tilde{\theta}(\mathbf{V}_{1,j}^\intercal\cdot\mathbf{D}_j^{-1}))\cdot\sqrt{md} = \sigma\cdot\sqrt{md}$ with overwhelming probability. Then, to bound the norm of $\mathbf{x}^\dagger$ we use triangle inequality

$$\left\|\mathbf{x}^\dagger\right\| \leq \|\mathbf{x}^*\| + \sum_{j\in\mathsf{act}^*}\|\mathbf{x}_j\| \leq \beta' + T\cdot\sigma\cdot\sqrt{md} \leq \alpha$$

holds with overwhelming probability. Finally, we show that

$$\Pr\left[\mathbf{x}^\dagger \notin \mathcal{K}\text{-span}(\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})\right] \geq 1 - \mathsf{negl}(\lambda).$$

Define the shorthand $\mathbf{U}_{\neq 0} \coloneqq (\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$. Suppose $\mathbf{x}^\dagger \notin \mathcal{K}\text{-span}\,(\mathbf{U}_{i^*}, \mathbf{U}_{\neq 0})$, then $\mathbf{x}^\dagger \notin \mathcal{K}\text{-span}\,(\mathbf{U}_{\neq 0})$ and we are done. In the rest consider $\mathbf{x}^\dagger \in \mathcal{K}\text{-span}\,(\mathbf{U}_{i^*}, \mathbf{U}_{\neq 0})$.

Any $\mathbf{x}_{i^*} \in \mathcal{K}\text{-span}\,(\mathbf{U}_{i^*})$ can be written as $\mathbf{x}_{i^*}^{(0)} + \mathbf{x}_{i^*}^{(\neq 0)}$, where $\mathbf{x}_{i^*}^{(0)}$ is orthogonal to $\mathcal{K}\text{-span}\,(\mathbf{U}_{\neq 0})$ and $\mathbf{x}_{i^*}^{(\neq 0)} \in \mathcal{K}\text{-span}\,(\mathbf{U}_{\neq 0})$. Recall $\mathbf{A} \cdot \mathbf{U}_{\neq 0} \equiv (\ldots, \mathbf{v}_j \otimes (\mathbf{C}_j, \mathbf{G}), \ldots)_{j \in \mathsf{cor}}$ up to permutation of columns, $\mathbf{A} \cdot \mathbf{U}_{i^*} = \mathbf{v}_{i^*} \otimes \mathbf{C}_{i^*} \bmod q$ for a surjective $\mathbf{C}_{i^*} \in \mathcal{R}_q^{n \times m}$, and $|\{i^*\} \cup \mathsf{cor}| = T$. Then by Proposition 6, we know that $\mathbf{x}_{i^*}^{(0)}$ is not trivial, i.e. lives in a space of dimension at least one. Since $\mathcal{D}_{\Lambda(\mathbf{U}_{i^*}),\sigma \cdot \mathbf{S}_{U_{i^*}}}$ is sphericalprojecting orthogonally to $\mathcal{K}\text{-span}\,(\mathbf{U}_{\neq 0})$ produces a Gaussian of the same width but in dimension $\dim\left(\mathcal{K}\text{-span}\,(\mathbf{U}_{i^*}) \setminus \mathcal{K}\text{-span}\,(\mathbf{U}_{\neq 0})\right) \geq 1$ over the ring, or $\geq d$ over the reals. Since $\sigma \cdot \mathbf{S}_{U_{i^*}} \geq \sigma \geq 2\,\eta_\varepsilon(\Lambda(\mathbf{U}_{i^*}))$ and the smoothing parameter of a projected sublattice can only decrease, we apply Lemma 31 and conclude that $\mathbf{x}_{i^*}^{(0)} = \mathbf{x}_{i^*} \bmod \mathbf{U}_{\neq 0}$ has min-entropy $\Omega(d)$. If $\mathbf{x}^\dagger = \mathbf{x}^* - \sum_{j \in \mathsf{act}^*} \mathbf{x}_j \in \mathcal{K}\text{-span}\,(\mathbf{U}_{\neq 0})$, it implies $\mathbf{x}_{i^*}^{(0)} = \mathbf{x}^* - \sum_{j \in \mathsf{act}^* \setminus \{i^*\}} \mathbf{x}_j \bmod \mathbf{U}_{\neq 0}$. However, it happens with probability at most $2^{-\Omega(d)} \leq \mathsf{negl}(\lambda)$.

Therefore, $\mathbf{x}^\dagger$ is a valid solution with overwhelming probability, which means that $\mathcal{B}^{\mathcal{A}}$ solves the $\mathsf{k\text{-}MSIS}$ problem where $k = (m+\tilde{m}) \cdot (T-1)$ with overwhelming probability. □

5.6 Security: Indistinguishability

Towards showing indistinguishability of our partial lattice trapdoor, we prove a version of leftover hash lemma concerning the k-uniform (see Definition 31) and discrete Gaussian distributions.

Definition 31 (k-Uniform Distribution[6])**.** *Let $d, k, m, \log q, \sigma \in \mathsf{poly}(\lambda)$ with $k \leq m$. For any $\mathbf{U} \in \mathcal{R}_q^{m \times k}$, denote $\mathbf{U}^\perp = \left\{\mathbf{x} \in \mathcal{R}_q^m : \mathbf{x}^\intercal \cdot \mathbf{U} = \mathbf{0}^\intercal \bmod q\right\}$. The k-uniform distribution with parameters $\mathbf{U}, \mathcal{R}, q, \sigma$ is*

$$\mathcal{U}(\mathbf{U}^\perp) + \mathcal{D}_{\mathcal{R},\sigma}^m \coloneqq \left\{\mathbf{b} \;\middle|\; \mathbf{x} \leftarrow \mathbf{U}^\perp;\ \mathbf{e} \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^m;\ \mathbf{b} \coloneqq \mathbf{x} + \mathbf{e}\right\}.$$

Lemma 100 (Leftover hash lemma for k-uniform distribution)**.** *Let $\mathcal{R}_q$ be a ring of degree d splitting into g fields. Let $m, k, \log q, \sigma, \sigma' \in \mathsf{poly}(\lambda)$, where $k < m$, and $\varepsilon', \varepsilon > 0$, $\sigma > 8d\sqrt{m-k} \cdot q^{1/(m-k)+2/d\cdot(m-k)}$. For any matrix $\mathbf{U} = (\mathbf{U}_0 || \ldots || \mathbf{U}_{g-1})$ where $\mathbf{U}_i \in \mathcal{R}_q^{m \times k}$ are surjective, it holds that*

$$\left\{\left[\mathbf{b}, v\right] \;\middle|\; \begin{array}{l} \mathbf{b} \leftarrow \mathcal{U}(\mathbf{U}^\perp) + \mathcal{D}_{\mathcal{R},\sigma'}^{g \cdot m} \\ \mathbf{r} \leftarrow \mathcal{D}_{\mathcal{R},\sigma}^{g \cdot m};\ v \coloneqq \mathbf{b}^\intercal \cdot \mathbf{r} \bmod q \end{array}\right\} \approx_s \left\{\left[\mathbf{b}, v\right] \;\middle|\; \begin{array}{l} \mathbf{b} \leftarrow \mathcal{U}(\mathbf{U}^\perp) + \mathcal{D}_{\mathcal{R},\sigma'}^{g \cdot m} \\ v \leftarrow \mathcal{R}_q \end{array}\right\},$$

where $\mathbf{U}^\perp = \left\{\mathbf{x} \in \mathcal{R}_q^{g \cdot m} \mid \mathbf{x}^\intercal \cdot \mathbf{U} = \mathbf{0} \bmod q\right\}$.

[6] In the sense of [LPSS14].

The key difference between Lemma 100 and standard versions of the leftover hash lemma for lattices is that the "matrix", $\mathbf{b}$, is not uniformly random but random in the kernel of $\mathbf{U}$. The proof handles this by effectively considering a basis of this kernel in normal form. Since $\mathbf{U}$ is full-rank, over the field somewhere in the normal form we have an identity matrix. Thus, the coefficients corresponding to this identity matrix in the basis are uniformly random as required. Then, since the entries of $\mathbf{r}$ are indexwise independent, the contribution of the non-pivot indices on the pivots does not affect the uniformity of the result. The proof takes into account that $\mathcal{R}_q$ may not be a field by running this argument in each "CRT slot" of $\mathcal{R}_q$, at the cost of an additional factor g in the dimension. Note that we place no restriction on σ' since the noise $\mathcal{D}^{g\cdot m}_{\mathcal{R},\sigma'}$ is not used in the proof[7].

Proof. For $\mathbf{x} \in \mathbf{U}^\perp$ we write $\mathbf{x} = (\mathbf{x}_0, \ldots, \mathbf{x}_{g-1})$, where $\mathbf{x}_i \in \mathcal{R}^m$. Then it holds that $\sum_{i=0}^{g-1} \mathbf{x}_i^\intercal \cdot \mathbf{U}_i = \mathbf{0} \bmod q$. For each set $\left\{\alpha_i \in \mathcal{R}^k\right\}_{i\in[g]}$ s.t. $\sum_{i=0}^{g-1} \alpha_i^\intercal = \mathbf{0} \bmod q$ we prove the statement for uniform $\mathbf{x}_i^\intercal$ satisfying $\forall i : \mathbf{x}_i^\intercal \cdot \mathbf{U}_i = \alpha_i^\intercal \bmod q$. Let $\langle q \rangle = \prod_{i=0}^{g-1} \mathfrak{q}_i$. W.l.o.g pick $i = 0$, note $\mathcal{R}_q/\mathfrak{q}_0 = \mathbb{F}_{(0)}$ is a field of size $q^{d/g}$. The matrix $\mathbf{U}_0$ is surjective in $\mathcal{R}_q$ which implies $\mathbf{U}_0 \bmod \mathfrak{q}_0$ contains a $k \times k$ invertible submatrix over $\mathbb{F}_{(0)}$. There exists a permutation matrix $\mathbf{T}_0 \in \mathcal{R}_q^{m\times m}$ such that $\mathbf{T}_0 \cdot \mathbf{U}_0 = [\mathbf{U}_{00} \| \mathbf{U}_{01}] \bmod \mathfrak{q}_0$ where $\mathbf{U}_{00} \in \mathbb{F}_{(0)}^{k\times k}$ is invertible and $\mathbf{U}_{01} \in \mathbb{F}_{(0)}^{(m-k)\times k}$. Write $\mathbf{x}_0^\intercal \cdot \mathbf{T}_0^{-1} \coloneqq (\mathbf{x}_{00}^\intercal, \mathbf{x}_{01}^\intercal) \in \mathcal{R}_q^k \times \mathcal{R}_q^{m-k}$. Then for any $\mathbf{x} \in \mathbf{U}^\perp$, $\mathbf{x}_{00} \bmod \mathfrak{q}_0$ is determined by the other variables as

$$\begin{aligned}
\mathbf{x}_0^\intercal \cdot \mathbf{T}_0^{-1} \cdot \mathbf{T}_0 \cdot \mathbf{U}_0 &= \alpha_0 \bmod \mathfrak{q}_0 \\
(\mathbf{x}_{00}^\intercal, \mathbf{x}_{01}^\intercal) \cdot \begin{bmatrix} \mathbf{U}_{00} \\ \mathbf{U}_{01} \end{bmatrix} &= \alpha_0 \bmod \mathfrak{q}_0 \\
\mathbf{x}_{00}^\intercal &= \alpha_0 - \mathbf{x}_{01}^\intercal \cdot \mathbf{U}_{01} \cdot \mathbf{U}_{00}^{-1} \bmod \mathfrak{q}_0.
\end{aligned}$$

Since $\mathbf{x}$ is uniform over $\mathbf{U}^\perp$ and the mapping $\mathbf{T}_0^{-1}$ is a bijection, we have that $\mathbf{x}_{01}$ is uniform over $\mathbb{F}_{(0)}^{m-k}$. Write $\mathbf{b} = \mathbf{x} + \mathbf{e}$ where $\mathbf{x} \leftarrow \mathcal{U}(\mathbf{U}^\perp)$, $\mathbf{e} = (\mathbf{e}_0, \ldots, \mathbf{e}_{g-1}) \leftarrow \mathcal{D}^{g\cdot m}_{\mathcal{R},\sigma'}$ and $\mathbf{e}_i^\intercal \cdot \mathbf{T}_i^{-1} =: (\mathbf{e}_{i0}^\intercal, \mathbf{e}_{i1}^\intercal)$. For $\mathbf{r} \leftarrow \mathcal{D}^m_{\mathcal{R},\sigma}$ we have

$$\begin{aligned}
\mathbf{b}^\intercal \cdot \mathbf{r} &= \sum_{i=0}^{g-1} (\mathbf{x}_i^\intercal + \mathbf{e}_i^\intercal) \cdot \mathbf{r}_i \\
&= \sum_{i=0}^{g-1} (\mathbf{x}_i^\intercal + \mathbf{e}_i^\intercal) \cdot \mathbf{T}_i^{-1} \cdot \mathbf{T}_i \cdot \mathbf{r}_i \\
&= \sum_{i=0}^{g-1} \underbrace{\left(\alpha_i - \mathbf{x}_{i1}^\intercal \cdot \mathbf{U}_{01} \cdot \mathbf{U}_{00}^{-1} + \mathbf{e}_{i0}^\intercal\right) \cdot \mathbf{r}_{i0}}_{\text{RHS}} + \underbrace{(\mathbf{x}_{i1}^\intercal + \mathbf{e}_{i1}^\intercal) \cdot \mathbf{r}_{i1}}_{\text{LHS}}.
\end{aligned}$$

[7] The statement is true without the noise $\mathcal{D}^{g\cdot m}_{\mathcal{R},\sigma'}$ and could be of independent interest.

where $\mathbf{T}_i \cdot \mathbf{r}_i =: (\mathbf{r}_{i0} || \mathbf{r}_{i1})$. In the following we adapt the classic leftover hash lemma (LHL) for the LHS with matrix $(\mathbf{x}_{i1}^\intercal + \mathbf{e}_{i1}^\intercal)$ and vector $\mathbf{r}_{i1}$. The RHS is a function of $\mathbf{x}_{i1}$ and parameters that are either public or independent of $\mathbf{r}_{i1}$. Hence, if the distribution of LHS is close to uniform and independent of $\mathbf{x}_{i1}^\intercal, \mathbf{e}_{i1}^\intercal$, the sum is also close to uniform. First, we modify the LHL to argue about the statistical distance between $(\mathbf{x}_{i1}, \mathbf{e}_{i1}, (\mathbf{x}_{i1} + \mathbf{e}_{i1})^\intercal \cdot \mathbf{r}_{i1} \bmod \mathfrak{q}_i)$ and $\left(\mathbf{x}_{i1}, \mathbf{e}_{i1}, \mathcal{U}(\mathbb{F}_{(i)})\right)$ for $\mathbf{x}_{i1} \leftarrow \mathbb{F}_{(i)}^{m-k}$ and $\mathbf{e}_{i1} \leftarrow \mathcal{D}_{\mathcal{R},\sigma'}^{m-k}$ where $\mathbf{x}_{i1}, \mathbf{e}_{i1}$ are revealed separately. Leaking the two values separately does not increase the statistical distance as they can be simulated from a single uniform vector as follows. For $\tilde{\mathbf{x}}_{i1} \leftarrow \mathbb{F}_{(i)}^{m-k}$

$$\mathrm{SD}\left(\left(\tilde{\mathbf{x}}_{i1},\ \tilde{\mathbf{x}}_{i1}^\intercal \cdot \mathbf{r}_{i1} \bmod \mathfrak{q}_i\right),\quad \left(\tilde{\mathbf{x}}_{i1},\ \mathcal{U}(\mathbb{F}_{(i)})\right)\right) \leq 2^{-\Omega(d/g)}$$

holds by leftover hash lemma from Lemma 57. Then we sample $\tilde{\mathbf{e}}_{i1} \leftarrow \mathcal{D}_{\mathcal{R},\sigma'}^{m-k}$ and set $\mathbf{e}_{i1} = \tilde{\mathbf{e}}_{i1}$ and $\mathbf{x}_{i1} = \tilde{\mathbf{x}}_{i1} - \tilde{\mathbf{e}}_{i1}$ and the statement follows. Then $\forall i = 0, \ldots, g-1$

$$\mathrm{SD}\left(\left(\{\mathbf{x}_{j1}, \mathbf{e}_{j1}\}_j,\ (\mathbf{x}_{i1} + \mathbf{e}_{i1})^\intercal \cdot \mathbf{r}_{i1} \bmod \mathfrak{q}_i\right),\ \left(\{\mathbf{x}_{j1}, \mathbf{e}_{j1}\}_j,\ \mathcal{U}(\mathbb{F}_{(i)})\right)\right) \leq \mathsf{negl}(\lambda).$$

Since $\{\mathbf{r}_{i1}\}_i$ are sampled independently, the random variables $(\mathbf{x}_{i1} + \mathbf{e}_{i1})^\intercal \cdot \mathbf{r}_{i1}$ are independent too. Then the sum $\sum_{i=1}^{g-1} (\mathbf{x}_{i1} + \mathbf{e}_{i1})^\intercal \cdot \mathbf{r}_{i1}$ is distributed uniformly modulo each $\mathfrak{q}_i$ hence it is uniform in the ring $\mathcal{R}_q$. □

The following is a direct corollary of Lemmas 94 and 100. We extend the LHL statement above to sampling the matrix $\mathbf{U}$ at random and prove that it satisfies the conditions of Lemma 100 with overwhelming probability.

Corollary 20. *Let $g, d, k, m, n, \log q, \chi, \chi' \in \mathsf{poly}(\lambda)$, $a \geq 1$ be a real number, and $\{\mathsf{diag}(\mathbf{\Sigma}_i, g)\}_{i\in[N]}$ be a set of k positive semi-definite matrices in $\mathbb{R}^{dgm \times dgm}$ satisfying the following constraints:*

- *$\mathcal{R}_q$ is a ring of degree d splitting into g fields,*
- *$n \leq m$, $k < m$,*
- *$d \cdot (m-k) \geq \Omega(\lambda)$,*
- *there exists $\varepsilon \leq \mathsf{negl}(\lambda)$ so that for all $i \in [N]$ it holds*

$$\max\left\{\eta_A, 2\sqrt{d} \cdot (a^k q^n)^{1/(m-k)}\right\} \leq \mathfrak{s}_{\min}\left(\sqrt{\mathbf{\Sigma}_i}\right),$$
$$\min\left\{q^{1/g}/\sqrt{m}, a \cdot \mathfrak{s}_{\min}\left(\sqrt{\mathbf{\Sigma}_i}\right)\right\} \geq \mathfrak{s}_{\max}\left(\sqrt{\mathbf{\Sigma}_i}\right),$$

where $\eta_A \geq 8\, d\sqrt{m} \cdot q^{n/m + 2/(d \cdot m)}$,

- $ng/q^{d\cdot(m-n+1)/g} \leq \mathsf{negl}(\lambda)$,
- $\chi > 8d\sqrt{m-k} \cdot q^{1/(m-k)+2/d\cdot(m-k)}$

It holds that $(\mathbf{A}, \mathbf{U}, \mathbf{b}, v)$ *sampled as*

$$\left\{\begin{array}{l} \mathbf{A} \leftarrow \mathcal{R}_q^{n\times gm} \\ \mathbf{u}_i \leftarrow \mathcal{D}_{\Lambda_q^\perp(\mathbf{A}),\sqrt{\mathsf{diag}(\mathbf{\Sigma}_i,g)}}\ \forall i \in [N] \\ \mathbf{U} \coloneqq (\mathbf{u}_i)_{i\in[N]} \\ \mathbf{b} \leftarrow \mathcal{U}(\mathbf{U}^\perp) + \mathcal{D}^{gm}_{\mathcal{R},\chi'} \\ \mathbf{r} \leftarrow \mathcal{D}^{gm}_{\mathcal{R},\chi};\ v \coloneqq \mathbf{b}^\mathsf{T}\cdot\mathbf{r} \bmod q \end{array}\right\} \textit{ and as } \left\{\begin{array}{l} \mathbf{A} \leftarrow \mathcal{R}_q^{n\times gm} \\ \mathbf{u}_i \leftarrow \mathcal{D}_{\Lambda_q^\perp(\mathbf{A}),\sqrt{\mathsf{diag}(\mathbf{\Sigma}_i,g)}}\ \forall i \in [N] \\ \mathbf{U} \coloneqq (\mathbf{u}_i)_{i\in[N]} \\ \mathbf{b} \leftarrow \mathcal{U}(\mathbf{U}^\perp) + \mathcal{D}^{gm}_{\mathcal{R},\chi'} \\ v \leftarrow \mathcal{R}_q \end{array}\right\}.$$

are statistically close.

Proof. Write $\mathbf{A} = [\mathbf{A}_0, \ldots, \mathbf{A}_{g-1}]$ where $\mathbf{A}_j \leftarrow \mathcal{R}_q^{n\times m}$ for $j \in [g]$. With overwhelming probability in λ, all $\mathbf{A}_j$ are surjective by Remark 1, and $\mathfrak{s}_{\min}(\sqrt{\mathsf{diag}(\mathbf{\Sigma}_i, g)}) = \mathfrak{s}_{\min}(\sqrt{\mathbf{\Sigma}_i}) > \eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}_j))$ for all $i \in [N]$ by our choice of parameters. Split $\mathbf{U} = [\mathbf{U}_0||\ldots||\mathbf{U}_{g-1}]$ where $\mathbf{U}_j =: (\mathbf{u}_{ij})_{i\in[N]}$. Then $\sum_{j=0}^{g-1}\mathbf{A}_j\cdot\mathbf{U}_j = \sum_{j=0}^{g-1}\alpha_j = 0 \bmod q$ for some $(\alpha_j)_{j\in[g]}$. Conditioned on the ith coordinate α_{ij} of α_j we have $\mathbf{u}_{ij} \leftarrow \mathcal{D}_{\Lambda_q^{\alpha_{ij}}(\mathbf{A}_j),\sqrt{\mathbf{\Sigma}_i}}$. The covariance matrix of the full vectors $\mathbf{u}_i$ is block diagonal so we split the preimages on g vectors as above. Conditioned on these, every $\mathbf{U}_j$ is surjective with overwhelming probability in λ by Lemma 94. Then the claim follows from Lemma 100. □

We are ready to prove that our partial trapdoor construction in Figure 5.2 has indistinguishability. The high level proof structure is similar to that of one-wayness (Theorem 13), with some challenges unique to k-MLWE, which we highlight below.

As in the proof of one-wayness, we observe that most components in the view of the adversary $\mathcal{A}$ in the indistinguishability experiments (see Figure 5.1) follow the distribution D_0 in Figure 5.3, except that the adversary $\mathcal{A}$ additionally obtains the challenge partial preimages and images $\left\{\mathbf{x}_j^*\right\}_{j\in\mathsf{act}^*\setminus\{i^*\}}, \mathbf{y}^*$ and the challenge LWE sample $(\mathbf{c}_0, c_1)$. Using Lemma 95, we switch to a pair of hybrid experiments $\mathsf{Hyb}_{b,4}$ where most components of the adversary's view are generated as in distribution D_4 in Figure 5.4. The challenge partial preimages and images are generated identically to responses of other preimage queries, while the challenge LWE sample is generated as detailed below.

It remains to show that this pair of hybrid experiments $\mathsf{Hyb}_{0,4}$ and $\mathsf{Hyb}_{1,4}$ are computationally indistinguishable under the k-MLWE assumption (see Definition 5). Given a k-MLWE instance $\left\{\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}, \mathbf{b}\right\}$, the components

$$\left\{\mathbf{A}, \left\{\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j}\right\}_{j\in\mathsf{cor}}, \left\{\mathbf{x}_{\ell,j}, \mathbf{y}_\ell\right\}_{\ell\in[L],j\in\mathsf{act}_\ell}, \left\{\mathbf{x}_j^*\right\}_{j\in\mathsf{act}^*\setminus\{i^*\}}\right\}$$

are simulated by our reduction as in distribution D_4 in Figure 5.4. The remaining component $\{\mathbf{y}^*, \mathbf{c}_0, c_1\}$ require some care.

Note that in the hybrid experiments $\mathbf{y}^*$ is computed from $\mathbf{x}^*_{i^*}$ sampled as a Gaussian distribution over $\Lambda(\mathbf{U}_{i^*})$. We show that replacing $\mathbf{x}^*_{i^*}$ by a Gaussian from the full module $\mathcal{R}^{2gmT}$ does not affect the distribution of $\mathbf{y}^*$. Then we argue the distribution of the challenge LWE sample c_1 via Lemma 100. In more detail, we show that if the hidden bit in the given $\mathsf{k\text{-}MLWE}$ instance is $b = 0$, i.e. $\mathbf{b}$ is an LWE sample, then our reduction perfectly simulates $\mathsf{Hyb}_{0,4}$. On the other hand, if the hidden bit is $b = 1$, then our reduction almost simulates $\mathsf{Hyb}_{1,4}$ in that c_1 is uniformly random due to Lemma 100, but $\mathbf{c}_0$ is of the form $\mathbf{c}_0 \leftarrow \mathcal{U}(\mathbf{U}^\perp) + \mathcal{D}^{2gmT}_{\mathcal{R},\chi_0} + \mathbf{s}^\intercal_{\neq 0} \cdot \tilde{\mathbf{A}}_{\neq 0} \bmod q$ instead of $\mathbf{s}^\intercal_0 \cdot \tilde{\mathbf{A}}_0 + \mathbf{s}^\intercal_{\neq 0} \cdot \tilde{\mathbf{A}}_{\neq 0} + \mathbf{e}^\intercal \bmod q$. We are therefore left with two cases: Either the above gap is computationally unnoticeable, in which case our reduction can solve the given $\mathsf{k\text{-}MLWE}$ instance $\left\{\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}, \mathbf{b}\right\}$ with non-negligible probability, or the gap is computationally noticeable, in which case we can still write down another reduction from $\mathsf{k\text{-}MLWE}$.

Theorem 14. *Let parameters be as in Tables 5.2 and 5.3. The partial lattice trapdoor scheme in Figure 5.2 has $(nT, 2gmT, \chi_0, \chi_1, \sigma)$-indistinguishability, if for $k = (m + \tilde{m}) \cdot (T - 1)$ the $\mathsf{k\text{-}MLWE}_{\mathcal{R}_q, n, 2gmT, \{\mathsf{diag}(\tilde{\mathbf{S}}_i, g)\}_i, \chi_0}$ assumption holds.*

Proof. Let $\mathcal{A}$ be a PPT adversary winning the indistinguishability experiment defined in Figure 5.1 with a non-negligible probability. We build a PPT algorithm $\mathcal{B}^{\mathcal{A}}$ solving the $\mathsf{k\text{-}MLWE}_{\mathcal{R}_q, n, 2gmT, \{\mathsf{diag}(\tilde{\mathbf{S}}_i, g)\}_i, \chi_0}$ problem with a non-negligible probability, where $k = (m + \tilde{m}) \cdot (T - 1)$. As before, we suppose that $\mathcal{A}$ has declared $\mathsf{cor} \subset [N]$ to be the set of $T - 1$ corrupt parties in the security experiment and w.l.o.g. assume that $0 \notin \mathsf{cor}$. Consider the following hybrid experiments[8] parametrised by a bit $b \in \{0, 1\}$:

$\mathsf{Hyb}_{b,0}$: The real indistinguishability experiments as defined in Figure 5.1. Note that for any possible sequence of queries $(\mathsf{act}_1, \ldots, \mathsf{act}_L, (\mathsf{act}^*, i^*))$ the view

$$\left\{\mathbf{A}, \left\{\mathsf{ptd}_j = (\mathbf{U}_j, \mathbf{C}_j, \mathsf{td}_{\mathbf{C}_j})\right\}_{j \in \mathsf{cor}}, \{\mathbf{x}_{\ell,j}, \mathbf{y}_\ell\}_{\ell,j}, \left\{\mathbf{x}^*_j\right\}_{j \in \mathsf{act}^* \setminus \{i^*\}}, \mathbf{y}^*, \mathbf{c}_0, c_1\right\} \tag{5.7}$$

of the adversary $\mathcal{A}$ follows the distribution D_0 defined in Figure 5.3, except

1. The adversary $\mathcal{A}$ obtains the challenge partial preimages $\left\{\mathbf{x}^*_j\right\}_{j \in \mathsf{act}^* \setminus \{i^*\}}$ and the corresponding image $\mathbf{y}^*$ sampled identically to the other oracle answers.

[8] We only define $\mathsf{Hyb}_{b,0}$ and $\mathsf{Hyb}_{b,4}$ since they naturally correspond to distributions D_0 and D_4 in Figures 5.3 and 5.4 respectively.

2. The adversary $\mathcal{A}$ additionally obtains the challenge LWE sample $(\mathbf{c}_0, c_1)$. For that we sample $\mathbf{s} \leftarrow \mathcal{R}_q^n$, $\mathbf{e} \leftarrow \chi_0^{2gmT}$, $e \leftarrow \chi_1$, and set $\mathbf{c}_0 \coloneqq \mathbf{s}^\intercal \cdot \mathbf{A} + \mathbf{e}^\intercal \bmod q$. Then we set $c_1 \coloneqq \mathbf{s}^\intercal \cdot \mathbf{y}^* + e \bmod q$ when $b = 0$ and $c_1 \leftarrow \mathcal{R}_q$ when $b = 1$.

$\mathsf{Hyb}_{b,4}$: Modified indistinguishability experiments where the view Equation (5.7) of $\mathcal{A}$, except for the $(\mathbf{c}_0, c_1)$ component, is instead generated as in distribution D_4 defined in Figure 5.4. The distribution of the partial preimages and image $\left\{\mathbf{x}_j^*\right\}_{j\in\mathsf{act}^*\setminus\{i^*\}}, \mathbf{y}^*$ nd the challenge LWE sample $(\mathbf{c}_0, c_1)$ does not change.

By Lemma 95, $\mathsf{Hyb}_{b,0}$ is statistically close in λ to $\mathsf{Hyb}_{b,4}$ for each $b = 0, 1$. Define $\mathsf{Hyb}_{b,5}$ where we change the distribution of hints $(\mathbf{x}_j^*)_{j\in\mathsf{act}^*\setminus\{i^*\}}$ and challenge $\mathbf{y}^*$. We now compute them as follows

- For $j \in \mathsf{act}^* \setminus \{i^*\}$, sample $\mathbf{x}_j^* \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}_j),\sigma\cdot\mathbf{S}_{U_j}}$ as before. Compute corresponding $\mathbf{z}_j$ from $\mathbf{A}\cdot\mathbf{x}_j^* = \mathbf{v}_j \otimes \mathbf{z}_j \bmod q$.
- Instead of $\mathbf{x}_{i^*}$ sample a Gaussian vector $\mathbf{r} \leftarrow \mathcal{D}_{\mathcal{R},\chi_2}^{2gmT}$, let $\mathbf{y}_0^* \coloneqq \tilde{\mathbf{A}}_0 \cdot \mathbf{r} \bmod q$.
- Solve linear equation for $(\mathbf{y}_{\neq 0}^*, \mathbf{z}_{i^*}) \in \mathcal{R}_q^{n(T-1)} \times \mathcal{R}_q^n$ satisfying

$$(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n) \begin{bmatrix} \mathbf{y}_0^* \\ \mathbf{y}_{\neq 0}^* \end{bmatrix} = \sum_{j\in\mathsf{act}^*\setminus\{i^*\}} \mathbf{v}_j \otimes \mathbf{z}_j + \mathbf{v}_{i^*} \otimes \mathbf{z}_{i^*} \bmod q. \tag{5.8}$$

- Let $\mathbf{y}^* \coloneqq (\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n) \begin{bmatrix} \mathbf{y}_0^* \\ \mathbf{y}_{\neq 0}^* \end{bmatrix} \bmod q$.

The distribution of partial preimages $\left\{\mathbf{x}_j^*\right\}_{j\in\mathsf{act}^*\setminus\{i^*\}}$ does not change compared to $\mathsf{Hyb}_{b,4}$. It remains to argue that $\mathbf{y}^*$ is also well-distributed. Firstly, using the guarantee that $i^* \notin \mathsf{cor}$ (as defined in the indistinguishability experiment Figure 5.1), we show that the system of linear equations in Equation (5.8) is solvable and has a unique solution. Note that $\mathbf{V}_{\{0\}\cup\mathsf{cor}} \in \mathcal{R}_q^{T\times T}$ is invertible since $0 \notin \mathsf{cor}$. Multiplying by $(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n)^{-1}$ on both sides and using properties of the tensor product, we obtain

$$\begin{bmatrix} \mathbf{y}_0^* \\ \mathbf{y}_{\neq 0}^* \end{bmatrix} = \underbrace{\sum_{j\in\mathsf{act}^*\setminus\{i^*\}} \mathbf{V}_{\{0\}\cup\mathsf{cor}}^{-1}\,\mathbf{v}_j \otimes \mathbf{z}_j}_{\mathbf{t}^* \coloneqq} + \underbrace{\mathbf{V}_{\{0\}\cup\mathsf{cor}}^{-1}\,\mathbf{v}_{i^*}}_{\mathbf{a}^* \coloneqq} \otimes \mathbf{z}_{i^*} = \begin{bmatrix} \mathbf{t}_0^* \\ \mathbf{t}_{\neq 0}^* \end{bmatrix} + \begin{bmatrix} a_0^* \\ \mathbf{a}_{\neq 0}^* \end{bmatrix} \otimes \mathbf{z}_{i^*} \bmod q$$

If $a_0^* \in \mathcal{R}_q^\times$, then $(\mathbf{y}_{\neq 0}^*, \mathbf{z}_{i^*})$ are uniquely determined as

$$\mathbf{z}_{i^*} = (\mathbf{y}_0^* - \mathbf{t}_0^*) \cdot (a_0^*)^{-1} \bmod q \qquad \text{and} \qquad \mathbf{y}_{\neq 0}^* = \mathbf{t}_{\neq 0}^* + \mathbf{a}_{\neq 0}^* \otimes \mathbf{z}_{i^*} \bmod q.$$

Suppose towards contradiction that $a_0^* \notin \mathcal{R}_q^\times$. Then there exists an ideal $I \mid q\mathcal{R}$ such that $a_0^* = 0 \bmod I$. Hence, after multiplying by $\mathbf{V}_{\{0\}\cup\mathsf{cor}}$ on both sides

$$\mathbf{V}_{\{0\}\cup\mathsf{cor}} \begin{bmatrix} 0 \\ \mathbf{a}^*_{\neq 0} \end{bmatrix} = \mathbf{V}_{\mathsf{cor}} \cdot \mathbf{a}^*_{\neq 0} = \mathbf{v}_{i^*} \bmod I.$$

Or equivalently $\mathbf{V}_{\{i^*\}\cup\mathsf{cor}} \cdot [-1 || \mathbf{a}^*_{\neq 0}] = 0 \bmod I$. However, since $i^* \notin \mathsf{cor}$, the matrix $\mathbf{V}_{\mathsf{cor}\cup\{i^*\}}$ is invertible and has an empty kernel modulo every ideal contradicting the conclusion above.

Secondly, since $\mathbf{y}_0^* = \tilde{\mathbf{A}}_0 \cdot \mathbf{r} \bmod q$ and $\chi_2 > 2d \exp(1/(d \cdot 2gmT)) \cdot q^{n/2gmT}$, by Theorem 3 and Lemma 57 $\mathbf{y}_0^*$ is statistically close to uniform over $\mathcal{R}_q^n$. From the analysis above we see that the relation between $\mathbf{y}_0^*$ and $\mathbf{z}_{i^*}$ is bijective, and thus a uniformly random $\mathbf{y}_0^*$ implies a uniformly random $\mathbf{z}_{i^*}$. Following the same steps as in Lemma 97, we conclude that $\mathbf{v}_{i^*} \otimes \mathbf{z}_{i^*} \bmod q$ with a uniformly random $\mathbf{z}_{i^*}$ is statistically close to $\mathbf{A} \cdot \mathbf{x}^*_{i^*} \bmod q$ for $\mathbf{x}^*_{i^*} \leftarrow \mathcal{D}_{\Lambda(\mathbf{U}_{i^*}),\sigma\cdot\mathbf{s}_U}$, as in $\mathsf{Hyb}_{b,4}$. Hence, $\mathbf{y}^*$ computed as the sum of well-distributed $\mathbf{v}_j \otimes \mathbf{z}_j$, $j \in \mathsf{act}^*$ is well-distributed.

Lastly, we define $\mathsf{Hyb}_{b,6}$ and modify the distributions of $(\mathbf{c}_0, c_1)$. Recall, that in k-MLWE instance $\left\{\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}, \mathbf{b}\right\}$ the values are distributed as $\tilde{\mathbf{A}}_0 \leftarrow \mathcal{R}_q^{n\times 2gmT}$, $(\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$ are Gaussian with varying Gaussian widths subject to $\tilde{\mathbf{A}}_0 \cdot (\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}) \equiv \mathbf{0}$, and $\mathbf{b}$ is either $\mathbf{b}^\intercal = \mathbf{s}_0^\intercal \cdot \tilde{\mathbf{A}}_0 + \mathbf{e}_0^\intercal \bmod q$ or $\mathbf{b} = \mathcal{U}(\mathbf{U}^\perp) + \mathbf{e}_0^\intercal$ for $\mathbf{U} \coloneqq (\mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}})$. To simulate $(\mathbf{c}_0, c_1)$ we sample $\mathbf{s}_{\neq 0} \leftarrow \mathcal{R}_q^{n(T-1)}$, $e \leftarrow \chi_1$ and let

$$\mathbf{c}_0^\intercal = \mathbf{b}^\intercal + \mathbf{s}_{\neq 0}^\intercal \cdot \tilde{\mathbf{A}}_{\neq 0} \bmod q \qquad \text{and} \qquad c_1 \coloneqq \mathbf{b}^\intercal \cdot \mathbf{r} + \mathbf{s}_{\neq 0}^\intercal \cdot \mathbf{y}^*_{\neq 0} + e \bmod q.$$

First, consider $b = 0$, then for $\mathbf{c}_0$ we have

$$\begin{aligned} \mathbf{c}_0^\intercal &= (\mathbf{s}_0^\intercal, \mathbf{s}_{\neq 0}^\intercal) \begin{bmatrix} \tilde{\mathbf{A}}_0 \\ \tilde{\mathbf{A}}_{\neq 0} \end{bmatrix} + \mathbf{e}^\intercal \bmod q \\ &= \underbrace{(\mathbf{s}_0^\intercal, \mathbf{s}_{\neq 0}^\intercal)(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n)^{-1}}_{\mathbf{s}^\intercal} (\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n) \begin{bmatrix} \tilde{\mathbf{A}}_0 \\ \tilde{\mathbf{A}}_{\neq 0} \end{bmatrix} + \mathbf{e}^\intercal = \mathbf{s}^\intercal \cdot \mathbf{A} + \mathbf{e} \bmod q, \end{aligned}$$

where $\mathbf{s}$ is uniformly distributed since the map $(\mathbf{V}_{\{0\}\cup\mathsf{cor}} \otimes \mathbf{I}_n)^{-1}$ is bijective. For c_1, we have

$$\mathbf{b}^\intercal \mathbf{r} + e = (\mathbf{s}_0^\intercal \cdot \tilde{\mathbf{A}}_0 + \mathbf{e}^\intercal) \cdot \mathbf{r} + e \approx_s \mathbf{s}_0^\intercal \cdot \tilde{\mathbf{A}}_0 \cdot \mathbf{r} + e = \mathbf{s}_0^\intercal \cdot \mathbf{y}_0^* + e \bmod q$$

where the statistical closeness follows from noise drowning (Lemma 36), since $\lambda^{\omega(1)} \chi_0 \chi_2 \cdot$

$2gmT \leq \chi_1$. Hence,

$$c_1 \approx_s \mathbf{s}_0^{\intercal} \cdot \mathbf{y}_0^* + \mathbf{s}_{\neq 0}^{\intercal} \cdot \mathbf{y}_{\neq 0}^* + e \bmod q$$
$$= (\mathbf{s}_0^{\intercal}, \mathbf{s}_{\neq 0}^{\intercal})(\mathbf{V}_{\{0\}\cup \mathsf{cor}} \otimes \mathbf{I}_n)^{-1}(\mathbf{V}_{\{0\}\cup \mathsf{cor}} \otimes \mathbf{I}_n) \begin{bmatrix} \tilde{\mathbf{y}}_0 \\ \mathbf{y}_{\neq 0} \end{bmatrix} + e = \mathbf{s}^{\intercal} \cdot \mathbf{y}^* + e \bmod q.$$

Jointly, $(\mathbf{c}_0, c_1)$ in $\mathsf{Hyb}_{0,6}$ is distributed statistically close to that in $\mathsf{Hyb}_{0,5}$. Next, consider $b = 1$ here we modify the corresponding distribution twice. Recall that in $\mathsf{Hyb}_{1,5}$ we have

$$\Big\{(\mathbf{c}_0, c_1) \mid \mathbf{c}_0^{\intercal} = \mathbf{s}^{\intercal} \cdot \mathbf{A} + \mathbf{e}^{\intercal} \bmod q,\ c_1 \leftarrow \mathcal{U}(\mathcal{R}_q)\Big\}.$$

First we move to $\mathsf{Hyb}_{1,6}$

$$\Big\{(\mathbf{c}_0, c_1) \mid \mathbf{c}_0^{\intercal} = \mathbf{s}^{\intercal} \cdot \mathbf{A} + \mathbf{e}^{\intercal} \bmod q,\ c_1 = \mathbf{b}^{\intercal} \cdot \mathbf{r} + \mathbf{s}_{\neq 0}^{\intercal} \cdot \mathbf{y}_{\neq 0}^* + e \bmod q\Big\},$$

for $\mathbf{b} \leftarrow \mathcal{U}(\mathbf{U}^{\perp}) + \mathbf{e}_0^{\intercal}$ and $\mathbf{s}_{\neq 0}, \mathbf{y}_{\neq 0}^*, e$ distributed as above. The two hybrids are statistically close by Corollary 20. Then we move to $\mathsf{Hyb}_{1,7}$ where

$$\Big\{(\mathbf{c}_0, c_1) \mid \mathbf{c}_0^{\intercal} = \mathbf{b}^{\intercal} + \mathbf{s}_{\neq 0}^{\intercal} \cdot \tilde{\mathbf{A}}_{\neq 0} \bmod q,\ c_1 = \mathbf{b}^{\intercal} \cdot \mathbf{r} + \mathbf{s}_{\neq 0}^{\intercal} \cdot \mathbf{y}_{\neq 0}^* + e \bmod q\Big\}.$$

The transition from $\mathsf{Hyb}_{1,6}$ to $\mathsf{Hyb}_{1,7}$ can be simulated efficiently using a k-MLWE instance $\Big\{\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}, \mathbf{b}\Big\}$. Note that the only difference in the distributions is that $\mathbf{c}_0$ is equal to either $\mathbf{s}_0^{\intercal} \cdot \tilde{\mathbf{A}}_0 + \mathbf{e}^{\intercal} + \mathbf{s}_{\neq 0}^{\intercal} \cdot \tilde{\mathbf{A}}_{\neq 0} \bmod q$ or $\mathcal{U}(\mathbf{U}^{\perp}) + \mathbf{e} + \mathbf{s}_{\neq 0}^{\intercal} \cdot \tilde{\mathbf{A}}_{\neq 0} \bmod q$ which is exactly the two distributions of the k-MLWE challenge $\mathbf{b}$. Hence, $\mathsf{Hyb}_{1,6}$ and $\mathsf{Hyb}_{1,7}$ are computationally indistinguishable based on the k-MLWE assumption. Lastly, the difference between the current pair of distributions in $\mathsf{Hyb}_{0,6}$ and $\mathsf{Hyb}_{1,7}$ is again exactly the distribution of the k-MLWE challenge $\mathbf{b}$ and they can be simulated efficiently using a fresh k-MLWE instance $\Big\{\tilde{\mathbf{A}}_0, \mathbf{U}_{\mathtt{Cor}}, \mathbf{U}_{\mathtt{Hon}}, \mathbf{b}\Big\}$. Hence, if $\mathcal{A}$ wins the game in Figure 5.1 game with non-negligible probability the reduction solves the k-MLWE problem with a close probability. □

5.7 Example Applications

We showcase simple constructions of threshold signatures and threshold identity-based encryption (IBE) as direct applications of partial trapdoors.

5.7.1 Threshold Signatures Construction

Let $n, m, T, N, \log q, \tilde{s}, \beta, \sigma \in \mathsf{poly}(\lambda)$ where $m = n \cdot (\lceil \log q \rceil + 1)$ and $T \leq N$. Let $\mathsf{H} : \mathcal{M} \to \mathcal{R}_q^{nT}$ be a hash function modelled by a random oracle, $(\mathsf{TrapGen}, \mathsf{SamplePreMP})$ a lattice

Table 5.2: Start of parameter table for the partial lattice trapdoor scheme in Figure 5.2.

d	$\in \mathsf{poly}(\lambda)$, $2\,dm \geq \Omega(\lambda)$	Ring degree, inequality used in Lem. 98
q	unramified prime s.t $q\mathcal{R} = \prod_{j\in[g]} \mathfrak{q}_j$, $g \geq 1$ $ng \cdot q^{-d\cdot(2mT-n+1)/g} \leq \mathsf{negl}(\lambda)$ (Lem. 98)	Modulus
δ	≥ 2	Base of $\mathbf{G}$
n	≥ 1	Matrix row parameter
$\tilde{m}$	$= n \cdot \lceil \log_\delta q \rceil$	Width of $\mathbf{G}_n$
m	$= \tilde{m} + n$	Matrix column parameter for Lem. 50
T, N	$T \leq N$	Threshold, Total users
k	$= (m + \tilde{m}) \cdot (T - 1)$	Number of hints for k-MSIS, k-MLWE
η_A	$\leq 2\exp(1/d2mT) \cdot d \cdot q^{n/2m}$	$\eta_\varepsilon(\Lambda_q^\perp(\mathbf{A}))$ bound
η_C	$\leq 2\exp(1/dm) \cdot d \cdot q^{n/m}$	$\eta_\varepsilon(\Lambda_q^\perp(\mathbf{C}_j))$ bound
η_G	$\leq \sqrt{5} \cdot \omega(\sqrt{\log(d \cdot \tilde{m})})$	$\eta_\varepsilon(\Lambda_q^\perp(\mathbf{G}))$ bound
B_A	$\leq \eta_A \cdot \sqrt{2\tilde{m} \cdot (2m - \tilde{m})T \cdot d}$	$\|\mathsf{td}_\mathbf{A}\|$ bound from Lem. 50
B_C	$\leq \eta_C \cdot \sqrt{2\tilde{m} \cdot (m - \tilde{m}) \cdot d}$	$\left\|\mathsf{td}_{\mathbf{C}_j}\right\|$ bound from Lem. 50
B_U	$\leq \mathfrak{s}_{\max}(\mathbf{S}_{\mathsf{Cor}}) \cdot \sqrt{2dm^2T}$	$\mathfrak{s}_{\max}(\tilde{\theta}(\mathbf{U}))$ bound from Lems. 24, 4. used in Lem. 97, Thm, 13

trapdoor scheme, PT the (T, N)-partial lattice trapdoor scheme in Figure 5.2. In Figure 5.5 we construct a (T, N)-signature scheme for the message space $\mathcal{M}$. Theorem 15 follows immediately from the construction, hence, we omit the proof.

Theorem 15. *The T-out-of-N threshold signature scheme* TSIG *in Figure 5.5 is correct, if the partial lattice trapdoor scheme* PT *is $(nT, 2mT, \tilde{s}, \beta, \sigma)$-correct.*

Theorem 16. *The T-out-of-N threshold signature scheme* TSIG *in Figure 5.5 is highly-selectively secure, if the partial lattice trapdoor scheme* PT *is $(nT, 2mT, \beta', \sigma)$-one-way and $\beta \leq \beta'$.*

Proof. Suppose there exists a PPT $\mathcal{A}$ against the highly-selective security of the threshold signature scheme. We construct a PPT $\mathcal{B}^\mathcal{A}$ against the β'-one-wayness of PT. Let $\mathcal{B}^\mathcal{A}$ proceed as follows:

- Initialise list H. Receive from $\mathcal{A}$ set $\mathsf{cor} \subset [N]$ of corrupt users, set $\{\mathsf{act}_\ell, \mathsf{msg}_\ell\}_{\ell\in[L]}$ of all signature queries, and challenge $(\mathsf{act}^*, i^*, \mathsf{msg}^*)$. W.l.o.g. assume $|\mathsf{cor}| = T - 1$.

Table 5.3: Table of parameters continuation.

$\mathbf{S}$	$= \mathbf{S}_{\mathtt{Cor}}$	In PTrapGen, Fig. 5.2
$\tilde{\mathbf{S}}_i$	$\tilde{\mathbf{S}}_i = \mathbf{S}_{\mathtt{Cor}} \ \ \forall i \in [m(T-1)]$, $\tilde{\mathbf{S}}_i = \mathbf{S}_{\mathtt{Hon}} \ \ \forall i \in [(m+\tilde{m})\cdot(T-1)] \setminus [m(T-1)]$	k-MSIS, k-MLWE hint Gaussian parameters
a	≥ 1	Thms. 13, 14
$\mathbf{S}_{\mathtt{Hon}}$	(k-MSIS$_{\mathcal{R}_q,n,2mT,\{\tilde{\mathbf{S}}_i\}_i,\alpha}$ is plausibly hard), (k-MLWE$_{\mathcal{R}_q,n,2gmT,\{\tilde{\mathbf{S}}_i\}_i,\chi_0}$ is plausibly hard), $\mathfrak{s}_{\min}(\mathbf{S}_{\mathtt{Hon}}) > \max\{\eta_A, 2\sqrt{d}\cdot a^{T-1}\cdot q^{n/(m+\tilde{m})}\}$, $\mathfrak{s}_{\max}(\mathbf{S}_{\mathtt{Hon}}) \leq \min\{q^{1/g}/\sqrt{2mT}, a\cdot\mathfrak{s}_{\min}(\mathbf{S}_{\mathtt{Hon}})\}$	Inequalities used in Lem. 98, assumptions used in Thms. 13, 14
$\mathbf{S}_{\mathtt{Cor}}$	(k-MSIS$_{\mathcal{R}_q,n,2mT,\{\tilde{\mathbf{S}}_i\}_i,\alpha}$ is plausibly hard), (k-MLWE$_{\mathcal{R}_q,n,2gmT,\{\tilde{\mathbf{S}}_i\}_i,\chi_0}$ is plausibly hard), $\mathfrak{s}_{\min}(\mathbf{S}_{\mathtt{Cor}}) \geq \delta\cdot B_A\cdot\omega(\sqrt{\log(2mTd)})$ (Lem. 96), $\mathfrak{s}_{\min}(\mathbf{S}_{\mathtt{Cor}}) \geq \eta_A$ (Lems. 97,99), $\mathfrak{s}_{\min}(\mathbf{S}_{\mathtt{Cor}}) > \max\{\eta_A, 2\sqrt{d}\cdot a^{T-1}\cdot q^{n/(m+\tilde{m})}\}$, $\mathfrak{s}_{\max}(\mathbf{S}_{\mathtt{Cor}}) \leq \min\{q^{1/g}/\sqrt{2mT}, a\cdot\mathfrak{s}_{\min}(\mathbf{S}_{\mathtt{Cor}})\}$ (Lem. 98) $\mathfrak{s}_{\min}(\mathbf{S}_{\mathtt{Cor}}) \geq \delta\cdot\mathfrak{s}_{\max}(\mathbf{S}_{\mathtt{Hon}})\sqrt{2dmT\cdot\tilde{m}\cdot(T-1)}\cdot\omega(\sqrt{\log((2mT+\tilde{m})\cdot d)})$ (Lem. 99)	Assumptions used in Thms. 13, 14
σ_G	$\geq \delta\cdot\mathfrak{s}_{\max}(\mathbf{S}_{\mathtt{Hon}})\sqrt{2dmT\cdot\tilde{m}\cdot(T-1)}\cdot\omega(\sqrt{\log((2mT+\tilde{m})\cdot d)})$	Parameter in Lem. 99
$\mathbf{S}_{\mathtt{BASIS}}$	$= \mathsf{diag}(\mathbf{S}_{\mathtt{Cor}}, \sigma_G\mathbf{I}_{\tilde{m}})$	Parameter in Lem. 99
σ	$\geq \delta\cdot B_C\cdot\omega(\sqrt{\log(md)})$ (Lem. 96), $\geq B_U\cdot\eta_C$ (Lem. 97), $\geq B_U\cdot 2\cdot\eta_\varepsilon^{(md)}$ (Thm. 13)	Gaussian parameter in PSampPre, Fig 5.2
β	$\geq T\cdot\sigma\cdot\sqrt{md}$	Used in Thm. 11
β'	$\leq \alpha - T\cdot\sigma\cdot\sqrt{md}$	Used in Thm. 13
α	(k-MSIS$_{\mathcal{R}_q,n,2mT,\{\tilde{\mathbf{S}}_i\}_i,\alpha}$ is plausibly hard)	Assumption parameter
χ_0	(k-MLWE$_{\mathcal{R}_q,n,2gmT,\{\tilde{\mathbf{S}}_i\}_i,\chi_0}$ is plausibly hard)	Assumption parameter
χ_2	$\geq 2\exp(1/d\,2gmT)\cdot d\cdot q^{n/2gmT}$, $> 2\exp(1/d(m+\tilde{m}))\cdot d\cdot q^{1/(m+\tilde{m})}$	Used in Thm. 14 and Cor. 20
χ_1	$\geq \lambda^{\omega(1)}\cdot\chi_0\cdot\chi_2\cdot 2gmT$	Used in Thm. 14

$\mathsf{TSIG.MKeygen}(1^\lambda)$

$(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^{nT}, 1^{2mT})$
return $(\mathsf{mpk}, \mathsf{msk}) \coloneqq (\mathbf{A}, \mathsf{td})$

$\mathsf{TSIG.Keygen}(\mathsf{mpk}, \mathsf{msk}, j)$

$\mathsf{ptd}_j \leftarrow \mathsf{PT.PTrapGen}(\mathbf{A}, \mathsf{td}, j)$
$\mathsf{sk}_j \coloneqq \mathsf{ptd}_j$; **return** sk_j

$\mathsf{TSIG.Combine}(\left(\mathsf{sig}_j\right)_{j \in \mathsf{act}})$

$\mathbf{x} \leftarrow \mathsf{PT.Combine}((\mathbf{x}_j)_{j \in \mathsf{act}})$; $\mathsf{sig} \coloneqq \mathbf{x}$; **return** sig

$\mathsf{TSIG.Sign}(\mathsf{mpk}, \mathsf{sk}_j, \mathsf{act}, \mathsf{msg})$

$\mathbf{y} \coloneqq \mathsf{H}(\mathsf{msg}) \in \mathcal{R}_q^{nT}$
$\mathbf{x}_j \leftarrow \mathsf{PT.PSampPre}(\mathbf{A}, \mathsf{ptd}_j, \mathsf{act}, \mathbf{y}, \sigma)$
$\mathsf{sig}_j \coloneqq \mathbf{x}_j$; **return** sig_j

$\mathsf{TSIG.Verify}(\mathsf{mpk}, \mathsf{sig}, \mathsf{msg})$

$\mathbf{y} \coloneqq \mathsf{H}(\mathsf{msg})$
$b_0 \coloneqq (\mathbf{A} \cdot \mathbf{x} = \mathbf{y} \bmod q)$
$b_1 \coloneqq (\|\mathbf{x}\| \leq \beta)$
return $b_0 \wedge b_1$

Figure 5.5: TSIG threshold signature construction.

- Pass cor to the PT security challenger, and receive $\mathbf{A}$ and partial trapdoors $\{\mathbf{U}_j\}_{j \in \mathsf{cor}}$. Let $\mathsf{mpk} \coloneqq \mathbf{A}$ and $\mathsf{sk}_j \coloneqq \mathbf{U}_j$ for all $j \in \mathsf{cor}$.
- For each $\ell \in [L]$, query $\mathsf{PSampPreO}(\mathsf{act}_\ell)$ to obtain $\left\{\{\mathbf{x}_{\ell,j}\}_{j \in \mathsf{act}_\ell}, \mathbf{y}_\ell\right\}$, where $\mathbf{y}_\ell$ is uniform. Let $\mathsf{sig}_{\ell,j} \coloneqq \mathbf{x}_{\ell,j}$. Program $\mathsf{H}[\mathsf{msg}_\ell] \coloneqq \mathbf{y}_\ell$.
- Pass (act^*, i^*) to PT challenger, and receive $\left\{\left\{\mathbf{x}_j^*\right\}_{j \in \mathsf{act}^* \setminus \{i^*\}}, \mathbf{y}^*\right\}$, where $\mathbf{y}^*$ is uniform. Let $\mathsf{sig}_j^* \coloneqq \mathbf{x}_j^*$. Program $\mathsf{H}[\mathsf{msg}^*] \coloneqq \mathbf{y}^*$.
- Upon any $\mathsf{H}[\mathsf{msg}]$ query from $\mathcal{A}$, if $\mathsf{H}[\mathsf{msg}] = \perp$, sample a random vector $\mathbf{y} \leftarrow \mathcal{R}_q^{nT}$, let $H[\mathsf{msg}] \coloneqq \mathbf{y}$. Answer with $H[\mathsf{msg}]$.
- Pass $\mathsf{mpk}, (\mathsf{sk}_j)_{j \in \mathsf{cor}}, \left\{\mathsf{sig}_{\ell,j}\right\}_{\ell \in [L], j \in \mathsf{act}_\ell}, \left\{\mathsf{sig}_j^*\right\}_{j \in \mathsf{act}^* \setminus \{i^*\}}$ to $\mathcal{A}$, then return the same output as $\mathcal{A}$.

It is straightforward that mpk, the corrupt signing keys $\left\{\mathsf{sk}_j\right\}_{j \in \mathsf{cor}}$, the answers to signature queries, and the challenge $\left\{\mathsf{sig}_j^*\right\}_{j \in \mathsf{act}^* \setminus i^*}$ are distributed identically to the real signature scheme by design. If $\mathcal{A}$ returns a valid forgery sig^*, this means $\mathbf{A} \cdot \mathsf{sig}^* = \mathbf{y}^* \bmod q$ and $\|\mathsf{sig}^*\| \leq \beta \leq \beta'$, hence $\mathcal{B}^{\mathcal{A}}$ breaks the β'-one-wayness of PT with sig^*. The success probability of $\mathcal{B}^{\mathcal{A}}$ is thus the same as $\mathcal{A}$. □

Remark 20. We note that a malicious party in the signing set act can always abort the protocol by going offline. Yet, we note that our scheme has identifiable aborts. The contribution of every party in the protocol is a short preimage $\mathbf{x}_j$ of $\mathbf{v}_j \otimes \mathbf{z}_j$ where the sets $\{\mathbf{v}_j\}$ and $\mathbf{z}_j$ can be computed by anyone who knows the signing set act. Therefore,

$\mathsf{TIBE.Setup}(1^\lambda)$

$(\mathbf{A}, \mathsf{td}) \leftarrow \mathsf{TrapGen}(1^\lambda, 1^{nT}, 1^{2mT})$
$\mathsf{ek} \coloneqq \mathbf{A}$
for $j \in [N]$:
  $\mathsf{ptd}_j \leftarrow \mathsf{PT.PTrapGen}(\mathbf{A}, \mathsf{td}, j)$
  $\mathsf{dk}_j \coloneqq \mathsf{ptd}_j$
return $(\mathsf{ek}, \{\mathsf{dk}_j\}_{j\in[N]})$

$\mathsf{TIBE.ShareExtract}_0(\mathsf{ek}, \mathsf{dk}_j, \mathsf{act}, \mathsf{id})$

$\mathbf{y} \coloneqq \mathsf{H}(\mathsf{id}) \in \mathcal{R}_q^{nT}$
$\mathbf{x}_j \leftarrow \mathsf{PT.PSampPre}(\mathbf{A}, \mathsf{dk}_j, \mathsf{act}, \mathbf{y}, \sigma)$
$\mathsf{sk}_j \coloneqq \mathbf{x}_j$; **return** sk_j

$\mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{id}, \mathsf{msg})$

$\mathbf{y} \coloneqq \mathsf{H}(\mathsf{id})$
$\mathbf{s} \leftarrow \mathcal{R}_q^{nT}$; $\mathbf{e}_0 \leftarrow \chi^{2mT}$; $e_1 \leftarrow \chi$
$\mathbf{c}_0^\mathsf{T} \coloneqq \mathbf{s}^\mathsf{T}\mathbf{A} + \mathbf{e}_0^\mathsf{T} \bmod q$
$c_1 \coloneqq \mathbf{s}^\mathsf{T}\mathbf{y} + e_1 + \lfloor q/2 \rfloor \mathsf{msg} \bmod q$
$\mathsf{ctxt} \coloneqq (\mathbf{c}_0, c_1)$; **return** ctxt

$\mathsf{TIBE.Combine}(\mathsf{ctxt}, \{\mathsf{sk}_j\}_{j\in\mathsf{act}})$

$\mathbf{x} \leftarrow \mathsf{PT.Combine}(\{\mathbf{x}_j\}_{j\in\mathsf{act}})$
$c \coloneqq c_1 - \mathbf{c}_0^\mathsf{T}\mathbf{x} \bmod q$
if $c > q/4$: **return** 1
return 0

Figure 5.6: TIBE threshold IBE construction.

to verify the contribution $\mathbf{x}_j$ of a participant, it is enough to check the norm bound and the image of $\mathbf{x}_j$. If the conditions are met then $\sum \mathbf{x}_j$ is guaranteed to be a valid signature.

5.7.2 Threshold Identity-Based Encryption Construction

Let $n, m, T, N, \log q, \tilde{s}, \beta, \sigma \in \mathsf{poly}(\lambda)$ where $m = n(\lceil \log q \rceil + 1)$ and $T \leq N$. Let χ_0, χ_1 be distributions parametrised by λ. Let $\mathsf{H} : \mathcal{ID} \to \mathcal{R}_q^{nT}$ be a hash function modelled by a random oracle, $(\mathsf{TrapGen}, \mathsf{SamplePreMP})$ a lattice trapdoor scheme, PT the (T, N) partial lattice trapdoor scheme in Figure 5.2. In Figure 5.6 we construct a T-out-of-N threshold IBE scheme for the identity space $\mathcal{ID}$ and message space $\mathcal{M} = \{0, 1\}$.

Theorem 17. *If the partial lattice trapdoor scheme* PT *is* $(nT, 2mgT, \tilde{s}, \beta, \sigma)$*-correct and* $q > 2(\chi_0 + \chi_1\beta\sqrt{2mgTd})$*, then the* T*-out-of-*N *threshold IBE scheme in Figure 5.6 is correct.*

Proof. During decryption, we obtain $\mathbf{x} \leftarrow \mathsf{PT.Combine}(\{\mathbf{x}_j\}_{j\in\mathsf{act}})$ which, by the β-correctness of PT, satisfies $\mathbf{Ax} = \mathbf{y} = \mathsf{H}(\mathsf{id}) \bmod q$ and $\|\mathbf{x}\| \leq \beta$. Then

$$\begin{aligned} c = c_1 - \mathbf{c}_0^\mathsf{T}\mathbf{x} &= \mathbf{s}^\mathsf{T} \cdot \mathbf{y} + e + \lfloor q/2 \rfloor \mathsf{msg} - (\mathbf{s}^\mathsf{T} \cdot \mathbf{A} + \mathbf{e}^\mathsf{T}) \cdot \mathbf{x} \bmod q \\ &= \mathbf{s}^\mathsf{T} \cdot \mathbf{y} + e + \lfloor q/2 \rfloor \mathsf{msg} - \mathbf{s}^\mathsf{T} \cdot \mathbf{y} - \mathbf{e}^\mathsf{T} \cdot \mathbf{x} \bmod q \\ &= e - \mathbf{e}^\mathsf{T} \cdot \mathbf{x} + \lfloor q/2 \rfloor \mathsf{msg} \bmod q. \end{aligned}$$

With overwhelming probability $\|e - \mathbf{e}^\mathsf{T} \cdot \mathbf{x}\| \leq \chi_0 + \chi_1\beta\sqrt{2mgTd}$ by Lemma 24, hence, decryption correctness is guaranteed as long as $q > 2(\chi_0 + \chi_1\beta\sqrt{2mgTd})$. □

Theorem 18. *The T-out-of-N threshold IBE scheme in Figure 5.6 is highly-selectively secure, if the partial lattice trapdoor scheme* PT *has* $(nT, 2mgT, \chi_0, \chi_1, \sigma)$*-indistinguishability.*

Proof. Define the following hybrids:

$\mathsf{Hyb}_{b,0}$: The real highly-selective experiment encrypting msg_b.

$\mathsf{Hyb}_{b,1}$: The ciphertext component c_1 is swapped to uniform over $\mathcal{R}_q$.

We have $\mathsf{Hyb}_{b,0} = \mathsf{Hyb}_{b,1}$ since c_1 is uniform in both hybrids. Below we show that $\mathsf{Hyb}_{b,0} \approx_c \mathsf{Hyb}_{b,1}$ if PT has (χ_0, χ_1)-indistinguishability. Suppose there exists a PPT $\mathcal{A}$ distinguishing $\mathsf{Hyb}_{b,0}$ and $\mathsf{Hyb}_{b,1}$ with non-negligible probability. We construct a PPT $\mathcal{B}^{\mathcal{A}}$ against the (χ_0, χ_1)-indistinguishability of PT. Let $\mathcal{B}^{\mathcal{A}}$ proceed as follows:

- Initialise list H. Receive from $\mathcal{A}$ set $\mathsf{cor} \subset [N]$ of corrupt users, set $\{\mathsf{act}_\ell, \mathsf{id}_\ell\}_{\ell \in [L]}$ of all secret key queries, and challenge $(\mathsf{act}^*, i^*, \mathsf{id}^*)$. W.l.o.g. assume $|\mathsf{cor}| = T - 1$.
- Pass cor to the PT indistinguishability challenger, and receive $\mathbf{A}$ and partial trapdoors $\{\mathbf{U}_j\}_{j \in \mathsf{cor}}$. Let $\mathsf{ek} := \mathbf{A}$ and $\mathsf{dk}_j := \mathbf{U}_j$ for all $j \in \mathsf{cor}$.
- For each $\ell \in [L]$, query $\mathsf{PSampPreO}(\mathsf{act}_\ell)$ to obtain $(\{\mathbf{x}_{\ell,j}\}_{j \in \mathsf{act}_\ell}, \mathbf{y}_\ell)$, where $\mathbf{y}_\ell$ is uniform. Let $\mathsf{sk}_{\ell,j} := \mathbf{x}_{\ell,j}$. Program $\mathsf{H}[\mathsf{id}_\ell] := \mathbf{y}_\ell$.
- Pass (act^*, i^*) to PT challenger, and receive $(\left\{\mathbf{x}^*_j\right\}_{j \in \mathsf{act}^* \setminus \{i^*\}}, \mathbf{y}^*, \mathbf{c}_0, \mathbf{c}_1)$, where $\mathbf{y}^*$ is uniform. Let $\mathsf{sk}^*_j := \mathbf{x}^*_j$. Program $\mathsf{H}[\mathsf{id}^*] := \mathbf{y}^*$.
- Upon $\mathsf{H}[\mathsf{id}]$ query from $\mathcal{A}$, if $\mathsf{H}[\mathsf{id}] = \perp$, sample a random vector $\mathbf{y} \leftarrow \mathcal{R}_q^{nT}$, let $H[\mathsf{id}] := \mathbf{y}$. Answer with $H[\mathsf{id}]$.
- Pass $\mathsf{ek}, (\mathsf{dk}_j)_{j \in \mathsf{cor}}, \left\{\mathsf{sk}_{\ell,j}\right\}_{\ell \in [L], j \in \mathsf{act}_\ell}, \left\{\mathsf{sk}^*_j\right\}_{j \in \mathsf{act}^* \setminus \{i^*\}}$ to $\mathcal{A}$, and receive $(\mathsf{msg}_0, \mathsf{msg}_1)$.
- Let $c_1 := c_1 + \lfloor q/2 \rfloor \mathsf{msg}_b \bmod q$. Pass $\mathsf{ctxt} := (\mathbf{c}_0, c_1)$ to $\mathcal{A}$, and return the same output as $\mathcal{A}$.

It is straightforward to verify that ek, the corrupt master secret keys $\{\mathsf{dk}_j\}_{j \in \mathsf{cor}}$, the answers to secret key queries, and the challenge $\left\{\mathsf{sk}^*_j\right\}_{j \in \mathsf{act}^* \setminus i^*}$ are distributed identically to corresponding values in $\mathsf{Hyb}_{b,0}$ and $\mathsf{Hyb}_{b,1}$. If c_1 from the PT challenger equals $\mathbf{s}^{\mathsf{T}}\mathbf{y}^* + e \bmod q$, then $\mathcal{B}^{\mathcal{A}}$ perfectly simulates c_1 in $\mathsf{Hyb}_{b,0}$. Else, c_1 is uniformly random, hence $\mathcal{B}^{\mathcal{A}}$ perfectly simulates c_1 in $\mathsf{Hyb}_{b,1}$. We conclude that the success probability of $\mathcal{B}^{\mathcal{A}}$ is same as $\mathcal{A}$. □

Chapter 6

A Simple IND-CCA Threshold KEM

While IND-CPA security constitutes a security baseline for KEMs and PKEs, the standard notion is the stronger IND-CCA security. Many "natural" PKE constructions achieve IND-CPA security out of the box, but not IND-CCA security. To resolve this issue, generic compilers have emerged which take as input an IND-CPA scheme, and output an IND-CCA one. The most popular CPA-to-CCA compiler is the Fujisaki-Okamoto transform [FO99, FO13], an efficient transform that only relies on hash functions modelled as random oracles, and is used in ML-KEM and the upcoming NIST standard HQC.

Threshold cryptosystems have many applications. E-voting protocols use them to guarantee tally confidentiality until the close of voting, see for example the work of [CGS97], or the lattice-based protocol of [ABGS23]. In the blockchain setting, [CGPP24] propose them as a way to enhance the privacy of pending and unconfirmed transactions. Moreover, NIST has released a call for threshold cryptosystems [BP25], and we expect it will foster more applications.

Building lattice-based threshold KEMs or PKEs has shown to be challenging. While there exist efficient IND-CPA constructions [BS23, MS24], achieving IND-CCA security in an efficient manner seems difficult. Indeed, thresholdising the Fujisaki-Okamoto transform requires expensive generic multi-party computation (MPC). Alternative IND-CCA lattice-based threshold constructions have been proposed, usually using the machinery of non-interactive zero-knowledge arguments (NIZKs) [DLN+21], threshold fully homomorphic encryption [BGG+18] or MPC [KLO+19, CCMS21], thus, inheriting the corresponding efficiency downsides.

Chapters contributions. In this chapter we propose a simple lattice-based threshold KEM that achieves IND-CCA security. At a high level, our construction is based on the Boneh-Canetti-Halevi-Katz transform [CHK04, BCHK07], or BCHK transform,

which has been shown by Boneh, Boyen and Halevi [BBH06] to allow constructing IND-CCA threshold PKEs (TPKEs) under suitable conditions. However unlike pairing-based constructions considered in [BBH06], lattice-based constructions do not natively allow for simple and private validity checks during the protocol. Hence, the BCHK transform cannot be used as-is, in particular it does not guarantee decryption/decapsulation consistency.

Our first contribution is to provide a variant of the BCHK transform tailored for lattices. This construction, which we call BCHK+, yields a threshold KEM (TKEM) with decapsulation consistency in addition to IND-CCA security. Equipped with our new BCHK+ transform, we build an IND-CCA TKEM. One of the key ingredients of the transform is a selectively secure threshold identity-based encryption (TIBE). In this chapter we construct a relatively efficient lattice-based TIBE with reduced public key and ciphertext sizes compared to Chapter 5. This construction is the second contribution of this chapter. A third contribution, which might be of independent interest, is the introduction of the Coset-Hint-MLWE assumption, a natural generalisation of the Hint-MLWE assumption [KLSS23]. This assumption underlies the security of our TIBE, and therefore of our TKEM. Finally, we show that the hardness of Hint-MLWE implies the hardness of Coset-Hint-MLWE.

Related work. Several threshold signatures schemes (TSS) based on lattices were recently proposed. In most of these, the base signature scheme is based on noise flooding and leverages Hint-MLWE assumption [KLSS23] for the security proof, see e.g. Raccoon [dPKPR24] and Plover [EEN+24]. The linear structure of these schemes facilitates their thresholdisation. Regarding the linearisation tools, 1-round TSS based on homomorphic encryption have been proposed [ASY22, GKS24]. However, most schemes construct bespoke signing algorithms, which give 4-round [KRT24, ENP24], 3-round [DKM+24, KRT24] or 2-round [EKT24, CATZ24, BKL+25] TSS.

Lattice threshold cryptosystems. Compared to TSS, few lattice threshold public key encryption schemes (TPKEs, TIBEs) and threshold KEMs (TKEMs) have been proposed. Two recent papers [BS23, MS24] propose IND-CPA TPKEs based on noise-flooded variants of the Dual Regev [LP11] and Regev [Reg09].

For lattice-based TIBEs the work of [BKP13] builds a threshold version of GPV IBE with a proof in UC and adaptive corruptions. It is the only robust scheme in the literature, and has potential for more compact sizes compared to our scheme. However, it requires generic MPC techniques for Gaussian sampling which typically has higher running time and round complexity. The only other TIBE construction is discussed in Chapter 5. It is expected to be relatively fast, since it only requires matrix vector

multiplications, it has identifiable aborts and only 1 round. However, it is proved secure under weaker highly-selective security where the adversary declares all oracle queries before receiving the public keys and the size of their master public key and ciphertexts are $O(T^2)$ and $O(T)$ respectively. In this chapter the ciphertext and public sizes only depend on T logarithmically.

IND-CCA constructions. IND-CCA security in a TPKE to TKEM remains challenging, as the usual reencryption approach by Fujisaki and Okamoto [FO99, FO13] does not thresholdise well. The universal thresholdiser by [BGG+18] uses ThFHE to thresholdise any functionality, and use it to build an INC-CCA TPKE. The scheme has identifiable aborts, has game-based selective security and only one round. However to our knowledge, this work remains theoretical, and no parameter set has been proposed. It mentions the slow runtime as an open problem.

Cong et al. [CCMS21] proposed an IND-CCA TKEM called Gladius. On one hand, it also has the potential for smaller public key and ciphertext sizes. On the other hand, their design requires to evaluate a hash function using MPC which results if a high number of rounds (136491 rounds for 3 users).

Finally, [DLN+21] propose a lattice-based INC-CCA TPKE achieving advanced notions such as adaptive security and identifiable aborts. However, it relies on heavy machinery such as lossy encryption schemes and NIZKs based on correlation-intractable hashes. Again, no parameter set has been proposed.

Compared to the above, our TIBE and TKEM are the only ones that simultaneously (i) do not require FHE/MPC/NIZK, (ii) have sizes essentially independent of T and N. As a result, they have small communication complexity and we expect that they would be fast when implemented. On the other hand, our schemes do not currently achieve robustness, have a trusted setup, and are less compact than [CCMS21]. We provide a summary of the related work in Table 6.1.

Open problems. The TKEM we build in this chapter only achieves weak decapsulation consistency. Intuitively, we prevent the adversary from decapsulating the same ciphertext into two different *valid* keys. However, the adversary is always able to make the protocol abort in one of the executions by providing an invalid decryption share. An interesting open question would be to build a share verification algorithm to improve the robustness

[1] In the 3-out-of-3 setting.
[2] Ignoring latency.

Table 6.1: Related work comparison for TIBE and TKEM. Sizes of |ek| and |ct| are given in KiB. N/A = Not Available, UC = Universal Composability, IA = Identifiable Aborts.

Reference	Security	Decrypt	|ek|	|ct|	Rnds	Rbst	Tools
TIBE							
[BKP13]	Adaptive, UC	N/A	N/A	N/A	N/A	Yes	MPC
Chapter 5	Highly-selective	N/A	$O(T^2)$	$O(T)$	1	IA	Direct
This chapter	Selective	N/A	50	450	3	No	Direct
IND-CCA TKEM							
[BGG+18]	Selective, UC	N/A	N/A	N/A	1	IA	FHE
[KLO+19]	Heuristic	4.342 sec.[1]	4	4	≥ 114	No	MPC
[DLN+21]	Adaptive	N/A	N/A	N/A	1	IA	NIZK
[CCMS21]	Selective, UC	4.99 sec.[2]	N/A	N/A	136 498	IA	MPC
This chapter	Selective	N/A	50	450	3	No	Direct

of our TKEM. A different future direction is optimising the underlying TIBE to improve the key and ciphertext sizes. We tackle both directions in Chapter 7.

Chapter structure. In Section 6.1 we define the hardness assumption underlying our TIBE construction, that we dub Coset-Hint-MLWE. We also give a reduction from the Hint-MLWE assumption with standard hardness to our Coset-Hint-MLWE. In Section 6.2 we discuss the BCHK+ transform and the design choices behind our TIBE. In Section 6.3 and Section 6.4 we prove selective security of the TIBE and decapsulation consistency of the TKEM respectively. Lastly in Section 6.5 we optimise the parameters of the scheme for smaller key and ciphertext sizes and present the results.

Chapter specific notations. We consider cyclotomic number fields $\mathcal{K} = \mathbb{Q}[\zeta_{2d}]$ for d a power-of-two. It is isomorphic to $\mathcal{K} = \mathbb{Q}[X]/(X^d + 1)$. Its ring of integers is of the form $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$. We choose q to be a prime modulus such that $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ is isomorphic to a direct product of two fields of order $q^{d/2}$. By Corollary 1 it suffices to pick $q = 5 \bmod 8$. We embed the ring element into real space using the coefficient embedding.

6.1 The Coset-Hint-MLWE Assumption

In this section, we formally define the Coset-Hint-MLWE assumption. Intuitively, the hints component $\mathbf{p}_{j,i}$ in this new problem is sampled from a lattice coset dependent on $\mathbf{c}_i$, as opposed to the whole ring. As a result, this assumption helps one simulate multiple

$\mathcal{P}_{\mathsf{cond}}(\mathbf{S}, \mathbf{E})$

$\forall i : \mathbf{u}_i \leftarrow \mathcal{R}_q^m$

$(\mathbf{c}_{0,i}, \mathbf{c}_{1,i}) \coloneqq \mathsf{Decomp}_\beta(\mathbf{u}_i)$

$(\mathbf{p}_i, \mathbf{x}_i) \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{r}_i - \mathbf{A}_0(\mathbf{B})\cdot\mathbf{T}\cdot\mathbf{c}_i}([\mathbf{A}_0(\mathbf{B})|\mathbf{A}_2]),\sigma_p}$

$\mathbf{z}_i = \mathbf{T}\cdot\mathbf{c}_i + \mathbf{p}_i$

return $\{\mathbf{c}_i = (\mathbf{c}_{0,i}, \mathbf{c}_{1,i}), (\mathbf{z}_i, \mathbf{x}_i)\}_i$

Figure 6.1: Hint distribution for Coset-Hint-MLWE

preimages of the same vector using techniques from standard model identity-based encryption literature.

One might think that restricting the noise distribution leaks more information about the MLWE secrets, but we prove that this is not the case by constructing reductions from standard assumptions to Coset-Hint-MLWE. Note that constructing the matrix $\mathbf{A}_0(\mathbf{B})$ in the definition below requires a multi-secret version of MLWE with $\mathbf{S} \in \mathcal{R}^{n\times m}$ and $\mathbf{E} \in \mathcal{R}^{m\times m}$. However, in our TKEM construction we use $n = m = 1$.

Definition 32 (Coset-Hint-MLWE Assumption)**.** *Let $0 < n \leq m \leq m'$ and $\ell \geq 0$ be integers, q an integer and $\sigma, \sigma_p, B > 0$. Let $\mathbf{A} \leftarrow \mathcal{R}_q^{m\times n}$, and $\{\mathbf{r}_i\}_{i=0}^{\ell-1}$ be arbitrary vectors in $\mathcal{R}_q^m$. Assume that there exists a PPT algorithm $(\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}) \leftarrow \mathsf{TrapGen}(\lambda)$ such that $\mathbf{A}_2 \in \mathcal{R}_q^{m\times m'}$, $\mathbf{T}_{\mathbf{A}_2} \in \mathcal{R}^{m'\times m'}$ is full-rank, $\mathbf{A}_2 \cdot \mathbf{T}_{\mathbf{A}_2} = \mathbf{0} \bmod q$ and $\|\mathbf{T}_{\mathbf{A}_2}\|_{GS} \leq B$. The $\mathsf{Coset\text{-}Hint\text{-}MLWE}^{\ell,\{\mathbf{r}_i\},\sigma_p,\beta}_{\mathcal{R},n,m,m',q,\sigma}$ problem requires the Adversary to distinguish $\mathcal{D}_{\mathsf{real}}$ and $\mathcal{D}_{\mathsf{ideal}}$, where $\mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}^{n\times m},\sigma}$, $\mathbf{E} \leftarrow \mathcal{D}_{\mathcal{R}^{m\times m},\sigma}$ and:*

$$\mathcal{D}_{\mathsf{real}} = \Big\{(\mathbf{A}, \mathbf{B} = \mathbf{A}\cdot\mathbf{S} + \mathbf{E}, \mathbf{A}_2, \{\mathbf{c}_i, (\mathbf{z}_i, \mathbf{x}_i)\}_{i=0}^{\ell-1}) \mid \{\mathbf{c}_i, (\mathbf{z}_i, \mathbf{x}_i)\}_i \leftarrow \mathcal{P}_{\mathsf{cond}}(\mathbf{S}, \mathbf{E})\Big\},$$
$$\mathcal{D}_{\mathsf{ideal}} = \Big\{(\mathbf{A}, \mathbf{B}, \mathbf{A}_2, \{\mathbf{c}_i, (\mathbf{z}_i, \mathbf{x}_i)\}_{i=0}^{\ell-1}) \mid \mathbf{B} \leftarrow \mathcal{R}_q^{m\times m}, \{\mathbf{c}_i, (\mathbf{z}_i, \mathbf{x}_i)\}_i \leftarrow \mathcal{P}_{\mathsf{cond}}(\mathbf{S}, \mathbf{E})\Big\},$$

where $\mathbf{A}_0(\mathbf{B}) = \Big[\mathbf{I}_m \;\Big|\; \mathbf{A} \;\Big|\; \beta\cdot\mathbf{I}_m - \mathbf{B}\Big]$, $\mathbf{T}^T = \begin{bmatrix}\mathbf{E}^T & \mathbf{S}^T & \mathbf{I}_m \\ \mathbf{I}_m & \mathbf{0} & \mathbf{0}\end{bmatrix}$, and the distribution $\mathcal{P}_{\mathsf{cond}}(\mathbf{S}, \mathbf{E})$ is described in Figure 6.1.

Remark 21. The hint distributions of Coset-Hint-MLWE and Hint-MLWE problems differ only in the sampling of $\mathbf{p}_i$. Now if we sample vectors $\{\mathbf{r}_i\}_i$ in Coset-Hint-MLWE uniformly at random we get distributions

$$\mathcal{D}_p \coloneqq \Big\{\{(\mathbf{p}_i, \mathbf{x}_i), \mathbf{r}_i\}_{i\in[\ell]} \mid \mathbf{r}_i \leftarrow \mathcal{R}_q^n, (\mathbf{p}_i, \mathbf{x}_i) \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{r}_i}([\mathbf{A}_0(\mathbf{B})|\mathbf{A}_2]),\sigma_p}\Big\}$$
$$\mathcal{D}'_p \coloneqq \left\{\{(\mathbf{p}_i, \mathbf{x}_i), \mathbf{r}_i\}_{i\in[\ell]} \mid (\mathbf{p}_i, \mathbf{x}_i) \leftarrow \mathcal{D}_{\mathcal{R}^{2m+n+m'},\sigma_p}, \mathbf{r}_i \equiv [\mathbf{A}_0(\mathbf{B})|\mathbf{A}_2]\cdot\begin{bmatrix}\mathbf{p}_i \\ \mathbf{x}_i\end{bmatrix}\right\}$$

for $\mathbf{p}_i$ in $\mathcal{P}_{\mathsf{cond}}$ and $\mathcal{P}$ accordingly. If $\sigma_p \geq \eta_\varepsilon(\Lambda_q^\perp([\mathbf{A}_0(\mathbf{B}) \mid \mathbf{A}_2]))$, by Leftover Hash Lemma in Theorem 3 the distributions $\mathcal{D}_p$ and $\mathcal{D}'_p$ are statistically close. Hence, we recover the usual Hint-MLWE assumption from Coset-Hint-MLWE up to a negligible statistical distance when $\{\mathbf{r}_i\}_i$ are uniform[3].

When $\{\mathbf{r}_i\}_i$ are fixed as in Definition 32 we need a more complex argument to reduce Coset-Hint-MLWE to standard assumptions.

Theorem 19 (Hint-MLWE $\leq$ Coset-Hint-MLWE)**.** *Let $0 < n \leq m \leq m'$, $\ell \geq 0$ and q be integers. Let $\sigma, B > 0$ and $\varepsilon \in (0, 1/2)$. Let $\sigma_p > B \cdot \eta_\varepsilon(\mathcal{R}^{m'})$. For any PPT adversary $\mathcal{A}$ against Coset-Hint-MLWE*

$$\mathsf{Adv}_{\mathcal{A}}(\mathcal{D}_{\mathsf{real}}, \mathcal{D}_{\mathsf{ideal}}) \leq m \cdot \mathsf{Adv}_{\mathcal{A}_2}^{\mathsf{Hint\text{-}MLWE}^{\ell,\sigma_p,\beta}_{\mathcal{R},n,m,q,\sigma}} + \ell \cdot O(\varepsilon)$$

where the algorithm $\mathcal{A}_2$ runs in approximately the same time as $\mathcal{A}$.

Remark 22. In our scheme $n = m = 1$, $m' = 3$. When instantiating the trapdoor sampling with $\mathsf{ModNTRU}$ trapdoors, by Remark 4 we have $B = 1.17 \cdot q^{1/3}$.

Proof. We consider a sequence of six hybrids. Differences between consecutive hybrids are highlighted in grey. $\mathbf{T}$ is defined as in Definition 32.

$\mathsf{Hyb}_0 = \mathcal{D}_{\mathsf{real}}$: The original distribution can be written as follows.

$$\begin{array}{ll}
\mathsf{Hyb}_0: & \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m},\ (\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}) \leftarrow \mathsf{TrapGen}(\lambda) \\
& \mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}^{n\times m},\sigma}, \mathbf{E} \leftarrow \mathcal{D}_{\mathcal{R}^{m\times m},\sigma}, \mathbf{B} = \mathbf{A}\cdot\mathbf{S} + \mathbf{E} \\
& \mathbf{A}_0 = \mathbf{A}_0(\mathbf{B}) \coloneqq \left[\mathbf{I}_m \ \mid \ \mathbf{A} \ \mid \ \beta\cdot\mathbf{I}_m - \mathbf{B}\right] \\
& \mathbf{F} \coloneqq [\mathbf{A}_0 \mid \mathbf{A}_2] \\
\forall i: & \mathbf{u}_i \leftarrow \mathcal{R}_q^m, (\mathbf{c}_{0,i}, \mathbf{c}_{1,i}) \coloneqq \mathsf{Decomp}_\beta(\mathbf{u}_i) \\
& (\mathbf{p}_i, \mathbf{x}_i) \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{r}_i - \mathbf{A}_0\cdot\mathbf{T}\cdot\mathbf{c}_i}(\mathbf{F}),\sigma_p} \\
& \mathbf{z}_i = \mathbf{T}\cdot\mathbf{c}_i + \mathbf{p}_i
\end{array}$$

Hyb_1 : We now sample $\mathbf{p}_i$ as a spherical Gaussian. Since $\sigma_p > \|\mathbf{T}_{\mathbf{A}_2}\|_{GS} \cdot \eta_\varepsilon(R^{m'})$, it follows from Lemma 21 and Lemma 35 that:

$$\mathrm{SD}(\mathsf{Hyb}_0, \mathsf{Hyb}_1) \leq 2\,\ell\,\varepsilon.$$

[3]The values $\mathbf{A}_2, \{\mathbf{x}_i\}$ do not contain any additional information on the secret anymore and can be dropped from $\mathcal{D}_{\mathsf{real}}$ and $\mathcal{D}_{\mathsf{ideal}}$.

$$
\begin{array}{ll}
\mathsf{Hyb}_1: & \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m}, (\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}) \leftarrow \mathsf{TrapGen}(\lambda) \\
& \mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}^{n\times m},\sigma}, \mathbf{E} \leftarrow \mathcal{D}_{\mathcal{R}^{m\times m},\sigma}, \mathbf{B} = \mathbf{A}\cdot\mathbf{S}+\mathbf{E} \\
& \mathbf{A}_0 = \mathbf{A}_0(\mathbf{B}) \coloneqq \left[\mathbf{I}_m \;\middle|\; \mathbf{A} \;\middle|\; \beta\cdot\mathbf{I}_m - \mathbf{B}\right] \\
& \mathbf{F} \coloneqq [\mathbf{A}_0 \mid \mathbf{A}_2] \\
\forall i: & \mathbf{u}_i \leftarrow \mathcal{R}_q^m, (\mathbf{c}_{0,i}, \mathbf{c}_{1,i}) \coloneqq \mathsf{Decomp}_\beta(\mathbf{u}_i) \\
& \mathbf{p}_i \leftarrow \mathcal{D}_{\mathcal{R}^{2m+n},\sigma_p}, \mathbf{x}_i \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{r}_i-\mathbf{A}_0\cdot(\mathbf{T}\cdot\mathbf{c}_i-\mathbf{p}_i)}(\mathbf{A}_2),\sigma_p} \\
& \mathbf{z}_i = \mathbf{T}\cdot\mathbf{c}_i + \mathbf{p}_i
\end{array}
$$

Hyb_2 : We now sample the $\mathbf{x}_i$ using the trapdoor for $\mathbf{A}_2$. Note that now all random variables in Hyb_2 can be sampled in polynomial time. By Remark 4 we have

$$\mathrm{SD}(\mathsf{Hyb}_1, \mathsf{Hyb}_2) \leq 2\ell\, O(\varepsilon).$$

$$
\begin{array}{ll}
\mathsf{Hyb}_2: & \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m}, (\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}) \leftarrow \mathsf{TrapGen}(\lambda) \\
& \mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}^{n\times m},\sigma}, \mathbf{E} \leftarrow \mathcal{D}_{\mathcal{R}^{m\times m},\sigma}, \mathbf{B} = \mathbf{A}\cdot\mathbf{S}+\mathbf{E} \\
& \mathbf{A}_0 = \mathbf{A}_0(\mathbf{B}) \coloneqq \left[\mathbf{I}_m \;\middle|\; \mathbf{A} \;\middle|\; \beta\cdot\mathbf{I}_m - \mathbf{B}\right] \\
& \mathbf{F} \coloneqq [\mathbf{A}_0 \mid \mathbf{A}_2] \\
\forall i: & \mathbf{u}_i \leftarrow \mathcal{R}_q^m, (\mathbf{c}_{0,i}, \mathbf{c}_{1,i}) \coloneqq \mathsf{Decomp}_\beta(\mathbf{u}_i) \\
& \mathbf{p}_i \leftarrow \mathcal{D}_{\mathcal{R}^{2m+n},\sigma_p} \\
& \mathbf{x}_i \leftarrow \mathsf{SamplePre}(\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}, \mathbf{r}_i - \mathbf{A}_0\cdot(\mathbf{T}\mathbf{c}_i + \mathbf{p}_i), \sigma_p) \\
& \mathbf{z}_i = \mathbf{T}\cdot\mathbf{c}_i + \mathbf{p}_i
\end{array}
$$

Hyb_3 : Between Hyb_2 and Hyb_3 we invoke $\mathsf{Hint\text{-}MLWE}^{\ell,\sigma_p,\beta}_{\mathcal{R},n,m,q,\sigma}$ m times.

$$
\begin{array}{ll}
\mathsf{Hyb}_3: & \mathbf{A} \leftarrow \mathcal{R}_q^{n\times m}, (\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}) \leftarrow \mathsf{TrapGen}(\lambda) \\
& \mathbf{B} \leftarrow \mathcal{R}_q^{m\times m}, \mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}^{n\times m},\sigma}, \mathbf{E} \leftarrow \mathcal{D}_{\mathcal{R}^{m\times m},\sigma} \\
& \mathbf{A}_0 = \mathbf{A}_0(\mathbf{B}) \coloneqq \left[\mathbf{I}_m \;\middle|\; \mathbf{A} \;\middle|\; \beta\cdot\mathbf{I}_m - \mathbf{B}\right] \\
& \mathbf{F} \coloneqq [\mathbf{A}_0 \mid \mathbf{A}_2] \\
\forall i: & \mathbf{u}_i \leftarrow \mathcal{R}_q^m, (\mathbf{c}_{0,i}, \mathbf{c}_{1,i}) \coloneqq \mathsf{Decomp}_\beta(\mathbf{u}_i) \\
& \mathbf{p}_i \leftarrow \mathcal{D}_{\mathcal{R}^{2m+n},\sigma_p} \\
& \mathbf{x}_i \leftarrow \mathsf{SamplePre}(\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}, \mathbf{r}_i - \mathbf{A}_0\cdot(\mathbf{T}\mathbf{c}_i + \mathbf{p}_i), \sigma_p) \\
& \mathbf{z}_i = \mathbf{T}\cdot\mathbf{c}_i + \mathbf{p}_i
\end{array}
$$

For the jth column of matrix $\mathbf{B}$, $j \in [m]$ we define the following transform. Before the transformation the jth column is equal to $\mathbf{b}^{(j)} = \mathbf{A}\cdot\mathbf{s}^{(j)} + \mathbf{e}^{(j)}$ where $\mathbf{s}^{(j)}$ and $\mathbf{e}^{(j)}$ are

the jth columns of $\mathbf{S}$ and $\mathbf{E}$ accordingly. After the transformation $\mathbf{b}^{(j)} \leftarrow \mathcal{R}_q^m$ and the remaining values stay unchanged.

This jump can be reduced to solving a $\mathsf{Hint\text{-}MLWE}^{\ell,\sigma_p,\beta}_{\mathcal{R},n,m,q,\sigma}$ instance. Assume that $\left\{\tilde{\mathbf{A}}, \tilde{\mathbf{b}}^{(j)}, \left\{(\tilde{\mathbf{c}}_i^j = (\tilde{c}_{0,i}^{(j)}, \tilde{c}_{1,i}^{(j)}), \tilde{\mathbf{z}}_i^{(j)} \in \mathcal{R}^2)\right\}_i\right\}$ is the $\mathsf{Hint\text{-}MLWE}$ instance received from a challenger. We set $\mathbf{A} \coloneqq \tilde{\mathbf{A}}, \mathbf{b}^{(j)} \coloneqq \tilde{\mathbf{b}}^{(j)}$, the first coordinates of $\mathbf{c}_{0,i}$ and $\mathbf{c}_{1,i}$ are now equal to $\tilde{c}_{0,i}^{(j)}$ and $\tilde{c}_{1,i}^{(j)}$ accordingly the remaining coordinates are sampled as before. For the matrix $\mathbf{T}$ we set the jth column to $\mathbf{0}$ and sample all other columns as before. Then $\mathbf{z}_i \coloneqq \mathbf{T} \cdot \mathbf{c}_i + \tilde{\mathbf{z}}_i^{(j)} + (\mathbf{0}^{m+n} || \mathbf{p}_2)$ with $\mathbf{p}_2 \leftarrow \mathcal{D}_{\mathcal{R}^m,\sigma_p}$. Following the definition of the $\mathsf{Hint\text{-}MLWE}$ problem if $\tilde{\mathbf{b}}^{(j)}$ is an LWE sample we have the distribution before the transformation and if $\tilde{\mathbf{b}}^{(j)}$ is uniform it is the distribution after. After m such transformations for $j \in [m]$ we get Hyb_3 distribution.

$$\mathsf{Adv}(\mathsf{Hyb}_2, \mathsf{Hyb}_3) \leq m \cdot \mathsf{Hint\text{-}MLWE}^{\ell,\sigma_p,\beta}_{\mathcal{R},n,m,q,\sigma}.$$

Hyb_4 **:** Lastly, we join the sampling of $(\mathbf{p}_i, \mathbf{x}_i)$ again to get back to $\mathsf{Hyb}_4 = \mathcal{D}_{\mathsf{ideal}}$.

Hyb_4 :	$\mathbf{A} \leftarrow \mathcal{R}_q^{n\times m}$, $(\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}) \leftarrow \mathsf{TrapGen}(\lambda)$
	$\mathbf{B} \leftarrow \mathcal{R}_q^{m\times m}$, $\mathbf{S} \leftarrow \mathcal{D}_{\mathcal{R}^{n\times m},\sigma}, \mathbf{E} \leftarrow \mathcal{D}_{\mathcal{R}^{m\times m},\sigma}$
	$\mathbf{A}_0 = \mathbf{A}_0(\mathbf{B}) \coloneqq \left[\mathbf{I}_m \mid \mathbf{A} \mid \beta \cdot \mathbf{I}_m - \mathbf{B}\right]$
	$\mathbf{F} \coloneqq [\mathbf{A}_0 \mid \mathbf{A}_2]$
$\forall i$:	$\mathbf{u}_i \leftarrow \mathcal{R}_q^m, (\mathbf{c}_{0,i}, \mathbf{c}_{1,i}) \coloneqq \mathsf{Decomp}_\beta(\mathbf{u}_i)$, $\alpha_i \coloneqq \mathbf{A}_0\mathbf{T}\mathbf{c}_i$
	$(\mathbf{p}_i, \mathbf{x}_i) \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{r}_i - \alpha_i}([\mathbf{A}_0(\mathbf{B})\mid\mathbf{A}_2]),\sigma_p}$
	$\mathbf{z}_i = \mathbf{T} \cdot \mathbf{c}_i + \mathbf{p}_i$

By the same argument, we have $\mathsf{Adv}(\mathsf{Hyb}_3, \mathsf{Hyb}_4) \leq 2\ell\, O(\varepsilon) + 2\ell\, \varepsilon$. □

6.2 Our TKEM Construction

In this section we describe the modified BCHK transform and its required building blocks. Our TIBE builds on the Agrawal-Boneh-Boyen IBE [ABB10a] and on the Plover signature [EEN+24]. For the one-time signature we can choose and post-quantum secure one-time signature such at WOTS^+ [HRS16] or Raccoon[dPKPR24]. The security proof of the TIBE uses techniques from [DKM+24] and improvements by Katsumata, Reichle and Takemure [KRT24].

6.2.1 The BCHK+ Transform

We describe the BCHK+ transform in Figure 6.2 and highlight the tweaks we make to the transform of [BBH06]. The tweaks are based on the Fujisaki-Okamoto transform [FO13] and its variants by Hofheinz, Hövelmanns and Kiltz [HHK17].

Intuitively, the BCHK part of our transform ensures that the adversary cannot tamper with received ciphertexts, since they do not own the corresponding signature keys. However, they are still able to craft their own malformed ciphertexts (e.g. sending uniform $(\mathbf{u}, v)$ instead of the LWE samples) which will not be detected by the honest parties. Hence, we additionally check the ciphertexts that correspond to valid messages using the FO part of our BCHK+ transform. Since this check is only performed after obtaining a valid message, it does not need to be thresholdised. We call this new transform the BCHK+ transform, and it can be summarised as:

$$\text{BCHK+} = \text{BCHK (IND-CCA)} + \text{FO (decapsulation consistency)}.$$

$\mathsf{TKEM.Keygen}(1^\lambda)$

$(\{\mathsf{dk}_i\}_{i\in[N]}, \mathsf{ek}) \leftarrow \mathsf{TIBE.Setup}(1^\lambda)$
return $(\{\mathsf{dk}_i\}_{i\in[N]}, \mathsf{ek})$

$\mathsf{TKEM.Encaps}(\mathsf{ek})$

$\mathsf{msg} \leftarrow \{0,1\}^d$
$\mathsf{rand} = G_{\mathsf{fo}}(\mathsf{msg})$
$(\mathsf{sk}, \mathsf{vk}) \leftarrow \mathsf{SIG.Keygen}(1^\lambda)$
$\mathsf{ct} = \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{id} = \mathsf{vk}, \mathsf{msg}; \mathsf{rand})$
$\mathsf{sig} \leftarrow \mathsf{SIG.Sign}(\mathsf{sk}, \mathsf{ct})$
$K = H_{\mathsf{fo}}(\mathsf{msg}||\mathsf{ct})$
return $(\mathsf{ct}, \mathsf{vk}, \mathsf{sig}), \mathrm{K}$

$\mathsf{TKEM.ShareDecaps}(\mathsf{dk}_i, (\mathsf{ct}, \mathsf{vk}, \mathsf{sig}))$

assert $\mathsf{SIG.Verify}((\mathsf{vk}, \mathsf{ct}, \mathsf{sig})) = 1$
for $j \in [r]$ **do**
  $\mathsf{contrib}_{i,j}^{(k)} = \mathsf{TIBE.ShareExtract}_j(\mathsf{dk}_i, \mathsf{id} = \mathsf{vk})$
endfor
return $\mathsf{contrib}_{i,j}$

$\mathsf{TKEM.Combine}(\{\mathsf{contrib}_{i,r}\}_{i\in\mathsf{act}}, (\mathsf{ct}, \mathsf{vk}, \mathsf{sig}))$

assert $\mathsf{SIG.Verify}((\mathsf{vk}, \mathsf{ct}, \mathsf{sig})) = 1$
$\mathsf{msg} = \mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{vk}, \mathsf{act}, \{\mathsf{contrib}_{i,r}\}_i)$
$\mathsf{rand} = G_{\mathsf{fo}}(\mathsf{msg})$
assert $\mathsf{ct} = \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{vk}, \mathsf{msg}; \mathsf{rand})$
$K = H_{\mathsf{fo}}(\mathsf{msg}||\mathsf{ct})$
return K

Figure 6.2: The BCHK+ transform. The major differences with the original BCHK transform [CHK04, BCHK07, BBH06] are highlighted. The parameter r denotes the number for rounds in the interactive decapsulation protocol.

We now prove that the modified transform still constructs an IND-CCA secure TKEM. The proof adaptation is slightly non-trivial since we move from a TPKE to a TKEM construction using H_{fo} and derandomise the TIBE encryption mechanism using G_{fo}. Hence, we need to redo the proof taking into account the H_{fo} and G_{fo} random oracles.

Theorem 20 (Adapted [BBH06, Theorem 2])**.** *Let* TIBE *be a threshold identity-based encryption scheme satisfying Definition 17. Let* $\mathcal{M}$ *and* $\mathcal{R}$ *denote the message space and randomness space of the* TIBE*, respectively. Let* SIG *be a one-time signature satisfying* $\varepsilon_{\mathsf{SIG}}$-sEU-CMA *and* $G_{\mathsf{fo}} : \{0,1\}^* \to \mathcal{R}$ *and* $H_{\mathsf{fo}} : \{0,1\}^* \to \mathcal{K}$ *for some set* $\mathcal{K}$ *modelled as random oracles. Then* TKEM *as described in Figure 6.2 is a CCA-secure TKEM with:*

$$\mathsf{Adv}^{\mathsf{CCA2}}_{\mathcal{A},\mathsf{TKEM}}(\lambda) \leq \varepsilon_{\mathsf{SIG}} + (Q_{G_{\mathsf{fo}}} + Q_{H_{\mathsf{fo}}}) \cdot \mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}IND}}_{\mathcal{A},\mathsf{TIBE}} + \frac{(Q_{G_{\mathsf{fo}}} + Q_{H_{\mathsf{fo}}})}{|\mathcal{M}|}.$$

Proof. First, we modify the CCA experiment in Figure 2.2c to add an abort condition. We denote the new adversary's advantage as ε_1. Here the encryption randomness $\mathsf{rand}^* = G_{\mathsf{fo}}(\mathsf{msg}^*)$ in the challenge ciphertext is sampled as $\mathsf{rand}^* \leftarrow \mathcal{R}$ instead. These distributions are exactly the same as long as msg^* has not been queried to the G_{fo} random oracle. Given ct^* the probability of querying msg^* to the G_{fo} oracle is equivalent to breaking sel-ID-OW security of the underlying TIBE. Hence, the advantage is bounded as

$$\mathsf{Adv}^{\mathsf{CCA2}}_{\mathcal{A},\mathsf{TKEM}}(\lambda) \leq \varepsilon_1 + Q_{G_{\mathsf{fo}}} \cdot \mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}OW}}_{\mathcal{A},\mathsf{TIBE}}.$$

We claim that $\varepsilon_1 \leq \varepsilon_{\mathsf{SIG}} + Q_{H_{\mathsf{fo}}} \cdot \mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}OW}}_{\mathcal{A},\mathsf{TIBE}}$. Using $\mathcal{A}$ we build an adversary $\mathcal{B}$ against TIBE. For the random oracles they are simulated by $\mathcal{B}$ that samples the answers at random and maintains a list of queried inputs. In the experiment $\mathcal{B}$ first samples $(\mathsf{sk}^*, \mathsf{vk}^*) \leftarrow \mathsf{SIG.Keygen}(1^\lambda)$ and sets vk^* as the challenge identity. Then $\mathcal{B}$ relays the messages between $\mathcal{A}$ and the Challenger on the corrupt parties and their keys with no changes. For every decryption query $(\mathsf{ct}, \mathsf{vk}, \mathsf{sig}), \mathsf{act}$ of $\mathcal{A}$ if $\mathsf{SIG.Verify}(\mathsf{vk}, \mathsf{ct}, \mathsf{sig}) = 0$ then $\mathcal{B}$ replies with $\perp$ to $\mathcal{A}$. Moreover, if a query has $\mathsf{vk} = \mathsf{vk}^*$, $\mathcal{B}$ aborts the experiment. Otherwise, $\mathcal{B}$ forwards $(\mathsf{ct}, \mathsf{id} = \mathsf{vk}, \mathsf{act})$ to the Challenger and returns the reply to $\mathcal{A}$.

When $\mathcal{A}$ is ready to receive the challenge $\mathcal{B}$ gets $\mathsf{ct}^* = \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{msg}^*)$ corresponding to some $\mathsf{msg}^* \leftarrow \mathcal{M}$ from the Challenger. Then $\mathcal{B}$ computes $\mathsf{ct}^*_{\mathsf{TKEM}} = (\mathsf{ct}^*, \mathsf{vk}^*, \mathsf{SIG.Sign}(\mathsf{sk}^*, \mathsf{ct}^*))$ and $K^* \leftarrow \mathcal{K}$ and sends $(\mathsf{ct}^*_{\mathsf{TKEM}}, K^*)$ to $\mathcal{A}$. The decryption queries after the challenge are simulated in the same way as before. Lastly, $\mathcal{A}$ returns a bit b then $\mathcal{B}$ checks the list of queries in H_{fo} and forwards msg^* to the Challenger if $(\mathsf{msg}^*, \mathsf{ct}^*)$ was queried.

We analyse the success probability of $\mathcal{B}$. The decryption queries are simulated perfectly unless $\mathcal{B}$ issues an abort. The first type of abort happens when $\mathcal{A}$ makes a decryption

query $(\mathsf{ct}, \mathsf{vk}, \mathsf{sig})$ for $\mathsf{vk} = \mathsf{vk}^*$ either before or after the challenge. It implies that sig is a valid signature for vk. In both cases this constitutes a one-time signature forgery. Hence, the probability of this type of abort is bounded by $\varepsilon_{\mathsf{SIG}}$.

If the abort condition does not occur the challenge value $\mathsf{ct}^*_{\mathsf{TKEM}}$ is simulated perfectly. For the value K^* the view of the Adversary is independent of whether $K^* = H_{\mathsf{fo}}(\mathsf{msg}^*||\mathsf{ct}^*)$ or $K^* \leftarrow \mathcal{K}$ as long as H_{fo} is not queried on $\mathsf{msg}^*, \mathsf{ct}^*$. Hence, winning the is equivalent to querying the pair $\mathsf{msg}^*, \mathsf{ct}^*$ and that happens with probability $Q_{H_{\mathsf{fo}}} \cdot \mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}OW}}_{\mathcal{A},\mathsf{TIBE}}$. Then

$$\varepsilon_1 \leq Q_{H_{\mathsf{fo}}} \cdot \mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}OW}}_{\mathcal{A},\mathsf{TIBE}} + \Pr(\mathbf{abort}) \leq Q_{H_{\mathsf{fo}}} \cdot \mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}OW}}_{\mathcal{A},\mathsf{TIBE}} + \varepsilon_{\mathsf{SIG}}$$

Lastly, we use a folklore fact that IND-CPA security implies OW-CPA by sampling the pair of challenge messages at random (see e.g. [Pas24, Tutorial 8]). This technique applies to TIBE directly hence $\mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}OW}}_{\mathcal{A},\mathsf{TIBE}} \leq \mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}IND}}_{\mathcal{A},\mathsf{TIBE}} + 1/|\mathcal{M}|$. Adding all advantage changes together we get the statement. □

It remains to prove that the additional integrity checks result in a construction with decapsulation consistency formally defined in Definition 21. To do that we require the underlying TIBE to be γ-spread in the sense of Definition 24 and use the properties of the random oracle G_{fo}.

Lemma 101 (Decapsulation Consistency)**.** *Let $G_{\mathsf{fo}} : \{0,1\}^* \to \{0,1\}^\lambda$ be modelled as a random oracle. Let* TKEM *be the scheme obtained by applying Figure 6.2. If the underlying* TIBE *is γ-spread, then for any PPT adversary $\mathcal{A}$:*

$$\mathsf{Adv}^{\mathsf{CD}}_{\mathcal{A},\mathsf{TKEM}} \leq \frac{Q_{G_{\mathsf{fo}}} \cdot (Q_{G_{\mathsf{fo}}} - 1)}{2} \cdot 2^{-\gamma}.$$

where $Q_{G_{\mathsf{fo}}}$ is the number of queries $\mathcal{A}$ makes to the G_{fo} random oracle.

Proof. If the Adversary wins the game in Figure 2.4a with a tuple $((\mathsf{ct}, \mathsf{vk}, \mathsf{sig}), \mathsf{act}, \mathsf{act}', S, S')$. Then the decapsulation returns two valid keys $K_0 \neq K_1$. If the messages reconstructed as $\mathsf{msg}_1 \coloneqq \mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{vk}, \mathsf{act}', S')$, $\mathsf{msg}_0 \coloneqq \mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{vk}, \mathsf{act}, S)$ are the same, by Figure 6.2 the corresponding keys are the same as well. Since $K_i = H_{\mathsf{fo}}(\mathsf{msg}_i||\mathsf{ct})$, this implies that there exists a pair of valid messages $\mathsf{msg}_0 \neq \mathsf{msg}_1$ for which the following holds:

$$\begin{aligned}
\mathsf{msg}_1 &\coloneqq \mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{vk}, \mathsf{act}', S') \\
\mathsf{msg}_0 &\coloneqq \mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{vk}, \mathsf{act}, S) \\
\mathsf{ct} &\coloneqq \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{vk}, \mathsf{msg}_0; G_{\mathsf{fo}}(\mathsf{msg}_0)) \\
\mathsf{ct} &\coloneqq \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{vk}, \mathsf{msg}_1; G_{\mathsf{fo}}(\mathsf{msg}_1)).
\end{aligned}$$

In particular it implies that $\mathcal{A}^{G_{\mathsf{fo}}}$ can compute $(\mathsf{ct}, \mathsf{vk}, \mathsf{msg}_0, \mathsf{msg}_1)$ such that

$$\begin{aligned} \mathsf{ct} &= \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{vk}, \mathsf{msg}_0; G_{\mathsf{fo}}(\mathsf{msg}_0)) \\ &= \mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{vk}, \mathsf{msg}_1; G_{\mathsf{fo}}(\mathsf{msg}_1)). \end{aligned} \tag{6.1}$$

We simplify the winning condition of the Adversary to only require $(\mathsf{ct}, \mathsf{vk}, \mathsf{msg}_0, \mathsf{msg}_1)$ that satisfies Equation (6.1), this can only increase their advantage. For a given tuple $(\mathsf{ct}, \mathsf{vk}, \mathsf{msg}_0, \mathsf{msg}_1)$, we consider two possible cases:

1. If either of the messages (e.g. msg_0) has not been queried to the G_{fo} oracle the winning probability is bounded as

$$\Pr(\mathsf{TIBE.Encrypt}(\mathsf{ek}, \mathsf{vk}, \mathsf{msg}_0; G_{\mathsf{fo}}(\mathsf{msg}_0)) = \mathsf{ct} \mid G_{\mathsf{fo}}(\mathsf{msg}_0) \leftarrow \{0,1\}^{\lambda}) \leq 2^{-\gamma}.$$

2. Otherwise, if both messages were queried before the probability that at least two of them have colliding ciphertexts satisfying Equation (6.1) is bounded by

$$\Pr(\exists i \neq j \in [Q_{G_{\mathsf{fo}}}] \text{ s.t. } \mathsf{ct}_i = \mathsf{ct}_j) \leq \sum_{i \in [Q_{G_{\mathsf{fo}}}]} i \cdot 2^{-\gamma} = \frac{Q_{G_{\mathsf{fo}}} \cdot (Q_{G_{\mathsf{fo}}} - 1)}{2} \cdot 2^{-\gamma}.$$

The Adversary's advantage is bounded by the maximum of both values. □

Remark 23. In Chapter 2 we define a stronger and a weaker decapsulation consistency property in Figure 2.4b and Figure 2.4a respectively. Here we consider the weaker notion where the Adversary only wins if they can decapsulate the ciphertext to two different **valid** keys. This is because our construction does not have a share verification mechanism, i.e. identifiable aborts. Hence, the adversary is always able to make the decryption fail by providing an invalid decryption share. We upgrade our construction to have stronger decapsulation consistency in Chapter 7.

6.2.2 Our TIBE construction

The next step is to instantiate the BCHK+ transform with a TIBE. For the threshold techniques we use the recent progress in lattice threshold signatures. A common pattern of these schemes is to replace Gaussian sampling by the threshold-friendlier *noise flooding*, and to quantify security using the recent Hint-MLWE assumption by [KLSS23]. For example, the Plover signature [EEN+24] and its threshold variant Pelican [ENP24] are noise-flooded variants of the trapdoor sampling-based Eagle [YJW23]. Given the correspondence between hash-then-sign and IBE schemes, a natural idea is to apply

noise flooding to an existing IBE in order to make it *threshold-friendly*, then apply thresholdisation techniques from e.g. [DKM+24, KRT24] to obtain a proper TIBE. Unfortunately, this is not necessarily secure.

Given an IBE, denote $\mathbf{A}$ the master public key, id an identity, and $\mathsf{sk}_{\mathsf{id}}$ a valid decryption key for id. IBEs such as the ones in [GPV08, DLP14] satisfy $\mathbf{A} \cdot \mathsf{sk}_{\mathsf{id}} = H(\mathsf{id})$. Thus an adversary could query two decryption keys $\mathsf{sk}^1_{\mathsf{id}}, \mathsf{sk}^2_{\mathsf{id}}$ for the same identity and compute a short $\mathbf{s} = \mathsf{sk}^1_{\mathsf{id}} - \mathsf{sk}^2_{\mathsf{id}}$ that satisfies $\mathbf{A} \cdot \mathbf{s} = \mathbf{0}$. In a classic construction this attack is avoided by de-randomising the preimage sampling of $\mathsf{sk}_{\mathsf{id}}$. However, in the threshold setting when $\mathsf{sk}_{\mathsf{id}}$ is sampled jointly we cannot rely on the corrupt parties always generating the same randomness unless we are willing to store the transcripts of every completed session.

Instead, we opt for the Agrawal-Boneh-Boyen IBE [ABB10a]. In this IBE, a valid identity key satisfies $[\mathbf{A} \mid \mathbf{B}(\mathsf{id})] \cdot \mathsf{sk}_{\mathsf{id}}$, where $\mathbf{B}(\mathsf{id})$ is a matrix that depends on id. While the same attack as before applies, it only gives a trapdoor for $[\mathbf{A} \mid \mathbf{B}(\mathsf{id})]$, which does not lead to a trapdoor for $\mathbf{A}$, and important properties such as selective ID security are preserved. Thus, we select this IBE and convert it into a *threshold-friendly* IBE via noise flooding. We then convert this threshold-friendly IBE into a proper TIBE using the thresholdisation techniques described in [DKM+24, KRT24], in particular the $\mathsf{TRaccoon}^{\mathsf{sel}}_{3-\mathsf{rnd}}$ scheme from [KRT24].

Public parameters. Let $H_{\mathsf{cmt}}(\cdot) : \mathcal{R}_q \to \{0,1\}^{2\lambda}$ and $H_{\mathsf{mask}}(\cdot) : \{0,1\}^* \to \mathcal{R}_q^2$ be hash functions modelled as random oracles. The public parameters include $(\mathbf{A}_0, \mathbf{A}_1, \mathbf{A}_2, \mathbf{G}, r, N, T)$, where N is the overall number of parties in the protocol and T is the threshold on the number of parties required for decryption. We explain how each matrix is used in more detail.

1. $\mathbf{A}_0$ is a matrix with trapdoor *à la* Eagle [YJW23] or Plover [EEN+24], multiplied by an invertible element of $\mathcal{R}_q$. We need d_0 since $\mathbf{A}_0$ is a part of the MLWE sample that masks messages and has to look uniformly random. The parameter $\beta \in \mathbb{Z}$ can be set arbitrarily and impacts the quality of the trapdoor.

$$\mathbf{A}_0 = \begin{bmatrix} 1 & a_0 & b_0 \end{bmatrix} \cdot d_0, \quad \text{where} \quad d_0, a_0 \leftarrow \mathcal{R}_q^\times \times \mathcal{R}_q, \quad s_a, e_a \leftarrow \mathcal{D}_{\mathcal{R},\sigma_a} \tag{6.2}$$
$$\text{and } b_0 = (a_0 \cdot s_a + e_a) - \beta \text{ for some } \beta \in \mathbb{Z}$$

Then the trapdoor with tag d_0 is $\mathbf{A}_0 \cdot \begin{bmatrix} e_a & s_a & 1 \\ 1 & 0 & 0 \end{bmatrix}^\intercal = d_0 \cdot \begin{bmatrix} \beta & 1 \end{bmatrix} \bmod q$.

2. $\mathbf{A}_1 \leftarrow \mathcal{R}_q^3$ is a uniformly random matrix. When constructing an identity-specific matrix we add a shift to $\mathbf{A}_1$ of the form $\mathbf{A}_1 + \mathsf{E}(\mathsf{id}) \cdot \mathbf{G}$ for E and $\mathbf{G}$ discussed below. This is to guarantee the selective ID security of the TIBE.

3. $\mathbf{A}_2$ is uniformly random with the first element equal to 1. This is because we embed a $\mathsf{ModNTRU}$ trapdoor in $\mathbf{A}_2$ in the proof. This additional matrix block with a $\mathsf{ModNTRU}$ trapdoor helps us incorporate the Plover preimage sampling into the ABB IBE security proof via Coset-Hint-MLWE. We further multiply $\mathbf{A}_2$ by a unit tag d_2 since $\mathbf{A}_2$ is also used as part of the MLWE sample that masks messages.

$$\mathbf{A}_2 = \begin{bmatrix} 1 & a_2 & b_2 \end{bmatrix} \cdot d_2, \quad \text{where} \quad d_2, a_2, b_2 \leftarrow \mathcal{R}_q^\times \times \mathcal{R}_q^2 \tag{6.3}$$

4. $\mathbf{G}$ is a gadget matrix as introduced by Micciancio and Peikert [MP12].

$$\mathbf{G} = \begin{bmatrix} 1 & g & g^2 \end{bmatrix}, \quad \text{where} \quad g = \left\lceil q^{1/3} \right\rceil \tag{6.4}$$

5. $r \leftarrow \mathcal{R}_q$ is used as a target for preimage sampling.

Identity Embedding. The set of identities of our TIBE scheme coincides with the set of signature verification keys of the corresponding TKEM we denote S_{vk}[4]. Let $\mathsf{E} : S_{\mathsf{vk}} \to \mathcal{R}_q$ denote an embedding map from the set of identities S_{vk} of the TIBE into ring elements. For the security proof to work this map has to satisfy $\forall \mathsf{vk}_0 \neq \mathsf{vk}_1 : \mathsf{E}(\mathsf{vk}_0) - \mathsf{E}(\mathsf{vk}_1) \in \mathcal{R}_q^\times$. Hence, we choose the ring $\mathcal{R}$ and modulus q such that $\mathcal{R}_q \sim \mathbb{F}_{d/2} \times \mathbb{F}_{d/2}$ has exponentially many unit elements. We denote the said isomorphism as $f : \mathcal{R}_q \to \mathbb{F}_{d/2} \times \mathbb{F}_{d/2}$.

Hence, we require that $|S_{\mathsf{vk}}| < q^{d/2}$, then we pick an arbitrary efficient embedding E_F from S_{vk} to $\mathbb{F}_{d/2} \setminus \{0\}$ e.g using binary representations of the polynomial coefficients. For $\mathsf{vk} \in S_{\mathsf{vk}}$ we define $\mathsf{E}(\mathsf{vk}) \coloneqq f^{-1}(E_F(\mathsf{vk}), E_F(\mathsf{vk}))$. Then $\forall \mathsf{vk}_0 \neq \mathsf{vk}_1 : E_F(\mathsf{vk}_0) - E_F(\mathsf{vk}_1) \neq 0$ implying that $\mathsf{E}(\mathsf{vk}_0) - \mathsf{E}(\mathsf{vk}_1)$ is a unit in $\mathcal{R}_q$.

Encryption procedure. In the encryption procedure, the encrypter constructs a public matrix $[\mathbf{F}_{\mathsf{id}} \mid r]$ for the identity id. The matrix is defined as $\mathbf{F}_{\mathsf{id}} \coloneqq [\mathbf{A}_0 \mid \mathbf{A}_1 - \mathsf{E}(\mathsf{id}) \cdot \mathbf{G} \mid \mathbf{A}_2]$. Then they generate a Dual Regev ciphertext $(\mathbf{u}, v)$ for the public key $[\mathbf{F}_{\mathsf{id}} \mid r]$ and the message $\mathsf{msg} \in \{0,1\}^d$. This is described in Algorithm 17.

[4]When instantiated with WOTS^+ the verification key is a binary string of fixed length.

Decryption procedure. In the decryption procedure, the parties jointly compute a vector $\bar{\mathbf{z}} = (\mathbf{z}, \mathbf{x}_0, \mathbf{x}_1)$ such that $\mathbf{F}_{\mathsf{id}} \cdot \bar{\mathbf{z}}^T = r \bmod q$. They then use $\bar{\mathbf{z}}$ to perform a Dual Regev style decryption procedure, by computing $v - \bar{\mathbf{z}} \cdot \mathbf{u}$ and decoding it to recover the message. This is described in Figure 6.4.

Threshold mechanism. To compute $\bar{\mathbf{z}}$ in the threshold way the master secret (s_a, e_a) is secret-shared using Shamir secret sharing. Then the parties execute a three round protocol similar to $\mathsf{TRaccoon}^{\mathsf{sel}}_{3-\mathsf{rnd}}$ scheme from [KRT24]. This can also be seen as a threshold version of [MP12] with noise flooding used instead of the convolution. We give an intuition of each round of the protocol for a given set of participants $\mathsf{act} \subset [N]$

1. In the first round the parties in act sample ephemeral noise $\mathbf{p}_i$ and the Gaussian vectors $\mathbf{x}_{i,0}, \mathbf{x}_{i,1}$ that will be multiplied with non-trapdoored parts of $\mathbf{F}_{\mathsf{id}}$. This changes the target image by the corresponding w_i. To avoid ROS-type attacks in this round they only broadcast a commitment H_{cmt} to their w_i.

2. In the second round they reveal the underlying value w_i.

3. In the third round they use the secret key shares to compute a preimage of $r - \sum_{i \in \mathsf{act}} w_i$. Taking into account the tag d_0 of $\mathbf{A}_0$ this becomes $d_0^{-1} \cdot r - \sum w_i$. Then they add back the noise $\mathbf{p}_i$ to "flood" the secret dependent information. However, this noise is insufficient for covering the relatively large Lagrange coefficients multiplied with the secret shares. Hence, they also add one-time masks $\mathbf{m}_i$ computed by hashing the protocol transcript so far and long-term shared seeds. By construction $\sum_{i \in \mathsf{act}} \mathbf{m}_i = 0$ so the masks cancel out after the reconstructing the final preimage.

The TIBE is formally described in Figures 6.3 and 6.4. For clarity of reading we do not specify the changes in the state of each party i, we assume that it contains (i) the secret key share of i, (ii) all contributions of all parties up to the current round, (iii) the values privately generated by i during previous rounds.

6.3 Threshold-CCA Security Proof

We use the matrix $\mathbf{R}$ sampled as below to introduce a Micciancio-Peikert trapdoor [MP12] into $[\mathbf{A}_0 \mid \mathbf{A}_1]$ as in Lemma 50. Hence, the quality of preimages sampled using $\mathbf{R}$ depends on its norm we upper bound below.

Algorithm 4 $\mathsf{TIBE.Setup}(1^\lambda) \to \mathsf{ek}, \{\mathsf{dk}_i\}_{i\in[N]}$

$d_0, a_0 \leftarrow \mathcal{R}_q^\times \times \mathcal{R}_q$
$(s_a, e_a) \leftarrow \mathcal{D}_{\mathcal{R}^2,\sigma_a}$
$\forall i, j \in [N], i \neq j : \mathsf{seed}_{i\to j} \leftarrow \{0,1\}^\lambda$
$\mathbf{A}_0 \coloneqq \begin{bmatrix} 1 & a_0 & b_0 \end{bmatrix} \cdot d_0$, where $b_0 = (a_0 \cdot s_a + e_a) - \beta$
$\mathbf{A}_1 \leftarrow \mathcal{R}_q^3$
$\mathbf{A}_2 \coloneqq \begin{bmatrix} 1 & a_2 & b_2 \end{bmatrix} \cdot d_2$, where $d_2, a_2, b_2 \leftarrow \mathcal{R}_q^\times \times \mathcal{R}_q^2$
$\mathbf{G} \coloneqq \begin{bmatrix} 1 & g & g^2 \end{bmatrix}$ where $g \approx q^{1/3}$
$r \leftarrow \mathcal{R}_q$
$\mathsf{ek} \coloneqq (\mathbf{A}_0, \mathbf{A}_1, \mathbf{A}_2, \mathbf{G}, r)$
$\forall i \in [N] : \mathsf{dk}_i \coloneqq (\llbracket s_a \rrbracket_i, \llbracket e_a \rrbracket_i, \{\mathsf{seed}_{i\to j}, \mathsf{seed}_{j\to i}\}_{j\in[N]\setminus\{i\}})$ ▷ $(T,[N])-$ Shamir sharing of s_a, e_a
return $\mathsf{ek}, \{\mathsf{dk}_i\}_{i\in[N]}$

Algorithm 5 $\mathsf{Encode}(\mathsf{msg} = (b_0, \ldots, b_{d-1})) \to M \in \mathcal{R}_q$

return $M = \sum_{i\in[d]} b_i \cdot \lfloor q/2 \rceil \cdot X^i$

Algorithm 6 $\mathsf{Decode}(M = \sum_{i\in[d]} \alpha_i \cdot X^i) \to \mathsf{msg} \in \{0,1\}^d/\perp$

return $\mathsf{msg} = \sum_{i\in[d]} \lfloor \alpha_i \rceil_q X^i$

Algorithm 7 $\mathsf{TIBE.Encrypt}(\mathsf{msg} \in \{0,1\}^d, \mathsf{id} = \mathsf{vk}, \mathsf{ek}) \to \mathsf{ct}$

$\mathbf{F}_{\mathsf{vk}} \coloneqq \begin{bmatrix} \mathbf{A}_0 & | & \mathbf{A}_1 - \mathsf{E}(\mathsf{vk}) \cdot \mathbf{G} & | & \mathbf{A}_2 \end{bmatrix}$ ▷ $\mathbf{F}_{\mathsf{vk}} \in \mathcal{R}_q^{1\times 9}$
$s \leftarrow D_{R,\sigma}$
$\mathbf{e} \leftarrow D_{R^3,\sigma} \times D_{R^3,\sigma'} \times D_{R^3,\sigma}$ ▷ $\mathbf{e} \in \mathcal{R}_q^{1\times 9}$
$e' \leftarrow D_{R,\sigma}$
$\mathbf{u} \coloneqq \mathbf{F}_{\mathsf{vk}}^T \cdot s + \mathbf{e}$ ▷ $\mathbf{u} \in \mathcal{R}_q^{1\times 9}$
$v \coloneqq r \cdot s + e' + \mathsf{Encode}(\mathsf{msg})$
return $\mathsf{ct} \coloneqq (\mathbf{u}, v)$

Figure 6.3: TIBE algorithms for key generation, encryption and message encoding.

Lemma 102 (Norm bound on $\mathbf{R}$.)**.** *Let* $\mathcal{R} = \mathbb{Z}[X]/(X^d+1)$*. Let*

$$\mathbf{R} = \begin{bmatrix} e_1 & e_2 & e_3 \\ s_1 & s_2 & s_3 \\ 0 & 0 & 0 \end{bmatrix} \text{ where } e_i, s_i \leftarrow \mathcal{D}_{\mathcal{R},\sigma_{\mathsf{RLWE}}}$$

Then:

$$\Pr(\mathfrak{s}_{\max}(\mathbf{R}) \geq \sqrt{6} \cdot d \cdot \sigma_{\mathsf{RLWE}}) \leq 6 \cdot 2^{-12d}.$$

Algorithm 8 $\mathsf{TIBE.ShareExtract}_0(\mathsf{ct}, \mathsf{vk}, \mathsf{act}, \mathsf{state}_i) \to \mathsf{contrib}_{i,0}/\perp$

$(\mathbf{x}_{i,0}, \mathbf{x}_{i,1}, \mathbf{p}_i) \leftarrow (D_{R^3,\sigma_p})^3$
$y_{i,0} = a_0 \cdot p_{i,0} + p_{i,1} \bmod q$
$y_{i,1} = (\mathbf{A}_1 - \mathsf{E}(\mathsf{vk}) \cdot \mathbf{G}) \cdot \mathbf{x}_{i,0} \bmod q$
$y_{i,2} = \mathbf{A}_2 \cdot \mathbf{x}_{i,1} \bmod q$
$w_i = y_{i,0} + y_{i,1} + y_{i,2}$
$\mathsf{cmt}_i = H_{\mathsf{cmt}}(w_i)$
return $\mathsf{contrib}_{i,0} \coloneqq \mathsf{cmt}_i$

Algorithm 9 $\mathsf{TIBE.ShareExtract}_1(\mathsf{ct}, \mathsf{vk}, \mathsf{act}, \mathsf{state}_i) \to \mathsf{contrib}_{i,1}/\perp$

return $\mathsf{contrib}_{i,1} \coloneqq w_i$

Algorithm 10 $\mathsf{TIBE.ShareExtract}_2(\mathsf{ct}, \mathsf{vk}, \mathsf{act}, \mathsf{state}_i) \to \mathsf{contrib}_{i,2}/\perp$

assert $\forall j \in \mathsf{act} \setminus \{i\} : H_{\mathsf{cmt}}(w_j) = \mathsf{cmt}_j$
$\mathsf{ctnt} = \mathsf{act}||\mathsf{ct}||\{\mathsf{cmt}_j, w_j\}_{j \in \mathsf{act}}$
$\mathbf{m}_i \coloneqq \sum_{j \in \mathsf{act}} H_{\mathsf{mask}}(\mathsf{seed}_{i \to j}, \mathsf{ctnt}) - \sum_{j \in \mathsf{act}} H_{\mathsf{mask}}(\mathsf{seed}_{j \to i}, \mathsf{ctnt})$
$w \coloneqq \sum_{j \in \mathsf{act}} w_j$
$(c_0, c_1) \coloneqq \mathsf{Decomp}_\beta(d_0^{-1} \cdot r - w)$
$z_{i,0} = p_{i,0} + c_0 \cdot \lambda_{i,\mathsf{act}} \cdot [\![e_a]\!]_i + m_{i,0}$
$z_{i,1} = p_{i,1} + c_0 \cdot \lambda_{i,\mathsf{act}} \cdot [\![s_a]\!]_i + m_{i,1}$
$z_{i,2} = p_{i,2}$
return $\mathsf{contrib}_{i,2} \coloneqq (\mathbf{z}_i, \mathbf{x}_{i,0}, \mathbf{x}_{i,1})$

Algorithm 11 $\mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{vk}, \mathsf{act}, \{(\mathbf{z}_i, \mathbf{x}_i)\}_{i \in \mathsf{act}}) \to \mathsf{msg}/\perp$

$z_0 = \sum_{i \in \mathsf{act}} z_{i,0} + c_1$
$z_1 = \sum_{i \in \mathsf{act}} z_{i,1}$
$z_2 = \sum_{i \in \mathsf{act}} z_{i,2} + c_0$
$\mathbf{x}_0 = \sum_{i \in \mathsf{act}} \mathbf{x}_{i,0}$
$\mathbf{x}_1 = \sum_{i \in \mathsf{act}} \mathbf{x}_{i,1}$
$\mathbf{F}_{\mathsf{vk}} \coloneqq \begin{bmatrix} \mathbf{A}_0 & | & \mathbf{A}_1 - \mathsf{E}(\mathsf{vk}) \cdot \mathbf{G} & | & \mathbf{A}_2 \end{bmatrix}$
assert $\mathbf{F}_{\mathsf{vk}} \cdot \begin{bmatrix} \mathbf{z} & \mathbf{x}_0 & \mathbf{x}_1 \end{bmatrix}^T = r \wedge \left\| \begin{bmatrix} \mathbf{z} & \mathbf{x}_0 & \mathbf{x}_1 \end{bmatrix} \right\|_\infty < B$
return $\mathsf{msg} \coloneqq \mathsf{Decode}(v - \begin{bmatrix} \mathbf{z} & \mathbf{x}_0 & \mathbf{x}_1 \end{bmatrix}^T \cdot \mathbf{u})$

Figure 6.4: TIBE algorithms for threshold decryption.

Proof. The spectral norm of $\mathbf{R}$ does not change if we remove the last all-zero row of the

matrix. Denote $\mathbf{R}' = \begin{bmatrix} e_1 & e_2 & e_3 \\ s_1 & s_2 & s_3 \end{bmatrix}$. Then:

$$\mathfrak{s}_{\max}(\mathbf{R}') \leq \|\mathrm{Rot}(\mathbf{R}')\|_F = \sqrt{d\left(\sum_{i=1}^{3}\|e_i\|^2 + \sum_{i=1}^{3}\|s_i\|^2\right)} \leq \sqrt{d \cdot \sigma_{\mathsf{RLWE}}^2 \cdot 6d} = \sqrt{6} \cdot d\,\sigma_{\mathsf{RLWE}}$$

The last inequality holds by Lemma 23 with probability at least $1 - 6 \cdot 2^{-12d}$. □

We prove the Selective-ID Security of the TIBE scheme in Section 7.1. Then the CCA Security of the TKEM follows by Theorem 20.

Theorem 21. *Let* TIBE *be the scheme defined in Figure 6.3 and Figure 6.4. Suppose that the space of identities is bounded as* $|S_{\mathsf{id}}| = |S_{\mathsf{vk}}| < q^{d/2}$. *Let* $\mathcal{R} = \mathbb{Z}[X]/(X^d + 1)$ *be a ring of degree* d, q *be a prime number s.t.* $\mathcal{R}_q \sim \mathbb{F}_{d/2} \times \mathbb{F}_{d/2}$. *Let* $\sigma, \sigma_{\mathsf{RLWE}}, \sigma_a, \beta > 0$, $\varepsilon \in (0, 1/2)$, $\delta > 0$ *s.t.* $2^{2\delta} \geq 2^{-d+1} + 6q^2$, $\sigma > \eta_\varepsilon(\mathcal{R}^3)$. *Suppose that:*

$$\sigma' > 2\sqrt{6} \cdot d \cdot \sigma_{\mathsf{RLWE}} \cdot \sigma \tag{6.5}$$

$$\sigma_p > 2d \cdot q^{1/3+1/3d} \tag{6.6}$$

$$\sigma_p > 1.17q^{1/3} \cdot \eta_\varepsilon(\mathcal{R}^3) \tag{6.7}$$

$$\sigma_p > q^{1/3} \cdot \sqrt{6} \cdot d \cdot \sigma_{\mathsf{RLWE}} \cdot \eta_\varepsilon(\mathcal{R}^6) \tag{6.8}$$

Then for any PPT adversary $\mathcal{A}$:

$$\begin{aligned}
\mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}IND}}_{\mathcal{A},\mathsf{TIBE}}(\lambda) \leq\ & 3 \cdot \mathsf{Adv}^{\mathsf{RLWE}_{\mathcal{R},1,q,\sigma_{\mathsf{RLWE}}}}_{\mathcal{A}_1}(\lambda) + 2\mathsf{Adv}^{\mathsf{ModNTRU}_{\mathcal{R},q,2}}_{\mathcal{A}_2}(\lambda) \\
& + \mathsf{Adv}^{\mathsf{Hint\text{-}RLWE}^{Q_{Dec},\sigma_p,\beta}_{\mathcal{R},1,q,\sigma_a}}_{\mathcal{A}_3}(\lambda) + \mathsf{Adv}^{\mathsf{RLWE}_{\mathcal{R},7,q,\sigma}}_{\mathcal{A}_4}(\lambda) \\
& + 6\varepsilon + Q_{Dec}\,O(\varepsilon) + 9 \cdot 2^{-d+\delta} \cdot Q_{Dec} + \frac{(Q_{Dec} + Q_{H_{\mathsf{cmt}}})^2}{2^{2\lambda}} \\
& + \frac{Q_{Dec}(Q_{Dec} + Q_{H_{\mathsf{cmt}}})}{2^{-d+\delta+2}} + 2 \cdot q^{-d/2} + \frac{Q_{H_{\mathsf{mask}}}}{2^\lambda} + 6 \cdot 2^{-12d}.
\end{aligned}$$

where adversaries $\mathcal{A}_4$ *to* $\mathcal{A}_1$ *run in approximately the same time as* $\mathcal{A}$.

We give an overview of the proof hybrid by hybrid.

Hyb_0 Identical to the real experiment.

$\mathsf{Hyb}_{1,2}$ In this transition, the challenger embeds a ModNTRU trapdoor into the matrix $\mathbf{A}_2$. They also change the distribution of $\mathbf{A}_1$ from uniformly random to $\mathbf{A}_1 = \mathbf{A}_0 \cdot \mathbf{R} + \mathsf{E}(\mathsf{vk}^*) \cdot \mathbf{G}$ based on the challenge identity vk^*. This introduces a gadget trapdoor for the matrix $[\mathbf{A}_0 \mid \mathbf{A}_1 - \mathsf{E}(\mathsf{vk}) \cdot \mathbf{G}]$ for every $\mathsf{vk} \neq \mathsf{vk}^*$ using the properties of the embedding function. This transition relies on $\mathsf{RLWE}_{\mathcal{R},1,q,\sigma_{\mathsf{RLWE}}}$.

Hyb_3 We pick an arbitrary honest party h which we call the "last acting" honest party. In this hybrid, the challenger delays sampling of $(\mathbf{p}_h, \mathbf{x}_{h,0}, \mathbf{x}_{h,1})$ for h until the end of $\mathsf{TIBE.ShareExtract}_1$. This is done by programming the random oracle $H_{\mathsf{cmt}}(w_h)$. The programming is possible since w_h has enough entropy, which is guaranteed by Equation (6.6) and Lemma 56. This transition is statistical.

$\mathsf{Hyb}_{4,5}$ The following hybrids change the distribution of the masks $\mathbf{m}_i$ to perfectly uniform for all honest parties except h. This transition relies on Lemma 103, that we borrow from [KRT24, Theorem 4.1]. The lemma requires all commitments to have a unique opening which we enforce in Hyb_4. We prove that the statistical distance between these hybrids is negligible.

Hyb_6 In this hybrid we change the simulation of honest party contributions in $\mathsf{TIBE.ShareExtract}_2$. The shares of regular honest parties are replaced with uniform values and the mask are computed accordingly. This relies on the distribution of the masks from Hyb_5. For the last acting honest party we rephrase its contributions as a function of the main secrets s_a, e_a removing the Shamir sharing. This change is syntactical and does not change the Adversary's advantage.

$\mathsf{Hyb}_{7\text{-}10}$ In $\mathsf{Hyb}_7 - \mathsf{Hyb}_{10}$, we remove the trapdoor $b = a_0 s_a + e_a$ from $\mathbf{A}_0$ using the Coset-Hint-MLWE assumption.

- In Hyb_7, we prepare the values dependent on (s_a, e_a) to follow the Coset-Hint-MLWE distributions exactly. To do that we change the order of sampling (c_0, c_1) and $(\mathbf{p}_h, \mathbf{x}_{h,0}, \mathbf{x}_{h,1})$ and sample the latter vector in two parts.
- In Hyb_8, we apply Coset-Hint-MLWE assumption and sample $b \leftarrow \mathcal{R}_q$. It is easy to see that the distributions in Hyb_7 and Hyb_8 can be simulated efficiently given a Coset-Hint-MLWE instance. Using Equation (6.7), and Theorem 19 we bound Coset-Hint-MLWE advantage by the Hint-MLWE advantage and negligible factors.
- In Hyb_9 we change the sampling of $(\mathbf{p}_h, \mathbf{x}_{h,0}, \mathbf{x}_{h,1})$ again to efficiently simulate the preimages only given the Gadget trapdoor $\mathbf{R}$ for every $\mathsf{vk} \neq \mathsf{vk}^*$. In this hybrid and the next one, we require $|S_{\mathsf{vk}}| < q^{d/2}$ as enforced in the conditions.
- In Hyb_{10} we remove the trapdoor from $\mathbf{A}_2$ that we no longer require for efficient simulation.

Note that now corresponding elements in matrices $\mathbf{A}_0, \mathbf{A}_2$ are perfectly random.

Hyb_{11} In this hybrid, thanks to $\sigma > \eta_\varepsilon(\mathcal{R}^3)$ and Equation (6.5), we can use Lemma 52 to change the noise distribution of the challenge ciphertext and embed an LWE sample with respect to $\mathbf{A}_0, \mathbf{A}_2$ and r. The norm of the new noise follows from Lemma 102. This change is statistical.

Hyb_{12} In the last hybrid we apply the RLWE assumption to the challenge ciphertext. Since now the challenge plaintext is masked with a uniformly random value the Adversary's advantage in this experiment is 0.

We conclude the argument with an upper bound on the $\mathcal{A}$'s advantage in the initial experiment by tracing the change in the advantage after every transition.

Proof. We consider a sequence of hybrids $(\mathsf{Hyb}_i)_i$. For each i, we note ε_i the advantage of an adversary against Hyb_i.

Hyb_0 **(sel-ID-IND game):** the original sel-ID-IND experiment (Figure 2.2b) against TIBE (Figures 6.3 and 6.4).

Hyb_1 **(Trapdoor for $\mathbf{A}_2$):** In the first hybrid we introduce the Module NTRU trapdoor for $\mathbf{A}_2$ as in Definition 7. All other values can be sampled efficiently using the trapdoor in $\mathbf{A}_0$. By definition:

$$\varepsilon - \varepsilon_1 \leq \mathsf{Adv}(\mathsf{Hyb}_0, \mathsf{Hyb}_1) \leq \mathsf{Adv}(\mathbf{A}_2, \begin{bmatrix} 1 & \mathcal{U}(\mathcal{R}_q^{1\times 2}) \end{bmatrix}) \leq \mathsf{Adv}^{\mathsf{ModNTRU}_{\mathcal{R}_q,2,\mathcal{D}}}_{\mathcal{A}_2}(\lambda).$$

Hyb_2 **(Rerandomisation of $\mathbf{A}_1$):** At the start of the sel-ID-IND experiment, the Challenger receives the challenge identity vk^* from the Adversary. We then change the $\mathsf{TIBE.Setup}(1^\lambda)$ algorithm as follows. Sample

$$\mathbf{R} \leftarrow \begin{bmatrix} e_1 & e_2 & e_3 \\ s_1 & s_2 & s_3 \\ 0 & 0 & 0 \end{bmatrix} \text{ where } (e_i, s_i) \leftarrow \mathcal{D}_{\mathcal{R}^2,\sigma_{\mathsf{RLWE}}}.$$

If $\mathfrak{s}_{\max}(\mathbf{R}) > \sqrt{6} \cdot d \cdot \sigma_{\mathsf{RLWE}}$, abort the experiment with a uniformly random answer $\{0,1\} \leftarrow \{1,0\}$. By Lemma 102, this event occurs with probability at most $6 \cdot 2^{-12d}$. Otherwise, set $\mathbf{A}_1 = \mathbf{A}_0 \cdot \mathbf{R} + \mathsf{E}(\mathsf{vk}^*) \cdot \mathbf{G}$. Now

$$\mathbf{A}_0 \cdot \mathbf{R} = d_0 \cdot \begin{bmatrix} a_0 \cdot s_1 + e_1, & a_0 \cdot s_2 + e_2, & a_0 \cdot s_3 + e_3 \end{bmatrix}$$

contains 3 RLWE samples instead of uniformly random elements. The preimage queries are still answered using the trapdoor (s_a, e_a) of $\mathbf{A}_0$. Therefore, the only difference between

Hyb_1 and Hyb_2 is the sampling of $\mathbf{A}_1$ and the abort condition. Then the adversary's advantage can only increase by

$$\varepsilon_1 - \varepsilon_2 \leq 6 \cdot 2^{-12d} + 3 \cdot \mathsf{Adv}_{\mathcal{A}_1}^{\mathsf{RLWE}_{\mathcal{R},1,q,\sigma_{\mathsf{RLWE}}}}(\lambda).$$

Hyb_3 **(Programming the commitment):** For each decryption query of the adversary, the set of active participants contains at least one honest party: $\mathsf{act} \cap \mathsf{honest} \neq \varnothing$. We designate $h \in \mathsf{act} \cap \mathsf{honest}$ as the "last acting" honest party. In this hybrid, in $\mathsf{TIBE.ShareExtract}_0$ we set $\mathsf{cmt}_h \leftarrow \{0,1\}^{2\lambda}$. We also delay the sampling of $(\mathbf{p}_h, \mathbf{x}_{h,0}, \mathbf{x}_{h,1}) \leftarrow \mathcal{D}_{\mathcal{R}^9,\sigma_p}$ to the end of $\mathsf{TIBE.ShareExtract}_1$ when other parties have already revealed their contributions $\{w_i\}_{i \in \mathsf{cor}}$, and we program the random oracle to $H_{\mathsf{cmt}}(\mathbf{F}_{\mathsf{vk}} \cdot (\mathbf{p}_h || \mathbf{x}_{h,0} || \mathbf{x}_{h,1})) := \mathsf{cmt}_h$.

This change is detected only if w_h has been queried to $H_{\mathsf{cmt}}(\cdot)$ before, we prove that the Adversary can only guess the value of w_h with negligible probability. By Lemma 56, with probability at least $1 - 2^{-d+\delta}$ it holds that

$$\sigma_p > 2d \cdot q^{1/3+1/3d} \geq \eta_{2^{-d+\delta}}(\Lambda_q^{\perp}(\begin{bmatrix} 1 & b_2 & d_2 \end{bmatrix})) \geq \eta_{2^{-d+\delta}}(\Lambda_q^{\perp}(\mathbf{F}_{\mathsf{vk}})).$$

Recall in Hyb_2 the commitment is generated honestly as $w_h \leftarrow \mathbf{F}_{\mathsf{vk}} \cdot \mathcal{D}_{\mathcal{R}^9,\sigma_p}$. By Theorem 3 and Lemma 56, its distribution is statistically close (for a term at most $2 \cdot 2^{-d+\delta}$) to the uniform distribution over $\mathcal{R}_q$. Then, the probability of sampling any fixed value of w_h is bounded by $q^{-d} + 2 \cdot 2^{-d+\delta} + 2^{-d+\delta} \leq 2^{-d+\delta+2}$ where the additional $2^{-d+\delta}$ comes from the smoothing condition on the matrix $\mathbf{F}_{\mathsf{vk}}$ described above. Hence, the probability that one of the Q_{Dec} honestly generated w_h collides with previous queries to $H_{\mathsf{cmt}}(\cdot)$ is at most:

$$\varepsilon_2 - \varepsilon_3 \leq 2^{-d+\delta+2} \cdot Q_{\mathsf{Dec}} \cdot (Q_{\mathsf{Dec}} + Q_{H_{\mathsf{cmt}}}).$$

Hyb_4 **(Unique commitment opening):** In this hybrid, the Challenger aborts if there is a collision in the output of the random oracle H_{cmt}. The number of queries to H_{cmt} is bounded by $Q_{\mathsf{Dec}} + Q_{H_{\mathsf{cmt}}}$, so the probability of the collision is bounded by $\frac{(Q_{\mathsf{Dec}}+Q_{H_{\mathsf{cmt}}})^2}{2^{2\lambda}}$. Then:

$$\varepsilon_3 - \varepsilon_4 \leq \frac{(Q_{\mathsf{Dec}} + Q_{H_{\mathsf{cmt}}})^2}{2^{2\lambda}}.$$

Hyb_5 **(Uniform masking):** We recall in Lemma 103 a fact proved within [KRT24, Theorem 4.1] in the transition from Game 3 to Game 8 in the authors' numbering. The proof idea of [KRT24] is to carefully trace the value of ctnt in the honest executions of the protocol and conclude that the adversary can distinguish the distributions only if it queries H_{mask} using a correct seed of one of the honest parties.

Lemma 103. *With the notations of Figure 6.4, let* $H_{\mathsf{mask}} : \{0,1\}^* \to \mathcal{R}_q^2$ *be modelled as a random oracle and* $\mathsf{seed}_{i,j} \leftarrow \{0,1\}^\lambda$ *be uniformly distributed. For every decryption query of the adversary, let* ctnt *be uniquely defined by* $\mathsf{ctnt}_w \coloneqq \mathsf{act}||\mathsf{ct}||\{\mathsf{cmt}_j\}_{j\in\mathsf{act}}$*. The adversary's advantage in distinguishing* $\mathcal{D}_m$ *and* $\mathcal{D}'_m$ *is at most* $Q_{H_{\mathsf{mask}}}/2^\lambda$*, where:*

$$\mathcal{D}_m = \left\{ \mathbf{m}_i = \sum_{j\in\mathsf{act}} H_{\mathsf{mask}}(\mathsf{seed}_{i\to j}, \mathsf{ctnt}) - \sum_{j\in\mathsf{act}} H_{\mathsf{mask}}(\mathsf{seed}_{j\to i}, \mathsf{ctnt}), i \in \mathsf{act} \right\}$$

$$\mathcal{D}'_m = \left\{ \begin{array}{l} \mathbf{m}_i = \sum_{j\in\mathsf{act}} H_{\mathsf{mask}}(\mathsf{seed}_{i\to j}, \mathsf{ctnt}) - \sum_{j\in\mathsf{act}} H_{\mathsf{mask}}(\mathsf{seed}_{j\to i}, \mathsf{ctnt}), i \in \mathsf{cor}, \\ \mathbf{m}_i \leftarrow \mathcal{R}_q^2, \quad i \in \mathsf{honest} \setminus \{h\} \\ \mathbf{m}_h = -\sum_{i\in\mathsf{act}\setminus\{h\}} \mathbf{m}_i \end{array} \right\}$$

The abort condition in Hyb_4 ensures that cmt_i corresponds to only one value of w_i. Hence, ctnt_w uniquely defines ctnt and the conditions of Lemma 103 are satisfied. As in $\mathcal{D}'_m$ we set $\mathbf{m}_j \leftarrow \mathcal{R}_q^2$ for $j \in \mathsf{honest} \setminus \{h\}$. For the last honest signer h we set for consistency $\mathbf{m}_h = -\sum_{j\in\mathsf{act}\setminus\{h\}} \mathbf{m}_j$. This jump is indistinguishable based on Lemma 103. The preimage value is not affected by the change, so we can still simulate all signing rounds using the $\mathbf{A}_0$ trapdoor. Therefore,

$$\varepsilon_4 - \varepsilon_5 \leq \frac{Q_{H_{\mathsf{mask}}}}{2^\lambda}.$$

Hyb_6 **(Secret sharing):** In this hybrid we change the simulation of partial decryption contributions. First, when the Adversary chooses the set cor of $T-1$ corrupt parties, the reduction samples $\mathsf{dk}_i = (\llbracket s_a \rrbracket_i, \llbracket e_a \rrbracket_i) \leftarrow \mathcal{R}_q^2$ for $i \in \mathsf{cor}$ and sends the keys dk_i and honestly sampled seeds $\{\mathsf{seed}_{i\to j}\}_{j\in[N]}$ to $i \in \mathsf{cor}$. This does not change the output distributions by the properties of Shamir secret sharing.

For all $j \in (\mathsf{honest} \cap \mathsf{act}) \setminus \{h\}$ we change the order of sampling $\mathbf{m}_j$ and $\mathbf{z}_j$. We sample $\mathbf{z}_j \leftarrow \mathcal{R}_q^2$ and then define:

$$\begin{bmatrix} m_{j,0} \\ m_{j,1} \end{bmatrix} = \begin{bmatrix} z_{j,0} \\ z_{j,1} \end{bmatrix} - \begin{bmatrix} p_{j,0} \\ p_{j,1} \end{bmatrix} - c_0 \begin{bmatrix} \lambda_{j,\mathsf{act}} \cdot \llbracket s_a \rrbracket_j \\ \lambda_{j\mathsf{act}} \cdot \llbracket e_a \rrbracket_j \end{bmatrix}$$

This change does not affect the distributions, since $\mathbf{m}_j$ and $\mathbf{z}_j$ are in bijective correspon-

dence. For the last honest signer h we write:

$$\begin{aligned}
z_{h,0} &= c_0 \cdot \lambda_{h,\mathsf{act}} \cdot [\![s_a]\!]_h + p_{h,0} + m_{h,0} \\
&= c_0(s_a - \sum_{j \in \mathsf{act} \setminus \{h\}} \lambda_{j,\mathsf{act}} \cdot [\![s_a]\!]_j) + p_{h,0} - \sum_{j \in \mathsf{act} \setminus \{h\}} m_{j,0} \qquad (6.9) \\
&= c_0(s_a - \sum_{j \in \mathsf{act} \setminus \{h\}} \lambda_{j,\mathsf{act}} \cdot [\![s_a]\!]_j) + p_{h,0} - \sum_{j \in \mathsf{cor} \cap \mathsf{act}} m_{j,0} \qquad (6.10) \\
&- \sum_{j \in (\mathsf{honest} \cap \mathsf{act}) \setminus \{h\}} (z_{j,0} - c_0 \cdot \lambda_{j,\mathsf{act}} \cdot [\![s_a]\!]_j - p_{j,0}) \\
&= (c_0 \cdot s_a + p_{h,0}) - \sum_{j \in \mathsf{cor} \cap \mathsf{act}} \lambda_{j,\mathsf{act}} \cdot [\![s_a]\!]_j - \sum_{j \in \mathsf{cor} \cap \mathsf{act}} m_{j,0} \\
&- \sum_{j \in (\mathsf{honest} \cap \mathsf{act}) \setminus \{h\}} (z_{j,0} - p_{j,0}).
\end{aligned}$$

Equation (6.9) uses the Shamir secret sharing definition and the transformation of $\mathbf{m}_h$ from Hyb_5. Equation (6.10) follows from changing the order of sampling $\mathbf{m}_j$ and $\mathbf{z}_j$ for $j \in (\mathsf{honest} \cap \mathsf{act}) \setminus \{h\}$ stated above. In the end, to simulate $z_{h,0}$ we only need $c_0 s_a + p_{h,0}$ and other known values. Similarly, for $z_{h,1}$ we only need $c_0 e_a + p_{h,1}$ and known values. The reconstructed preimage looks like

$$\begin{aligned}
z_0 &= \underbrace{p_{h,0} + c_0 s_a + c_1}_{\tilde{z}_0} + \sum_{j \in \mathsf{act} \setminus \{h\}} p_{j,0} \\
z_1 &= \underbrace{p_{h,1} + c_0 e_a}_{\tilde{z}_1} + \sum_{j \in \mathsf{act} \setminus \{h\}} p_{j,1} \\
z_2 &= \underbrace{p_{h,2} + c_0}_{\tilde{z}_2} + \sum_{j \in \mathsf{act} \setminus \{h\}} p_{j,2} \\
\mathbf{x} &= \mathbf{x}_h + \sum_{j \in \mathsf{act} \setminus \{h\}} \mathbf{x}_j
\end{aligned}$$

and by construction $\mathbf{A}_0 \cdot \tilde{\mathbf{z}} = c_0 \beta + c_1 \bmod q$. Hence,

$$\begin{aligned}
w_h &= d_0^{-1} \cdot r - \sum_{i \in \mathsf{act} \setminus \{h\}} w_i - \mathbf{A}_0 \cdot \tilde{\mathbf{z}} \\
&= d_0^{-1} \cdot r - \sum_{i \in \mathsf{act} \setminus \{h\}} w_i - c_0 \beta - c_1.
\end{aligned}$$

So w_h can be simulated using public values as well. These changes in the distributions are only syntactical. However, they allow us to simulate the protocol execution only using the LWE sample $a_0 s_a + e_a$ for the matrix $\mathbf{A}_0$ and Q_{Dec} hints of the form $(c_0, c_1), (c_0 s_a + p_{h,0}, c_0 e_a + p_{h,1})$. For convenience, we define an offset target vector $r_1 := d_0^{-1} \cdot r - \sum_{i \in \mathsf{act} \setminus \{h\}} w_i$. Since the described changes do not affect any distributions, the advantage of the adversary does not change $\varepsilon_6 = \varepsilon_5$.

Hyb_7 **(Coset-Hint-LWE preparation):** At this point, public parameters are sampled as:

$$\begin{cases} a_0 \leftarrow \mathcal{R}_q, (s_a, e_a) \leftarrow \mathcal{D}_{\mathcal{R}^2,\sigma_a}, b = a_0 \cdot s_a + e_a \\ \mathbf{A}_0 \coloneqq \begin{bmatrix} 1 & a_0 & \beta - b \end{bmatrix}, \mathbf{A}_1 \coloneqq \mathbf{A}_0 \cdot \mathbf{R} + \mathsf{E}(\mathsf{vk}^*) \cdot \mathbf{G} \\ (\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}) \leftarrow \mathsf{TrapGenNTRU}(\lambda), \mathbf{T}^T = \begin{bmatrix} e_a & s_a & 1 \\ 1 & 0 & 0 \end{bmatrix} \end{cases}$$

Denote $\mathbf{A}'_1 = \mathbf{A}_1 - \mathsf{E}(\mathsf{vk}) \cdot \mathbf{G}$. Then $\mathbf{F}_{\mathsf{vk}} = [\mathbf{A}_0 \mid \mathbf{A}'_1 \mid \mathbf{A}_2]$, and the response to a decryption query with respect to $\mathsf{id} = \mathsf{vk}$ looks as follows.

$\mathbf{p}_h \leftarrow \mathcal{D}_{\mathcal{R}^3,\sigma_p}, \mathbf{x}_{h,1} \leftarrow \mathcal{D}_{\mathcal{R}^3,\sigma_p}, \mathbf{x}_{h,2} \leftarrow \mathcal{D}_{\mathcal{R}^3,\sigma_p}$

$w_h = \mathbf{A}_0 \cdot \mathbf{p}_h + \mathbf{A}'_1 \cdot \mathbf{x}_{h,1} + \mathbf{A}_2 \cdot \mathbf{x}_{h,2}$

$(c_0, c_1) \coloneqq \mathsf{Decomp}_\beta(r_1 - w_h)$

$\tilde{\mathbf{z}} = \mathbf{T} \cdot \mathbf{c} + \mathbf{p}_h$

return $(\tilde{\mathbf{z}}, \mathbf{x}_h)$

In Hyb_7, for each decryption query we do a statistical jump to the distribution where we sample (c_0, c_1) as independent variables, and adjust the distribution of $(\mathbf{p}_h, \mathbf{x}_h)$ to satisfy the image constraint.

$u \leftarrow \mathcal{R}_q, (c_0, c_1) \coloneqq \mathsf{Decomp}_\beta(u)$

$\mathbf{x}_{h,1} \leftarrow \mathcal{D}_{\mathcal{R}^3,\sigma_p}, (\mathbf{p}_h, \mathbf{x}_{h,2}) \leftarrow \mathcal{D}_{\Lambda_q^{r_1-u-\mathbf{A}'_1\cdot\mathbf{x}_{h,1}}([\mathbf{A}_0|\mathbf{A}_2]),\sigma_p}$

$\tilde{\mathbf{z}} = \mathbf{T} \cdot \mathbf{c} + \mathbf{p}_h$

return $(\tilde{\mathbf{z}}, \mathbf{x}_h)$

By construction $w_h = r_1 - u$, hence the value $\mathbf{c}$ satisfies the original formula $\beta c_0 + c_1 = r_1 - w = u$. By Lemma 56, with probability $\geq 1 - 2^{-d+\delta}$ it holds that:

$$\sigma_p > 2d \cdot q^{1/3+1/3d} \geq \eta_{2^{-d+\delta}}(\Lambda_q^\perp(\begin{bmatrix} 1 & b_2 & d_2 \end{bmatrix})) \geq \eta_{2^{-d+\delta}}(\Lambda_q^\perp(\mathbf{F}_{\mathsf{vk}})).$$

Then in Hyb_6 by Theorem 3 $u' = \mathbf{F}_{\mathsf{vk}} \cdot \begin{bmatrix} \mathbf{p}_h & \mathbf{x}_h \end{bmatrix}^T = w_h \bmod q$ is distributed within $3 \cdot 2^{-d+\delta}$ of uniform. Therefore, $u = d_0^{-1} \cdot r - u' \bmod q$ is statistically close to $u \leftarrow \mathcal{R}_q$ in Hyb_7.

The distribution of $\mathbf{p}_h, \mathbf{x}_h$ conditioned on u is exactly equal to $\mathcal{D}_{\Lambda_q^{r_1-u}(\mathbf{F}_{\mathsf{vk}}),\sigma_p}$. We sample it by parts using Lemma 35 and the smoothing condition. We get $\mathrm{SD}(D_1, D_2) \leq 2 \cdot 2^{-d+\delta}$,

where:

$$\begin{cases} D_1 = \left\{ \mathbf{x}_{h,1} \leftarrow \mathcal{D}_{\mathcal{R}^3,\sigma_p}, (\mathbf{p}_h, \mathbf{x}_{h,2}) \leftarrow \mathcal{D}_{\Lambda_q^{r_1-u-\mathbf{A}_1' \cdot \mathbf{x}_{h,1}}([\mathbf{A}_0|\mathbf{A}_2]),\sigma_p} \right\} \\ D_2 = \left\{ (\mathbf{p}_h, \mathbf{x}_{h,1}, \mathbf{x}_{h,2}) \leftarrow \mathcal{D}_{\Lambda_q^{r_1-u}(\mathbf{F}_{\mathsf{vk}}),\sigma_p} \right\} \end{cases}$$

Hence, overall:

$$\varepsilon_6 - \varepsilon_7 \leq 5 \cdot 2^{-d+\delta} \cdot Q_{\mathsf{Dec}}.$$

Hyb_8 **(Hint-LWE jump):** In Hyb_7 the distribution of $(\tilde{\mathbf{z}}_h, \mathbf{x}_{h,2})$ with target vector $r_2 \coloneqq r_1 - \mathbf{A}_1' \cdot \mathbf{x}_{h,1}$ is equal to $\mathcal{D}_{\mathsf{real}}$ of Coset-Hint-MLWE, therefore in Hyb_8 we transition to the distribution that contains $\mathcal{D}_{\mathsf{ideal}}$ as:

$a_0 \leftarrow \mathcal{R}_q$, $b \leftarrow \mathcal{R}_q$, $(s_a, e_a) \leftarrow \mathcal{D}_{\mathcal{R}^2,\sigma_a}$

$\mathbf{A}_0 \coloneqq \begin{bmatrix} 1 & a_0 & \beta - b \end{bmatrix}$, $\mathbf{A}_1 \coloneqq \mathbf{A}_0 \cdot \mathbf{R} + \mathsf{E}(\mathsf{vk}^*) \cdot \mathbf{G}$

$(\mathbf{A}_2, \mathbf{T}_{\mathbf{A}_2}) \leftarrow \mathsf{TrapGenNTRU}(\lambda)$, $\mathbf{T}^T = \begin{bmatrix} e_a & s_a & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$u \leftarrow \mathcal{R}_q, (c_0, c_1) \coloneqq \mathsf{Decomp}_\beta(u), \alpha \coloneqq \mathbf{A}_0\mathbf{T}\mathbf{c}$

$\mathbf{x}_{h,1} \leftarrow \mathcal{D}_{\mathcal{R}^3,\sigma_p}, (\mathbf{p}_h, \mathbf{x}_{h,2}) \leftarrow \mathcal{D}_{\Lambda_q^{r_2-\alpha}([\mathbf{A}_0|\mathbf{A}_2]),\sigma_p}$

$\tilde{\mathbf{z}} = \mathbf{T} \cdot \mathbf{c} + \mathbf{p}_h$

return $(\tilde{\mathbf{z}}, \mathbf{x}_h)$

The advantage of the adversary changes according to Theorem 19:

$$\varepsilon_7 - \varepsilon_8 \leq \mathsf{Adv}_{\mathcal{A}_3}^{\mathsf{Hint\text{-}RLWE}_{\mathcal{R},1,q,\sigma_a}^{Q_{\mathsf{Dec}},\sigma_p,\beta}}(\lambda) + Q_{\mathsf{Dec}} \cdot O(\varepsilon)$$

Hyb_9 **(Hint-LWE post-processing):** To get back to a distribution we can efficiently simulate we change the order of sampling again to:

$u \leftarrow \mathcal{R}_q, (c_0, c_1) \coloneqq \mathsf{Decomp}_\beta(u), \alpha \coloneqq \mathbf{A}_0\mathbf{T}\mathbf{c}$

$\mathbf{x}_{h,2} \leftarrow \mathcal{D}_{\mathcal{R}^3,\sigma_p}, (\mathbf{p}_h, \mathbf{x}_{h,1}) \leftarrow \mathcal{D}_{\Lambda_q^{r_1-\alpha-\mathbf{A}_2 \cdot \mathbf{x}_{h,2}}([\mathbf{A}_0|\mathbf{A}_1']),\sigma_p}$

$\tilde{\mathbf{z}} = \mathbf{T} \cdot \mathbf{c} + \mathbf{p}_h$

return $(\tilde{\mathbf{z}}, \mathbf{x}_h)$

This distribution can be simulated efficiently since for all $\mathsf{vk} \neq \mathsf{vk}^*$ we have $\mathsf{E}(\mathsf{vk}) - \mathsf{E}(\mathsf{vk}^*) \in \mathcal{R}_q^\times$, hence, by Lemma 50 and Equation (6.8) the matrix $\mathbf{R}$ is sufficient to sample preimages

of $[\mathbf{A}_0 \mid \mathbf{A}'_1] = \left[\mathbf{A}_0 \mid -\mathbf{A}_0 \cdot \mathbf{R} - (\mathsf{E}(\mathsf{vk}) - \mathsf{E}(\mathsf{vk}^*)) \cdot \mathbf{G}\right]$. The advantage changes by another

$$\varepsilon_8 - \varepsilon_9 \leq 3 \cdot 2^{-d+\delta} \cdot Q_{\mathsf{Dec}}.$$

Hyb_{10} **(Removing $\mathbf{A}_2$ Trapdoor):** As stated in the previous hybrid the distributions can now be simulated efficiently using $\mathbf{R}$. Hence, we remove the trapdoor related to $\mathbf{A}_2$ and sample the corresponding elements uniformly at random. The advantage of the adversary changes at most as

$$\varepsilon_9 - \varepsilon_{10} \leq \mathsf{Adv}^{\mathsf{ModNTRU}_{\mathcal{R}_q,2,\mathcal{D}}}_{\mathcal{A}_2}(\lambda).$$

Note that in this hybrid we have $\mathbf{F}_{\mathsf{vk}^*} = [\mathbf{A}_0 \mid -\mathbf{A}_0 \cdot \mathbf{R} \mid \mathbf{A}_2]$ and the challenge ciphertext is of the form

$$\mathbf{u}^* = \mathbf{F}^T_{\mathsf{vk}^*} \cdot s^* + \mathbf{e}^* = \begin{bmatrix} \mathbf{A}_0^T \cdot s^* + \mathbf{e}_0^* \\ -\mathbf{R}^T \mathbf{A}_0^T \cdot s^* + \mathbf{e}_1^* \\ \mathbf{A}_2^T \cdot s^* + \mathbf{e}_2^* \end{bmatrix} \tag{6.11}$$

$$v^* = r \cdot s^* + e_3^* + \mathbf{M}_b \cdot \left\lfloor \frac{q}{2} \right\rfloor \tag{6.12}$$

The distributions of errors are $(\mathbf{e}_0^*, \mathbf{e}_2^*, e_3^*) \leftarrow \mathcal{D}_{\mathcal{R}^7,\sigma}$, $\mathbf{e}_1^* \leftarrow \mathcal{D}_{\mathcal{R}^3,\sigma'}$.

Hyb_{11} **(Noise convolution):** The distribution of the challenge ciphertext in Hyb_{10} is statistically close to

$$\mathbf{u}^* = \mathbf{F}^T_{ID^*} \cdot s^* + \mathbf{e}^* = \begin{bmatrix} \mathbf{A}_0^T \cdot s^* + \mathbf{e}_1^* \\ \mathsf{ReRand}(-\mathbf{R}^T, \mathbf{A}_0^T \cdot s^* + \mathbf{e}_1^*, \sigma', \sigma) \\ \mathbf{A}_2^T \cdot s^* + \mathbf{e}_2^* \end{bmatrix}.$$

The advantage of the adversary can only increase by the negligible distance between distributions determined in Lemma 52 as $\varepsilon_{10} - \varepsilon_{11} \leq 6\varepsilon$, which applies since $\sigma > \eta_\varepsilon(\mathcal{R}^3)$, and σ' as constructed in Hyb_1 and constrained in Equation (6.5), satisfy the conditions of Lemma 52. The challenge ciphertext now includes RLWE samples $\mathbf{A}_0^T \cdot s^* + \mathbf{e}_0^*$, $\mathbf{A}_2^T \cdot s^* + \mathbf{e}_2^*$ and $r \cdot s^* + e_3^*$.

Hyb_{12} **(LWE for the ciphertext):** Finally, the challenge ciphertext is sampled as $\mathsf{ct}^* = (\mathbf{u}^*, v^*)$ where $\mathbf{u}^* = (\mathbf{b}_0, \mathsf{ReRand}(-\mathbf{R}^T, \mathbf{b}_0, \sigma', \sigma), \mathbf{b}_2)$ and $v^* \leftarrow b_3 + \mathsf{Encode}(\mathsf{msg}_b)$ where $(\mathbf{b}_0, \mathbf{b}_2, b_3) \leftarrow \mathcal{R}_q^7$. Since the message is masked by a uniformly random value, the advantage of the adversary is $\varepsilon_{12} = 0$.

If the adversary can distinguish Hybrids 11 and 12 they can break $\mathsf{RLWE}_{\mathcal{R},7,q,\sigma}$ conditioned on the first matrix element being a unit. The advantage of the adversary in LWE with a unit in the challenge matrix can decrease at most by

$$\Pr(a \notin \mathcal{R}_q^* \mid a \leftarrow \mathcal{R}_q) = \frac{2q^{d/2}-1}{q^d} \leq \frac{2}{q^{d/2}}$$

Hence, the advantage changes as $\varepsilon_{11} - \varepsilon_{12} \leq \mathsf{Adv}_{\mathcal{A}_4}^{\mathsf{RLWE}_{\mathcal{R},7,q,\sigma}}(\lambda) + 2 \cdot q^{-d/2}$. □

The main security theorem is a direct corollary of **??** and Theorems 20 and 21.

Theorem 22. *The TKEM defined in Figures 6.2 to 6.4 is CCA secure with*

$$\begin{aligned}\mathsf{Adv}_{\mathcal{A}}^{\mathsf{CCA2}}(\lambda) \leq 2^{-128} &+ 3 \cdot \mathsf{Adv}_{\mathcal{A}_1}^{\mathsf{RLWE}_{\mathcal{R},1,q,\sigma_{\mathsf{RLWE}}}}(\lambda) + 2\mathsf{Adv}_{\mathcal{A}_2}^{\mathsf{ModNTRU}_{\mathcal{R},q,2}}(\lambda) \\ &+ \mathsf{Adv}_{\mathcal{A}_3}^{\mathsf{Hint\text{-}RLWE}_{\mathcal{R},1,q,\sigma_a}^{Q_{Dec},\sigma_p,\beta}}(\lambda) + \mathsf{Adv}_{\mathcal{A}_4}^{\mathsf{RLWE}_{\mathcal{R},7,q,\sigma}}(\lambda) + 6\varepsilon \\ &+ Q_{Dec}\, O(\varepsilon) + 9 \cdot 2^{-d+\delta} \cdot Q_{Dec} + \frac{(Q_{Dec}+Q_{H_{\mathsf{cmt}}})^2}{2^{2\lambda}} \\ &+ \frac{Q_{Dec}(Q_{Dec}+Q_{H_{\mathsf{cmt}}})}{2^{-d+\delta+2}} + 2 \cdot q^{-d/2} + \frac{Q_{H_{\mathsf{mask}}}}{2^{\lambda}} + 6 \cdot 2^{-12d}\end{aligned}$$

where adversaries $\mathcal{A}_4$ to $\mathcal{A}_1$ run in approximately the same time as $\mathcal{A}$.

Remark 24. (On adaptive corruptions.) We believe the proof above can be modified to consider adaptive corruptions. The main reason is that simulating communications of honest parties is simple thanks to the one-time masks issued by a random oracle. When simulating key extraction shares for honest parties, we sample random $\mathbf{z}^{(i)}$, conditioned on the sum of all shares, namely $\mathbf{z}$, to fulfil $\begin{bmatrix}1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix} \cdot \mathbf{z} = r$. One can again use the notion of the "last acting" honest party to achieve this. This simulation does not reveal the masks $\mathbf{m}_i$. When an honest party later gets corrupted, one can program the random oracle that controls the masks $\mathbf{m}_i$ according to the now corrupted key share $\mathbf{s}^{(i)}$. Note that this is possible, as $\mathbf{m}_i$ changes with every key extraction query.

6.4 Decapsulation Consistency

In a TKEM, we expect that if a valid message was recovered from the decapsulation algorithm, it is equal to the message that was encapsulated in the given ciphertext. There are 4 possible configurations for proving this property: 1) Honest ciphertext, and honest shares; 2) Honest ciphertext, and malicious shares; 3) Malicious ciphertext, and honest shares; 4) Malicious ciphertext, and malicious shares. Note that since our scheme is does

not have identifiable aborts, in any of the malicious settings (Cases 2-4) the Adversary can force the decapsulation procedure to output $\perp$. When both the ciphertext and the shares are honest (Case 1), we recover the correctness property from Definition 20. We prove that the TIBE in Figures 6.3 and 6.4 fulfils this property with overwhelming probability.

Theorem 23 (Correctness)**.** *Let* $B = d\,\sigma_a\,\sqrt{d}\left\lceil\frac{q-1}{2\beta}\right\rceil + \beta/2 + \sigma_p\,\sqrt{T\,d}$*, and* $\sigma' = 2\sqrt{6}\cdot d\cdot \sigma_{\mathsf{RLWE}}\cdot\sigma$ *and* $\varepsilon \in (0, 1/2)$*. If:*

$$\sigma_p \geq \sqrt{1 + 1/(T-1)}\cdot\eta_\varepsilon(\mathcal{R}) \tag{6.13}$$

$$\sigma\sqrt{d} + d\sqrt{d}\,\left(\sigma\,B + (\sigma+\sigma')\,\sigma_p\sqrt{T\,d}\right) \leq q/4, \tag{6.14}$$

then the TIBE *in Figures 6.3 and 6.4 is correct except with probability* $O(d\,e^{-\pi\,d} + (T-1)\cdot 8\varepsilon)$*.*

Proof. For any message msg and honestly generated ciphertext ct:

$$\mathsf{ct} = (\mathbf{u} \coloneqq \mathbf{F}_{\mathsf{vk}}\cdot s + \mathbf{e}, v \coloneqq r\cdot s + e' + \mathsf{Encode}(\mathsf{msg}), \mathsf{vk}, \mathsf{sig}),$$

at the end of the honest decryption procedure (Figure 6.4), TIBE.Combine computes a vector $\mathbf{y} = (z_0, z_1, z_2, \mathbf{x}) \in R^9$ where:

$$z_0 = c_0\cdot s_a + c_1 + \sum_{j\in\mathsf{act}} p_{j,0}, \quad z_1 = c_0\cdot e_a + \sum_{j\in\mathsf{act}} p_{j,1}, \quad z_2 = c_0 + \sum_{j\in\mathsf{act}} p_{j,2}$$

$$\mathbf{x} = \sum_{j\in\mathsf{act}} \mathbf{x}_j.$$

Recall from Figures 6.3 and 6.4 that $(s_a, e_a) \sim \mathcal{D}_{\mathcal{R}^2,\sigma_a}$ and $(\mathbf{x}_{i,0}, \mathbf{x}_{i,1}, \mathbf{p}_i) \sim (D_{R^3,\sigma_p})^3$. It follows from (i) the triangle inequality and coefficient embedding properties, (ii) Lemma 25 and Lemma 34 for the sum for $p_{j,0}$ that:

$$\|z_0\|_\infty \leq d\cdot\|c_0\|_\infty\cdot\|s_a\|_\infty \;+\; \|c_1\|_\infty \;+\; \left\|\textstyle\sum_{j\in\mathsf{act}} p_{j,0}\right\|_\infty \tag{6.15}$$

$$\leq d\,\sigma_a\,\sqrt{d}\left\lceil\frac{q-1}{2\beta}\right\rceil \;+\; \beta/2 \;+\; \sigma_p\,\sqrt{T\,d} = B \tag{6.16}$$

except with probability $\leq 4\,d\,e^{-\pi\,d} + (T-1)\cdot 8\varepsilon$. The same tail bound holds for $\|z_1\|_\infty$ and $\|z_2\|_\infty$. Similarly, by Lemmas 25 and 34 we obtain $\|\mathbf{x}\|_\infty \leq \sigma_p\sqrt{T\,d}$ with the same probability. Then for the decryption, by applying the triangle inequality in Equation (6.17),

followed by Lemma 25 in Equation (6.18), we get:

$$\begin{aligned}
\left\| v - \mathbf{u}^T \cdot \mathbf{y} - \mathsf{Encode}(\mathsf{msg}) \right\|_\infty &= \left\| e' - \mathbf{e}^T \cdot \mathbf{y} \right\|_\infty \\
&\leq \|e'\|_\infty + \left\| \mathbf{e}^T \cdot \mathbf{y} \right\|_\infty && (6.17) \\
&\leq \|e'\|_\infty + \left\| \mathbf{e}_0^T \cdot \mathbf{z} \right\|_\infty + \left\| \mathbf{e}_1^T \cdot \mathbf{x}_0 \right\|_\infty + \left\| \mathbf{e}_2^T \cdot \mathbf{x}_1 \right\|_\infty \\
&\leq \sigma\sqrt{d} + d\sqrt{d}\,\left(\sigma\, B + (\sigma + \sigma')\,\sigma_p \sqrt{T\,d}\right) && (6.18) \\
&\leq q/4
\end{aligned}$$

except with probability $O(d\, e^{-\pi\, d} + (T-1)\cdot 8\varepsilon)$. Equation (6.14) allows us to conclude. □

To cover the setting of Case 2 we slightly modify Theorem 23. Indeed, the checks of $\mathsf{TKEM.Combine}$ imply that the shares reconstruct to $\mathbf{y}$ s.t. $\mathbf{F}_{\mathsf{vk}} \cdot \mathbf{y} = r \bmod q$ and $\|\mathbf{y}\|_\infty \leq B$. Hence, we have the same bound on $\|\mathbf{z}\|_\infty$, but $\mathbf{x}_i$ in $\mathbf{x} = \sum_i \mathbf{x}_i$ may not be independent. Then we cannot use Lemma 34 and the bound becomes $\sigma\sqrt{d} + d\sqrt{d}\,(2\sigma + \sigma')\,B \leq q/4$. Otherwise, Theorem 23 guarantees correct decryption with overwhelming probability.

The Cases 3 and 4 translate into the decapsulation consistency formally defined in Figure 2.4a. To prove it we lower bound the min-entropy of the honest ciphertext in Lemma 104. Then decapsulation consistency of the scheme follows from Lemma 101.

Lemma 104 (Honest Ciphertext Min-Entropy)**.** *Let* TIBE *be the scheme described in Figures 6.3 and 6.4. If* $\sigma, \sigma' \geq \eta_\varepsilon(\mathcal{R})$*, it is* γ*-spread with* $\gamma = \log(\sigma^{7d} \cdot \sigma'^{3d} \cdot (1-\varepsilon)^{10})$*.*

Proof. For a given $\mathsf{msg} \in \{0,1\}^d$, $\mathsf{vk} \in S_{\mathsf{vk}}$ and the public parameters ek, the ciphertext is of the form $(\mathbf{u} = \mathbf{F}_{\mathsf{vk}}^T \cdot s + \mathbf{e}, v = r \cdot s + e' + \mathsf{Encode}(\mathsf{msg}))$, where $s \leftarrow D_{R,\sigma}$, $\mathbf{e} \leftarrow D_{R^3,\sigma} \times D_{R^3,\sigma'} \times D_{R^3,\sigma}$, $e' \leftarrow D_{R,\sigma}$ are sampled at random. Let $p \coloneqq \Pr_{\mathbf{e},s,e'}((\mathbf{u}, v) = \mathsf{TIBE.Encrypt}(\mathsf{msg}, \mathsf{vk}, \mathsf{ek}))$. For any $(\mathbf{u}, v) \in \mathcal{R}_q^9 \times \mathcal{R}_q$ over the randomness of $\mathbf{e}, s, e'$ we have

$$\begin{aligned}
p &= \sum_{s \in \mathcal{R}} \Pr_{\mathbf{e},e'}(\mathbf{e} = \mathbf{u} - \mathbf{F}_{\mathsf{vk}}^T \cdot s, e' = v - r \cdot s - \mathsf{Encode}(\mathsf{msg})) \cdot \Pr(s \leftarrow D_{R,\sigma}) \\
&\leq \sum_{s \in \mathcal{R}} \Pr_{\mathbf{e},e'}(\mathbf{e} = \mathbf{0}, e' = 0) \cdot \Pr(s \leftarrow D_{R,\sigma}) = \Pr_{\mathbf{e},e'}(\mathbf{e} = \mathbf{0}, e' = 0) \\
&= \frac{1}{\rho_\sigma(\mathcal{R})^7 \cdot \rho_{\sigma'}(\mathcal{R})^3} \leq \frac{1}{\sigma^{7d} \cdot \sigma'^{3d} \cdot (1-\varepsilon)^{10}}
\end{aligned}$$

where the last inequality comes from $\det(\mathcal{R}) = 1$ and Corollary 6. □

6.5 Instantiation and Parameter Sets

Assuming the conditions of Theorem 21 are met, we recall that $\mathsf{Adv}^{\mathsf{CCA2}}_{\mathcal{A}}(\lambda)$ is upper-bounded as in Theorem 22. We now study each term independently. We denote a shorthand $a \asymp b$ for the Landau symbol $a = \Theta(b)$.

RLWE. The hardnesses of $\mathsf{RLWE}_{\mathcal{R},1,q,\sigma_{\mathsf{RLWE}}}$ and $\mathsf{RLWE}_{\mathcal{R},7,q,\sigma}$ can estimated in a straightforward way using standard tools such as the lattice estimator. We take $\sigma \asymp 1$ and $\sigma_{\mathsf{RLWE}} \asymp 1$, we can then later set $d = \Omega(\lambda \log q)$. Theorem 21 also requires $\sigma > \eta_\varepsilon(\mathcal{R}^3)$, but for concrete parameters $\eta_\varepsilon(\mathcal{R}^3)$ remains small.

Rerandomisation. Equation 6.5 in Theorem 21 requires $\sigma' \asymp d \cdot \sigma_{\mathsf{RLWE}} \cdot \sigma \asymp d$ in order to apply the rerandomisation lemma of [KY16].

Hint-MLWE. Next, we estimate the hardness of $\mathsf{Hint\text{-}MLWE}^{Q_{\mathsf{Dec}},\sigma_p,\beta}_{\mathcal{R},1,1,q,\sigma_a}$. Following Lemma 48, $\mathsf{Hint\text{-}MLWE}^{Q_{\mathsf{Dec}},\sigma_p,\beta}_{\mathcal{R},1,1,q,\sigma_a} \geq \mathsf{RLWE}_{\mathcal{R},1,q,\sigma_0}$, for $\frac{1}{\sigma_0^2} \sim 2\left(\frac{1}{\sigma_a^2} + \frac{B_{\mathsf{hint}}}{\sigma_p^2}\right)$ and $B_{\mathsf{hint}} \sim \frac{Q_{\mathsf{Dec}}\, d\, q^2}{12\,\beta^2}$. The hardness of RLWE requires $\sigma_0 = \Omega(1)$, therefore we set $\sigma_a \asymp 1$ and $\sigma_p \asymp (q/\beta) \cdot \sqrt{Q_{\mathsf{Dec}}\, d}$. Maximising the number of decryption queries $Q_{\mathsf{Dec}} = O\left(\frac{\beta^2\sigma_p^2}{d\,q^2}\right)$ requires us to take a very large β. There are two other conditions on σ_p, in Equations 6.6 and 6.7 of Theorem 21, but both are subsumed by the one above.

ModNTRU. The best known attack against $\mathsf{ModNTRU}_{\mathcal{R},q,m=2}$ is described in [CPS+20]. Due to the extremely large modulus q, we would like to check if overstretched NTRU attacks [ABD16, KF17, DvW21] apply. A straightforward application to the attacks would not work, since existing works focused on the NTRU case ($m = 1$). Assessing the applicability of this class of attacks to our case would require additional work, which we believe to interesting and relevant future work.

Table 6.2: Parameter sets.

λ	T	Q_{Dec}	q	d	β	σ_a	σ	σ_p	σ_{RLWE}	$\lvert\mathsf{ek}\rvert$	$\lvert\mathsf{ct}\rvert$
128	32	2^{40}	2^{100}	4096	2^{77}	8	4	2^{47}	4	50 KiB	450 KiB

Correctness. Finally, we study correctness via Theorem 23. Since the Hint-RLWE reduction forces β to be very large, it holds that $B \sim \beta/2$. For fixed d, the term σ'

is essentially fixed by Equation 6.5. Since σ is also fixed (see RLWE paragraph), we may simply increase σ_q and T as long as Equation 6.14 is satisfied. As a result, we get $\beta \asymp d^2 \sqrt{q\,T}\, Q_{\mathsf{Dec}}^{1/4}$, in which case Equation 6.14 simplifies to $q \asymp d^5 \sqrt{Q_{\mathsf{Dec}}\, T}$.

Putting everything together gives us the parameter sets in Table 6.2. Asymptotically, for the modulus we obtain the constraint $q \asymp d^5 \sqrt{Q_{\mathsf{Dec}}\, T}$.

Chapter 7

IND-CCA Threshold KEM under 30 KiB

The main contribution of this chapter is an IND-CCA lattice TKEM with significantly reduced sizes of the public key and the ciphertext compared to the construction in Chapter 6. The BCHK+ transform from Chapter 6 requires a threshold identity-based encryption scheme (TIBE) with suitable properties. For a threshold T and a number Q of decryption queries, the TIBE in Chapter 6 required sending 10 ring elements, for a ring $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^d + 1)$ with q scaling roughly in $\Theta(d^5 \sqrt{QT})$. In this chapter we design a TIBE that is much simpler and more efficient requiring only 4 ring elements with q scaling roughly in $\Theta(d^{3/2} \sqrt{QT})$. As a result, for a similar $T = 32$ and $Q = 2^{45}$, our public encapsulation key (resp. ciphertexts) is around 6.6 KiB (resp. 30 KiB), which is about 7 times (resp. 18 times) shorter than in Chapter 6.

Our second contribution is a TIBE construction which provides robustness for almost identical parameters, although it supports a smaller number of decapsulation queries $Q = 2^{25}$. This feature is underpinned by our use of Vandermonde secret sharing [BCdP+25], a type of secret sharing that guarantees short shares, as opposed to Shamir secret sharing used in Chapter 6. Using Vandermonde secret sharing allows identifying misbehaving parties via simple procedures (checking a linear identity and a norm bound) as opposed to using a non-interactive zero-knowledge proof system, etc.

A more efficient TIBE. In this chapter, we take a more efficient TIBE construction, the "ROHIBE" from [CHKP10, Section 5.3] as a starting point, and heavily optimise it. We start with a reminder on the ABB [ABB10a] and CHKP [CHKP10] (H)IBEs, which underpin the TIBE of both Chapters 6 and 7. In both ABB and CHKP IBEs the master

public key is a pseudorandom matrix $\mathbf{A}$ and a uniform vector t. A trapdoor for $\mathbf{A}$ allows, for any identity id, to compute a short preimage $\mathbf{z}$ such that $\mathbf{F}_{\mathsf{id}} \cdot \mathbf{z} \approx t$, where $\mathbf{F}_{\mathsf{id}}$ is a matrix related to $\mathbf{A}$ and the identity string id. Encryption computes a ciphertext:

$$\begin{bmatrix}\mathbf{u}^T & v\end{bmatrix} = r \cdot \begin{bmatrix}\mathbf{F}_{\mathsf{id}} & t\end{bmatrix} + \begin{bmatrix}\mathbf{e}^T & e'\end{bmatrix} + \begin{bmatrix}0 & \mathsf{Encode}(\mathsf{msg})\end{bmatrix}. \tag{7.1}$$

Finally, decryption computes $v - \mathbf{u}^T \cdot \mathbf{z} \approx \mathsf{Encode}(\mathsf{msg})$.

In [ABB10a], we have $\mathbf{F}_{\mathsf{id}} = [\mathbf{A}_0 \mid \mathbf{A}_1 + E(\mathsf{id}) \cdot \mathbf{B}]$, where $\mathbf{A}_0$ is a (trapdoored) pseudorandom matrix, and $\mathbf{A}_1$ is uniformly random and $\mathbf{B}$ is a gadget matrix. In Chapter 6, $\mathbf{F}_{\mathsf{id}} = [\mathbf{A}_0 \mid \mathbf{A}_1 + E(\mathsf{id}) \cdot \mathbf{B} \mid \mathbf{A}_2]$ is expanded to the right with an additional matrix $\mathbf{A}_2$, and each matrix $\mathbf{A}_i$ is in $\mathcal{R}_q^{1\times 3}$. $\mathbf{A}_0$ embeds the master trapdoor à la Eagle/Plover [YJW23, EEN+24] (essentially, a RLWE sample), used in the preimage sampling procedure. $\mathbf{A}_1 + E(\mathsf{id}) \cdot \mathbf{B}$ is used as the identity-dependent element for the selective-identity security of the TIBE. Lastly, $\mathbf{A}_2$ is used to embed a module-NTRU trapdoor in the security proof, which helps us integrate Plover-like preimages into the ABB framework. Hence, the ciphertext in Equation (7.1) has $3 \cdot 3 + 1 = 10$ ring elements. In this chapter we start with the ROHIBE framework that has the following advantages.

1. *Simpler embedding.* The main difference of the ROHIBE construction compared to the ABB IBE is that identities are embedded via $\mathbf{F}_{\mathsf{id}} = [\mathbf{A}_0 \mid H(\mathsf{id})]$, where the matrix $\mathbf{A}_0$ still embeds a Plover trapdoor and H is modelled as a random oracle. Since, $H(\mathsf{id})$ only needs one ring element to embed an NTRU trapdoor, we save two ring elements compared to the embedding $\mathbf{A}_1 + E(\mathsf{id}) \cdot \mathbf{B}$ from Chapter 6.

   In addition, in [ABB10a], the identity is embedded as a "tag" of the gadget matrix. The difference of any two tags has to be invertible for preimage sampling to work, so it requires picking a low-splitting ring that has exponentially many unit elements. In the new construction, this restriction is no longer required.

2. *Fewer matrix blocks.* Our second major design optimisation compared to Chapter 6 is that we do not have a matrix $\mathbf{A}_2$. The simpler embedding implies that $H(\mathsf{id})$ can fulfil the role of $\mathbf{A}_2$ and $\mathbf{A}_1 + E(\mathsf{id}) \cdot \mathbf{B}$ at the same time. In the previous construction the matrices $\mathbf{F}_{\mathsf{id}}$ and $\mathbf{F}_{\mathsf{id}'}$ for $\mathsf{id} \neq \mathsf{id}'$ were related by an additive shift so constructing a trapdoor that works for every shift was more complex and incompatible with Hint-LWE leakage. In the current construction each id has its independent matrix extension and its own NTRU trapdoor embedded in $H(\mathsf{id})$. This trapdoor can be used both for Hint-LWE and for the IBE security.

3. *Smaller modulus.* Our simpler design also allows reducing the modulus. Our parameter selection process (Section 7.5) allows taking $q = \Theta(d^{3/2}\sqrt{QT}\log\delta)$, where δ is an upper bound on the probability that decryption fails. Carrying out a similar analysis for the Chapter 6 TIBE would result in a much larger $q = \Theta(d^5\sqrt{QT})$. Thus, our construction allows a smaller q, which in turn allows decreasing d, compounding bandwidth gains.

4. *From module-NRTU to NTRU trapdoors.* As a result of the previous improvements, an NTRU trapdoor becomes tight enough for the reduction to go through. The security of NTRU trapdoors is better-understood than module-NTRU trapdoors, which is especially important in our setting since q is large and therefore overstretched NTRU attacks need to be considered [ABD16, KF17]. In our parameter regimes such attack does not apply (Section 7.5).

5. *Additional optimisations.* We perform a few more optimisations. Recall that Chapter 6 takes $\mathbf{A}_0 = \begin{bmatrix}1 & a_0 & b_0\end{bmatrix} \cdot d_0$, where the role of d_0 was to randomise the first entry of $\mathbf{A}_0$. Instead we simply set $d_0 = 1$ and drop the first coefficient of $\mathbf{A}_0$ when generating a ciphertext.

6. *Bit dropping.* While it is known that we can perform heavy bit-dropping on the v part of the ciphertext in Equation (7.1), we also notice that we can do the same on two coefficient of $\mathbf{u}$. The reason for that is more subtle: decryption computes $\mathbf{u}^T \cdot \mathbf{z}$, where $\mathbf{z} \in \mathcal{R}^4$ is heavily unbalanced, with the first two coefficients being very large, the third very small, and the last in-between. We adapt the bit-dropping of $\mathbf{u}$ accordingly to gain in ciphertext size without impacting correctness.

To summarise, we save six ring elements: two in Item 1, three in Item 2 and one in Item 5. Our modulus q and ring degree d are smaller as well (Item 3) due to the simpler construction, and bit-dropping (Item 6) provides further gains.

Obtaining share robustness. Let us recall the joint interactive key extraction of Chapter 6: in the first round, each party i computes their image offset w_i and sends a commitment $\mathsf{cmt}_i = H_{\mathsf{cmt}}(w_i)$ to the other parties. Upon receiving the commitments of the others, in the next round the parties reveal their value w_i. In the last round, each party computes their preimage contribution z_i. This response contains each party's Shamir key share scaled by the corresponding Lagrange coefficient. Directly giving out such scalings would be vulnerable to key-recovery attacks as discussed in [DKM+24].

To mitigate this, Chapter 6 copies the strategy of Threshold Raccoon [DKM+24] and adds blinding one-time masks m_i, that sum to zero during reconstruction. While simple from a proof perspective, such masks come with two main drawbacks. First, they require the parties to set up and store mutual secret seeds. More importantly, they make it difficult to detect which party is misbehaving when a failure occurs during reconstruction. Put differently, a party can simply send random ring elements in the last round, causing a systematic failure in the combine stage, without ever being detected. This problem has been addressed for Threshold Raccoon by either relying on non-interactive zero-knowledge proof systems [PKN+25] or by using different types of secret sharing [BCdP+25].

In this work, we follow the latter approach and replace the Shamir secret sharing used in Chapter 6 by the Vandermonde secret sharing (VSS) [BCdP+25], a lattice-friendly adaptation of a secret sharing by Desmedt et al. [DDB95]. This technique allows for sharing short values into short shares and with short reconstruction coefficients. Its security relies on the Hint-RLWE problem. As the VSS allows for short responses z_i, we can now verify the correctness of each party's response z_i, detecting misbehaving parties. This leads to what we call extraction share robustness: if the adversary maliciously generates the key extraction shares that passes the share verification function, they cannot cause the decapsulation of an honest ciphertext to fail. We actually achieve a slightly stronger notion, where this is true even for ciphertexts generated with biased randomness. Lastly, note that the security proof of the resulting TIBE now needs to handle the leakage coming from the VSS. To make the proof simpler, we blend the Coset-Hint-RLWE to Hint-RLWE reduction from Chapter 6 directly into the security proof. Hence, the Coset-Hint-RLWE only appears indirectly in this chapter.

Stronger decapsulation consistency. Finally, we realise that the extraction share robustness of the underlying TIBE allows us to generically reach a stronger notion of decapsulation consistency in the resulting TKEM compared to Chapter 6. Our notion corresponds to the one defined in [DLN+21] in the context of threshold public-key encryption schemes. If Chapter 6 only counted decapsulation inconsistencies from two unequal but *valid* keys $K \neq K'$, we now also consider the inconsistency from one of the reconstructed keys is *invalid*, i.e. leeds to $\perp$. Let us give some intuition on why robustness of the TIBE and consistency of the TKEM are related. If only one of K or K' is equal to $\perp$, we can use the valid key and the BCHK+ transform to recover the underlying message and encryption randomness used to generated the TIBE ciphertext. Then, the shares associated to the other (invalid) key define a forgery for the extraction share robustness

game of the underlying TIBE. If neither K nor K' are equal to $\perp$, we fall back to the case considered in Chapter 6. We provide more intuition on the different decapsulation consistency notions in Section 7.4.

Open problems. An interesting open problem would be to step away from the BCHK+ transform and consider primitives other than the TIBE that allow us to link the ciphertext to the signature verification key. In addition, we believe it is possible to further optimise the size of the ciphertext and the round complexity of the scheme by replacing the Hint-MLWE with NTRU with hints assumption [EEN+24, Appendix A] or replacing the commitment round with a random linear combination of committed values.

Chapter structure. In Section 7.1 we present our improved TIBE construction and discuss the design choices behind it. Then in Section 7.2 we present a more fine-grained analysis of the correctness and share robustness of our scheme. In Section 7.3 we prove the selective ID security of the TIBE required by the BCHK+ and Section 7.4 we prove a stronger decapsulation consistency for the corresponding TKEM. Lastly, in Section 7.5 we optimise the key and ciphertext sizes of our improved TKEM.

7.1 Improved TIBE Construction From Lattices

Let $\mathsf{H}_{\mathsf{cmt}}(\cdot) : \mathcal{R}_q \to \{0,1\}^{2\lambda}$ and $\mathsf{H}_{\mathsf{id}}(\cdot) : S_{\mathsf{id}} \to \mathcal{R}_q$ be hash functions modelled as random oracles.

Public parameters. These include $(\mathbf{A}, t, N, T)$, where N is the overall number of parties in the protocol and T is the threshold on the number of parties required for decryption.

1. The public matrix $\mathbf{A}$ is a matrix with trapdoor *à la* Eagle [YJW23] or Plover [EEN+24]. The parameter $\beta \in \mathbb{Z}$ can be set arbitrarily between 2 and $q/2$ and impacts the quality of the trapdoor.

$$\mathbf{A} = \begin{bmatrix} 1 & a & b \end{bmatrix}, \quad \text{where} \quad a \leftarrow \mathcal{R}_q, \quad s, s' \leftarrow \mathcal{D}_{\mathcal{R},\sigma_{\mathbf{s}}}, \quad b = \beta - (a \cdot s + s') \text{ for some } \beta \in \mathbb{Z} \tag{7.2}$$

   Then the trapdoor is constructed as follows $\mathbf{A} \cdot \begin{bmatrix} s' & s & 1 \\ 1 & 0 & 0 \end{bmatrix}^{\mathsf{T}} = \begin{bmatrix} \beta & 1 \end{bmatrix} \bmod q$.

2. $t \leftarrow \mathcal{R}_q$ is used as a target for preimage sampling.

Encryption procedure. In the encryption procedure, the encrypter constructs a public key $\begin{bmatrix} a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) & t \end{bmatrix}$ for the identity id. Then they generate a Dual Regev ciphertext $(\mathbf{u}, v)$ for this public key and the message $\mathsf{msg} \in \{0,1\}^d$. This is formally described in Algorithm 17.

Decryption procedure. In the decryption procedure, the parties jointly compute a vector $\mathbf{z} = \begin{bmatrix} x & \mathbf{z}' \end{bmatrix}$ such that $\begin{bmatrix} 1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) \end{bmatrix} \cdot \mathbf{z} \approx t$, where t is a part of the public key discussed above. Lastly, they use $\mathbf{z}$ for a Dual Regev style decryption procedure, by computing $v - \mathbf{u}^T \cdot \mathbf{z}'$ and decoding it to recover the message. The remaining x component of the product is incorporated into the noise we remove during the decoding procedure. The decryption is formally described in Algorithm 22.

Threshold mechanisms. The structure for extracting a key $\mathbf{z}$ for identity id in a distributed way differs from Chapter 6 in two aspects. First, we replace the Shamir sharing with VSS defined in Algorithms 1 and 2. Second, we add $\mathsf{TIBE.ShareVerify}$ that we run as part of the $\mathsf{TIBE.Combine}$ procedure. $\mathsf{TIBE.ShareVerify}$ takes as input a user identifier, contributions of all parties from a concluded session, the public key and the identity id. The algorithm returns 1 if the user's contribution is valid and $\perp$ otherwise. We compute partial public keys of the form $b^{(i)} = a \cdot s^{(i)} + s'^{(i)}$ for every share of (s, s') in Shares dictionary during the setup. Then using the fact that for each $\mathbf{z}^{(i)}$ the value $\begin{bmatrix} 1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) \end{bmatrix} \cdot \mathbf{z}^{(i)}$ only depends on public $w^{(i)}$ and $b^{(i)}$ we verify the share by checking a linear equation and a norm bound of $\mathbf{z}^{(i)}$. Intuitively, one can see $\mathbf{z}^{(i)}$ as a signature for the public key $b^{(i)}$. The procedure in formally described in Figure 7.2.

Small optimisations. We add other small optimisations. For example the third coordinate of $\mathbf{p}^{(i)}$ is set to 0 since the third coordinate of $\mathbf{z}^{(i)}$ does not contain sensitive values. The infinity norm and the image check of $\mathbf{z}$ during $\mathsf{TIBE.Combine}$ is replaced with the an image check in $\mathsf{TIBE.ShareVerify}$ and an ℓ_2 norm check of each share.

State. The TIBE is formally described in Figures 7.1 and 7.2. For clarity of reading, we do not specify it explicitly, but we assume that each party i maintains a state throughout the session. The state of party i contains (i) the secret key share of i, (ii) all contributions of all parties up to the current round, (iii) the values privately generated by i during previous rounds. In order to avoid ROS-style attacks, it is also important to ensure that the participants' views are consistent, by checking their internal state against received

contributions, see for example [KRT24, Figure 3]. We omit the consistency checks from our description and refer to [KRT24] for more details.

7.2 TIBE Correctness and Share Robustness

We now study the correctness and share robustness of the TIBE described in Figures 7.1 and 7.2. We start be providing useful equations that hold by construction. Let us note $\mathbf{p}^{(i)} = \begin{bmatrix} p_0^{(i)} & p_1^{(i)} & 0 & p_3^{(i)} \end{bmatrix}^\intercal$, $\mathbf{p} = \sum_i \mathbf{p}^{(i)}$ and $p_j = \sum_{i \in \mathsf{act}} p_j^{(i)}$ for $j \in [4]$. Recall that $b = \beta - \sum_{i \in \mathsf{act}} b^{(i)}$, $\mathbf{s} = \begin{bmatrix} \sum_i s^{(i)} & \sum_i s'^{(i)} & 1 & 0 \end{bmatrix}^\intercal$, $w = \sum_i w^{(i)}$ and $\mathbf{z} = \sum_i \mathbf{z}^{(i)}$. Then it is easy to verify that

$$\forall\, \mathsf{id}, \begin{bmatrix} 1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) \end{bmatrix} \cdot \mathbf{s} = \beta \qquad \text{(Algorithm 12, Line 4)} \tag{7.3}$$

$$\forall\, \mathsf{id}, \forall i \in \mathsf{act}, \begin{bmatrix} 1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) \end{bmatrix} \cdot \mathbf{s}^{(i)} = b^{(i)} \qquad \text{(Algorithm 13, Line 6)} \tag{7.4}$$

$$\forall\, \mathsf{id}, \forall i \in \mathsf{act}, \begin{bmatrix} 1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) \end{bmatrix} \cdot \mathbf{p}^{(i)} = w^{(i)} \qquad \text{(Algorithm 18, Line 2)} \tag{7.5}$$

$$c_0 \cdot \beta + c_1 = t - w \qquad \text{(Algorithm 20, Line 3)} \tag{7.6}$$

Extraction Share Correctness We start by proving correctness of the key extraction shares formally stated in Definition 22.

Lemma 105. *Let δ' be the target correctness probability of share verification in Algorithm 21. Given the* TIBE *described in Figures 7.1 and 7.2 with $\tau \geq 1 + \sqrt{\frac{4\lambda \ln 2}{2d}}$, $\sigma_{\mathbf{s}} \geq \sqrt{2}\eta_\varepsilon(\mathcal{R}^2)$ for $\varepsilon \in (0, 1/2)$ and norm bound B_{ind} given as*

$$B_{\mathsf{ind}} = \tau \left(\sqrt{2\,\sigma_{\mathbf{p}}^2 + {\sigma'_{\mathbf{p}}}^2}\, \sqrt{(2\pi)^{-1} d} + \frac{d\,q}{\beta} \cdot \sqrt{\pi\, d\, (\log N + 1)}\, \sigma_{\mathbf{s}} \right),$$

where d is the ring degree, ($\sigma_{\mathbf{p}}$,$\sigma'_{\mathbf{p}}$) (resp. $\sigma_{\mathbf{s}}$) is the Gaussian width of the noise $\mathbf{p}^{(i)}$ (resp. secret $\mathbf{s}^{(i)}$), q is the modulus, β is the decomposition parameter and N is the number of parties. Then, TIBE *has $8\varepsilon \cdot (\log N + 1)\varepsilon + \delta' \simeq \delta'$ extraction share correctness.*

Proof. Note that TIBE.ShareVerify described in Algorithm 21 outputs 1 if both **assert** checks pass. First, for $i \in \mathsf{act}$, let us write $\mathbf{z}^{(i)}$ and $\mathbf{z}$ explicitly:

$$\mathbf{z}^{(i)} = \begin{bmatrix} p_0^{(i)} + c_0 \cdot s'^{(i)} \\ p_1^{(i)} + c_0 \cdot s^{(i)} \\ 0 \\ p_3^{(i)} \end{bmatrix}, \quad \mathbf{z} = \left(\sum_{i \in \mathsf{act}} \mathbf{z}^{(i)} \right) + \begin{bmatrix} c_1 \\ 0 \\ c_0 \\ 0 \end{bmatrix} = \begin{bmatrix} p_0 + c_0 \cdot s' + c_1 \\ p_1 + c_0 \cdot s \\ c_0 \\ p_3 \end{bmatrix}. \tag{7.7}$$

Algorithm 12 $\mathsf{TIBE.KG}(1^\lambda) \to \mathsf{ek}, \mathsf{dk}$

$t, a \leftarrow \mathcal{R}_q \times \mathcal{R}_q$
$(s, s') \leftarrow \mathcal{D}_{\mathcal{R}^2, \sigma_{\mathbf{s}}}$
$b = \beta - (a \cdot s + s')$
$\mathbf{s} = \begin{bmatrix} s' & s & 1 & 0 \end{bmatrix}^T$ ▷ $\mathbf{s} \in \mathcal{R}^4$, and $\forall \mathsf{id}, \begin{bmatrix} 1 & a & b & H(\mathsf{id}) \end{bmatrix} \cdot \mathbf{s} = \beta$
return $\mathsf{ek} \coloneqq (a, b, t), \mathsf{dk} \coloneqq (s, s')$

Algorithm 13 $\mathsf{TIBE.Share}(\mathsf{ek}, \mathsf{dk}) \to \mathsf{aux}, \left\{\mathsf{dk}^{(i)}\right\}_{i \in [N]}$

$\mathsf{Shares} \leftarrow \mathsf{VandShare}((s, s'), [N], T, \mathbf{i} = \{\})$
$\mathsf{pek} \coloneqq \mathsf{Dict}\{\}$ ▷ pek lists all partial encryption keys (initially empty)
for $i \in [N]$ **do**
  $\mathsf{dk}^{(i)} \coloneqq \mathsf{Shares}[i]$ ▷ Each user i has a dictionary $\{\mathbf{i} : \mathbf{s}^{(\mathbf{i})}\}$of private keys.
  for $\mathbf{i} \in \mathsf{Shares}[i]$ **do** ▷ Compute all partial public keys
    $\mathsf{pek}[\mathbf{i}] \coloneqq b^{(\mathbf{i})} = a \cdot s^{(\mathbf{i})} + s'^{(\mathbf{i})}$
$\mathsf{ek} \coloneqq (a, b, t, \mathsf{pek})$
return $\mathsf{aux} \coloneqq \mathsf{pek}, \left\{\mathsf{dk}^{(i)}\right\}_{i \in [N]}$

Algorithm 14 $\mathsf{TIBE.Setup}(1^\lambda) \to \mathsf{ek}, \left\{\mathsf{dk}^{(i)}\right\}_{i \in [N]}$

$(\mathsf{ek}_0, \mathsf{dk}) \leftarrow \mathsf{TIBE.KG}(1^\lambda)$
$(\mathsf{ek}_1, \left\{\mathsf{dk}^{(i)}\right\}_{i \in [N]}) \leftarrow \mathsf{TIBE.Share}(\mathsf{ek}, \mathsf{dk})$
return $\mathsf{ek} \coloneqq (\mathsf{ek}_0, \mathsf{ek}_1), \left\{\mathsf{dk}^{(i)}\right\}_{i \in [N]}$

Algorithm 15 $\mathsf{Encode}(\mathsf{msg} = (\alpha_0, \ldots, \alpha_{d-1})) \to M \in \mathcal{R}_q$

return $M = \sum_{i \in [d]} \alpha_i \cdot \lfloor q/2 \rceil \cdot X^i$

Algorithm 16 $\mathsf{Decode}(M = \sum_{i \in [d]} \alpha_i \cdot X^i) \to \mathsf{msg} \in \{0,1\}^d / \perp$

return $\mathsf{msg} = \sum_{i \in [d]} \lfloor \alpha_i \rceil_q X^i$

Algorithm 17 $\mathsf{TIBE.Encrypt}(\mathsf{msg} \in \{0,1\}^d, \mathsf{id}, \mathsf{ek}) \to \mathsf{ct}$

$(r, \mathbf{e}, e') \leftarrow D_{R, \sigma_{\mathbf{r}}} \times D_{R^3, \sigma_{\mathbf{r}}} \times D_{R, \sigma_{\mathbf{r}}}$
$\begin{bmatrix} \mathbf{u}^T & v \end{bmatrix} \coloneqq r \cdot \begin{bmatrix} a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) & t \end{bmatrix} + \begin{bmatrix} \mathbf{e}^T & e' \end{bmatrix} + \begin{bmatrix} \mathbf{0}^{1\times 3} & \mathsf{Encode}(\mathsf{msg}) \end{bmatrix}$ ▷ $\begin{bmatrix} \mathbf{u} \\ v \end{bmatrix} \in \mathcal{R}_q^4$
return $\mathsf{ct} \coloneqq (\mathbf{u}, v)$

Figure 7.1: TIBE algorithms for key generation, encryption and message encoding.

Algorithm 18 $\mathsf{TIBE.Extract}_0(\mathsf{id}, \mathsf{act}, \mathsf{dk}^{(i)}, \mathsf{state}^{(i)}) \to \mathsf{contrib}_0^{(i)}/\perp$

$\mathbf{p}^{(i)} \leftarrow D_{\mathcal{R}^2,\sigma_\mathbf{p}} \times \{0\} \times D_{\mathcal{R},\sigma'_\mathbf{p}}$
$w^{(i)} = \begin{bmatrix}1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix} \cdot \mathbf{p}^{(i)}$
$\mathsf{cmt}^{(i)} = \mathsf{H}_{\mathsf{cmt}}(w^{(i)})$
return $\mathsf{contrib}_0^{(i)} := \mathsf{cmt}^{(i)}$

Algorithm 19 $\mathsf{TIBE.Extract}_1(\mathsf{id}, \mathsf{act}, \mathsf{dk}^{(i)}, \mathsf{state}^{(i)}) \to \mathsf{contrib}_1^{(i)}/\perp$

return $\mathsf{contrib}_1^{(i)} := w^{(i)}$

Algorithm 20 $\mathsf{TIBE.Extract}_2(\mathsf{id}, \mathsf{act}, \mathsf{dk}^{(i)}, \mathsf{state}^{(i)}) \to \mathsf{contrib}_2^{(i)}/\perp$

assert $\forall j \in \mathsf{act} \setminus \{i\} : \mathsf{H}_{\mathsf{cmt}}(w^{(j)}) = \mathsf{cmt}^{(j)}$
$w := \sum_{j\in\mathsf{act}} w^{(j)}$
$(c_0, c_1) := \mathsf{Decomp}_\beta(t - w)$ ▷ $c_0 \cdot \beta + c_1 = t - w$
$\mathsf{Indices} := \mathsf{VandRecover}([N], \mathsf{act})$ ▷ i is assigned one index $\mathsf{Indices}[i]$
$s^{(i)}, s'^{(i)} := \mathsf{dk}^{(i)}[\mathsf{Indices}[i]]$
$\mathbf{s}^{(i)} := \begin{bmatrix}s'^{(i)} & s^{(i)} & 0 & 0\end{bmatrix}^T$ ▷ $\mathbf{s}^{(i)} \in \mathcal{R}^4$ is the partial signing key assigned to i
$\mathbf{z}^{(i)} = \mathbf{p}^{(i)} + c_0 \cdot \mathbf{s}^{(i)}$
return $\mathsf{contrib}_2^{(i)} := \mathbf{z}^{(i)}$

Algorithm 21 $\mathsf{TIBE.ShareVerify}(\mathsf{ek}, \mathsf{id}, \mathsf{act}, \left\{(w^{(j)}, \mathbf{z}^{(j)})\right\}_{j\in\mathsf{act}}, i) \to \{1, \perp\}$

$w := \sum_{j\in\mathsf{act}} w^{(j)}$
$(c_0, c_1) := \mathsf{Decomp}_\beta(t - w)$
$\mathsf{Indices} := \mathsf{VandRecover}([N], \mathsf{act})$ ▷ User i is assigned one index $\mathsf{Indices}[i]$
$b^{(i)} := \mathsf{pek}[\mathsf{Indices}[i]]$
assert $\left\{\begin{bmatrix}1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix} \cdot \mathbf{z}^{(i)} = w^{(i)} + c_0 \cdot b^{(i)}\right\}$ ▷ Equality
assert $\left\{\left\|\mathbf{z}^{(i)}\right\| \le B_{\mathsf{ind}}\right\}$ ▷ Shortness
return 1 ▷ Accept if both asserts pass

Algorithm 22 $\mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{ek}, \mathsf{id}, \mathsf{act}, \left\{(w^{(i)}, \mathbf{z}^{(i)})\right\}_{i\in\mathsf{act}}) \to \mathsf{msg}/\perp$

$\mathsf{Indices} := \mathsf{VandRecover}([N], \mathsf{act})$ ▷ Each user i is assigned one index $\mathsf{Indices}[i]$
for $i \in \mathsf{act}$ **do**
  $b^{(i)} := \mathsf{pek}[\mathsf{Indices}[i]]$
  assert $\mathsf{TIBE.ShareVerify}(\mathsf{ek}, \mathsf{id}, \mathsf{act}, \left\{(w^{(j)}, \mathbf{z}^{(j)})\right\}_{j\in\mathsf{act}}, i)$
$\mathbf{z} := \left(\sum_{i\in\mathsf{act}} \mathbf{z}^{(i)}\right) + \begin{bmatrix}c_1 & 0 & c_0 & 0\end{bmatrix}^T$
$\mathbf{z}' := \begin{bmatrix}z_1 & z_2 & z_3\end{bmatrix}^T$ ▷ $\mathbf{z}'$ drops the first coefficient of $\mathbf{z}$
return $\mathsf{msg} := \mathsf{Decode}\left(v - \mathbf{u}^T \cdot \mathbf{z}'\right)$

Figure 7.2: TIBE algorithms for threshold decryption.

Recall that the third coefficient of $\mathbf{p}^{(i)}$ is 0. It is immediate from Equations (7.4) to (7.7) that:

$$\begin{aligned}\begin{bmatrix}1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix} \cdot \mathbf{z}^{(i)} &= \begin{bmatrix}1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix} \left(\mathbf{p}^{(i)} + c_0 \cdot \mathbf{s}^{(i)}\right) \\ &= \mathbf{w}^{(i)} + c_0 \cdot b^{(i)}. \end{aligned} \tag{7.8}$$

Therefore, the first **assert** checking equality in Algorithm 21 for honest shares is always satisfied. Next, we study the norm of $\mathbf{z}^{(i)}$. For an honestly generated $\mathbf{z}^{(i)}$, except with probability $\delta' := 8\varepsilon \cdot (\log N + 1) + 2^{-\lambda}$ and for $\tau \geq 1 + \sqrt{\frac{4\lambda \ln 2}{2d}}$ it holds:

$$\left\|\mathbf{z}^{(i)}\right\| \leq \left\|\mathbf{p}^{(i)}\right\| + \left\|c_0 \cdot \mathbf{s}^{(i)}\right\| \tag{7.9}$$

$$\leq \sum_{j=0}^{3} \left\|\mathbf{p}_j^{(i)}\right\| + d \cdot \|c_0\|_\infty \cdot \left\|\mathbf{s}^{(i)}\right\| \tag{7.10}$$

$$\leq \tau \cdot \sqrt{2\,\sigma_{\mathbf{p}}^2 + {\sigma'_{\mathbf{p}}}^2}\,\sqrt{(2\pi)^{-1} d} + \frac{d\,q}{\beta} \cdot \tau \sqrt{(4\pi)^{-1}\, d\,(\log N + 1)}\,\sigma_{\mathbf{s}}, \tag{7.11}$$

Here we applied the triangle inequality in Equation (7.9), then once again in Equation (7.10), and finally Lemma 27, Lemma 49 and $\|c_0\|_\infty \leq \left\lceil \frac{q-1}{2\beta} \right\rceil$ from Section 2.6 in Equation (7.11). Therefore setting:

$$B_{\mathsf{ind}} = \tau \left(\sqrt{2\,\sigma_{\mathbf{p}}^2 + {\sigma'_{\mathbf{p}}}^2}\,\sqrt{(2\pi)^{-1} d} + \frac{d\,q}{\beta} \cdot \sqrt{(4\pi)^{-1}\, d\,(\log N + 1)}\,\sigma_{\mathbf{s}} \right) \tag{7.12}$$

guarantees that the shortness **assert** in Alg. 22 passes except with probability δ'. □

Decryption Correctness We now study the decryption correctness of our TIBE following Definition 19.

Lemma 106. *Let λ be the security parameter. Given* TIBE *as described in Figures 7.1 and 7.2 with $\tau \geq 1 + \sqrt{\frac{4\lambda \ln 2}{d}}$, $\min(\sigma_{\mathbf{p}}, \sigma_{\mathbf{p}'}) \geq \sqrt{1 + 1/(T-1)} \cdot \eta_\varepsilon(\mathcal{R})$ and norm bounds B_z and B_r given as*

$$\begin{aligned} B_z &= \tau\sqrt{2\,\sigma_{\mathbf{p}}^2 + {\sigma'_{\mathbf{p}}}^2}\,\sqrt{(2\pi)^{-1}\, d\, T} + \frac{d\,q}{2\,\beta}\left(2\tau\sqrt{(4\pi)^{-1}\, d}\,\sigma_{\mathbf{s}} + 1\right) + B_1, \\ B_r &= \sqrt{\pi^{-1}\,(-\ln \delta' + \ln(4d))}\,\sigma_{\mathbf{r}}, \end{aligned}$$

where T is the number of parties, d is the ring degree, $\sigma_{\mathbf{p}}$ (resp. $\sigma_{\mathbf{s}}$ and $\sigma_{\mathbf{r}}$) is the Gaussian width of the masks (resp. secret and encryption randomness), q is the modulus, β is the decomposition parameter and B_1 is an ℓ_2 norm bound on c_1. Furthermore assume that TIBE *is δ' extraction share correct. If $(B_z + 1) \cdot B_r \leq \lfloor q/4 \rfloor$, then,* TIBE *has $\delta := T(\delta' + 1) + 2 \cdot 2^{-\lambda} + 8\varepsilon \cdot (T - 1)$ decryption correctness.*

We highlight that B_z is mainly impacted by $\sqrt{T}\sigma_{\mathbf{p}}$.

Proof. First we characterize the global preimage $\mathbf{z}$. By Lemma 105, the check in Algorithm 22, Line 4 passes for a given $i \in \mathsf{act}$ with probability δ' and by a union bound we arrive at the next line with probability $\geq 1 - T\delta'$. Hence, we assume in the following that equality and shortness checks have passed for all $i \in \mathsf{act}$.

We first aggregate Equation (7.8) over all $i \in \mathsf{act}$, then observe that $\sum_i b^{(i)} = \beta - b$, and finally use Equation (7.6), which gives:

$$\begin{aligned}\begin{bmatrix}1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix} \cdot \mathbf{z} &= \begin{bmatrix}1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix} \left(\sum \mathbf{z}^{(i)} + \begin{bmatrix}c_1 & 0 & c_0 & 0\end{bmatrix}^{\intercal}\right) \\ &= \sum_i \left(w^{(i)} + c_0 \cdot b^{(i)}\right) + c_1 + c_0 \cdot b \\ &= w + c_0 \sum_i b^{(i)} + c_1 + c_0 \cdot b \\ &= w + c_0 \cdot \beta - c_0 \cdot b + c_1 + c_0 \cdot b = t \bmod q \end{aligned} \tag{7.13}$$

Equation (7.13) is essentially the same as Equation (7.8), but for the global trapdoor instead of a partial trapdoor. It states that $\mathbf{z} = [z_0 \mid \mathbf{z}']$ is a valid preimage for the matrix $\begin{bmatrix}1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix}$ and the target t. We now explain why the scheme decrypts correctly. Recall that the honest decryption procedure computes:

$$v - \mathbf{u} \cdot \mathbf{z}' = (r \cdot t + e' + \mathsf{Encode}(\mathsf{msg})) - (r \cdot \begin{bmatrix}a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id})\end{bmatrix} + \mathbf{e}^{\intercal}) \cdot \mathbf{z}' \tag{7.14}$$

$$= \mathsf{Encode}(\mathsf{msg}) + r \cdot t + e' - r(t - z_0) - \mathbf{e}^{\intercal} \cdot \mathbf{z}' \tag{7.15}$$

$$= \mathsf{Encode}(\mathsf{msg}) + (e' + [r \mid -\mathbf{e}^{\intercal}] \cdot \mathbf{z}). \tag{7.16}$$

As long as it holds for the residual noise that $\|e' + [r \mid -\mathbf{e}^{\intercal}] \cdot \mathbf{z}\|_\infty \leq \lfloor q/4 \rfloor$, decryption will be correct. In practice, the impact of e' in the residual noise is negligible, the dominating term being $[r \mid -\mathbf{e}^{\intercal}] \cdot \mathbf{z}$.

Bounding $\|\mathbf{z}\|$. By Equation (7.7), after honest execution of the key share extraction, it holds $\mathbf{z} = \mathbf{p} + c_0 \cdot \mathbf{s} + \begin{bmatrix}c_1 & 0 & 0 & 0\end{bmatrix}^{\intercal}$ where $\mathbf{s} = \begin{bmatrix}s & s' & 1 & 0\end{bmatrix}^{\intercal}$. By the triangle inequality, except with probability $2 \cdot 2^{-\lambda} + 8\varepsilon \cdot (T-1)$, it holds[1]

$$\|\mathbf{z}\| \leq \left\|\mathbf{p} + \begin{bmatrix}c_1 & 0 & 0 & 0\end{bmatrix}^{\intercal}\right\| + d \cdot \|c_0\|_\infty \cdot \|\mathbf{s}\| \tag{7.17}$$

$$\leq \|\mathbf{p}\| + \|c_1\| + d \cdot \|c_0\|_\infty \cdot \|\mathbf{s}\| \tag{7.18}$$

$$\leq \tau\sqrt{2\,\sigma_{\mathbf{p}}^2 + {\sigma'_{\mathbf{p}}}^2}\,\sqrt{(2\pi)^{-1}\,d\,T} + \frac{d\,q}{2\,\beta}\left(2\tau\sqrt{(4\pi)^{-1}\,d}\,\sigma_{\mathbf{s}} + 1\right) + B_1 := B_z, \tag{7.19}$$

[1]This analysis is tighter compared to $\|\mathbf{z}\| \leq T \cdot \left\|\mathbf{z}^{(i)}\right\| \leq T \cdot B_{\mathsf{ind}}$ from above.

where B_1 is a bound on the norm of c_1. Here, we used that $\mathbf{s}$ is a discrete Gaussian and $\mathbf{p}$ is a sum of T discrete Gaussian vectors $\mathbf{p}^{(i)}$. Their norms are bounded by Lemma 27 and Lemma 34. In practice, $\|\mathbf{p}\|$ dominates $d \cdot \|c_0\|_\infty \cdot \|\mathbf{s}\|$, but does not necessarily dominate $\|c_1\|$.

Bounding $\|[r \mid -\mathbf{e}^\mathsf{T}] \cdot \mathbf{z}\|_\infty$. Recall that we assume the ciphertext is honestly generated, including the noise $[r \mid -\mathbf{e}^\mathsf{T}]$. Then by Lemma 28 with probability $\geq 1 - \delta'/2$ it holds

$$\left\|[r \mid -\mathbf{e}^\mathsf{T}] \cdot \mathbf{z}\right\|_\infty \leq \|\mathbf{z}\| \cdot \sqrt{\pi^{-1}\left(-\ln \delta' + \ln(4d)\right)} \cdot \sigma_\mathbf{r}. \tag{7.20}$$

Similarly, $\|e'\|_\infty$ is upper bounded by $B_r := \sqrt{\pi^{-1}\left(-\ln \delta' + \ln(4d)\right)}\,\sigma_\mathbf{r}$ with probability $\geq 1 - \delta'/2$. Overall, except with probability at most δ' it holds

$$\begin{aligned}\left\|e' + [r \mid -\mathbf{e}^\mathsf{T}] \cdot \mathbf{z}\right\|_\infty &\leq \sqrt{\pi^{-1}\left(-\ln \delta' + \ln(4d)\right)} \cdot \sigma_\mathbf{r} + \sqrt{\pi^{-1}\left(-\ln \delta' + \ln(4d)\right)} \cdot \sigma_\mathbf{r} \cdot B_z \\ &\leq B_r(1 + B_z) \leq \lfloor q/4 \rfloor,\end{aligned}$$

concluding the proof. □

Extraction Share Robustness Lastly, we analyse whether the TIBE decrypts correctly, when the adversary maliciously generates the key extraction shares that still pass the share verification algorithm. In practice this means that $\mathbf{p}^{(i)}$ are no longer independent and we cannot use the convolution lemma.

Note that we still assume a quasi-honestly generated ciphertext. Concretely, we require the encryption randomness of the ciphertext to come from RSpace defined below (honestly generated randomness lies in RSpace with overwhelming probability). We call this property the extraction share robustness, as formally defined in Definition 23.

For our TIBE the encryption randomness comes from $\mathsf{RDistr} = D_{\mathcal{R},\sigma_\mathbf{r}} \times D_{\mathcal{R}^{1\times 3},\sigma_\mathbf{r}} \times D_{\mathcal{R},\sigma_\mathbf{r}}$. Hence, we set $\mathsf{RSpace} = \left\{(r, \mathbf{e}, e') \in \mathcal{R}^{1\times 5} \colon \|(r, \mathbf{e}, e')^\mathsf{T}\|_2 \leq \tau\,\sigma_\mathbf{r}\sqrt{(2\pi)^{-1} \cdot 5d}\right\}$, for $\tau \geq 1 + \sqrt{\frac{4\lambda \ln 2}{5d}}$. Then by Lemma 27 honestly sampled encryption randomness from RDistr lies in RSpace with probability $\geq 1 - 2^{-\lambda}$.

Lemma 107. *Given* TIBE *as described in Figures 7.1 and 7.2 with $\tau \geq 1 + \sqrt{\frac{4\lambda \ln 2}{5d}}$ and norm bounds B_m and B given as*

$$B_m = TB_\mathsf{ind} + B_0 + B_1, \text{ and } B = \tau\,\sigma_\mathbf{r}\sqrt{(2\pi)^{-1} \cdot 5d}$$

where T is the number of parties, $\sigma_\mathbf{r}$ is the Gaussian width of the encryption randomness, B_0 (resp. B_1 and B_ind) is a norm bound on c_0 (resp. c_1 and $\mathbf{z}^{(i)}$). Furthermore assume

that TIBE *is* δ' *extraction share correct. If* $B \cdot (1 + B_m) \leq \lfloor q/4 \rfloor$*, then, the advantage of any PPT adversary* $\mathcal{A}$ *against the extraction share robustness game for* TIBE *is at most* $T\delta'$*.*

We remark that B_m is dominated by $T\sigma_{\mathbf{p}}$, as opposed to $\sqrt{T}\sigma_{\mathbf{p}}$ in the honest case considered for correctness in Lemma 106.

Proof. Informally, the three main differences between the extraction share robustness game and the decryption correctness, is that (i) the $\mathbf{z}^{(i)}$ are malicious, (ii) the encryption randomness might not follow the honest distribution, but is only guaranteed to lie in RSpace and (iii) the $\mathbf{z}^{(i)}$ might be colinear to the encryption randomness. This requires worse norm bounds to bound the norm of $\mathbf{z}$ and the inner product between $\mathbf{z}$ and encryption randomness.

More formally, note that Equation (7.13) holds for every share that passes the **assert** in Algorithm 22. Since we assume quasi-honest ciphertexts, then Equation (7.16) holds too. Let B_0 (resp. B_1) be a bound on the norm of c_0 (resp. c_1). Note that the adversary has no control over (c_0, c_1), so these bounds are independent whether we are looking at semi-honest or malicious parties.

Bounding $\|\mathbf{z}\|$. Here, we deviate from the analysis in the honest setting. By the equality and shortness checks in Algorithm 21 and the definition of $\mathbf{z}$, we have

$$\|\mathbf{z}\| = \left\| \left(\sum_{i \in \mathsf{act}} \mathbf{z}^{(i)} \right) + \begin{bmatrix} c_1 & 0 & c_0 & 0 \end{bmatrix}^{\mathsf{T}} \right\| \tag{7.21}$$

$$\leq T\, B_{\mathsf{ind}} + B_1 + B_0 =: B_m. \tag{7.22}$$

This reflects the fact that corrupted parties may collude and generate $\mathbf{z}^{(i)}$ that are as colinear as possible.

Bounding $\|[r \mid -\mathbf{e}^{\mathsf{T}}] \cdot \mathbf{z}\|_\infty$. In the malicious setting we cannot apply Lemma 26, as we cannot guarantee that the ciphertext encryption randomness $[r \mid -\mathbf{e}^{\mathsf{T}}]$ was sampled as a discrete Gaussian. We only know that it lies in RSpace. Hence, we use Cauchy-Schwarz instead:

$$\begin{aligned} \left\| e' + [r \mid -\mathbf{e}^{\mathsf{T}}] \cdot \mathbf{z} \right\|_\infty &= \left\| [e' \mid r \mid -\mathbf{e}^{\mathsf{T}}] \cdot [1 || \mathbf{z}] \right\|_\infty \\ &\leq \|(e', r, -\mathbf{e})\|_2 \cdot \|(1, \mathbf{z})\|_2 \\ &\leq B \cdot (1 + B_m), \end{aligned}$$

concluding the proof. □

$\mathsf{Exp}^{\mathsf{KG\text{-}sim}}_{\mathcal{A},\mathsf{KG},\mathsf{Share},\mathsf{Eval}}(1^\lambda)$

$\mathsf{cor} \leftarrow \mathcal{A}(1^\lambda)$
assert $|\mathsf{cor}| = T - 1$
$(\mathsf{ek}, \mathsf{dk}) \leftarrow \mathsf{KG}(1^\lambda)$, $b \leftarrow \{0,1\}$
if $b = 0$: $\left(\left\{\mathsf{dk}^{(i)}\right\}_{i\in[N]}, \mathsf{aux}\right) \leftarrow \mathsf{Share}(\mathsf{dk})$
  $b' \leftarrow \mathcal{A}^{\mathsf{Eval}}\left(\mathsf{ek}, \left\{\mathsf{dk}^{(i)}\right\}_{i\in\mathsf{cor}}\right)$
else $\left(\left\{\mathsf{dk}^{(i)}\right\}_{i\in\mathsf{cor}}, \mathsf{aux}\right) \leftarrow \mathsf{SimShare}(\mathsf{ek}, \mathsf{cor})$
  $b' \leftarrow \mathcal{A}^{\mathsf{SimEval}}\left(\mathsf{ek}, \left\{\mathsf{dk}^{(i)}\right\}_{i\in\mathsf{cor}}\right)$
return $b = b'$

Figure 7.3: Security experiment for Functional Simulatability.

Regarding the ℓ_2 norm bounds B_0 (resp. B_1) on uniform bounded c_0 (resp. c_1), w.o.p. it holds

$$B_0 = \frac{q}{\beta}\sqrt{\frac{d}{12}}(1 + o(1)), \qquad B_1 = \beta\sqrt{\frac{d}{12}}(1 + o(1)) \tag{7.23}$$

7.3 TIBE Security

Before starting with our security proof, we recall the definition of functional simulation of a pair of key generation and sharing algorithms. It is a simplified version of the one in [dPENP25], tailored to a trusted setup.

Definition 33 ([dPENP25])**.** *A key generation and sharing algorithms* KG *and* Share *together with a function oracle* Eval *are called functionally simulatable if there exist efficient functions* SimShare *and* SimEval *s.t. no PPT adversary* $\mathcal{A}$ *has a non-negligible advantage in the experiment defined in Figure 7.3. We denote the advantage of an adversary* $\mathcal{A}$ *in* $\mathsf{Exp}^{\mathsf{KG\text{-}sim}}_{\mathcal{A},\mathsf{KG},\mathsf{Share},\mathsf{Eval}}(1^\lambda)$ *as* $\mathsf{Adv}^{\mathsf{KG\text{-}sim}}_{\mathcal{A},\mathsf{KG},\mathsf{Share},\mathsf{Eval}}$.

Lemma 108 ([BCdP+25, Lemma B.2])**.** *Let* KG *and* Share *be as defined in Algorithm 12, Algorithm 13 and let* $\mathsf{Eval} = \{\mathsf{ShareExtract}_i\}_{i\in[2]}$*, be as in Figure 7.2. Then there exists a constant* $Q_v = Q_v(T, N, Q_{Dec})$ *(see [BBD+25] Theorem 1) such that*

$$\mathsf{Adv}^{\mathsf{KG\text{-}sim}}_{\mathcal{A},\mathsf{KG},\mathsf{Share},\{\mathsf{ShareExtract}_i\}_i} \leq \mathsf{Hint\text{-}RLWE}^{Q_{dec}+Q_v,\sqrt{\sigma_{\mathbf{p}}^2-{\sigma'_{\mathbf{p}}}^2},\sigma_{\mathbf{s}},\beta}_{\mathcal{R},q,m,\sigma_{\mathbf{s}}}.$$

Theorem 24. *Let* TIBE *be the scheme defined in Figure 7.1 and Figure 7.2. Let* $\mathcal{R} = \mathbb{Z}[X]/(X^d+1)$ *be a ring of degree* d*, and* q *be a prime number. Let* $\sigma, \sigma_{\mathbf{s}}, \sigma_{\mathbf{p}}, \sigma'_{\mathbf{p}}, \beta > 0$*,* $\varepsilon \in (0,1)$*. Suppose that*

$$\min(\sigma_{\mathbf{p}}, \sigma'_{\mathbf{p}}) > 1.17 q^{1/2} \cdot \eta_\varepsilon(\mathcal{R}^2), \tag{7.24}$$

$$\frac{\sigma'_{\mathbf{p}} \cdot \sqrt{\sigma_{\mathbf{p}}^2 - \sigma_{\mathbf{p}}'^2}}{\sigma_{\mathbf{p}}} \geq \eta_\varepsilon(\mathcal{R}). \tag{7.25}$$

Then for any PPT adversary $\mathcal{A}$*:*

$$\begin{aligned}
\mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}IND}}_{\mathcal{A},\mathsf{TIBE}}(\lambda) \leq\ & (Q_{Dec} + Q_{\mathsf{H_{id}}}) \cdot \mathsf{Adv}^{\mathsf{NTRU}_{\mathcal{R},q}}_{\mathcal{A}_1}(\lambda) \\
& + \mathsf{Adv}^{\mathsf{Hint\text{-}RLWE}^{Q_{dec}+Q_v, \sqrt{\sigma_{\mathbf{p}}^2 - \sigma_{\mathbf{p}}'^2}, \sigma_{\mathbf{s}}, \beta}_{\mathcal{R},q,m,\sigma_{\mathbf{s}}}}_{\mathcal{A}_2}(\lambda) + \mathsf{Adv}^{\mathsf{RLWE}_{\mathcal{R},4,q,\sigma_{\mathbf{r}}}}_{\mathcal{A}_3}(\lambda) \\
& + Q_{Dec} \cdot O(\varepsilon) + Q_{Dec} \cdot (T\, Q_{Dec} + Q_{H_{\mathsf{cmt}}}) \cdot (q^{-d} + \mathsf{negl}(\lambda)) \\
& + \frac{(T \cdot Q_{Dec} + Q_{H_{\mathsf{cmt}}})^2 + Q_{H_{\mathsf{cmt}}} \cdot T \cdot Q_{Dec}}{2^{2\lambda}}
\end{aligned}$$

where adversaries $\mathcal{A}_3$ *to* $\mathcal{A}_1$ *run in approximately the same time as* $\mathcal{A}$*.*

Proof. Consider a sequence of hybrids.

Hyb_0 **(**$\mathsf{sel\text{-}ID\text{-}IND}$ **game):** Identical to the real experiment.

Hyb_1 **(Delay** w_h**):** In this second hybrid we postpone sampling the noise $\mathbf{p}^{(i)}$ from $\mathsf{TIBE.Extract}_0$ to the end of $\mathsf{TIBE.Extract}_1$ for one of the honest parties. For each key extraction query $(\mathsf{id}, \mathsf{act})$ of the adversary we pick an arbitrary honest party in act as "last acting" honest party and denote its index as h. Now in $\mathsf{TIBE.Extract}_0$ the party h samples $\mathsf{cmt}_h \leftarrow \{0,1\}^{2\lambda}$. And in $\mathsf{TIBE.Extract}_1$ after every other party in act has revealed their values $\mathbf{w}^{(i)}$ the last honest party samples $\mathbf{p}^{(h)}$ honestly, computes $\mathbf{w}^{(h)} = \begin{bmatrix}1 & a & b & \mathsf{H_{id}}(\mathsf{id})\end{bmatrix} \cdot \mathbf{p}^{(h)}$ and programs $\mathsf{H_{cmt}}(\mathbf{w}^{(h)}) := \mathsf{cmt}_h$. The advantage of the adversary in this hybrid can increase only if $\mathbf{w}^{(h)}$ has been queried to $\mathsf{H_{cmt}}(\cdot)$ before. However, $\mathbf{w}^{(h)}$ computed above has high entropy. This is because by Corollary 8 and Lemma 56 for the vector $[1\ a\ \mathsf{H_{id}}(\mathsf{id})]$ and selected coordinates of $\mathbf{p}^{(h)}$, using $\min(\sigma_{\mathbf{p}}, \sigma'_{\mathbf{p}}) \geq 2d \cdot q^{1/3+2/3d}$ we have $\mathrm{SD}(\mathbf{w}^{(h)}, \mathcal{U}(\mathcal{R}_q)) = \mathsf{negl}(\lambda)$. Hence, the probability of querying a fixed $\mathbf{w}^{(h)}$ is bounded by $q^{-d} + \mathsf{negl}(\lambda)$ and the number of queries to $\mathsf{H_{cmt}}$ is bounded by $T \cdot Q_{\mathsf{Dec}} + Q_{H_{\mathsf{cmt}}}$ where $Q_{H_{\mathsf{cmt}}}$ denotes the number of direct queries to $\mathsf{H_{cmt}}$ oracle and $T \cdot Q_{\mathsf{Dec}}$ is a bound on queries made by the Challenger when answering the decryption queries. Then by the union, when we apply this change to every decryption query we get

$$\varepsilon_0 - \varepsilon_1 \leq (q^{-d} + \mathsf{negl}(\lambda)) \cdot Q_{\mathsf{Dec}} \cdot (T \cdot Q_{\mathsf{Dec}} + Q_{H_{\mathsf{cmt}}}).$$

Hyb_2 **(Trapdoors in the hash):** In this hybrid we change the answers of the $\mathsf{H}_{\mathsf{id}}(\cdot)$ oracle for all $\mathsf{id} \neq \mathsf{id}^*$. Now on query id instead of sampling a fresh uniformly random value the oracle samples a pair $(h_{\mathsf{id}}, \mathbf{T}_h) \leftarrow \mathsf{TrapGenNTRU}(1^\lambda)$ and replies with $H_{id}(\mathsf{id}) = h_{\mathsf{id}}$. The rest of the experiment is run efficiently as in the original scheme. The total number of queries to $\mathsf{H}_{\mathsf{id}}(\cdot)$ is $Q_{\mathsf{Dec}} + Q_{\mathsf{H}_{\mathsf{id}}}$, hence, the adversary's advantage can only increase as

$$|\varepsilon_1 - \varepsilon_2| \leq (Q_{\mathsf{Dec}} + Q_{\mathsf{H}_{\mathsf{id}}}) \cdot \mathsf{Adv}_{\mathcal{A}_1}^{\mathsf{NTRU}_{\mathcal{R},q}}(\lambda)$$

where $\mathcal{A}_1$ runs in approximately the same time as $\mathcal{A}$.

Hyb_3 **(Hash collisions):** In this hybrid, we ensure that $\mathsf{H}_{\mathsf{cmt}}$ has no collisions and that the adversary cannot find preimages of hashes cmt_i after passing them to the second key extraction oracle. We introduce a table Cmt that records every cmt sampled in $\mathsf{H}_{\mathsf{cmt}}$ or in $\mathsf{TIBE.Extract}_0$. Whenever a new hash cmt is sampled, we ensure that it was not previously added to Cmt, or the challenger aborts the game. The view of the adversary differs in Hyb_3 if it was previously able to (i) find a preimage of a cmt_i from one of the decryption queries, or (ii) if a given cmt was sampled twice. We can bound the probability of (i) due to the preimage resistance of the random oracle $\mathsf{H}_{\mathsf{cmt}}$. This gives a bound $Q_{H_{\mathsf{cmt}}} \cdot (T \cdot Q_{\mathsf{Dec}} \cdot 2^{-2\lambda})$. As for (ii), since $\mathsf{H}_{\mathsf{cmt}}$ is a random oracle and the commitments are sampled uniformly from $\{0,1\}$, we can bound the probability of (ii) by $(Q_{H_{\mathsf{cmt}}} + T \cdot Q_{\mathsf{Dec}})^2 \cdot 2^{-2\lambda}$. We conclude that,

$$|\varepsilon_2 - \varepsilon_3| \leq \frac{(T \cdot Q_{\mathsf{Dec}} + Q_{H_{\mathsf{cmt}}})^2 + Q_{H_{\mathsf{cmt}}} \cdot T \cdot Q_{\mathsf{Dec}}}{2^{2\lambda}}$$

Hyb_4 **(Uniform** (c_0, c_1)**):** In this hybrid, for each decryption we want to set the "challenge" $u = t - w$ to be uniform and then set the honest $\mathbf{z}^{(i)}$ accordingly. We first sample $u \leftarrow \mathcal{R}_q$, and set $w^{(h)} = t - u - \sum_{i \in \mathsf{act} \setminus h} w^{(i)}$, as shown in Hyb_1 the distribution of $w^{(h)}$ is close to uniform so this change is statistical. The behaviour of all parties except h is unchanged, party h however needs to still be able to sample $\mathbf{p}^{(h)}$ from the appropriate distribution. Note that sampling $\mathbf{p}^{(h)} \leftarrow D_{\mathcal{R}^2,\sigma_{\mathbf{p}}} \times \{0\} \times D_{\mathcal{R},\sigma'_{\mathbf{p}}}$ and setting $w^{(h)} = \begin{bmatrix} 1 & a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) \end{bmatrix} \cdot \mathbf{p}^{(h)}$ is the same as sampling $w^{(h)}$ from the appropriate distribution (statistically close to uniform) and sampling $\mathbf{p}^{(h)} \leftarrow \mathcal{D}_{\Lambda_q^{w^{(h)}}(\mathbf{F}_{\mathsf{id}}),\Sigma}$ with $\Sigma = \mathsf{diag}(\sigma_{\mathbf{p}}, \sigma_{\mathbf{p}}, \sigma'_{\mathbf{p}})$ and $\mathbf{F}_{\mathsf{id}} = \begin{bmatrix} 1 & a & \mathsf{H}_{\mathsf{id}}(\mathsf{id}) \end{bmatrix}$. Using Lemma 35 this is statistically close to sampling $\tilde{\mathbf{p}} = (\mathbf{p}_0^{(h)}, \mathbf{p}_1^{(h)}) \leftarrow \mathcal{D}_{\mathcal{R}^2,\sigma_{\mathbf{p}}}$ and $\mathbf{p}_3^{(h)} \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{w}^{(h)} - \mathbf{F}\tilde{\mathbf{p}}}(\mathsf{H}_{\mathsf{id}}(\mathsf{id})),\sigma'_{\mathbf{p}}}$, with $\mathbf{F} = [1\ a]$. We want to use the NTRU trapdoor introduced in the previous hybrid to sample this preimage. Since the trapdoor is for the matrix $[1\ \mathsf{H}_{\mathsf{id}}(\mathsf{id})]$, we consider the second

transformation. We now sample $\tilde{\mathbf{p}} = (\mathbf{p}_0^{(h)}, \mathbf{p}_1^{(h)}) \leftarrow \mathcal{D}_{\mathcal{R},\sqrt{\sigma_{\mathbf{p}}^2 - \sigma_{\mathbf{p}}'^2}} \times \mathcal{D}_{\mathcal{R},\sigma_{\mathbf{p}}}$, and then sample $(\mathbf{x}, \mathbf{p}_3^{(h)}) \leftarrow \mathcal{D}_{\Lambda_q^{\mathbf{w}^{(h)} - \mathbf{F}\tilde{\mathbf{p}}}([1\ \mathsf{H}_{\mathsf{id}}(\mathsf{id})]),\sigma_{\mathbf{p}}'}$, finally return $(\mathbf{p}_0^{(h)} + \mathbf{x}, \mathbf{p}_1^{(h)}, \mathbf{p}_3^{(h)})$. By Lemma 34, as long as the smoothing condition in Equation (7.25) holds, this modification is statistically unnoticeable. The NTRU trapdoor can sample such a distribution according to Lemma 47 if $\sigma_{\mathbf{p}} > \sigma_{\mathbf{p}}' \geq 1.17\, q^{1/2}\, \eta_\varepsilon(\mathcal{R}^2)$. Gathering the bounds, the advantage of the adversary can increase by at most

$$|\varepsilon_3 - \varepsilon_4| \leq (\mathsf{negl}(\lambda) + 2\,\varepsilon) \cdot Q_{\mathsf{Dec}}.$$

Hyb_5 **(Secret Sharing):** Let SimShare and SimEval be as per Lemma 108. In this hybrid we sample the public key without a trapdoor (i.e. $b \leftarrow \mathcal{R}_q$) and use SimShare to simulate the corrupt key shares. Using Lemma 108 we observe that all honest partial secrets $(s^{(i)}, s^{(i)\prime})$ can be written either as a linear combination of the leaked secrets, or as (s, s') plus a linear combination of the leaked secrets. Meaning that there are $Q_v(T, N)$ hints of the form $\mathbf{s} + \mathbf{r}$ (corresponding to the leakages) and at most $T \cdot Q_{\mathsf{Dec}}$ hints of the form $\mathbf{s} \cdot c + \mathbf{p}$. Since we do not have access to the secret keys, the partial decryption oracles are simulated using the corresponding SimEval. Then by Lemma 108 the advantage changes as

$$|\varepsilon_4 - \varepsilon_5| \leq \mathsf{Adv}^{\mathsf{KG\text{-}sim}}_{\mathcal{A}_2,\mathsf{KG},\mathsf{Share},\{\mathsf{ShareExtract}_i\}_i} \leq \mathsf{Adv}^{\mathsf{Hint\text{-}RLWE}^{Q_v+Q_{dec},\sqrt{\sigma_{\mathbf{p}}^2-\sigma_{\mathbf{p}}'^2},\sigma_{\mathbf{s}},\beta}_{\mathcal{R},q,m,\sigma_{\mathbf{s}}}}_{\mathcal{A}_2}.$$

Hyb_6 **(Ciphertext RLWE):** Note that now $\mathbf{F}_{\mathsf{id}^*} = \begin{bmatrix} a & b & \mathsf{H}_{\mathsf{id}}(\mathsf{id}^*) \end{bmatrix}$ and t are perfectly random, however, the experiment can be simulated efficiently. Hence, we apply the RLWE assumption to the challenge ciphertext. As a result, the challenge plaintext is masked by a uniformly random value and the advantage in this experiment is $\varepsilon_5 = 0$. The advantage change can also be bounded as

$$|\varepsilon_5 - \varepsilon_6| \leq \mathsf{Adv}^{\mathsf{RLWE}_{\mathcal{R},4,q,\sigma_{\mathbf{r}}}(\lambda)}_{\mathcal{A}_3}.$$

where $\mathcal{A}_3$ runs in approximately the same time as $\mathcal{A}$. □

On achieving adaptive corruptions. In all security notions we consider in this work, the adversary has to decide which $T - 1$ parties to corrupt at the very beginning of the game. This is usually called *static* corruptions. Closer to the real world is the concept of *adaptive* corruptions where the adversary can decide during the game which parties they want to corrupt, in particular after already having seen partial key extraction shares. In general, the latter is significantly harder to achieve as it requires the shares of the

yet-honest parties given to the adversary during key extraction queries to stay consistent even after the set of corrupted parties changes.

Our Vandermonde secret sharing seems difficult to use in an adaptive corruption context. Intuitively, in the security proof of VSS, we simulate the honest parties' communication as a function of hints and corrupted key shares. Thus, changing the set of corrupted parties would lead to inconsistent simulation.

If one prioritises security against adaptive corruptions over share robustness, one can modify our scheme as follows (to get closer again to Chapter 6): set $\mathbf{z}^{(i)} = \mathbf{p}^{(i)} + c_0 \cdot \mathbf{s}^{(i)} + \mathbf{m}_i$, and proceed with the proof as in Remark 24. We leave it as an interesting future work to simultaneously achieve security against adaptive corruptions *and* extraction share robustness for our TIBE.

7.4 Stronger TKEM Decapsulation Consistency

We now prove that the resulting TKEM from the BCHK+ transform of Chapter 6 fulfils a stronger decapsulation consistency due to extraction share robustness of our TIBE.

7.4.1 Definition

The decryption consistency experiment used in Chapter 6 is defined in Figure 2.4a. In Figure 2.4b we recall the decryption consistency notion of [DLN+21, Def. A.3], that we achieve in this chapter. Note that the new experiment leads to a stronger security guarantee than one in Figure 2.4a as the adversary wins even if one of the resulting keys is equal to $\perp$. To accommodate for this change, we require the decryption shares to pass the share verification algorithm.

Another change is that we give out all secret key shares to the adversary, whereas [LP25, Def. 7] only gives out corrupted key shares and unrestricted access to TKEM.ShareDecaps. In general the latter is weaker than the first. However, this limitation was not needed for the proof of Chapter 6, there we could also give all secret key shares to the adversary.

To give some more intuition on why unrestricted access to TKEM.ShareDecaps is not as powerful as having the honest secret key shares, we can look at our concrete lattice constructions from Section 7.1: the adversary can query TKEM.ShareDecaps on the same ciphertext many times, providing them with many short preimages of the same matrix $\begin{bmatrix} 1 & a & b & H_{\mathsf{id}}(\mathsf{id} = \mathsf{vk}) \end{bmatrix}$ for a fixed target t. This can be seen as a trapdoor for

this identity-depending matrix. However, this is not enough to recover honest secret key shares, which are small preimages of the matrix $\begin{bmatrix}1 & a & b\end{bmatrix}$.

7.4.2 Proof

Lemma 109. *Let $G_{\mathsf{fo}}\colon \{0,1\}^* \to \{0,1\}^\lambda$ be modelled as a random oracle and let $Q_{G_{\mathsf{fo}}} \geq 2$ denote the number of queries to it. Let* TKEM *be the scheme obtained as detailed in Figure 6.2. If the underlying* TIBE *is γ-spread and extraction share robust, the resulting* TKEM *fulfils decapsulation consistency. More concretely, for any PPT adversary $\mathcal{A}$ against $\mathsf{Exp}^{\mathsf{strong\text{-}CD}}_{\mathcal{A},\mathsf{TKEM}}$ there exists a PPT adversary $\mathcal{B}$ against $\mathsf{Exp}^{\mathsf{Robust}}_{\mathcal{B},\mathsf{TIBE}}$ such that*

$$\mathsf{Adv}^{\mathsf{strong\text{-}CD}}_{\mathcal{A},\mathsf{TKEM}} \leq \frac{Q_{F_{\mathsf{fo}}} \cdot (Q_{F_{\mathsf{fo}}} - 1)}{2}\, 2^{-\gamma} + \mathsf{Adv}^{\mathsf{Robust}}_{\mathcal{B},\mathsf{TIBE}}.$$

Proof. Let $\mathcal{B}$ be the adversary we build against $\mathsf{Exp}^{\mathsf{Robust}}_{\mathcal{B},\mathsf{TIBE}}$ and $\mathcal{A}$ an adversary in $\mathsf{Exp}^{\mathsf{strong\text{-}CD}}_{\mathcal{A},\mathsf{TKEM}}$. Upon receiving the public evaluation key ek and all secret key shares $(\mathsf{dk}_i)_{i\in[N]}$, the reduction $\mathcal{B}$ forwards it to $\mathcal{A}$. Now, assume $\mathcal{A}$ wins in $\mathsf{Exp}^{\mathsf{strong\text{-}CD}}_{\mathcal{A},\mathsf{TKEM}}$. That is, they output $(\mathsf{ct}, \mathsf{act}, \mathsf{act}', S, S')$ such that all decapsulation shares verify for both (act, S) and (act', S'), and $K \neq K'$, where $K = \mathsf{TKEM.Combine}(\mathsf{ct}, \mathsf{act}, S)$ and $K' = \mathsf{TKEM.Combine}(\mathsf{ct}, \mathsf{act}', S')$. We distinguish two cases. In the first case neither K nor K' equals $\perp$. This case is equivalent to the one considered in Chapter 6. In the second case, exactly one of the keys equals $\perp$.

Case 1. As neither K nor K' equals $\perp$, all assert checks in $\mathsf{TKEM.Combine}$ from the BCHK+ transform detailed in Figure 6.2 have to pass. In particular, when running $\mathsf{TKEM.Combine}$, the reduction $\mathcal{B}$ recovers $\mathsf{msg}, \mathsf{msg}'$ where $K = H_{\mathsf{fo}}(\mathsf{msg}\|\mathsf{ct})$ and $K' = H_{\mathsf{fo}}(\mathsf{msg}'\|\mathsf{ct})$ but $\mathsf{msg} \neq \mathsf{msg}'$. At the same time, both messages lead to ct, i.e., $\mathsf{TKEM.Encaps}(\mathsf{ek}, \mathsf{vk}, \mathsf{msg}; G_{\mathsf{fo}}(\mathsf{msg})) = \mathsf{TKEM.Encaps}(\mathsf{ek}, \mathsf{vk}, \mathsf{msg}'; G_{\mathsf{fo}}(\mathsf{msg}')) = \mathsf{ct}$. As done in Chapter 6, we distinguish two subcases. Either both msg and msg' have already been queried to G_{fo} or one of them has not been queried to G_{fo}. Using the γ-spreadness of TIBE, we upper bound the probability in both subcases by $\frac{Q_{F_{\mathsf{fo}}} \cdot (Q_{F_{\mathsf{fo}}} - 1)}{2}\, 2^{-\gamma}$. Note that the argument is purely probabilistic and does not depend on $\mathcal{A}$ having access to only corrupted (as in Chapter 6) or all secret key shares.

Case 2. W.l.o.g. we assume $K' = \perp$ and $K \neq \perp$. Since, $K \neq \perp$ we know as above that the assert checks in $\mathsf{TKEM.Combine}$ pass and hence $K = H_{\mathsf{fo}}(\mathsf{msg}\|\mathsf{ct})$ and $\mathsf{rand} = G_{\mathsf{fo}}(\mathsf{msg})$. The reduction $\mathcal{B}$ then forwards the tuple $(\mathsf{msg}, \mathsf{vk}, \mathsf{rand}, \mathsf{ct}, \mathsf{act}', S')$ as its output for the robustness game of the TIBE defined in Figure 2.2e. We now argue that this output is a valid forgery in the robustness game of the TIBE. First, note that

$\mathsf{rand} = G_{\mathsf{fo}}(\mathsf{msg})$, hence rand follows the randomness distribution RDistr and belongs to RSpace with overwhelming probability. Second, by the winning condition of the decapsulation consistency game, the checks of $\mathsf{TKEM.ShareVerify}$, the signature and the re-encryption have to succeed. Lastly, we have $\mathsf{TIBE.Combine}(\mathsf{ct}, \mathsf{vk}, \mathsf{act}', S') \neq \mathsf{msg}$ since otherwise the keys K' and K would be the same. Consequently, in this case the difference in winning probabilities of $\mathcal{B}$ and $\mathcal{A}$ is at most negligible. □

7.5 Instantiation

Parameter selection for our scheme with semi-honest correctness (resp. malicious correctness) is performed by ensuring that both Theorem 24 and Lemma 106 (resp. Lemma 107) provide negligible bounds for the adversary's advantage and the decryption failure probability.

Recall from Theorem 24 that the adversary's advantage is upper-bounded as:

$$\mathsf{Adv}^{\mathsf{sel\text{-}ID\text{-}IND}}_{\mathcal{A},\mathsf{TIBE}}(\kappa) \leq \overbrace{\mathsf{Adv}^{\mathsf{Hint\text{-}RLWE}^{Q_{\mathsf{Dec}}+Q_v,\sqrt{\sigma_{\mathbf{p}}^2-{\sigma'_{\mathbf{p}}}^2},\sigma_{\mathbf{s}},\beta}_{\mathcal{R},q,m,\sigma_{\mathbf{s}}}}_{\mathcal{A}_2}(\kappa)}^{\mathsf{ek\ is\ pseudorandom}} + \overbrace{\mathsf{Adv}^{\mathsf{RLWE}_{\mathcal{R},4,q,\sigma_{\mathbf{r}}}}_{\mathcal{A}_3}(\kappa)}^{\mathsf{ct\ is\ pseudorandom}} \tag{7.26}$$

$$(Q_{\mathsf{H_{id}}} + Q_{\mathsf{Dec}}) \cdot \underbrace{\mathsf{Adv}^{\mathsf{NTRU}_{\mathcal{R},q}}_{\mathcal{A}_1}(\kappa)}_{\text{Embed trapdoor}} + \mathsf{negl}(\lambda). \tag{7.27}$$

We study each term in one paragraph, then we study correctness and conclude.

RLWE. The pseudorandomness of the ciphertext ct is quantified by the hardness of $\mathsf{RLWE}_{\mathcal{R},4,q,\sigma_{\mathbf{r}}}$. We estimate it using standard tools such as the lattice estimator of [APS15].

Hint-RLWE. Following the reduction of Lemma 48, $\mathsf{Hint\text{-}RLWE}^{Q_{\mathsf{Dec}},Q_v,\sqrt{\sigma_{\mathbf{p}}^2-{\sigma'_{\mathbf{p}}}^2},\sigma_{\mathbf{s}},\beta}_{\mathcal{R},q,m,\sigma_{\mathbf{s}}}$ is no easier than $\mathsf{RLWE}_{\mathcal{R},q,1,\sigma_0}$, where $\frac{1}{\sigma_0^2} \leq 2\left(\frac{1+Q_v(T,N)}{\sigma_{\mathbf{s}}^2} + \frac{B_{\mathsf{hint}}(1+o(1))}{\sigma_{\mathbf{p}}^2}\right)$, $B_{\mathsf{hint}} = \frac{Q_{\mathsf{Dec}}\, d\, M^2}{12}$, $M = 2\left\lceil\frac{q-1}{2\beta}\right\rceil + 1$, and Q_v as in [BBD+25, Theorem 1]. As typical in other works, we ignore the factor 2 in $1/\sigma_0^2$ which seems to be a proof artefact. We also set $\sigma_{\mathbf{s}} = \Omega(\sqrt{Q_v(T,N)}) = \omega(\sqrt{B_{\mathsf{hint}}} \cdot \sigma_{\mathbf{p}})$, so that it has a negligible influence on σ_0. All in all, we have

$$\sigma_0 \approx \sigma_{\mathbf{p}}/\sqrt{B_{\mathsf{hint}}} \approx \sigma_{\mathbf{p}}\sqrt{\frac{12}{Q_{\mathsf{Dec}}\, d\, M^2}} \approx \sigma_{\mathbf{p}}\frac{\beta}{q}\sqrt{\frac{12}{Q_{\mathsf{Dec}}\, n}}. \tag{7.28}$$

Equation (7.28) incentivises us to take $\sigma_{\mathbf{p}}$ large. This condition directly subsumes the condition $\sigma_{\mathbf{p}} > 1.17q^{1/2} \cdot \eta_\varepsilon(\mathcal{R}^2)$. of Theorem 24. Again, we estimate the hardness of $\mathsf{Hint\text{-}RLWE}_{\mathcal{R},q,\sigma_0}$ using standard tools of [APS15].

NTRU. The multiple instances of NTRU allow us to embed a trapdoor in $H_{\mathsf{id}}(\mathsf{id})$ (one for each identity queried). The hardness of NTRU is well-studied, and tools for estimating it are included in the lattice estimator [APS15].

Note that our choices of q are very large. This raises the question of whether overstretched NTRU attacks [ABD16, KF17, DvW21] apply. Our experiments with [APS15] show that it is not the case. This is not surprising since we can take a large standard deviation σ for the secrets f, g, namely $\sigma \geq \sqrt{q/4\pi\, d}$ as in Falcon [PFH+22]. The state-of-the-art attacks [HSS25] states that on overstretched NTRU attacks become relevant when:

$$0.0058 \cdot 2\pi \cdot \sigma^2 \cdot d^{2.484}/q < 1. \tag{7.29}$$

In our case, the left side of Equation (7.29) is no smaller than $0.0029 \cdot d^{1.484}$, which gives us a comfortable margin.

Correctness. In the semi-honest (resp. malicious) setting, we apply Lemma 106 (resp. Lemma 107) and therefore set our parameters so that their requirements apply. In practice, $B_{\mathsf{ind}} \approx \tau\sqrt{6\pi\, d}\,\sigma_{\mathbf{p}}$. Note that Lemmas 106 and 107 guarantee decryption correctness with probability at least $1 - \delta'$. In our case it suffices to set δ' to be small (say, 2^{-30}), not negligible. Indeed, the BCHK$^+$ transform guarantees CCA security even in the presence of decryption failures.

Bit dropping. We apply bit dropping on four ring elements: the encryption key b, the first ciphertext v, the two last elements of the second ciphertext (u_1, u_2). This has the benefit of decreasing the public key size and ciphertext size. We denote $\delta_b, \delta_v, \delta_{u_1}, \delta_{u_2}$ the (deterministic) errors incurred by bit-dropping. This error addition has no negative impact on the security analysis, however it impacts the correctness since the decryption equation becomes:

$$v - \mathbf{u}\cdot\mathbf{z}' = \mathsf{Encode}(\mathsf{msg}) + (e' + \delta_v) + r\cdot(z_0 + c_0\cdot\delta_b) \tag{7.30}$$

$$- (e_0\cdot z_1 + (e_1 + \delta_{u_1})\cdot c_0 + (e_2 + \delta_{u_2})\cdot p_3) \tag{7.31}$$

Hence, it has a small impact on correctness when $\|\delta_v + c_0\,(r\cdot\delta_b - \delta_{u_1}) - \delta_{u_2}\cdot p_3\|_\infty = o(q)$ w.o.p. We denote $\nu_b, \nu_{u_1}, \nu_{u_2}, \nu_v$ the number of bits dropped in b, u_1, u_2, v respectively.

Coefficient dropping. Since we encapsulate a key of λ bits, we only need to send λ coefficients of v. We can therefore drop the remaining coefficients of v, which further decreases the ciphertext size. With overwhelming probability, the noise incurred by rounding is no larger than $\sqrt{4\pi\,(\lambda + \log_2\lambda + 1)\ln 2}\cdot\sigma_{\mathsf{bitdrop}}$.

How to select parameters. Selecting parameters is a multi-dimensional optimisation process, and it can seem obscure. We explain here our selection process. We recall Hardy's asymptotic notation: $a \asymp b$ is shorthand for $a = \Theta(b)$.

We select $\sigma_{\mathbf{r}} \asymp \sigma_0 \asymp \sqrt{2\pi}$ (recall that σ_0 is found in the RLWE instance reduced from $\mathsf{Hint\text{-}RLWE}$ reduction). We find that resulting RLWE instances have λ bits of security against the common attacks as long as $d \asymp \lambda \cdot \ln q$. This seems somewhat optimal, as setting $\sigma_{\mathbf{r}} = o(1)$ opens the door to hybrid attacks, while setting $\sigma_{\mathbf{r}} = \omega(1)$ degrades correctness (and similarly for σ_0).

From Equation (7.28), this implies $\sigma_{\mathbf{p}} \asymp \frac{q}{\beta}\sqrt{Q_{\mathsf{Dec}}\, d \cdot 2\pi/3}$ and $\sigma_{\mathbf{s}} = \omega(1)$. We also set $\sigma_{\mathbf{s}} = o\left(\frac{\beta}{d\,q\sqrt{\ln N}}\sigma_{\mathbf{p}}\right)$, so that Equation (7.12) gives $B_{\mathsf{ind}} \asymp \sqrt{3d/2\pi} \cdot \sigma_{\mathbf{p}} \asymp \frac{q\,d}{\beta}\sqrt{Q_{\mathsf{Dec}}}$.

Next, we set β. Recall that $B_1 \asymp \beta\sqrt{d}$. Setting β as high as possible is good for the $\mathsf{Hint\text{-}RLWE}$ reduction (see Equation (7.28)) but may increase $\|\mathbf{z}\|$, see Equations (7.19) and (7.22). We set $B_1 \asymp \sqrt{T}\, B_{\mathsf{ind}}$ for the honest (that is, no robustness) setting, and $B_1 \asymp T\, B_{\mathsf{ind}}$ for the robust extraction shares setting, so that B_1 only increases $\|\mathbf{z}\|$ by a constant factor. This gives:

$$\textit{Non-Robust setting:} \quad \beta^2 \asymp q\sqrt{d\,T\,Q_{\mathsf{Dec}}} \tag{7.32}$$

$$\textit{Robust setting:} \quad \beta^2 \asymp q\,T\sqrt{d\,Q_{\mathsf{Dec}}} \tag{7.33}$$

Finally, we study correctness, and try to minimise the modulus q. In the non-robust setting, Lemma 106 guarantees correctness if $q \asymp \sqrt{T}\, B_{\mathsf{ind}}\sqrt{\lambda/2\pi}\,\sigma_{\mathbf{r}} \asymp \frac{q\,d}{\beta}\sqrt{Q_{\mathsf{Dec}}\,T\,\lambda}$, which implies $\beta \asymp d\sqrt{Q_{\mathsf{Dec}}\,T\,\lambda}$. In the robust setting, Lemma 107 guarantee correctness if $q \asymp T\, B_{\mathsf{ind}}\sqrt{d/2\pi}\,\sigma_{\mathbf{r}} \asymp \frac{q\,T\,d}{\beta}\sqrt{Q_{\mathsf{Dec}}\,d}$, which implies $\beta \asymp d\,T\sqrt{Q_{\mathsf{Dec}}\,d}$. Putting everything together:

$$\textit{Non-Robust setting:} \quad \beta^2 \asymp q\sqrt{d\,T\,Q_{\mathsf{Dec}}} \asymp d^2\,Q_{\mathsf{Dec}}\,T\,\lambda \quad \Rightarrow \quad q \asymp d^{3/2}\,\lambda\sqrt{Q_{\mathsf{Dec}}\,T} \tag{7.34}$$

$$\textit{Robust setting:} \quad \beta^2 \asymp q\,T\sqrt{d\,Q_{\mathsf{Dec}}} \asymp d^3\,T^2\,Q_{\mathsf{Dec}} \quad \Rightarrow \quad q \asymp d^{5/2}\,T\sqrt{Q_{\mathsf{Dec}}} \tag{7.35}$$

This gives us an asymptotic formula to determine a minimal viable q. We increase d until the underlying RLWE instances are hard. Finally, bit-dropping has a small impact on correctness as long as $\sigma_{\mathsf{bitdrop}} = O(q/\sqrt{\lambda})$, so we may take $2^{\nu_v} \asymp \sqrt{d}\frac{q}{\beta}2^{\nu_{u_1}} \asymp \sqrt{d\,q}2^{\nu_{u_2}} \asymp \frac{d\,q}{\beta}2^{\nu_b} \asymp q/\sqrt{\lambda}$.

Parameter sets. Putting everything together, we obtain parameter sets as detailed in Table 7.1. The encryption key and ciphertext sizes (in bytes) for our TIBE are given as:

$$|\mathsf{ek}| = \left(2 \cdot \lambda + d\left(\ln q - \nu_b\right)\right)/8 \tag{7.36}$$

$$|\mathsf{ct}| = \left(d\left(3\ln q - \nu_{u_1} - \nu_{u_2}\right) + \lambda(\ln q - \nu_v)\right)/8 \tag{7.37}$$

For the TKEM construction, the ciphertext additionally contains a verification key vk and a signature sig. For ML-DSA at NIST security levels 2, and 5, the sum $|\mathsf{vk}| + |\mathsf{sig}|$ is equal to 3732, and 7219, respectively. For Falcon [PFH+22] at NIST security levels 1 and 5, the sum $|\mathsf{vk}| + |\mathsf{sig}|$ is equal to 1563, and 3073, respectively.

Table 7.1: λ is the bit-security, T (resp. Q_{Dec}) is an upper bound on the decryption threshold (resp. number of decryption queries), R = "Extraction Share Robustness", λ_{ek} (resp. λ_{ct}) quantifies the pseudorandomness of the encryption key (resp. ciphertext) in the CoreSVP metric, and $|\mathsf{ek}|, |\mathsf{ct}|$ are the sizes in bytes of the TIBE encryption key and ciphertext. Across all parameter sets, we take $T = 32$, $q \approx 2^{50}$, $d = 2048$, $\sigma_{\mathbf{s}}/\sqrt{2\pi} = 2^{15}$, $\sigma'_{\mathbf{p}}/\sqrt{2\pi} = 2^{27}$ and $\nu_{u_2} = 10$.

Goals			Scheme parameters							Security		Sizes	
λ	R	Q_{Dec}	β	$\sigma_{\mathbf{s}}/\sqrt{2\pi}$	$\sigma_{\mathbf{p}}/\sqrt{2\pi}$	$\sigma_{\mathbf{r}}/\sqrt{2\pi}$	ν_b	ν_{u_1}	ν_v	λ_{ek}	λ_{ct}	$\lvert\mathsf{ek}\rvert$	$\lvert\mathsf{ct}\rvert$
128	No	2^{46}	2^{41}	2^{15}	2^{35}	1	24	29	42	128	133	6 688	28 544
128	Yes	2^{25}	2^{37}	2^{15}	2^{29}	1	21	27	44	132	133	7 456	29 056
256	No	2^{46}	2^{40}	2^{15}	2^{36}	1/4	26	30	44	263	253	12 352	56 512
256	Yes	2^{25}	2^{36}	2^{15}	2^{29}	1/4	22	26	44	257	253	14 400	58 560

Chapter 8

Conclusion

In this work we made progress on a number of curious topics in lattice-based cryptography. In order to achieve the results in this thesis we had to consider a broad range of proof techniques in lattice-based cryptography for module lattices. In Chapter 3 we studied the relationship between the distribution of a linear transform and the input's entropy modulo ideals. In Chapter 4 we linked the surjectivity of Gaussian matrices with the prime zeta function of the ring or its the smallest prime ideal, we also linked the map's shortest kernel basis and the number of roots of unity the ring contains. We have studied discrete Gaussians in rank-deficient lattices in Chapter 5 and established useful definitions, smoothing statements and preimage sampling algorithms. Lastly, in Chapters 6 and 7 we took a more applied approach and explored efficient parameter setting and design choices for lattice-based threshold primitives.

We identified a number of topics that warrant further research. For example, our analysis of distribution's entropy modulo ideals in Chapter 3 requires extensive computations for every parameter choice. Techniques for analysing existing distributions or constructing short distributions that behave well modulo ideals remain a challenge. We believe that our bound on the smoothing parameter of the kernel lattice in Chapter 4 contains numerous proof artefacts. In particular, more work is required to design better sets of short ring elements allowing to construct short kernel vectors. A promising research direction also lies in optimising the proofs specifically to power-of-two cyclotomic rings.

The construction in Chapter 5 can be improved from a number of angles too. We can potentially remove the linear dependency on T in the signature length by replacing the signature with a zero-knowledge proof of the corresponding statement. Otherwise, we could improve the "naive" preimage sampling algorithm in Figure 5.2 to a randomised-rounding approach and improve the conditions on the Gaussian parameter we can sample

with.

For Chapters 6 and 7 it remains unknown if the BCHK+ transform can be improved to further simplify the integrity checks on the ciphertext or the underlying building blocks. On the other hand, it is clear that the TIBE construction from Chapter 7 could be optimised by switching to a two-round protocol similar to [BKL+25], or an NTRU-based trapdoor for the main matrix as in [EEN+24, Appendix A]. However, the most interesting direction for us would be finding a threshold technique that achieves adaptive corruptions and the share robustness properties at the same time. We hope that these open question inspire further research into threshold cryptography and proof techniques for module lattices.

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Appendix A

Code

A.1 Parametrising singular value inequalities

```
from sage.symbolic.relation import solve

"""
    This script simplifies the tail bounds on Chi random
       ↪ variables defined in Section 6, Lemma 6.1 of the paper.

    We first write down the expressions obtain from the
       ↪ Gershgorin circle theorem that dominate other values in
       ↪  the maximum (resp. minimum). They are stated as a
       ↪ comment. Then we manually write down the corresponding
       ↪ tail bounds defined in variables f1 to f7.

    Next we simplify the tail bounds only making them larger (
       ↪ resp. smaller). Lastly, we plot the difference between
       ↪ our values and the desired upper (resp. lower) bound 1
       ↪ + 1/x (resp. 1 - 1/x).

    We check that the inequalities work by plotting them. The
       ↪ larger we make m the closer the values are to the bound
       ↪ , hence we parametrise it with c and give the user the
       ↪ option to plot for different values of c. In the paper
       ↪ we use c = 7 and x \geq 4 since it satisfies all
       ↪ inequalities.
```

```
    This script has a range of different plots we draw via show(
       ↪ plot(...)). To change the plot displayed please comment
       ↪  out the current plot and uncomment the one required.
       ↪ In all plots the bound is satisfied whenever the line
       ↪ is above zero.
"""

var("m,k,r,c,x")
r = k   # We set r = k for simplicity, in the paper we set m = c
   ↪ ^2 * x^2 * max(r,k).
m = c^2 * x^2 * k

# We only need the inequality plots for positive parameters.
assume(c>1)
assume(x>1)

# THE UPPER TAIL

# The random variables we analyse:

# x_(m-1)^2 + y_(r-1)^2 + x_(m) * y_(r-1) + x_(m-1) * y_(r-2)

# Their norm divided by m is smaller than:

f1 = (m - 1)/m + 2 * sqrt((m-1)*k/m^2) + 2*k/m + (r - 1)/m + 2 *
   ↪  sqrt((r-1)*k/m^2) + 2*k/m + sqrt(1 + 2* sqrt(k/m) + 2*k/m)
   ↪  * sqrt((r - 1)/m + 2 * sqrt((r-1)*k/m^2) + 2*k/m) + sqrt((
   ↪ m-1)/m + 2 * sqrt((m-1)*k/m^2) + 2*k/m) * sqrt((r - 2)/m +
   ↪ 2 * sqrt((r-2)*k/m^2) + 2*k/m)

# We simplify the expression making the bound looser

f1 = f1.subs({(m-1):m})
f1 = f1.subs({(k-1):k})
f1 = f1.subs({(k-2):k})
f1 = f1.canonicalize_radical()
```

```
# Next random variables we analyse:

# x_(m)^2 / m + x_(m) * y_(r-1) / m <=

f2 = 1 + 2 * sqrt(k/m) + 2*k/m + sqrt(1 + 2* sqrt(k/m) + 2*k/m)
   ↪ * sqrt((r - 1)/m + 2 * sqrt((r-1)*k/m^2) + 2*k/m)

# We simplify the expression

f2 = f2.subs({(k-1):k})
f2 = f2.canonicalize_radical()

# The plot of f1 and f2 compared with 1 + 1/x.

show(plot(((1 + 1/x) - f2).subs(c=7), 1, 10) + plot(((1 + 1/x) -
   ↪  f1).subs(c=7), 1, 10, color = "green"))

# THE LOWER TAIL

"""
    Here O \cdot Z \cdot Z^\transpose \cdot O^\transpose has
       ↪ different form depending on r so we consider cases r >=
       ↪  4, r = 3 and r = 2. Otherwise the lower tail is
       ↪ analysed in the same way.
"""

# r >= 4

# Random variable we analyse:

# x_(m)^2 /m - x_(m) * y_(r-1) /m >=

f3 = 1 - 2 * sqrt(k/m) - sqrt(1 + 2 * sqrt(k/m) + 2 * k/m) *
   ↪ sqrt((r - 1)/m + 2 * sqrt((r-1)*k/m^2) + 2 * k/m)

# We simplify the expression
```

```
f3 = f3.subs({(k-1):k})
f3 = f3.canonicalize_radical()

# Random variable we analyse:

# x_(m - r + 2)^2 / m + y_(2)^2 / m - x_(m - r + 2)  * y_(1) / m
   ↪  - x_(m - r + 3) * y_(2) / m >=
# we expect this one to be the smallest

f4 = 1 - (r - 4)/m - 2 * sqrt((m - r + 4)*k/m^2) - sqrt((m - r +
   ↪  2)/m + 2 * sqrt((m - r + 2)*k/m^2) + 2 * k/m) * sqrt(1/m +
   ↪  2 * sqrt(k/m^2) + 2 * k/m) - sqrt((m - r + 3)/m + 2 * sqrt
   ↪ ((m - r + 3)*k/m^2) + 2 * k/m) * sqrt(2/m + 2 * sqrt(2*k/m
   ↪ ^2) + 2 * k/m)

# We simplify the expression

f4 = f4.subs({(k-4):k})
f4 = f4.subs({(m - r + 4):m})
f4 = f4.subs({(m - r + 2):m})
f4 = f4.subs({(m - r + 3):m})
f4 = f4.subs(k = 4)
f4 = f4.canonicalize_radical()

# Random variable we analyse:

# x_(m - r + 1)^2 / m + y_(1)^2 / m - x_(m - r + 2) * y_(1) / m
   ↪ >=

f5 = 1 - (r - 2)/m - 2 * sqrt((m - r + 2)*k/m^2) - sqrt((m - r+
   ↪ 2)/m + 2 * sqrt((m - r + 2)*k/m^2) + 2 * k/m) * sqrt(1/m +
   ↪ 2 * sqrt(k/m^2) + 2 * k/m)

# We simplify the expression

f5 = f5.subs({(k-2):k})
```

```
f5 = f5.subs({(m - r + 2):m})
f5 = f5.subs(k = 4)
f5 = f5.canonicalize_radical()

# c = 6 works for x > 4

# Uncomment to show the plot one at a time.

#show(plot((f3 - (1 - 1/x)).subs(c=6), 1, 10) + plot((f4 - (1 -
   ↪ 1/x)).subs(c = 6), 1, 10, color = "green") + plot((f5 - (1
   ↪ - 1/x)).subs(c=6), 1, 10, color = "red"))

# special case for r = 3 (since we have m - r + 4) f6 = f4 but
   ↪ the bounds after are different

f6 = (m - r + 4)/m - 2 * sqrt((m - r + 4)*k/m^2) - sqrt((m - r+
   ↪ 2)/m + 2 * sqrt((m - r + 2)*k/m^2) + 2 * k/m) * sqrt(1/m +
   ↪ 2 * sqrt(k/m^2) + 2 * k/m) - sqrt((m - r + 3)/m + 2 * sqrt
   ↪ ((m-r+3)*k/m^2) + 2 * k/m) * sqrt(2/m + 2 * sqrt(2*k/m^2) +
   ↪  2 * k/m)

# We simplify the expression

f6 = f6.subs({(m - r + 4):(m+1)})
f6 = f6.subs({(m - r + 2):m})
f6 = f6.subs({(m - r + 3):m})
f6 = f6.subs(k = 3)
f6 = f6.canonicalize_radical()

# since k = 3 now the combined effect makes c = 6 require x > 8
   ↪ which is a bit worse than for r > 4

# Uncomment to show the plot one at a time.

#show(plot((f3 - (1 - 1/x)).subs(c=7), 1, 10) + plot((f6 - (1 -
   ↪ 1/x)).subs(c=7), 1, 10, color = "green") + plot((f5 - (1 -
   ↪ 1/x)).subs(c=7), 1, 10, color = "red"))
```

```
# for r = 2 the matrix looks simpler and we only get x_(m)^2 / m
   ↪  - x_(m) * y_(r-1)) / m >=

f7 = 1 - 2 * sqrt(k/m) - sqrt( 1 + 2 * sqrt(k/m) + 2 * k/m) *
   ↪ sqrt((r - 1)/m + 2 * sqrt((r-1)*k/m^2) + 2 * k/m)

# We simplify the expression

f7 = f7.subs({(k-1):k})
f7 = f7.canonicalize_radical()

# Uncomment to show the plot one at a time.

#show(plot((f7 - (1 - 1/x)).subs(c=5), 1, 10))
```